Was this page helpful? Yes No

convert basic

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Converts an empty dynamic disk to a basic disk. A dynamic disk must be selected for this operation to succeed. Use the select disk command to select a dynamic disk and shift the focus to it.

> **Important**
>
> The disk must be empty to convert it to a basic disk. Back up your data, and then delete all partitions or volumes before converting the disk.

> **Note**
>
> For instructions regarding how to use this command, see Change a Dynamic Disk Back to a Basic Disk).

Syntax

```
convert basic [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To convert the selected dynamic disk to basic, type:

```
convert basic
```

Related links

- Command-Line Syntax Key
- convert command

Feedback

Was this page helpful? Yes No

convert dynamic

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Converts a basic disk into a dynamic disk. A basic disk must be selected for this operation to succeed. Use the select disk command to select a basic disk and shift the focus to it.

> **Note**
>
> For instructions regarding how to use this command, see Change a Dynamic Disk Back to a Basic Disk).

Syntax

```
convert dynamic [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Remarks

- Any existing partitions on the basic disk become simple volumes.

Examples

To convert a basic disk into a dynamic disk, type:

```
convert dynamic
```

Related links

- Command-Line Syntax Key
- convert command

Feedback

Was this page helpful? Yes No

convert gpt

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Converts an empty basic disk with the master boot record (MBR) partition style into a basic disk with the GUID partition table (GPT) partition style. A basic MBR disk must be selected for this operation to succeed. Use the select disk command to select a basic disk and shift the focus to it.

> **ⓘ Important**
>
> The disk must be empty to convert it to a basic disk. Back up your data, and then delete all partitions or volumes before converting the disk. The required minimum disk size for conversion to GPT is 128 megabytes.

> **ⓘ Note**
>
> For instructions regarding how to use this command, see Change a Master Boot Record Disk into a GUID Partition Table Disk.

Syntax

```
convert gpt [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To convert a basic disc from MBR partition style to GPT partition style, type:

```
convert gpt
```

Related links

- Command-Line Syntax Key
- convert command

Feedback

Was this page helpful? Yes No

convert mbr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Converts an empty basic disk with the GUID Partition Table (GPT) partition style into a basic disk with the master boot record (MBR) partition style. A basic disk must be selected for this operation to succeed. Use the select disk command to select a basic disk and shift the focus to it.

> **Important**
>
> The disk must be empty to convert it to a basic disk. Back up your data, and then delete all partitions or volumes before converting the disk.

> **Note**
>
> For instructions regarding how to use this command, see Change a GUID Partition Table Disk into a Master Boot Record Disk.

Syntax

```
convert mbr [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To convert a basic disc from GPT partition style to MBR partition style, type>:

```
convert mbr
```

Related links

- Command-Line Syntax Key
- convert command

Feedback

Was this page helpful? Yes No

create

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Creates a partition or shadow on a disk, a volume on one or more disks, or a virtual hard disk (VHD). If you're using this command to create a volume on the shadow disk, you must already have at least one volume in the shadow copy set.

Syntax

```
create partition
create volume
```

Parameters

Expand table

Parameter	Description
create partition primary command	Creates a primary partition on the basic disk with focus.
create partition efi command	Creates an Extensible Firmware Interface (EFI) system partition on a GUID Partition Table (gpt) disk on Itanium-based computers.
create partition extended command	Creates an extended partition on the disk with focus.
create partition logical command	Creates a logical partition in an existing extended partition.
create partition msr command	Creates a Microsoft Reserved (MSR) partition on a GUID partition table (gpt) disk.
create volume simple command	Creates a simple volume on the specified dynamic disk.
create volume mirror command	Creates a volume mirror by using the two specified dynamic disks.
create volume raid	Creates a RAID-5 volume using three or more specified dynamic disks.

Parameter	Description
command	
create volume stripe command	Creates a striped volume using two or more specified dynamic disks.

Examples

You run these commands in the DiskPart interpreter. To start it, type `diskpart` at an elevated command prompt. Each example creates a 1000-megabyte (MB) object; omit the `size` parameter to use the remaining available space. For the full parameter list and behavior of each command, see the linked child article.

Before you create a partition, run `list disk` to identify the intended basic disk, then use the select disk command to give it focus. In the following example, replace `<disk-number>` with the number of the basic disk you want. To create a 1000-MB primary partition on the disk with focus:

```
select disk <disk-number>
create partition primary size=1000
```

Volume commands create volumes on dynamic disks that you specify by number. To create a 1000-MB simple volume on dynamic disk 1:

```
create volume simple size=1000 disk=1
```

To create a 1000-MB mirrored volume across dynamic disks 1 and 2:

```
create volume mirror size=1000 disk=1,2
```

To create a 1000-MB striped volume across dynamic disks 1 and 2:

```
create volume stripe size=1000 disk=1,2
```

To create a 1000-MB RAID-5 volume across dynamic disks 1, 2, and 3:

```
create volume raid size=1000 disk=1,2,3
```

Related links

- Command-Line Syntax Key

Last updated on 09/08/2026

delete

Article • 10/16/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a partition or a volume. It also deletes a dynamic disk from the list of disks.

Syntax

```
delete disk
delete partition
delete volume
```

Parameters

Expand table

Parameter	Description
Delete disk	Deletes a missing dynamic disk from the list of disks.
Delete partition	Deletes a partition.
Delete volume	Deletes a volume.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

delete disk

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a missing dynamic disk from the list of disks.

> **Note**
>
> For detailed instructions about how to use this command, see Remove a Missing Dynamic Disk.

Syntax

```
delete disk [noerr] [override]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.
override	Enables DiskPart to delete all simple volumes on the disk. If the disk contains half of a mirrored volume, the half of the mirror on the disk is deleted. The delete disk override command fails if the disk is a member of a RAID-5 volume.

Examples

To delete a missing dynamic disk from the list of disks, type:

```
delete disk
```

Related links

- Command-Line Syntax Key
- delete command

Feedback

Was this page helpful? Yes No

delete partition

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes the partition with focus. Before you begin, you must select a partition for this operation to succeed. Use the select partition command to select a partition and shift the focus to it.

> ⚠ **Warning**
>
> Deleting a partition on a dynamic disk can delete all dynamic volumes on the disk, destroying any data and leaving the disk in a corrupt state.
>
> You can't delete the system partition, boot partition, or any partition that contains the active paging file or crash dump information.

Syntax

```
delete partition [noerr] [override]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.
override	Enables DiskPart to delete any partition regardless of type. Typically, DiskPart only permits you to delete known data partitions.

Remarks

- To delete a dynamic volume, always use the delete volume command instead.

- Partitions can be deleted from dynamic disks, but they shouldn't be created. For example, it's possible to delete an unrecognized GUID Partition Table (GPT) partition on a dynamic GPT disk. Deleting such a partition doesn't cause the resulting free space to become available. Instead, This command is intended to allow you to reclaim space on a corrupted offline dynamic disk in an emergency situation where the clean command in DiskPart can't be used.

Examples

To delete the partition with focus, type:

```
delete partition
```

Related links

- Command-Line Syntax Key
- select partition
- delete command
- delete volume command
- clean command

Feedback

Was this page helpful? Yes No

delete volume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes the selected volume. Before you begin, you must select a volume for this operation to succeed. Use the select volume command to select a volume and shift the focus to it.

> **ⓘ Important**
>
> You can't delete the system volume, boot volume, or any volume that contains the active paging file or crash dump (memory dump).

Syntax

```
delete volume [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To delete the volume with focus, type:

```
delete volume
```

Related links

- Command-Line Syntax Key
- select volume
- delete command

Feedback

Was this page helpful? Yes No

detach vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Stops the selected virtual hard disk (VHD) from appearing as a local hard disk drive on the host computer. When a VHD is detached, you can copy it to other locations. Before you begin, you must select a VHD for this operation to succeed. Use the select vdisk command to select a VHD and shift the focus to it.

Syntax

```
detach vdisk [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To detach the selected VHD, type:

```
detach vdisk
```

Related links

- Command-Line Syntax Key

- attach vdisk command
- compact vdisk command
- detail vdisk command
- expand vdisk command
- Merge vdisk command
- select vdisk command
- list command

Feedback

Was this page helpful? Yes No

detail

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about the selected disk, partition, volume, or virtual hard disk (VHD).

Syntax

```
detail disk
detail partition
detail volume
detail vdisk
```

Parameters

Expand table

Parameter	Description
Detail disk	Displays the properties of the selected disk and the volumes on that disk.
Detail partition	Displays the properties of the selected partition.
Detail volume	Displays the disks on which the current volume resides.
Detail vdisk	Displays the properties of the selected VHD.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

detail disk

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the properties of the selected disk and the volumes on that disk. Before you begin, you must select a disk for this operation to succeed. Use the select disk command to select a disk and shift the focus to it. If you select a virtual hard disk (VHD), this command will show the disk's bus type as *Virtual*.

Syntax

```
detail disk
```

Examples

To see the properties of the selected disk, and information about the volumes in the disk, type:

```
detail disk
```

Related links

- Command-Line Syntax Key
- detail command

Feedback

Was this page helpful? Yes No

detail partition

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the properties of the selected partition. Before you begin, you must select a partition for this operation to succeed. Use the select partition command to select a partition and shift the focus to it.

Syntax

```
detail partition
```

Examples

To see the properties of the selected partition, type:

```
detail partition
```

Related links

- Command-Line Syntax Key
- detail command

Feedback

Was this page helpful? Yes No

detail vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the properties of the selected virtual hard disk (VHD). Before you begin, you must select a VHD for this operation to succeed. Use the select vdisk command to select a VHD and shift the focus to it.

Syntax

```
detail vdisk
```

Examples

To see details about the selected VHD, type:

```
detail vdisk
```

Related links

- Command-Line Syntax Key
- detail command
- attach vdisk command
- compact vdisk command
- detach vdisk command
- expand vdisk command
- merge vdisk command
- select vdisk

- list command

Feedback

Was this page helpful? Yes No

detail volume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the disks on which the current volume resides. Before you begin, you must select a volume for this operation to succeed. Use the select volume command to select a volume and shift the focus to it. The volume details aren't applicable to read-only volumes, such as a DVD-ROM or CD-ROM drive.

Syntax

```
detail volume
```

Examples

To see all the disks in which the current volume resides, type:

```
detail volume
```

Related links

- Command-Line Syntax Key
- select volume
- detail command

Feedback

Was this page helpful? Yes No

exit

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Exits the command interpreter or the current batch script.

Syntax

```
exit [/b] [<exitcode>]
```

Parameters

Expand table

Parameter	Description
/b	Exits the current batch script instead of exiting Cmd.exe. If executed from outside a batch script, exits Cmd.exe.
`<exitcode>`	Specifies a numeric number. If **/b** is specified, the ERRORLEVEL environment variable is set to that number. If you are quitting the command interpreter, the process exit code is set to that number.
/?	Displays help at the command prompt.

Examples

To close the command interpreter, type:

```
exit
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

expand vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Expands a virtual hard disk (VHD) to a specified size.

A VHD must be selected and detached for this operation to succeed. Use the select vdisk command to select a volume and shift the focus to it.

Syntax

```
expand vdisk maximum=<n>
```

Parameters

Expand table

Parameter	Description
maximum=`<n>`	Specifies the new size for the VHD in megabytes (MB).

Examples

To expand the selected VHD to 20 GB, type:

```
expand vdisk maximum=20000
```

Related links

- Command-Line Syntax Key
- select vdisk command
- attach vdisk command

- compact vdisk command
- detach vdisk command
- detail vdisk command
- merge vdisk command
- list command

Feedback

Was this page helpful? Yes No

extend

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Extends the volume or partition with focus and its file system into free (unallocated) space on a disk.

Syntax

```
extend [size=<n>] [disk=<n>] [noerr]
extend filesystem [noerr]
```

Parameters

Expand table

Parameter	Description
size=`<n>`	Specifies the amount of space in megabytes (MB) to add to the current volume or partition. If no size is given, all of the contiguous free space that is available on the disk is used.
disk=`<n>`	Specifies the disk on which the volume or partition is extended. If no disk is specified, the volume or partition is extended on the current disk.
filesystem	Extends the file system of the volume with focus. For use only on disks where the file system was not extended with the volume.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Remarks

- On basic disks, the free space must be on the same disk as the volume or partition with focus. It must also immediately follow the volume or partition with focus (that is, it must start at the next sector offset).

- On dynamic disks with simple or spanned volumes, a volume can be extended to any free space on any dynamic disk. Using this command, you can convert a simple dynamic volume into a spanned dynamic volume. Mirrored, RAID-5 and striped volumes cannot be extended.
- If the partition was previously formatted with the NTFS file system, the file system is automatically extended to fill the larger partition and no data loss will occur.
- If the partition was previously formatted with a file system other than NTFS, the command fails with no change to the partition.
- If the partition was not previously formatted with a file system, the partition will still be extended.
- The partition must have an associated volume before it can be extended.

Examples

To extend the volume or partition with focus by 500 megabytes, on disk 3, type:

```
extend size=500 disk=3
```

To extend the file system of a volume after it was extended, type:

```
extend filesystem
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

filesystems

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about the current file system of the volume with focus and lists the file systems that are supported for formatting the volume.

A volume must be selected for this operation to succeed. Use the select volume command to select a volume and shift the focus to it.

Syntax

```
filesystems
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Format

Article • 09/28/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The **format** command formats a drive to accept Windows files. You must be a member of the Administrators group to format a hard drive.

> **Note**
>
> You can also use the **format** command, with different parameters, from the Recovery Console. For more information about the Recovery Console, see Windows Recovery Environment (Windows RE).

Syntax

```
format volume [/FS:file-system] [/V:label] [/Q] [/L[:state]] [/A:size] [/C]
[/I:state] [/X] [/P:passes] [/S:state]
format volume [/V:label] [/Q] [/F:size] [/P:passes]
format volume [/V:label] [/Q] [/T:tracks /N:sectors] [/P:passes]
format volume [/V:label] [/Q] [/P:passes]
format volume [/Q]
```

Parameters

Expand table

Parameter	Description
`<volume>`	Specifies the mount point, volume name, or drive letter (followed by a colon) of the drive that you want to format. If you don't specify any of the following command-line options, **format** uses the volume type to determine the default format for the disk.
/FS:`<filesystem>`	Specifies the type of file system (FAT, FAT32, NTFS, exFAT, ReFS, or UDF).
/V:`<label>`	Specifies the volume label. If you omit the **/V** command-line option or use it without specifying a volume label, **format** prompts you for the volume label after the formatting is complete. Use the syntax **/V:** to prevent the

Parameter	Description
	prompt for a volume label. If you use a single **format** command to format more than one disk, all of the disks are given the same volume label.
/Q	Performs a quick format. Deletes the file table and the root directory of a previously formatted volume, but doesn't perform a sector-by-sector scan for bad areas. You should use the **/Q** command-line option to format only previously formatted volumes that you know are in good condition. **/Q** overrides **/P**.
/C	**NTFS Only**: Files created on the new volume are compressed by default.
/X	Forces the volume to dismount, if necessary, before it's formatted. Any open handles to the volume are no longer be valid.
/R	**NTFS Only**: Files created on the new volume are compressed by default.
/D	UDF 2.50 only. Metadata is duplicated.
/L:`<state>`	NTFS only. Overrides the default size of file record. By default, a nontiered volume is formatted with small size file records and a tiered volume is formatted with large size file records. **/L** and **/L:enable** forces format to use large size file records and **/L:disable** forces format to use small size file records.
/A:`<size>`	Specifies the allocation unit size to use on FAT, FAT32, NTFS, exFAT, or ReFS volumes. If you don't specify *unit size*, it's chosen based on volume size. Default settings are recommended for general use. The following list presents valid values for each type of file system *unit size*: • **FAT and FAT32**: `512`, `1024`, `2048`, `4096`, `8192`, `16K`, `32K`, `64K`. Also `128K` and `256K` for a sector size greater than 512 bytes. • **NTFS**: `512`, `1024`, `2048`, `4096`, `8192`, `16K`, `32K`, `64K`, `128K`, `256K`, `512K`, `1M`, `2M` • **exFAT**: `12`, `1024`, `2048`, `4096`, `8192`, `16K`, `32K`, `64K`, `128K`, `256K`, `512K`, `1M`, `2M`, `4M`, `8M`, `16M`, `32M` • **ReFS**: `4096`, `64K`
/F:`<size>`	Specifies the size of the floppy disk to format. When possible, use this command-line option instead of the **/T** and **/T** command-line options. Windows accepts the following values for size: • `1440` or `1440k` or `1440kb` • `1.44` or `1.44m` or `1.44mb` • `1.44-MB`, `double-sided`, `quadruple-density`, `3.5-inch disk`
/T:`<tracks>`	Specifies the number of tracks on the disk. When possible, use the **/F** command-line option instead. If you use the **/T** option, you must also use the **/N** option. These options together provide an alternative method of

Parameter	Description
	specifying the size of the disk that's being formatted. This option isn't valid with the **/F** option.
/N:`<sectors>`	Specifies the number of sectors per track. When possible, use the **/F** command-line option instead of **/N**. If you use **/N**, you must also use **/T**. These two options together provide an alternative method of specifying the size of the disk that's being formatted. This option isn't valid with the **/F** option.
/P:`<count>`	Zero every sector on the volume. After that, the volume will be overwritten **count** times using a different random number each time. If **count** is zero, no other overwrites are made after zeroing every sector. This switch is ignored when **/Q** is specified.
/S:`<state>`	Specifies support for short filenames. State is either **enable** or **disable**. Short names are disabled by default.
/TXF:`<state>`	Specifies TxF is enabled/disabled. State is either **enable** or **disable**. TxF is enabled by default
/I:`<state>`	**ReFS Only**: Specifies whether integrity should be enabled on the new volume. State is either **enable** or **disable**. Integrity is enabled on storage that supports data redundancy by default.
/DAX:`<state>`	**NTFS Only**: Enable direct access storage (DAX) mode for this volume. In DAX mode, the volume is accessed via the memory bus, boosting IO performance. A volume can be formatted with DAX mode only if the hardware is DAX capable. State is either **enable** or **disable**. **/DAX** is considered the same as **/DAX:enable**.
/LogSize::`<size>`	**NTFS Only**: Specifies the size for NTFS log file in kilobytes. The minimum supported size is 2MB, so specifying a size smaller than 2MB results in a 2MB log file. Zero indicates the default value. The default value generally depends on the volume size.
/NoRepairLogs	**NTFS Only**: Disables NTFS repair logs. If the **spotfix** flag for chkdsk is specified, then the **/NoReairLogs** parameter doesn't work.
/NoTrim	Skips sending trim (delete notification) during a format.
/DevDrv	**ReFS Only**: Formats the volume as a dev drive. A dev drive or a developer volume is a volume optimized for performance of developer scenarios. Gives administrators control over what mini-filters are attached to this volume.
/SHA256Checksums	**ReFS Only**: Uses SHA-256 in all operations involving checksums.
/Y	Doesn't prompt to force the volume to dismount and assumes an empty label when no label is specified.

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- The **format** command creates a new root directory and file system for the disk. It can also check for bad areas on the disk, and it can delete all data on the disk. To be able to use a new disk, you must first use this command to format the disk.
- After formatting a floppy disk, **format** displays the following message:

  `Volume label (11 characters, ENTER for none)?`

  To add a volume label, type up to 11 characters (including spaces). If you don't want to add a volume label to the disk, press ENTER.
- When you use the **format** command to format a hard disk, a warning message similar to the following displays:

  ```
  WARNING, ALL DATA ON NON-REMOVABLE DISK
  DRIVE x: WILL BE LOST!
  Proceed with Format (Y/N)? _
  ```

  To format the hard disk, press **Y**. If you don't want to format the disk, press **N**.
- FAT file systems restrict the number of clusters to no more than 65526. FAT32 file systems restrict the number of clusters to between 65527 and 4177917.
- NTFS compression isn't supported for allocation unit sizes above 4096.

  > ⓘ Note
  >
  > **Format** will immediately stop processing if it determines that the previous requirements can't be met using the specified cluster size.
- When formatting is complete, **format** displays messages that show the total disk space, the spaces marked as defective, and the space available for your files.
- You can speed up the formatting process by using the **/q** command-line option. Use this option only if there are no bad sectors on your hard disk.

- You shouldn't use the **format** command on a drive that was prepared by using the **subst** command. You can't format disks over a network.
- The following table lists each exit code and a brief description of its meaning.

Expand table

Exit code	Description
0	The format operation was successful.
1	Incorrect parameters were supplied.
4	A fatal error occurred (which is any error other than 0, 1, or 5).
5	The user pressed N in response to the prompt "Proceed with Format (Y/N)?" to stop the process.

You can check these exit codes by using the ERRORLEVEL environment variable with the **if** batch command.

Examples

To format a new floppy disk in drive A using the default size, type:

```
format a:
```

To perform a quick format operation on a previously formatted floppy disk in drive A, type:

```
format a: /q
```

To format a floppy disk in drive A and assign it the volume label *DATA*, type:

```
format a: /v:DATA
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

gpt

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

On basic GUID partition table (gpt) disks, this command assigns the gpt attribute(s) to the partition with focus. Gpt partition attributes give additional information about the use of the partition. Some attributes are specific to the partition type GUID.

You must choose a basic gpt partition for this operation to succeed. Use the select partition command to select a basic gpt partition and shift the focus to it.

> **⊗ Caution**
>
> Changing the gpt attributes might cause your basic data volumes to fail to be assigned drive letters, or to prevent the file system from mounting. We strongly recommend that you don't change the gpt attributes unless you're an original equipment manufacturer (OEM) or an IT professional who's experienced with gpt disks.

Syntax

```
gpt attributes=<n>
```

Parameters

Expand table

Parameter	Description
attributes=`<n>`	Specifies the value for the attribute that you want to apply to the partition with focus. The gpt attribute field is a 64-bit field that contains two subfields. The higher field is interpreted only in the context of the partition ID, while the lower field is common to all partition IDs. Accepted values include: • **0x0000000000000001** - Specifies that the partition is required by the computer to function properly.

Parameter	Description
	• **0x8000000000000000** - Specifies that the partition won't receive a drive letter by default when the disk is moved to another computer, or when the disk is seen for the first time by a computer. • **0x4000000000000000** - Hides a partition's volume so it's not detected by the mount manager. • **0x2000000000000000** - Specifies that the partition is a shadow copy of another partition. • **0x1000000000000000** - Specifies that the partition is read-only. This attribute prevents the volume from being written to. For more information about these attributes, see the attributes section at create_PARTITION_PARAMETERS Structure.

Remarks

- The EFI System partition contains only those binaries necessary to start the operating system. This makes it easy for OEM binaries or binaries specific to an operating system to be placed in other partitions.

Examples

To prevent the computer from automatically assigning a drive letter to the partition with focus, while moving a gpt disk, type:

```
gpt attributes=0x8000000000000000
```

Related links

- Command-Line Syntax Key
- select partition command
- create_PARTITION_PARAMETERS Structure

Feedback

Was this page helpful? Yes No

import (diskpart)

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Imports a foreign disk group into the disk group of the local computer. This command imports every disk that is in the same group as the disk with focus.

> [IMPORTANT] Before you can use this command, you must use the select disk command to select a dynamic disk in a foreign disk group and shift the focus to it.

Syntax

```
import [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To import every disk that is in the same disk group as the disk with focus into the disk group of the local computer, type:

```
import
```

Related links

- Command-Line Syntax Key
- diskpart command

Feedback

Was this page helpful? Yes No

inactive

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Marks the system partition or boot partition with focus as inactive on basic master boot record (MBR) disks.

An active system or boot partition must be selected for this operation to succeed. Use the select partition command command to select the active partition and shift the focus to it.

> **⊗ Caution**
>
> Your computer might not start without an active partition. Don't mark a system or boot partition as inactive unless you are an experienced user with a thorough understanding of the Windows family of operating systems.
>
> If you're unable to start your computer after marking the system or boot partition as inactive, insert the Windows Setup CD in the CD-ROM drive, restart the computer, and then repair the partition using the **fixmbr** and **fixboot** commands in the Recovery Console.
>
> After you mark the system partition or boot partition as inactive, your computer starts from the next option specified in the BIOS, such as the CD-ROM drive or a Pre-Boot eXecution Environment (PXE).

Syntax

```
inactive
```

Examples

```
inactive
```

Related links

- Command-Line Syntax Key
- select partition command
- Advanced troubleshooting for Windows boot problems

Feedback

Was this page helpful? Yes No

list

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays a list of disks, of partitions in a disk, of volumes in a disk, or of virtual hard disks (VHDs).

Syntax

```
list { disk | partition | volume | vdisk }
```

Parameters

Expand table

Parameter	Description
disk	Displays a list of disks and information about them, such as their size, amount of available free space, whether the disk is a basic or dynamic disk, and whether the disk uses the master boot record (MBR) or GUID partition table (GPT) partition style.
partition	Displays the partitions listed in the partition table of the current disk.
volume	Displays a list of basic and dynamic volumes on all disks.
vdisk	Displays a list of the VHDs that are attached and/or selected. This command lists detached VHDs if they are currently selected; however, the disk type is set to Unknown until the VHD is attached. The VHD marked with an asterisk (*) has focus.

Remarks

- When listing partitions on a dynamic disk, the partitions might not correspond to the dynamic volumes on the disk. This discrepancy occurs because dynamic disks contain entries in the partition table for the system volume or boot volume (if present on the disk). They also contain a partition that occupies the remainder of the disk in order to reserve the space for use by dynamic volumes.
- The object marked with an asterisk (*) has focus.

- When listing disks, if a disk is missing, its disk number is prefixed with M. For example, the first missing disk is numbered *M0*.

Examples

```
list disk
list partition
list volume
list vdisk
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

merge vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Merges a differencing virtual hard disk (VHD) with its corresponding parent VHD. The parent VHD will be modified to include the modifications from the differencing VHD. This command modifies the parent VHD. As a result, other differencing VHDs that are dependent on the parent will no longer be valid.

> **ⓘ Important**
>
> You must choose and detach a VHD for this operation to succeed. Use the **select vdisk** command to select a VHD and shift the focus to it.

Syntax

```
merge vdisk depth=<n>
```

Parameters

Expand table

Parameter	Description
depth=`<n>`	Indicates the number of parent VHD files to merge together. For example, `depth=1` indicates that the differencing VHD will be merged with one level of the differencing chain.

Examples

To merge a differencing VHD with its parent VHD, type:

```
merge vdisk depth=1
```

Related links

- Command-Line Syntax Key
- attach vdisk command
- compact vdisk command
- detail vdisk command
- detach vdisk command
- expand vdisk command
- select vdisk command
- list command

Feedback

Was this page helpful? Yes No

offline

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Takes an online disk or volume to the offline state.

Syntax

```
offline disk
offline volume
```

Parameters

Expand table

Parameter	Description
offline disk	Takes the online disk with focus to the offline state.
offline volume	Takes the online volume with focus to the offline state.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

offline disk

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Takes the online disk with focus to the offline state. If a dynamic disk in a disk group is taken offline, the status of the disk changes to **missing** and the group shows a disk that's offline. The missing disk is moved to the invalid group. If the dynamic disk is the last disk in the group, then the status of the disk changes to **offline**, and the empty group is removed.

> **Note**
>
> A disk must be selected for the **offline disk** command to succeed. Use the select disk command to select a disk and shift the focus to it.
>
> This command also works on disks in SAN online mode by changing the SAN mode to offline.

Syntax

```
offline disk [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To take the disk with focus offline, type:

```
offline disk
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

offline volume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Takes the online volume with focus to the offline state.

> **Note**
>
> A volume must be selected for the **offline volume** command to succeed. Use the select volume command to select a disk and shift the focus to it.

Syntax

```
offline volume [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To take the disk with focus offline, type:

```
offline volume
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

online

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Takes an offline disk or volume to the online state.

Syntax

```
online disk
online volume
```

Parameters

Expand table

Parameter	Description
online disk	Takes the offline disk with focus to the online state.
online volume	Takes the offline volume with focus to the online state.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

online disk

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Takes the offline disk to the online state. For basic disks, this command attempts to bring online the selected disk and all volumes on that disk. For dynamic disks, this command attempts to bring online all disks that are not marked as foreign on the local computer. It also attempts to bring online all volumes on the set of dynamic disks.

If a dynamic disk in a disk group is brought online and it's the only disk in the group, then the original group is recreated and the disk is moved to that group. If there are other disks in the group and they're online, then the disk is simply added back into the group. If the group of a selected disk contains mirrored or RAID-5 volumes, this command also resynchronizes these volumes.

> **ⓘ Note**
>
> A disk must be selected for the **online disk** command to succeed. Use the select disk command to select a disk and shift the focus to it.

> **ⓘ Important**
>
> This command will fails if it's used on a read-only disk.

Syntax

```
online disk [noerr]
```

Parameters

For instructions about using this command, see Reactivate a Missing or Offline Dynamic Disk.

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To take the disk with focus online, type:

```
online disk
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

online volume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Takes the offline volume to the online state. This command works on volumes that have failed, are failing, or are in failed redundancy state.

> **Note**
>
> A volume must be selected for the **online volume** command to succeed. Use the select volume command to select a volume and shift the focus to it.

> **Important**
>
> This command will fails if it's used on a read-only disk.

Syntax

```
online volume [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To take the volume with focus online, type:

```
online volume
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

recover

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Recovers readable information from a bad or defective disk. This command reads a file, sector-by-sector, and recovers data from the good sectors. Data in bad sectors is lost. Because all data in bad sectors is lost when you recover a file, you should recover only one file at a time.

Bad sectors reported by the **chkdsk** command were marked as bad when your disk was prepared for operation. They pose no danger, and **recover** does not affect them.

Syntax

```
recover [<drive>:][<path>]<filename>
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]<filename>`	Specifies the file name (and the location of the file if it is not in the current directory) you want to recover. *Filename* is required and wildcards aren't supported.
/?	Displays help at the command prompt.

Examples

To recover the file *story.txt* in the *\fiction* directory on drive D, type:

```
recover d:\fiction\story.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rem

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Records comments in a script, batch, or config.sys file. If no comment is specified, **rem** adds vertical spacing.

> **Note**
>
> This command is internal to the command-line interpreter, cmd.exe.

Syntax

```
rem [<comment>]
```

Parameters

Expand table

Parameter	Description
`<comment>`	Specifies a string of characters to include as a comment.
/?	Displays help at the command prompt.

Remarks

- The **rem** command doesn't display comments on the screen. To display comments on the screen, you must include the **echo on** command in your file.
- You can't use a redirection character (`<` or `>`) or pipe (`|`) in a batch file comment.
- Although you can use **rem** without a comment to add vertical spacing to a batch file, you can also use blank lines. Blank lines are ignored when a batch program is processed.

Examples

To add vertical spacing through batch file comments, type:

```
@echo off
rem  This batch program formats and checks new disks.
rem  It is named Checknew.bat.
rem
rem echo Insert new disk in Drive B.
pause
format b: /v chkdsk b:
```

To include an explanatory comment before the **prompt** command in a config.sys file, type:

```
rem Set prompt to indicate current directory
prompt $p$g
```

To provide a comment about what a script does, type:

```
rem The commands in this script set up 3 drives.
rem The first drive is a primary partition and is
rem assigned the letter D. The second and third drives
rem are logical partitions, and are assigned letters
rem E and F.
create partition primary size=2048
assign d:
create partition extended
create partition logical size=2048
assign e:
create partition logical
assign f:
```

For multi-line comments, use conditional execution:

```
  Rem/||(
    The REM statement evaluates to success,
    so these lines will never be executed.
    Keep in mind that you will need to escape closing parentheses
```

```
    within multi-line comment blocks like shown in this example. ^)
)
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

remove

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a drive letter or mount point from the volume with focus. If the all parameter is used, all current drive letters and mount points are removed. If no drive letter or mount point is specified, then DiskPart removes the first drive letter or mount point it encounters.

The remove command can also be used to change the drive letter associated with a removable drive. You can't remove the drive letters on system, boot, or paging volumes. In addition, you can't remove the drive letter for an OEM partition, any GPT partition with an unrecognized GUID, or any of the special, non-data, GPT partitions such as the EFI system partition.

> **Note**
>
> A volume must be selected for the **remove** command to succeed. Use the select volume command to select a disk and shift the focus to it.

Syntax

```
remove [{letter=<drive> | mount=<path> [all]}] [noerr]
```

Parameters

Expand table

Parameter	Description
letter= `<drive>`	The drive letter to remove.
mount= `<path>`	The mount point path to remove.

Parameter	Description
all	Removes all current drive letters and mount points.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To remove the d:\ drive, type:

```
remove letter=d
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

repair

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Repairs the RAID-5 volume with focus by replacing the failed disk region with the specified dynamic disk.

A volume in a RAID-5 array must be selected for this operation to succeed. Use the **select volume** command to select a volume and shift the focus to it.

Syntax

```
repair disk=<n> [align=<n>] [noerr]
```

Parameters

Expand table

Parameter	Description
disk=`<n>`	Specifies the dynamic disk that will replace the failed disk region. Where *n* must have free space greater than or equal to the total size of the failed disk region in the RAID-5 volume.
align=`<n>`	Aligns all volume or partition extents to the closest alignment boundary. Where *n* is the number of kilobytes (KB) from the beginning of the disk to the closest alignment boundary.
noerr	for scripting only. When an error is encountered, DiskPart continues to process commands as if the error didn't occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To replace the volume with focus by replacing it with dynamic disk 4, type:

```
repair disk=4
```

Related links

- Command-Line Syntax Key
- select volume command

Feedback

Was this page helpful? Yes No

rescan

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Using the diskpart command interpreter, you can locate new disks added to your computer.

Syntax

```
rescan
```

Related links

- Command-Line Syntax Key
- Diskpart command

Feedback

Was this page helpful? Yes No

retain

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Prepares an existing simple dynamic volume for use as a boot or system volume. If you use a master boot record (MBR) dynamic disk, this command creates a partition entry in the master boot record. If you use a GUID partition table (GPT) dynamic disk, this command creates a partition entry in the GUID partition table.

Syntax

```
retain
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

san

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays or sets the storage area network (san) policy for the operating system. If used without parameters, the current san policy is displayed.

Syntax

```
san [policy={onlineAll | offlineAll | offlineShared}] [noerr]
```

Parameters

Expand table

Parameter	Description
policy={onlineAll|offlineAll|offlineShared}]	Sets the san policy for the currently booted operating system. The san policy determines whether a newly discovered disk is brought online or remains offline, and whether it becomes read/write or remains read-only. When a disk is offline, the disk layout can be read, but no volume devices are surfaced through Plug and Play. When offline no file system can be mounted on the disk. When a disk is online, one or more volume devices are installed for the disk. The following parameter can be set: • **onlineAll**. Specifies that all newly discovered disks are online and made read/write. **IMPORTANT:** Specifying **onlineAll** on a server that shares disks could lead to data corruption. Therefore, you shouldn't set this policy if disks are shared among servers unless the server is part of a cluster. • **offlineAll**. Specifies that all newly discovered disks except the startup disk are offline and read-only by default. • **offlineShared**. Specifies that all newly discovered disks that don't reside on a shared bus (such as SCSI and iSCSI) are brought online and made read-write. Disks that are left offline are read-only by default.

Parameter	Description
	For more information, see VDS_san_POLICY Enumeration.
noerr	Used for scripting only. When an error is encountered, DiskPart continues to process commands as if the error didn't occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To view the current policy, type:

```
san
```

To make all newly discovered disks, except the startup disk, offline and read-only by default, type:

```
san policy=offlineAll
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

select commands

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Shifts the focus to a disk, partition, volume, or virtual hard disk (VHD).

Syntax

```
select disk
select partition
select vdisk
select volume
```

Parameters

Expand table

Parameter	Description
Select disk	Shifts the focus to a disk.
Select partition	Shifts the focus to a partition.
Select vdisk	Shifts the focus to a VHD.
Select volume	Shifts the focus to a volume.

Remarks

- If a volume is selected with a corresponding partition, the partition will be automatically selected.
- If a partition is selected with a corresponding volume, the volume will be automatically selected.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

select disk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Selects the specified disk and shifts the focus to it.

Syntax

```
select disk={<n>|<disk path>|system|next}
```

Parameters

Expand table

Parameter	Description
`<n>`	Specifies the number of the disk to receive focus. You can view the numbers for all the disks on the computer by using the **list disk** command in DiskPart. **NOTE** When configuring systems with multiple disks, don't use **select disk=0** to specify the system disk. The computer may reassign disk numbers when you reboot, and different computers with the same disk configuration can have different disk numbers.
`<disk path>`	Specifies the location of the disk to receive focus, for example, `PCIROOT(0)#PCI(0F02)#atA(C00T00L00)`. To view the location path of a disk, select it and then type **detail disk**.
system	On BIOS computers, this option specifies that disk 0 receives focus. On EFI computers, the disk containing the EFI system partition (ESP), used for the current boot, receives focus. On EFI computers, the command will fail if there's no ESP, if there's more than one ESP, or if the computer is booted from Windows Preinstallation Environment (Windows PE).
next	After a disk is selected, this option iterates over all disks in the disk list. When you run this option, the next disk in the list receives focus.

Examples

To shift the focus to disk 1, type:

```
select disk=1
```

To select a disk by using its location path, type:

```
select disk=PCIROOT(0)#PCI(0100)#atA(C00T00L01)
```

To shift the focus to the system disk, type:

```
select disk=system
```

To shift the focus to the next disk on the computer, type:

```
select disk=next
```

Related links

- Command-Line Syntax Key
- select partition command
- select vdisk command
- select volume command

Feedback

Was this page helpful? Yes No

select partition

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Selects the specified partition and shifts the focus to it. This command can also be used to display the partition that currently has the focus in the selected disk.

Syntax

```
select partition=<n>
```

Parameters

Expand table

Parameter	Description
partition=`<n>`	The number of the partition to receive the focus. You can view the numbers for all partitions on the disk currently selected by using the **list partition** command in DiskPart.

Remarks

- Before you can select a partition you must first select a disk using the **select disk** command.
    - If no partition number is specified, this option displays the partition that currently has the focus in the selected disk.
    - If a volume is selected with a corresponding partition, the partition is automatically selected.
    - If a partition is selected with a corresponding volume, the volume is automatically selected.

Examples

To shift the focus to *partition 3*, type:

```
select partitition=3
```

To display the partition that currently has the focus in the selected disk, type:

```
select partition
```

Related links

- Command-Line Syntax Key
- create partition efi command
- create partition extended command
- create partition logical command
- create partition msr command
- create partition primary command
- delete partition command
- detail partition command
- select disk command
- select vdisk command
- select volume command

Feedback

Was this page helpful? Yes No

select vdisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Selects the specified virtual hard disk (VHD) and shifts the focus to it.

Syntax

```
select vdisk file=<full path> [noerr]
```

Parameters

Expand table

Parameter	Description
file=`<full path>`	Specifies the full path and file name of an existing VHD file.
noerr	Used for scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To shift the focus to the VHD named *c:\test\test.vhd*, type:

```
select vdisk file=c:\test\test.vhd
```

Related links

- Command-Line Syntax Key
- attach vdisk

- compact vdisk
- detach vdisk
- detail vdisk
- expand vdisk
- merge vdisk
- list
- select disk command
- select partition command
- select volume command

Feedback

Was this page helpful? Yes No

select volume

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Selects the specified volume and shifts the focus to it. This command can also be used to display the volume that currently has the focus in the selected disk.

Syntax

```
select volume={<n>|<d>}
```

Parameters

Expand table

Parameter	Description
`<n>`	The number of the volume to receive the focus. You can view the numbers for all volumes on the disk currently selected by using the **list volume** command in DiskPart.
`<d>`	The drive letter or mount point path of the volume to receive the focus.

Remarks

- If no volume is specified, this command displays the volume that currently has the focus in the selected disk.
- On a basic disk, selecting a volume also gives the focus to the corresponding partition.
  - If a volume is selected with a corresponding partition, the partition will be automatically selected.
  - If a partition is selected with a corresponding volume, the volume will be automatically selected.

Examples

To shift the focus to *volume 2*, type:

```
select volume=2
```

To shift the focus to *Drive C*, type:

```
select volume=c
```

To shift the focus to the volume mounted on a folder named *c:\mountpath*, type:

```
select volume=c:\mountpath
```

To display the volume that currently has the focus in the selected disk, type:

```
select volume
```

Related links

- Command-Line Syntax Key
- add volume command
- attributes volume command
- create volume mirror command
- create volume raid command
- create volume simple command
- create volume stripe command
- delete volume command

- detail volume command
- fsutil volume command
- list volume command
- offline volume command
- onine volume command
- select disk command
- select partition command
- select vdisk command

Feedback

Was this page helpful? Yes No

set id (Diskpart)

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the partition type field for the partition with focus. This command doesn't work on dynamic disks or on Microsoft Reserved partitions.

> **Important**
>
> This command is intended for use by original equipment manufacturers (OEMs) only. Changing partition type fields with this parameter might cause your computer to fail or be unable to boot. Unless you are an OEM or experienced with gpt disks, you should not change partition type fields on gpt disks by using this parameter. Instead, always use the **create partition efi** command to create EFI system partitions, the **create partition msr** command to create Microsoft Reserved partitions, and the **create partition primary** command without the ID parameter to create primary partitions on gpt disks.

Syntax

```
set id={ <byte> | <GUID> } [override] [noerr]
```

Parameters

Expand table

Parameter	Description
`<byte>`	For master boot record (MBR) disks, specifies the new value for the type field, in hexadecimal form, for the partition. Any partition type **byte** can be specified with this parameter except for type 0x42, which specifies an LDM partition. Note that the leading 0x is omitted when specifying the hexadecimal partition type.
`<GUID>`	For GUID partition table (gpt) disks, specifies the new GUID value for the type field for the partition. Recognized GUIDs include: • **EFI system partition:** c12a7328-f81f-11d2-ba4b-00a0c93ec93b

Parameter	Description
	• **Basic data partition:** ebd0a0a2-b9e5-4433-87c0-68b6b72699c7 Any partition type GUID can be specified with this parameter except the following: • **Microsoft Reserved partition:** e3c9e316-0b5c-4db8-817d-f92df00215ae • **LDM metadata partition on a dynamic disk:** 5808c8aa-7e8f-42e0-85d2-e1e90434cfb3 • **LDM data partition on a dynamic disk:** af9b60a0-1431-4f62-bc68-3311714a69ad • **Cluster metadata partition:** db97dba9-0840-4bae-97f0-ffb9a327c7e1
override	forces the file system on the volume to dismount before changing the partition type. When you run the **set id** command, DiskPart attempts to lock and dismount the file system on the volume. If **override** isn't specified, and the call to lock the file system fails (for example, because there is an open handle), the operation fails. If **override** is specified, DiskPart forces the dismount even if the call to lock the file system fails, and any open handles to the volume will stop being valid.
noerr	Used for scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Remarks

- Other than the limitations previously mentioned, DiskPart doesn't check the validity of the value that you specify (except to ensure that it is a byte in hexadecimal form or a GUID).

Examples

To set the type field to *0x07* and force the file system to dismount, type:

```
set id=0x07 override
```

To set the type field to be a basic data partition, type:

```
set id=ebd0a0a2-b9e5-4433-87c0-68b6b72699c7
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

shrink

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

The Diskpart shrink command reduces the size of the selected volume by the amount you specify. This command makes free disk space available from the unused space at the end of the volume.

A volume must be selected for this operation to succeed. Use the **select volume** command to select a volume and shift the focus to it.

> **Note**
>
> This command works on basic volumes, and on simple or spanned dynamic volumes. It doesn't work on original equipment manufacturer (OEM) partitions, Extensible Firmware Interface (EFI) system partitions, or recovery partitions.

Syntax

```
shrink [desired=<n>] [minimum=<n>] [nowait] [noerr]
shrink querymax [noerr]
```

Parameters

Expand table

Parameter	Description
desired=`<n>`	Specifies the desired amount of space in megabytes (MB) to reduce the size of the volume by.
minimum=`<n>`	Specifies the minimum amount of space in MB to reduce the size of the volume by.
querymax	Returns the maximum amount of space in MB by which the volume can be reduced. This value may change if applications are currently accessing the volume.

Parameter	Description
nowait	Forces the command to return immediately while the shrink process is still in progress.
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Remarks

- You can reduce the size of a volume only if it is formatted using the NTFS file system or if it does not have a file system.
- If a desired amount isn't specified, the volume is reduced by the minimum amount (if specified).
- If a minimum amount isn't specified, the volume is reduced by the desired amount (if specified).
- If neither a minimum amount nor a desired amount is specified, the volume is reduced by as much as possible.
- If a minimum amount is specified, but not enough free space is available, the command fails.

Examples

To reduce the size of the selected volume by the largest possible amount between 250 and 500 megabytes, type:

```
shrink desired=500 minimum=250
```

To display the maximum number of MB that the volume can be reduced by, type:

```
shrink querymax
```

Related links

- Command-Line Syntax Key
- Resize-Partition

Feedback

Was this page helpful? Yes No

uniqueid

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or sets the GUID partition table (GPT) identifier or master boot record (MBR) signature for the basic or dynamic disk with focus. A basic or dynamic disk must be selected for this operation to succeed. Use the select disk command to select a disk and shift the focus to it.

Syntax

```
uniqueid disk [id={<dword> | <GUID>}] [noerr]
```

Parameters

Expand table

Parameter	Description
id=`{<dword> | <GUID>}`	For MBR disks, this parameter specifies a 4-byte (DWORD) value in hexadecimal form for the signature. For GPT disks, this parameter specifies a GUID for the identifier.
noerr	For scripting only. When an error occurs, DiskPart continues to process commands as if the error didn't occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To display the signature of the MBR disk with focus, type:

```
uniqueid disk
```

To set the signature of the MBR disk with focus to the DWORD value *5f1b2c36*, type:

```
uniqueid disk id=5f1b2c36
```

To set the identifier of the GPT disk with focus to the GUID value baf784e7-6bbd-4cfb-aaac-e86c96e166ee, type:

```
uniqueid disk id=baf784e7-6bbd-4cfb-aaac-e86c96e166ee
```

Related links

- Command-Line Syntax Key
- select disk command

Feedback

Was this page helpful? Yes No

diskperf

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The **diskperf** command remotely enables or disables physical or logical disk performance counters on computers running Windows.

Syntax

```
diskperf [-y[d|v] | -n[d|v]] [\\computername]
```

Options

Expand table

Option	Description
-y	Starts all disk performance counters when the computer restarts.
-yd	Enables disk performance counters for physical drives when the computer restarts.
-yv	Enables disk performance counters for logical drives or storage volumes when the computer restarts.
-n	Disables all disk performance counters when the computer restarts.
-nd	Disable disk performance counters for physical drives when the computer restarts.
-nv	Disable disk performance counters for logical drives or storage volumes when the computer restarts.
`\\ <computername>`	Specifies the name of the computer where you want to enable or disable disk performance counters.
-?	Displays context sensitive help.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Diskraid

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Diskraid is a command-line tool that enables you to configure and manage redundant array of independent (or inexpensive) disks (RAID) storage subsystems.

RAID is typically used on servers to standardize and categorize fault-tolerant disk systems. RAID levels provide various mixes of performance, reliability, and cost. Some servers provide three of the RAID levels: Level 0 (striping), Level 1 (mirroring), and Level 5 (striping with parity).

A hardware RAID subsystem distinguishes physically addressable storage units from one another by using a Logical Unit Number (LUN). A LUN object must have at least one plex, and can have any number of additional plexes. Each plex contains a copy of the data on the LUN object. Plexes can be added to and removed from a LUN object.

Most Diskraid commands operate on a specific host bus adapter (HBA) port, initiator adapter, initiator portal, provider, subsystem, controller, port, drive, LUN, target portal, target, or target portal group. You use the **SELECT** command to select an object. The selected object is said to have focus. Focus simplifies common configuration tasks, such as creating multiple LUNs within the same subsystem.

> ⓘ **Note**
>
> The Diskraid command-line tool works only with storage subsystems that support Virtual Disk Service (VDS).

Diskraid commands

The following commands are available from within the Diskraid tool.

add

Adds an existing LUN to the currently selected LUN, or adds an iSCSI target portal to the currently selected iSCSI target portal group.

Syntax

```
add plex lun=n [noerr]
add tpgroup tportal=n [noerr]
```

Parameters

Expand table

Parameter	Description
plex lun=`<n>`	Specifies the LUN number to add as a plex to the currently selected LUN. CAUTION: All data on the LUN being added as a plex will be deleted.
tpgroup tportal=`<n>`	Specifies the iSCSI target portal number to add to the currently selected iSCSI target portal group.
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

associate

Sets the specified list of controller ports as active for the currently selected LUN (other controller ports are made inactive), or adds the specified controller ports to the list of existing active controller ports for the currently selected LUN, or associates the specified iSCSI target for the currently selected LUN.

Syntax

```
associate controllers [add] <n>[,<n> [,…]]
associate ports [add] <n-m>[,<n-m>[,…]]
associate targets [add] <n>[,<n> [,…]]
```

Parameters

Expand table

Parameter	Description
controller	Adds to or replaces the list of controllers that are associated with the currently selected LUN. Use only with VDS 1.0 providers.
ports	Adds to or replaces the list of controller ports that are associated with the currently selected LUN. Use only with VDS 1.1 providers.
targets	Adds to or replaces the list of iSCSI targets that are associated with the currently selected LUN. Use only with VDS 1.1 providers.
add	**If using VDS 1.0 providers:** Adds the specified controllers to the existing list of controllers associated with the LUN. If this parameter is not specified, the list of controllers replaces the existing list of controllers associated with this LUN. **If using VDS 1.1 providers:** Adds the specified controller ports to the existing list of controller ports associated with the LUN. If this parameter is not specified, the list of controller ports replaces the existing list of controller ports associated with this LUN.
`<n>[,<n> [,...]]`	Use with the **controllers** or **targets** parameter. Specifies the numbers of the controllers or iSCSI targets to set to active or associate.
`<n-m>[,<n-m>[,…]]`	Use with the **ports** parameter. Specifies the controller ports to set active using a controller number (*n*) and port number (*m*) pair.

Example

To associate and add ports to a LUN that uses a VDS 1.1 provider:

```
DISKRAID> SEL LUN 5
LUN 5 is now the selected LUN.

DISKRAID> ASSOCIATE PORTS 0-0,0-1
Controller port associations changed.
(Controller ports active after this command: Ctlr 0 Port 0, Ctlr 0 Port 1)

DISKRAID> ASSOCIATE PORTS ADD 1-1
Controller port associations changed.
(Controller ports active after this command: Ctlr 0 Port 0, Ctlr 0 Port 1,
Ctlr 1 Port 1)
```

automagic

Sets or clears flags that give hints to providers on how to configure a LUN. Used with no parameters, the **automagic** operation displays a list of flags.

Syntax

```
automagic {set | clear | apply} all <flag=value> [<flag=value> [...]]
```

Parameters

Expand table

Parameter	Description
set	Sets the specified flags to the specified values.
clear	Clears the specified flags. The **all** keyword clears all the automagic flags.
apply	Applies the current flags to the selected LUN.
`<flag>`	Flags are identified by three-letter acronyms, including: • **FCR** - Fast Crash Recovery Required • **FTL** - Fault Tolerant • **MSR** - Mostly Reads • **MXD** - Maximum drives • **MXS** - Maximum Size Expected • **ORA** - Optimal Read Alignment • **ORS** - Optimal Read Size • **OSR** - Optimize for Sequential Reads • **OSW** - Optimize for Sequential Writes • **OWA** - Optimal Write Alignment • **OWS** - Optimal Write Size • **RBP** - Rebuild Priority • **RBV** - Read Back Verify Enabled • **RMP** - Remap Enabled • **STS** - Strip Size • **WTC** - Write-Through Caching Enabled • **YNK** - Removable

break

Removes the plex from the currently selected LUN. The plex and the data it contained are not retained, and the drive extents may be reclaimed.

> ⊗ **Caution**

You must first select a mirrored LUN before using this command. All data on the plex will be deleted. All data contained on the original LUN is not guaranteed to be consistent.

Syntax

```
break plex=<plex_number> [noerr]
```

Parameters

Expand table

Parameter	Description
plex	Specifies the number of the plex to remove. The plex and the data it contained will not be retained, and the resources used by this plex will be reclaimed. The data contained on the LUN is not guaranteed to be consistent. If you want to retain this plex, use the Volume Shadow Copy Service (VSS).
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

chap

Sets the Challenge Handshake Authentication Protocol (CHAP) shared secret so that iSCSI initiators and iSCSI targets can communicate with one another.

Syntax

```
chap initiator set secret=[<secret>] [target=<target>]
chap initiator remember secret=[<secret>] target=<target>
chap target set secret=[<secret>] [initiator=<initiatorname>]
chap target remember secret=[<secret>] initiator=<initiatorname>
```

Parameters

Expand table

Parameter	Description
initiator set	Sets the shared secret in the local iSCSI initiator service used for mutual CHAP authentication when the initiator authenticates the target.
initiator remember	Communicates the CHAP secret of an iSCSI target to the local iSCSI initiator service so that the initiator service can use the secret in order to authenticate itself to the target during CHAP authentication.
target set	Sets the shared secret in the currently selected iSCSI target used for CHAP authentication when the target authenticates the initiator.
target remember	Communicates the CHAP secret of an iSCSI initiator to the current in-focus iSCSI target so that the target can use the secret in order to authenticate itself to the initiator during mutual CHAP authentication.
secret	Specifies the secret to use. If empty the secret will be cleared.
target	Specifies a target in the currently selected subsystem to associate with the secret. This is optional when setting a secret on the initiator and leaving it out indicates that the secret will be used for all targets that do not already have an associated secret.
initiatorname	Specifies an initiator iSCSI name to associate with the secret. This is optional when setting a secret on a target and leaving it out indicates that the secret will be used for all initiators that do not already have an associated secret.

create

Creates a new LUN or iSCSI target on the currently selected subsystem, or creates a target portal group on the currently selected target. You can view the actual binding using the **Diskraid list** command.

Syntax

```
create lun simple [size=<n>] [drives=<n>] [noerr]
create lun stripe [size=<n>] [drives=<n, n> [,...]]  [stripesize=<n>]
[noerr]
create lun raid [size=<n>] [drives=<n, n> [,...]] [stripesize=<n>] [noerr]
create lun mirror [size=<n>] [drives=<n, n> [,...]] [stripesize=<n>] [noerr]
create lun automagic size=<n> [noerr]
create target name=<name> [iscsiname=<iscsiname>] [noerr]
create tpgroup [noerr]
```

Parameters

Expand table

Parameter	Description
simple	Creates a simple LUN.
stripe	Creates a striped LUN.
raid	Creates a striped LUN with parity.
mirror	Creates a mirrored LUN.
automagic	Creates a LUN using the *automagic* hints currently in effect. For more info, see the **automagic** sub-command in this article.
size=	Specifies the total LUN size in megabytes. Either the **size**= or the **drives**= parameter must be specified. They can also be used together. If the **size=** parameter is not specified, the LUN created will be the largest possible size allowed by all the specified drives. A provider typically creates a LUN at least as big as the requested size, but the provider may have to round up to the next largest size in some cases. For example, if size is specified as .99 GB and the provider can only allocate GB disk extents, the resulting LUN would be 1 GB. To specify the size using other units, use one of the following recognized suffixes immediately after the size: • **B** - byte • **KB** - kilobyte • **MB** - megabyte • **GB** - gigabyte • **TB** - terabyte • **PB** - petabyte.
drives=	Specifies the *drive_number* for the drives to use to create a LUN. Either the **size**= or the **drives**= parameter must be specified. They can also be used together. If the **size=** parameter is not specified, the LUN created is the largest possible size allowed by all the specified drives. If the **size=** parameter is specified, providers will select drives from the specified drive list to create the LUN. Providers will attempt to use the drives in the order specified when possible.
stripesize=	Specifies the size in megabytes for a *stripe* or *raid* LUN. The stripesize cannot be changed after the LUN is created. To specify the size using other units, use one of the following recognized suffixes immediately after the size: • **B** - byte • **KB** - kilobyte • **MB** - megabyte • **GB** - gigabyte

Parameter	Description
	• **TB** - terabyte • **PB** - petabyte.
target	Creates a new iSCSI target on the currently selected subsystem.
name	Supplies the friendly name for the target.
iscsiname	Supplies the iSCSI name for the target and can be omitted to have the provider generate a name.
tpgroup	Creates a new iSCSI target portal group on the currently selected target.
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

delete

Deletes the currently selected LUN, iSCSI target (as long as there are not any LUNs associated with the iSCSI target) or iSCSI target portal group.

Syntax

```
delete lun [uninstall] [noerr]
delete target [noerr]
delete tpgroup [noerr]
```

Parameters

Expand table

Parameter	Description
lun	Deletes the currently selected LUN and all data on it.
uninstall	Specifies that the disk on the local system associated with the LUN will be cleaned up before the LUN is deleted.
target	Deletes the currently selected iSCSI target if no LUNs are associated with the target.
tpgroup	Deletes the currently selected iSCSI target portal group.

Parameter	Description
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

detail

Displays detailed information about the currently selected object of the specified type.

Syntax

```
detail {hbaport | iadapter | iportal | provider | subsystem | controller |
port | drive | lun | tportal | target | tpgroup} [verbose]
```

Parameters

Expand table

Parameter	Description
hbaport	Lists detailed information about the currently selected host bus adapter (HBA) port.
iadapter	Lists detailed information about the currently selected iSCSI initiator adapter.
iportal	Lists detailed information about the currently selected iSCSI initiator portal.
provider	Lists detailed information about the currently selected provider.
subsystem	Lists detailed information about the currently selected subsystem.
controller	Lists detailed information about the currently selected controller.
port	Lists detailed information about the currently selected controller port.
drive	Lists detailed information about the currently selected drive, including the occupying LUNs.
lun	Lists detailed information about the currently selected LUN, including the contributing drives. The output differs slightly depending on whether the LUN is part of a Fibre Channel or iSCSI subsystem. If the Unmasked Hosts list contains only an asterisk, this means that the LUN is unmasked to all hosts.
tportal	Lists detailed information about the currently selected iSCSI target portal.

Parameter	Description
target	Lists detailed information about the currently selected iSCSI target.
tpgroup	Lists detailed information about the currently selected iSCSI target portal group.
verbose	For use only with the LUN parameter. Lists additional information, including its plexes.

dissociate

Sets specified list of controller ports as inactive for the currently selected LUN (other controller ports are not affected), or dissociates the specified list of iSCSI targets for the currently selected LUN.

Syntax

```
dissociate controllers <n> [,<n> [,...]]
dissociate ports <n-m>[,<n-m>[,…]]
dissociate targets <n> [,<n> [,…]]
```

Parameter

Expand table

Parameter	Description
controllers	Removes controllers from the list of controllers that are associated with the currently selected LUN. Use only with VDS 1.0 providers.
ports	Removes controller ports from the list of controller ports that are associated with the currently selected LUN. Use only with VDS 1.1 providers.
targets	Removes targets from the list of iSCSI targets that are associated with the currently selected LUN. Use only with VDS 1.1 providers.
`<n> [,<n> [,…]]`	For use with the **controllers** or **targets** parameter. Specifies the numbers of the controllers or iSCSI targets to set as inactive or dissociate.
`<n-m>[,<n-m> [,…]]`	For use with the **ports** parameter. Specifies the controller ports to set as inactive by using a controller number (*n*) and port number (*m*) pair.

Example

```
DISKRAID> SEL LUN 5
LUN 5 is now the selected LUN.

DISKRAID> ASSOCIATE PORTS 0-0,0-1
Controller port associations changed.
(Controller ports active after this command: Ctlr 0 Port 0, Ctlr 0 Port 1)

DISKRAID> ASSOCIATE PORTS ADD 1-1
Controller port associations changed.
(Controller ports active after this command: Ctlr 0 Port 0, Ctlr 0 Port 1,
Ctlr 1 Port 1)

DISKRAID> DISSOCIATE PORTS 0-0,1-1
Controller port associations changed.
(Controller ports active after this command: Ctlr 0 Port 1)
```

exit

Exits Diskraid.

Syntax

```
exit
```

extend

Extends the currently selected LUN by adding sectors to the end of the LUN. Not all providers support extending LUNs. Does not extend any volumes or file systems contained on the LUN. After you extend the LUN, you should extend the associated on-disk structures using the **DiskPart extend** command.

Syntax

```
extend lun [size=<LUN_size>] [drives=<drive_number>, [<drive_number>, ...]]
[noerr]
```

Parameters

Expand table

Parameter	Description
size	Specifies the size in megabytes to extend the LUN. Either the *size* or the `<drive>` parameter must be specified. They can also be used together. If the **size=** parameter is not specified, the LUN is extended by the largest possible size allowed by all the specified drives. If the **size=** parameter is specified, providers select drives from the list specified by the **drives=** parameter to create the LUN. To specify the size using other units, use one of the following recognized suffixes immediately after the size: • **B** - byte • **KB** - kilobyte • **MB** - megabyte • **GB** - gigabyte • **TB** - terabyte • **PB** - petabyte.
drives=	Specifies the `<drive_number>` for the drives to use when creating a LUN. Either the *size* or the `<drive>` parameter must be specified. They can also be used together. If the **size=** parameter is not specified, the LUN created is the largest possible size allowed by all the specified drives. Providers use the drives in the order specified when possible.
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

flushcache

Clears the cache on the currently selected controller.

Syntax

```
flushcache controller
```

help

Displays a list of all Diskraid commands.

Syntax

```
help
```

importtarget

Retrieves or sets the current Volume Shadow Copy Service (VSS) import target that is set for the currently selected subsystem.

Syntax

```
importtarget subsystem [set target]
```

Parameter

Expand table

Parameter	Description
set target	If specified, sets the currently selected target to the VSS import target for the currently selected subsystem. If not specified, the command retrieves the current VSS import target that is set for the currently selected subsystem.

initiator

Retrieves information about the local iSCSI initiator.

Syntax

```
initiator
```

invalidatecache

Invalidates the cache on the currently selected controller.

Syntax

```
invalidatecache controller
```

lbpolicy

Sets the load balance policy on the currently selected LUN.

Syntax

```
lbpolicy set lun type=<type> [paths=<path>-{primary | <weight>}[,<path>-
{primary | <weight>}[,…]]]
lbpolicy set lun paths=<path>-{primary | <weight>}[,<path>-{primary |
<weight>}[,…]]
```

Parameters

Expand table

Parameter	Description
type	Specifies the load balance policy. If the type is not specified, then the **path** parameter must be specified. Type can be one of the following: • **FAILOVER** - Uses one primary path with other paths being backup paths. • **ROUNDROBIN** - Uses all paths in round-robin fashion, which tries each path sequentially. • **SUBSETROUNDROBIN** - Uses all primary paths in round-robin fashion; backup paths are used only if all primary paths fail. • **DYNLQD** - Uses the path with the least number of active requests. • • **WEIGHTED** - Uses the path with the least weight (each path must be assigned a weight). • **LEASTBLOCKS** - Uses the path with the least blocks. • **VENDORSPECIFIC** - Uses a vendor-specific policy.
path	Specifies whether a path is **primary** or has a particular `<weight>`. Any paths not specified are implicitly set as backup. Any paths listed must be one of the currently selected LUN's paths.

list

Displays a list of objects of the specified type.

Syntax

```
list {hbaports | iadapters | iportals | providers | subsystems | controllers
| ports | drives | LUNs | tportals | targets | tpgroups}
```

Parameters

Expand table

Parameter	Description
hbaports	Lists summary information about all HBA ports known to VDS. The currently selected HBA port is marked by an asterisk (*).
iadapters	Lists summary information about all iSCSI initiator adapters known to VDS. The currently selected initiator adapter is marked by an asterisk (*).
iportals	Lists summary information about all iSCSI initiator portals in the currently selected initiator adapter. The currently selected initiator portal is marked by an asterisk (*).
providers	Lists summary information about each provider known to VDS. The currently selected provider is marked by an asterisk (*).
subsystems	Lists summary information about each subsystem in the system. The currently selected subsystem is marked by an asterisk (*).
controllers	Lists summary information about each controller in the currently selected subsystem. The currently selected controller is marked by an asterisk (*).
ports	Lists summary information about each controller port in the currently selected controller. The currently selected port is marked by an asterisk (*).
drives	Lists summary information about each drive in the currently selected subsystem. The currently selected drive is marked by an asterisk (*).
luns	Lists summary information about each LUN in the currently selected subsystem. The currently selected LUN is marked by an asterisk (*).
tportals	Lists summary information about all iSCSI target portals in the currently selected subsystem. The currently selected target portal is marked by an asterisk (*).

Parameter	Description
targets	Lists summary information about all iSCSI targets in the currently selected subsystem. The currently selected target is marked by an asterisk (*).
tpgroups	Lists summary information about all iSCSI target portal groups in the currently selected target. The currently selected portal group is marked by an asterisk (*).

login

Logs the specified iSCSI initiator adapter into the currently selected iSCSI target.

Syntax

```
login target iadapter=<iadapter> [type={manual | persistent | boot}] [chap=
{none | oneway | mutual}] [iportal=<iportal>] [tportal=<tportal>] [<flag>
[<flag> […]]]
```

Parameters

Expand table

Parameter	Description
type	Specifies the type of login to perform: **manual** or **persistent**. If unspecified, a manual login will be performed.
manual	Login manually. There's also a **boot** option that is intended for future development and isn't currently used.
persistent	Automatically use the same login when the computer is restarted.
chap	Specifies the type of CHAP authentication to use: **none**, **oneway** CHAP, or **mutual** CHAP; if unspecified, no authentication will be used.
tportal	Specifies an optional target portal in the currently selected subsystem to use for the log in.
iportal	Specifies an optional initiator portal in the specified initiator adapter to use for the log in.
`<flag>`	Identified by three-letter acronyms: • **IPS** - Require IPsec • **EMP** - Enable multipath

Parameter	Description
	• **EHD** - Enable header digest • **EDD** - Enable data digest

logout

Logs the specified iSCSI initiator adapter out of the currently selected iSCSI target.

Syntax

```
logout target iadapter= <iadapter>
```

Parameters

Expand table

Parameter	Description
iadapter	Specifies the initiator adapter with a login session to logout from.

maintenance

Performs maintenance operations on the currently selected object of the specified type.

Syntax

```
maintenance <object operation> [count=<iteration>]
```

Parameters

Expand table

Parameter	Description
`<object>`	Specifies the type of object on which to perform the operation. The *object* type can be a **subsystem**, **controller**, **port, drive** or **LUN**.
`<operation>`	Specifies the maintenance operation to perform. The *operation* type can be **spinup**, **spindown**, **blink**, **beep** or **ping**. An *operation* must be specified.
count=	Specifies the number of times to repeat the *operation*. This is typically used with **blink**, **beep**,or **ping**.

name

Sets the friendly name of the currently selected subsystem, LUN, or iSCSI target to the specified name.

Syntax

```
name {subsystem | lun | target} [<name>]
```

Parameter

Expand table

Parameter	Description
`<name>`	Specifies a name for the subsystem, LUN, or target. The name must be less than 64 characters in length. If no name is supplied, the existing name, if any, is deleted.

offline

Sets the state of the currently selected object of the specified type to **offline**.

Syntax

```
offline <object>
```

Parameter

Expand table

Parameter	Description
`<object>`	Specifies the type of object on which to perform this operation. The type can be: **subsystem**, **controller**, **drive**, **LUN**, or **tportal**.

online

Sets the state of the selected object of the specified type to **online**. If object is **hbaport**, changes the status of the paths to the currently selected HBA port to **online**.

Syntax

```
online <object>
```

Parameter

Expand table

Parameter	Description
`<object>`	Specifies the type of object on which to perform this operation. The type can be: **hbaport**, **subsystem**, **controller**, **drive**, **LUN**, or **tportal**.

recover

Performs operations necessary, such as resynchronization or hot sparing, to repair the currently selected fault-tolerant LUN. For example, RECOVER might cause a hot spare to be bound to a RAID set that has a failed disk or other disk extent reallocation.

Syntax

```
recover <lun>
```

reenumerate

Reenumerates objects of the specified type. If you use the extend LUN command, you must use the refresh command to update the disk size before using the reenumerate command.

Syntax

```
reenumerate {subsystems | drives}
```

Parameters

Expand table

Parameter	Description
subsystems	Queries the provider to discover any new subsystems that were added in the currently selected provider.
drives	Queries the internal I/O buses to discover any new drives that were added in the currently selected subsystem.

refresh

Refreshes internal data for the currently selected provider.

Syntax

```
refresh provider
```

rem

Used to comment scripts.

Syntax

```
```

```
Rem <comment>
```

remove

Removes the specified iSCSI target portal from the currently selected target portal group.

Syntax

```
remove tpgroup tportal=<tportal> [noerr]
```

Parameter

Expand table

Parameter	Description
tpgroup tportal=`<tportal>`	Specifies the iSCSI target portal to remove.
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

replace

Replaces the specified drive with the currently selected drive. The specified drive may not be the currently selected drive.

Syntax

```
replace drive=<drive_number>
```

Parameter

Expand table

Parameter	Description
drive=	Specifies the `<drive_number>` for the drive to be replaced.

reset

Resets the currently selected controller or port.

Syntax

```
reset {controller | port}
```

Parameters

Expand table

Parameter	Description
controller	Resets the controller.
port	Resets the port.

select

Displays or changes the currently selected object.

Syntax

```
select {hbaport | iadapter | iportal | provider | subsystem | controller |
port | drive | lun | tportal | target | tpgroup } [<n>]
```

Parameters

Expand table

Parameter	Description
object	Specifies the type of object to select, including: **provider**, **subsystem**, **controller**, **drive**, or **LUN**.
hbaport `[<n>]`	Sets the focus to the specified local HBA port. If no HBA port is specified, the command displays the currently selected HBA port (if any). Specifying an invalid HBA port index results in no in-focus HBA port. Selecting an HBA port deselects any selected initiator adapters and initiator portals.
iadapter `[<n>]`	Sets the focus to the specified local iSCSI initiator adapter. If no initiator adapter is specified, the command displays the currently selected initiator adapter (if any). Specifying an invalid initiator adapter index results in no in-focus initiator adapter. Selecting an initiator adapter deselects any selected HBA ports and initiator portals.
iportal `[<n>]`	Sets the focus to the specified local iSCSI initiator portal within the selected iSCSI initiator adapter. If no initiator portal is specified, the command displays the currently selected initiator portal (if any). Specifying an invalid initiator portal index results in no selected initiator portal.
provider `[<n>]`	Sets the focus to the specified provider. If no provider is specified, the command displays the currently selected provider (if any). Specifying an invalid provider index results in no in-focus provider.
subsystem `[<n>]`	Sets the focus to the specified subsystem. If no subsystem is specified, the command displays the subsystem with focus (if any). Specifying an invalid subsystem index results in no in-focus subsystem. Selecting a subsystem implicitly selects its associated provider.
controller `[<n>]`	Sets the focus to the specified controller within the currently selected subsystem. If no controller is specified, the command displays the currently selected controller (if any). Specifying an invalid controller index results in no in-focus controller. Selecting a controller deselects any selected controller ports, drives, LUNs, target portals, targets, and target portal groups.
port `[<n>]`	Sets the focus to the specified controller port within the currently selected controller. If no port is specified, the command displays the currently selected port (if any). Specifying an invalid port index results in no selected port.
drive `[<n>]`	Sets the focus to the specified drive, or physical spindle, within the currently selected subsystem. If no drive is specified, the command displays the currently selected drive (if any). Specifying an invalid drive index results in no in-focus drive. Selecting a drive deselects any selected controllers, controller ports, LUNs, target portals, targets, and target portal groups.
lun `[<n>]`	Sets the focus to the specified LUN within the currently selected subsystem. If no LUN is specified, the command displays the currently selected LUN (if any). Specifying an invalid LUN index results in no selected LUN. Selecting a LUN deselects any selected controllers, controller ports, drives, target portals, targets, and target portal groups.

Parameter	Description
tportal `[<n>]`	Sets the focus to the specified iSCSI target portal within the currently selected subsystem. If no target portal is specified, the command displays the currently selected target portal (if any). Specifying an invalid target portal index results in no selected target portal. Selecting a target portal deselects any controllers, controller ports, drives, LUNs, targets, and target portal groups.
target `[<n>]`	Sets the focus to the specified iSCSI target within the currently selected subsystem. If no target is specified, the command displays the currently selected target (if any). Specifying an invalid target index results in no selected target. Selecting a target deselects any controllers, controller ports, drives, LUNs, target portals, and target portal groups.
tpgroup `[<n>]`	Sets the focus to the specified iSCSI target portal group within the currently selected iSCSI target. If no target portal group is specified, the command displays the currently selected target portal group (if any). Specifying an invalid target portal group index results in no in-focus target portal group.
`[<n>]`	Specifies the `<object number>` to select. If the `<object number>` specified is not valid, any existing selections for objects of the specified type are cleared. If no `<object number>` is specified, the current object is displayed.

setflag

Sets the currently selected drive as a hot spare. Hot spares can't be used for ordinary LUN binding operations. They're reserved for fault handling only. The drive must not be currently bound to any existing LUN.

Syntax

```
setflag drive hotspare={true | false}
```

Parameters

Expand table

Parameter	Description
true	Selects the currently selected drive as a hot spare.
false	Unselects the currently selected drive as a hot spare.

shrink

Reduces the size of the selected LUN.

Syntax

```
shrink lun size=<n> [noerr]
```

Parameters

Expand table

Parameter	Description
size	Specifies the desired amount of space in megabytes (MB) to reduce the size of the LUN by. To specify the size using other units, use one of the following recognized suffixes immediately after the size: • **B** - byte • **KB** - kilobyte • **MB** - megabyte • **GB** - gigabyte • **TB** - terabyte • **PB** - petabyte.
noerr	For scripting only. When an error is encountered, Diskraid continues to process commands as if the error did not occur.

standby

Changes the status of the paths to the currently selected host bus adapter (HBA) port to STANDBY.

Syntax

```
standby hbaport
```

Parameters

Expand table

Parameter	Description
hbaport	Changes the status of the paths to the currently selected host bus adapter (HBA) port to STANDBY.

unmask

Makes the currently selected LUNs accessible from the specified hosts.

Syntax

```
unmask lun {all | none | [add] wwn=<hexadecimal_number> [;
<hexadecimal_number> [;…]] | [add] initiator=<initiator>[;<initiator>[;…]]}
[uninstall]
```

Parameters

Expand table

Parameter	Description
all	Specifies that the LUN should be made accessible from all hosts. However, you cannot unmask the LUN to all targets in an iSCSI subsystem. You must logout of the target before you run the `unmask lun all` command.
none	Specifies that the LUN should not be accessible to any host. You must logout of the target before you run the `unmask lun none` command.
add	Specifies that the hosts specified must be added to the existing list of hosts that this LUN is accessible from. If this parameter is not specified, the list of hosts supplied replaces the existing list of hosts that this LUN is accessible from.
wwn=	Specifies a list of hexadecimal numbers representing world-wide names from which the LUN or hosts should be made accessible. To mask/unmask to a specific set of hosts in a Fibre Channel subsystem, you can type a semicolon-separated list of WWN's for the ports on the host machines of interest.
initiator=	Specifies a list of iSCSI initiators to which the currently selected LUN should be made accessible. To mask/unmask to a specific set of hosts in an iSCSI subsystem, you can type a semicolon-separated list of iSCSI initiator names for the initiators on the host computers of interest.

Parameter	Description
uninstall	If specified, uninstalls the disk associated with the LUN on the local system before the LUN is masked.

Scripting Diskraid

Diskraid can be scripted on any computer running a supported version of Windows Server, with an associated VDS hardware provider. To invoke a Diskraid script, at the command prompt type:

```
diskraid /s <script.txt>
```

By default, Diskraid stops processing commands and returns an error code if there is a problem in the script. To continue running the script and ignore errors, include the **noerr** parameter on the command. This permits such useful practices as using a single script to delete all the LUNs in a subsystem regardless of the total number of LUNs. Not all commands support the **noerr** parameter. Errors are always returned on command-syntax errors, regardless of whether you included the **noerr** parameter.

Diskraid error codes

Expand table

Error Code	Error Description
0	No error occurred. The entire script ran without failure.
1	A fatal exception occurred.
2	The arguments specified on a Diskraid command line were incorrect.
3	Diskraid was unable to open the specified script or output file.
4	One of the services Diskraid uses returned a failure.
5	A command syntax error occurred. The script failed because an object was improperly selected or was invalid for use with that command.

Example

To view the status of subsystem 0 on your computer, type:

```
diskraid
```

Press ENTER and output similar to the following is displayed:

```
Microsoft Diskraid version 5.2.xxxx
Copyright (©) 2003 Microsoft Corporation
On computer: COMPUTER_NAME
```

To select subsystem 0, type the following at the Diskraid prompt:

```
select subsystem 0
```

Press ENTER and output similar to the following is displayed:

```
Subsystem 0 is now the selected subsystem.

DISKRAID> list drives

  Drive ###  Status      Health          Size      Free    Bus  Slot  Flags
  ---------  ----------  ------------  --------  --------  ---  ----  -----
  Drive 0    Online      Healthy         107 GB    107 GB    0     1
  Drive 1    Offline     Healthy          29 GB     29 GB    1     0
  Drive 2    Online      Healthy         107 GB    107 GB    0     2
  Drive 3    Not Ready   Healthy          19 GB     19 GB    1     1
```

To exit Diskraid, type the following at the Diskraid prompt:

```
exit
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Diskshadow

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Diskshadow.exe is a tool that exposes the functionality offered by the volume shadow copy Service (VSS). By default, Diskshadow uses an interactive command interpreter similar to that of Diskraid or Diskpart. Diskshadow also includes a scriptable mode.

> **Note**
>
> Membership in the local Administrators group, or equivalent, is the minimum required to run Diskshadow.

Syntax

For interactive mode, type the following at the command prompt to start the Diskshadow command interpreter:

```
diskshadow
```

For script mode, type the following, where *script.txt* is a script file containing Diskshadow commands:

```
diskshadow -s script.txt
```

Parameters

You can run the following commands in the Diskshadow command interpreter or through a script file. At a minimum, only **add** and **create** are necessary to create a shadow copy. However, this forfeits the context and option settings, will be a copy backup, and creates a shadow copy with no backup execution script.

Expand table

Command	Description
set command	Sets the context, options, verbose mode, and metadata file for creating shadow copies.
load metadata command	Loads a metadata .cab file prior to importing a transportable shadow copy or loads the writer metadata in the case of a restore.
writer command	verifies that a writer or component is included or excludes a writer or component from the backup or restore procedure.
add command	Adds volumes to the set of volumes that are to be shadow copied, or adds aliases to the alias environment.
create command	Starts the shadow copy creation process, using the current context and option settings.
exec command	Executes a file on the local computer.
begin backup command	Starts a full backup session.
end backup command	Ends a full backup session and issues a **backupcomplete** event with the appropriate writer state, if needed.
begin restore command	Starts a restore session and issues a **prerestore** event to involved writers.
end restore command	Ends a restore session and issues a **postrestore** event to involved writers.
reset command	Resets Diskshadow to the default state.
list command	Lists writers, shadow copies, or currently registered shadow copy providers that are on the system.
delete shadows command	Deletes shadow copies.
import command	Imports a transportable shadow copy from a loaded metadata file into the system.
mask command	Removes hardware shadow copies that were imported by using the **import** command.
expose command	Exposes a persistent shadow copy as a drive letter, share, or mount point.
unexpose command	Unexposes a shadow copy that was exposed by using the **expose** command.

Command	Description
break command	Disassociates a shadow copy volume from VSS.
revert command	Reverts a volume back to a specified shadow copy.
exit command	Exits the command interpreter or script.

Examples

This is a sample sequence of commands that will create a shadow copy for backup. It can be saved to file as script.dsh, and executed using `diskshadow /s script.dsh`.

Assume the following:

- You have an existing directory called c:\diskshadowdata.
- Your system volume is C: and your data volume is D:.
- You have a backupscript.cmd file in c:\diskshadowdata.
- Your backupscript.cmd file will perform the copy of shadow data p: and q: to your backup drive.

You can enter these commands manually or script them:

```
#Diskshadow script file
set context persistent nowriters
set metadata c:\diskshadowdata\example.cab
set verbose on
begin backup
add volume c: alias systemvolumeshadow
add volume d: alias datavolumeshadow

create

expose %systemvolumeshadow% p:
expose %datavolumeshadow% q:
exec c:\diskshadowdata\backupscript.cmd
end backup
#End of script
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

add (diskshadow)

Article • 10/13/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds volumes to the set of volumes that are to be shadow copied, or adds aliases to the alias environment. If used without subcommands, **add** lists the current volumes and aliases.

> **Note**
>
> Aliases are not added to the alias environment until the shadow copy is created. Aliases that you need immediately should be added by using **add alias**.

Syntax

```
add
add volume <volume> [provider <providerid>]
add alias <aliasname> <aliasvalue>
```

Parameters

Expand table

Parameter	Description
volume	Adds a volume to the Shadow Copy Set, which is the set of volumes to be shadow copied. See add volume for syntax and parameters.
alias	Adds the given name and value to the alias environment. See add alias for syntax and parameters.
/?	Displays help at the command line.

Examples

To display the volumes added and the aliases that are currently in the environment, type:

```
add
```

The following output shows that drive C has been added to the Shadow Copy Set:

```
Volume c: alias System1    GUID \\?\Volume{XXXXXXXX-XXXX-XXXX-XXXX-
XXXXXXXXXXXX}\
1 volume in Shadow Copy Set.
No Diskshadow aliases in the environment.
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

add alias

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds aliases to the alias environment. If used without parameters, **add alias** displays help at the command prompt. Aliases are saved in the metadata file and will be loaded with the **load metadata** command.

Syntax

```
add alias <aliasname> <aliasvalue>
```

Parameters

Expand table

Parameter	Description
`<aliasname>`	Specifies the name of the alias.
`<aliasvalue>`	Specifies the value of the alias.
`?`	Displays help at the command prompt.

Examples

To list all shadows, including their aliases, type:

```
list shadows all
```

The following excerpt shows a shadow copy to which the default alias, *VSS_SHADOW_x*, has been assigned:

```
* Shadow Copy ID = {ff47165a-1946-4a0c-b7f4-80f46a309278}
%VSS_SHADOW_1%
```

To assign a new alias with the name *System1* to this shadow copy, type:

```
add alias System1 %VSS_SHADOW_1%
```

Alternatively, you can assign the alias by using the shadow copy ID:

```
add alias System1 {ff47165a-1946-4a0c-b7f4-80f46a309278}
```

Related links

- Command-Line Syntax Key
- load metadata command

Feedback

Was this page helpful? Yes No

add volume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds volumes to the Shadow Copy Set, which is the set of volumes to be shadow copied. When a shadow copy is created, an environment variable links the alias to the shadow ID, so the alias can then be used for scripting.

Volumes are added one at a time. Each time a volume is added, it's checked to make sure VSS supports shadow copy creation for that volume. This check can be invalidated by later use of the **set context** command.

This command is necessary to create shadow copies. If used without parameters, **add volume** displays help at the command prompt.

Syntax

```
add volume <volume> [provider <providerid>]
```

Parameters

Expand table

Parameter	Description
`<volume>`	Specifies a volume to add to the Shadow Copy Set. At least one volume is required for shadow copy creation.
`[provider \ <providerid>]`	Specifies the Provider ID for a registered provider to use to create the shadow copy. If **provider** is not specified, the default provider is used.

Examples

To view the current list of registered providers, at the `diskshadow>` prompt, type:

```
list providers
```

The following output displays a single provider, which will be used by default:

```
* ProviderID: {b5946137-7b9f-4925-af80-51abd60b20d5}
        Type: [1] VSS_PROV_SYSTEM
        Name: Microsoft Software Shadow Copy provider 1.0
        Version: 1.0.0.7
        CLSID: {65ee1dba-8ff4-4a58-ac1c-3470ee2f376a}
1 provider registered.
```

To add drive C: to the Shadow Copy Set and assign an alias named *System1*, type:

```
add volume c: alias System1
```

Related links

- Command-Line Syntax Key
- set context command

Feedback

Was this page helpful? Yes No

begin backup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts a full backup session. This command overrides the default copy backup setting.

Syntax

```
begin backup
```

Related links

- Command-Line Syntax Key
- begin restore command

Feedback

Was this page helpful? Yes No

begin restore

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts a restore session and issues a **PreRestore** event to involved writers.

Syntax

```
begin restore
```

Related links

- Command-Line Syntax Key
- begin backup command
- Prepare for restore

Feedback

Was this page helpful? Yes No

break (diskshadow)

Article • 10/12/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Disassociates a shadow copy volume from VSS and makes it accessible as a regular volume. The volume can then be accessed using a drive letter (if assigned) or volume name. If used without parameters, **break** displays help at the command prompt.

> **Note**
>
> This command is relevant only for hardware shadow copies after import.
>
> Exposed volumes, like the shadow copies they originate from, are read-only by default. Access to the volume is made directly to the hardware provider without record of the volume having been a shadow copy.

Syntax

```
break [writable] <setid>
```

Parameters

Expand table

Parameter	Description
writable	Enables read/write access on the volume.
`<setid>`	Specifies the ID of the shadow copy set. The alias of the shadow copy ID, which is stored as an environment variable by the **load metadata** command, can be used in the *SetID* parameter.

Examples

To make a shadow copy using the alias name Alias1 accessible as a writable volume in the operating system:

```
break writable %Alias1%
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

create (diskshadow)

Article • 10/16/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Starts the shadow copy creation process using the current context and option settings. Requires at least one volume in the Shadow Copy Set.

Syntax

```
create
```

Remarks

- You must add at least one volume with the add volume command before you can use the create command.
- You can use the begin backup command to specify a full backup rather than a copy backup.
- After you run the create command, you can use the exec command to run a duplication script for backup from the shadow copy.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

delete shadows

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes shadow copies.

Syntax

```
delete shadows [all | volume <volume> | oldest <volume> | set <setID> | id <shadowID> | exposed {<drive> | <mountpoint>}]
```

Parameters

Expand table

Parameter	Description
all	Deletes all shadow copies.
volume `<volume>`	Deletes all shadow copies of the given volume.
oldest `<volume>`	Deletes the oldest shadow copy of the given volume.
set `<setID>`	Deletes the shadow copies in the Shadow Copy Set of the given ID. You can specify an alias by using the % symbol if the alias exists in the current environment.
id `<shadowID>`	Deletes a shadow copy of the given ID. You can specify an alias by using the % symbol if the alias exists in the current environment.
exposed {`<drive>` | `<mountpoint>`}	Deletes shadow copies exposed at the specified drive or mount point.

Related links

- Command-Line Syntax Key
- delete command

Feedback

Was this page helpful? Yes No

end backup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Ends a full backup session and issues a **BackupComplete** event with the appropriate writer state, if needed.

Syntax

```
end backup
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

end restore

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Ends a restore session and issues a **PostRestore** event to involved writers.

Syntax

```
end restore
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

exec

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Runs a script file on the local computer. This command also duplicates or restores data as part of a backup or restore sequence. If the script fails, an error is returned and DiskShadow quits.

The file can be a **cmd** script.

Syntax

```
exec <scriptfile.cmd>
```

Parameters

Expand table

Parameter	Description
`<scriptfile.cmd>`	Specifies the script file to run.

Related links

- Command-Line Syntax Key
- diskshadow command

Feedback

Was this page helpful? Yes No

exit (diskshadow)

Article • 10/16/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Exits the DiskShadow command line interpreter.

Syntax

```
exit
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

expose

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Exposes a persistent shadow copy as a drive letter, share, or mount point.

Syntax

```
expose <shadowID> {<drive:> | <share> | <mountpoint>}
```

Parameters

Expand table

Parameter	Description
shadowID	Specifies the shadow ID of the shadow copy you want to expose. You can also use an existing alias or an environment variable in place of *shadowID*. Use **add** without parameters to see existing aliases.
`<drive:>`	Exposes the specified shadow copy as a drive letter (for example, `p:`).
`<share>`	Exposes the specified shadow copy at a share (for example, `\\machinename`).
`<mountpoint>`	Exposes the specified shadow copy to a mount point (for example, `C:\shadowcopy`).

Examples

To expose the persistent shadow copy associated with the VSS_SHADOW_1 environment variable as drive X, type:

```
expose %vss_shadow_1% x:
```

Related links

- Command-Line Syntax Key
- diskshadow command

Feedback

Was this page helpful? Yes No

import (diskshadow)

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Imports a transportable shadow copy from a loaded metadata file into the system.

> [IMPORTANT] Before you can use this command, you must use the load metadata command to load a DiskShadow metadata file.

Syntax

```
import
```

Remarks

- Transportable shadow copies aren't stored on the system immediately. Their details are stored in a Backup Components Document XML file, which DiskShadow automatically requests and saves in a .cab metadata file in the working directory. Use the set metadata command to change the path and name of this XML file.

Examples

The following is a sample DiskShadow script that demonstrates the use of the **import** command:

```
#Sample DiskShadow script demonstrating IMPORT
SET CONTEXT PERSISTENT
SET CONTEXT TRANSPORTABLE
SET METADATA transHWshadow_p.cab
#P: is the volume supported by the Hardware Shadow Copy provider
ADD VOLUME P:
CREATE
END BACKUP
#The (transportable) shadow copy is not in the system yet.
#You can reset or exit now if you wish.
```

```
LOAD METADATA transHWshadow_p.cab
IMPORT
#The shadow copy will now be loaded into the system.
```

Related links

- Command-Line Syntax Key
- diskshadow command

Feedback

Was this page helpful? Yes No

list (diskshadow)

Article • 10/16/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists writers, shadow copies, or currently registered shadow copy providers that are on the system. If used without parameters, list displays help at the command prompt.

For examples of how to use this command, see Examples.

Syntax

```
list writers [metadata | detailed | status]
list shadows {all | set <SetID> | id <ShadowID>}
list providers
```

Parameters

Expand table

Parameter	Description
writers	Lists writers. See List writers for syntax and parameters.
shadows	Lists persistent and existing non-persistent shadow copies. See List shadows for syntax and parameters.
providers	Lists currently registered shadow copy providers. See List providers for syntax and parameters.

Examples

To list all shadow copies, type:

```
list shadows all
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

list providers

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists shadow copy providers that are currently registered on the system.

Syntax

```
list providers
```

Examples

To list the currently registered shadow copy providers, type:

```
list providers
```

Output that is similar to the following displays:

```
* ProviderID: {b5946137-7b9f-4925-af80-51abd60b20d5}
        Type: [1] VSS_PROV_SYSTEM
        Name: Microsoft Software Shadow Copy provider 1.0
        Version: 1.0.0.7
        CLSID: {65ee1dba-8ff4-4a58-ac1c-3470ee2f376a}
1 provider registered.
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

list shadows

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists persistent and existing non-persistent shadow copies that are on the system.

Syntax

```
list shadows {all | set <setID> | id <shadowID>}
```

Parameters

Expand table

Parameter	Description
all	Lists all shadow copies.
set `<setID>`	Lists shadow copies that belong to the specified Shadow Copy Set ID.
id `<shadowID>`	Lists any shadow copy with the specified shadow copy ID.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

list writers

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists writers that are on the system. If used without parameters, **list** displays the output for **list metadata** by default.

Syntax

```
list writers [metadata | detailed | status]
```

Parameters

Expand table

Parameter	Description
metadata	Lists the identity and status of writers, and displays metadata such as component details and excluded files. This is the default parameter.
detailed	Lists the same information as **metadata**, but also includes the full file list for all components.
status	Lists only the identity and status of registered writers.

Examples

To list only the identity and status of writers, type:

```
list writers status
```

Output that is similar to the following displays:

```
Listing writer status ...
* WRITER System Writer
        - Status: 5 (VSS_WS_WAITING_FOR_BACKUP_COMPLETE)
        - Writer Failure code: 0x00000000 (S_OK)
        - Writer ID: {e8132975-6f93-4464-a53e-1050253ae220}
        - Instance ID: {7e631031-c695-4229-9da1-a7de057e64cb}
* WRITER Shadow Copy Optimization Writer
        - Status: 1 (VSS_WS_STABLE)
        - Writer Failure code: 0x00000000 (S_OK)
        - Writer ID: {4dc3bdd4-ab48-4d07-adb0-3bee2926fd7f}
        - Instance ID: {9e362607-9794-4dd4-a7cd-b3d5de0aad20}
* WRITER Registry Writer
        - Status: 1 (VSS_WS_STABLE)
        - Writer Failure code: 0x00000000 (S_OK)
        - Writer ID: {afbab4a2-367d-4d15-a586-71dbb18f8485}
        - Instance ID: {e87ba7e3-f8d8-42d8-b2ee-c76ae26b98e8}
8 writers listed.
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Load metadata

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Loads a metadata .cab file prior to importing a transportable shadow copy or loads the writer metadata in the case of a restore. If used without parameters, **load metadata** displays help at the command prompt.

Syntax

```
load metadata [<drive>:][<path>]<metadata.cab>
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]`	Specifies the location of the metadata file.
metadata.cab	Specifies the metadata .cab file to load.

Remarks

- You can use the **import** command to import a transportable shadow copy based on the metadata specified by **load metadata**.
- You must run this command before the **begin restore** command, to load the selected writers and components for the restore.

Examples

To load a metadata file called metafile.cab from the default location, type:

```
load metadata metafile.cab
```

Related links

- Command-Line Syntax Key
- import diskshadow command
- begin restore command

Feedback

Was this page helpful? Yes No

mask

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes hardware shadow copies that were imported by using the **import** command.

Syntax

```
mask <shadowsetID>
```

Parameters

Expand table

Parameter	Description
shadowsetID	Removes shadow copies that belong to the specified Shadow Copy Set ID.

Remarks

- You can use an existing alias or an environment variable in place of *ShadowSetID*. Use **add** without parameters to see existing aliases.

Examples

To remove the imported shadow copy *%Import_1%*, type:

```
mask %Import_1%
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reset

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Resets DiskShadow.exe to the default state. This command is especially useful in separating compound DiskShadow operations, such as **create**, **import**, **backup**, or **restore**.

> [!IMPORTANT After you run this command, you will lose state information from commands, such as **add**, **set**, **load**, or **writer**. This command also releases IVssBackupComponent interfaces and loses non-persistent shadow copies.

Syntax

```
reset
```

Related links

- Command-Line Syntax Key
- create command
- import command
- backup command
- restore command
- add command
- set command
- load command
- writer command

Feedback

Was this page helpful? Yes No

reset session

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables you to reset (delete) a session on a Remote Desktop Session Host server. You should reset a session only when it malfunctions or appears to have stopped responding.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
reset session {<sessionname> | <sessionID>} [/server:<servername>] [/v]
```

Parameters

Expand table

Parameter	Description
`<sessionname>`	Specifies the name of the session that you want to reset. To determine the name of the session, use the query session command.
`<sessionID>`	Specifies the ID of the session to reset.
/server: `<servername>`	Specifies the terminal server containing the session that you want to reset. Otherwise, it uses the current Remote Desktop Session Host server. This parameter is required only if you use this command from a remote server.
/v	Displays information about the actions being performed.
/?	Displays help at the command prompt.

Remarks

- You can always reset your own sessions, but you must have **Full Control** access permission to reset another user's session. Be aware that resetting a user's session without warning the user can result in the loss of data at the session.

Examples

To reset the session designated *rdp-tcp#6*, type:

```
reset session rdp-tcp#6
```

To reset the session that uses *session ID 3*, type:

```
reset session 3
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services Command Reference

Feedback

Was this page helpful? Yes No

revert

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Reverts a volume back to a specified shadow copy. This is supported only for shadow copies in the CLIENTACCESSIBLE context. These shadow copies are persistent and can only be made by the system provider. If used without parameters, **revert** displays help at the command prompt.

Syntax

```
revert <shadowcopyID>
```

Parameters

Expand table

Parameter	Description
`<shadowcopyID>`	Specifies the shadow copy ID to revert the volume to. If you don't use this parameter, the command displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

set commands (shadow copy creation)

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the context, options, verbose mode, and metadata file for shadow copy creation. If used without parameters, **set** lists all current settings.

Syntax

```
set
set context
set option
set verbose
set metadata
```

Parameters

Expand table

Parameters	Description
set context	Sets the context for shadow copy creation.
set metadata	Sets the name and location of the shadow creation metadata file.
set option	Sets options for shadow copy creation.
set verbose	Turns the verbose output mode on or off.
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Set context

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the context for shadow copy creation. If used without parameters, **set context** displays help at the command prompt.

Syntax

```
set context {clientaccessible | persistent [nowriters] | volatile [nowriters]}
```

Parameters

Expand table

Parameter	Description
clientaccessible	Specifies that the shadow copy is usable by client versions of Windows. This context is persistent by default.
persistent	Specifies that the shadow copy persists across program exit, reset, or restart.
volatile	Deletes the shadow copy on exit or reset.
nowriters	Specifies that all writers are excluded.

Examples

To prevent shadow copies from being deleted when you exit DiskShadow, type:

```
set context persistent
```

Related links

- Command-Line Syntax Key
- set metadata command
- set option command
- set verbose command

Feedback

Was this page helpful? Yes No

set metadata

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the name and location of the shadow creation metadata file used to transfer shadow copies from one computer to another. If used without parameters, **set metadata** displays help at the command prompt.

Syntax

```
set metadata [<drive>:][<path>]<metadata.cab>
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]`	Specifies the location to create the metadata file.
`<metadata.cab>`	Specifies the name of the cab file to store shadow creation metadata.

Related links

- Command-Line Syntax Key
- set context command
- set option command
- set verbose command

Feedback

Was this page helpful? Yes No

set option

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the options for shadow copy creation. If used without parameters, **set option** displays help at the command prompt.

Syntax

```
set option {[differential | plex] [transportable] [[rollbackrecover] [txfrecover] | [noautorecover]]}
```

Parameters

Expand table

Parameter	Description
[differential]	Specifies to create a point-in-time snapshot of specified volumes.
[plex]	Specifies to create a point-in-time clone copy of the data on a specified volume.
[transportable]	Specifies that the shadow copy is not to be imported yet. The metadata .cab file can later be used to import the shadow copy to the same or a different computer.
[rollbackrecover]	Signals writers to use *autorecover* during the **PostSnapshot** event. This is useful if the shadow copy will be used for rollback (for example, with data mining).
[txfrecover]	Requests VSS to make the shadow copy transactionally consistent during creation.
[noautorecover]	Stops writers and the file system from performing any recovery changes to the shadow copy to a transactionally consistent state. **Noautorecover** can't be used with **txfrecover** or **rollbackrecover**.

Related links

- Command-Line Syntax Key
- set context command
- set metadata command
- set verbose command

Feedback

Was this page helpful? Yes No

Set verbose

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Specifies whether verbose output is provided during shadow copy creation. If used without parameters, **set verbose** displays help at the command prompt.

Syntax

```
set verbose {on | off}
```

Parameters

Expand table

Parameter	Description
on	Turns on verbose output logging during the shadow copy creation process. If verbose mode is on, **set** provides details of writer inclusion or exclusion and details of metadata compression and extraction.
off	Turns off verbose output logging during the shadow copy creation process.

Related links

- Command-Line Syntax Key
- set context command
- set metadata command
- set option command

Feedback

Was this page helpful? Yes No

Simulate restore

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Tests whether writer involvement in restore sessions will be successful on the computer without issuing **PreRestore** or **PostRestore** events to writers.

> **Note**
>
> A DiskShadow metadata file must be selected for the **simulate restore** command to succeed. Use the load metadata command to load the selected writers and components for the restore.

Syntax

```
simulate restore
```

Related links

- Command-Line Syntax Key
- load metadata command

Feedback

Was this page helpful? Yes No

unexpose

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Unexposes a shadow copy that was exposed by using the expose command. The exposed shadow copy can be specified by its Shadow ID, drive letter, share, or mount point.

Syntax

```
unexpose {<shadowID> | <drive:> | <share> | <mountpoint>}
```

Parameters

Expand table

Parameter	Description
`<shadowID>`	Displays the shadow copy specified by the given Shadow ID. You can use an existing alias or an environment variable in place of `<shadowID>`. Use the add command without parameters to see all existing aliases.
`<drive:>`	Displays the shadow copy associated with the specified drive letter (for example, drive P).
`<share>`	Displays the shadow copy associated with the specified share (for example, `\\MachineName`).
`<mountpoint>`	Displays the shadow copy associated with the specified mount point (for example, `C:\shadowcopy\`).
add	Used without parameters will show you the existing aliases.

Examples

To unexpose the shadow copy associated with *drive P:*, type:

```
unexpose P:
```

Related links

- Command-Line Syntax Key
- add command
- expose command

Feedback

Was this page helpful? Yes No

writer

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Verifies that a writer or component is included or excludes a writer or component from the backup or restore procedure. If used without parameters, **writer** displays help at the command prompt.

Syntax

```
writer verify [writer> | <component>]
writer exclude [<writer> | <component>]
```

Parameters

Expand table

Parameter	Description
verify	Verifies that the specified writer or component is included in the backup or restore procedure. The backup or restore procedure will fail if the writer or component is not included.
exclude	Excludes the specified writer or component from the backup or restore procedure.

Examples

To verify a writer by specifying its GUID (for this example, 4dc3bdd4-ab48-4d07-adb0-3bee2926fd7f), type:

```
writer verify {4dc3bdd4-ab48-4d07-adb0-3bee2926fd7f}
```

To exclude a writer with the name *System Writer*, type:

```
writer exclude System Writer
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

dispdiag

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Logs display information to a file.

Syntax

```
dispdiag [-testacpi] [-d] [-delay <seconds>] [-out <filepath>]
```

Parameters

Expand table

Parameter	Description
- testacpi	Runs hotkey diagnostics test. Displays the key name, code and scan code for any key pressed during the test.
-d	Generates a dump file with test results.
-delay `<seconds>`	Delays the collection of data by specified time in *seconds*.
-out `<filepath>`	Specifies path and filename to save collected data. This must be the last parameter.
-?	Displays available command parameters and provides help for using them.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Dnscmd

Article • 03/21/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

A command-line interface for managing DNS servers. This utility is useful in scripting batch files to help automate routine DNS management tasks, or to perform simple unattended setup and configuration of new DNS servers on your network.

Syntax

```
dnscmd <servername> <command> [<command parameters>]
```

Parameters

Expand table

Parameter	Description
`<servername>`	The IP address or host name of a remote or local DNS server.

dnscmd /ageallrecords command

Sets the current time on a time stamp on resource records at a specified zone or node on a DNS server.

Syntax

```
dnscmd [<servername>] /ageallrecords <zonename>[<nodename>] | [/tree]|[/f]
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the DNS server that the administrator plans to manage, represented by IP address, fully qualified domain name (FQDN), or Host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the FQDN of the zone.
`<nodename>`	Specifies a specific node or subtree in the zone, using the following: • @ for root zone or FQDN • The FQDN of a node (the name with a period (.) at the end) • A single label for the name relative to the zone root.
/tree	Specifies that all child nodes also receive the time stamp.
/f	Runs the command without asking for confirmation.

Remarks

- The **ageallrecords** command is for backward compatibility between the current version of DNS and previous releases of DNS in which aging and scavenging weren't supported. It adds a time stamp with the current time to resource records that don't have a time stamp, and it sets the current time on resource records that do have a time stamp.
- Record scavenging doesn't occur unless the records are time stamped. Name server (NS) resource records, start of authority (SOA) resource records, and Windows Internet Name Service (WINS) resource records aren't included in the scavenging process, and they aren't time stamped even when the **ageallrecords** command runs.
- This command fails unless scavenging is enabled for the DNS server and the zone. For information about how to enable scavenging for the zone, see the **aging** parameter, within the syntax of the `dnscmd /config` command in this article.
- The addition of a time stamp to DNS resource records makes them incompatible with DNS servers that run on operating systems other than Windows Server. A time stamp added by using the **ageallrecords** command can't be reversed.
- If none of the optional parameters are specified, the command returns all resource records at the specified node. If a value is specified for at least one of the optional parameters, **dnscmd** enumerates only the resource records that correspond to the value or values that are specified in the optional parameter or parameters.

Examples

Example 1: Set the current time on a time stamp to resource records.

dnscmd /clearcache command

Clears the DNS cache memory of resource records on the specified DNS server.

Syntax

```
dnscmd [<servername>] /clearcache
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.

Example

```
dnscmd dnssvr1.contoso.com /clearcache
```

dnscmd /config command

Changes values in the registry for the DNS server and individual zones. This command also modifies the configuration of the specified server. Accepts server-level and zone-level settings.

> **⊗ Caution**
>
> Don't edit the registry directly unless you have no alternative. The registry editor bypasses standard safeguards, allowing settings that can degrade performance,

damage your system, or even require you to reinstall Windows. You can safely alter most registry settings by using the programs in Control Panel or Microsoft Management Console (mmc). If you must edit the registry directly, back it up first. Read the registry editor help for more information.

Server-level syntax

```
dnscmd [<servername>] /config <parameter>
```

Parameters

> **Note**
>
> This article contains references to the term slave, a term that Microsoft no longer uses. When the term is removed from the software, we'll remove it from this article.

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server that you're planning to manage, represented by local computer syntax, IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<parameter>`	Specify a setting and, as an option, a value. Parameter values use this syntax: *parameter* [*value*].
/addressanswerlimit `[0|5-28]`	Specifies the maximum number of host records that a DNS server can send in response to a query. The value can be zero (0), or it can be in the range of 5 through 28 records. The default value is zero (0).
/bindsecondaries `[0|1]`	Changes the format of the zone transfer so that it can achieve maximum compression and efficiency. Accepts the values: • **0** - Uses maximum compression and is compatible with BIND versions 4.9.4 and later only • **1** - Sends only one resource record per message to non-Microsoft DNS servers and is compatible with

Parameters	Description
	BIND versions earlier than 4.9.4. This is the default setting.
/bootmethod `[0|1|2|3]`	Determines the source from which the DNS server gets its configuration information. Accepts the values: • **0** - Clears the source of configuration information. • **1** - Loads from the BIND file that is located in the DNS directory, which is `%systemroot%\System32\DNS` by default. • **2** - Loads from the registry. • **3** - Loads from AD DS and the registry. This is the default setting.
/defaultagingstate `[0|1]`	Determines whether the DNS scavenging feature is enabled by default on newly created zones. Accepts the values: • **0** - Disables scavenging. This is the default setting. • **1** - Enables scavenging.
/defaultnorefreshinterval `[0x1-0xFFFFFFFF|0xA8]`	Sets a period of time in which no refreshes are accepted for dynamically updated records. Zones on the server inherit this value automatically. To change the default value, type a value in the range of **0x1-0xFFFFFFFF**. The default value from the server is **0xA8**.
/defaultrefreshinterval `[0x1-0xFFFFFFFF|0xA8]`	Sets a period of time that is allowed for dynamic updates to DNS records. Zones on the server inherit this value automatically. To change the default value, type a value in the range of **0x1-0xFFFFFFFF**. The default value from the server is **0xA8**.
/disableautoreversezones `[0|1]`	Enables or disables the automatic creation of reverse lookup zones. Reverse lookup zones provide resolution of Internet Protocol (IP) addresses to DNS domain names. Accepts the values: • **0** - Enables the automatic creation of reverse lookup zones. This is the default setting. • **1** - Disables the automatic creation of reverse lookup zones.

Parameters	Description
/disablensrecordsautocreation `[0|1]`	Specifies whether the DNS server automatically creates name server (NS) resource records for zones that it hosts. Accepts the values: • **0** - Automatically creates name server (NS) resource records for zones that the DNS server hosts. • **1** - Doesn't automatically create name server (NS) resource records for zones that the DNS server hosts.
/dspollinginterval `<seconds>`	Specifies in seconds how often the DNS server polls AD DS for changes in the AD integrated zones. The minimum accepted value is **30** seconds. If a value isn't specified after this parameter, the default value is set to **0xB4** (3 minutes or 180 seconds).
/dstombstoneinterval `<seconds>`	The amount of time in seconds to retain deleted records in AD DS. This value should be limited to the range from **0x3F480** (3 days or 259,200 seconds) to **0x49D400** (8 weeks or 4,147,200 seconds). The default value should be **0x127500** (14 days or 1,209,600 seconds) if no value is specified for the tombstoneLifetime attribute of the Directory Services object.
/ednscachetimeout `[3600-15724800]`	Specifies the number of seconds that extended DNS (EDNS) information is cached. The minimum value is **3600**, and the maximum value is **15,724,800**. The default value is **604,800** seconds (one week).
/enableednsprobes `[0|1]`	Enables or disables the server to probe other servers to determine if they support EDNS. Accepts the values: • **0** - Disables active support for EDNS probes. • **1** - Enables active support for EDNS probes.
/enablednssec `[0|1]`	Enables or disables support for DNS Security Extensions (DNSSEC). Accepts the values: • **0** - Disables DNSSEC. • **1** - Enables DNSSEC.
/enableglobalnamessupport `[0|1]`	Enables or disables support for the GlobalNames zone. The GlobalNames zone supports resolution of single-label DNS names across a forest. Accepts the values: • **0** - Disables support for the GlobalNames zone. When you set the value of this command to 0, the DNS Server service doesn't resolve single-label names in the GlobalNames zone. • **1** - Enables support for the GlobalNames zone. When you set the value of this command to 1, the DNS Server

Parameters	Description
	service resolves single-label names in the GlobalNames zone.
/enableglobalqueryblocklist `[0|1]`	Enables or disables support for the global query block list that blocks name resolution for names in the list. The DNS Server service creates and enables the global query block list by default when the service starts the first time. To view the current global query block list, use the dnscmd /info **/globalqueryblocklist** command. Accepts the values: • **0** - Disables support for the global query block list. When you set the value of this command to 0, the DNS Server service responds to queries for names in the block list. • **1** - Enables support for the global query block list. When you set the value of this command to 1, the DNS Server service doesn't respond to queries for names in the block list.
/eventloglevel `[0|1|2|4]`	Determines which events are logged in the DNS server log in Event Viewer. Accepts the values: • **0** - Logs no events. • **1** - Logs only errors. • **2** - Logs only errors and warnings. • **4** - Logs errors, warnings, and informational events. This is the default setting.
/forwarddelegations `[0|1]`	Determines how the DNS server handles a query for a delegated subzone. These queries can be sent either to the subzone that is referred to in the query or to the list of forwarders that is named for the DNS server. Entries in the setting are used only when forwarding is enabled. Accepts the values: • **0** - Automatically sends queries that refer to delegated subzones to the appropriate subzone. This is the default setting. • **1** - Forwards queries that refer to the delegated subzone to the existing forwarders.
/forwardingtimeout `[<seconds>]`	Determines how many seconds (**0x1-0xFFFFFFFF**) a DNS server waits for a forwarder to respond before trying another forwarder. The default value is **0x5**, which is 5 seconds.
/globalneamesqueryorder `[0|1]`	Specifies whether the DNS Server service looks first in the GlobalNames zone or local zones when it resolves names. Accepts the values:

Parameters	Description
	• **0** - The DNS Server service attempts to resolve names by querying the GlobalNames zone before it queries the zones for which it's authoritative. • **1** - The DNS Server service attempts to resolve names by querying the zones for which it's authoritative before it queries the GlobalNames zone.
/globalqueryblocklist `[[<name> [<name>]...]`	Replaces the current global query block list with a list of the names that you specify. If you don't specify any names, this command clears the block list. By default, the global query block list contains the following items: • isatap • wpad The DNS Server service can remove either or both of these names when it starts the first time, if it finds these names in an existing zone.
/isslave `[0|1]`	Determines how the DNS server responds when queries that it forwards receive no response. Accepts the values: • **0** - Specifies that the DNS server isn't a subordinate. If the forwarder doesn't respond, the DNS server attempts to resolve the query itself. This is the default setting. • **1** - Specifies that the DNS server is a subordinate. If the forwarder doesn't respond, the DNS server terminates the search and sends a failure message to the resolver.
/localnetpriority `[0|1]`	Determines the order in which host records are returned when the DNS server has multiple host records for the same name. Accepts the values: • **0** - Returns the records in the order in which they're listed in the DNS database. • **1** - Returns the records that have similar IP network addresses first. This is the default setting.
/logfilemaxsize `[<size>]`	Specifies the maximum size in bytes (**0x10000-0xFFFFFFFF**) of the Dns.log file. When the file reaches its maximum size, DNS overwrites the oldest events. The default size is **0x400000**, which is 4 megabytes (MB).
/logfilepath `[<path+logfilename>]`	Specifies the path of the Dns.log file. The default path is `%systemroot%\System32\Dns\Dns.log`. You can specify a different path by using the format `path+logfilename`.

Parameters	Description
/logipfilterlist `<IPaddress> [, <IPaddress>...]`	Specifies which packets are logged in the debug log file. The entries are a list of IP addresses. Only packets going to and from the IP addresses in the list are logged.
/loglevel `[<eventtype>]`	Determines which types of events are recorded in the Dns.log file. Each event type is represented by a hexadecimal number. If you want more than one event in the log, use hexadecimal addition to add the values, and then enter the sum. Accepts the values: • **0x0** - The DNS server doesn't create a log. This is the default entry. • **0x10** - Logs queries and notifications. • **0x20** - Logs updates. • **0xFE** - Logs nonquery transactions. • **0x100** - Logs question transactions. • **0x200** - Logs answers. • **0x1000** - Logs send packets. • **0x2000** - Logs receive packets. • **0x4000** - Logs User Datagram Protocol (UDP) packets. • **0x8000** - Logs Transmission Control Protocol (TCP) packets. • **0xFFFF** - Logs all packets. • **0x10000** - Logs active directory write transactions. • **0x20000** - Logs active directory update transactions. • **0x1000000** - Logs full packets. • **0x80000000** - Logs write-through transactions. •
/maxcachesize	Specifies the maximum size, in kilobytes (KB), of the DNS server s memory cache.
/maxcachettl `[<seconds>]`	Determines how many seconds (**0x0-0xFFFFFFFF**) a record is saved in the cache. If the **0x0** setting is used, the DNS server doesn't cache records. The default setting is **0x15180** (86,400 seconds or 1 day).
/maxnegativecachettl `[<seconds>]`	Specifies how many seconds (**0x1-0xFFFFFFFF**) an entry that records a negative answer to a query remains stored in the DNS cache. The default setting is **0x384** (900 seconds).
/namecheckflag `[0|1|2|3]`	Specifies which character standard is used when checking DNS names. Accepts the values: • **0** - Uses ANSI characters that comply with Internet Engineering Task force (IETF) Request for Comments (Rfcs). • **1** - Uses ANSI characters that don't necessarily comply with IETF Rfcs.

Parameters	Description
	• **2** - Uses multibyte UCS Transformation format 8 (UTF-8) characters. This is the default setting. • **3** - Uses all characters.
/norecursion `[0|1]`	Determines whether a DNS server performs recursive name resolution. Accepts the values: • **0** - The DNS server performs recursive name resolution if it's requested in a query. This is the default setting. • **1** - The DNS server doesn't perform recursive name resolution.
/notcp	This parameter is obsolete, and it has no effect in current versions of Windows Server.
/recursionretry `[<seconds>]`	Determines the number of seconds (**0x1-0xFFFFFFFF**) that a DNS server waits before again trying to contact a remote server. The default setting is **0x3** (three seconds). This value should be increased when recursion occurs over a slow wide area network (WAN) link.
/recursiontimeout `[<seconds>]`	Determines the number of seconds (**0x1-0xFFFFFFFF**) that a DNS server waits before discontinuing attempts to contact a remote server. The settings range from **0x1** through **0xFFFFFFFF**. The default setting is **0xF** (15 seconds). This value should be increased when recursion occurs over a slow WAN link.
/roundrobin `[0|1]`	Determines the order in which host records are returned when a server has multiple host records for the same name. Accepts the values: • **0** - The DNS server doesn't use round robin. Instead, it returns the first record to every query. • **1** - The DNS server rotates among the records that it returns from the top to the bottom of the list of matching records. This is the default setting.
/rpcprotocol `[0x0|0x1|0x2|0x4|0xFFFFFFFF]`	Specifies the protocol that remote procedure call (RPC) uses when it makes a connection from the DNS server. Accepts the values: • **0x0** - Disables RPC for DNS. • **0x01** - Uses TCP/IP • **0x2** - Uses named pipes. • **0x4** - Uses local procedure call (LPC). • **0xFFFFFFFF** - All protocols. This is the default setting.

Parameters	Description
/scavenginginterval `[<hours>]`	Determines whether the scavenging feature for the DNS server is enabled, and sets the number of hours (**0x0-0xFFFFFFFF**) between scavenging cycles. The default setting is **0x0**, which disables scavenging for the DNS server. A setting greater than **0x0** enables scavenging for the server and sets the number of hours between scavenging cycles.
/secureresponses `[0|1]`	Determines whether DNS filters records that are saved in a cache. Accepts the values: • **0** - Saves all responses to name queries to a cache. This is the default setting. • **1** - Saves only the records that belong to the same DNS subtree to a cache.
/sendport `[<port>]`	Specifies the port number (**0x0-0xFFFFFFFF**) that DNS uses to send recursive queries to other DNS servers. The default setting is **0x0**, which means that the port number is selected randomly.
/serverlevelplugindll `[<dllpath>]`	Specifies the path of a custom plug-in. When Dllpath specifies the fully qualified path name of a valid DNS server plug-in, the DNS server calls functions in the plug-in to resolve name queries that are outside the scope of all locally hosted zones. If a queried name is out of the scope of the plug-in, the DNS server performs name resolution using forwarding or recursion, as configured. If Dllpath isn't specified, the DNS server ceases to use a custom plug-in if a custom plug-in was previously configured.
/strictfileparsing `[0|1]`	Determines a DNS server's behavior when it encounters an erroneous record while loading a zone. Accepts the values: • **0** - The DNS server continues to load the zone even if the server encounters an erroneous record. The error is recorded in the DNS log. This is the default setting. • **1** - The DNS server stops loading the zone, and it records the error in the DNS log.
/updateoptions `<RecordValue>`	Prohibits dynamic updates of specified types of records. If you want more than one record type to be prohibited in the log, use hexadecimal addition to add the values, and then enter the sum. Accepts the values: • **0x0** - Doesn't restrict any record types. • **0x1** - Excludes start of authority (SOA) resource records. • **0x2** - Excludes name server (NS) resource records. • **0x4** - Excludes delegation of name server (NS) resource records. • **0x8** - Excludes server host records.

Parameters	Description
	• **0x100** - During secure dynamic update, excludes start of authority (SOA) resource records. • **0x200** - During secure dynamic update, excludes root name server (NS) resource records. • **0x30F** - During standard dynamic update, excludes name server (NS) resource records, start of authority (SOA) resource records, and server host records. During secure dynamic update, excludes root name server (NS) resource records and start of authority (SOA) resource records. Allows delegations and server host updates. • **0x400** - During secure dynamic update, excludes delegation name server (NS) resource records. • **0x800** - During secure dynamic update, excludes server host records. • **0x1000000** - Excludes delegation signer (DS) records. • **0x80000000** - Disables DNS dynamic update.
/writeauthorityns `[0|1]`	Determines when the DNS server writes name server (NS) resource records in the Authority section of a response. Accepts the values: • **0** - Writes name server (NS) resource records in the Authority section of referrals only. This setting complies with Rfc 1034, Domain names concepts and facilities, and with Rfc 2181, Clarifications to the DNS Specification. This is the default setting. • **1** - Writes name server (NS) resource records in the Authority section of all successful authoritative responses.
/xfrconnecttimeout `[<seconds>]`	Determines the number of seconds (**0x0-0xFFFFFFFF**) a primary DNS server waits for a transfer response from its secondary server. The default value is **0x1E** (30 seconds). After the time-out value expires, the connection is terminated.

Zone-level syntax

Modifies the configuration of the specified zone. The zone name must be specified only for zone-level parameters.

```
dnscmd /config <parameters>
```

Parameters

Expand table

Parameters	Description
`<parameter>`	Specify a setting, a zone name, and, as an option, a value. Parameter values use this syntax: `zonename parameter [value]`.
/aging `<zonename>`	Enables or disables scavenging in a specific zone.
/allownsrecordsautocreation `<zonename>` `[value]`	Overrides the DNS server's name server (NS) resource record autocreation setting. Name server (NS) resource records that were previously registered for this zone aren't affected. Therefore, you must remove them manually if you don't want them.
/allowupdate `<zonename>`	Determines whether the specified zone accepts dynamic updates.
/forwarderslave `<zonename>`	Overrides the DNS server **/isslave** setting.
/forwardertimeout `<zonename>`	Determines how many seconds a DNS zone waits for a forwarder to respond before trying another forwarder. This value overrides the value that is set at the server level.
/norefreshinterval `<zonename>`	Sets a time interval for a zone during which no refreshes can dynamically update DNS records in a specified zone.
/refreshinterval `<zonename>`	Sets a time interval for a zone during which refreshes can dynamically update DNS records in a specified zone.
/securesecondaries `<zonename>`	Determines which secondary servers can receive zone updates from the primary server for this zone.

dnscmd /createbuiltindirectorypartitions command

Creates a DNS application directory partition. When DNS is installed, an application directory partition for the service is created at the forest and domain levels. Use this command to create DNS application directory partitions that were deleted or never created. With no parameter, this command creates a built-in DNS directory partition for the domain.

Syntax

```
dnscmd [<servername>] /createbuiltindirectorypartitions [/forest]
[/alldomains]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
/forest	Creates a DNS directory partition for the forest.
/alldomains	Creates DNS partitions for all domains in the forest.

dnscmd /createdirectorypartition command

Creates a DNS application directory partition. When DNS is installed, an application directory partition for the service is created at the forest and domain levels. This operation creates additional DNS application directory partitions.

Syntax

```
dnscmd [<servername>] /createdirectorypartition <partitionFQDN>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<partitionFQDN>`	The FQDN of the DNS application directory partition that will be created.

dnscmd /deletedirectorypartition command

Removes an existing DNS application directory partition.

Syntax

```
dnscmd [<servername>] /deletedirectorypartition <partitionFQDN>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<partitionFQDN>`	The FQDN of the DNS application directory partition that will be removed.

dnscmd /directorypartitioninfo command

Lists information about a specified DNS application directory partition.

Syntax

```
dnscmd [<servername>] /directorypartitioninfo <partitionFQDN> [/detail]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<partitionFQDN>`	The FQDN of the DNS application directory partition.
/detail	Lists all information about the application directory partition.

dnscmd /enlistdirectorypartition command

Adds the DNS server to the specified directory partition's replica set.

Syntax

```
dnscmd [<servername>] /enlistdirectorypartition <partitionFQDN>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<partitionFQDN>`	The FQDN of the DNS application directory partition.

dnscmd /enumdirectorypartitions command

Lists the DNS application directory partitions for the specified server.

Syntax

```
dnscmd [<servername>] /enumdirectorypartitions [/custom]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
/custom	Lists only user-created directory partitions.

dnscmd /enumrecords command

Lists the resource records of a specified node in a DNS zone.

Syntax

```
dnscmd [<servername>] /enumrecords <zonename> <nodename> [/type <rrtype>
<rrdata>] [/authority] [/glue] [/additional] [/node | /child |
/startchild<childname>] [/continue | /detail]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
/enumrecords	Lists resource records in the specified zone.
`<zonename>`	Specifies the name of the zone to which the resource records belong.
`<nodename>`	Specifies the name of the node of the resource records.
`[/type <rrtype> <rrdata>]`	Specifies the type of resource records to be listed and the type of data that is expected. Accepts the values: • `<rrtype>` - Specifies the type of resource records to be listed. • `<rrdata>` - Specifies the type of data that is expected record.
/authority	Includes authoritative data.
/glue	Includes glue data.
/additional	Includes all additional information about the listed resource records.
/node	Lists only the resource records of the specified node.
/child	Lists only the resource records of a specified child domain.
/startchild `<childname>`	Begins the list at the specified child domain.
/continue	Lists only the resource records with their type and data.
/detail	Lists all information about the resource records.

Example

```
dnscmd /enumrecords test.contoso.com test /additional
```

dnscmd /enumzones command

Lists the zones that exist on the specified DNS server. The **enumzones** parameters act as filters on the list of zones. If no filters are specified, a complete list of zones is returned. When a filter is specified, only the zones that meet that filter's criteria are included in the returned list of zones.

Syntax

```
dnscmd [<servername>] /enumzones [/primary | /secondary | /forwarder | /stub
| /cache | /auto-created] [/forward | /reverse | /ds | /file]
[/domaindirectorypartition | /forestdirectorypartition |
/customdirectorypartition | /legacydirectorypartition | /directorypartition
<partitionFQDN>]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
/primary	Lists all zones that are either standard primary zones or active directory integrated zones.
/secondary	Lists all standard secondary zones.
/forwarder	Lists zones that forward unresolved queries to another DNS server.
/stub	Lists all stub zones.
/cache	Lists only the zones that are loaded into the cache.

Parameters	Description
/auto-created]	Lists the zones that were created automatically during the DNS server installation.
/forward	Lists forward lookup zones.
/reverse	Lists reverse lookup zones.
/ds	Lists active directory integrated zones.
/file	Lists zones that are backed by files.
/domaindirectorypartition	Lists zones that are stored in the domain directory partition.
/forestdirectorypartition	Lists zones that are stored in the forest DNS application directory partition.
/customdirectorypartition	Lists all zones that are stored in a user-defined application directory partition.
/legacydirectorypartition	Lists all zones that are stored in the domain directory partition.
/directorypartition `<partitionFQDN>`	Lists all zones that are stored in the specified directory partition.

Examples

- Example 2: Display a complete list of zones on a DNS server)
- Example 3: Display a list of autocreated zones on a DNS server

dnscmd /exportsettings command

Creates a text file that lists the configuration details of a DNS server. The text file is named *DnsSettings.txt*. It's located in the `%systemroot%\system32\dns` directory of the server. You can use the information in the file that **dnscmd /exportsettings** creates to troubleshoot configuration problems or to ensure that you have configured multiple servers identically.

Syntax

```
dnscmd [<servername>] /exportsettings
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.

dnscmd /info command

Displays settings from the DNS section of the registry of the specified server `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\DNS\Parameters`. To display zone-level registry settings, use the `dnscmd zoneinfo` command.

Syntax

```
dnscmd [<servername>] /info [<settings>]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<settings>`	Any setting that the **info** command returns can be specified individually. If a setting isn't specified, a report of common settings is returned.

Example

- Example 4: Display the IsSlave setting from a DNS server
- Example 5: Display the RecursionTimeout setting from a DNS server

dnscmd /ipvalidate command

Tests whether an IP address identifies a functioning DNS server or whether the DNS server can act as a forwarder, a root hint server, or a primary server for a specific zone.

Syntax

```
dnscmd [<servername>] /ipvalidate <context> [<zonename>] [[<IPaddress>]]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<context>`	Specifies the type of test to perform. You can specify any of the following tests: • **/dnsservers** - Tests that the computers with the addresses that you specify are functioning DNS servers. • **/forwarders** - Tests that the addresses that you specify identify DNS servers that can act as forwarders. • **/roothints** - Tests that the addresses that you specify identify DNS servers that can act as root hint name servers. • **/zonemasters** - Tests that the addresses that you specify identify DNS servers that are primary servers for *zonename*.
`<zonename>`	Identifies the zone. Use this parameter with the **/zonemasters** parameter.
`<IPaddress>`	Specifies the IP addresses that the command tests.

Examples

```
nscmd dnssvr1.contoso.com /ipvalidate /dnsservers 10.0.0.1 10.0.0.2
dnscmd dnssvr1.contoso.com /ipvalidate /zonemasters corp.contoso.com
10.0.0.2
```

dnscmd /nodedelete command

Deletes all records for a specified host.

Syntax

```
dnscmd [<servername>] /nodedelete <zonename> <nodename> [/tree] [/f]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone.
`<nodename>`	Specifies the host name of the node to delete.
/tree	Deletes all the child records.
/f	Executes the command without asking for confirmation.

Example

Example 6: Delete the records from a node.

dnscmd /recordadd command

Adds a record to a specified zone in a DNS server.

Syntax

```
dnscmd [<servername>] /recordadd <zonename> <nodename> <rrtype> <rrdata>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the zone in which the record resides.
`<nodename>`	Specifies a specific node in the zone.
`<rrtype>`	Specifies the type of record to be added.
`<rrdata>`	Specifies the type of data that is expected.

> **Note**
>
> After you add a record, make sure that you use the correct data type and data format. For a list of resource record types and the appropriate data types, see Dnscmd Examples.

Examples

```
dnscmd dnssvr1.contoso.com /recordadd test A 10.0.0.5
dnscmd /recordadd test.contoso.com test MX 10 mailserver.test.contoso.com
```

dnscmd /recorddelete command

Deletes a resource record to a specified zone.

Syntax

```
dnscmd [<servername>] /recorddelete <zonename> <nodename> <rrtype> <rrdata>
[/f]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the zone in which the resource record resides.
`<nodename>`	Specifies a name of the host.
`<rrtype>`	Specifies the type of resource record to be deleted.
`<rrdata>`	Specifies the type of data that is expected.
/f	Executes the command without asking for confirmation. Because nodes can have more than one resource record, this command requires you to be specific about the type of resource record that you want to delete. If you specify a data type and you don't specify a type of resource record data, all records with that specific data type for the specified node are deleted.

Examples

```
dnscmd /recorddelete test.contoso.com test MX 10 mailserver.test.contoso.com
```

dnscmd /resetforwarders command

Selects or resets the IP addresses to which the DNS server forwards DNS queries when it can't resolve them locally.

Syntax

```
dnscmd [<servername>] /resetforwarders <IPaddress> [,<IPaddress>]...]
[/timeout <timeout>] [/slave | /noslave]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<IPaddress>`	Lists the IP addresses to which the DNS server forwards unresolved queries.
/timeout `<timeout>`	Sets the number of seconds that the DNS server waits for a response from the forwarder. By default, this value is five seconds.
/slave	Prevents the DNS server from performing its own iterative queries if the forwarder fails to resolve a query.
/noslave	Allows the DNS server to perform its own iterative queries if the forwarder fails to resolve a query. This is the default setting.
/f	Executes the command without asking for confirmation. Because nodes can have more than one resource record, this command requires you to be specific about the type of resource record that you want to delete. If you specify a data type and you don't specify a type of resource record data, all records with that specific data type for the specified node are deleted.

Remarks

- By default, a DNS server performs iterative queries when it can't resolve a query.
- Setting IP addresses by using the **resetforwarders** command causes the DNS server to perform recursive queries to the DNS servers at the specified IP addresses. If the forwarders don't resolve the query, the DNS server can then perform its own iterative queries.
- If the **/slave** parameter is used, the DNS server doesn't perform its own iterative queries. This means that the DNS server forwards unresolved queries only to the DNS servers in the list, and it doesn't attempt iterative queries if the forwarders don't resolve them. It's more efficient to set one IP address as a forwarder for a DNS server. You can use the **resetforwarders** command for internal servers in a network to forward their unresolved queries to one DNS server that has an external connection.
- Listing a forwarder's IP address twice causes the DNS server to attempt to forward to that server twice.

Examples

```
```

```
dnscmd dnssvr1.contoso.com /resetforwarders 10.0.0.1 /timeout 7 /slave
dnscmd dnssvr1.contoso.com /resetforwarders /noslave
```

dnscmd /resetlistenaddresses command

Specifies the IP addresses on a server that listens for DNS client requests. By default, all IP addresses on a DNS server listen for client DNS requests.

Syntax

```
dnscmd [<servername>] /resetlistenaddresses <listenaddress>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<listenaddress>`	Specifies an IP address on the DNS server that listens for DNS client requests. If no listen address is specified, all IP addresses on the server listen for client requests.

Examples

```
dnscmd dnssvr1.contoso.com /resetlistenaddresses 10.0.0.1
```

dnscmd /startscavenging command

Tells a DNS server to attempt an immediate search for stale resource records in a specified DNS server.

Syntax

```
dnscmd [<servername>] /startscavenging
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.

Remarks

- Successful completion of this command starts a scavenge immediately. If the scavenge fails, no warning message appears.
- Although the command to start the scavenge appears to complete successfully, the scavenge doesn't start unless the following preconditions are met:
  - Scavenging is enabled for both the server and the zone.
  - The zone is started.
  - The resource records have a time stamp.
- For information about how to enable scavenging for the server, see the **scavenginginterval** parameter under **Server-level syntax** in the **/config** section.
- For information about how to enable scavenging for the zone, see the **aging** parameter under **Zone-level syntax** in the **/config** section.
- For information about how to restart a paused zone, see the **zoneresume** parameter in this article.
- For information about how to check resource records for a time stamp, see the **ageallrecords** parameter in this article.

Examples

```
dnscmd dnssvr1.contoso.com /startscavenging
```

dnscmd /statistics command

Displays or clears data for a specified DNS server.

Syntax

```
dnscmd [<servername>] /statistics [<statid>] [/clear]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<statid>`	Specifies which statistic or combination of statistics to display. The **statistics** command displays counters that begin on the DNS server when it's started or resumed. An identification number is used to identify a statistic. If no statistic ID number is specified, all statistics display. The numbers that can be specified, along with the corresponding statistic that displays, can include: • **00000001** - Time • **00000002** - Query • **00000004** - Query2 • **00000008** - Recurse • **00000010** - Master • **00000020** - Secondary • **00000040** - WINS • **00000100** - Update • **00000200** - SkwanSec • **00000400** - Ds • **00010000** - Memory • **00100000** - PacketMem • **00040000** - Dbase • **00080000** - Records • **00200000** - NbstatMem • **/clear** - Resets the specified statistics counter to zero.

Examples

- Example 7:
- Example 8: Display NbstatMem statistics for a DNS server

dnscmd /unenlistdirectorypartition command

Removes the DNS server from the specified directory partition's replica set.

Syntax

```
dnscmd [<servername>] /unenlistdirectorypartition <partitionFQDN>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<partitionFQDN>`	The FQDN of the DNS application directory partition that will be removed.

dnscmd /writebackfiles command

Checks the DNS server memory for changes, and writes them to persistent storage. The **writebackfiles** command updates all dirty zones or a specified zone. A zone is dirty when there are changes in memory that haven't yet been written to persistent storage. This is a server-level operation that checks all zones. You can specify one zone in this operation or you can use the **zonewriteback** operation.

Syntax

```
dnscmd [<servername>] /writebackfiles <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be updated.

Examples

```
dnscmd dnssvr1.contoso.com /writebackfiles
```

dnscmd /zoneadd command

Adds a zone to the DNS server.

Syntax

```
dnscmd [<servername>] /zoneadd <zonename> <zonetype> [/dp <FQDN> | {/domain
| enterprise | legacy}]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone.
`<zonetype>`	Specifies the type of zone to create. Specifying a zone type of **/forwarder** or **/dsforwarder** creates a zone that performs conditional forwarding. Each zone type has different required parameters:

Parameters	Description
	• **/dsprimary** - Creates an active directory integrated zone. • **/primary /file** `<filename>` - Creates a standard primary zone, and specifies the name of the file that will store the zone information. • **/secondary** `<masterIPaddress> [<masterIPaddress>...]` - Creates a standard secondary zone. • **/stub** `<masterIPaddress> [<masterIPaddress>...]` **/file** `<filename>` - Creates a file-backed stub zone. • **/dsstub** `<masterIPaddress> [<masterIPaddress>...]` - Creates an active directory integrated stub zone. • **/forwarder** `<masterIPaddress> [<masterIPaddress>]` **... /file** `<filename>` - Specifies that the created zone forwards unresolved queries to another DNS server. • **/dsforwarder** - Specifies that the created active directory integrated zone forwards unresolved queries to another DNS server.
`<FQDN>`	Specifies FQDN of the directory partition.
/domain	Stores the zone on the domain directory partition.
/enterprise	Stores the zone on the enterprise directory partition.
/legacy	Stores the zone on a legacy directory partition.

Examples

```
dnscmd dnssvr1.contoso.com /zoneadd test.contoso.com /dsprimary
dnscmd dnssvr1.contoso.com /zoneadd secondtest.contoso.com /secondary
10.0.0.2
```

dnscmd /zonechangedirectorypartition command

Changes the directory partition on which the specified zone resides.

Syntax

```
dnscmd [<servername>] /zonechangedirectorypartition <zonename>
{[<newpartitionname>] | [<zonetype>]}
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	The FQDN of the current directory partition on which the zone resides.
`<newpartitionname>`	The FQDN of the directory partition that the zone will be moved to.
`<zonetype>`	Specifies the type of directory partition that the zone will be moved to.
/domain	Moves the zone to the built-in domain directory partition.
/forest	Moves the zone to the built-in forest directory partition.
/legacy	Moves the zone to the directory partition that is created for pre active directory domain controllers. These directory partitions aren't necessary for native mode.

dnscmd /zonedelete command

Deletes a specified zone.

Syntax

```
dnscmd [<servername>] /zonedelete <zonename> [/dsdel] [/f]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be deleted.
/dsdel	Deletes the zone from Azure Directory Domain Services (AD DS).

Parameters	Description
/f	Runs the command without asking for confirmation.

Examples

- Example 9: Delete a zone from a DNS server

dnscmd /zoneexport command

Creates a text file that lists the resource records of a specified zone. The **zoneexport** operation creates a file of resource records for an active directory integrated zone for troubleshooting purposes. By default, the file that this command creates is placed in the DNS directory, which is by default the `%systemroot%/System32/Dns` directory.

Syntax

```
dnscmd [<servername>] /zoneexport <zonename> <zoneexportfile>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone.
`<zoneexportfile>`	Specifies the name of the file to create.

Examples

- Example 10: Export zone resource records list to a file

dnscmd /zoneinfo

Displays settings from the section of the registry of the specified zone: `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\DNS\Parameters\Zones\<zonename>`

Syntax

```
dnscmd [<servername>] /zoneinfo <zonename> [<setting>]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone.
`<setting>`	You can individually specify any setting that the **zoneinfo** command returns. If you don't specify a setting, all settings are returned.

Remarks

- To display server-level registry settings, use the **/info** command.
- To see a list of settings that you can display with this command, see the **/config** command.

Examples

- Example 11: Display RefreshInterval setting from the registry
- Example 12: Display Aging setting from the registry

dnscmd /zonepause command

Pauses the specified zone, which then ignores query requests.

Syntax

```
dnscmd [<servername>] /zonepause <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be paused.

Remarks

- To resume a zone and make it available after it has been paused, use the **/zoneresume** command.

Examples

```
dnscmd dnssvr1.contoso.com /zonepause test.contoso.com
```

dnscmd /zoneprint command

Lists the records in a zone.

Syntax

```
dnscmd [<servername>] /zoneprint <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be listed.

dnscmd /zonerefresh command

Forces a secondary DNS zone to update from the master zone.

Syntax

```
dnscmd [<servername>] /zonerefresh <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be refreshed.

Remarks

- The **zonerefresh** command forces a check of the version number in the primary server s start of authority (SOA) resource record. If the version number on the primary server is higher than the secondary server's version number, a zone transfer is initiated that updates the secondary server. If the version number is the same, no zone transfer occurs.
- The forced check occurs by default every 15 minutes. To change the default, use the `dnscmd config refreshinterval` command.

Examples

```
dnscmd dnssvr1.contoso.com /zonerefresh test.contoso.com
```

dnscmd /zonereload command

Copies zone information from its source.

Syntax

```
dnscmd [<servername>] /zonereload <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be reloaded.

Remarks

- If the zone is active directory integrated, it reloads from Active Directory Domain Services (AD DS).
- If the zone is a standard file-backed zone, it reloads from a file.

Examples

```
dnscmd dnssvr1.contoso.com /zonereload test.contoso.com
```

dnscmd /zoneresetmasters command

Resets the IP addresses of the primary server that provides zone transfer information to a secondary zone.

Syntax

```
dnscmd [<servername>] /zoneresetmasters <zonename> [/local] [<IPaddress>
[<IPaddress>]...]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to be reset.
/local	Sets a local master list. This parameter is used for active directory integrated zones.
`<IPaddress>`	The IP addresses of the primary servers of the secondary zone.

Remarks

- This value is originally set when the secondary zone is created. Use the **zoneresetmasters** command on the secondary server. This value has no effect if it's set on the master DNS server.

Examples

```
dnscmd dnssvr1.contoso.com /zoneresetmasters test.contoso.com 10.0.0.1
dnscmd dnssvr1.contoso.com /zoneresetmasters test.contoso.com /local
```

dnscmd /zoneresetscavengeservers command

Changes the IP addresses of the servers that can scavenge the specified zone.

Syntax

```
dnscmd [<servername>] /zoneresetscavengeservers <zonename> [/local]
[<IPaddress> [<IPaddress>]...]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the zone to scavenge.
/local	Sets a local master list. This parameter is used for active directory integrated zones.
`<IPaddress>`	Lists the IP addresses of the servers that can perform the scavenge. If this parameter is omitted, all servers that host this zone can scavenge it.

Remarks

- By default, all servers that host a zone can scavenge that zone.
- If a zone is hosted on more than one DNS server, you can use this command to reduce the number of times a zone is scavenged.
- Scavenging must be enabled on the DNS server and zone that is affected by this command.

Examples

```
dnscmd dnssvr1.contoso.com /zoneresetscavengeservers test.contoso.com
```

```
10.0.0.1 10.0.0.2
```

dnscmd /zoneresetsecondaries command

Specifies a list of IP addresses of secondary servers to which a primary server responds when it's asked for a zone transfer.

Syntax

```
dnscmd [<servername>] /zoneresetsecondaries <zonename> {/noxfr | /nonsecure
| /securens | /securelist <securityIPaddresses>} {/nonotify | /notify |
/notifylist <notifyIPaddresses>}
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone that will have its secondary servers reset.
/local	Sets a local master list. This parameter is used for active directory integrated zones.
/noxfr	Specifies that no zone transfers are allowed.
/nonsecure	Specifies that all zone transfer requests are granted.
/securens	Specifies that only the server that is listed in the name server (NS) resource record for the zone is granted a transfer.
/securelist	Specifies that zone transfers are granted only to the list of servers. This parameter must be followed by an IP address or addresses that the primary server uses.
`<securityIPaddresses>`	Lists the IP addresses that receive zone transfers from the primary server. This parameter is used only with the **/securelist** parameter.
/nonotify	Specifies that no change notifications are sent to secondary servers.
/notify	Specifies that change notifications are sent to all secondary servers.

Parameters	Description
/notifylist	Specifies that change notifications are sent to only the list of servers. This command must be followed by an IP address or addresses that the primary server uses.
`<notifyIPaddresses>`	Specifies the IP address or addresses of the secondary server or servers to which change notifications are sent. This list is used only with the **/notifylist** parameter.

Remarks

- Use the **zoneresetsecondaries** command on the primary server to specify how it responds to zone transfer requests from secondary servers.

Examples

```
dnscmd dnssvr1.contoso.com /zoneresetsecondaries test.contoso.com /noxfr
/nonotify
dnscmd dnssvr1.contoso.com /zoneresetsecondaries test.contoso.com
/securelist 11.0.0.2
```

dnscmd /zoneresettype command

Changes the type of the zone.

Syntax

```
dnscmd [<servername>] /zoneresettype <zonename> <zonetype> [/overwrite_mem |
/overwrite_ds]
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Identifies the zone on which the type will be changed.
`<zonetype>`	Specifies the type of zone to create. Each type has different required parameters, including: • **/dsprimary** - Creates an active directory integrated zone. • **/primary /file** `<filename>` - Creates a standard primary zone. • **/secondary** `<masterIPaddress> [,<masterIPaddress>...]` - Creates a standard secondary zone. • **/stub** `<masterIPaddress>[,<masterIPaddress>...]` **/file** `<filename>` - Creates a file-backed stub zone. • **/dsstub** `<masterIPaddress>[,<masterIPaddress>...]` - Creates an active directory integrated stub zone. • **/forwarder** `<masterIPaddress[,<masterIPaddress>]` **... /file** `<filename>` - Specifies that the created zone forwards unresolved queries to another DNS server. • **/dsforwarder** - Specifies that the created active directory integrated zone forwards unresolved queries to another DNS server.
/overwrite_mem	Overwrites DNS data from data in AD DS.
/overwrite_ds	Overwrites existing data in AD DS.

Remarks

- Setting the zone type as **/dsforwarder** creates a zone that performs conditional forwarding.

Examples

```
dnscmd dnssvr1.contoso.com /zoneresettype test.contoso.com /primary /file
test.contoso.com.dns
dnscmd dnssvr1.contoso.com /zoneresettype second.contoso.com /secondary
10.0.0.2
```

dnscmd /zoneresume command

Starts a specified zone that was previously paused.

Syntax

```
dnscmd [<servername>] /zoneresume <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to resume.

Remarks

- You can use this operation to restart from the **/zonepause** operation.

Examples

```
dnscmd dnssvr1.contoso.com /zoneresume test.contoso.com
```

dnscmd /zoneupdatefromds command

Updates the specified active directory integrated zone from AD DS.

Syntax

```
dnscmd [<servername>] /zoneupdatefromds <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.
`<zonename>`	Specifies the name of the zone to update.

Remarks

- Active directory integrated zones perform this update by default every five minutes. To change this parameter, use the `dnscmd config dspollinginterval` command.

Examples

```
dnscmd dnssvr1.contoso.com /zoneupdatefromds
```

dnscmd /zonewriteback command

Checks DNS server memory for changes that are relevant to a specified zone, and writes them to persistent storage.

Syntax

```
dnscmd [<servername>] /zonewriteback <zonename>
```

Parameters

Expand table

Parameters	Description
`<servername>`	Specifies the DNS server to manage, represented by IP address, FQDN, or host name. If this parameter is omitted, the local server is used.

Parameters	Description
`<zonename>`	Specifies the name of the zone to update.

Remarks

- This is a zone-level operation. You can update all zones on a DNS server by using the **/writebackfiles** operation.

Examples

```
dnscmd dnssvr1.contoso.com /zonewriteback test.contoso.com
```

Related links

- Command-Line Syntax Key

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doskey

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Calls Doskey.exe, which recalls previously entered command-line commands, edits command lines, and creates macros.

Syntax

```
doskey [/reinstall] [/listsize=<size>] [/macros:[all | <exename>] [/history]
[/insert | /overstrike] [/exename=<exename>] [/macrofile=<filename>]
[<macroname>=[<text>]]
```

Parameters

Expand table

Parameter	Description
/reinstall	Installs a new copy of Doskey.exe and clears the command history buffer.
/listsize=`<size>`	Specifies the maximum number of commands in the history buffer.
/macros	Displays a list of all **doskey** macros. You can use the redirection symbol (`>`) with **/macros** to redirect the list to a file. You can abbreviate **/macros** to **/m**.
/macros:all	Displays **doskey** macros for all executables.
/macros: `<exename>`	Displays **doskey** macros for the executable specified by *exename*.
/history	Displays all commands that are stored in memory. You can use the redirection symbol (`>`) with **/history** to redirect the list to a file. You can abbreviate **/history** as **/h**.
/insert	Specifies that new text you type is inserted in old text.
/overstrike	Specifies that new text overwrites old text.
/exename=	Specifies the program (that is, executable) in which the **doskey** macro runs.

Parameter	Description
`<exename>`	
/macrofile= `<filename>`	Specifies a file that contains the macros that you want to install.
`<macroname>` = [`<text>`]	Creates a macro that carries out the commands specified by *Text*. *MacroName* specifies the name you want to assign to the macro. *Text* specifies the commands you want to record. If *Text* is left blank, *MacroName* is cleared of any assigned commands.
/?	Displays help at the command prompt.

Remarks

- Certain character-based, interactive programs, such as program debuggers or file transfer programs (FTP) automatically use Doskey.exe. To use Doskey.exe, a program must be a console process and use buffered input. Program key assignments override **doskey** key assignments. For example, if the program uses the F7 key for a function, you cannot get a **doskey** command history in a pop-up window.
- You can use Doskey.exe to edit the current command line, but you can't use the command-line options from a program's command prompt. You must run **doskey** command-line options before you start a program. If you use Doskey.exe within a program, that program's key assignments take precedence and some Doskey.exe editing keys might not work.
- With Doskey.exe, you can maintain a command history for each program that you start or repeat. You can edit previous commands at the program's prompt, and start **doskey** macros created for the program. If you exit and then restart a program from the same Command Prompt window, the command history from the previous program session is available.
- To recall a command, you can use any of the following keys after you start Doskey.exe:

Expand table

Key	Description
UP ARROW	Recalls the command that you used before the one that is displayed.
DOWN ARROW	Recalls the command that you used after the one that is displayed.

Key	Description
PAGE UP	Recalls the first command that you used in the current session.
PAGE DOWN	Recalls the most recent command that you used in the current session.

- The following table lists **doskey** editing keys and their functions:

Expand table

Key or key combination	Description
LEFT ARROW	Moves the insertion point back one character.
RIGHT ARROW	Moves the insertion point forward one character.
CTRL+LEFT ARROW	Moves the insertion point back one word.
CTRL+RIGHT ARROW	Moves the insertion point forward one word.
HOME	Moves the insertion point to the beginning of the line.
END	Moves the insertion point to the end of the line.
ESC	Clears the command from the display.
F1	Copies one character from a column in the template to the same column in the Command Prompt window. (The template is a memory buffer that holds the last command you typed.)
F2	Searches forward in the template for the next key that you type after you press F2. Doskey.exe inserts the text from the template—up to, but not including, the character you specify.
F3	Copies the remainder of the template to the command line. Doskey.exe begins copying characters from the position in the template that corresponds to the position indicated by the insertion point on the command line.
F4	Deletes all characters from the current insertion point position up to, but not including, the next occurrence of the character that you type after you press F4.
F5	Copies the template into the current command line.
F6	Places an end-of-file character (CTRL+Z) at the current insertion point position.

Key or key combination	Description
F7	Displays (in a dialog box) all commands for this program that are stored in memory. Use the UP ARROW key and the DOWN ARROW key to select the command you want, and press ENTER to run the command. You can also note the sequential number in front of the command and use this number in conjunction with the F9 key.
ALT+F7	Deletes all commands stored in memory for the current history buffer.
F8	Displays all commands in the history buffer that start with the characters in the current command.
F9	Prompts you for a history buffer command number, and then displays the command associated with the number that you specify. Press ENTER to run the command. To display all the numbers and their associated commands, press F7.
ALT+F10	Deletes all macro definitions.

- If you press the INSERT key, you can type text on the **doskey** command line in the midst of existing text without replacing the text. However, after you press ENTER, Doskey.exe returns your keyboard to **Replace** mode. You must press INSERT again to return to **Insert** mode.
- The insertion point changes shape when you use the INSERT key to change from one mode to the other.
- If you want to customize how Doskey.exe works with a program and create **doskey** macros for that program, you can create a batch program that modifies Doskey.exe and starts the program.
- You can use Doskey.exe to create macros that carry out one or more commands. The following table lists special characters that you can use to control command operations when you define a macro.

Expand table

Character	Description
`$G` or `$g`	Redirects output. Use either of these special characters to send output to a device or a file instead of to the screen. This character is equivalent to the redirection symbol for output (`>`).
`$G$G` or `$g$g`	Appends output to the end of a file. Use either of these double characters to append output to an existing file instead of replacing the data in the file.

Character	Description
	These double characters are equivalent to the append redirection symbol for output (`>>`).
`$L` or `$l`	Redirects input. Use either of these special characters to read input from a device or a file instead of from the keyboard. This character is equivalent to the redirection symbol for input (`<`).
`$B` or `$b`	Sends macro output to a command. These special characters are equivalent to using the pipe `(` and `*`.
`$T` or `$t`	Separates commands. Use either of these special characters to separate commands when you create macros or type commands on the **doskey** command line. These special characters are equivalent to using the ampersand (`&`) on a command line.
`$$`	Specifies the dollar-sign character (`$`).
`$1` through `$9`	Represent any command-line information you want to specify when you run the macro. The special characters `$1` through `$9` are batch parameters that enable you to use different data on the command line each time you run the macro. The `$1` character in a **doskey** command is similar to the `%1` character in a batch program.
`$*`	Represents all the command-line information that you want to specify when you type the macro name. The special character `$*` is a replaceable parameter that is similar to the batch parameters `$1` through `$9`, with one important difference: everything you type on the command line after the macro name is substituted for the `$*` in the macro.

- To run a macro, type the macro name at the command prompt, starting at the first position. If the macro was defined with `$*` or any of the batch parameters `$1` through `$9`, use a space to separate the parameters. You cannot run a **doskey** macro from a batch program.
- If you always use a particular command with specific command-line options, you can create a macro that has the same name as the command. To specify whether you want to run the macro or the command, follow these guidelines:
  - To run the macro, type the macro name at the command prompt. Do not add a space before the macro name.
  - To run the command, insert one or more spaces at the command prompt, and then type the command name.

Examples

The **/macros** and **/history** command-line options are useful for creating batch programs to save macros and commands. For example, to store all current **doskey** macros, type:

```
doskey /macros > macinit
```

To use the macros stored in Macinit, type:

```
doskey /macrofile=macinit
```

To create a batch program named Tmp.bat that contains recently used commands, type:

```
doskey /history> tmp.bat
```

To define a macro with multiple commands, use `$t` to separate commands, as follows:

```
doskey tx=cd temp$tdir/w $*
```

In the preceding example, the TX macro changes the current directory to Temp and then displays a directory listing in wide display format. You can use `$*` at the end of the macro to append other command-line options to **dir** when you run the tx option.

The following macro uses a batch parameter for a new directory name:

```
doskey mc=md $1$tcd $1
```

The macro creates a new directory and then changes to the new directory from the current directory.

To use the preceding macro to create and change to a directory named *Books*, type:

```
mc books
```

To create a **doskey** macro for a program called *Ftp.exe*, include **/exename** as follows:

```
doskey /exename=ftp.exe go=open 172.27.1.100$tmget *.TXT c:\reports$tbye
```

To use the preceding macro, start FTP. At the FTP prompt, type:

```
go
```

FTP runs the **open**, **mget**, and **bye** commands.

To create a macro that quickly and unconditionally formats a disk, type:

```
doskey qf=format $1 /q /u
```

To quickly and unconditionally format a disk in drive A, type:

```
qf a:
```

To delete a macro called *vlist*, type:

```
doskey vlist =
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

driverquery

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables an administrator to display a list of installed device drivers and their properties. If used without parameters, **driverquery** runs on the local computer.

Syntax

```
driverquery [/s <system> [/u [<domain>\]<username> [/p <password>]]] [/fo {table | list | csv}] [/nh] [/v | /si]
```

Parameters

Expand table

Parameter	Description
/s `<system>`	Specifies the name or IP address of a remote computer. Do not use backslashes. The default is the local computer.
/u `[<domain>]` `<username>`	Runs the command with the credentials of the user account as specified by *user* or *domain\user*. By default, */s* uses the credentials of the user who is currently logged on to the computer that is issuing the command. **/u** can't be used unless **/s** is specified.
/p `<password>`	Specifies the password of the user account that is specified in the **/u** parameter. **/p** cannot be used unless **/u** is specified.
/fo table	Formats the output as a table. This is the default.
/fo list	Formats the output as a list.
/fo csv	Formats the output with comma-separated values.
/nh	Omits the header row from the displayed driver information. Not valid if the **/fo** parameter is set to **list**.
/v	Displays verbose output. **/v** is not valid for signed drivers.

Parameter	Description
/si	Provides information about signed drivers.
/?	Displays help at the command prompt.

Examples

To display a list of installed device drivers on the local computer, type:

```
driverquery
```

To display the output in a comma-separated values (CSV) format, type:

```
driverquery /fo csv
```

To hide the header row in the output, type:

```
driverquery /nh
```

To use the **driverquery** command on a remote server named *server1* using your current credentials on the local computer, type:

```
driverquery /s server1
```

To use the **driverquery** command on a remote server named *server1* using the credentials for *user1* on the domain *maindom*, type:

```
driverquery /s server1 /u maindom\user1 /p p@ssw3d
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

DTrace

Article • 04/19/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Beginning with Windows Server 2025, `dtrace` is included as a built-in tool. DTrace is a command-line utility that allows users to monitor and debug their system performance in real-time. With `dtrace`, users can dynamically instrument the kernel and user-space code without modifying the code itself.

This powerful tool supports various data collection and analysis techniques, including aggregations, histograms, and tracing of user-level events. Probes can be specified in `dtrace` scripts where the script defines the probes to be monitored and the actions to be taken when the probes fire. A probe is a specific point in the code where data can be collected to perform these actions.

> **Note**
>
> This built-in port of `dtrace` differs from the MSI installer of DTrace for Windows surrounding parameters and other capabilities. To learn more about DTrace for Windows, see DTrace on Windows.
>
> For a comprehensive guide on using DTrace, see the Dynamic Tracing Guide.

Enable DTrace

Before `dtrace` can be used, it must first be enabled. To enable `dtrace`, open an elevated command prompt or PowerShell as admin and run:

```
bcdedit /set dtrace on
```

A reboot is required for this change to take effect.

Syntax

```
dtrace [-BCeFhlqSvVwYZ] [-b bufsz] [-c cmd] [-D name[=def]] [-I path]
[-L path] [-o output] [-p pid] [-s script] [-U name] [-x opt[=val]] [-X
a|c|s|t]
[-y symbol path]

[-P provider [[ predicate ] action ]] [-m [ provider: ]
module [[ predicate ] action ]] [-f [[ provider: ] module: ]
func [[ predicate ] action ]] [-n [[[ provider: ] module: ] func: ]
name [[ predicate ] action ]] [-i probe-id [[ predicate ] action ]] [ args
... ]
```

> **Note**
>
> The parameters for dtrace.exe are *case-sensitive*. Make sure to use the correct casing when specifying parameters to avoid any unexpected behavior.

Expand table

Parameter	Description
-b `<bufsz>`	Sets the size of the buffer used to store trace data, where *bufsz* is the desired size of the buffer in bytes, kilobytes (**k**), megabytes (**m**), or gigabytes (**g**).
-c `<cmd>`	Runs the specified command and exits once it completes, where *cmd* is the command that you want to run before starting the trace. If more than one instance of **-c** is used, dtrace exits when all commands finish running and reports the exit status for each child process.
-C	Runs the ucpp preprocessor on script files before running the trace.
-D `<name>` = `<def>`	Defines a symbol when invoking the preprocessor, where *name* is the name of the symbol to define, and *def* is an optional value to assign to the symbol.
-e	Exits after compiling a request but before enabling probes.
-f	Enables or lists probes that match the specified function name.
-F	Combines the trace output by function, making it easier to analyze.
-h	Generates a header file with definitions for static probes.
-i `<probe-id>`	Enables or lists probes that match the specified probe ID, where *probe-id* is the identifier for the probe to be traced.
-I `<path>`	Adds the specified directory to the preprocessor search path, where *path* is the directory that you want to add that contain the *#include* files.

Parameter	Description
-l	Lists probes that match specified criteria based on the **-P**, **-m**, **-f**, **-n**, **-i**, and **-s** parameters. If these parameters aren't specified, all probes are listed.
-L `<path>`	Adds the specified directory to the library search path, where *path* is the library directory that you want to add that contain common definitions.
-m	Enables or lists probes that match the specified module name in an argument using the format *provider:module* or *module*. If qualifiers aren't specified besides the module name, all probes with that module name are matched.
-n	Enables or lists probes that match the specified probe name in an argument using the format *provider:module:function:name*, *module:function:name*, *function:name*, or *name*. If qualifiers aren't specified besides the probe name, all probes with that name are matched.
-o `<output>`	Sets the output file for the trace data, where *output* is the name of the file that you want to use for the trace data.
-p `<pid>`	Grabs the specified process-ID (PID) and caches its symbol tables, which can be used to analyze the program's behavior.
-P `<provider>`	Enables or lists probes that match the specified provider name, where *provider* is the name of the provider. More than one instance of the **-P** parameter can be used at the same time.
-q	Sets quiet mode, which only outputs explicitly traced data.
-s `<script>`	Enables or lists probes according to the specified D script, where *script* is the name of the script that you want to run. If **-e** is specified, the program is compiled but no data collection is performed. If **-l** is specified, the program is compiled and the list of probes matched are displayed, but no data collection is performed. If **-e** or **-l** aren't specified, the program is compiled, data collection is performed according to the probes specified and tracing begins.
-S	Prints the D-language compiler intermediate code for debugging to *stderr*.
-U `<name>`	Undefines a symbol when invoking the preprocessor, where *name* is the name of the symbol that you want to undefine.
-v	Sets verbose mode, which reports stability attributes and arguments.
-V	Displays the version of the dtrace API.
-w	Permits destructive actions when specified with the **-s**, **-P**, **-m**, **-f**, **-n**, or **-i** parameters. Destructive actions can include actions such as modifying kernel variables, changing the behavior of system calls, or crashing the system.
-x `<opt>` = `<val>`	Enables or modifies the compiler and tracing options, where *opt* is the name of the option that you want to enable or modify, and *val* is an optional value.

Parameter	Description
-X `<a|c|s|t>`	Controls how strict the C code being compiled adheres to the ISO C standard when invoking the cpp. The available arguments are: • `-Xa` (default): provides ISO C plus K&R compatibility extensions with semantic changes required by ISO C. The predefined macro __STDC__ has a value of **0** when cpp is invoked. • `-Xc` (conformance): Provides a strict conformant of ISO C without K&R C compatibility extensions. The predefined macro __STDC__ has a value of **1** when cpp is invoked. • `-Xs` (K&R C): Provides K&R C only, and the __STDC__ macro isn't defined when cpp is invoked. • `-Xt` (transition): Provides ISO C plus K&R C compatibility extensions without semantic changes required by ISO C. The predefined macro __STDC__ has a value of **0** when cpp is invoked.
-y `<symbol path>`	Sets the symbol search path for the dtrace script to resolve, where *symbol path* is the path to the shared library or directory that contains the symbols. To learn more, see Symbol Paths.
-Y	Uses the **default** symbol search path for the dtrace script.
-Z	Permits probe descriptions that match zero probes for debugging.

The following list describes the remaining descriptions:

- **Predicate**: The predicate is enclosed in forward slashes (`/ /`) and is a D-expression, which is a boolean expression that can reference variables, constants, and functions. Predicates can be used to filter the output of `dtrace` based on these events. This expression is evaluated each time a probe fires. If the predicate evaluates to true, the associated action is executed.
- **Action**: The action is enclosed in curly braces (`{ }`) and is a set of D-language statements that are executed when a probe fires and its associated predicate, if any, evaluates to true. Actions can be used to print output statements, record data, or perform other operations, such as sending a signal or modifying a variable.
- **Module**: A component of a provider that contains a set of related probes. Modules can be specified in `dtrace` scripts to limit the scope of the script to a specific module or set of modules.
- **Func**: A function name that is associated with a probe. For example, the `syscall::NtReadFile` probe is associated with the `read` function. Functions can be specified in `dtrace` scripts to limit the scope of the script to a specific function or set of functions.

- **Args**: The arguments that are passed to the action expression when a probe fires and its associated predicate, if any, evaluates to true. Args can be used to capture data from the probe context, such as the values of function arguments or system call return values. Args can also be used to pass data between probes or to modify the behavior of the script.

Examples

To trace all system calls, print the name of the executable and the name of the system call being made, run:

Windows Command Prompt

```
dtrace -n 'syscall:::entry { printf("%s called syscall %s", execname,
probefunc); }'
```

This command traces the "function_name" function in the process with the specified PID and prints the name of the function, the PID of the process calling it, and the name of the executable.

Windows Command Prompt

```
dtrace -n 'pid$target::function_name:entry { printf("Function %s called by
process %d (%s)", probefunc, pid, execname); }' -p <PID>
```

For further examples on using `dtrace`, see Getting started with DTrace - One line commands.

See also

- DTrace Programming
- DTrace Code Samples
- DTrace Event Tracing for Windows
- DTrace Live Dump

Feedback

Was this page helpful? Yes No

echo

Article • 02/16/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays messages or turns on or off the command echoing feature. If used without parameters, **echo** displays the current echo setting.

Syntax

```
echo [<message>]
echo [on | off]
```

Parameters

Expand table

Parameter	Description
[on | off]	Turns on or off the command echoing feature. Command echoing is on by default.
`<message>`	Specifies the text to display on the screen.
/?	Displays help at the command prompt.

Remarks

- The `echo <message>` command is more useful when **echo** is turned off. To display a message that is several lines long without displaying any commands, you can include several `echo <message>` commands after the **echo off** command in your batch program.
- After **echo** is turned off, the command prompt doesn't appear in the Command Prompt window. To display the command prompt, type **echo on.**
- If used in a batch file, **echo on** and **echo off** don't affect the setting at the command prompt.

- If there's an empty variable in a batch file while using **echo**, it displays "ECHO is off". To prevent seeing this message, and produce a blank line instead, place a colon (`:`) between **echo** and the variable. For example, `echo:%var%`.
- To prevent echoing a particular command in a batch file, insert an (`@`) sign in front of the command. To prevent echoing all commands in a batch file, include the **echo off** command at the beginning of the file.
- To display an exclamation mark (`!`) in batch scripts, wrap the word or phrase in double quotes followed by a caret before the exclamation mark (`"Hello World^!"`). Alternatively, a double caret (`^^`) can be used without the need for double quotes (`Hello World^^!`).
- To display a pipe (`|`), ampersand (`&`) or redirection character (`<` or `>`) when you're using **echo**, use a caret (`^`) immediately before that character. For example, `^|`, `^&`, `^>`, or `^<`. To display a caret, type two carets in succession (`^^`).
- When inside a block terminated by parentheses (`()`), both opening and closing parentheses must also be escaped using the caret (`^`) immediately before each one. For example, `This is ^(now^) correct` will correctly display `This is (now) correct`.

Examples

To display the current **echo** setting, type:

```
echo
```

To echo a blank line on the screen, type:

```
echo.
```

> ⓘ **Note**
>
> Don't include a space before the period. Otherwise, the period appears instead of a blank line.

To prevent echoing commands at the command prompt, type:

```
echo off
```

> ⓘ **Note**
>
> When **echo** is turned off, the command prompt doesn't appear in the Command Prompt window. To display the command prompt again, type **echo on**.

To prevent all commands in a batch file (including the **echo off** command) from displaying on the screen, on the first line of the batch file type:

```
@echo off
```

You can use the **echo** command as part of an **if** statement. For example, to search the current directory for any file with the .rpt file name extension, and to echo a message if such a file is found, type:

```
if exist *.rpt echo The report has arrived.
```

The following batch file searches the current directory for files with the .txt file name extension, and displays a message indicating the results of the search:

```
@echo off
if not exist *.txt (
echo This directory contains no text files.
) else (
   echo This directory contains the following text file^(s^):
   echo.
   dir /b *.txt
   )
```

If no .txt files are found when the batch file is run, the following message displays:

```
```

```
This directory contains no text files.
```

If .txt files are found when the batch file is run the following output displays (for this example, assume the files File1.txt, File2.txt, and File3.txt exist):

```
This directory contains the following text file(s):

File1.txt
File2.txt
File3.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

edit

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Starts the MS-DOS Editor, which creates and changes ASCII text files.

Syntax

```
edit [/b] [/h] [/r] [/s] [/<nnn>] [[<drive>:][<path>]<filename> [<filename2> [...]]]
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]<filename> [<filename2> [...]]`	Specifies the location and name of one or more ASCII text files. If the file doens't exist, MS-DOS Editor creates it. If the file exists, MS-DOS Editor opens it and displays its contents on the screen. The *filename* option can contain wildcard characters (* and ?). Separate multiple file names with spaces.
/b	Forces monochrome mode, so that MS-DOS Editor displays in black and white.
/h	Displays the maximum number of lines possible for the current monitor.
/r	Loads file(s) in read-only mode.
/s	Forces the use of short filenames.
`<nnn>`	Loads binary file(s), wrapping lines to *nnn* characters wide.
/?	Displays help at the command prompt.

Remarks

- For additional help, open MS-DOS Editor, and then press the F1 key.

- Some monitors don't support the display of shortcut keys by default. If your monitor doesn't display shortcut keys, use **/b**.

Examples

To open MS-DOS Editor, type:

```
edit
```

To create and edit a file named *newtextfile.txt* in the current directory, type:

```
edit newtextfile.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

endlocal

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Ends localization of environment changes in a batch file, and restores environment variables to their values before the corresponding **setlocal** command was run.

Syntax

```
endlocal
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- The **endlocal** command has no effect outside a script or batch file.
- There is an implicit **endlocal** command at the end of a batch file.
- If command extensions are enabled (command extensions are enabled by default), the **endlocal** command restores the state of command extensions (that is, enabled or disabled) to what it was before the corresponding **setlocal** command was run.

> **Note**
>
> For more information about enabling and disabling command extensions, see the Cmd command.

Examples

You can localize environment variables in a batch file. For example, the following program starts the *superapp* batch program on the network, directs the output to a file, and displays the file in Notepad:

```
@echo off
setlocal
path=g:\programs\superapp;%path%
call superapp>c:\superapp.out
endlocal
start notepad c:\superapp.out
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

erase

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes one or more files. If you use **erase** to delete a file from your disk, you can't retrieve it.

> **Note**
>
> This command is the same as the del command.

Syntax

```
erase [/p] [/f] [/s] [/q] [/a[:]<attributes>] <names>
del [/p] [/f] [/s] [/q] [/a[:]<attributes>] <names>
```

Parameters

Expand table

Parameter	Description
`<names>`	Specifies a list of one or more files or directories. Wildcards may be used to delete multiple files. If a directory is specified, all files within the directory will be deleted.
/p	Prompts for confirmation before deleting the specified file.
/f	Forces deletion of read-only files.
/s	Deletes specified files from the current directory and all subdirectories. Displays the names of the files as they are being deleted.
/q	Specifies quiet mode. You are not prompted for delete confirmation.
/a[:] `<attributes>`	Deletes files based on the following file attributes: • **r** Read-only files • **h** Hidden files • **i** Not content indexed files

Parameter	Description
	• **s** System files • **a** Files ready for archiving • **l** Reparse points • **-** Used as a prefix meaning 'not'
/?	Displays help at the command prompt.

Remarks

- If you use the `erase /p` command, you'll see the following message:

  `FileName, Delete (Y/N)?`

  To confirm the deletion, press **Y**. To cancel the deletion and to display the next file name (if you specified a group of files), press **N**. To stop the **erase** command, press CTRL+C.

- If you disable command extension, the **/s** parameter will display the names of any files that weren't found ,instead of displaying the names of files that are being deleted.

- If you specify specific folders in the `<names>` parameter, all of the included files will also be deleted. For example, if you want to delete all of the files in the *\work* folder, type:

  ```
  erase \work
  ```

- You can use wildcards (***** and **?**) to delete more than one file at a time. However, to avoid deleting files unintentionally, you should use wildcards cautiously. For example, if you type the following command:

  ```
  erase *.*
  ```

  The **erase** command displays the following prompt:

  `Are you sure (Y/N)?`

To delete all of the files in the current directory, press **Y** and then press ENTER. To cancel the deletion, press **N** and then press ENTER.

> **Note**
>
> Before you use wildcard characters with the **erase** command, use the same wildcard characters with the **dir** command to list all the files that will be deleted.

Examples

To delete all the files in a folder named Test on drive C, type either of the following:

```
erase c:\test
erase c:\test\*.*
```

To delete all files with the .bat file name extension from the current directory, type:

```
erase *.bat
```

To delete all read-only files in the current directory, type:

```
erase /a:r *.*
```

Related links

- Command-Line Syntax Key
- del command

Feedback

Was this page helpful? Yes No

eventcreate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables an administrator to create a custom event in a specified event log.

> **ⓘ Important**
>
> Custom events can't be written to the security log.

Syntax

```
eventcreate [/s <computer> [/u <domain\user> [/p <password>]] {[/l
{APPLICATION|SYSTEM}]|[/so <srcname>]} /t
{ERROR|WARNING|INFORMATION|SUCCESSAUDIT|FAILUREAUDIT} /id <eventID> /d
<description>
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default is the local computer.
/u `<domain\user>`	Runs the command with the account permissions of the user specified by `<user>` or `<domain\user>`. The default is the permissions of the current logged on user on the computer issuing the command.
/p `<password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/l `{APPLICATION | SYSTEM}`	Specifies the name of the event log where the event will be created. The valid log names are **APPLICATION** or **SYSTEM**.
/so `<srcname>`	Specifies the source to use for the event. A valid source can be any string and should represent the application or component

Parameter	Description
	that is generating the event.
/t `{ERROR | WARNING | INFORMATION | SUCCESSAUDIT | FAILUREAUDIT}`	Specifies the type of event to create. The valid types are **ERROR**, **WARNING**, **INFORMATION**, **SUCCESSAUDIT**, and **FAILUREAUDIT**.
/id `<eventID>`	Specifies the event ID for the event. A valid ID is any number from 1 to 1000.
/d `<description>`	Specifies the description to use for the newly created event.
/?	Displays help at the command prompt.

Examples

The following examples show how you can use the **eventcreate** command:

```
eventcreate /t ERROR /id 100 /l application /d "Create event in application
log"
eventcreate /t INFORMATION /id 1000 /d "Create event in WinMgmt source"
eventcreate /t ERROR /id 201 /so winword /l application /d "New src Winword
in application log"
eventcreate /s server /t ERROR /id 100 /l application /d "Remote machine
without user credentials"
eventcreate /s server /u user /p password /id 100 /t ERROR /l application /d
"Remote machine with user credentials"
eventcreate /s server1 /s server2 /u user /p password /id 100 /t ERROR /d
"Creating events on Multiple remote machines"
eventcreate /s server /u user /id 100 /t WARNING /d "Remote machine with
partial user credentials"
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

evntcmd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configures the translation of events to traps, trap destinations, or both based on information in a configuration file.

Syntax

```
evntcmd [/s <computername>] [/v <verbositylevel>] [/n] <filename>
```

Parameters

Expand table

Parameter	Description
/s `<computername>`	Specifies, by name, the computer on which you want to configure the translation of events to traps, trap destinations, or both. If you do not specify a computer, the configuration occurs on the local computer.
/v `<verbositylevel>`	Specifies which types of status messages appear as traps and trap destinations are configured. This parameter must be an integer between 0 and 10. If you specify 10, all types of messages appear, including tracing messages and warnings about whether trap configuration was successful. If you specify 0, no messages appear.
/n	Specifies that the SNMP service should not be restarted if this computer receives trap configuration changes.
`<filename>`	Specifies, by name, the configuration file that contains information about the translation of events to traps and trap destinations you want to configure.
/?	Displays help at the command prompt.

Remarks

- if you want to configure traps but not trap destinations, you can create a valid configuration file by using Event to Trap Translator, which is a graphical utility. If you have the SNMP service installed, you can start Event to Trap Translator by typing **evntwin** at a command prompt. After you have defined the traps you want, click **Export** to create a file suitable for use with **evntcmd**. You can use Event to Trap Translator to easily create a configuration file and then use the configuration file with **evntcmd** at the command prompt to quickly configure traps on multiple computers.

- The syntax for configuring a trap is as follows:

  ```
  #pragma add <eventlogfile> <eventsource> <eventID> [<count> [<period>]]
  ```

  Where the text following is true:

  - **#pragma** must appear at the beginning of every entry in the file.

  - The parameter **add** specifies that you want to add an event to trap configuration.

  - The parameters **eventlogfile**, **eventsource**, and **eventID** are required, and where **eventlogfile** specifies the file in which the event is recorded, **eventsource** specifies the application that generates the event and **eventID** specifies the unique number that identifies each event.

  To determine what values correspond to each event, start the Event to Trap Translator by typing **evntwin** at a command prompt. Click **Custom**, and then click **edit**. Under **Event Sources**, browse the folders until you locate the event you want to configure, click it, and then click **add**. Information about the event source, the event log file, and the event ID appear under **Source, Log**, and **Trap specific ID**, respectively.

  - The **count** parameter is optional, and it specifies how many times the event must occur before a trap message is sent. If you don't use this parameter, the trap message is sent after the event occurs once.

  - The **period** parameter is optional, but it requires you to use the **count** parameter. The **period** parameter specifies a length of time (in seconds) during which the event must occur the number of times specified with the **count** parameter before a trap message is sent. If you don't use this parameter, a trap message is sent after the event occurs the number of times specified with the ***count*** parameter, no matter how much time elapses between occurrences.

- The syntax for removing a trap is as follows:

  ```
  #pragma delete <eventlogfile> <eventsource> <eventID>
  ```

  Where the text following is true:

  - **#pragma** must appear at the beginning of every entry in the file.
  - The parameter **delete** specifies that you want to remove an event to trap configuration.
  - The parameters **eventlogfile**, **eventsource**, and **eventID** are required, and where **eventlogfile** specifies the file in which the event is recorded, **eventsource** specifies the application that generates the event and **eventID** specifies the unique number that identifies each event.

  To determine what values correspond to each event, start the Event to Trap Translator by typing **evntwin** at a command prompt. Click **Custom**, and then click **edit**. Under **Event Sources**, browse the folders until you locate the event you want to configure, click it, and then click **add**. Information about the event source, the event log file, and the event ID appear under **Source, Log**, and **Trap specific ID**, respectively.

- The syntax for configuring a trap destination is as follows:

  ```
  #pragma add_TRAP_DEST <communityname> <hostID>
  ```

  Where the text following is true:

  - **#pragma** must appear at the beginning of every entry in the file.
  - The parameter **add_TRAP_DEST** specifies that you want trap messages to be sent to a specified host within a community.
  - The parameter **communityname** specifies, by name, the community in which trap messages are sent.
  - The parameter **hostID** specifies, by name or IP address, the host to which you want trap messages to be sent.

- The syntax for removing a trap destination is as follows:

```
#pragma delete_TRAP_DEST <communityname> <hostID>
```

Where the text following is true:

- **#pragma** must appear at the beginning of every entry in the file.
- The parameter **delete_TRAP_DEST** specifies that you do not want trap messages to be sent to a specified host within a community.
- The parameter **communityname** specifies, by name, the community to which trap messages shouldn't be sent.
- The parameter **hostID** specifies, by name or IP address, the host to which you don't want trap messages to be sent.

Examples

The following examples illustrate entries in the configuration file for the **evntcmd** command. They are not designed to be typed at a command prompt.

To send a trap message if the Event Log service is restarted, type:

```
#pragma add System Eventlog 2147489653
```

To send a trap message if the Event Log service is restarted twice in three minutes, type:

```
#pragma add System Eventlog 2147489653 2 180
```

To stop sending a trap message whenever the Event Log service is restarted, type:

```
#pragma delete System Eventlog 2147489653
```

To send trap messages within the community named *Public* to the host with the IP address *192.168.100.100*, type:

```
#pragma add_TRAP_DEST public 192.168.100.100
```

To send trap messages within the community named *Private* to the host named *Host1*, type:

```
#pragma add_TRAP_DEST private Host1
```

To stop sending trap messages within the community named *Private* to the same computer on which you are configuring trap destinations, type:

```
#pragma delete_TRAP_DEST private localhost
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

expand

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> Applies to: Windows Server 2022, Windows Server 2019, Windows Server 2016, Windows Server 2012 R2, Windows Server 2012

Expands one or more compressed files. You can also use this command to retrieve compressed files from distribution disks.

The **expand** command can also run from the Windows Recovery Console, using different parameters. For more information, see Windows Recovery Environment (WinRE).

Syntax

```
expand [-r] <source> <destination>
expand -r <source> [<destination>]
expand -i <source> [<destination>]
expand -d <source>.cab [-f:<files>]
expand <source>.cab -f:<files> <destination>
```

Parameters

Expand table

Parameter	Description
-r	Renames expanded files.
source	Specifies the files to expand. *Source* can consist of a drive letter and colon, a directory name, a file name, or a combination of these. You can use wildcards (* or ?).
destination	Specifies where files are to be expanded. If *source* consists of multiple files and you don't specify **-r**, the *destination* must be a directory that already exists. *Destination* can consist of a drive letter and colon, a directory name, a file name, or a combination of these. Destination `file | path` specification.

Parameter	Description
-i	Renames expanded files but ignores the directory structure.
-d	Displays a list of files in the source location. Doesn't expand or extract the files.
-f:`<files>`	Specifies the files in a cabinet (.cab) file that you want to expand. You can use wildcards (* or ?).
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

extract / extrac32

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Extracts files from a cabinet or source.

> **Note**
>
> On Windows Server 2016 and newer, and on Windows 10, the program file Extract.exe is neither provided nor supported. It is replaced by Extrac32.exe, originally part of Internet Explorer, now part of the operating system.

Syntax

Extract.exe

```
extract [/y] [/a] [/d | /e] [/l dir] cabinet [filename ...]
extract [/y] source [newname]
extract [/y] /c source destination
```

Parameters

Expand table

Parameter	Description
cabinet	Use if you want to extract two or more files.
filename	Name of the file to extract from the cabinet. Wild cards and multiple filenames (separated by blanks) may be used.
source	Compressed file (a cabinet with only one file).
newname	New filename to give the extracted file. If not supplied, the original name is used.
/a	Process ALL cabinets. Follows cabinet chain starting in first cabinet mentioned.

Parameter	Description
/c	Copy source file to destination (to copy from DMF disks).
/d	Display cabinet directory (use with filename to avoid extract).
/e	Extract (use instead of . to extract all files).
/l dir	Location to place extracted files (default is current directory).
/y	Don't prompt before overwriting an existing file.

Extrac32.exe

> **Note**
>
> Extrac32.exe can be used from the command line, but does not display any output on the console. Redirect the help output through the more command, like this: `extrac32.exe /? | more`

```
Extrac32 [/Y] [/A] [/D | /E] [/L dir] cabinet [filename ...]
Extrac32 [/Y] source [newname]
Extrac32 [/Y] /C source destination
```

Parameters

Expand table

Parameter	Description
cabinet	Cabinet file (contains two or more files).
filename	Name of the file to extract from the cabinet. Wild cards and multiple filenames (separated by blanks) may be used.
source	Compressed file (a cabinet with only one file).
newname	New filename to give the extracted file. If not supplied, the original name is used.
/A	Process ALL cabinets. Follows cabinet chain starting in first cabinet mentioned.
/C	Copy source file to destination (to copy from DMF disks).

Parameter	Description
/D	Display cabinet directory (use with filename to avoid extract).
/E	Extract (use instead of . to extract all files).
/L dir	Location to place extracted files (default is current directory).
/Y	Do not prompt before overwriting an existing file.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

fc

Article • 01/07/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Compares two files or sets of files and displays the differences between them.

Syntax

```
fc /a [/c] [/l] [/lb<n>] [/n] [/off[line]] [/t] [/u] [/w] [/<nnnn>]
[<drive1>:][<path1>]<filename1> [<drive2>:][<path2>]<filename2>
fc /b [<drive1:>][<path1>]<filename1> [<drive2:>][<path2>]<filename2>
```

Parameters

Expand table

Parameter	Description
/a	Abbreviates the output of an ASCII comparison. Instead of displaying all of the lines that are different, **fc** displays only the first and last line for each set of differences.
/b	Compares the two files in binary mode, byte by byte, and does not attempt to resynchronize the files after finding a mismatch. This is the default mode for comparing files that have the following file extensions: .exe, .com, .sys, .obj, .lib, or .bin.
/c	Ignores the letter case.
/l	Compares the files in ASCII mode, line-by-line, and attempts to resynchronize the files after finding a mismatch. This is the default mode for comparing files, except files with the following file extensions: .exe, .com, .sys, .obj, .lib, or .bin.
/lb`<n>`	Sets the number of lines for the internal line buffer to *N*. The default length of the line buffer is 100 lines. If the files that you are comparing have more than 100 consecutive differing lines, **fc** cancels the comparison.
/n	Displays the line numbers during an ASCII comparison.
/off[line]	Doesn't skip files that have the offline attribute set.

Parameter	Description
/t	Prevents **fc** from converting tabs to spaces. The default behavior is to treat tabs as spaces, with stops at each eighth character position.
/u	Compares files as Unicode text files.
/w	Compresses white space (that is, tabs and spaces) during the comparison. If a line contains many consecutive spaces or tabs, **/w** treats these characters as a single space. When used with **/w**, **fc** ignores white space at the beginning and end of a line.
/`<nnnn>`	Specifies the number of consecutive lines that must match following a mismatch, before **fc** considers the files to be resynchronized. If the number of matching lines in the files is less than *nnnn*, **fc** displays the matching lines as differences. The default value is 2.
`[<drive1>:] [<path1>] <filename1>`	Specifies the location and name of the first file or set of files to compare. *filename1* is required.
`[<drive2>:] [<path2>] <filename2>`	Specifies the location and name of the second file or set of files to compare. *filename2* is required.
/?	Displays help at the command prompt.

Exit codes

The following table lists each `fc.exe` exit code and a description.

Expand table

Exit code	Description
0	Files are identical.
1	Files are different.
2	An error occurred during comparison.

Remarks

- This command is implemented by c:\WINDOWS\fc.exe. You can use this command within PowerShell, but be sure to spell out the full executable (fc.exe) since 'fc' is also an alias for Format-Custom.

- When you use **fc** for an ASCII comparison, **fc** displays the differences between two files in the following order:
  - Name of the first file
  - Lines from *filename1* that differ between the files
  - First line to match in both files
  - Name of the second file
  - Lines from *filename2* that differ
  - First line to match
- **/b** displays mismatches that are found during a binary comparison in the following syntax:

  ```
  \<XXXXXXXX: YY ZZ>
  ```

  The value of *XXXXXXXX* specifies the relative hexadecimal address for the pair of bytes, measured from the beginning of the file. Addresses start at 00000000. The hexadecimal values for *YY* and *ZZ* represent the mismatched bytes from *filename1* and *filename2*, respectively.
- You can use wildcard characters (***** and **?**) in *filename1* and *filename2*. If you use a wildcard in *filename1*, **fc** compares all the specified files to the file or set of files specified by *filename2*. If you use a wildcard in *filename2*, **fc** uses the corresponding value from *filename1*.
- When comparing ASCII files, **fc** uses an internal buffer (large enough to hold 100 lines) as storage. If the files are larger than the buffer, **fc** compares what it can load into the buffer. If **fc** doesn't find a match in the loaded portions of the files, it stops and displays the following message:

  ```
  Resynch failed. Files are too different.
  ```

  When comparing binary files that are larger than the available memory, **fc** compares both files completely, overlaying the portions in memory with the next portions from the disk. The output is the same as that for files that fit completely in memory.

Examples

To make an ASCII comparison of two text files, *monthly.rpt* and *sales.rpt*, and display the results in abbreviated format, type:

```
fc /a monthly.rpt sales.rpt
```

To make a binary comparison of two batch files, *profits.bat* and *earnings.bat*, type:

```
fc /b profits.bat earnings.bat
```

Results similar to the following appear:

```
00000002: 72 43
00000004: 65 3A
0000000E: 56 92
000005E8: 00 6E
FC: earnings.bat longer than profits.bat
```

If the profits.bat and earnings.bat files are identical, **fc** displays the following message:

```
Comparing files profits.bat and earnings.bat
FC: no differences encountered
```

To compare every .bat file in the current directory with the file *new.bat*, type:

```
fc *.bat new.bat
```

To compare the file *new.bat* on drive C with the file *new.bat* on drive D, type:

```
fc c:new.bat d:*.bat
```

To compare each batch file in the root directory on drive C to the file with the same name in the root directory on drive D, type:

```
fc c:*.bat d:*.bat
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

find

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Searches for a string of text in a file or files, and displays lines of text that contain the specified string.

Syntax

```
find [/v] [/c] [/n] [/i] [/off[line]] <"string"> [[<drive>:][<path>]
<filename>[...]]
```

Parameters

Expand table

Parameter	Description
/v	Displays all lines that don't contain the specified `<string>`.
/c	Counts the lines that contain the specified `<string>` and displays the total.
/n	Precedes each line with the file's line number.
/i	Specifies that the search is not case-sensitive.
[/off[line]]	Doesn't skip files that have the offline attribute set.
`<"string">`	Required. Specifies the group of characters (enclosed in quotation marks) that you want to search for.
`[<drive>:][<path>] <filename>`	Specifies the location and name of the file in which to search for the specified string.
/?	Displays help at the command prompt.

Exit codes

Expand table

Exit code	Description
0	The searched string was found
1	Searched string not found
2	Searched file not found or invalid command line switch was given

Remarks

- If you don't use **/i**, this command searches for exactly what you specify for *string*. For example, this command treats the characters `a` and `A` differently. If you use **/i**, however, the search becomes case insensitive, and it treats `a` and `A` as the same character.
- If the string you want to search for contains quotation marks, you must use double quotation marks for each quotation mark contained within the string (for example, """This string contains quotation marks""").
- If you omit a file name, this command acts as a filter, taking input from the standard input source (usually the keyboard, a pipe (|), or a redirected file) and then displays any lines that contain *string*.
- To exit the console search use `CTRL-X` or `CTRL-z`.
- You can type parameters and command-line options for the **find** command in any order.
- You can't use wildcards (***** and **?**) in the searched string. To search for a string with wild cards and regex patterns, you can use the **FINDSTR** command.
- If you use **/c** and **/v** in the same command line, this command displays a count of the lines that don't contain the specified string. If you specify **/c** and **/n** in the same command line, **find** ignores **/n**.
- This command doesn't recognize carriage returns. When you use this command to search for text in a file that includes carriage returns, you must limit the search string to text that can be found between carriage returns (that is, a string that is not likely to be interrupted by a carriage return). For example, this command doesn't report a match for the string tax file if a carriage return occurs between the words tax and file.

- The command accepts wildcards for file names. When searching in file (or files) it will print the file of the processed file preceded by ten dashes.
- **Find** command cannot read alternate data streams. For searching in alternate data streams use **findstr**, **more** or **for /f** commands.

Examples

To display all lines from *pencil.md* that contain the string *pencil sharpener*, type:

```
find "pencil sharpener" pencil.md
```

To find the text, "*The scientists labeled their paper for discussion only. It is not a final report.*" (including the quotes) in the *report.txt* file, type:

```
find """The scientists labeled their paper for discussion only. It is not a
final report.""" < report.txt
```

To search for a set of files, you can use wildcards. To search the current directory for files that have the extension *.bat* and that contain the string *PROMPT* ignoring the case, type:

```
find /i "PROMPT" *.bat
```

To find files names in a directory that contain the string *CPU*, use the pipe (|) to direct the output of the *dir* command to the find command as follows:

```
dir c:\temp /s /b | find "CPU"
```

Find all running processes that do NOT contain *agent*:

```
tasklist | find /v /i "agent"
```

Check if a service is running:

```
sc query  Winmgmt | find "RUNNING" >nul 2>&1 && (echo service is started) ||
(echo service is stopped)
```

Related links

- Command-Line Syntax Key
- findstr command

Feedback

Was this page helpful? Yes No

findstr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Searches for patterns of text in files.

Syntax

```
findstr [/b] [/e] [/l | /r] [/s] [/i] [/x] [/v] [/n] [/m] [/o] [/p] [/f:<file>] [/c:<string>] [/g:<file>] [/d:<dirlist>] [/a:<colorattribute>] [/off[line]] <strings> [<drive>:][<path>]<filename>[ ...]
```

Parameters

Expand table

Parameter	Description
/b	Matches the text pattern if it is at the beginning of a line.
/e	Matches the text pattern if it is at the end of a line.
/l	Processes search strings literally.
/r	Processes search strings as regular expressions. This is the default setting.
/s	Searches the current directory and all subdirectories.
/i	Ignores the case of the characters when searching for the string.
/x	Prints lines that match exactly.
/v	Prints only lines that don't contain a match.
/n	Prints the line number of each line that matches.
/m	Prints only the file name if a file contains a match.
/o	Prints character offset before each matching line.

Parameter	Description
/p	Skips files with non-printable characters.
/off[line]	Does not skip files that have the offline attribute set.
/f:`<file>`	Gets a file list from the specified file.
/c:`<string>`	Uses the specified text as a literal search string.
/g:`<file>`	Gets search strings from the specified file.
/d:`<dirlist>`	Searches the specified list of directories. Each directory must be separated with a semicolon (;), for example `dir1;dir2;dir3`.
/a:`<colorattribute>`	Specifies color attributes with two hexadecimal digits. Type `color /?` for additional information.
`<strings>`	Specifies the text to search for in *filename*. Required.
`[\<drive>:][<path>] <filename>[...]`	Specifies the location and file or files to search. At least one file name is required.
/?	Displays Help at the command prompt.

Remarks

- All **findstr** command-line options must precede *strings* and *filename* in the command string.
- Regular expressions use both literal characters and meta-characters to find patterns of text, rather than exact strings of characters.
  - A literal character is a character that doesn't have a special meaning in the regular-expression syntax; instead, it matches an occurrence of that character. For example, letters and numbers are literal characters.
  - A meta-character is a symbol with special meaning (an operator or delimiter) in the regular-expression syntax.

    The accepted meta-characters are:

    Expand table

Meta-character	Value
`.`	**Wildcard** - Any character

Meta-character	Value
`*`	**Repeat** - Zero or more occurrences of the previous character or class.
`^`	**Beginning line position** - Beginning of the line.
`$`	**Ending line position** - End of the line.
`[class]`	**Character class** - Any one character in a set.
`[^class]`	**Inverse class** - Any one character not in a set.
`[x-y]`	**Range** - Any characters within the specified range.
`\x`	**Escape** - Literal use of a meta-character.
`\<string`	**Beginning word position** - Beginning of the word.
`string\>`	**Ending word position** - End of the word.

The special characters in regular expression syntax have the most power when you use them together. For example, use the combination of the wildcard character (`.`) and repeat (`*`) character to match any string of characters: `.*`

Use the following expression as part of a larger expression to match any string beginning with *b* and ending with *ing*: `b.*ing`

- To search for multiple strings in a set of files, you must create a text file that contains each search criterion on a separate line.
- Use spaces to separate multiple search strings unless the argument is prefixed with **/c**.

Examples

To search for *hello* or *there* in file *x.y*, type:

```
findstr hello there x.y
```

To search for *hello there* in file *x.y*, type:

```
findstr /c:"hello there" x.y
```

To find all occurrences of the word *Windows* (with an initial capital letter W) in the file *proposal.txt*, type:

```
findstr Windows proposal.txt
```

To search every file in the current directory and all subdirectories that contained the word *Windows*, regardless of the letter case, type:

```
findstr /s /i Windows *.*
```

To find all occurrences of lines that begin with *FOR* and are preceded by zero or more spaces (as in a computer program loop), and to display the line number where each occurrence is found, type:

```
findstr /b /n /r /c:^ *FOR *.bas
```

To list the exact files that you want to search in a text file, use the search criteria in the file *stringlist.txt*, to search the files listed in *filelist.txt*, and then to store the results in the file *results.out*, type:

```
findstr /g:stringlist.txt /f:filelist.txt > results.out
```

To list every file containing the word *computer* within the current directory and all subdirectories, regardless of case, type:

```
findstr /s /i /m \<computer\> *.*
```

To list every file containing the word computer and any other words that begin with comp, (such as compliment and compete), type:

```
```

```
findstr /s /i /m \<comp.* *.*
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

finger

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about users on a specified remote computer (typically a computer running UNIX) that is running the finger service or daemon. The remote computer specifies the format and output of the user information display. Used without parameters, **finger** displays help.

> **ⓘ Important**
>
> This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network Connections.

Syntax

```
finger [-l] [<user>] [@<host>] [...]
```

Parameters

Expand table

Parameter	Description
-l	Displays user information in long list format.
`<user>`	Specifies the user about which you want information. If you omit the *user* parameter, this command displays information about all users on the specified computer.
`@<host>`	Specifies the remote computer running the finger service where you are looking for user information. You can specify a computer name or IP address.
/?	Displays help at the command prompt.

Remarks

- You must prefix **finger** parameters with a hyphen (-) rather than a slash (/).
- Multiple `user@host` parameters can be specified.

Examples

To display information for *user1* on the computer *users.microsoft.com*, type:

```
finger user1@users.microsoft.com
```

To display information for *all users* on the computer *users.microsoft.com*, type:

```
finger @users.microsoft.com
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

flattemp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables or disables flat temporary folders. You must have administrative credentials to run this command.

> **Note**
>
> This command is only available if you have installed the Remote Desktop Session Host role service.

Syntax

```
flattemp {/query | /enable | /disable}
```

Parameters

Expand table

Parameter	Description
/query	Queries the current setting.
/enable	Enables flat temporary folders. Users will share the temporary folder unless the temporary folder resides in the user's home folder.
/disable	Disables flat temporary folders. Each user's temporary folder will reside in a separate folder (determined by the user's Session ID).
/?	Displays help at the command prompt.

Remarks

- After each user has a unique temporary folder, use `flattemp /enable` to enable flat temporary folders.

- The default method for creating temporary folders for multiple users (usually pointed to by the TEMP and TMP environment variables) is to create subfolders in the **\Temp** folder, by using the logonID as the subfolder name. For example, if the TEMP environment variable points to C:\Temp, the temporary folder assigned to the user logonID 4 is C:\Temp\4.

  Using **flattemp**, you can point directly to the \Temp folder and prevent subfolders from forming. This is useful when you want the user temporary folders to be contained in home folders, whether on an Remote Desktop Session Host server local drive or on a shared network drive. You should use the `flattemp /enable*` command only when each user has a separate temporary folder.

- You might encounter app errors if the user's temporary folder is on a network drive. This occurs when the shared network drive becomes momentarily inaccessible on the network. Because the temporary files of the app are either inaccessible or out of synchronization, it responds as if the disk has stopped. Moving the temporary folder to a network drive is not recommended. The default is to keep temporary folders on the local hard disk. If you experience unexpected behavior or disk-corruption errors with certain applications, stabilize your network or move the temporary folders back to the local hard disk.

- If you disable using separate temporary folders per-session, **flattemp** settings are ignored. This option is set in the Remote Desktop Services Configuration tool.

Examples

To display the current setting for flat temporary folders, type:

```
flattemp /query
```

To enable flat temporary folders, type:

```
flattemp /enable
```

To disable flat temporary folders, type:

```
```

```
flattemp /disable
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

fondue

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables Windows optional features by downloading required files from Windows Update or another source specified by Group Policy. The manifest file for the feature must already be installed in your Windows image.

Syntax

```
fondue.exe /enable-feature:<feature_name> [/caller-name:<program_name>]
[/hide-ux:{all | rebootrequest}]
```

Parameters

Expand table

Parameter	Description
/enable-feature: `<feature_name>`	Specifies the name of the Windows optional feature you want to enable. You can only enable one feature per command line. To enable multiple features, use fondue.exe for each feature.
/caller-name: `<program_name>`	Specifies the program or process name when you call fondue.exe from a script or batch file. You can use this option to add the program name to the SQM report if there is an error.
/hide-ux: `{all | rebootrequest}`	Use **all** to hide all messages to the user including progress and permission requests to access Windows Update. If permission is required, the operation will fail. Use **rebootrequest** to only hide user messages asking for permission to reboot the computer. Use this option if you have a script that controls reboot requests.

Examples

To enable Microsoft .NET Framework 4.8, type:

```
fondue.exe /enable-feature:NETFX4
```

To enable Microsoft .NET Framework 4.8, add the program name to the SQM report, and not display messages to the user, type:

```
fondue.exe /enable-feature:NETFX4 /caller-name:Admin.bat /hide-ux:all
```

Related links

- Command-Line Syntax Key
- Microsoft .NET Framework 4.8 Download

Feedback

Was this page helpful? Yes No

for

Article • 10/05/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Runs a specified command for each file, within a set of files.

Syntax

```
for {%% | %}<variable> in (<set>) do <command> [<commandlineoptions>]
```

Parameters

Expand table

Parameter	Description
`{%% | %}<variable>`	Required. Represents a replaceable parameter. Use a single percent sign (`%`) to carry out the **for** command at the command prompt. Use double percent signs (`%%`) to carry out the **for** command within a batch file. Variables are case sensitive, and they must be represented with an alphabetical value such as %**a**, %**b**, or %**c**.
(`<set>`)	Required. Specifies one or more files, directories, or text strings, or a range of values on which to run the command. The parentheses are required.
`<command>`	Required. Specifies the command that you want to carry out on each file, directory, or text string, or on the range of values included in *set*.
`<commandlineoptions>`	Specifies any command-line options that you want to use with the specified command.
/?	Displays help at the command prompt.

Remarks

- You can use this command within a batch file or directly from the command prompt.

- The following attributes apply to the **for** command:
  - This command replaces `% variable` or `%% variable` with each text string in the specified set until the specified command processes all of the files.
  - Variable names are case sensitive, global, and no more than 52 can be active at a time.
  - To avoid confusion with the batch parameters, `%0` through `%9`, you can use any character for *variable* except the numerals **0** through **9**. For simple batch files, a single character such as `%%f` will work.
  - You can use multiple values for *variable* in complex batch files to distinguish different replaceable variables.
- The *set* parameter can represent a single group of files or several groups of files. You can use wildcard characters (***** and **?**) to specify a file set. The following are valid file sets:

```
(*.doc)
(*.doc *.txt *.me)
(jan*.doc jan*.rpt feb*.doc feb*.rpt)
(ar??1991.* ap??1991.*)
```

- When you use this command, the first value in *set* replaces `% variable` or `%% variable`, and then the specified command processes this value. This continues until all of the files (or groups of files) that correspond to the *set* value are processed.
- **In** and **do** aren't parameters, but you must use them with this command. If you omit either of these keywords, an error message appears.
- If command extensions are enabled (that is the default), the following additional forms of **for** are supported:
  - **Directories only:** If *set* contains wildcard characters (***** or **?**), the specified *command* executes for each directory (instead of a set of files in a specified directory) that matches *set*. The syntax is:

```
for /d {%%|%}<variable> in (<set>) do <command>
[<commandlineoptions>]
```

- **Recursive:** Walks the directory tree that is rooted at *drive*:*path* and executes the **for** statement in each directory of the tree. If no directory is specified after **/r**, the current directory is used as the root directory. If *set* is just a single period (.), it only enumerates the directory tree. The syntax is:

```
for /r [[<drive>:]<path>] {%%|%}<variable> in (<set>) do <command>
[<commandlineoptions>]
```

- **Iterating a range of values:** Use an iterative variable to set the starting value (*start#*) and then step through a set range of values until the value exceeds the set ending value (*end#*). **/l** executes the iterative by comparing *start#* with *end#*. If *start#* is less than *end#* the command executes. When the iterative variable exceeds *end#*, the command shell exits the loop. You can also use a negative *step#* to step through a range in decreasing values. For example, (1,1,5) generates the sequence 1 2 3 4 5 and (5,-1,1) generates the sequence 5 4 3 2 1. The syntax is:

```
for /l {%%|%}<variable> in (<start#>,<step#>,<end#>) do <command>
[<commandlineoptions>]
```

- **Iterating and file parsing:** Use file parsing to process command output, strings, and file content. Use iterative variables to define the content or strings that you want to examine, and use the various *parsingkeywords* options to further modify the parsing. Use the *parsingkeywords* token option to specify which tokens should be passed as iterative variables. When used without the token option, **/f** will only examine the first token.

  File parsing consists of reading the output, string, or file content, and then breaking it into individual lines of text and parsing each line into zero or more tokens. The **for** loop is then called with the iterative variable value set to the token. By default, **/f** passes the first blank separated token from each line of each file. Blank lines are skipped.

  The syntaxes are:

```
for /f [<parsingkeywords>] {%%|%}<variable> in (<set>) do <command>
[<commandlineoptions>]
for /f [<parsingkeywords>] {%%|%}<variable> in (<literalstring>) do
```

```
<command> [<commandlineoptions>]
for /f [<parsingkeywords>] {%%|%}<variable> in ('<command>') do
<command> [<commandlineoptions>]
```

The *set* argument specifies one or more file names. Each file is opened, read, and processed before moving to the next file in *set*. To override the default parsing behavior, specify *parsingkeywords*. This is a quoted string that contains one or more keywords to specify different parsing options.

If you use the **usebackq** option, use one of the following syntaxes:

```
for /f [usebackq <parsingkeywords>] {%%|%}<variable> in (<set>) do
<command> [<commandlineoptions>]
for /f [usebackq <parsingkeywords>] {%%|%}<variable> in
('<literalstring>') do <command> [<commandlineoptions>]
for /f [usebackq <parsingkeywords>] {%%|%}<variable> in
(`<command>`) do <command> [<commandlineoptions>]
```

The following table lists the parsing keywords that you can use for *parsingkeywords*.

Expand table

Keyword	Description
eol=`<c>`	Specifies an end of line character (just one character).
skip=`<n>`	Specifies the number of lines to skip at the beginning of the file.
delims=`<xxx>`	Specifies a delimiter set. This replaces the default delimiter set of space and tab.
tokens=`<x,y,m-n>`	Specifies which tokens from each line are to be passed to the **for** loop for each iteration. As a result, additional variable names are allocated. *m-n* specifies a range, from the *m*th through the *n*th tokens. If the last character in the **tokens=** string is an asterisk (*), an additional variable is allocated, and it receives the remaining text on the line after the last token that is parsed.
usebackq	Specifies to run a back-quoted string as a command, use a single-quoted string as a literal string, or, for long file names that contain spaces, allow file names in `<set>`, to each be enclosed in double-quotation marks.

- **Variable substitution:** The following table lists optional syntax (for any variable **I**):

Expand table

Variable with modifier	Description
`%~I`	Expands `%I` which removes any surrounding quotation marks.
`%~fI`	Expands `%I` to a fully qualified path name.
`%~dI`	Expands `%I` to a drive letter only.
`%~pI`	Expands `%I` to a path only.
`%~nI`	Expands `%I` to a file name only.
`%~xI`	Expands `%I` to a file name extension only.
`%~sI`	Expands path to contain short names only.
`%~aI`	Expands `%I` to the file attributes of file.
`%~tI`	Expands `%I` to the date and time of file.
`%~zI`	Expands `%I` to the size of the file.
`%~$PATH:I`	Searches the directories listed in the PATH environment variable and expands `%I` to the fully qualified name of the first directory found. If the environment variable name isn't defined or the file isn't found by the search, this modifier expands to the empty string.

The following table lists modifier combinations that you can use to get compound results.

Expand table

Variable with combined modifiers	Description
`%~dpI`	Expands `%I` to a drive letter and path only.
`%~nxI`	Expands `%I` to a file name and extension only.
`%~fsI`	Expands `%I` to a full path name with short names only.
`%~dp$PATH:I`	Searches the directories that are listed in the PATH environment variable for `%I` and expands to the drive letter and path of the first one found.
`%~ftzaI`	Expands `%I` to an output line that is like **dir**.

In the above examples, you can replace `%I` and PATH with other valid values. A valid **for** variable name ends the %~ syntax.

By using uppercase variable names such as `%I`, you can make your code more readable and avoid confusion with the modifiers, which aren't case sensitive.

- **Parsing a string:** You can use the `for /f` parsing logic on an immediate string by wrapping `<literalstring>` in either: double quotes (*without* usebackq) or in single quotes (*with* usebackq) --for example, (MyString) or ('MyString'). `<literalstring>` is treated as a single line of input from a file. When parsing `<literalstring>` in double-quotes, command symbols such as (`\ & | > < ^`) are treated as ordinary characters.

- **Parsing output:** You can use the `for /f` command to parse the output of a command by placing a back-quoted `<command>` between the parentheses. It's treated as a command line, which is passed to a child Cmd.exe. The output is captured into memory and parsed as if it's a file.

Examples

To use **for** in a batch file, use the following syntax:

```
for {%%|%}<variable> in (<set>) do <command> [<commandlineoptions>]
```

To display the contents of all the files in the current directory that have the extension .doc or .txt by using the replaceable variable **%f**, type:

```
for %f in (*.doc *.txt) do type %f
```

In the preceding example, each file that has the .doc or .txt extension in the current directory is substituted for the **%f** variable until the contents of every file are displayed. To use this command in a batch file, replace every occurrence of **%f** with **%%f**. Otherwise, the variable is ignored and an error message is displayed.

To parse a file, ignoring commented lines, type:

```
```

```
for /f eol=; tokens=2,3* delims=, %i in (myfile.txt) do @echo %i %j %k
```

This command parses each line in *myfile.txt*. It ignores lines that begin with a semicolon and passes the second and third token from each line to the **for** body (tokens are delimited by commas or spaces). The body of the **for** statement references **%i** to get the second token, **%j** to get the third token, and **%k** to get all of the remaining tokens. If the file names that you supply contain spaces, use quotation marks around the text (for example, File Name). To use quotation marks, you must use **usebackq**. Otherwise, the quotation marks are interpreted as defining a literal string to parse.

%i is explicitly declared in the **for** statement. **%j** and **%k** are implicitly declared by using **tokens=**. You can use **tokens=** to specify up to 26 tokens, if it doesn't cause an attempt to declare a variable higher than the letter z or Z.

To parse the output of a command by placing *set* between the parentheses, type:

```
for /f "usebackq delims==" %i in (`set`) do @echo %i
```

To perform a recursive loop of all files in a directory, including subdirectories and echo their full path, when they were last modified, and their file size, type:

```
for /r "C:\My Dir\" %A in (*.*) do echo %~ftzA
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

forfiles

10/17/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Selects and runs a command on a file or set of files. This command is most commonly used in batch files.

Syntax

```
forfiles [/P pathname] [/M searchmask] [/S] [/C command] [/D [+ | -] [{<date> | <days>}]]
```

Parameters

Expand table

Parameter	Description
/P `<pathname>`	Specifies the path from which to start the search. By default, searching starts in the current working directory.
/M `<searchmask>`	Searches files according to the specified search mask. The default searchmask is `*`.
/S	Instructs the **forfiles** command to search in subdirectories recursively.
/C `<command>`	Runs the specified command on each file. Command strings should be wrapped in double quotes. The default command is `"cmd /c echo @file"`.
/D `[{+|-}] [{<date> | <days>}]`	Selects files with a last modified date within the specified time frame: • Selects files with a last modified date later than or equal to (+) or earlier than or equal to (-) the specified date, where *date* is in the same regional format as your Windows device, for example MM/DD/YYYY for English (United States). • Selects files with a last modified date later than or equal to (+) the current date plus the number of days specified, or earlier than or equal to (-) the current date minus the number of days specified. • Valid values for *days* include any number in the range 0-32,768. If no sign is specified, + is used by default.
/?	Displays the help text in the cmd window.

Remarks

- The `forfiles /S` command is similar to `dir /S`.
- You can use the following variables in the command string as specified by the **/C** command-line option:

Expand table

Variable	Description
@FILE	File name.
@FNAME	File name without extension.
@EXT	File name extension.
@PATH	Full path of the file.
@RELPATH	Relative path of the file.
@ISDIR	Evaluates to TRUE if a file type is a directory. Otherwise, this variable evaluates to FALSE.
@FSIZE	File size, in bytes.
@FDATE	Last modified date stamp on the file.
@FTIME	Last modified time stamp on the file.

- The **forfiles** command lets you run a command on or pass arguments to multiple files. For example, you could run the **type** command on all files in a tree with the .txt file name extension. Or you could execute every batch file (*.bat) on drive C, with the file name Myinput.txt as the first argument.
- This command can:
  - Select files by an absolute date or a relative date by using the **/d** parameter.
  - Build an archive tree of files by using variables such as @FSIZE and @FDATE.
  - Differentiate files from directories by using the @ISDIR variable.
  - Include special characters in the command line by using the hexadecimal code for the character, in 0x*HH* format (for example, 0x09 for a tab).
- This command works by implementing the `recurse subdirectories` flag on tools that are designed to process only a single file.

Examples

To list all of the batch files on drive C, type:

```
forfiles /P c:\ /S /M *.bat /C "cmd /c echo @file is a batch file"
```

To list all of the directories on drive C, type:

```
forfiles /P c:\ /S /M * /C "cmd /c if @isdir==TRUE echo @file is a directory"
```

To list all of the files in the current directory that are at least one year old, type:

```
forfiles /S /M *.* /D -365 /C "cmd /c echo @file is at least one year old."
```

To display the text *file* is outdated for each of the files in the current directory that are older than January 1, 2007, type:

```
forfiles /S /M *.* /D -01/01/2007 /C "cmd /c echo @file is outdated."
```

To list the file name extensions of all the files in the current directory in column format, and add a tab before the extension, type:

```
forfiles /S /M *.* /C "cmd /c echo The extension of @file is 0x09@ext"
```

Related links

- Command-Line Syntax Key

freedisk

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Checks to see if the specified amount of disk space is available before continuing with an installation process.

Syntax

```
freedisk [/s <computer> [/u [<domain>\]<user> [/p [<password>]]]] [/d
<drive>] [<value>]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default is the local computer. This parameter applies to all files and folders specified in the command.
/u [`<domain>`\] `<user>`	Runs the script with the permissions of the specified user account. The default is system permissions.
/p [`<password>`]	Specifies the password of the user account that is specified in **/u**.
/d `<drive>`	Specifies the drive for which you want to find out the availability of free space. You must specify `<drive>` for a remote computer.
`<value>`	Checks for a specific amount of free disk space. You can specify `<value>` in bytes, KB, MB, GB, TB, PB, EB, ZB, or YB.

Remarks

- Using the **/s**, **/u**, and **/p** command-line options are available only when you use **/s**. You must use **/p** with **/u**to provide the user's password.

- For unattended installations, you can use **freedisk** in installation batch files to check for the prerequisite amount free space before continuing with the installation.
- When you use **freedisk** in a batch file, it returns a **0** if there's enough space and a **1** if there's not enough space.

Examples

To determine whether there are at least 50 MB of free space available on drive C, type:

```
freedisk 50mb
```

Output similar to the following example appears on the screen:

```
INFO: The specified 52,428,800 byte(s) of free space is available on current
drive.
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

fsutil

Article • 03/05/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> Applies to: Windows Server 2022, Windows Server 2019, Windows Server 2016, Windows 10, Windows Server 2012 R2, Windows 8.1, Windows Server 2012, Windows 8, Windows Server 2008 R2, Windows 7

Performs tasks related to FAT and NTFS file systems, such as managing reparse points, handling sparse files, or dismounting a volume. When used without parameters, `fsutil` displays a list of supported subcommands.

> You must log on as an administrator or a member of the Administrators group to use `fsutil`. Only advanced users with a thorough understanding of Windows operating systems should use this powerful command.

Parameters

Expand table

Subcommand	Description
fsutil 8dot3name	Queries or changes the settings for short name behavior on the system, for example, generates 8.3 character-length file names. Removes short names for all files within a directory. Scans a directory and identifies registry keys that might be impacted if short names were stripped from the files in the directory.
fsutil clfs	Creates or corrects authentication codes for Common Log File System (CLFS) logfiles.
fsutil devdrv	Manages dev drive, which is a volume tuned for performance of developer scenarios. Dev drive also lets an administrator of the device control the file system minifilters that are attached to the volume.
fsutil dirty	Queries whether the volume's dirty bit is set or sets a volume's dirty bit. When a volume's dirty bit is set, **autochk** automatically checks the volume for errors the next time the computer is restarted.
fsutil file	Finds a file by user name (if Disk Quotas are enabled), queries allocated ranges for a file, sets a file's short name, sets a file's valid data length, sets zero data for

Subcommand	Description
	a file, creates a new file of a specified size, finds a file ID if given the name, or finds a file link name for a specified file ID.
fsutil fsinfo	Lists all drives and queries the drive type, volume information, NTFS-specific volume information, or file system statistics.
fsutil hardlink	Lists hard links for a file, or creates a hard link (a directory entry for a file). Every file can be considered to have at least one hard link. On NTFS volumes, each file can have multiple hard links, so a single file can appear in many directories (or even in the same directory, with different names). Because all of the links reference the same file, programs can open any of the links and modify the file. A file is deleted from the file system only after all links to it are deleted. After you create a hard link, programs can use it like any other file name.
fsutil objectid	Manages object identifiers, which are used by the Windows operating system to track objects such as files and directories.
fsutil quota	Manages disk quotas on NTFS volumes to provide more precise control of network-based storage. Disk quotas are implemented on a per-volume basis and enable both hard- and soft-storage limits to be implemented on a per-user basis.
fsutil repair	Queries or sets the self-healing state of the volume. Self-healing NTFS attempts to correct corruptions of the NTFS file system online without requiring `chkdsk.exe` to be run. Includes initiating on-disk verification and waiting for repair completion.
fsutil reparsepoint	Queries or deletes reparse points (NTFS file system objects that have a definable attribute containing user-controlled data). Reparse points are used to extend functionality in the input/output (I/O) subsystem. They're used for directory junction points and volume mount points. They're also used by file system filter drivers to mark certain files as special to that driver.
fsutil resource	Creates a Secondary Transactional Resource Manager, starts or stops a Transactional Resource Manager, displays information about a Transactional Resource Manager, or modifies its behavior.
fsutil sparse	Manages sparse files. A sparse file is a file with one or more regions of unallocated data in it. A program sees these unallocated regions as containing bytes with the value zero, but no disk space is used to represent these zeros. All meaningful or nonzero data is allocated, whereas all non-meaningful data (large strings of data composed of zeros) isn't allocated. When a sparse file is read, allocated data is returned as stored, and unallocated data is returned as zeros (by default in accordance with the C2 security requirement specification). Sparse file support allows data to be deallocated from anywhere in the file.
fsutil tiering	Enables management of storage tier functions, such as setting and disabling flags and listing of tiers.

Subcommand	Description
fsutil transaction	Commits a specified transaction, rolls back a specified transaction, or displays info about the transaction.
fsutil usn	Manages the update sequence number (USN) change journal, which provides a persistent log of all changes made to files on the volume.
fsutil volume	Manages a volume. Dismounts a volume, queries to see how much free space is available on a disk, or finds a file that is using a specified cluster.
fsutil wim	Provides functions to discover and manage WIM-backed files.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

fsutil 8dot3name

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Queries or changes the settings for short name (8dot3 name) behavior, which includes:

- Querying the current setting for the short name behavior.
- Scanning the specified directory path for registry keys that might be impacted if short names were stripped from the specified directory path.
- Changing the setting that controls the short name behavior. This setting can be applied to a specified volume or to the default volume setting.
- Removing the short names for all files within a directory.

> **Important**
>
> Permanently removing 8dot3 file names and not modifying registry keys that point to the 8dot3 file names may lead to unexpected application failures, including the inability to uninstall an application. It is recommended you first back up your directory or volume before you attempt to remove 8dot3 file names.

Syntax

```
fsutil 8dot3name [query] [<volumepath>]
fsutil 8dot3name [scan] [/s] [/l [<log file>] ] [/v] <directorypath>
fsutil 8dot3name [set] { <defaultvalue> | <volumepath> {1|0}}
fsutil 8dot3name [strip] [/t] [/s] [/f] [/l [<log file.] ] [/v] <directorypath>
```

Parameters

Expand table

Parameter	Description
query `[<volumepath>]`	Queries the file system for the state of the 8dot3 short name creation behavior. If a *volumepath* isn't specified as a parameter, the default 8dot3name creation behavior setting for all volumes is displayed.

Parameter	Description
scan `<directorypath>`	Scans the files that are located in the specified *directorypath* for registry keys that might be impacted if 8dot3 short names were stripped from the file names.
set `{<defaultvalue> | <volumepath>}`	Changes the file system behavior for 8dot3 name creation in the following instances: • When *defaultvalue* is specified, the registry key, **HKLM\System\CurrentControlSet\Control\FileSystem\NtfsDisable8dot3NameCreation**, is set to the *defaultvalue*. The *DefaultValue* can have the following values: ○ **0**: Enables 8dot3 name creation for all volumes on the system. ○ **1**: Disables 8dot3 name creation for all volumes on the system. ○ **2**: Sets 8dot3 name creation on a per volume basis. ○ **3**: Disables 8dot3 name creation for all volumes except the system volume. • When a *volumepath* is specified, the specified volumes on disk flag 8dot3name properties are set to enable 8dot3 name creation for a specified volume (**0**) or set to disable 8dot3 name creation on the specified volume (**1**). You must set the default file system behavior for 8dot3 name creation to the value **2** before you can enable or disable 8dot3 name creation for a specified volume.
strip `<directorypath>`	Removes the 8dot3 file names for all files that are located in the specified *directorypath*. The 8dot3 file name is not removed for any files where the *directorypath* combined with the file name contains more than 260 characters. This command lists, but does not modify the registry keys that point to the files that had 8dot3 file names permanently removed.
`<volumepath>`	Specifies the drive name followed by a colon or the GUID in the format `volume{GUID}`.
/f	Specifies that all files that are located in the specified *directorypath* have the 8dot3 file names removed even if there are registry keys that point to files using the 8dot3 file name. In this case, the operation removes the 8dot3 file names, but does not modify any registry keys that point to the files that are using the 8dot3 file names. **Warning:** It's recommended that you back up your directory or volume prior to using the **/f** parameter because it may lead to unexpected application failures, including the inability to uninstall programs.
/l `[<log file>]`	Specifies a log file where information is written. If the **/l** parameter isn't specified, all information is written to the default log file: `%temp%\8dot3_removal_log@(GMT YYYY-MM-DD HH-MM-SS)`.log**
/s	Specifies that the operation should be applied to the subdirectories of the specified *directorypath*.
/t	Specifies that the removal of 8dot3 file names should be run in test mode. All operations except the actual removal of the 8dot3 file names are performed. You can use test mode to discover which registry keys point to files that use the 8dot3 file names.
/v	Specifies that all information that is written to the log file is also displayed on the command-line.

Examples

To query for the disable 8dot3 name behavior for a disk volume that is specified with the GUID, {928842df-5a01-11de-a85c-806e6f6e6963}, type:

```
fsutil 8dot3name query volume{928842df-5a01-11de-a85c-806e6f6e6963}
```

You can also query the 8dot3 name behavior by using the **behavior** subcommand.

To remove 8dot3 file names in the *D:\MyData* directory and all subdirectories, while writing the information to the log file that is specified as *mylogfile.log*, type:

```
fsutil 8dot3name strip /l mylogfile.log /s d:\MyData
```

Related links

- Command-Line Syntax Key
- fsutil
- fsutil behavior

Feedback

Was this page helpful? Yes No

fsutil behavior

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Queries or sets NTFS volume behavior, which includes:

- Creating the 8.3 character-length file names.
- Extending character use in 8.3 character-length short file names on NTFS volumes.
- Updating of the **Last Access Time** stamp when directories are listed on NTFS volumes.
- The frequency with which quota events are written to the system log and to NTFS paged pool and NTFS non-paged pool memory cache levels.
- The size of the master file table zone (MFT Zone).
- Silent deletion of data when the system encounters corruption on an NTFS volume.
- File-delete notification (also known as trim or unmap).

Syntax

```
fsutil behavior query {allowextchar | bugcheckoncorrupt | disable8dot3
[<volumepath>] | disablecompression | disablecompressionlimit |
disableencryption | disablefilemetadataoptimization | disablelastaccess |
disablespotcorruptionhandling | disabletxf | disablewriteautotiering |
encryptpagingfile | mftzone | memoryusage | quotanotify | symlinkevaluation
| disabledeletenotify}

fsutil behavior set {allowextchar {1|0} | bugcheckoncorrupt {1|0} |
disable8dot3 [ <value> | [<volumepath> {1|0}] ] | disablecompression {1|0} |
disablecompressionlimit {1|0} | disableencryption {1|0} |
disablefilemetadataoptimization {1|0} | disablelastaccess {1|0} |
disablespotcorruptionhandling {1|0} | disabletxf {1|0} |
disablewriteautotiering {1|0} | encryptpagingfile {1|0} | mftzone <Value> |
memoryusage <Value> | quotanotify <frequency> | symlinkevaluation
<symboliclinktype> | disabledeletenotify {1|0}}
```

Parameters

Expand table

Parameter	Description
query	Queries the file system behavior parameters.
set	Changes the file system behavior parameters.
allowextchar `{1|0}`	Allows (**1**) or disallows (**0**) characters from the extended character set (including diacritic characters) to be used in 8.3 character-length short file names on NTFS volumes. You must restart your computer for this parameter to take effect.
Bugcheckoncorrupt `{1|0}`	Allows (**1**) or disallows (**0**) generation of a bug check when there is corruption on an NTFS volume. This feature can be used to prevent NTFS from silently deleting data when used with the Self-Healing NTFS feature. You must restart your computer for this parameter to take effect.
disable8dot3 [`<volumepath>`] `{1|0}`	Disables (**1**) or enables (**0**) the creation of 8.3 character-length file names on FAT- and NTFS-formatted volumes. Optionally, prefix with the *volumepath* specified as a drive name followed by a colon or GUID.
disablecompression `{1|0}`	Disables **(1)** or enables **(0)** NTFS compression. You must restart your computer for this parameter to take effect.
disablecompressionlimit `{1|0}`	Disables **(1)** or enables **(0)** NTFS compression limit on NTFS volume. When a compressed file reaches a certain level of fragmentation, rather than failing to extend the file, NTFS stops compressing additional extents of the file. This was done to allow compressed files to be larger than they normally would be. Setting this value to **TRUE** disables this feature which limits the size of compressed files on the system. We don't recommend disabling this feature. You must restart your computer for this parameter to take effect.
disableencryption `{1|0}`	Disables **(1)** or enables **(0)** the encryption of folders and files on NTFS volumes. You must restart your computer for this parameter to take effect.

Parameter	Description
disablefilemetadataoptimization `{1|0}`	Disables **(1)** or enables **(0)** file metadata optimization. NTFS has a limit on how many extents a given file can have. Compressed and sparse files can become very fragmented. By default, NTFS periodically compacts its internal metadata structures to allow for more fragmented files. Setting this value to **TRUE** disables this internal optimization. We don't recommend disabling this feature. You must restart your computer for this parameter to take effect.
disablelastaccess `{1|0}`	Disables (**1**) or enables (**0**) updates to the Last Access Time stamp on each directory when directories are listed on an NTFS volume. You must restart your computer for this parameter to take effect.
disablespotcorruptionhandling `{1|0}`	Disables **(1)** or enables **(0)** spot corruption handling. Also allows system administrators to run CHKDSK to analyze the state of a volume without taking it offline. We don't recommend disabling this feature. You must restart your computer for this parameter to take effect.
disabletxf `{1|0}`	Disables **(1)** or enables **(0)** txf on the specified NTFS volume. TxF is an NTFS feature that provides transaction like semantics to file system operations. TxF is presently deprecated, but the functionality is still available. We don't recommend disabling this feature on the C: volume. You must restart your computer for this parameter to take effect.
disablewriteautotiering `{1|0}`	Disables ReFS v2 auto tiering logic for tiered volumes. You must restart your computer for this parameter to take effect.
encryptpagingfile `{1|0}`	Encrypts (**1**) or doesn't encrypt (**0**) the memory paging file in the Windows operating system. You must restart your computer for this parameter to take effect.
mftzone `<value>`	Sets the size of the MFT Zone, and is expressed as a multiple of 200MB units. Set *value* to a number from **1** (default is 200 MB) to **4** (maximum is 800 MB).

Parameter	Description
	You must restart your computer for this parameter to take effect.
memoryusage `<value>`	Configures the internal cache levels of NTFS paged-pool memory and NTFS nonpaged-pool memory. Set to **1** or **2**. When set to **1** (the default), NTFS uses the default amount of paged-pool memory. When set to **2**, NTFS increases the size of its lookaside lists and memory thresholds. (A lookaside list is a pool of fixed-size memory buffers that the kernel and device drivers create as private memory caches for file system operations, such as reading a file.) You must restart your computer for this parameter to take effect.
quotanotify `<frequency>`	Configures how frequently NTFS quota violations are reported in the system log. Valid values for are in the range **0 – 4294967295**. The default frequency is **3600** seconds (one hour). You must restart your computer for this parameter to take effect.
symlinkevaluation `<symboliclinktype>`	Controls the kind of symbolic links that can be created on a computer. Valid choices are: • **1** - Local to local symbolic links, `L2L:{0|1}` • **2** - Local to remote symbolic links, `L2R:{1|0}` • **3** - Remote to local symbolic links, `R2L:{1|0}` • **4** - Remote to remote symbolic links, `R2R:{1|0}`
disabledeletenotify	Disables (**1**) or enables (**0**) delete notifications. Delete notifications (also known as trim or unmap) is a feature that notifies the underlying storage device of clusters that have been freed due to a file delete operation. In addition: • For systems using ReFS v2, trim is disabled by default. • For systems using ReFS v1, trim is enabled by default. • For systems using NTFS, trim is enabled by default unless an administrator disables it. • If your hard disk drive or SAN reports that it doesn't support trim, then your hard disk drive and SANs don't get trim notifications. • Enabling or disabling doesn't require a restart. • Trim is effective when the next unmap command is issued. • Existing inflight IO are not impacted by the registry change.

Parameter	Description
	• Doesn't require any service restart when you enable or disable trim.

Remarks

- The MFT Zone is a reserved area that enables the master file table (MFT) to expand as needed to prevent MFT fragmentation. If the average file size on the volume is 2 KB or less, it can be beneficial to set the **mftzone** value to **2**. If the average file size on the volume is 1 KB or less, it can be beneficial to set the **mftzone** value to **4**.
- When **disable8dot3** is set to **0**, every time you create a file with a long file name, NTFS creates a second file entry that has an 8.3 character-length file name. When NTFS creates files in a directory, it must look up the 8.3 character-length file names that are associated with the long file names. This parameter updates the **HKLM\SYSTEM\CurrentControlSet\Control\FileSystem\NtfsDisable8dot3NameCreation** registry key.
- The **allowextchar** parameter updates the **HKLM\SYSTEM\CurrentControlSet\Control\FileSystem\NtfsAllowExtendedCharacterIn8dot3Name** registry key.
- The **disablelastaccess** parameter reduces the impact of logging updates to the **Last Access Time** stamp on files and directories. Disabling the **Last Access Time** feature improves the speed of file and directory access. This parameter updates the **HKLM\SYSTEM\CurrentControlSet\Control\FileSystem\NtfsDisableLastAccessUpdate** registry key.

  Notes:

  - File-based **Last Access Time** queries are accurate even if all on-disk values aren't current. NTFS returns the correct value on queries because the accurate value is stored in memory.
  - One hour is the maximum amount of time that NTFS can defer updating **Last Access Time** on disk. If NTFS updates other file attributes such as **Last Modify Time**, and a **Last Access Time** update is pending, NTFS updates **Last Access Time** with the other updates without additional performance impact.
  - The **disablelastaccess** parameter can affect programs such as Backup and Remote Storage, which rely on this feature.

- Increasing the physical memory doesn't always increase the amount of paged pool memory available to NTFS. Setting **memoryusage** to **2** raises the limit of paged pool memory. This might improve performance if your system is opening and closing many files in the same file set and is not already using large amounts of system memory for other apps or for cache memory. If your computer is already using large amounts of system memory for other apps or for cache memory, increasing the limit of NTFS paged and non-paged pool memory reduces the available pool memory for other processes. This might reduce overall system performance. This parameter updates the **HKLM\SYSTEM\CurrentControlSet\Control\FileSystem\NtfsMemoryUsage** registry key.

- The value specified in the **mftzone** parameter is an approximation of the initial size of the MFT plus the MFT Zone on a new volume, and it is set at mount time for each file system. As space on the volume is used, NTFS adjusts the space reserved for future MFT growth. If the MFT Zone is already large, the full MFT Zone size is not reserved again. Because the MFT Zone is based on the contiguous range past the end of the MFT, it shrinks as the space is used.

  The file system doesn't determine the new MFT Zone location until the current MFT Zone is completely used. Note that this never occurs on a typical system.

- Some devices may experience performance degradation when the delete notification feature is turned on. In this case, use the **disabledeletenotify** option to turn off the notification feature.

Examples

To query for the disable 8dot3 name behavior for a disk volume specified with the GUID, {928842df-5a01-11de-a85c-806e6f6e6963}, type:

```
fsutil behavior query disable8dot3 volume{928842df-5a01-11de-a85c-
806e6f6e6963}
```

You can also query the 8dot3 name behavior by using the **8dot3name** subcommand.

To query the system to see if TRIM is enabled or not, type:

```
fsutil behavior query DisableDeleteNotify
```

This yields an output similar to this:

```
NTFS DisableDeleteNotify = 1
ReFS DisableDeleteNotify is not currently set
```

To override the default behavior for TRIM (disabledeletenotify) for ReFS v2, type:

```
fsutil behavior set disabledeletenotify ReFS 0
```

To override the default behavior for TRIM (disabledeletenotify) for NTFS and ReFS v1, type:

```
fsutil behavior set disabledeletenotify 1
```

Related links

- Command-Line Syntax Key
- fsutil
- fsutil 8dot3name

Feedback

Was this page helpful? Yes No

fsutil clfs

Article • 03/05/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `fsutil clfs` command adds authentication support to a Common Log File System (CLFS) logfile that has either invalid or missing authentication codes. Authentication codes are written to the Base Logfile (`.blf`) and all containers associated with the logfile. When authentication is enforced, CLFS opens a logfile if it has valid authentication codes.

Syntax

```
fsutil clfs authenticate [<BLF Logfile path>]
```

Parameters

Expand table

Parameter	Description
authenticate	Adds authentication codes to an existing CLFS logfile.

Example

The following example adds authentication codes to the "C:\example_log" logfile:

```
fsutil clfs authenticate "C:\example_log.blf"
```

Related links

- Fsutil
- Introduction to the Common Log File System

- Common Log File System API
- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

fsutil devdrv

Article • 09/06/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Dev drive or a developer volume is a volume that is tuned for performance of developer scenarios. The command also lets an administrator of the device control the file system minifilters that are attached to the volume.

Devdrv commands supported

Expand table

Command	Description
query	Query dev drive information.
enable	Enable dev drive support on this system.
disable	Disable dev drive support on this system.
trust	Trust the given dev drive.
untrust	Untrust the given dev drive.
setFiltersAllowed	Set the list of allowed filters for dev drive.
clearFiltersAllowed	Clear the list of allowed filters for dev drive.

Syntax

```
fsutil devdrv query [<volume path>]

fsutil devdrv enable [/allowAv|/disallowAv]

fsutil devdrv disable

fsutil devdrv trust [/f] <volume path>

fsutil devdrv untrust [/f] <volume path>

fsutil devdrv setFiltersAllowed [[/f] /volume <volume path>] "filter1,
filter 2, ..."
```

```
fsutil devdrv clearFiltersAllowed [[/f] <volume path>]
```

fsutil devdrv query

Queries information about dev drives or the given dev drive. The command shows if the given volume is a dev drive and if it's trusted. The command also lists the filters allowed to attach and currently attached to the dev drive.

fsutil devdrv query examples

The following example shows the output of the command for the query command.

Example 1: Query dev drive information

The command queries information about dev drives on this system.

```
fsutil devdrv query
```

Example 2: Query dev drive information for a specified volume

The following command queries dev drive related information for `D:`. The command lists the filters currently attached to `D:` if it's a dev drive.

```
fsutil devdrv query D:
```

fsutil devdrv enable

Enable dev drive support on this system.

fsutil devdrv enable examples

The following example shows the output of the command for the enable command.

Example 1: Enable dev drive support

This command enables the dev drive and let the system decide whether to allow antivirus filter to attach on a dev drive.

```
fsutil devdrv enable
```

Example 2: Enable dev drive support and allow antivirus filter to attach

In this command, the dev drive is enabled while allowing antivirus filter to be attached on a dev drive.

```
fsutil devdrv enable /allowAv
```

Example 3: Enable dev drive support and disallow antivirus filter to attach

The following command enables dev drive without allowing antivirus filter to be attached on a dev drive.

```
fsutil devdrv enable /disallowAv
```

fsutil devdrv disable

Disable dev drive support on this system.

> ⓘ **Note**
>
> Existing dev drives will function as a regular volume i.e., all filters are allowed to attach.

fsutil devdrv trust

Trust the specified dev drive. The dev drive filter attach policy is honored only for a trusted dev drive.

fsutil devdrv trust parameters

Expand table

Parameter	Description
/F	Force dismount the volume for the change to take effect immediately even if the volume is in use. Otherwise the volume is dismounted only if it isn't in use.

fsutil devdrv trust examples

The following examples show the output of the command for the trust command.

Example 1: Trust a dev drive for a specified volume

This command trusts the volume `D:`. The volume won't be dismounted if it is in use, the change takes effect next time the volume mounts.

```
fsutil devdrv trust D:
```

Example 2: Trust a dev drive for a specified volume immediately

In this command, the volume is trusted and is dismounted forcefully for the change to take effect immediately.

```
fsutil devdrv trust /f D:
```

fsutil devdrv untrust

Untrust the given dev drive. The dev drive filter attach policy isn't honored for an untrusted dev drive.

fsutil devdrv untrust parameters

Expand table

Parameter	Description
/F	Force dismount the volume for the change to take effect immediately even if the volume is in use. Otherwise the volume is dismounted only if it isn't in use.

fsutil devdrv untrust examples

The following examples show the output of the command for the untrust command.

Example 1: Untrust a dev drive for a specified volume

The following command untrusts the volume `D:`. The volume won't be dismounted if it is in use and the change would take effect next time the volume mounts.

```
fsutil devdrv untrust D:
```

Example 2: Untrust a dev drive for a specified volume immediately

The command untrusts the volume `D:` and dismounts it forcefully for the change to take effect immediately.

```
fsutil devdrv untrust /f D:
```

fsutil devdrv setFiltersAllowed

Set the list of allowed filters for the specified dev drive or any dev drive on this system.

fsutil devdrv setFiltersAllowed parameters

Expand table

Parameter	Description
/F	Force dismount the volume for the change to take effect immediately even if the volume is in use. Otherwise the volume, if specified, is dismounted only if it isn't in

Parameter	Description
	use. Note: the parameter is available only when `<volume path>` is provided.
/Volume	If specified, the allowed list of filters is set only for this volume.

fsutil devdrv setFiltersAllowed examples

The following examples show the output of the command for the setFiltersAllowed command.

Example 1: Set the list of allowed filters for any dev drive on the system

The command allows `filter1` and `filter2` to be attached to any dev drive on the system. The change will take effect next time a dev drive is mounted.

```
fsutil devdrv setFiltersAllowed "filter1, filter 2"
```

Example 2: Set the list of allowed filters for a specified volume

The command, `filter1` and `filter2` are allowed to attach to dev drive `D:`. The volume won't be dismounted if it is in use and the change would take effect next time the volume mounts.

```
fsutil devdrv setFiltersAllowed /volume D: filter1,filter2
```

Example 3: Set the list of allowed filters for a specified volume immediately

The following command allows `filter1` and `filter2` to attach to dev drive `D:`. The command forcefully dismounts the volume for the change to take effect immediately.

```
fsutil devdrv setFiltersAllowed /f /volume D: filter1,filter2
```

fsutil devdrv clearFiltersAllowed

Clear the list of allowed filters for the specified dev drive or any dev drive on this system.

fsutil devdrv clearFiltersAllowed parameters

Expand table

Parameter	Description
/F	Force dismount the volume for the change to take effect immediately even if the volume is in use. Otherwise the volume, if specified, is dismounted only if it isn't in use. Note: the parameter is available only when `<volume path>` is provided.
/Volume	If specified, the allowed list of filters is set only for this volume.

fsutil devdrv clearFiltersAllowed examples

The following examples show the output of the command for the clearFiltersAllowed command.

Example 1: Clear the list of allowed filters for any dev drive on this system

This command clears the list of allowed filters for any dev drive on this system. The change will take effect next time a dev drive is mounted on this system.

```
fsutil devdrv clearFiltersAllowed
```

Example 2: Clear the list of allowed filters for a specified volume

The following command clears the list of allowed filters `D:`. The volume won't be dismounted if it is in use, the change takes effect next time the volume mounts.

```
fsutil devdrv clearFiltersAllowed D:
```

> **Note**
>
> `D:` must be a dev drive.

Example 3: Clear the list of allowed filters for a specified volume immediately

The following command clears the list of allowed filters on `D:`. The command forcefully dismounts the volume for the change to take effect immediately.

```
fsutil devdrv clearFiltersAllowed /f D:
```

> **Note**
>
> `D:` must be a dev drive.

Related links

- Command-Line Syntax Key
- fsutil
- Dev Drive

Feedback

Was this page helpful? Yes No

fsutil dirty

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Queries or sets a volume's dirty bit. When a volume's dirty bit is set, **autochk** automatically checks the volume for errors the next time the computer is restarted.

Syntax

```
fsutil dirty {query | set} <volumepath>
```

Parameters

Expand table

Parameter	Description
query	Queries the specified volume's dirty bit.
set	Sets the specified volume's dirty bit.
`<volumepath>`	Specifies the drive name followed by a colon or GUID in the following format: `volume{GUID}`.

Remarks

- A volume's dirty bit indicates that the file system may be in an inconsistent state. The dirty bit can be set because:
  - The volume is online and it has outstanding changes.
  - Changes were made to the volume and the computer was shut down before the changes were committed to the disk.
  - Corruption was detected on the volume.
- If the dirty bit is set when the computer restarts, **chkdsk** runs to verify the file system integrity and to attempt to fix any issues with the volume.

Examples

To query the dirty bit on drive C, type:

```
fsutil dirty query c:
```

- If the volume is dirty, the following output displays: `Volume C: is dirty`
- If the volume isn't dirty, the following output displays: `Volume C: is not dirty`

To set the dirty bit on drive C, type:

```
fsutil dirty set C:
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil file

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Finds a file by user name (if Disk Quotas are enabled), queries allocated ranges for a file, sets a file's short name, sets a file's valid data length, sets zero data for a file, or creates a new file.

Syntax

```
fsutil file [createnew] <filename> <length>
fsutil file [findbysid] <username> <directory>
fsutil file [optimizemetadata] [/A] <filename>
fsutil file [queryallocranges] offset=<offset> length=<length> <filename>
fsutil file [queryextents] [/R] <filename> [<startingvcn> [<numvcns>]]
fsutil file [queryfileid] <filename>
fsutil file [queryfilenamebyid] <volume> <fileid>
fsutil file [queryoptimizemetadata] <filename>
fsutil file [queryvaliddata] [/R] [/D] <filename>
fsutil file [seteof] <filename> <length>
fsutil file [setshortname] <filename> <shortname>
fsutil file [setvaliddata] <filename> <datalength>
fsutil file [setzerodata] offset=<offset> length=<length> <filename>
```

Parameters

Expand table

Parameter	Description
createnew	Creates a file of the specified name and size, with content that consists of zeroes.
`<length>`	Specifies the file's valid data length.
findbysid	Finds files that belong to a specified user on NTFS volumes where Disk Quotas are enabled.
`<username>`	Specifies the user's user name or logon name.

Parameter	Description
`<directory>`	Specifies the full path to the directory, for example C:\users.
optimizemetadata	This performs an immediate compaction of the metadata for a given file.
/a	Analyze file metadata before and after optimization.
queryallocranges	Queries the allocated ranges for a file on an NTFS volume. Useful for determining whether a file has sparse regions.
offset=`<offset>`	Specifies the start of the range that should be set to zeroes.
length=`<length>`	Specifies the length of the range (in bytes).
queryextents	Queries extents for a file.
/r	If `<filename>` is a reparse point, open it rather than its target.
`<startingvcn>`	Specifies first VCN to query. If omitted, start at VCN 0.
`<numvcns>`	Number of VCNs to query. If omitted or 0, query until EOF.
queryfileid	Queries the file ID of a file on an NTFS volume.
`<volume>`	Specifies the volume as drive name followed by a colon.
queryfilenamebyid	Displays a random link name for a specified file ID on an NTFS volume. Since a file can have more than one link name pointing to that file, it is not guaranteed which file link will be provided as a result of the query for the file name.
`<fileid>`	Specifies the ID of the file on an NTFS volume.
queryoptimizemetadata	Queries the metadata state of a file.
queryvaliddata	Queries the valid data length for a file.
/d	Display detailed valid data information.
seteof	Sets the EOF of the given file.
setshortname	Sets the short name (8.3 character-length file name) for a file on an NTFS volume.
`<shortname>`	Specifies the file's short name.
setvaliddata	Sets the valid data length for a file on an NTFS volume.
`<datalength>`	Specifies the length of the file in bytes.

Parameter	Description
setzerodata	Sets a range (specified by *offset* and *length*) of the file to zeroes, which empties the file. If the file is a sparse file, the underlying allocation units are decommitted.

Remarks

- In NTFS, there are two important concepts of file length: the end-of-file (EOF) marker and the Valid Data Length (VDL). The EOF indicates the actual length of the file. The VDL identifies the length of valid data on disk. Any reads between VDL and EOF automatically return 0 to preserve the C2 object reuse requirement.
- The **setvaliddata** parameter is only available for administrators because it requires the Perform volume maintenance tasks (SeManageVolumePrivilege) privilege. This feature is only required for advanced multimedia and system area network scenarios. The **setvaliddata** parameter must be a positive value that is greater than the current VDL, but less than the current file size.

  It is useful for programs to set a VDL when:

  - Writing raw clusters directly to disk through a hardware channel. This allows the program to inform the file system that this range contains valid data that can be returned to the user.
  - Creating large files when performance is an issue. This avoids the time it takes to fill the file with zeroes when the file is created or extended.

Examples

To find files that are owned by *scottb* on drive C, type:

```
fsutil file findbysid scottb c:\users
```

To query the allocated ranges for a file on an NTFS volume, type:

```
fsutil file queryallocranges offset=1024 length=64 c:\temp\sample.txt
```

To optimize metadata for a file, type:

```
fsutil file optimizemetadata C:\largefragmentedfile.txt
```

To query the extents for a file, type:

```
fsutil file queryextents C:\Temp\sample.txt
```

To set the EOF for a file, type:

```
fsutil file seteof C:\testfile.txt 1000
```

To set the short name for the file, *longfilename.txt* on drive C to *longfile.txt*, type:

```
fsutil file setshortname c:\longfilename.txt longfile.txt
```

To set the valid data length to *4096 bytes* for a file named *testfile.txt* on an NTFS volume, type:

```
fsutil file setvaliddata c:\testfile.txt 4096
```

To set a range of a file on an NTFS volume to zeros to empty it, type:

```
fsutil file setzerodata offset=100 length=150 c:\temp\sample.txt
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil fsinfo

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists all drives, queries the drive type, queries volume information, queries NTFS-specific volume information, or queries file system statistics.

Syntax

```
fsutil fsinfo [drives]
fsutil fsinfo [drivetype] <volumepath>
fsutil fsinfo [ntfsinfo] <rootpath>
fsutil fsinfo [statistics] <volumepath>
fsutil fsinfo [volumeinfo] <rootpath>
```

Parameters

Expand table

Parameter	Description
drives	Lists all drives in the computer.
drivetype	Queries a drive and lists its type, for example CD-ROM drive.
ntfsinfo	Lists NTFS specific volume information for the specified volume, such as the number of sectors, total clusters, free clusters, and the start and end of the MFT Zone.
sectorinfo	Lists information about the hardware's sector size and alignment.
statistics	Lists file system statistics for the specified volume, such as metadata, log file, and MFT reads and writes.
volumeinfo	Lists information for the specified volume, such as the file system, and whether the volume supports case-sensitive file names, unicode in file names, disk quotas, or is a DirectAccess (DAX) volume.
`<volumepath>:`	Specifies the drive letter (followed by a colon).

Parameter	Description
`<rootpath>:`	Specifies the drive letter (followed by a colon) of the root drive.

Examples

To list all of the drives in the computer, type:

```
fsutil fsinfo drives
```

Output similar to the following displays:

```
Drives: A:\ C:\ D:\ E:\
```

To query the drive type of drive C, type:

```
fsutil fsinfo drivetype c:
```

Possible results of the query include:

```
Unknown Drive
No such Root Directory
Removable Drive, for example floppy
Fixed Drive
Remote/Network Drive
CD-ROM Drive
Ram Disk
```

To query the volume information for volume E, type:

```
fsutil fsinfo volumeinfo e:
```

Output similar to the following displays:

```
Volume Name : Volume
Serial Number : 0xd0b634d9
Max Component Length : 255
File System Name : NTFS
Supports Named Streams
Is DAX Volume
```

To query drive F for NTFS-specific volume information, type:

```
fsutil fsinfo ntfsinfo f:
```

Output similar to the following displays:

```
NTFS Volume Serial Number : 0xe660d46a60d442cb
Number Sectors : 0x0000000010ea04f
Total Clusters : 0x000000000021d409
Mft Zone End : 0x0000000000004700
```

To query the file system's underlying hardware for sector information, type:

```
fsutil fsinfo sectorinfo d:
```

Output similar to the following displays:

```
D:\>fsutil fsinfo sectorinfo d:
LogicalBytesPerSector : 4096
PhysicalBytesPerSectorForAtomicity : 4096
Trim Not Supported
DAX capable
```

To query the file system statistics for drive E, type:

```
fsutil fsinfo statistics e:
```

Output similar to the following displays:

```
File System Type : NTFS
Version : 1
UserFileReads : 75021
UserFileReadBytes : 1305244512
LogFileWriteBytes : 180936704
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil hardlink

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a hard link between an existing file and a new file. A hard link is a directory entry for a file. Every file can be considered to have at least one hard link.

On NTFS volumes, each file can have multiple hard links, so a single file can appear in many directories (or even in the same directory with different names). Because all of the links reference the same file, programs can open any of the links and modify the file. A file is deleted from the file system only after all links to it have been deleted. After you create a hard link, programs can use it like any other file name.

Syntax

```
fsutil hardlink create <newfilename> <existingfilename>
fsutil hardlink list <filename>
```

Parameters

Expand table

Parameter	Description
create	Establishes an NTFS hard link between an existing file and a new file. (An NTFS hard link is similar to a POSIX hard link.)
<newfilename>	Specifies the file that you want to create a hard link to.
<existingfilename>	Specifies the file that you want to create a hard link from.
list	Lists the hard links to *filename*.

Related links

- Command-Line Syntax Key

- fsutil

Feedback

Was this page helpful? Yes No

fsutil objectid

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Manages object identifiers (OIDs), which are internal objects used by the Distributed Link Tracking (DLT) Client service and File Replication Service (FRS), to track other objects such as files, directories, and links. Object identifiers are invisible to most programs and should never be modified.

> ⚠ **Warning**
>
> Don't delete, set, or otherwise modify an object identifier. Deleting or setting an object identifier can result in the loss of data from portions of a file, up to and including entire volumes of data. In addition, you might cause adverse behavior in the Distributed Link Tracking (DLT) Client service and File Replication Service (FRS).

Syntax

```
fsutil objectid [create] <filename>
fsutil objectid [delete] <filename>
fsutil objectid [query] <filename>
fsutil objectid [set] <objectID> <birthvolumeID> <birthobjectID> <domainID>
<filename>
```

Parameters

Expand table

Parameter	Description
create	Creates an object identifier if the specified file does not already have one. If the file already has an object identifier, this subcommand is equivalent to the **query** subcommand.
delete	Deletes an object identifier.
query	Queries an object identifier.

Parameter	Description
set	Sets an object identifier.
`<objectID>`	Sets a file-specific 16 byte hexadecimal identifier that is guaranteed to be unique within a volume. The object identifier is used by the Distributed Link Tracking (DLT) Client service and the File Replication Service (FRS) to identify files.
`<birthvolumeID>`	Indicates the volume on which the file was located when it first obtained an object identifier. This value is a 16-byte hexadecimal identifier that is used by the DLT Client service.
`<birthobjectID>`	Indicates the file's original object identifier (The *objectID* may change when a file is moved). This value is a 16-byte hexadecimal identifier that is used by the DLT Client service.
`<domainID>`	16-byte hexadecimal domain identifier. This value isn't currently used and must be set to all zeros.
`<filename>`	Specifies the full path to the file including the file name and extension, for example *C:\documents\filename.txt*.

Remarks

- Any file that has an object identifier also has a birth volume identifier, a birth object identifier, and a domain identifier. When you move a file, the object identifier may change, but the birth volume and birth object identifiers remain the same. This behavior enables the Windows operating system to always find a file, no matter where it has been moved.

Examples

To create an object identifier, type:

```
fsutil objectid create c:\temp\sample.txt
```

To delete an object identifier, type:

```
fsutil objectid delete c:\temp\sample.txt
```

To query an object identifier, type:

```
fsutil objectid query c:\temp\sample.txt
```

To set an object identifier, type:

```
fsutil objectid set 40dff02fc9b4d4118f120090273fa9fc
f86ad6865fe8d21183910008c709d19e 40dff02fc9b4d4118f120090273fa9fc
00000000000000000000000000000000 c:\temp\sample.txt
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil quota

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Manages disk quotas on NTFS volumes to provide more precise control of network-based storage.

Syntax

```
fsutil quota [disable] <volumepath>
fsutil quota [enforce] <volumepath>
fsutil quota [modify] <volumepath> <threshold> <limit> <username>
fsutil quota [query] <volumepath>
fsutil quota [track] <volumepath>
fsutil quota [violations]
```

Parameters

Expand table

Parameter	Description
disable	Disables quota tracking and enforcement on the specified volume.
enforce	Enforces quota usage on the specified volume.
modify	Modifies an existing disk quota or creates a new quota.
query	Lists existing disk quotas.
track	Tracks disk usage on the specified volume.
violations	Searches the system and application logs and displays a message to indicate that quota violations have been detected or that a user has reached a quota threshold or quota limit.
`<volumepath>`	Required. Specifies the drive name followed by a colon or the GUID in the format `volume{GUID}`.
`<threshold>`	Sets the limit (in bytes) at which warnings are issued. This parameter is required for the `fsutil quota modify` command.

Parameter	Description
`<limit>`	Sets the maximum allowed disk usage (in bytes). This parameter is required for the `fsutil quota modify` command.
`<username>`	Specifies the domain or user name. This parameter is required for the `fsutil quota modify` command.

Remarks

- Disk quotas are implemented on a per-volume basis, and they enable both hard and soft storage limits to be implemented on a per-user basis.
- You can use write scripts that use **fsutil quota** to set the quota limits every time you add a new user or to automatically track quota limits, compile them into a report, and automatically send them to the system administrator in e-mail.

Examples

To list existing disk quotas for a disk volume that is specified with the GUID, {928842df-5a01-11de-a85c-806e6f6e6963}, type:

```
fsutil quota query volume{928842df-5a01-11de-a85c-806e6f6e6963}
```

To list existing disk quotas for a disk volume that is specified with the drive letter, **C:**, type:

```
fsutil quota query C:
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil repair

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Administers and monitors NTFS self-healing repair operations. Self-healing NTFS attempts to correct corruptions of the NTFS file system online, without requiring **Chkdsk.exe** to be run. For more information, see Self-healing NTFS.

Syntax

```
fsutil repair [enumerate] <volumepath> [<logname>]
fsutil repair [initiate] <volumepath> <filereference>
fsutil repair [query] <volumepath>
fsutil repair [set] <volumepath> <flags>
fsutil repair [wait][<waittype>] <volumepath>
```

Parameters

Expand table

Parameter	Description
enumerate	Enumerates the entires of a volume's corruption log.
`<logname>`	Can be `$corrupt`, the set of confirmed corruptions in the volume or `$verify`, a set of potential, unverified corruptions in the volume.
initiate	Initiates NTFS self-healing.
`<filereference>`	Specifies the NTFS volume-specific file ID (file reference number). The file reference includes the segment number of the file.
query	Queries the self-healing state of the NTFS volume.
set	Sets the self-healing state of the volume.
`<flags>`	Specifies the repair method to be used when setting the self-healing state of the volume. This parameter can be set to three values:

Parameter	Description
	• **0x01** - Enables general repair. • **0x09** - Warns about potential data loss without repair. • **0x00** - Disables NTFS self-healing repair operations.
state	Queries the corruption state of the system or for a given volume.
wait	Waits for repair(s) to complete. If NTFS has detected a problem on a volume on which it is performing repairs, this option allows the system to wait until the repair is complete before it runs any pending scripts.
`[waittype {0|1}]`	Indicates whether to wait for the current repair to complete or to wait for all repairs to complete. The *waittype* parameter can be set to the following values: • **0** - Waits for all repairs to complete. (default value) • **1** - Waits for the current repair to complete.

Examples

To enumerate the confirmed corruptions of a volume, type:

```
fsutil repair enumerate C: $Corrupt
```

To enable self-healing repair on drive C, type:

```
fsutil repair set c: 1
```

To disable self-healing repair on drive C, type:

```
fsutil repair set c: 0
```

Related links

- Command-Line Syntax Key
- fsutil

- Self-healing NTFS

Feedback

Was this page helpful? Yes No

fsutil reparsepoint

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Queries or deletes reparse points. The **fsutil reparsepoint** command is typically used by support professionals.

Reparse points are NTFS file system objects that have a definable attribute, which contains user-defined data. They're used to:

- Extend functionality in the input/output (I/O) subsystem.
- Act as directory junction points and volume mount points.
- Mark certain files as special to a file system filter driver.

Syntax

```
fsutil reparsepoint [query] <filename>
fsutil reparsepoint [delete] <filename>
```

Parameters

Expand table

Parameter	Description
query	Retrieves the reparse point data that is associated with the file or directory identified by the specified handle.
delete	Deletes a reparse point from the file or directory that is identified by the specified handle, but does not delete the file or directory.
`<filename>`	Specifies the full path to the file including the file name and extension, for example *C:\documents\filename.txt*.

Remarks

- When a program sets a reparse point, it stores this data, plus a reparse tag, which uniquely identifies the data it is storing. When the file system opens a file with a reparse point, it attempts to find the associated file system filter. If the file system filter is found, the filter processes the file as directed by the reparse data. If no file system filter is found, the **File open** operation fails.

Examples

To retrieve reparse point data associated with *c:\server*, type:

```
fsutil reparsepoint query c:\server
```

To delete a reparse point from a specified file or directory, use the following format:

```
fsutil reparsepoint delete c:\server
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil resource

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a Secondary Transactional Resource Manager, starts or stops a Transactional Resource Manager, or displays information about a Transactional Resource Manager, and modifies the following behavior:

- Whether a default Transactional Resource Manager cleans its transactional metadata at the next mount.
- The specified Transactional Resource Manager to prefer consistency over availability.
- The specified Transaction Resource Manager to prefer availability over consistency.
- The characteristics of a running Transactional Resource Manager.

Syntax

```
fsutil resource [create] <rmrootpathname>
fsutil resource [info] <rmrootpathname>
fsutil resource [setautoreset] {true|false} <Defaultrmrootpathname>
fsutil resource [setavailable] <rmrootpathname>
fsutil resource [setconsistent] <rmrootpathname>
fsutil resource [setlog] [growth {<containers> containers|<percent> percent}
<rmrootpathname>] [maxextents <containers> <rmrootpathname>] [minextents
<containers> <rmrootpathname>] [mode {full|undo} <rmrootpathname>] [rename
<rmrootpathname>] [shrink <percent> <rmrootpathname>] [size <containers>
<rmrootpathname>]
fsutil resource [start] <rmrootpathname> [<rmlogpathname> <tmlogpathname>
fsutil resource [stop] <rmrootpathname>
```

Parameters

Expand table

Parameter	Description
create	Creates a secondary Transactional Resource Manager.
`<rmrootpathname>`	Specifies the full path to a Transactional Resource Manager root directory.
info	Displays the specified Transactional Resource Manager's information.
setautoreset	Specifies whether a default Transactional Resource Manager will clean the transactional metadata on the next mount. • **true** - Specifies that the Transaction Resource Manager will clean the transactional metadata on the next mount, by default. • **false** - Specifies that the Transaction Resource Manager will not clean the transactional metadata on the next mount, by default.
`<defaultrmrootpathname>`	Specifies the drive name followed by a colon.
setavailable	Specifies that a Transactional Resource Manager will prefer availability over consistency.
setconsistent	Specifies that a Transactional Resource Manager will prefer consistency over availability.
setlog	Changes the characteristics of a Transactional Resource Manager that is already running.
growth	Specifies the amount by which the Transactional Resource Manager log can grow. The growth parameter can be specified as follows: • Number of containers, using the format: `<containers> containers` • Percentage, using the format: `<percent> percent`
`<containers>`	Specifies the data objects that are used by the Transactional Resource Manager.
maxextent	Specifies the maximum number of containers for the specified Transactional Resource Manager.
minextent	Specifies the minimum number of containers for the specified Transactional Resource Manager.
mode `{full|undo}`	Specifies whether all transactions are logged (**full**) or only rolled back events are logged (**undo**).
rename	Changes the GUID for the Transactional Resource Manager.

Parameter	Description
shrink	Specifies percentage by which the Transactional Resource Manager log can automatically decrease.
size	Specifies the size of the Transactional Resource Manager as a specified number of *containers*.
start	Starts the specified Transactional Resource Manager.
stop	Stops the specified Transactional Resource Manager.

Examples

To set the log for the Transactional Resource Manager that is specified by *c:\test*, to have an automatic growth of five containers, type:

```
fsutil resource setlog growth 5 containers c:test
```

To set the log for the Transactional Resource Manager that is specified by *c:\test*, to have an automatic growth of two percent, type:

```
fsutil resource setlog growth 2 percent c:test
```

To specify that the default Transactional Resource Manager will clean the transactional metadata on the next mount on drive C, type:

```
fsutil resource setautoreset true c:\
```

Related links

- Command-Line Syntax Key
- fsutil
- Transactional NTFS

Feedback

Was this page helpful? Yes No

fsutil sparse

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Manages sparse files. A sparse file is a file with one or more regions of unallocated data in it.

A program sees these unallocated regions as containing bytes with a zero value and that there's no disk space representing these zeros. When a sparse file is read, allocated data is returned as stored, and unallocated data is returned, by default, as zeros, in accordance with the C2 security requirement specification. Sparse file support allows data to be deallocated from anywhere in the file.

Syntax

```
fsutil sparse [queryflag] <filename>
fsutil sparse [queryrange] <filename>
fsutil sparse [setflag] <filename>
fsutil sparse [setrange] <filename> <beginningoffset> <length>
```

Parameters

Expand table

Parameter	Description
queryflag	Queries sparse.
queryrange	Scans a file and searches for ranges that may contain nonzero data.
setflag	Marks the indicated file as sparse.
setrange	Fills a specified range of a file with zeros.
`<filename>`	Specifies the full path to the file including the file name and extension, for example *C:\documents\filename.txt*.
`<beginningoffset>`	Specifies the offset within the file to mark as sparse.

Parameter	Description
`<length>`	Specifies the length of the region in the file to be marked as sparse (in bytes).

Remarks

- All meaningful or nonzero data is allocated, whereas all non-meaningful data (large strings of data that is composed of zeros) is not allocated.
- In a sparse file, large ranges of zeroes may not require disk allocation. Space for nonzero data is allocated as needed when the file is written.
- Only compressed or sparse files can have zeroed ranges known to the operating system.
- If the file is sparse or compressed, NTFS may de-allocate disk space within the file. This sets the range of bytes to zeroes without extending the file size.

Examples

To mark a file named *sample.txt* in the *c:\temp* directory as sparse, type:

```
fsutil sparse setflag c:\temp\sample.txt
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil tiering

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables management of storage tier functions, such as setting and disabling flags and listing of tiers.

Syntax

```
fsutil tiering [clearflags] <volume> <flags>
fsutil tiering [queryflags] <volume>
fsutil tiering [regionlist] <volume>
fsutil tiering [setflags] <volume> <flags>
fsutil tiering [tierlist] <volume>
```

Parameters

Expand table

Parameter	Description
clearflags	Disables the tiering behavior flags of a volume.
`<volume>`	Specifies the volume.
/trnh	For volumes with tiered storage, causes Heat gathering to be disabled. Applies to NTFS and ReFS only.
queryflags	Queries the tiering behavior flags of a volume.
regionlist	Lists the tiered regions of a volume and their respective storage tiers.
setflags	Enables the tiering behavior flags of a volume.
tierlist	Lists the storage tiers associated with a volume.

Examples

To query the flags on volume C, type:

```
fsutil tiering queryflags C:
```

To set the flags on volume C, type:

```
fsutil tiering setflags C: /trnh
```

To clear the flags on volume C, type:

```
fsutil tiering clearflags C: /trnh
```

To list the regions of volume C and their respective storage tiers, type:

```
fsutil tiering regionlist C:
```

To list the tiers of volume C, type:

```
fsutil tiering tierlist C:
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil transaction

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Manages NTFS transactions.

Syntax

```
fsutil transaction [commit] <GUID>
fsutil transaction [fileinfo] <filename>
fsutil transaction [list]
fsutil transaction [query] [{files | all}] <GUID>
fsutil transaction [rollback] <GUID>
```

Parameters

Expand table

Parameter	Description
commit	Marks the end of a successful implicit or explicit specified transaction.
`<GUID>`	Specifies the GUID value that represents a transaction.
fileinfo	Displays transaction information for the specified file.
`<filename>`	Specifies full path and file name.
list	Displays a list of currently running transactions.
query	Displays information for the specified transaction. • If `fsutil transaction query files` is specified, the file information is displayed only for the specified transaction. • If `fsutil transaction query all` is specified, all information for the transaction will be displayed.
rollback	Rolls back a specified transaction to the beginning.

Examples

To display transaction information for file *c:\test.txt*, type:

```
fsutil transaction fileinfo c:\test.txt
```

Related links

- Command-Line Syntax Key
- fsutil
- Transactional NTFS

Feedback

Was this page helpful? Yes No

fsutil usn

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Manages the update sequence number (USN) change journal. The USN change journal provides a persistent log of all changes made to files on the volume. As files, directories, and other NTFS objects are added, deleted, and modified, NTFS enters records into the USN change journal, one for each volume on the computer. Each record indicates the type of change and the object changed. New records are appended to the end of the stream.

Syntax

```
fsutil usn [createjournal] m=<maxsize> a=<allocationdelta> <volumepath>
fsutil usn [deletejournal] {/d | /n} <volumepath>
fsutil usn [enablerangetracking] <volumepath> [options]
fsutil usn [enumdata] <fileref> <lowUSN> <highUSN> <volumepath>
fsutil usn [queryjournal] <volumepath>
fsutil usn [readdata] <filename>
fsutil usn [readjournal] [c= <chunk-size> s=<file-size-threshold>]
<volumepath>
```

Parameters

Expand table

Parameter	Description
createjournal	Creates a USN change journal.
m=`<maxsize>`	Specifies the maximum size, in bytes, that NTFS allocates for the change journal.
a=`<allocationdelta>`	Specifies the size, in bytes, of memory allocation that is added to the end and removed from the beginning of the change journal.
`<volumepath>`	Specifies the drive letter (followed by a colon).
deletejournal	Deletes or disables an active USN change journal.

Parameter	Description
	CAUTION: Deleting the change journal impacts the File Replication Service (FRS) and the Indexing Service, because it requires these services to perform a complete (and time-consuming) scan of the volume. This in turn negatively impacts FRS SYSVOL replication and replication between DFS link alternates while the volume is being rescanned.
/d	Disables an active USN change journal, and returns input/output (I/O) control while the change journal is being disabled.
/n	Disables an active USN change journal and returns I/O control only after the change journal is disabled.
enablerangetracking	Enables USN write range tracking for a volume.
c=`<chunk-size>`	Specifies the chunk size to track on a volume.
s=`<file-size-threshold>`	Specifies the file size threshold for range tracking.
enumdata	Enumerates and lists the change journal entries between two specified boundaries.
`<fileref>`	Specifies the ordinal position within the files on the volume at which the enumeration is to begin.
`<lowUSN>`	Specifies the lower boundary of the range of USN values used to filter the records that are returned. Only records whose last change journal USN is between or equal to the *lowUSN* and *highUSN* member values are returned.
`<highUSN>`	Specifies the upper boundary of the range of USN values used to filter the files that are returned.
queryjournal	Queries a volume's USN data to gather information about the current change journal, its records, and its capacity.
readdata	Reads the USN data for a file.
`<filename>`	Specifies the full path to the file, including the file name and extension For example: *C:\documents\filename.txt*.
readjournal	Reads the USN records in the USN journal.
minver=`<number>`	Minimum Major Version of USN_RECORD to return. Default = 2.
maxver=`<number>`	Maximum Major Version of USN_RECORD to return. Default = 4.
startusn=`<USN number>`	USN to start reading the USN journal from. Default = 0.

Remarks

- Programs can consult the USN change journal to determine all the modifications made to a set of files. The USN change journal is much more efficient than checking time stamps or registering for file notifications. The USN change journal is enabled and used by the Indexing Service, File Replication Service (FRS), Remote Installation Services (RIS), and Remote Storage.
- If a change journal already exists on a volume, the **createjournal** parameter updates the change journal's **maxsize** and **allocationdelta** parameters. This enables you to expand the number of records that an active journal maintains without having to disable it.
- The change journal can grow larger than this target value, but the change journal is truncated at the next NTFS checkpoint to less than this value. NTFS examines the change journal and trims it when its size exceeds the value of **maxsize** plus the value of **allocationdelta**. At NTFS checkpoints, the operating system writes records to the NTFS log file that enable NTFS to determine what processing is required to recover from a failure.
- The change journal can grow to more than the sum of the values of **maxsize** and **allocationdelta** before being trimmed.
- Deleting or disabling an active change journal is very time consuming, because the system must access all the records in the master file table (MFT) and set the last USN attribute to 0 (zero). This process can take several minutes, and it can continue after the system restarts, if a restart is necessary. During this process, the change journal is not considered active, nor is it disabled. While the system is disabling the journal, it cannot be accessed, and all journal operations return errors. You should use extreme care when disabling an active journal, because it adversely affects other applications that are using the journal.

Examples

To create a USN change journal on drive C, type:

```
fsutil usn createjournal m=1000 a=100 c:
```

To delete an active USN change journal on drive C, type:

```
```

```
fsutil usn deletejournal /d c:
```

To enable range tracking with a specified chunk-size and file-size-threshold, type:

```
fsutil usn enablerangetracking c=16384 s=67108864 C:
```

To enumerate and list the change journal entries between two specified boundaries on drive C, type:

```
fsutil usn enumdata 1 0 1 c:
```

To query USN data for a volume on drive C, type:

```
fsutil usn queryjournal c:
```

To read the USN data for a file in the \Temp folder on drive C, type:

```
fsutil usn readdata c:\temp\sample.txt
```

To read the USN journal with a specific start USN, type:

```
fsutil usn readjournal startusn=0xF00
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

fsutil volume

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Dismounts a volume, or queries the hard disk drive to determine how much free space is currently available on the hard disk drive or which file is using a particular cluster.

Syntax

```
fsutil volume [allocationreport] <volumepath>
fsutil volume [diskfree] <volumepath>
fsutil volume [dismount] <volumepath>
fsutil volume [filelayout] <volumepath> <fileID>
fsutil volume [list]
fsutil volume [querycluster] <volumepath> <cluster> [<cluster>] … …
```

Parameters

Expand table

Parameter	Description
allocationreport	Displays information about how storage is used on a given volume.
`<volumepath>`	Specifies the drive letter (followed by a colon).
diskfree	Queries the hard disk drive to determine the amount of free space on it.
dismount	Dismounts a volume.
filelayout	Displays NTFS metadata for the given file.
`<fileID>`	Specifies the file id.
list	Lists all of the volumes on the system.
querycluster	Finds which file is using a specified cluster. You can specify multiple clusters with the **querycluster** parameter.
`<cluster>`	Specifies the logical cluster number (LCN).

Examples

To display an allocated clusters report, type:

```
fsutil volume allocationreport C:
```

To dismount a volume on drive C, type:

```
fsutil volume dismount c:
```

To query the amount of free space of a volume on drive C, type:

```
fsutil volume diskfree c:
```

To display all the information about a specified file(s), type:

```
fsutil volume C: *
fsutil volume C:\Windows
fsutil volume C: 0x00040000000001bf
```

To list the volumes on disk, type:

```
fsutil volume list
```

To find the file(s) that are using the clusters, specified by the logical cluster numbers 50 and 0x2000, on drive C, type:

```
fsutil volume querycluster C: 50 0x2000
```

Related links

- Command-Line Syntax Key
- fsutil
- How NTFS Works

Feedback

Was this page helpful? Yes No

fsutil wim

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Provides functions to discover and manage Windows Image (WIM)-backed files.

Syntax

```
fsutil wim [enumfiles] <drive name> <data source>
fsutil wim [enumwims] <drive name>
fsutil wim [queryfile] <filename>
fsutil wim [removewim] <drive name> <data source>
```

Parameters

Expand table

Parameter	Description
enumfiles	Enumerates WIM backed files.
`<drive name>`	Specifies the drive name.
`<data source>`	Specifies the data source.
enumwims	Enumerates backing WIM files.
queryfile	Queries if the file is backed by WIM, and if so, displays details about the WIM file.
`<filename>`	Specifies the filename.
removewim	Removes a WIM from backing files.

Examples

To enumerate the files for drive C: from data source 0, type:

```
fsutil wim enumfiles C: 0
```

To enumerate backing WIM files for drive C:, type:

```
fsutil wim enumwims C:
```

To see if a file is backed by WIM, type:

```
fsutil wim queryFile C:\Windows\Notepad.exe
```

To remove the WIM from backing files for volume C: and data source 2, type:

```
fsutil wim removewims C: 2
```

Related links

- Command-Line Syntax Key
- fsutil

Feedback

Was this page helpful? Yes No

ftp

10/17/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Transfers files to and from a computer running a File Transfer Protocol (ftp) server service. This command can be used interactively or in batch mode by processing ASCII text files.

Syntax

```
ftp [-v] [-d] [-i] [-n] [-g] [-s:<filename>] [-a] [-A] [-x:<sendbuffer>] [-r:
<recvbuffer>] [-b:<asyncbuffers>][-w:<windowssize>][<host>] [-?]
```

Parameters

Expand table

Parameter	Description
-v	Suppresses display of remote server responses.
-d	Enables debugging, displaying all commands passed between the FTP client and FTP server.
-i	Disables interactive prompting during multiple file transfers.
-n	Suppresses auto-login upon initial connection.
-g	Disables file name globbing. **Glob** permits the use of the asterisk (*) and question mark (?) as wildcard characters in local file and path names.
-s:`<filename>`	Specifies a text file that contains **ftp** commands. These commands run automatically after **ftp** starts. This parameter allows no spaces. Use this parameter instead of redirection (`<`). **Note:** In Windows 8 and Windows Server 2012 or later operating systems, the text file must be written in UTF-8.
-a	Specifies that any local interface can be used when binding the ftp data connection.
-A	Logs onto the ftp server as anonymous.
-x:`<sendbuffer>`	Overrides the default SO_SNDBUF size of 8192.
-r:`<recvbuffer>`	Overrides the default SO_RCVBUF size of 8192.

Parameter	Description
-b: `<asyncbuffers>`	Overrides the default async buffer count of 3.
-w: `<windowsize>`	Specifies the size of the transfer buffer. The default window size is 65535 bytes.
`<host>`	Specifies the computer name, IP address, or IPv6 address of the ftp server to which to connect. The host name or address, if specified, must be the last parameter on the line.
-?	Displays help at the command prompt.

Remarks

- The **ftp** command-line parameters are case-sensitive.
- This command is available only if the **Internet Protocol (TCP/IP)** protocol is installed as a component in the properties of a network adapter in Network Connections.
- The **ftp** command can be used interactively. After it is started, **ftp** creates a sub-environment in which you can use **ftp** commands. You can return to the command prompt by typing the **quit** command. When the **ftp** sub-environment is running, it is indicated by the `ftp >` command prompt. For more information, see the **ftp** commands.
- The **ftp** command supports the use of IPv6 when the IPv6 protocol is installed.

Examples

To log on to the ftp server named `ftp.example.microsoft.com`, type:

```
ftp ftp.example.microsoft.com
```

To log on to the ftp server named `ftp.example.microsoft.com` and run the **ftp** commands contained in a file named *resync.txt*, type:

```
ftp -s:resync.txt ftp.example.microsoft.com
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance
- IP version 6
- IPv6 applications

ftp append

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Appends a local file to a file on the remote computer using the current file type setting.

Syntax

```
append <localfile> [remotefile]
```

Parameters

Expand table

Parameter	Description
`<localfile>`	Specifies the local file to add.
[remotefile]	Specifies the file on the remote computer to which `<localfile>` is added. If you don't use this parameter, the `<localfile>` name is used in place of the remote file name.

Examples

To append *file1.txt* to *file2.txt* on the remote computer, type:

```
append file1.txt file2.txt
```

To append the local *file1.txt* to a file named *file1.txt* on the remote computer.

```
append file1.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp ascii

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets the file transfer type to ASCII. The **ftp** command supports both ASCII (default) and binary image file transfer types, but we recommend using ASCII when transferring text files. In ASCII mode, character conversions to and from the network standard character set are performed. For example, end-of-line characters are converted as necessary, based on the target operating system.

Syntax

```
ascii
```

Examples

To set the file transfer type to ASCII, type:

```
ascii
```

Related links

- Command-Line Syntax Key
- ftp binary command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp bell

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles an audible sound to occur after each file transfer command is completed. By default, this command is toggled off.

Syntax

```
bell
```

Examples

To toggle an audible sound to occur after each file transfer command is completed, type:

```
bell
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp binary

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets the file transfer type to binary. The **ftp** command supports both ASCII (default) and binary image file transfer types, but we recommend using binary when transferring executable files. In binary mode, files are transferred in one-byte units.

Syntax

```
binary
```

Examples

To set the file transfer type to binary, type:

```
binary
```

Related links

- Command-Line Syntax Key
- ftp ascii command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp bye

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Ends the ftp session on the remote computer, and then exits.

> **Note**
>
> This command is the same as the **ftp quit command**.

Syntax

```
bye
```

Examples

To end the ftp session with the remote computer and exit, type:

```
bye
```

Related links

- Command-Line Syntax Key
- ftp quit command
- Additional FTP guidance

Feedback

Was this page helpful?

Yes No

ftp cd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the working directory on the remote computer.

Syntax

```
cd <remotedirectory>
```

Parameters

Expand table

Parameter	Description
`<remotedirectory>`	Specifies the directory on the remote computer to which you want to change.

Examples

To change the directory on the remote computer to *Docs*, type:

```
cd Docs
```

To change the directory on the remote computer to *May Videos*, type:

```
cd  May Videos
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp close

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Ends the ftp session with the remote server and remains at the `ftp>` prompt.

Syntax

```
close
```

Examples

To end the ftp session with the remote server and remain at the `ftp>` prompt, type:

```
close
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp debug

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles Debugging mode. By default, Debugging mode is turned off. If Debugging mode is turned on, you'll see each command sent to the remote computer, preceeded by the `>` character.

Syntax

```
debug
```

Examples

To toggle debug mode on and off, type:

```
debug
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp delete

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Deletes files on remote computers.

Syntax

```
delete <remotefile>
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the file to delete.

Examples

To delete the *test.txt* file on the remote computer, type:

```
delete test.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp dir

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays a list of directory files and subdirectories on a remote computer.

Syntax

```
dir [<remotedirectory>] [<localfile>]
```

Parameters

Expand table

Parameter	Description
`[<remotedirectory>]`	Specifies the directory for which you want to see a listing. If no directory is specified, the current working directory on the remote computer is used.
`[<localfile>]`	Specifies a local file in which to store the directory listing. If a local file is not specified, results are displayed on the screen.

Examples

To display a directory listing for *dir1* on the remote computer, type:

```
dir dir1
```

To save a list of the current directory on the remote computer in the local file *dirlist.txt*, type:

```
dir . dirlist.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp disconnect

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Disconnects from the remote computer and remains at the `ftp>` prompt.

Syntax

```
disconnect
```

Examples

To disconnect from the remote computer and remains at the `ftp>` prompt, type:

```
disconnect
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp get

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies a remote file to the local computer using the current file transfer type.

> ⓘ Note
>
> This command is the same as the ftp recv command.

Syntax

```
get <remotefile> [<localfile>]
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the remote file to copy.
`[<localfile>]`	Specifies the name of the file to use on the local computer. If *localfile* isn't specified, the file is given the name of the *remotefile*.

Examples

To copy *test.txt* to the local computer using the current file transfer, type:

```
get test.txt
```

To copy *test.txt* to the local computer as *test1.txt* using the current file transfer, type:

```
get test.txt test1.txt
```

Related links

- Command-Line Syntax Key
- ftp recv command
- ftp ascii command
- ftp binary command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp glob

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles allowing wildcard expansion for local file names. By default, globbing (wildcard expansion) is turned on. If globbing is turned on, you'll be able to use the asterisk (*) and question mark (?) as wildcard characters in local file or path names.

Syntax

```
glob
```

Examples

To toggle whether to allow wildcard expansion of local file names, type:

```
glob
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp hash

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles number sign (#) printing for each transferred data block. By default, the hash command is turned off. The size of a data block is 2048 bytes.

Syntax

```
hash
```

Examples

To toggle number sign (#) printing for each data block that is transferred, type:

```
hash
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp lcd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the working directory on the local computer. By default, the working directory is the directory in which the **ftp** command was started.

Syntax

```
lcd [<directory>]
```

Parameters

Expand table

Parameter	Description
`[<directory>]`	Specifies the directory on the local computer to which to change. If *directory* isn't specified, the current working directory is changed to the default directory.

Examples

To change the working directory on the local computer to *c:\dir1*, type:

```
lcd c:\dir1
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp literal

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sends verbatim arguments to the remote ftp server. A single ftp reply code is returned.

> **Note**
>
> This command is the same as the ftp quote command.

Syntax

```
literal <argument> [ ]
```

Parameters

Expand table

Parameter	Description
`<argument>`	Specifies the argument to send to the ftp server.

Examples

To send a **quit** command to the remote ftp server, type:

```
literal quit
```

Related links

- Command-Line Syntax Key

- ftp quote command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp ls

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays an abbreviated list of files and subdirectories from the remote computer.

Syntax

```
ls [<remotedirectory>] [<localfile>]
```

Parameters

Expand table

Parameter	Description
`[<remotedirectory>]`	Specifies the directory for which you want to see a listing. If no directory is specified, the current working directory on the remote computer is used.
`[<localfile>]`	Specifies a local file in which to store the listing. If a local file is not specified, results are displayed on the screen.

Examples

To display an abbreviated list of files and subdirectories from the remote computer, type:

```
ls
```

To get an abbreviated directory listing of *dir1* on the remote computer and save it in a local file called *dirlist.txt*, type:

```
ls dir1 dirlist.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp mget

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies remote files to the local computer using the current file transfer type.

Syntax

```
mget <remotefile>[ ]
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the remote files to copy to the local computer.

Examples

To copy remote files *a.exe* and *b.exe* to the local computer using the current file transfer type, type:

```
mget a.exe b.exe
```

Related links

- Command-Line Syntax Key
- ftp ascii command
- ftp binary command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp mkdir

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a directory on the remote computer.

Syntax

```
mkdir <directory>
```

Parameters

Expand table

Parameter	Description
`<directory>`	Specifies the name of the new remote directory.

Examples

To create a directory called *dir1* on the remote computer, type:

```
mkdir dir1
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp mls

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays an abbreviated list of files and subdirectories in a remote directory.

Syntax

```
mls <remotefile>[ ] <localfile>
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the file for which you want to see a listing. When specifying *remotefiles*, use a hyphen to represent the current working directory on the remote computer.
`<localfile>`	Specifies a local file in which to store the listing. When specifying *localfile*, use a hyphen to display the listing on the screen.

Examples

To display an abbreviated list of files and subdirectories for *dir1* and *dir2*, type:

```
mls dir1 dir2 -
```

To save an abbreviated list of files and subdirectories for *dir1* and *dir2* in the local file *dirlist.txt*, type:

```
mls dir1 dir2 dirlist.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp mput

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies local files to the remote computer using the current file transfer type.

Syntax

```
mput <localfile>[ ]
```

Parameters

Expand table

Parameter	Description
`<localfile>`	Specifies the local file to copy to the remote computer.

Examples

To copy *Program1.exe* and *Program2.exe* to the remote computer using the current file transfer type, type:

```
mput Program1.exe Program2.exe
```

Related links

- Command-Line Syntax Key
- ftp ascii command
- ftp binary command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp open

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Connects to the specified ftp server.

Syntax

```
open <computer> [<port>]
```

Parameters

Expand table

Parameter	Description
`<computer>`	Specifies the remote computer to which you are trying to connect. You can use an IP address or computer name (in which case a DNS server or Hosts file must be available).
`[<port>]`	Specifies a TCP port number to use to connect to an ftp server. By default, TCP port 21 is used.

Examples

To connect to the ftp server at *ftp.microsoft.com*, type:

```
open ftp.microsoft.com
```

To connect to the ftp server at *ftp.microsoft.com* that is listening on TCP port *755*, type:

```
open ftp.microsoft.com 755
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp prompt

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles Prompt mode on and off. By default, Prompt mode is turned on. If Prompt mode is turned on, the ftp command prompts during multiple file transfers to allow you to selectively retrieve or store files.

> **Note**
>
> You can use the **ftp mget** and **ftp mput** commands to transfer all files when Prompt mode is turned off.

Syntax

```
prompt
```

Examples

To toggle Prompt mode on and off, type:

```
prompt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp put

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies a local file to the remote computer using the current file transfer type.

> **Note**
>
> This command is the same as the ftp send command.

Syntax

```
put <localfile> [<remotefile>]
```

Parameters

Expand table

Parameter	Description
`<localfile>`	Specifies the local file to copy.
`[<remotefile>]`	Specifies the name to use on the remote computer. If you don't specify a *remotefile*, the file is give the *localfile* name.

Examples

To copy the local file *test.txt* and name it *test1.txt* on the remote computer, type:

```
put test.txt test1.txt
```

To copy the local file *program.exe* to the remote computer, type:

```
put program.exe
```

Related links

- Command-Line Syntax Key
- ftp ascii command
- ftp binary command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp pwd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the current remote computer directory.

Syntax

```
pwd
```

Examples

To display the current remote computer directory, type:

```
pwd
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp quit

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Ends the ftp session with the remote computer, and then exits.

> **Note**
>
> This command is the same as the ftp bye command.

Syntax

```
quit
```

Examples

To end the ftp session with the remote computer and return to the operating system command prompt, type:

```
quit
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp quote

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sends verbatim arguments to the remote ftp server. A single ftp reply code is returned.

> **Note**
>
> This command is the same as the ftp literal command.

Syntax

```
quote <argument>[ ]
```

Parameters

Expand table

Parameter	Description
`<argument>`	Specifies the argument to send to the ftp server.

Examples

To send a **quit** command to the remote ftp server, type:

```
quote quit
```

Related links

- Command-Line Syntax Key

- ftp literal command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp recv

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies a remote file to the local computer using the current file transfer type.

> **Note**
>
> This command is the same as the ftp get command.

Syntax

```
recv <remotefile> [<localfile>]
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the remote file to copy.
`[<localfile>]`	Specifies the name of the file to use on the local computer. If *localfile* isn't specified, the file is given the name of the *remotefile*.

Examples

To copy *test.txt* to the local computer using the current file transfer, type:

```
recv test.txt
```

To copy *test.txt* to the local computer as *test1.txt* using the current file transfer, type:

```
```

```
recv test.txt test1.txt
```

Related links

- Command-Line Syntax Key
- ftp get command
- ftp ascii command
- ftp binary command
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp remotehelp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays help for remote commands.

Syntax

```
remotehelp [<command>]
```

Parameters

Expand table

Parameter	Description
`[<command>]`	Specifies the name of the command about which you want help. If `<command>` isn't specified, this command displays a list of all remote commands. You can also run remote commands using ftp quote or ftp literal.

Examples

To display a list of remote commands, type:

```
remotehelp
```

To display the syntax for the *feat* remote command, type:

```
remotehelp feat
```

Related links

- Command-Line Syntax Key
- ftp quote
- ftp literal
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp rename

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Renames remote files.

Syntax

```
rename <filename> <newfilename>
```

Parameters

Expand table

Parameter	Description
`<filename>`	Specifies the file that you want to rename.
`<newfilename>`	Specifies the new file name.

Examples

To rename the remote file *example.txt* to *example1.txt*, type:

```
rename example.txt example1.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp rmdir

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Deletes a remote directory.

Syntax

```
rmdir <directory>
```

Parameters

Expand table

Parameter	Description
`<directory>`	Specifies the name of the remote directory to delete.

Examples

To delete the *pictures* remote directory, type:

```
rmdir pictures
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp send

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies a local file to the remote computer using the current file transfer type.

> **Note**
>
> This command is the same as the ftp put command.

Syntax

```
send <localfile> [<remotefile>]
```

Parameters

Expand table

Parameter	Description
`<localfile>`	Specifies the local file to copy.
`<remotefile>`	Specifies the name to use on the remote computer. If you don't specify a *remotefile*, the file will get the *localfile* name.

Examples

To copy the local file *test.txt* and name it *test1.txt* on the remote computer, type:

```
send test.txt test1.txt
```

To copy the local file *program.exe* to the remote computer, type:

```
send program.exe
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp status

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the current status of ftp connections.

Syntax

```
status
```

Examples

To display the current status of ftp connections, type:

```
status
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp trace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles packet tracing. This command also displays the series of internal FTP function calls when running a **ftp** command.

Syntax

```
trace
```

Examples

Toggle tracing on and off, type:

```
trace
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp type

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets or displays the file transfer type. The **ftp** command supports both ASCII (default) and binary image file transfer types:

- We recommend using ASCII when transferring text files. In ASCII mode, character conversions to and from the network standard character set are performed. For example, end-of-line characters are converted as necessary, based on the target operating system.
- We recommend using binary when transferring executable files. In binary mode, files are transferred in one-byte units.

Syntax

```
type [<typename>]
```

Parameters

Expand table

Parameter	Description
`[<typename>]`	Specifies the file transfer type. If you don't specify this parameter, the current type is displayed.

Examples

To set the file transfer type to ASCII, type:

```
type ascii
```

To set the transfer file type to binary, type:

```
type binary
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp user

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Specifies a user to the remote computer.

Syntax

```
user <username> [<password>] [<account>]
```

Parameters

Expand table

Parameter	Description
`<username>`	Specifies a user name with which to log on to the remote computer.
`[<password>]`	Specifies the password for *username*. If a password is not specified but is required, the **ftp** command prompts for the password.
`[<account>]`	Specifies an account with which to log on to the remote computer. If an *account* isn't specified but is required, the **ftp** command prompts for the account.

Examples

To specify *User1* with the password *Password1*, type:

```
user User1 Password1
```

Related links

- Command-Line Syntax Key

- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp verbose

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Toggles Verbose mode. By default, Verbose mode is turned on. When Verbose mode is on, all **ftp** command responses are displayed. When a file transfer is completed, statistics regarding the efficiency of the transfer are also displayed.

Syntax

```
verbose
```

Examples

To toggle Verbose mode on and off, type:

```
verbose
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp mdelete

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Deletes files on the remote computer.

Syntax

```
mdelete <remotefile>[...]
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the remote file to delete.

Examples

To delete remote files *a.exe* and *b.exe*, type:

```
mdelete a.exe b.exe
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftp mdir

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays a directory list of files and subdirectories in a remote directory.

Syntax

```
mdir <remotefile>[...] <localfile>
```

Parameters

Expand table

Parameter	Description
`<remotefile>`	Specifies the directory or file for which you want to see a listing. You can specify multiple *remotefiles*. Type a hyphen (-) to use the current working directory on the remote computer.
`<localfile>`	Specifies a local file to store the listing. This parameter is required. Type a hyphen (-) to display the listing on the screen.

Examples

To display a directory listing of *dir1* and *dir2* on the screen, type:

```
mdir dir1 dir2 -
```

To save the combined directory listing of *dir1* and *dir2* in a local file called *dirlist.txt*, type:

```
```

```
mdir dir1 dir2 dirlist.txt
```

Related links

- Command-Line Syntax Key
- Additional FTP guidance

Feedback

Was this page helpful? Yes No

ftype

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or modifies file types that are used in file name extension associations. If used without an assignment operator (=), this command displays the current open command string for the specified file type. If used without parameters, this command displays the file types that have open command strings defined.

> **Note**
>
> This command is only supported within cmd.exe and is not available from PowerShell. Though you can use `cmd /c ftype` as a workaround.

Syntax

```
ftype [<filetype>[=[<opencommandstring>]]]
```

Parameters

Expand table

Parameter	Description
`<filetype>`	Specifies the file type to display or change.
`<opencommandstring>`	Specifies the open command string to use when opening files of the specified file type.
/?	Displays help at the command prompt.

Remarks

The following table describes how **ftype** substitutes variables within an open command string:

Expand table

Variable	Replacement value
`%0` or `%1`	Gets substituted with the file name being launched through the association.
`%*`	Gets all of the parameters.
`%2`, `%3`, ...	Gets the first parameter (`%2`), the second parameter (`%3`), and so on.
`%~<n>`	Gets all of the remaining parameters starting with the *n*th parameter, where *n* can be any number from 2 to 9.

Examples

To display the current file types that have open command strings defined, type:

```
ftype
```

To display the current open command string for the *txtfile* file type, type:

```
ftype txtfile
```

This command produces output similar to the following:

```
txtfile=%SystemRoot%\system32\NOTEPAD.EXE %1
```

To delete the open command string for a file type called *example*, type:

```
ftype example=
```

To associate the .pl file name extension with the PerlScript file type and enable the PerlScript file type to run PERL.EXE, type the following commands:

```
assoc .pl=PerlScript
ftype PerlScript=perl.exe %1 %*
```

To eliminate the need to type the .pl file name extension when invoking a Perl script, type:

```
set PATHEXT=.pl;%PATHEXT%
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

fveupdate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

FveUpdate is an internal tool, used by the setup program when a computer is upgraded. It updates the metadata associated with BitLocker to the latest version. This tool cannot be run independently.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

getmac

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Returns the media access control (MAC) address and list of network protocols associated with each address for all network cards in each computer, either locally or across a network. This command is particularly useful either when you want to enter the MAC address into a network analyzer, or when you need to know what protocols are currently in use on each network adapter on a computer.

Syntax

```
getmac[.exe][/s <computer> [/u <domain\<user> [/p <password>]]][/fo {table | list | csv}][/nh][/v]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default is the local computer.
/u `<domain>\<user>`	Runs the command with the account permissions of the user specified by *user* or *domain\user*. The default is the permissions of the current logged on user on the computer issuing the command.
/p `<password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/fo {table | list | csv}	Specifies the format to use for the query output. Valid values are **table**, **list**, and **csv**. The default format for output is **table**.
/nh	Suppresses column header in output. Valid when the **/fo** parameter is set to **table** or **csv**.
/v	Specifies that the output display verbose information.
/?	Displays help at the command prompt.

Examples

The following examples show how you can use the **getmac** command:

```
getmac /fo table /nh /v
```

```
getmac /s srvmain
```

```
getmac /s srvmain /u maindom\hiropln
```

```
getmac /s srvmain /u maindom\hiropln /p p@ssW23
```

```
getmac /s srvmain /u maindom\hiropln /p p@ssW23 /fo list /v
```

```
getmac /s srvmain /u maindom\hiropln /p p@ssW23 /fo table /nh
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

gettype

Article • 05/22/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The `gettype` command is deprecated and may not be supported in future releases of Windows.

`gettype` is included in Windows Server 2003. For more information, see gettype.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

goto

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Directs cmd.exe to a labeled line in a batch program. Within a batch program, this command directs command processing to a line that is identified by a label. When the label is found, processing continues starting with the commands that begin on the next line.

Syntax

```
goto <label>
```

Parameters

Expand table

Parameter	Description
`<label>`	Specifies a text string that is used as a label in the batch program.
/?	Displays help at the command prompt.

Remarks

- If command extensions are enabled (the default), and you use the **goto** command with a target label of **:EOF**, you transfer control to the end of the current batch script file and exit the batch script file without defining a label. When you use this command with the **:EOF** label, you must insert a colon before the label. For example: `goto:EOF`.
- You can use spaces in the *label* parameter, but you can't include other separators (for example, semicolons (;) or equal signs (=)).
- The *label* value that you specify must match a label in the batch program. The label within the batch program must begin with a colon (:). If a line begins with a colon,

it's treated as a label and any commands on that line are ignored. If your batch program doesn't contain the label that you specify in the *label* parameter, then the batch program stops and displays the following message: `Label not found`.

- You can use **goto** with other commands to perform conditional operations. For more information about using **goto** for conditional operations, see the if command.

Examples

The following batch program formats a disk in drive A as a system disk. If the operation is successful, the **goto** command directs processing to the **:end** label:

```
echo off
format a: /s
if not errorlevel 1 goto end
echo An error occurred during formatting.
:end
echo End of batch program.
```

Related links

- Command-Line Syntax Key
- cmd command
- if command

Feedback

Was this page helpful? Yes No

gpfixup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Fixes domain name dependencies in Group Policy Objects and Group Policy links after a domain rename operation. To use this command, you must install Group Policy Management as a feature through Server Manager.

Syntax

```
gpfixup [/v]
[/olddns:<olddnsname> /newdns:<newdnsname>]
[/oldnb:<oldflatname> /newnb:<newflatname>]
[/dc:<dcname>] [/sionly]
[/user:<username> [/pwd:{<password>|*}]] [/?]
```

Parameters

Expand table

Parameter	Description
/v	Displays detailed status messages. If this parameter isn't used, only error messages or a summary status message stating, **SUCCESS** or **FAILURE** appears.
/olddns: `<olddnsname>`	Specifies the old DNS name of the renamed domain as `<olddnsname>` when the domain rename operation changes the DNS name of a domain. You can use this parameter only if you also use the **/newdns** parameter to specify a new domain DNS name.
/newdns: `<newdnsname>`	Specifies the new DNS name of the renamed domain as `<newdnsname>` when the domain rename operation changes the DNS name of a domain. You can use this parameter only if you also use the **/olddns** parameter to specify the old domain DNS name.
/oldnb: `<oldflatname>`	Specifies the old NetBIOS name of the renamed domain as `<oldflatname>` when the domain rename operation changes the NetBIOS name of a domain. You can use this parameter only if you use the **/newnb** parameter to specify a new domain NetBIOS name.

Parameter	Description
/newnb: `<newflatname>`	Specifies the new NetBIOS name of the renamed domain as `<newflatname>` when the domain rename operation changes the NetBIOS name of a domain. You can use this parameter only if you use the **/oldnb** parameter to specify the old domain NetBIOS name.
/dc:`<dcname>`	Connect to the domain controller named `<dcname>` (a DNS name or a NetBIOS name). `<dcname>` must host a writable replica of the domain directory partition as indicated by one of the following: • The DNS name `<newdnsname>` by using **/newdns** • The NetBIOS name `<newflatname>` by using **/newnb** If this parameter isn't used, you can connect to any domain controller in the renamed domain indicated by `<newdnsname>` or `<newflatname>`.
/sionly	Performs only the Group Policy fix that relates to managed software installation (the Software Installation extension for Group Policy). Skip the actions that fix Group Policy links and the SYSVOL paths in GPOs.
/user: `<username>`	Runs this command in the security context of the user `<username>`, where `<username>` is in the format domain\user. If this parameter isn't used, this command runs as the logged in user.
/pwd: `{<password> | *}`	Specifies the password for the user.
/?	Displays Help at the command prompt.

Examples

This example assumes that you have already performed a domain rename operation in which you changed the DNS name from **MyOldDnsName** to **MyNewDnsName**, and the NetBIOS name from **MyOldNetBIOSName** to **MyNewNetBIOSName**.

In this example, you use the **gpfixup** command to connect to the domain controller named **MyDcDnsName** and repair GPOs and Group Policy links by updating the old domain name embedded in the GPOs and links. Status and error output is saved to a file that is named **gpfixup.log**.

```
gpfixup /olddns: MyOldDnsName /newdns:MyNewDnsName /oldnb:MyOldNetBIOSName
/newnb:MyNewNetBIOSName /dc:MyDcDnsName 2>&1 >gpfixup.log
```

This example is the same as the previous one, except that it assumes the NetBIOS name of the domain was not changed during the domain rename operation.

```
gpfixup /olddns: MyOldDnsName /newdns:MyNewDnsName /dc:MyDcDnsName 2>&1
>gpfixup.log
```

Related links

- Command-Line Syntax Key
- Administering Active Directory Domain Rename

Feedback

Was this page helpful? Yes No

gpresult

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the Resultant Set of Policy (RSoP) information for a remote user and computer. To use RSoP reporting for remotely targeted computers through the firewall, you must have firewall rules that enable inbound network traffic on the ports.

Syntax

```
gpresult [/s <system> [/u <username> [/p [<password>]]]] [/user
[<targetdomain>\]<targetuser>] [/scope {user | computer}] {/r | /v | /z |
[/x | /h] <filename> [/f] | /?}
```

> **Note**
>
> Except when using **/?**, you must include an output option, **/r**, **/v**, **/z**, **/x**, or **/h**.

Parameters

Expand table

Parameter	Description
/s `<system>`	Specifies the name or IP address of a remote computer. Don't use backslashes. The default is the local computer.
/u `<username>`	Uses the credentials of the specified user to run the command. The default user is the user who is signed in to the computer that issues the command.
/p `[<password>]`	Specifies the password of the user account that is provided in the **/u** parameter. If **/p** is omitted, **gpresult** prompts for the password. The **/p** parameter can't be used with **/x** or **/h**.
/user `[<targetdomain>\]` `<targetuser>]`	Specifies the remote user whose RSoP data is to be displayed.

Parameter	Description
/scope `{user | computer}`	Displays RSoP data for either the user or the computer. If **/scope** is omitted, **gpresult** displays RSoP data for both the user and the computer.
`[/x | /h] <filename>`	Saves the report in either XML (**/x**) or HTML (**/h**) format at the location and with the file name that is specified by the *filename* parameter. Can't be used with **/u**, **/p**, **/r**, **/v**, or **/z**.
/f	Forces **gpresult** to overwrite the file name that is specified in the **/x** or **/h** option.
/r	Displays RSoP summary data.
/v	Displays verbose policy information. This includes detailed settings that were applied with a precedence of 1.
/z	Displays all available information about Group Policy. This includes detailed settings that were applied with a precedence of 1 and higher.
/?	Displays help at the command prompt.

Remarks

- Group Policy is the primary administrative tool for defining and controlling how programs, network resources, and the operating system operate for users and computers in an organization. In an active directory environment, Group Policy is applied to users or computers based on their membership in sites, domains, or organizational units.
- Because you can apply overlapping policy settings to any computer or user, the Group Policy feature generates a resulting set of policy settings when the user signs in. The **gpresult** command displays the resulting set of policy settings that were enforced on the computer for the specified user when the user signed in.
- Because **/v** and **/z** produce much information, it's useful to redirect output to a text file (for example, `gpresult/z >policy.txt`).
- On ARM64 versions of Windows, only the `gpresult` in SysWow64 works with the `/h` parameter.

Examples

To retrieve RSoP data for only the remote user *maindom**targetuser*, on the computer *srvmain*, enter:

```
gpresult /s srvmain /user maindom\targetuser /scope user /r
```

To save all available information about Group Policy to a file named *policy.txt* for only the remote user *maindom\targetuser*, on the computer *srvmain*, enter:

```
gpresult /s srvmain /user maindom\targetuser /z > policy.txt
```

To display RSoP data for the signed-in user, *maindom\hiropln* with the password *p@ssW23*, for the computer *srvmain*, enter:

```
gpresult /s srvmain /u maindom\hiropln /p p@ssW23 /r
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

gpupdate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Updates Group Policy settings.

Syntax

```
gpupdate [/target:{computer | user}] [/force] [/wait:<VALUE>] [/logoff] [/boot] [/sync] [/?]
```

Parameters

Expand table

Parameter	Description
/target: `{computer|user}`	Specifies that only User or only Computer policy settings are updated. By default, both User and Computer policy settings are updated.
/force	Reapplies all policy settings. By default, only policy settings that have changed are applied.
/wait: `<VALUE>`	Sets the number of seconds to wait for policy processing to finish before returning to the command prompt. When the time limit is exceeded, the command prompt appears, but policy processing continues. The default value is 600 seconds. The value **0** means not to wait. The value **-1** means to wait indefinitely. In a script, by using this command with a time limit specified, you can run **gpupdate** and continue with commands that do not depend upon the completion of **gpupdate**. Alternatively, you can use this command with no time limit specified to let **gpupdate** finish running before other commands that depend on it are run.
/logoff	Causes a logoff after the Group Policy settings are updated. This is required for those Group Policy client-side extensions that do not process policy on a background update cycle but do process policy when a user logs on. Examples include user-targeted Software Installation and Folder Redirection.

Parameter	Description
	This option has no effect if there are no extensions called that require a logoff.
/boot	Causes a computer restart after the Group Policy settings are applied. This is required for those Group Policy client-side extensions that do not process policy on a background update cycle but do process policy at computer startup. Examples include computer-targeted Software Installation. This option has no effect if there are no extensions called that require a restart.
/sync	Causes the next foreground policy application to be done synchronously. Foreground policy is applied at computer boot and user logon. You can specify this for the user, computer, or both, by using the **/target** parameter. The **/force** and **/wait** parameters are ignored if you specify them.
/?	Displays Help at the command prompt.

Examples

To force a background update of all Group Policy settings, regardless of whether they've changed, type:

```
gpupdate /force
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

graftabl

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables Windows operating systems to display an extended character set in graphics mode. If used without parameters, **graftabl** displays the previous and the current code page.

> **Important**
>
> The **graftabl** command is a legacy command, and therefore outdated. It is normally not installed in modern Windows versions. Please see the **chcp** page for codepage handling.

Syntax

```
graftabl <codepage>
graftabl /status
```

Parameters

Expand table

Parameter	Description
`<codepage>`	Specifies a code page to define the appearance of extended characters in graphics mode. Valid code page identification numbers are: • **437** - United States • **850** - Multilingual (Latin I) • **852** - Slavic (Latin II) • **855** - Cyrillic (Russian) • **857** - Turkish • **860** - Portuguese • **861** - Icelandic • **863** - Canadian-French • **865** - Nordic • **866** - Russian

Parameter	Description
	• **869** - Modern Greek
/status	Displays the current code page being used by this command.
/?	Displays help at the command prompt.

Remarks

- The **graftabl** command affects only the monitor display of extended characters of the code page that you specify. It doesn't change the actual console input code page. To change the console input code page, use the mode or chcp command.
- Each exit code and a brief description of it:

Expand table

Exit code	Description
0	Character set was loaded successfully. No previous code page was loaded.
1	An incorrect parameter was specified. No action was taken.
2	A file error occurred.

- You can use the ERRORLEVEL environment variable in a batch program to process exit codes that are returned by **graftabl**.

Examples

To view the current code page used by **graftabl**, type:

```
graftabl /status
```

To load the graphics character set for code page 437 (United States) into memory, type:

```
graftabl 437
```

To load the graphics character set for code page 850 (multilingual) into memory, type:

```
graftabl 850
```

Related links

- Command-Line Syntax Key
- freedisk command
- mode command
- chcp command

Feedback

Was this page helpful? Yes No

help

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays a list of the available commands or detailed help information on a specified command. If used without parameters, **help** lists and briefly describes every system command.

Syntax

```
help [<command>]
```

Parameters

Expand table

Parameter	Description
`<command>`	Specifies the command for which to display detailed help information.

Examples

To view information about the **robocopy** command, type:

```
help robocopy
```

To display a list of all commands available in DiskPart, type:

```
help
```

To display detailed help information about how to use the **create partition primary** command in DiskPart, type:

```
help create partition primary
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

helpctr

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The helpctr command has been deprecated, and isn't guaranteed to be supported in Windows.

This tool is included in Windows Server 2003. For more information, see Helpctr.

Feedback

Was this page helpful? Yes No

hostname

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the host name portion of the full computer name of the computer.

> **ⓘ Important**
>
> This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network.

Syntax

```
hostname
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Any parameter different than `/?` produces an error message and sets the errorlevel to 1.

Notes

- Environment variable `%COMPUTERNAME%` usually will print the same string as `hostname`, but in uppercase.
- If environment variable `_CLUSTER_NETWORK_NAME_` is defined, `hostname` will print its value.

Examples

- To display the name of the computer, type:

```shell
hostname
```

- To display the name of the computer in uppercase:

```shell
echo %COMPUTERNAME%
```

- To alter the hostname output:

```shell
set "_CLUSTER_NETWORK_NAME_=Altered Computer Name"
hostname
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

icacls

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Displays or modifies discretionary access control lists (DACLs) on specified files and applies stored DACLs to files in specified directories.

> **Note**
>
> This command replaces the deprecated cacls command.

Syntax

```
icacls name [/save aclfile] [/setowner user] [/findsid Sid] [/verify] [/reset]
[/T] [/C] [/L] [/Q]
icacls name [/grant[:r] Sid:perm[...]] [/deny Sid:perm [...]] [/remove[:g|:d]]
Sid[...]]] [/setintegritylevel Level:policy[...]] [/T] [/C] [/L] [/Q]
icacls directory [/substitute SidOld SidNew [...]] [/restore aclfile] [/C] [/L]
[/Q]
```

Parameters

Expand table

Parameter	Description
`<name>`	Specifies the file for which to display or modify DACLs.
`<directory>`	Specifies the directory for which to display or modify DACLs.
/t	Performs the operation on all specified files in the current directory and its subdirectories.
/c	Continues the operation even if file errors occur. Error messages are still shown.
/l	Performs the operation on a symbolic link instead of its destination.
/q	Suppresses success messages.
/save `<ACLfile>`	Stores DACLs for all matching files into an access control list (ACL) file for later

Parameter	Description
	use with `/restore`.
/setowner `<user>`	Changes the owner of all matching files to the specified user.
/findsid `<sid>`	Finds all matching files that contain a DACL explicitly mentioning the specified security identifier (SID).
/verify	Finds all files with ACLs that aren't canonical or have lengths inconsistent with access control entry (ACE) counts.
/reset	Replaces ACLs with default inherited ACLs for all matching files.
/grant[:r] `<sid>`: `<perm>`	Grants specified user access rights. Permissions replace previously granted explicit permissions. Not adding **:r** means that permissions are added to any previously granted explicit permissions.
/deny `<sid>`:`<perm>`	Explicitly denies specified user access rights. An explicit deny ACE is added for the stated permissions and the same permissions in any explicit grant are removed.
/remove: g | d `<sid>`	Removes all occurrences of the specified SID from the DACL. This command can also use: • **g** - Removes all occurrences of granted rights to the specified SID • **d** - Removes all occurrences of denied rights to the specified SID
/setintegritylevel `<perm><level>`	Explicitly adds an integrity ACE to all matching files. The level can be specified as: • **l** - Low • **m** - Medium • **h** - High Inheritance options for the integrity ACE may precede the level and are applied only to directories.
/substitute `<sidold>` `<sidnew>`	Replaces an existing SID (*sidold*) with a new SID (*sidnew*). Requires using with the `<directory>` parameter.
/restore `<ACLfile>` /c | /l | /q	Applies stored DACLs from `<ACLfile>` to files in the specified directory. Requires using with the `<directory>` parameter.
/inheritancelevel: e | d | r	Sets the inheritance level, which can be: • **e** - Enables inheritance • **d** - Disables inheritance and copies the ACEs • **r** - Disables inheritance and removes only inherited ACEs

Remarks

- SIDs may be in either numerical or friendly name form. If you use a numerical form, affix the wildcard character * to the beginning of the SID.
- This command preserves the canonical order of ACE entries as:
  - Explicit denials
  - Explicit grants
  - Inherited denials
  - Inherited grants
- The `<perm>` option is a permission mask that can be specified for basic rights, advanced rights, or inheritance rights:
  - A sequence of simple rights (basic permissions) without the need to use parenthesis:
    - **N** - No access
    - **F** - Full access
    - **M** - Modify access
    - **RX** - Read and execute access
    - **R** - Read-only access
    - **W** - Write-only access
    - **D** - Delete access
  - A comma-separated list of specific rights (advanced permissions) which must use parenthesis:
    - **DE** - Delete
    - **RC** - Read control (read permissions)
    - **WDAC** - Write DAC (change permissions)
    - **WO** - Write owner (take ownership)
    - **S** - Synchronize
    - **AS** - Access system security
    - **MA** - Maximum allowed
    - **GR** - Generic read
    - **GW** - Generic write
    - **GE** - Generic execute
    - **GA** - Generic all
    - **RD** - Read data/list directory
    - **WD** - Write data/add file
    - **AD** - Append data/add subdirectory
    - **REA** - Read extended attributes
    - **WEA** - Write extended attributes

    - **X** - Execute/traverse
    - **DC** - Delete child
    - **RA** - Read attributes
    - **WA** - Write attributes

- A sequence of inheritance rights which must use parenthesis:
    - **(I)** - Inherit. ACE inherited from the parent container.
    - **(OI)** - Object inherit. Objects in this container inherits this ACE. Applies only to directories.
    - **(CI)** - Container inherit. Containers in this parent container inherits this ACE. Applies only to directories.
    - **(IO)** - Inherit only. ACE inherited from the parent container, but doesn't apply to the object itself. Applies only to directories.
    - **(NP)** - Don't propagate inherit. ACE inherited by containers and objects from the parent container, but doesn't propagate to nested containers. Applies only to directories.

Examples

To save the DACLs for all files in the C:\Windows directory and its subdirectories to the ACLFile file, type:

```
icacls c:\windows\* /save aclfile /t
```

To restore the DACLs for every file within ACLFile that exists in the C:\Windows directory and its subdirectories, type:

```
icacls c:\windows\ /restore aclfile
```

To grant the user User1 Delete and Write DAC permissions to a file named Test1, type:

```
icacls test1 /grant User1:(d,wdac)
```

To grant the user defined by SID S-1-1-0 Delete and Write DAC permissions to a file named TestFile, type:

```
icacls TestFile /grant *S-1-1-0:(d,wdac)
```

To apply a *high* integrity level to a directory and ensure that both its files and subdirectories inherit this level, type:

```
icacls "myDirectory" /setintegritylevel (CI)(OI)H
```

Related links

- Command-Line Syntax Key

if

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Performs conditional processing in batch programs.

Syntax

```
if [not] ERRORLEVEL <number> <command> [else <expression>]
if [not] <string1>==<string2> <command> [else <expression>]
if [not] exist <filename> <command> [else <expression>]
```

If command extensions are enabled, use the following syntax:

```
if [/i] <string1> <compareop> <string2> <command> [else <expression>]
if cmdextversion <number> <command> [else <expression>]
if defined <variable> <command> [else <expression>]
```

Parameters

Expand table

Parameter	Description
not	Specifies that the command should be carried out only if the condition is false.
errorlevel `<number>`	Specifies a true condition only if the previous program run by Cmd.exe returned an exit code equal to or greater than *number*.
`<command>`	Specifies the command that should be carried out if the preceding condition is met.
`<string1>==<string2>`	Specifies a true condition only if *string1* and *string2* are the same. These values can be literal strings or batch variables (for example, `%1`). You do not need to enclose literal strings in quotation marks.
exist	Specifies a true condition if the specified file name exists.

Parameter	Description
`<filename>`	
`<compareop>`	Specifies a three-letter comparison operator, including: • **EQU** - Equal to • **NEQ** - Not equal to • **LSS** - Less than • **LEQ** - Less than or equal to • **GTR** - Greater than • **GEQ** - Greater than or equal to
/i	Forces string comparisons to ignore case. You can use **/i** on the `string1==string2` form of **if**. These comparisons are generic, in that if both *string1* and *string2* are comprised of numeric digits only, the strings are converted to numbers and a numeric comparison is performed.
cmdextversion `<number>`	Specifies a true condition only if the internal version number associated with the command extensions feature of Cmd.exe is equal to or greater than the number specified. The first version is 1. It increases by increments of one when significant enhancements are added to the command extensions. The **cmdextversion** conditional is never true when command extensions are disabled (by default, command extensions are enabled).
defined `<variable>`	Specifies a true condition if *variable* is defined.
`<expression>`	Specifies a command-line command and any parameters to be passed to the command in an **else** clause.
/?	Displays help at the command prompt.

Remarks

- If the condition specified in an **if** clause is true, the command that follows the condition is carried out. If the condition is false, the command in the **if** clause is ignored and the command executes any command that is specified in the **else** clause.
- When a program stops, it returns an exit code. To use exit codes as conditions, use the **errorlevel** parameter.
- If you use **defined**, the following three variables are added to the environment: **%errorlevel%**, **%cmdcmdline%**, and **%cmdextversion%**.
    - **%errorlevel%**: Expands into a string representation of the current value of the ERRORLEVEL environment variable. This variable assumes that there isn't already

an existing environment variable with the name ERRORLEVEL. If there is, you'll get that ERRORLEVEL value instead.

  - **%cmdcmdline%**: Expands into the original command line that was passed to Cmd.exe prior to any processing by Cmd.exe. This assumes that there isn't already an existing environment variable with the name CMDCMDLINE. If there is, you'll get that CMDCMDLINE value instead.

  - **%cmdextversion%**: Expands into the string representation of the current value of **cmdextversion**. This assumes that there isn't already an existing environment variable with the name CMDEXTVERSION. If there is, you'll get that CMDEXTVERSION value instead.

- You must use the **else** clause on the same line as the command after the **if**.

Examples

To display the message **Cannot find data file if the file Product.dat cannot be found**, type:

```
if not exist product.dat echo Cannot find data file
```

To format a disk in drive A and display an error message if an error occurs during the formatting process, type the following lines in a batch file:

```
:begin
@echo off
format a: /s
if not errorlevel 1 goto end
echo An error occurred during formatting.
:end
echo End of batch program.
```

To delete the file Product.dat from the current directory or display a message if Product.dat is not found, type the following lines in a batch file:

```
IF EXIST Product.dat (
del Product.dat
) ELSE (
```

```
echo The Product.dat file is missing.
)
```

> **Note**
>
> These lines can be combined into a single line as follows:
>
> ```
> IF EXIST Product.dat (del Product.dat) ELSE (echo The Product.dat file
> is missing.)
> ```

To echo the value of the ERRORLEVEL environment variable after running a batch file, type the following lines in the batch file:

```
goto answer%errorlevel%
:answer1
echo The program returned error level 1
goto end
:answer0
echo The program returned error level 0
goto end
:end
echo Done!
```

To go to the okay label if the value of the ERRORLEVEL environment variable is less than or equal to 1, type:

```
if %errorlevel% LEQ 1 goto okay
```

Related links

- Command-Line Syntax Key
- goto command

Feedback

Was this page helpful? Yes No

ipconfig

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings. Used without parameters, **ipconfig** displays Internet Protocol version 4 (IPv4) and IPv6 addresses, subnet mask, and default gateway for all adapters.

Syntax

```
ipconfig [/allcompartments] [/all] [/renew [<adapter>]] [/release
[<adapter>]] [/renew6[<adapter>]] [/release6 [<adapter>]] [/flushdns]
[/displaydns] [/registerdns] [/showclassid <adapter>] [/setclassid <adapter>
[<classID>]]
```

Parameters

Expand table

Parameter	Description
/all	Displays the full TCP/IP configuration for all adapters. Adapters can represent physical interfaces, such as installed network adapters, or logical interfaces, such as dial-up connections.
/displaydns	Displays the contents of the DNS client resolver cache, which includes both entries preloaded from the local Hosts file and any recently obtained resource records for name queries resolved by the computer. The DNS Client service uses this information to resolve frequently queried names quickly, before querying its configured DNS servers.
/flushdns	Flushes and resets the contents of the DNS client resolver cache. During DNS troubleshooting, you can use this procedure to discard negative cache entries from the cache, as well as any other entries that have been added dynamically.
/registerdns	Initiates manual dynamic registration for the DNS names and IP addresses that are configured at a computer. You can use this parameter to troubleshoot a failed DNS name registration or resolve a dynamic

Parameter	Description
	update problem between a client and the DNS server without rebooting the client computer. The DNS settings in the advanced properties of the TCP/IP protocol determine which names are registered in DNS.
/release `[<adapter>]`	Sends a DHCPRELEASE message to the DHCP server to release the current DHCP configuration and discard the IP address configuration for either all adapters (if an adapter is not specified) or for a specific adapter if the *adapter* parameter is included. This parameter disables TCP/IP for adapters configured to obtain an IP address automatically. To specify an adapter name, type the adapter name that appears when you use **ipconfig** without parameters.
/release6`[<adapter>]`	Sends a DHCPRELEASE message to the DHCPv6 server to release the current DHCP configuration and discard the IPv6 address configuration for either all adapters (if an adapter is not specified) or for a specific adapter if the *adapter* parameter is included. This parameter disables TCP/IP for adapters configured to obtain an IP address automatically. To specify an adapter name, type the adapter name that appears when you use **ipconfig** without parameters.
/renew `[<adapter>]`	Renews DHCP configuration for all adapters (if an adapter is not specified) or for a specific adapter if the *adapter* parameter is included. This parameter is available only on computers with adapters that are configured to obtain an IP address automatically. To specify an adapter name, type the adapter name that appears when you use **ipconfig** without parameters.
/renew6 `[<adapter>]`	Renews DHCPv6 configuration for all adapters (if an adapter is not specified) or for a specific adapter if the *adapter* parameter is included. This parameter is available only on computers with adapters that are configured to obtain an IPv6 address automatically. To specify an adapter name, type the adapter name that appears when you use **ipconfig** without parameters.
/setclassid `<adapter> [<classID>]`	Configures the DHCP class ID for a specified adapter. To set the DHCP class ID for all adapters, use the asterisk (*) wildcard character in place of *adapter*. This parameter is available only on computers with adapters that are configured to obtain an IP address automatically. If a DHCP class ID is not specified, the current class ID is removed.
/showclassid `<adapter>`	Displays the DHCP class ID for a specified adapter. To see the DHCP class ID for all adapters, use the asterisk (*) wildcard character in place of *adapter*. This parameter is available only on computers with adapters that are configured to obtain an IP address automatically.
/?	Displays Help at the command prompt.

Remarks

- This command is most useful on computers that are configured to obtain an IP address automatically. This enables users to determine which TCP/IP configuration values have been configured by DHCP, Automatic Private IP Addressing (APIPA), or an alternate configuration.
- If the name you supply for *adapter* contains any spaces, use quotation marks around the adapter name (for example, "adapter name").
- For adapter names, **ipconfig** supports the use of the asterisk (*) wildcard character to specify either adapters with names that begin with a specified string or adapters with names that contain a specified string. For example, `Local*` matches all adapters that start with the string Local and `*Con*` matches all adapters that contain the string Con.

Examples

To display the basic TCP/IP configuration for all adapters, type:

```
ipconfig
```

To display the full TCP/IP configuration for all adapters, type:

```
ipconfig /all
```

To renew a DHCP-assigned IP address configuration for only the Local Area Connection adapter, type:

```
ipconfig /renew Local Area Connection
```

To flush the DNS resolver cache when troubleshooting DNS name resolution problems, type:

```
ipconfig /flushdns
```

To display the DHCP class ID for all adapters with names that start with Local, type:

```
ipconfig /showclassid Local*
```

To set the DHCP class ID for the Local Area Connection adapter to TEST, type:

```
ipconfig /setclassid Local Area Connection TEST
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

ipxroute

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays and modifies information about the routing tables used by the IPX protocol. Used without parameters, **ipxroute** displays the default settings for packets that are sent to unknown, broadcast, and multicast addresses.

Syntax

```
ipxroute servers [/type=x]
ipxroute ripout <network>
ipxroute resolve {guid | name} {GUID | <adaptername>}
ipxroute board= N [def] [gbr] [mbr] [remove=xxxxxxxxxxxx]
ipxroute config
```

Parameters

Expand table

Parameter	Description
servers `[/type=x]`	Displays the Service Access Point (SAP) table for the specified server type. **x** must be an integer. For example, `/type=4` displays all file servers. If you don't specify **/type**, `ipxroute servers` displays all types of servers, listing them by server name.
resolve `{GUID | name}` `{GUID | adaptername}`	Resolves the name of the GUID to its friendly name, or the friendly name to its GUID.
board= *n*	Specifies the network adapter for which to query or set parameters.
def	Sends packets to the ALL ROUTES broadcast. If a packet is transmitted to a unique Media Access Card (MAC) address that is not in the source routing table, **ipxroute** sends the packet to the SINGLE ROUTES broadcast by default.
gbr	Sends packets to the ALL ROUTES broadcast. If a packet is transmitted to the broadcast address (FFFFFFFFFFFF), **ipxroute** sends the packet to the SINGLE ROUTES broadcast by default.

Parameter	Description
mbr	Sends packets to the ALL ROUTES broadcast. If a packet is transmitted to a multicast address (C000xxxxxxxx), **ipxroute** sends the packet to the SINGLE ROUTES broadcast by default.
remove=*xxxxxxxxxxxx*	removes the given node address from the source routing table.
config	Displays information about all of the bindings for which IPX is configured.
/?	Displays help at the command prompt.

Examples

To display the network segments that the workstation is attached to, the workstation node address, and frame type being used, type:

```
ipxroute config
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

irftp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sends files over an infrared link.

> **ⓘ Important**
>
> Make sure the devices intended to communicate over an infrared link have infrared functionality enabled and are working correctly. Also make sure an infrared link is established between the devices.

Syntax

```
irftp [<drive>:\] [[<path>] <filename>] [/h][/s]
```

Parameters

Expand table

Parameter	Description
`<drive>:\`	Specifies the drive that contains the files that you want to send over an infrared link.
`[path] <filename>`	Specifies the location and name of the file or set of files that you want to send over an infrared link. If you specify a set of files, you must specify the full path for each file.
/h	Specifies hidden mode. When hidden mode is used, the files are sent without displaying the Wireless Link dialog box.
/s	Opens the **Wireless Link** dialog box, so that you can select the file or set of files that you want to send without using the command line to specify the drive, path, and file names. The **Wireless Link** dialog box also opens if you use this command without any parameters.

Examples

To send *c:\example.txt* over the infrared link, type:

```
irftp c:\example.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

jetpack

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Compacts a Windows Internet Name Service (WINS) or Dynamic Host Configuration Protocol (DHCP) database. We recommend you compact the WINS database whenever it approaches 30 MB.

Jetpack.exe compacts the database by:

1. Copying the database information to a temporary database file.
2. Deleting the original database file, either WINS or DHCP.
3. Renames the temporary database files to the original filename.

Syntax

```
jetpack.exe <database_name> <temp_database_name>
```

Parameters

Expand table

Parameter	Description
`<database_name>`	Specifies the name of the original database file.
`<temp_database_name>`	Specifies the name of the temporary database file to be created by jetpack.exe. Note: This temporary file is removed when the compact process is complete. For this command to work properly, you must make sure your temp file name is unique and that a file with that name doesn't already exist.
/?	Displays help at the command prompt.

Examples

To compact the WINS database, where **Tmp.mdb** is a temporary database and **Wins.mdb** is the WINS database, type:

```
cd %SYSTEMROOT%\SYSTEM32\WINS
NET STOP WINS
jetpack Wins.mdb Tmp.mdb
NET start WINS
```

To compact the DHCP database, where **Tmp.mdb** is a temporary database and **Dhcp.mdb** is the DHCP database, type:

```
cd %SYSTEMROOT%\SYSTEM32\DHCP
NET STOP DHCPSERVER
jetpack Dhcp.mdb Tmp.mdb
NET start DHCPSERVER
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

klist

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays a list of currently cached Kerberos tickets.

> **ⓘ Important**
>
> You must be at least a **Domain Admin**, or equivalent, to run all the parameters of this command.

Syntax

```
klist [-lh <logonID.highpart>] [-li <logonID.lowpart>] tickets | tgt | purge | sessions | kcd_cache | get | add_bind | query_bind | purge_bind
```

Parameters

Expand table

Parameter	Description
-lh	Denotes the high part of the user's locally unique identifier (LUID), expressed in hexadecimal. If neither **–lh** nor **–li** are present, the command defaults to the LUID of the user who is currently signed in.
-li	Denotes the low part of the user's locally unique identifier (LUID), expressed in hexadecimal. If neither **–lh** nor **–li** are present, the command defaults to the LUID of the user who is currently signed in.
tickets	Lists the currently cached ticket-granting-tickets (TGTs), and service tickets of the specified logon session. This is the default option.
tgt	Displays the initial Kerberos TGT.
purge	Allows you to delete all the tickets of the specified logon session.
sessions	Displays a list of logon sessions on this computer.

Parameter	Description
kcd_cache	Displays the Kerberos constrained delegation cache information.
get	Allows you to request a ticket to the target computer specified by the service principal name (SPN).
add_bind	Allows you to specify a preferred domain controller for Kerberos authentication.
query_bind	Displays a list of cached preferred domain controllers for each domain that Kerberos has contacted.
purge_bind	Removes the cached preferred domain controllers for the domains specified.
kdcoptions	Displays the Key Distribution Center (KDC) options specified in RFC 4120.
/?	Displays Help for this command.

Remarks

- If no parameters are provided, **klist** retrieves all the tickets for the currently logged on user.
- The parameters display the following information:
  - **tickets** - Lists the currently cached tickets of services that you have authenticated to since logon. Displays the following attributes of all cached tickets:
    - **LogonID:** The LUID.
    - **Client:** The concatenation of the client name and the domain name of the client.
    - **Server:** The concatenation of the service name and the domain name of the service.
    - **KerbTicket Encryption Type:** The encryption type that is used to encrypt the Kerberos ticket.
    - **Ticket Flags:** The Kerberos ticket flags.
    - **Start Time:** The time from which the ticket is valid.
    - **End Time:** The time the ticket becomes no longer valid. When a ticket is past this time, it can no longer be used to authenticate to a service or be used for renewal.

- **Renew Time:** The time that a new initial authentication is required.
- **Session Key Type:** The encryption algorithm that is used for the session key.

- **tgt** - Lists the initial Kerberos TGT and the following attributes of the currently cached ticket:
  - **LogonID:** Identified in hexadecimal.
  - **ServiceName:** krbtgt
  - **TargetName** `<SPN>`**:** krbtgt
  - **DomainName:** Name of the domain that issues the TGT.
  - **TargetDomainName:** Domain that the TGT is issued to.
  - **AltTargetDomainName:** Domain that the TGT is issued to.
  - **Ticket Flags:** Address and target actions and type.
  - **Session Key:** Key length and encryption algorithm.
  - **StartTime:** Local computer time that the ticket was requested.
  - **EndTime:** Time the ticket becomes no longer valid. When a ticket is past this time, it can no longer be used to authenticate to a service.
  - **RenewUntil:** Deadline for ticket renewal.
  - **TimeSkew:** Time difference with the Key Distribution Center (KDC).
  - **EncodedTicket:** Encoded ticket.
- **purge** - Allows you to delete a specific ticket. Purging tickets destroys all tickets that you have cached, so use this attribute with caution. It might stop you from being able to authenticate to resources. If this happens, you'll have to log off and log on again.
  - **LogonID:** Identified in hexadecimal.
- **sessions** - Allows you to list and display the information for all logon sessions on this computer.
  - **LogonID:** If specified, displays the logon session only by the given value. If not specified, displays all the logon sessions on this computer.
- **kcd_cache** - Allows you to display the Kerberos constrained delegation cache information.

    - **LogonID:** If specified, displays the cache information for the logon session by the given value. If not specified, displays the cache information for the current user's logon session.

- **get** - Allows you to request a ticket to the target that is specified by the SPN.
    - **LogonID:** If specified, requests a ticket by using the logon session by the given value. If not specified, requests a ticket by using the current user's logon session.
    - **kdcoptions:** Requests a ticket with the given KDC options
- **add_bind** - Allows you to specify a preferred domain controller for Kerberos authentication.
- **query_bind** - Allows you to display cached, preferred domain controllers for the domains.
- **purge_bind** - Allows you to remove cached, preferred domain controllers for the domains.
- **kdcoptions** - For the current list of options and their explanations, see RFC 4120.

Examples

To query the Kerberos ticket cache to determine if any tickets are missing, if the target server or account is in error, or if the encryption type is not supported due to an Event ID 27 error, type:

```
klist
```

```
klist -li 0x3e7
```

To learn about the specifics of each ticket-granting-ticket that is cached on the computer for a logon session, type:

```
```

```
klist tgt
```

To purge the Kerberos ticket cache, log off, and then log back on, type:

```
klist purge
```

```
klist purge –li 0x3e7
```

To diagnose a logon session and to locate a logonID for a user or a service, type:

```
klist sessions
```

To diagnose Kerberos constrained delegation failure, and to find the last error that was encountered, type:

```
klist kcd_cache
```

To diagnose if a user or a service can get a ticket to a server, or to request a ticket for a specific SPN, type:

```
klist get host/%computername%
```

To diagnose replication issues across domain controllers, you typically need the client computer to target a specific domain controller. To target the client computer to the specific domain controller, type:

```
klist add_bind CONTOSO KDC.CONTOSO.COM
```

```
```

```
klist add_bind CONTOSO.COM KDC.CONTOSO.COM
```

To query which domain controllers were recently contacted by this computer, type:

```
klist query_bind
```

To rediscover domain controllers, or to flush the cache before creating new domain controller bindings with `klist add_bind`, type:

```
klist purge_bind
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

ksetup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Performs tasks related to setting up and maintaining Kerberos protocol and the Key Distribution Center (KDC) to support Kerberos realms. Specifically, this command is used to:

- Change the computer settings for locating Kerberos realms. In non-Microsoft, Kerberos–based implementations, this information is usually kept in the Krb5.conf file. In Windows Server operating systems, it's kept in the registry. You can use this tool to modify these settings. These settings are used by workstations to locate Kerberos realms and by domain controllers to locate Kerberos realms for cross-realm trust relationships.
- Initialize registry keys that the Kerberos Security Support Provider (SSP) uses to locate a KDC for the Kerberos realm, if the computer is isn't a member of a Windows domain. After configuration, the user of a client computer running the Windows operating system can log on to accounts in the Kerberos realm.
- Search the registry for the domain name of the user's realm and then resolves the name to an IP address by querying a DNS server. The Kerberos protocol can use DNS to locate KDCs by using only the realm name, but it must be specially configured to do so.

Syntax

```
ksetup
[/setrealm <DNSdomainname>]
[/mapuser <principal> <account>]
[/addkdc <realmname> <KDCname>]
[/delkdc <realmname> <KDCname>]
[/addkpasswd <realmname> <KDCPasswordName>]
[/delkpasswd <realmname> <KDCPasswordName>]
[/server <servername>]
[/setcomputerpassword <password>]
[/removerealm <realmname>]
[/domain <domainname>]
[/changepassword <oldpassword> <newpassword>]
[/listrealmflags]
```

```
[/setrealmflags <realmname> [sendaddress] [tcpsupported] [delegate]
[ncsupported] [rc4]]
[/addrealmflags <realmname> [sendaddress] [tcpsupported] [delegate]
[ncsupported] [rc4]]
[/delrealmflags [sendaddress] [tcpsupported] [delegate] [ncsupported] [rc4]]
[/dumpstate]
[/addhosttorealmmap] <hostname> <realmname>]
[/delhosttorealmmap] <hostname> <realmname>]
[/setenctypeattr] <domainname> {DES-CBC-CRC | DES-CBC-MD5 | RC4-HMAC-MD5 |
AES128-CTS-HMAC-SHA1-96 | AES256-CTS-HMAC-SHA1-96}
[/getenctypeattr] <domainname>
[/addenctypeattr] <domainname> {DES-CBC-CRC | DES-CBC-MD5 | RC4-HMAC-MD5 |
AES128-CTS-HMAC-SHA1-96 | AES256-CTS-HMAC-SHA1-96}
[/delenctypeattr] <domainname>
```

Parameters

Expand table

Parameter	Description
ksetup setrealm	Makes this computer a member of a Kerberos realm.
ksetup addkdc	Defines a KDC entry for the given realm.
ksetup delkdc	Deletes a KDC entry for the realm.
ksetup addkpasswd	Adds a kpasswd server address for a realm.
ksetup delkpasswd	Deletes a kpasswd server address for a realm.
ksetup server	Allows you to specify the name of a Windows computer on which to apply the changes.
ksetup setcomputerpassword	Sets the password for the computer's domain account (or host principal).
ksetup removerealm	Deletes all information for the specified realm from the registry.
ksetup domain	Allows you to specify a domain (if the `<domainname>` hasn't already been set by the **/domain** parameter).
ksetup changepassword	Allows you to use the kpasswd to change the logged on user's password.
ksetup listrealmflags	Lists the available realm flags that **ksetup** can detect.
ksetup setrealmflags	Sets realm flags for a specific realm.
ksetup addrealmflags	Adds additional realm flags to a realm.

Parameter	Description
ksetup delrealmflags	Deletes realm flags from a realm.
ksetup dumpstate	Analyzes the Kerberos configuration on the given computer. Adds a host to realm mapping to the registry.
ksetup addhosttorealmmap	Adds a registry value to map the host to the Kerberos realm.
ksetup delhosttorealmmap	Deletes the registry value that mapped the host computer to the Kerberos realm.
ksetup setenctypeattr	Sets one or more encryption types trust attributes for the domain.
ksetup getenctypeattr	Gets the encryption types trust attribute for the domain.
ksetup addenctypeattr	Adds encryption types to the encryption types trust attribute for the domain.
ksetup delenctypeattr	Deletes the encryption types trust attribute for the domain.
/?	Displays Help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

ksetup addenctypeattr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds the encryption type attribute to the list of possible types for the domain. A status message is displayed upon successful or failed completion.

Syntax

```
ksetup /addenctypeattr <domainname> {DES-CBC-CRC | DES-CBC-MD5 | RC4-HMAC-MD5 | AES128-CTS-HMAC-SHA1-96 | AES256-CTS-HMAC-SHA1-96}
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Name of the domain to which you want to establish a connection. Use the fully qualified domain name or a simple form of the name, such as corp.contoso.com or contoso.
encryption type	Must be one of the following supported encryption types: • DES-CBC-CRC • DES-CBC-MD5 • RC4-HMAC-MD5 • AES128-CTS-HMAC-SHA1-96 • AES256-CTS-HMAC-SHA1-96

Remarks

- You can set or add multiple encryption types by separating the encryption types in the command with a space. However, you can only do so for one domain at a time.

Examples

To view the encryption type for the Kerberos ticket-granting ticket (TGT) and the session key, type:

```
klist
```

To set the domain to corp.contoso.com, type:

```
ksetup /domain corp.contoso.com
```

To add the encryption type *AES-256-CTS-HMAC-SHA1-96* to the list of possible types for the domain *corp.contoso.com*, type:

```
ksetup /addenctypeattr corp.contoso.com AES-256-CTS-HMAC-SHA1-96
```

To set the encryption type attribute to *AES-256-CTS-HMAC-SHA1-96* for the domain *corp.contoso.com*, type:

```
ksetup /setenctypeattr corp.contoso.com AES-256-CTS-HMAC-SHA1-96
```

To verify that the encryption type attribute was set as intended for the domain, type:

```
ksetup /getenctypeattr corp.contoso.com
```

Related links

- Command-Line Syntax Key
- klist command
- ksetup command
- ksetup domain command

- ksetup setenctypeattr command
- ksetup getenctypeattr command
- ksetup delenctypeattr command

Feedback

Was this page helpful? Yes No

ksetup addhosttorealmmap

Article • 08/31/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a service principal name (SPN) mapping between the stated host and the realm. This command also allows you to map a host or multiple hosts that are sharing the same DNS suffix to the realm.

The mapping is stored in the registry, under **HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\HostToRealm**.

Syntax

```
ksetup /addhosttorealmmap <hostname> <realmname>
```

Parameters

Expand table

Parameter	Description
`<hostname>`	The host name is the computer name, and it can be stated as the computer's fully qualified domain name.
`<realmname>`	The realm name is stated as an uppercase DNS name, such as CORP.CONTOSO.COM.

Examples

To map the host computer *IPops897* to the *CONTOSO* realm, type:

```
ksetup /addhosttorealmmap IPops897 CONTOSO
```

Check the registry to make sure the mapping occurred as intended.

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup delhosttorealmmap command

Feedback

Was this page helpful? Yes No

ksetup addkdc

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a Key Distribution Center (KDC) address for the given Kerberos realm

The mapping is stored in the registry, under **HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\LSA\Kerberos\Domains** and the computer must be restarted before the new realm setting will be used.

> **Note**
>
> To deploy Kerberos realm configuration data to multiple computers, you must use the **Security Configuration Template** snap-in and policy distribution, explicitly on individual computers. You can't use this command.

Syntax

```
ksetup /addkdc <realmname> [<KDCname>]
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM. This value also appears as the default realm when **ksetup** is run, and is the realm to which you want to add the other KDC.
`<KDCname>`	Specifies the case-insensitive, fully-qualified domain name, such as mitkdc.contoso.com. If the KDC name is omitted, DNS will locate KDCs.

Examples

To configure a non-Windows KDC server and the realm that the workstation should use, type:

```
ksetup /addkdc CORP.CONTOSO.COM mitkdc.contoso.com
```

To set the local computer account password to p@sswrd1% on the same computer as in the previous example, and then to restart the computer, type:

```
ksetup /setcomputerpassword p@sswrd1%
```

To verify the default realm name for the computer or to verify that this command worked as intended, type:

```
ksetup
```

Check the registry to make sure the mapping occurred as intended.

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup setcomputerpassword command

Feedback

Was this page helpful? Yes No

ksetup addkpasswd

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a Kerberos password (kpasswd) server address for a realm.

Syntax

```
ksetup /addkpasswd <realmname> [<kpasswdname>]
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM, and is listed as the default realm or **Realm=** when **ksetup** is run.
`<kpasswdname>`	Specifies the Kerberos password server. It's stated as a case-insensitive, fully-qualified domain name, such as mitkdc.contoso.com. If the KDC name is omitted, DNS might be used to locate KDCs.

Remarks

- If the Kerberos realm that the workstation will be authenticating to supports the Kerberos change password protocol, you can configure a client computer running the Windows operating system to use a Kerberos password server.
- You can add additional KDC names one at a time.

Examples

To configure the CORP.CONTOSO.COM realm to use the non-Windows KDC server, mitkdc.contoso.com, as the password server, type:

```
ksetup /addkpasswd CORP.CONTOSO.COM mitkdc.contoso.com
```

To verify the KDC name is set, type `ksetup` and then view the output, looking for the text, **kpasswd =**. If you don't see the text, it means the mapping hasn't been configured.

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup delkpasswd command

Feedback

Was this page helpful? Yes No

ksetup addrealmflags

Article • 05/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to Windows Server (All supported versions)

The ksetup addrealmflags command adds other realm flags to a specified realm.

Syntax

```
ksetup /addrealmflags <realmname> [sendaddress] [tcpsupported] [delegate]
[ncsupported] [rc4]
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as `CORP.CONTOSO.COM`.

Remarks

- Realm flags are stored in the registry under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\Domains\<realmname>`. This entry doesn't exist in the registry by default. You can use the ksetup addrealmflags command to populate the registry.
- The realm flags specify other features of a Kerberos realm that aren't based on the Windows Server operating system. Computers running Windows Server can use a Kerberos server to administer authentication in the Kerberos realm instead of using a domain running a Windows Server operating system. This registry entry establishes the features of the realm and is as follows:

Expand table

Value	Realm flag	Description
`0xF`	`All`	All realm flags are set.
`0x00`	`None`	No realm flags are set and no other features are enabled.
`0x01`	`sendaddress`	The IP address is included within the ticket-granting tickets.
`0x02`	`tcpsupported`	Both the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP) are supported in this realm.
`0x04`	`delegate`	Everyone in this realm is trusted for delegation.
`0x08`	`ncsupported`	This realm supports name canonicalization, which allows for DNS and Realm naming standards.
`0x80`	`rc4`	This realm supports RC4 encryption to enable cross-realm trust, which allows for the use of TLS.

- You can see the available and set realm flags by viewing the output of **ksetup** or `ksetup /dumpstate`.

Examples

To list the available realm flags for the realm CONTOSO, type:

```
ksetup /listrealmflags
```

To set two flags to the CONTOSO realm, type:

```
ksetup /setrealmflags CONTOSO ncsupported delegate
```

To add one more flag that isn't currently in the set, type:

```
ksetup /addrealmflags CONTOSO SendAddress
```

To verify the realm flag is set, type `ksetup` and then view the output, looking for the text, **Realm flags** =. If you don't see the text, it means that the flag hasn't been set.

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup listrealmflags command
- ksetup setrealmflags command
- ksetup delrealmflags command
- ksetup dumpstate command

Feedback

Was this page helpful? Yes No

ksetup changepassword

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Uses the Key Distribution Center (KDC) password (kpasswd) value to change the password of the logged-on user. The output of the command informs you of the success or failure status.

You can check whether the **kpasswd** is set, by running the `ksetup /dumpstate` command and viewing the output.

Syntax

```
ksetup /changepassword <oldpassword> <newpassword>
```

Parameters

Expand table

Parameter	Description
`<oldpassword>`	Specifies the logged-on user's existing password.
`<newpassword>`	Specifies the logged on user's new password. This password must meet all the password requirements set on this computer.

Remarks

- If the user account isn't found in the current domain, the system will ask you to supply the domain name where the user account resides.
- If you want to force a password change at next logon, this command allows the use of the asterisk (*) so the user will be prompted for a new password.
-

Examples

To change the password of a user who is currently logged on to this computer in this domain, type:

```
ksetup /changepassword Pas$w0rd Pa$$w0rd
```

To change the password of a user who is currently logged on in the Contoso domain, type:

```
ksetup /domain CONTOSO /changepassword Pas$w0rd Pa$$w0rd
```

To force the currently logged on user to change the password at the next logon, type:

```
ksetup /changepassword Pas$w0rd *
```

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup dumpstate command
- ksetup addkpasswd command
- ksetup delkpasswd command
- ksetup dumpstate command

Feedback

Was this page helpful? Yes No

ksetup delenctypeattr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes the encryption type attribute for the domain. A status message is displayed upon successful or failed completion.

You can view the encryption type for the Kerberos ticket-granting ticket (TGT) and the session key, by running the **klist** command and viewing the output. You can set the domain to connect to and use, by running the `ksetup /domain <domainname>` command.

Syntax

```
ksetup /delenctypeattr <domainname>
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Name of the domain to which you want to establish a connection. You can use either the fully-qualified domain name or a simple form of the name, such as corp.contoso.com or contoso.

Examples

To determine the current encryption types that are set on this computer, type:

```
klist
```

To set the domain to mit.contoso.com, type:

```
ksetup /domain mit.contoso.com
```

To verify what the encryption type attribute is for the domain, type:

```
ksetup /getenctypeattr mit.contoso.com
```

To remove the set encryption type attribute for the domain mit.contoso.com, type:

```
ksetup /delenctypeattr mit.contoso.com
```

Related links

- Command-Line Syntax Key
- klist command
- ksetup command
- ksetup domain command
- ksetup addenctypeattr command
- ksetup setenctypeattr command

Feedback

Was this page helpful? Yes No

ksetup delhosttorealmmap

Article • 12/18/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a service principal name (SPN) mapping between the stated host and the realm. This command also removes any mapping between a host to realm (or multiple hosts to realm).

The mapping is stored in the registry, under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\HostToRealm`. After running this command, we recommend making sure the mapping appears in the registry.

Syntax

```
ksetup /delhosttorealmmap <hostname> <realmname>
```

Parameters

Expand table

Parameter	Description
`<hostname>`	Specifies the fully-qualified domain name of the computer.
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM.

Examples

To change the configuration of the realm CONTOSO, and to delete the mapping of the host computer IPops897 to the realm, type:

```
ksetup /delhosttorealmmap IPops897 CONTOSO
```

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup addhosttorealmmap command

Feedback

Was this page helpful? Yes No

ksetup delkdc

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes instances of Key Distribution Center (KDC) names for the Kerberos realm.

The mapping is stored in the registry, under `HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\LSA\Kerberos\Domains`. After running this command, we recommend making sure the KDC was removed and no longer appears in the list.

> **Note**
>
> To remove realm configuration data from multiple computers, use the **Security Configuration Template** snap-in with policy distribution, instead of using the **ksetup** command explicitly on individual computers.

Syntax

```
ksetup /delkdc <realmname> <KDCname>
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM. This is the default realm that appears when you run the **ksetup** command, and it's the realm from which you want to delete the KDC.
`<KDCname>`	Specifies the case-sensitive, fully-qualified domain name, such as mitkdc.contoso.com.

Examples

To view all of the associations between the Windows realm and the non-Windows realm, and to determine which ones to remove, type:

```
ksetup
```

To remove the association, type:

```
ksetup /delkdc CORP.CONTOSO.COM mitkdc.contoso.com
```

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup addkdc command

Feedback

Was this page helpful? Yes No

ksetup delkpasswd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes a Kerberos password server (kpasswd) for a realm.

Syntax

```
ksetup /delkpasswd <realmname> <kpasswdname>
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM, and is listed as the default realm or **Realm=** when **ksetup** is run.
`<kpasswdname>`	Specifies the Kerberos password server. It's stated as a case-insensitive, fully-qualified domain name, such as mitkdc.contoso.com. If the KDC name is omitted, DNS might be used to locate KDCs.

Examples

To make sure the realm CORP.CONTOSO.COM uses the non-Windows KDC server mitkdc.contoso.com as the password server, type:

```
ksetup /delkpasswd CORP.CONTOSO.COM mitkdc.contoso.com
```

To make sure the realm CORP.CONTOSO.COM is not mapped to a Kerberos password server (the KDC name), type `ksetup` on the Windows computer and then view the output.

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup delkpasswd command

Feedback

Was this page helpful? Yes No

ksetup delrealmflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes realm flags from the specified realm.

Syntax

```
ksetup /delrealmflags <realmname> [sendaddress] [tcpsupported] [delegate] [ncsupported] [rc4]
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM, and is listed as the default realm or **Realm**= when **ksetup** is run.

Remarks

- The realm flags specify additional features of a Kerberos realm that aren't based on the Windows Server operating system. Computers that are running Windows Server, can use a Kerberos server to administer authentication in the Kerberos realm, instead of using a domain running a Windows Server operating system. This entry establishes the features of the realm, and are as follows:

Expand table

Value	Realm flag	Description
0xF	All	All realm flags are set.
0x00	None	No realm flags are set, and no additional features are enabled.
0x01	sendaddress	The IP address will be included within the ticket-granting tickets.

Value	Realm flag	Description
0x02	tcpsupported	Both the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP) are supported in this realm.
0x04	delegate	Everyone in this realm is trusted for delegation.
0x08	ncsupported	This realm supports name canonicalization, which allows for DNS and Realm naming standards.
0x80	rc4	This realm supports RC4 encryption to enable cross-realm trust, which allows for the use of TLS.

- Realm flags are stored in the registry under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\Domains\<realmname>`. This entry doesn't exist in the registry by default. You can use the ksetup addrealmflags command to populate the registry.
- You can see the available and set realm flags by viewing the output of **ksetup** or `ksetup /dumpstate`.

Examples

To list the available realm flags for the realm CONTOSO, type:

```
ksetup /listrealmflags
```

To remove two flags currently in the set, type:

```
ksetup /delrealmflags CONTOSO ncsupported delegate
```

To verify the realm flags have been removed, type `ksetup` and then view the output, looking for the text, **Realm flags** =.

Related links

- Command-Line Syntax Key
- ksetup command

- ksetup listrealmflags command
- ksetup setrealmflags command
- ksetup addrealmflags command
- ksetup dumpstate command

Feedback

Was this page helpful? Yes No

ksetup domain

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the domain name for all Kerberos operations.

Syntax

```
ksetup /domain <domainname>
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Name of the domain to which you want to establish a connection. Use the fully-qualified domain name or a simple form of the name, such as contoso.com or contoso.

Examples

To establish a connection to a valid domain, such as Microsoft, by using the `ksetup /mapuser` subcommand, type:

```
ksetup /mapuser principal@realm domain-user /domain domain-name
```

After a successful connection, you'll receive a new TGT or an existing TGT will be refreshed.

Related links

- Command-Line Syntax Key

- ksetup command
- ksetup mapuser command

Feedback

Was this page helpful? Yes No

ksetup dumpstate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the current state of realm settings for all realms that are defined on the computer. This command displays the same output as the **ksetup** command.

Syntax

```
ksetup /dumpstate
```

Remarks

- The output of this command includes the default realm (the domain that the computer is a member of) and all the realms that are defined on this computer. The following is included for each realm:
  - All the Key Distribution Centers (KDCs) that are associated with this realm.
  - All the **set realm** flags for this realm.
  - The KDC password.
- This command doesn't display the domain name specified by DNS detection or by the command `ksetup /domain`.
- This command doesn't display the computer password set by using the command `ksetup /setcomputerpassword`.

Examples

To locate the Kerberos realm configurations on a computer, type:

```
ksetup /dumpstate
```

Related links

- Command-Line Syntax Key
- ksetup command

Feedback

Was this page helpful? Yes No

ksetup getenctypeattr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves the encryption type attribute for the domain. A status message is displayed upon successful or failed completion.

You can view the encryption type for the Kerberos ticket-granting ticket (TGT) and the session key, by running the **klist** command and viewing the output. You can set the domain to connect to and use, by running the `ksetup /domain <domainname>` command.

Syntax

```
ksetup /getenctypeattr <domainname>
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Name of the domain to which you want to establish a connection. Use the fully-qualified domain name or a simple form of the name, such as corp.contoso.com or contoso.

Examples

To verify the encryption type attribute for the domain, type:

```
ksetup /getenctypeattr mit.contoso.com
```

Related links

- Command-Line Syntax Key
- klist command
- ksetup command
- ksetup domain command
- ksetup addenctypeattr command
- ksetup setenctypeattr command
- ksetup delenctypeattr command

Feedback

Was this page helpful? Yes No

ksetup listrealmflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists the available realm flags that can be reported by **ksetup**.

Syntax

```
ksetup /listrealmflags
```

Remarks

- The realm flags specify additional features of a Kerberos realm that aren't based on the Windows Server operating system. Computers that are running Windows Server, can use a Kerberos server to administer authentication in the Kerberos realm, instead of using a domain running a Windows Server operating system. This entry establishes the features of the realm, and are as follows:

Expand table

Value	Realm flag	Description
0xF	All	All realm flags are set.
0x00	None	No realm flags are set, and no additional features are enabled.
0x01	sendaddress	The IP address will be included within the ticket-granting tickets.
0x02	tcpsupported	Both the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP) are supported in this realm.
0x04	delegate	Everyone in this realm is trusted for delegation.
0x08	ncsupported	This realm supports name canonicalization, which allows for DNS and Realm naming standards.
0x80	rc4	This realm supports RC4 encryption to enable cross-realm trust, which allows for the use of TLS.

- Realm flags are stored in the registry under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\Domains\<realmname>`. This entry doesn't exist in the registry by default. You can use the ksetup addrealmflags command to populate the registry.

Examples

To list the known realm flags on this computer, type:

```
ksetup /listrealmflags
```

To set the available realm flags that **ksetup** doesn't know, type:

```
ksetup /setrealmflags CORP.CONTOSO.COM sendaddress tcpsupported delete
ncsupported
```

-OR-

```
ksetup /setrealmflags CORP.CONTOSO.COM 0xF
```

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup addrealmflags command
- ksetup setrealmflags command
- ksetup delrealmflags command

Feedback

Was this page helpful? Yes No

ksetup mapuser

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Maps the name of a Kerberos principal to an account.

Syntax

```
ksetup /mapuser <principal> <account>
```

Parameters

Expand table

Parameter	Description
`<principal>`	Specifies the fully-qualified domain name of any principal user. For example, mike@corp.CONTOSO.COM. If you don't specify an account parameter, mapping is deleted for the specified principal.
`<account>`	Specifies any account or security group name that exists on this computer, such as **Guest**, **Domain Users**, or **Administrator**. If this parameter is omitted, mapping is deleted for the specified principal.

Remarks

- An account can be specifically identified, such as **Domain Guests**, or you can use a wildcard character (*) to include all accounts.
- The computer only authenticates the principals of the given realm if they present valid Kerberos tickets.
- Whenever changes are made to the external Key Distribution Center (KDC) and the realm configuration, a restart of the computer where the setting was changed is required.

Examples

To see the current mapped settings and the default realm, type:

```
ksetup
```

To map Mike Danseglio's account within the Kerberos realm CONTOSO to the guest account on this computer, granting him all the privileges of a member of the built-in Guest account without having to authenticate to this computer, type:

```
ksetup /mapuser mike@corp.CONTOSO.COM guest
```

To remove the mapping of Mike Danseglio's account to the guest account on this computer to prevent him from authenticating to this computer with his credentials from CONTOSO, type:

```
ksetup /mapuser mike@corp.CONTOSO.COM
```

To map Mike Danseglio's account within the CONTOSO Kerberos realm to any existing account on this computer, type:

```
ksetup /mapuser mike@corp.CONTOSO.COM *
```

> **Note**
>
> If only the Standard User and Guest accounts are active on this computer, Mike's privileges are set to those.

To map all accounts within the CONTOSO Kerberos realm to any existing account of the same name on this computer, type:

```
ksetup /mapuser * *
```

Related links

- Command-Line Syntax Key
- ksetup command

Feedback

Was this page helpful? Yes No

ksetup removerealm

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes all information for the specified realm from the registry.

The realm name is stored in the registry under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos` and `HKEY_LOCAL_MACHINE\SYSTEM\ControlSet001\Control\Lsa\Kerberos`. This entry doesn't exist in the registry by default. You can use the ksetup addrealmflags command to populate the registry.

> **ⓘ Important**
>
> You can't remove the default realm name from the domain controller because this resets its DNS information, and removing it might make the domain controller unusable.

Syntax

```
ksetup /removerealm <realmname>
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM, and is listed as the default realm or **Realm**= when **ksetup** is run.

Examples

To remove an erroneous realm name (.CON instead of .COM) from the local computer, type:

```
ksetup /removerealm CORP.CONTOSO.CON
```

To verify the removal, you can run the **ksetup** command and review the output.

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup setrealm command

Feedback

Was this page helpful? Yes No

ksetup server

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Allows you to specify a name for a computer running the Windows operating system, so changes made by the **ksetup** command update the target computer.

The target server name is stored in the registry under `HKEY_LOCAL_MACHINE\SYSTEM\ControlSet001\Control\LSA\Kerberos`. This entry isn't reported when you run the **ksetup** command.

> **ⓘ Important**
>
> There's no way to remove the targeted server name. Instead, you can change it back to the local computer name, which is the default.

Syntax

```
ksetup /server <servername>
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the full computer name on which the configuration will be effective, such as *IPops897.corp.contoso.com*. If an incomplete fully-qualified domain computer name is specified, the command will fail.

Examples

To make your **ksetup** configurations effective on the *IPops897* computer, which is connected on the Contoso domain, type:

```
ksetup /server IPops897.corp.contoso.com
```

Related links

- Command-Line Syntax Key
- ksetup command

Feedback

Was this page helpful? Yes No

ksetup setcomputerpassword

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the password for the local computer. This command affects the computer account only and requires a restart for the password change to take effect.

> **ⓘ Important**
>
> The computer account password isn't displayed in the registry or as output from the **ksetup** command.

Syntax

```
ksetup /setcomputerpassword <password>
```

Parameters

Expand table

Parameter	Description
`<password>`	Specifies the supplied password to set the computer account on the local computer. The password can only be set by using an account with administrative privileges, and the password must be from 1 to 156 alphanumeric or special characters.

Examples

To change the computer account password on the local computer from *IPops897* to *IPop$897!*, type:

```
ksetup /setcomputerpassword IPop$897!
```

Related links

- Command-Line Syntax Key
- ksetup command

Feedback

Was this page helpful? Yes No

ksetup setenctypeattr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the encryption type attribute for the domain. A status message is displayed upon successful or failed completion.

You can view the encryption type for the Kerberos ticket-granting ticket (TGT) and the session key, by running the **klist** command and viewing the output. You can set the domain to connect to and use, by running the `ksetup /domain <domainname>` command.

Syntax

```
ksetup /setenctypeattr <domainname> {DES-CBC-CRC | DES-CBC-MD5 | RC4-HMAC-MD5 | AES128-CTS-HMAC-SHA1-96 | AES256-CTS-HMAC-SHA1-96}
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Name of the domain to which you want to establish a connection. Use the fully qualified domain name or a simple form of the name, such as corp.contoso.com or contoso.
encryption type	Must be one of the following supported encryption types: • DES-CBC-CRC • DES-CBC-MD5 • RC4-HMAC-MD5 • AES128-CTS-HMAC-SHA1-96 • AES256-CTS-HMAC-SHA1-96

Remarks

- You can set or add multiple encryption types by separating the encryption types in the command with a space. However, you can only do so for one domain at a time.

Examples

To view the encryption type for the Kerberos ticket-granting ticket (TGT) and the session key, type:

```
klist
```

To set the domain to corp.contoso.com, type:

```
ksetup /domain corp.contoso.com
```

To set the encryption type attribute to AES-256-CTS-HMAC-SHA1-96 for the domain corp.contoso.com, type:

```
ksetup /setenctypeattr corp.contoso.com AES-256-CTS-HMAC-SHA1-96
```

To verify that the encryption type attribute was set as intended for the domain, type:

```
ksetup /getenctypeattr corp.contoso.com
```

Related links

- Command-Line Syntax Key
- klist command
- ksetup command
- ksetup domain command
- ksetup addenctypeattr command
- ksetup getenctypeattr command
- ksetup delenctypeattr command

Feedback

Was this page helpful? Yes No

ksetup setrealm

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the name of a Kerberos realm.

> **ⓘ Important**
>
> Setting the Kerberos realm on a domain controller isn't supported. Attempting to do so causes a warning and a command failure.

Syntax

```
ksetup /setrealm <DNSdomainname>
```

Parameters

Expand table

Parameter	Description
`<DNSdomainname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM. You can use the fully-qualified domain name or a simple form of the name. If you don't use uppercase for the DNS name, you'll be asked for verification to continue.

Examples

To set the realm of this computer to a specific domain name, and to restrict access by a non-domain controller just to the CONTOSO Kerberos realm, type:

```
ksetup /setrealm CONTOSO
```

Related links

- Command-Line Syntax Key
- ksetup command
- ksetup removerealm

Feedback

Was this page helpful? Yes No

ksetup setrealmflags

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets realm flags for the specified realm.

Syntax

```
ksetup /setrealmflags <realmname> [sendaddress] [tcpsupported] [delegate] [ncsupported] [rc4]
```

Parameters

Expand table

Parameter	Description
`<realmname>`	Specifies the uppercase DNS name, such as CORP.CONTOSO.COM.

Remarks

- The realm flags specify additional features of a Kerberos realm that aren't based on the Windows Server operating system. Computers that are running Windows Server, can use a Kerberos server to administer authentication in the Kerberos realm, instead of using a domain running a Windows Server operating system. This entry establishes the features of the realm, and are as follows:

Expand table

Value	Realm flag	Description
0xF	All	All realm flags are set.
0x00	None	No realm flags are set, and no additional features are enabled.
0x01	sendaddress	The IP address will be included within the ticket-granting tickets.

Value	Realm flag	Description
0x02	tcpsupported	Both the Transmission Control Protocol (TCP) and the User Datagram Protocol (UDP) are supported in this realm.
0x04	delegate	Everyone in this realm is trusted for delegation.
0x08	ncsupported	This realm supports name canonicalization, which allows for DNS and Realm naming standards.
0x80	rc4	This realm supports RC4 encryption to enable cross-realm trust, which allows for the use of TLS.

- Realm flags are stored in the registry under `HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Control\Lsa\Kerberos\Domains\<realmname>`. This entry doesn't exist in the registry by default. You can use the ksetup addrealmflags command to populate the registry.
- You can see the available and set realm flags by viewing the output of **ksetup** or `ksetup /dumpstate`.

Examples

To list the available, and to set realm flags for the realm CONTOSO, type:

```
ksetup
```

To set two flags that aren't currently set, type:

```
ksetup /setrealmflags CONTOSO ncsupported delegate
```

To verify the realm flag is set, type `ksetup` and then view the output, looking for the text, **Realm flags** =. If you don't see the text, it means that the flag hasn't been set.

Related links

- Command-Line Syntax Key
- ksetup command

- ksetup listrealmflags command
- ksetup addrealmflags command
- ksetup delrealmflags command
- ksetup dumpstate command

Feedback

Was this page helpful? Yes No

ktmutil

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Starts the Kernel Transaction Manager utility. If used without parameters, **ktmutil** displays available subcommands.

Syntax

```
ktmutil list tms
ktmutil list transactions [{TmGUID}]
ktmutil resolve complete {TmGUID} {RmGUID} {EnGUID}
ktmutil resolve commit {TxGUID}
ktmutil resolve rollback {TxGUID}
ktmutil force commit {GUID}
ktmutil force rollback {GUID}
ktmutil forget
```

Examples

To force an Indoubt transaction with GUID 311a9209-03f4-11dc-918f-00188b8f707b to commit, type:

```
ktmutil force commit {311a9209-03f4-11dc-918f-00188b8f707b}
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

ktpass

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configures the server principal name for the host or service in Active Directory Domain Services (AD DS) and generates a .keytab file that contains the shared secret key of the service. The .keytab file is based on the Massachusetts Institute of Technology (MIT) implementation of the Kerberos authentication protocol. The ktpass command-line tool allows non-Windows services that support Kerberos authentication to use the interoperability features provided by the Kerberos Key Distribution Center (KDC) service.

Syntax

```
ktpass
[/out <filename>]
[/princ <principalname>]
[/mapuser <useraccount>]
[/mapop {add|set}] [{-|+}desonly] [/in <filename>]
[/pass {password|*|{-|+}rndpass}]
[/minpass]
[/maxpass]
[/crypto {DES-CBC-CRC|DES-CBC-MD5|RC4-HMAC-NT|AES256-SHA1|AES128-SHA1|All}]
[/itercount]
[/ptype {KRB5_NT_PRINCIPAL|KRB5_NT_SRV_INST|KRB5_NT_SRV_HST}]
[/kvno <keyversionnum>]
[/answer {-|+}]
[/target]
[/rawsalt] [{-|+}dumpsalt] [{-|+}setupn] [{-|+}setpass <password>]  [/?
|/h|/help]
```

Parameters

Expand table

Parameter	Description
/out `<filename>`	Specifies the name of the Kerberos version 5 .keytab file to generate. **Note:** This is the .keytab file you transfer to a computer that isn't running the Windows operating system,

Parameter	Description
	and then replace or merge with your existing .keytab file, */Etc/Krb5.keytab*.
/princ `<principalname>`	Specifies the principal name in the form host/computer.contoso.com@CONTOSO.COM. **Warning:** This parameter is case-sensitive.
/mapuser `<useraccount>`	Maps the name of the Kerberos principal, which is specified by the **princ** parameter, to the specified domain account.
/mapop `{add|set}`	Specifies how the mapping attribute is set. • **Add** - Adds the value of the specified local user name. This is the default. • **Set** - Sets the value for Data Encryption Standard (DES)-only encryption for the specified local user name.
`{-|+}`desonly	DES-only encryption is set by default. • + Sets an account for DES-only encryption. • - Releases restriction on an account for DES-only encryption. **Important:** Windows doesn't support DES by default.
/in `<filename>`	Specifies the .keytab file to read from a host computer that is not running the Windows operating system.
/pass `{password|*|{-|+}rndpass}`	Specifies a password for the principal user name that is specified by the **princ** parameter. Use `*` to prompt for a password.
/minpass	Sets the minimum length of the random password to 15 characters.
/maxpass	Sets the maximum length of the random password to 256 characters.
/crypto `{DES-CBC-CRC|DES-CBC-MD5|RC4-HMAC-NT|AES256-SHA1|AES128-SHA1|All}`	Specifies the keys that are generated in the keytab file: • **DES-CBC-CRC** - Used for compatibility. • **DES-CBC-MD5** - Adheres more closely to the MIT implementation and is used for compatibility. • **RC4-HMAC-NT** - Employs 128-bit encryption.

Parameter	Description
	• **AES256-SHA1** - Employs AES256-CTS-HMAC-SHA1-96 encryption. • **AES128-SHA1** - Employs AES128-CTS-HMAC-SHA1-96 encryption. • **All** - States that all supported cryptographic types can be used. **Note:** Because the default settings are based on older MIT versions, you should always use the `/crypto` parameter.
/itercount	Specifies the iteration count that is used for AES encryption. The default ignores **itercount** for non-AES encryption and sets AES encryption to 4,096.
/ptype `{KRB5_NT_PRINCIPAL|KRB5_NT_SRV_INST|KRB5_NT_SRV_HST}`	Specifies the principal type. • **KRB5_NT_PRINCIPAL** - The general principal type (recommended). • **KRB5_NT_SRV_INST** - The user service instance • **KRB5_NT_SRV_HST** - The host service instance
/kvno `<keyversionnum>`	Specifies the key version number. The default value is 1.
/answer `{-|+}`	Sets the background answer mode: • **-** Answers reset password prompts automatically with **NO**. • **+** Answers reset password prompts automatically with **YES**.
/target	Sets which domain controller to use. The default is for the domain controller to be detected, based on the principal name. If the domain controller name doesn't resolve, a dialog box will prompt for a valid domain controller.
/rawsalt	forces ktpass to use the rawsalt algorithm when generating the key. This parameter is optional.
`{-|+}dumpsalt`	The output of this parameter shows the MIT salt algorithm that is being used to generate the key.

Parameter	Description
`{-|+}setupn`	Sets the user principal name (UPN) in addition to the service principal name (SPN). The default is to set both in the .keytab file.
`{-|+}setpass <password>`	Sets the user's password when supplied. If rndpass is used, a random password is generated instead.
/?	Displays Help for this command.

Remarks

- Services running on systems that aren't running the Windows operating system can be configured with service instance accounts in AD DS. This allows any Kerberos client to authenticate to services that are not running the Windows operating system by using Windows KDCs.
- The **/princ** parameter isn't evaluated by ktpass and is used as provided. There's no check to see if the parameter matches the exact case of the **userPrincipalName** attribute value when generating the Keytab file. Case-sensitive Kerberos distributions using this Keytab file might have problems if there's no exact case match, and could even fail during pre-authentication. To check and retrieve the correct **userPrincipalName** attribute value from a LDifDE export file. For example:

```
ldifde /f keytab_user.ldf /d CN=Keytab
User,OU=UserAccounts,DC=contoso,DC=corp,DC=microsoft,DC=com /p base /l
samaccountname,userprincipalname
```

Examples

To create a Kerberos .keytab file for a host computer that isn't running the Windows operating system, you must map the principal to the account and set the host principal password.

1. Use the active directory **User and computers** snap-in to create a user account for a service on a computer that is not running the Windows operating system. For example, create an account with the name *User1*.
2. Use the **ktpass** command to set up an identity mapping for the user account by typing:

```
ktpass /princ host/User1.contoso.com@CONTOSO.COM /mapuser User1 /pass
MyPas$w0rd /out machine.keytab /crypto all /ptype KRB5_NT_PRINCIPAL
/mapop set
```

> **Note**
>
> You cannot map multiple service instances to the same user account.

3. Merge the .keytab file with the */Etc/Krb5.keytab* file on a host computer that isn't running the Windows operating system.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

label

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates, changes, or deletes the volume label (that is, the name) of a disk. If used without parameters, the **label** command changes the current volume label or deletes the existing label.

Syntax

```
label [/mp] [<volume>] [<label>]
```

Parameters

Expand table

Parameter	Description
/mp	Specifies that the volume should be treated as a mount point or volume name.
`<volume>`	Specifies a drive letter (followed by a colon), mount point, or volume name. If a volume name is specified, the **/mp** parameter is unnecessary.
`<label>`	Specifies the label for the volume.
/?	Displays help at the command prompt.

Remarks

- Windows displays the volume label and serial number (if it has one) as part of the directory listing.
- An NTFS volume label can be up to 32 characters in length, including spaces. NTFS volume labels retain and display the case that was used when the label was created.

Examples

To label a disk in drive A that contains sales information for July, type:

```
label a:sales-july
```

To view and delete the current label for drive C, follow these steps:

1. At the command prompt, type:

   ```
   label
   ```

   Output similar to the following should be displayed:

   ```
   Volume in drive C: is Main Disk
   Volume Serial Number is 6789-ABCD
   Volume label (32 characters, ENTER for none)?
   ```

2. Press ENTER. The following prompt should be displayed:

   ```
   Delete current volume label (Y/N)?
   ```

3. Press **Y** to delete the current label, or **N** if you want to keep the existing label.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

lodctr

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Allows you to register or save performance counter name and registry settings in a file and designate trusted services.

Syntax

```
lodctr <filename> [/s:<filename>] [/r:<filename>] [/t:<servicename>]
```

Parameters

Expand table

Parameter	Description
`<filename>`	Specifies the name of the initialization file that registers the performance counter name settings and explanatory text.
/s:`<filename>`	Specifies the name of the file to which the performance counter registry settings and explanatory text are saved.
/r	Restores counter registry settings and explanatory text from current registry settings and cached performance files related to the registry.
/r:`<filename>`	Specifies the name of the file that restores the performance counter registry settings and explanatory text. **Warning:** If you use this command, you'll overwrite all performance counter registry settings and explanatory text, replacing them with the configuration defined in the specified file.
/t: `<servicename>`	Indicates that service `<servicename>` is trusted.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "file name 1").

- An exit code of **0** indicates the command line syntax was correct. It doesn't mean the operation was successful. Check the output from the command for potential error messages.

Examples

To save the current performance registry settings and explanatory text to file "*perf backup1.txt*", type:

```
lodctr /s:"perf backup1.txt"
```

Related links

- Command-Line Syntax Key

Last updated on 10/30/2025

logman

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates and manages Event Trace Session and Performance logs and supports many functions of Performance Monitor from the command line.

Syntax

```
logman [create | query | start | stop | delete| update | import | export | /?] [options]
```

Parameters

Expand table

Parameter	Description
logman create	Creates a counter, trace, configuration data collector, or API.
logman query	Queries data collector properties.
logman start | stop	Starts or stops data collection.
logman delete	Deletes an existing data collector.
logman update	Updates the properties of an existing data collector.
logman import | export	Imports a data collector set from an XML file or export a data collector set to an XML file.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

logman create

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a counter, trace, configuration data collector, or API.

Syntax

```
logman create <counter | trace | alert | cfg | api> <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
logman create counter	Creates a counter data collector.
logman create trace	Creates a trace data collector.
logman create alert	Creates an alert data collector.
logman create cfg	Creates a configuration data collector.
logman create api	Creates an API tracing data collector.

Related links

- Command-Line Syntax Key
- logman command

Feedback

Was this page helpful? Yes No

logman create alert

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates an alert data collector.

Syntax

```
logman create alert <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data collectors.

Parameter	Description
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-cf `<filename>`	Specifies the file listing performance counters to collect. The file should contain one performance counter name per line.
-[-]el	Enables or disables Event Log reporting.
-th `<threshold [threshold [...]]>`	Specify counters and their threshold values for an alert.
-[-]rdcs `<name>`	Specifies the Data Collector Set to start when an alert fires.
-[-]tn `<task>`	Specifies the task to run when an alert fires.
-[-]targ `<argument>`	Specifies the task arguments to be used with the task specified using -tn.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To create a new alert called, *new_alert*, which fires when the performance counter % Processor time in the Processor(_Total) counter group exceeds the counter value of 50, type:

```
logman create alert new_alert -th \Processor(_Total)\% Processor time>50
```

> **Note**
>
> The defined threshold value is based on the value collected by the counter, so in this example, the value of 50 equates to 50% Processor time.

Related links

- Command-Line Syntax Key
- logman update alert command
- logman command

Feedback

Was this page helpful? Yes No

logman create api

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates an API tracing data collector.

Syntax

```
logman create api <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-f `<bin|bincirc>`	Specifies the log format for the data collector.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering a `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changed to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Run the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begin collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	End data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data

Parameter	Description
	collectors.
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeat the data collector daily at the specified begin and end times.
-[-]a	Append an existing log file.
-[-]ow	Overwrite an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Run the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answer yes to all questions without prompting.
-mods `<path [path [...]]>`	Specifies the list of modules to log API calls from.
-inapis `<module!api [module!api [...]]>`	Specifies the list of API calls to include in logging.
-exapis `<module!api [module!api [...]]>`	Specifies the list of API calls to exclude from logging.
-[-]ano	Log (-ano) API names only, or do not log only (-ano) API names.
-[-]recursive	Log (-recursive) or do not log (-recursive) APIs recursively beyond the first layer.
-exe `<value>`	Specifies the full path to an executable for API Tracing.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To create an API trace counter called trace_notepad, for the executable file c:\windows\notepad.exe, and putting the results in the file c:\notepad.etl, type:

```
logman create api trace_notepad -exe c:\windows\notepad.exe -o
c:\notepad.etl
```

To create an API trace counter called trace_notepad, for the executable file c:\windows\notepad.exe, collecting values produced by the module at c:\windows\system32\advapi32.dll, type:

```
logman create api trace_notepad -exe c:\windows\notepad.exe -mods
c:\windows\system32\advapi32.dll
```

To create an API trace counter called trace_notepad, for the executable file c:\windows\notepad.exe, excluding the API call TlsGetValue produced by the module kernel32.dll, type:

```
logman create api trace_notepad -exe c:\windows\notepad.exe -exapis
kernel32.dll!TlsGetValue
```

Related links

- Command-Line Syntax Key
- logman update api command
- logman command

Feedback

Was this page helpful? Yes No

logman create cfg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a configuration data collector.

Syntax

```
logman create cfg <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering a * for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data collectors.

Parameter	Description
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-[-]ni	Enables (-ni) or disable (-ni) network interface query.
-reg `<path [path [...]]>`	Specifies registry value(s) to collect.
-mgt `<query [query [...]]>`	Specifies WMI object(s) to collect using SQL query language.
-ftc `<path [path [...]]>`	Specifies the full path to the file(s) to collect.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To create a configuration data collector called cfg_log, using the registry key `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\Currentverion\`, type:

```
logman create cfg cfg_log -reg HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
```

```
NT\Currentverion\\
```

To create a configuration data collector called cfg_log, which records all WMI objects from `root\wmi` in the database column `MSNdis_Vendordriverversion`, type:

```
logman create cfg cfg_log -mgt root\wmi:select * FROM
MSNdis_Vendordriverversion
```

Related links

- Command-Line Syntax Key
- logman update cfg command
- logman command

Feedback

Was this page helpful? Yes No

logman create counter

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a counter data collector.

Syntax

```
logman create counter <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-f `<bin|bincirc|csv|tsv|sql>`	Specifies the log format for the data collector. The maximum log file size will be limited to 2 GB if `csv` is specified
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data collectors.

Parameter	Description
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, create a new file when the specified time has elapsed. When time is not specified, create a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-cf `<filename>`	Specifies the file listing performance counters to collect. The file should contain one performance counter name per line.
-c `<path [path [ ]]>`	Specifies performance counter(s) to collect.
-sc `<value>`	Specifies the maximum number of samples to collect with a performance counter data collector.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To create a counter called *perf_log* using the % Processor time counter from the Processor(_Total) counter category, type:

```
logman create counter perf_log -c \Processor(_Total)\% Processor time
```

To create a counter called *perf_log* using the % Processor time counter from the Processor(_Total) counter category, creating a log file with a maximum size of 10 MB, and collecting data for 1 minute and 0 seconds, type:

```
logman create counter perf_log -c \Processor(_Total)\% Processor time -max
10 -rf 01:00
```

Related links

- Command-Line Syntax Key
- logman update counter command
- logman command

Feedback

Was this page helpful? Yes No

logman create trace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Create an event trace data collector.

Syntax

```
logman create trace <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
-ets	Sends commands to Event Trace Sessions directly without saving or scheduling.
[-n] `<name>`	Name of the target object.
-f `<bin|bincirc>`	Specifies the log format for the data collector.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.

Parameter	Description
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-ct `<perf|system|cycle>`	Specifies the Event Trace Session clock type.
-ln `<logger_name>`	Specifies the logger name for Event Trace Sessions.
-ft `<[[hh:]mm:]ss>`	Specifies the Event Trace Session flush timer.
-[-]p `<provider [flags [level]]>`	Specifies a single Event Trace provider to enable.
-pf `<filename>`	Specifies a file listing multiple Event Trace providers to enable. The file should be a text file containing one provider per line.
-[-]rt	Runs the Event Trace Session in real-time mode.
-[-]ul	Runs the Event Trace Session in user.
-bs `<value>`	Specifies the Event Trace Session buffer size in kb.

Parameter	Description
-nb `<min max>`	Specifies the number of Event Trace Session buffers.
-mode `<globalsequence|localsequence|pagedmemory>`	Specifies the event trace session logger mode, including: • **Globalsequence** - Specifies that the event tracer add a sequence number to every event it receives irrespective of which trace session received the event. • **Localsequence** - Specifies that the event tracer add sequence numbers for events received at a specific trace session. When this option is used, duplicate sequence numbers can exist across all sessions but will be unique within each trace session. • **Pagedmemory** - Specifies that the event tracer use paged memory rather than the default non-paged memory pool for its internal buffer allocations.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To create an event trace data collector called *trace_log*, using no fewer than 16 and no more than 256 buffers, with each buffer being 64kb in size, putting the results in c:\logfile, type:

```
logman create trace trace_log -nb 16 256 -bs 64 -o c:\logfile
```

Related links

- Command-Line Syntax Key
- logman update trace command

- logman command

Feedback

Was this page helpful? Yes No

logman delete

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Deletes an existing data collector.

Syntax

```
logman delete <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-ets	Sends commands to Event Trace Sessions directly without saving or scheduling.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering a * for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
/?	Displays context-sensitive help.

Examples

To delete the data collector *perf_log*, type:

```
logman delete perf_log
```

Related links

- Command-Line Syntax Key
- logman command

Feedback

Was this page helpful? Yes No

logman import and logman export

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Imports a Data Collector Set from an XML file, or exports a Data Collector Set to an XML file.

Syntax

```
logman import <[-n] <name> <-xml <name> [options]
logman export <[-n] <name> <-xml <name> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-xml `<name>`	Name of the XML file to import or export.
-ets	Sends commands to Event Trace Sessions directly without saving or scheduling.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-y	Answers yes to all questions without prompting.
/?	Displays context-sensitive help.

Examples

To import the XML file *c:\windows\perf_log.xml* from the computer *server_1* as a data collector set called *perf_log*, type:

```
logman import perf_log -s server_1 -xml c:\windows\perf_log.xml
```

Related links

- Command-Line Syntax Key
- logman command

Feedback

Was this page helpful? Yes No

logman query

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Queries data collector or data collector set properties.

Syntax

```
logman query [providers|Data Collector Set name] [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-ets	Sends commands to Event Trace Sessions directly without saving or scheduling.
/?	Displays context-sensitive help.

Examples

To list all Data Collector Sets configured on the target system, type:

```
logman query
```

To list the data collectors contained in the Data Collector Set named *perf_log*, type:

```
logman query perf_log
```

To list all available providers of data collectors on the target system, type:

```
logman query providers
```

Related links

- Command-Line Syntax Key
- logman command

Feedback

Was this page helpful? Yes No

logman start and logman stop

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

The **logman start** command starts a data collector and sets the begin time to manual. The **logman stop** command stops a Data Collector Set and sets the end time to manual.

Syntax

```
logman start <[-n] <name>> [options]
logman stop <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Specifies the name of the target object.
-ets	Sends commands to Event Trace Sessions directly, without saving or scheduling.
-as	Performs the requested operation asynchronously.
-?	Displays context-sensitive help.

Examples

To start the data collector *perf_log*, on the remote computer *server_1*, type:

```
logman start perf_log -s server_1
```

Related links

- Command-Line Syntax Key
- logman command

Feedback

Was this page helpful? Yes No

logman update

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Updates an existing data collector.

Syntax

```
logman update <counter | trace | alert | cfg | api> <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
logman update counter	Updates a counter data collector.
logman update alert	Updates an alert data collector.
logman update cfg	Updates a configuration data collector.
logman update api	Updates an API tracing data collector.

Related links

- Command-Line Syntax Key
- logman command

Feedback

Was this page helpful? Yes No

logman update alert

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Updates the properties of an existing alert data collector.

Syntax

```
logman update alert <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data collectors.

Parameter	Description
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-cf `<filename>`	Specifies the file listing performance counters to collect. The file should contain one performance counter name per line.
-[-]el	Enables or disables Event Log reporting.
-th `<threshold [threshold [...]]>`	Specify counters and their threshold values for an alert.
-[-]rdcs `<name>`	Specifies the Data Collector Set to start when an alert fires.
-[-]tn `<task>`	Specifies the task to run when an alert fires.
-[-]targ `<argument>`	Specifies the task arguments to be used with the task specified using -tn.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To update the existing alert called *new_alert*, setting the threshold value for the counter % Processor time in the Processor(_Total) counter group to 40%, type:

```
logman update alert new_alert -th \Processor(_Total)\% Processor time>40
```

Related links

- Command-Line Syntax Key
- logman create alert command
- logman command

Feedback

Was this page helpful? Yes No

logman update api

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Updates the properties of an existing API tracing data collector.

Syntax

```
logman update api <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-f `<bin|bincirc>`	Specifies the log format for the data collector.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering a `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changed to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Run the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begin collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	End data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data

Parameter	Description
	collectors.
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeat the data collector daily at the specified begin and end times.
-[-]a	Append an existing log file.
-[-]ow	Overwrite an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Run the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answer yes to all questions without prompting.
-mods `<path [path [...]]>`	Specifies the list of modules to log API calls from.
-inapis `<module!api [module!api [...]]>`	Specifies the list of API calls to include in logging.
-exapis `<module!api [module!api [...]]>`	Specifies the list of API calls to exclude from logging.
-[-]ano	Log (-ano) API names only, or do not log only (-ano) API names.
-[-]recursive	Log (-recursive) or do not log (-recursive) APIs recursively beyond the first layer.
-exe `<value>`	Specifies the full path to an executable for API Tracing.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To update an existing API trace counter called *trace_notepad*, for the executable file c:\windows\notepad.exe, by excluding the API call TlsGetValue produced by the module kernel32.dll, type:

```
logman update api trace_notepad -exe c:\windows\notepad.exe -exapis
kernel32.dll!TlsGetValue
```

Related links

- Command-Line Syntax Key
- logman create api command
- logman command

Feedback

Was this page helpful? Yes No

logman update cfg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Updates the properties of an existing configuration data collector.

Syntax

```
logman update cfg <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering a * for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data collectors.

Parameter	Description
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-[-]ni	Enables (-ni) or disable (-ni) network interface query.
-reg `<path [path [...]]>`	Specifies registry value(s) to collect.
-mgt `<query [query [...]]>`	Specifies WMI object(s) to collect using SQL query language.
-ftc `<path [path [...]]>`	Specifies the full path to the file(s) to collect.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To update a configuration data collector called *cfg_log*, to collect the registry key `HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows NT\Currentverion\`, type:

```
logman update cfg cfg_log -reg HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows
```

```
NT\Currentverion\
```

Related links

- Command-Line Syntax Key
- logman create cfg command
- logman command

Feedback

Was this page helpful? Yes No

logman update counter

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Updates an existing counter data collector's properties.

Syntax

```
logman update counter <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Perform the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
[-n] `<name>`	Name of the target object.
-f `<bin|bincirc>`	Specifies the log format for the data collector.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval for performance counter data

Parameter	Description
	collectors.
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, create a new file when the specified time has elapsed. When time is not specified, create a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-cf `<filename>`	Specifies the file listing performance counters to collect. The file should contain one performance counter name per line.
-c `<path [path [ ]]>`	Specifies performance counter(s) to collect.
-sc `<value>`	Specifies the maximum number of samples to collect with a performance counter data collector.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To create a counter called *perf_log* using the % Processor time counter from the Processor(_Total) counter category, type:

```
logman create counter perf_log -c \Processor(_Total)\% Processor time
```

To update an existing counter called *perf_log*, changing the sample interval to 10, the log format to CSV, and adding versioning to the log file name in the format mmddhhmm, type:

```
logman update counter perf_log -si 10 -f csv -v mmddhhmm
```

Related links

- Command-Line Syntax Key
- logman create counter command
- logman command

Feedback

Was this page helpful? Yes No

logman update trace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Updates the properties of an existing event trace data collector.

Syntax

```
logman update trace <[-n] <name>> [options]
```

Parameters

Expand table

Parameter	Description
-s `<computer name>`	Performs the command on the specified remote computer.
-config `<value>`	Specifies the settings file containing command options.
-ets	Sends commands to Event Trace Sessions directly without saving or scheduling.
[-n] `<name>`	Name of the target object.
-f `<bin|bincirc>`	Specifies the log format for the data collector.
-[-]u `<user [password]>`	Specifies the user to Run As. Entering an `*` for the password produces a prompt for the password. The password is not displayed when you type it at the password prompt.
-m `<[start] [stop] [[start] [stop] [...]]>`	Changes to manual start or stop instead of a scheduled begin or end time.
-rf `<[[hh:]mm:]ss>`	Runs the data collector for the specified period of time.

Parameter	Description
-b `<M/d/yyyy h:mm:ss[AM|PM]>`	Begins collecting data at the specified time.
-e `<M/d/yyyy h:mm:ss[AM|PM]>`	Ends data collection at the specified time.
-o `<path|dsn!log>`	Specifies the output log file or the DSN and log set name in a SQL database.
-[-]r	Repeats the data collector daily at the specified begin and end times.
-[-]a	Appends an existing log file.
-[-]ow	Overwrites an existing log file.
-[-]v `<nnnnnn|mmddhhmm>`	Attaches file versioning information to the end of the log file name.
-[-]rc `<task>`	Runs the command specified each time the log is closed.
-[-]max `<value>`	Maximum log file size in MB or maximum number of records for SQL logs.
-[-]cnf `<[[hh:]mm:]ss>`	When time is specified, creates a new file when the specified time has elapsed. When time is not specified, creates a new file when the maximum size is exceeded.
-y	Answers yes to all questions without prompting.
-ct `<perf|system|cycle>`	Specifies the Event Trace Session clock type.
-ln `<logger_name>`	Specifies the logger name for Event Trace Sessions.
-ft `<[[hh:]mm:]ss>`	Specifies the Event Trace Session flush timer.
-[-]p `<provider [flags [level]]>`	Specifies a single Event Trace provider to enable.
-pf `<filename>`	Specifies a file listing multiple Event Trace providers to enable. The file should be a text file containing one provider per line.
-[-]rt	Runs the Event Trace Session in real-time mode.
-[-]ul	Runs the Event Trace Session in user.
-bs `<value>`	Specifies the Event Trace Session buffer size in kb.

Parameter	Description
-nb `<min max>`	Specifies the number of Event Trace Session buffers.
-mode `<globalsequence|localsequence|pagedmemory>`	Specifies the event trace session logger mode, including: • **Globalsequence** - Specifies that the event tracer add a sequence number to every event it receives irrespective of which trace session received the event. • **Localsequence** - Specifies that the event tracer add sequence numbers for events received at a specific trace session. When this option is used, duplicate sequence numbers can exist across all sessions but will be unique within each trace session. • **Pagedmemory** - Specifies that the event tracer use paged memory rather than the default non-paged memory pool for its internal buffer allocations.
/?	Displays context-sensitive help.

Remarks

- Where [-] is listed, adding an extra hyphen (-) negates the option.

Examples

To update an existing event trace data collector called *trace_log*, changing the maximum log size to 10 MB, updating the log file format to CSV, and appending file versioning in the format mmddhhmm, type:

```
logman update trace trace_log -max 10 -f csv -v mmddhhmm
```

Related links

- Command-Line Syntax Key
- logman create trace command

- logman command

Feedback

Was this page helpful? Yes No

logoff

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Logs off a user from a session on a Remote Desktop Session Host server and deletes the session.

Syntax

```
logoff [<sessionname> | <sessionID>] [/server:<servername>] [/v]
```

Parameters

Expand table

Parameter	Description
`<sessionname>`	Specifies the name of the session. This must be an active session.
`<sessionID>`	Specifies the numeric ID which identifies the session to the server.
/server: `<servername>`	Specifies the Remote Desktop Session Host server that contains the session whose user you want to log off. If unspecified, the server on which you are currently active is used.
/v	Displays information about the actions being performed.
/?	Displays help at the command prompt.

Remarks

- You can always log off yourself from the session to which you are currently logged on. You must, however, have **Full Control** permission to log off users from other sessions.
- Logging off a user from a session without warning can result in loss of data at the user's session. You should send a message to the user by using the **msg** command to warn the user before taking this action.

- If `<sessionID>` or `<sessionname>` isn't specified, **logoff** logs the user off from the current session.
- After you log off a user, all processes end and the session is deleted from the server.
- You can't log off a user from the console session.

Examples

To log off a user from the current session, type:

```
logoff
```

To log off a user from a session by using the session's ID, for example *session 12*, type:

```
logoff 12
```

To log off a user from a session by using the name of the session and server, for example session *TERM04* on *Server1*, type:

```
logoff TERM04 /server:Server1
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

lpq

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the status of a print queue on a computer running Line printer Daemon (LPD).

Syntax

```
lpq -S <servername> -P <printername> [-l]
```

Parameters

Expand table

Parameter	Description
-S `<servername>`	Specifies (by name or IP address) the computer or printer sharing device that hosts the LPD print queue with a status that you want to display. This parameter is required and must be capitalized.
-P `<Printername>`	Specifies (by name) the printer for the print queue with a status that you want to display. This parameter is required and must be capitalized.
-l	Specifies that you want to display details about the status of the print queue.
/?	Displays help at the command prompt.

Examples

To display the status of the *Laserprinter1* printer queue on an LPD host at *10.0.0.45*, type:

```
lpq -S 10.0.0.45 -P Laserprinter1
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

lpr

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sends a file to a computer or printer sharing device running the Line printer Daemon (LPD) service in preparation for printing.

Syntax

```
lpr [-S <servername>] -P <printername> [-C <bannercontent>] [-J <jobname>]
[-o | -o l] [-x] [-d] <filename>
```

Parameters

Expand table

Parameter	Description
-S `<servername>`	Specifies (by name or IP address) the computer or printer sharing device that hosts the LPD print queue with a status that you want to display. This parameter is required and must be capitalized.
-P `<printername>`	Specifies (by name) the printer for the print queue with a status that you want to display. To find the name of the printer, open the **Printers** folder. This parameter is required and must be capitalized.
-C `<bannercontent>`	Specifies the content to print on the banner page of the print job. If you don't include this parameter, the name of the computer from which the print job was sent appears on the banner page. This parameter must be capitalized.
-J `<jobname>`	Specifies the print job name that will be printed on the banner page. If you don't include this parameter, the name of the file being printed appears on the banner page. This parameter must be capitalized.
`[-o | -o l]`	Specifies the type of file that you want to print. The parameter **-o** specifies that you want to print a text file. The parameter **-o l** specifies that you want to print a binary file (for example, a PostScript file).
-d	Specifies that the data file must be sent before the control file. Use this parameter if your printer requires the data file to be sent first. For more

Parameter	Description
	information, see your printer documentation.
-x	Specifies that the **lpr** command must be compatible with the Sun Microsystems operating system (referred to as SunOS) for releases up to and including 4.1.4_u1.
`<filename>`	Specifies (by name) the file to be printed. This parameter is required.
/?	Displays help at the command prompt.

Examples

To print the *Document.txt* text file to the *Laserprinter1* printer queue on an LPD host at *10.0.0.45*, type:

```
lpr -S 10.0.0.45 -P Laserprinter1 -o Document.txt
```

To print the *PostScript_file.ps* Adobe PostScript file to the *Laserprinter1* printer queue on an LPD host at *10.0.0.45*, type:

```
lpr -S 10.0.0.45 -P Laserprinter1 -o l PostScript_file.ps
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

macfile

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Manages File Server for Macintosh servers, volumes, directories, and files. You can automate administrative tasks by including a series of commands in batch files and starting them manually or at predetermined times.

Modify directories in Macintosh-accessible volumes

To change the directory name, location, owner, group, and permissions for Macintosh-accessible volumes.

Syntax

```
macfile directory[/server:\\<computername>] /path:<directory> [/owner:
<ownername>] [/group:<groupname>] [/permissions:<permissions>]
```

Parameters

Expand table

Parameter	Description
/server:`\\<computername>`	Specifies the server on which to change a directory. If omitted, the operation is performed on the local computer.
/path:`<directory>`	Specifies the path to the directory that you want to change. This parameter is required. **Note:** The directory must exist, using **macfile directory** won't create directories.
/owner:`<ownername>`	Changes the owner of the directory. If omitted, the owner name won't change.
/group:`<groupname>`	Specifies or changes the Macintosh primary group that is associated with the directory. If omitted, the primary group remains unchanged.

Parameter	Description
/permissions: `<permissions>`	Sets permissions on the directory for the owner, primary group, and world (everyone). This must be an 11-digit number, where the number 1 grants permission and 0 revokes permission (for example, 11111011000). If this parameter is omitted, permissions remain unchanged.
/?	Displays help at the command prompt.

Position of permissions digit

The position of the permissions digit determines which permission is set, including:

Expand table

Position	Sets Permission
First	OwnerSeeFiles
Second	OwnerSeeFolders
Third	OwnerMakechanges
Fourth	GroupSeeFiles
Fifth	GroupSeeFolders
Sixth	GroupMakechanges
Seventh	WorldSeeFiles
Eighth	WorldSeeFolders
Ninth	WorldMakechanges
Tenth	The directory can't be renamed, moved, or deleted.
Eleventh	The changes apply to the current directory and all subdirectories.

Remarks

- If the information that you supply contains spaces or special characters, use quotation marks around the text (for example, "`<computer name>`").
- Use **macfile directory** to make an existing directory in a Macintosh-accessible volume available to Macintosh users. The **macfile directory** command doesn't create directories.

- Use File Manager, the command prompt, or the **macintosh new folder** command to create a directory in a Macintosh-accessible volume before you use the **macfile directory** command.

Examples

To assign *See Files*, *See Folders*, and *Make changes* permissions to the owner, to set *See Folder* permissions to all other users, and to prevent the directory from being renamed, moved, or deleted, type:

```
macfile directory /path:e:\statistics\may sales /permissions:11111011000
```

Where the subdirectory is *May sales*, located in the Macintosh-accessible volume *Statistics*, on the E:\ drive of the local server.

Join a Macintosh file's data and resource forks

To specify the server on which to join files, who created the file, the type of file, where the data fork is located, where the resource fork is located, and where the output file should be located.

Syntax

```
macfile forkize[/server:\\<computername>] [/creator:<creatorname>] [/type:
<typename>]  [/datafork:<filepath>] [/resourcefork:<filepath>] /targetfile:
<filepath>
```

Parameters

Expand table

Parameter	Description
/server:`\\ <computername>`	Specifies the server on which to join files. If omitted, the operation is performed on the local computer.

Parameter	Description
/creator: `<creatorname>`	Specifies the creator of the file. The Macintosh finder uses the **/creator** command-line option to determine the application that created the file.
/type: `<typename>`	Specifies the type of file. The Macintosh finder uses the **/type** command-line option to determine the file type within the application that created the file.
/datafork: `<filepath>`	Specifies the location of the data fork that is to be joined. You can specify a remote path.
/resourcefork: `<filepath>`	Specifies the location of the resource fork that is to be joined. You can specify a remote path.
/targetfile: `<filepath>`	Specifies the location of the file that's created by joining a data fork and a resource fork, or specifies the location of the file whose type or creator you are changing. The file must be on the specified server. This parameter is required.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces or special characters, use quotation marks around the text (for example, "`<computer name>`").

Examples

To create the file *tree_app* on the Macintosh-accessible volume *D:\Release*, using the resource fork *C:\Cross\Mac\Appcode*, and to make this new file appear to Macintosh clients as an application (Macintosh applications use the type *APPL*) with the creator (signature) set to *MAGNOLIA*, type:

```
macfile forkize /resourcefork:c:\cross\mac\appcode /type:APPL
/creator:MAGNOLIA /targetfile:D:\Release\tree_app
```

To change the file creator to *Microsoft Word 5.1*, for the file *Word.txt* in the directory *D:\Word documents\Group files*, on the server *\ServerA*, type:

```
macfile forkize /server:\\ServerA /creator:MSWD /type:TEXT
/targetfile:d:\Word documents\Group files\Word.txt
```

Change the sign-in message and limit sessions

To change the sign on message that appears when a user signs in to the File Server for Macintosh server and to limit the number of users who can simultaneously use File and print Servers for Macintosh.

Syntax

```
macfile server [/server:\\<computername>] [/maxsessions:{number | unlimited}] [/loginmessage:<message>]
```

Parameters

Expand table

Parameter	Description
/server:`\\<computername>`	Specifies the server on which to change parameters. If omitted, the operation is performed on the local computer.
/maxsessions:`{number | unlimited}`	Specifies the maximum number of users who can simultaneously use File and print Servers for Macintosh. If omitted, the **maxsessions** setting for the server remains unchanged.
/loginmessage:`<message>`	Changes the message Macintosh users see when signing in to the File Server for Macintosh server. The maximum number of characters for the sign-in message is 199. If omitted, the **loginmessage** message for the server remains unchanged. To remove an existing sign-in message, include the **/loginmessage** parameter, but leave the *message* variable blank.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces or special characters, use quotation marks around the text (for example, "`<computer name>`").

Examples

To change the number of permitted File and print Server for Macintosh sessions on the local server to five sessions, and to add the sign-in message "Sign off from Server for

Macintosh when you are finished", type:

```
macfile server /maxsessions:5 /loginmessage:Sign off from Server for
Macintosh when you are finished
```

Add, change, or remove Macintosh-accessible volumes

To add, change, or remove a Macintosh-accessible volume.

Syntax

```
macfile volume {/add|/set} [/server:\\<computername>] /name:
<volumename>/path:<directory>[/readonly:{true | false}] [/guestsallowed:
{true | false}] [/password:<password>] [/maxusers:{<number>>|unlimited}]
macfile volume /remove[/server:\\<computername>] /name:<volumename>
```

Parameters

Expand table

Parameter	Description
`{/add | /set}`	Required when adding or changing a Macintosh-accessible volume. Adds or changes the specified volume.
/server: `\\ <computername>`	Specifies the server on which to add, change, or remove a volume. If omitted, the operation is performed on the local computer.
/name: `<volumename>`	Required. Specifies the volume name to be added, changed, or removed.
/path: `<directory>`	Required and valid only when you are adding a volume. Specifies the path to the root directory of the volume to be added.
/readonly: `{true | false}`	Specifies whether users can change files in the volume. Use **True** to specify that users can't change files in the volume. Use **False** to specify that users can change files in the volume. If omitted when adding a volume, changes to files are allowed. If omitted when changing a volume, the **readonly** setting for the volume remains unchanged.

Parameter	Description
/guestsallowed: `{true | false}`	Specifies whether users who log on as guests can use the volume. Use **True** to specify that guests can use the volume. Use **False** to specify that guests can't use the volume. If omitted when adding a volume, guests can use the volume. If omitted when changing a volume, the **guestsallowed** setting for the volume remains unchanged.
/password: `<password>`	Specifies a password that will be required to access the volume. If omitted when adding a volume, no password is created. If omitted when changing a volume, the password remains unchanged.
/maxusers: `{<number>> | unlimited}`	Specifies the maximum number of users who can simultaneously use the files on the volume. If omitted when adding a volume, an unlimited number of users can use the volume. If omitted when changing a volume, the **maxusers** value remains unchanged.
/remove	Required when you are removing a Macintosh-accessible volume. removes the specified volume.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces or special characters, use quotation marks around the text (for example, "`<computer name>`").

Examples

To create a volume called *US Marketing Statistics* on the local server, using the *Stats* directory in the E drive, and to specify that the volume cannot be accessed by guests, type:

```
macfile volume /add /name:US Marketing Statistics /guestsallowed:false
/path:e:\Stats
```

To change the volume created above to be read-only, to require a password, and to set the number of maximum users to five, type:

```
macfile volume /set /name:US Marketing Statistics /readonly:true
/password:saturn /maxusers:5
```

To add a volume called *Landscape Design*, on the server *Magnolia*, using the *trees* directory in the E drive, and to specify that the volume can be accessed by guests, type:

```
macfile volume /add /server:\\Magnolia /name:Landscape Design /path:e:\trees
```

To remove the volume called *Sales Reports* on the local server, type:

```
macfile volume /remove /name:Sales Reports
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

makecab

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Package existing files into a cabinet (.cab) file.

> **Note**
>
> This command is the same as the diantz command.

Syntax

```
makecab [/v[n]] [/d var=<value> ...] [/l <dir>] <source> [<destination>]
makecab [/v[<n>]] [/d var=<value> ...] /f <directives_file> [...]
```

Parameters

Expand table

Parameter	Description
`<source>`	File to compress.
`<destination>`	File name to give compressed file. If omitted, the last character of the source file name is replaced with an underscore (_) and used as the destination.
/f `<directives_file>`	A file with **makecab** directives (may be repeated).
/d var=`<value>`	Defines variable with specified value.
/l `<dir>`	Location to place destination (default is current directory).
/v[`<n>`]	Set debugging verbosity level (0=none,...,3=full).
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- diantz command
- Microsoft Cabinet format

Feedback

Was this page helpful? Yes No

manage-bde

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Turns on or turns off BitLocker, specifies unlock mechanisms, updates recovery methods, and unlocks BitLocker-protected data drives.

> **Note**
>
> This command-line tool can be used in place of the **BitLocker Drive Encryption** Control Panel item.

Syntax

```
manage-bde [-status] [-on] [-off] [-pause] [-resume] [-lock] [-unlock] [-autounlock]
[-protectors] [-tpm]
[-setidentifier] [-forcerecovery] [-changepassword] [-changepin] [-changekey] [-
keypackage] [-upgrade] [-wipefreespace] [{-?|/?}] [{-help|-h}]
```

> **Note**
>
> Each subcommand has its own syntax, parameters, and examples. Select a subcommand link in the following table to view its detailed usage information.

Parameters

Expand table

Parameter	Description
manage-bde status	Provides information about all drives on the computer, whether or not they are BitLocker-protected.
manage-bde on	Encrypts the drive and turns on BitLocker.
manage-bde off	Decrypts the drive and turns off BitLocker. All key protectors are removed when decryption is complete.
manage-bde pause	Pauses encryption or decryption.

Parameter	Description
manage-bde resume	Resumes encryption or decryption.
manage-bde lock	Prevents access to BitLocker-protected data.
manage-bde unlock	Allows access to BitLocker-protected data with a recovery password or a recovery key.
manage-bde autounlock	Manages automatic unlocking of data drives.
manage-bde protectors	Manages protection methods for the encryption key.
manage-bde tpm	Configures the computer's Trusted Platform Module (TPM). This command isn't supported on computers running Windows 8 or **win8_server_2**. To manage the TPM on these computers, use either the TPM Management MMC snap-in or the TPM Management cmdlets for Windows PowerShell.
manage-bde setidentifier	Sets the drive identifier field on the drive to the value specified in the **Provide the unique identifiers for your organization** Group Policy setting.
manage-bde ForceRecovery	Forces a BitLocker-protected drive into recovery mode on restart. This command deletes all TPM-related key protectors from the drive. When the computer restarts, only a recovery password or recovery key can be used to unlock the drive.
manage-bde changepassword	Modifies the password for a data drive.
manage-bde changepin	Modifies the PIN for an operating system drive.
manage-bde changekey	Modifies the startup key for an operating system drive.
manage-bde KeyPackage	Generates a key package for a drive.
manage-bde upgrade	Upgrades the BitLocker version.
manage-bde WipeFreeSpace	Wipes the free space on a drive.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Related links

- Command-Line Syntax Key
- Enabling BitLocker by Using the Command Line

Last updated on 12/15/2025

manage-bde status

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Provides information about all drives on the computer; whether or not they are BitLocker-protected, including:

- Size
- BitLocker version
- Conversion status
- Percentage encrypted
- Encryption method
- Protection status
- Lock status
- Identification field
- Key protectors

Syntax

```
manage-bde -status [<drive>] [-protectionaserrorlevel] [-computername
<name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.

Parameter	Description
-protectionaserrorlevel	Causes the manage-bde command-line tool to send the return code of **0** if the volume is protected and **1** if the volume is unprotected; most commonly used for batch scripts to determine if a drive is BitLocker-protected. You can also use **-p** as an abbreviated version of this command.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To display the status of drive C, type:

```
manage-bde –status C:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde on

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Encrypts the drive and turns on BitLocker.

Syntax

```
manage-bde –on <drive> {[-recoverypassword <numericalpassword>]|[-
recoverykey <pathtoexternaldirectory>]|[-startupkey
<pathtoexternalkeydirectory>]|[-certificate]|
[-tpmandpin]|[-tpmandpinandstartupkey <pathtoexternalkeydirectory>]|[-
tpmandstartupkey <pathtoexternalkeydirectory>]|[-password]|[-
ADaccountorgroup <domain\account>]}
[-usedspaceonly][-encryptionmethod
{aes128_diffuser|aes256_diffuser|aes128|aes256}] [-skiphardwaretest] [-
discoveryvolumetype <filesystemtype>] [-forceencryptiontype <type>] [-
removevolumeshadowcopies][-computername <name>]
[{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-recoverypassword	Adds a numerical password protector. You can also use **-rp** as an abbreviated version of this command.
`<numericalpassword>`	Represents the recovery password.
-recoverykey	Adds an external key protector for recovery. You can also use **-rk** as an abbreviated version of this command.
`<pathtoexternaldirectory>`	Represents the directory path to the recovery key.
-startupkey	Adds an external key protector for startup. You can also use **-sk** as an abbreviated version of this command.

Parameter	Description
`<pathtoexternalkeydirectory>`	Represents the directory path to the startup key.
-certificate	Adds a public key protector for a data drive. You can also use **-cert** as an abbreviated version of this command.
-tpmandpin	Adds a Trusted Platform Module (TPM) and personal identification number (PIN) protector for the operating system drive. You can also use **-tp** as an abbreviated version of this command.
-tpmandstartupkey	Adds a TPM and startup key protector for the operating system drive. You can also use **-tsk** as an abbreviated version of this command.
-tpmandpinandstartupkey	Adds a TPM, PIN, and startup key protector for the operating system drive. You can also use **-tpsk** as an abbreviated version of this command.
-password	Adds a password key protector for the data drive. You can also use **-pw** as an abbreviated version of this command.
-ADaccountorgroup	Adds a SID-based identity protector for the volume. The volume will automatically unlock if the user or computer has the proper credentials. When specifying a computer account, append a $ to the computer name and specify **–service** to indicate that the unlock should happen in the content of the BitLocker server instead of the user. You can also use **-sid** as an abbreviated version of this command.
-usedspaceonly	Sets the encryption mode to Used Space Only encryption. The sections of the volume containing used space will be encrypted but the free space will not. If this option is not specified, all used space and free space on the volume will be encrypted. You can also use **-used** as an abbreviated version of this command.
-encryptionMethod	Configures the encryption algorithm and key size. You can also use **-em** as an abbreviated version of this command.
-skiphardwaretest	Begins encryption without a hardware test. You can also use **-s** as an abbreviated version of this command.
-discoveryvolumetype	Specifies the file system to use for the discovery data drive. The discovery data drive is a hidden drive added to a FAT-formatted, BitLocker-protected removable data drive that contains the BitLocker To Go Reader.
-forceencryptiontype	Forces BitLocker to use either software or hardware encryption. You can specify either **Hardware** or **Software** as the encryption type. If the **hardware** parameter is selected, but the drive doesn't

Parameter	Description
	support hardware encryption, manage-bde returns an error. If Group Policy settings forbids the specified encryption type, manage-bde returns an error. You can also use **-fet** as an abbreviated version of this command.
-removevolumeshadowcopies	Force deletion of Volume Shadow Copies for the volume. You won't be able to restore this volume using previous system restore points after running this command. You can also use **-rvsc** as an abbreviated version of this command.
`<filesystemtype>`	Specifies which file systems can be used with discovery data drives: FAT32, default, or none.
-computername	Specifies that manage-bde is being used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To turn on BitLocker for drive C, and to add a recovery password to the drive, type:

```
manage-bde –on C: -recoverypassword
```

To turn on BitLocker for drive C, add a recovery password to the drive, and to save a recovery key to drive E, type:

```
manage-bde –on C: -recoverykey E:\ -recoverypassword
```

To turn on BitLocker for drive C, using an external key protector (such as a USB key) to unlock the operating system drive, type:

```
```

```
manage-bde -on C: -startupkey E:\
```

> **Important**
>
> This method is required if you are using BitLocker with computers that don't have a TPM.

To turn on BitLocker for data drive E, and to add a password key protector, type:

```
manage-bde –on E: -pw
```

To turn on BitLocker for operating system drive C, and to use hardware-based encryption, type:

```
manage-bde –on C: -fet hardware
```

Related links

- Command-Line Syntax Key
- manage-bde off command
- manage-bde pause command
- manage-bde resume command
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde off

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Decrypts the drive and turns off BitLocker. All key protectors are removed when decryption is complete.

Syntax

```
manage-bde -off [<volume>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<volume>`	Specifies a drive letter followed by a colon, a volume GUID path, or a mounted volume.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To turn off BitLocker on drive C, type:

```
manage-bde -off C:
```

Related links

- Command-Line Syntax Key
- manage-bde on command
- manage-bde pause command
- manage-bde resume command
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde -pause

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Pauses BitLocker encryption or decryption.

Syntax

```
manage-bde -pause [<volume>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<volume>`	Specifies a drive letter followed by a colon, a volume GUID path, or a mounted volume.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To pause BitLocker encryption on drive C, type:

Output

```
manage-bde -pause C:
```

Related links

- Command-Line Syntax Key
- manage-bde on command
- manage-bde off command
- manage-bde resume command
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde resume

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Resumes BitLocker encryption or decryption after it has been paused.

Syntax

```
manage-bde -resume [<drive>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To resume BitLocker encryption on drive C, type:

```
manage-bde -resume C:
```

Related links

- Command-Line Syntax Key
- manage-bde on command
- manage-bde off command
- manage-bde pause command
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde lock

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Locks a BitLocker-protected drive to prevent access to it unless the unlock key is provided.

Syntax

```
manage-bde -lock [<drive>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To lock data drive D, type:

```
manage-bde -lock D:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde unlock

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Unlocks a BitLocker-protected drive by using a recovery password or a recovery key.

Syntax

```
manage-bde -unlock {-recoverypassword -password | -recoverykey
<pathtoexternalkeyfile>} <drive> [-certificate {-cf pathtocertificatefile |
-ct certificatethumbprint} {-pin}] [-password] [-computername <name>] [{-?
|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
-recoverypassword	Specifies that a recovery password will be used to unlock the drive. You can also use **-rp** as an abbreviated version of this command. This is the recovery key that's saved to a text file and must be written exactly as shown including dashes.
-password	Represents the recovery password that can be used to unlock the drive that either you or your administrator has set.
-recoverykey	Specifies that an external recovery key file will be used to unlock the drive. You can also use **-rk** as an abbreviated version of this command. This method searches for the **.bek** recovery key file that is saved to a USB drive.
`<pathtoexternalkeyfile>`	Represents the external recovery key file that can be used to unlock the drive.
`<drive>`	Represents a drive letter followed by a colon.
-certificate	The local user certificate for a BitLocker certificate to unlock the volume is located in the local user certificate store. You can also use **-cert** as an abbreviated version of this command.

Parameter	Description
-cf `<pathtocertificatefile>`	Path to the certificate file.
-ct `<certificatethumbprint>`	Certificate thumbprint which may optionally include the PIN (-pin).
-password	Presents a prompt for the password to unlock the volume. You can also use **-pw** as an abbreviated version of this command.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To unlock drive E with a password, type:

Windows Command Prompt

```
manage-bde -unlock E: -password
```

To unlock drive E with a recovery password, type:

Windows Command Prompt

```
manage-bde -unlock E: -recoverypassword xxxxxx-xxxxxx-xxxxxx-xxxxxx-xxxxxx-
xxxxxx-xxxxxx-xxxxxx
```

To unlock drive E with a recovery key file that's been saved to a backup folder on another drive, type:

Windows Command Prompt

```
manage-bde -unlock E: -recoverykey F:\Backupkeys\recoverykey.bek
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde autounlock

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Manages the automatic unlocking of BitLocker-protected data drives.

Syntax

```
manage-bde -autounlock [{-enable|-disable|-clearallkeys}] <drive> [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
-enable	Enables automatic unlocking for a data drive.
-disable	Disables automatic unlocking for a data drive.
-clearallkeys	Removes all stored external keys on the operating system drive.
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To enable automatic unlocking of data drive E, type:

```
manage-bde –autounlock -enable E:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde protectors

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Manages the protection methods used for the BitLocker encryption key.

Syntax

```
manage-bde -protectors [{-get|-add|-delete|-disable|-enable|-adbackup|-aadbackup}] <drive> [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
-get	Displays all the key protection methods enabled on the drive and provides their type and identifier (ID).
-add	Adds key protection methods as specified by using additional **-add** parameters.
-delete	Deletes key protection methods used by BitLocker. All key protectors will be removed from a drive unless the optional **-delete** parameters are used to specify which protectors to delete. When the last protector on a drive is deleted, BitLocker protection of the drive is disabled to ensure that access to data is not lost inadvertently.
-disable	Disables protection, which will allow anyone to access encrypted data by making the encryption key available unsecured on drive. No key protectors are removed. Protection will be resumed the next time Windows is booted unless the optional **-disable** parameters are used to specify the reboot count.
-enable	Enables protection by removing the unsecured encryption key from the drive. All configured key protectors on the drive will be enforced.
-adbackup	Backs up recovery information for the drive specified to Active Directory Domain Services (AD DS). Append the **-id** parameter and specify the ID of a specific recovery key to back up. The **-id** parameter is required.

Parameter	Description
-aadbackup	Backs up all recovery information for the drive specified to Microsoft Entra ID. Append the **-id** parameter and specify the ID of a specific recovery key to back up. The **-id** parameter is required.
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief help at the command prompt.
-help or -h	Displays complete help at the command prompt.

Additional -add parameters

The -add parameter can also use these valid additional parameters.

```
manage-bde -protectors -add [<drive>] [-forceupgrade] [-recoverypassword
<numericalpassword>] [-recoverykey <pathtoexternalkeydirectory>]
[-startupkey <pathtoexternalkeydirectory>] [-certificate {-cf
<pathtocertificatefile>|-ct <certificatethumbprint>}] [-tpm] [-tpmandpin]
[-tpmandstartupkey <pathtoexternalkeydirectory>] [-tpmandpinandstartupkey
<pathtoexternalkeydirectory>] [-password][-adaccountorgroup
<securityidentifier> [-computername <name>]
[{-?|/?}] [{-help|-h}]
```

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-recoverypassword	Adds a numerical password protector. You can also use **-rp** as an abbreviated version of this command.
`<numericalpassword>`	Represents the recovery password.
-recoverykey	Adds an external key protector for recovery. You can also use **-rk** as an abbreviated version of this command.

Parameter	Description
`<pathtoexternalkeydirectory>`	Represents the directory path to the recovery key.
-startupkey	Adds an external key protector for startup. You can also use **-sk** as an abbreviated version of this command.
`<pathtoexternalkeydirectory>`	Represents the directory path to the startup key.
-certificate	Adds a public key protector for a data drive. You can also use **-cert** as an abbreviated version of this command.
-cf	Specifies that a certificate file will be used to provide the public key certificate.
`<pathtocertificatefile>`	Represents the directory path to the certificate file.
-ct	Specifies that a certificate thumbprint will be used to identify the public key certificate
`<certificatethumbprint>`	Specifies the value of the thumbprint property of the certificate you want to use. For example, a certificate thumbprint value of a9 09 50 2d d8 2a e4 14 33 e6 f8 38 86 b0 0d 42 77 a3 2a 7b should be specified as a909502dd82ae41433e6f83886b00d4277a32a7b.
-tpmandpin	Adds a Trusted Platform Module (TPM) and personal identification number (PIN) protector for the operating system drive. You can also use **-tp** as an abbreviated version of this command.
-tpmandstartupkey	Adds a TPM and startup key protector for the operating system drive. You can also use **-tsk** as an abbreviated version of this command.
-tpmandpinandstartupkey	Adds a TPM, PIN, and startup key protector for the operating system drive. You can also use **-tpsk** as an abbreviated version of this command.
-password	Adds a password key protector for the data drive. You can also use **-pw** as an abbreviated version of this command.
-adaccountorgroup	Adds a security identifier(SID)-based identity protector for the volume. You can also use **-sid** as an abbreviated version of this command. **IMPORTANT:** By default, you can't add an ADaccountorgroup protector remotely using either WMI or manage-bde. If your deployment requires the ability to add this protector remotely, you must enable constrained delegation.
-computername	Specifies that manage-bde is being used to modify BitLocker protection on a different computer. You can also use **-cn** as an

Parameter	Description
	abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief help at the command prompt.
-help or -h	Displays complete help at the command prompt.

Additional -delete parameters

```
manage-bde -protectors -delete <drive> [-type
{recoverypassword|externalkey|certificate|tpm|tpmandstartupkey|tpmandpin|tpm
andpinandstartupkey|Password|Identity}]
[-id <keyprotectorID>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-type	Identifies the key protector to delete. You can also use **-t** as an abbreviated version of this command.
recoverypassword	Specifies that any recovery password key protectors should be deleted.
externalkey	Specifies that any external key protectors associated with the drive should be deleted.
certificate	Specifies that any certificate key protectors associated with the drive should be deleted.
tpm	Specifies that any TPM-only key protectors associated with the drive should be deleted.
tpmandstartupkey	Specifies that any TPM and startup key based key protectors associated with the drive should be deleted.
tpmandpin	Specifies that any TPM and PIN based key protectors associated with the drive should be deleted.

Parameter	Description
tpmandpinandstartupkey	Specifies that any TPM, PIN, and startup key based key protectors associated with the drive should be deleted.
password	Specifies that any password key protectors associated with the drive should be deleted.
identity	Specifies that any identity key protectors associated with the drive should be deleted.
-ID	Identifies the key protector to delete by using the key identifier. This parameter is an alternative option to the **-type** parameter.
`<keyprotectorID>`	Identifies an individual key protector on the drive to delete. Key protector IDs can be displayed by using the **manage-bde -protectors -get** command.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief help at the command prompt.
-help or -h	Displays complete help at the command prompt.

Additional -disable parameters

```
manage-bde -protectors -disable <drive> [-rebootcount <integer 0 - 15>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
rebootcount	Specifies that protection of the operating system volume has been suspended and will resume after Windows has been restarted the number of times specified in the **rebootcount** parameter. Specify **0** to suspend protection indefinitely. If this parameter isn't specified, BitLocker protection automatically

Parameter	Description
	resumes after Windows is restarted. You can also use **-rc** as an abbreviated version of this command.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief help at the command prompt.
-help or -h	Displays complete help at the command prompt.

Examples

To add a certificate key protector, identified by a certificate file, to drive E, type:

```
manage-bde -protectors -add E: -certificate -cf c:\File Folder\Filename.cer
```

To add an **adaccountorgroup** key protector, identified by domain and user name, to drive E, type:

```
manage-bde -protectors -add E: -sid DOMAIN\user
```

To disable protection until the computer has rebooted 3 times, type:

```
manage-bde -protectors -disable C: -rc 3
```

To delete all TPM and startup keys-based key protectors on drive C, type:

```
manage-bde -protectors -delete C: -type tpmandstartupkey
```

To list all key protectors for drive C, type:

```
manage-bde -protectors -get C:
```

To back up all recovery information for drive C to AD DS, type (where `-id` is the ID of the specific key protector to back up):

```
manage-bde -protectors -adbackup C: -id '{00000000-0000-0000-0000-000000000000}'
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde tpm

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Configures the computer's Trusted Platform Module (TPM).

Syntax

```
manage-bde -tpm [-turnon] [-takeownership <ownerpassword>] [-computername
<name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
-turnon	Enables and activates the TPM, allowing the TPM owner password to be set. You can also use **-t** as an abbreviated version of this command.
-takeownership	Takes ownership of the TPM by setting an owner password. You can also use **-o** as an abbreviated version of this command.
`<ownerpassword>`	Represents the owner password that you specify for the TPM.
-computername	Specifies that manage-bde.exe is used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To turn on the TPM, type:

```
manage-bde -tpm -turnon
```

To take ownership of the TPM and set the owner password to *0wnerP@ss*, type:

```
manage-bde -tpm -takeownership 0wnerP@ss
```

Related links

- Command-Line Syntax Key
- TPM Management cmdlets for Windows PowerShell
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde setidentifier

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the drive identifier field on the drive to the value specified in the **Provide the unique identifiers for your organization** Group Policy setting.

Syntax

```
manage-bde –setidentifier <drive> [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To set BitLocker drive identifier field for C, type:

```
manage-bde –setidentifier C:
```

Related links

- Command-Line Syntax Key
- manage-bde command
- BitLocker Recovery Guide

Feedback

Was this page helpful? Yes No

manage-bde forcerecovery

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Forces a BitLocker-protected drive into recovery mode on restart. This command deletes all Trusted Platform Module (TPM)-related key protectors from the drive. When the computer restarts, only a recovery password or recovery key can be used to unlock the drive.

Syntax

```
manage-bde –forcerecovery <drive> [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To cause BitLocker to start in recovery mode on drive C, type:

```
manage-bde -forcerecovery C:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde changepassword

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Modifies the password for a data drive. The user is prompted for a new password.

Syntax

```
manage-bde -changepassword [<drive>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To change the password used to unlock BitLocker on data drive D, type:

```
manage-bde –changepassword D:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde changepin

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Modifies the PIN for an operating system drive. The user is prompted to enter a new PIN.

Syntax

```
manage-bde -changepin [<drive>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To change the PIN used with BitLocker on drive C, type:

```
manage-bde –changepin C:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde changekey

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Modifies the startup key for an operating system drive.

Syntax

```
manage-bde -changekey [<drive>] [<pathtoexternalkeydirectory>] [-
computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To create a new startup key on drive E, to use with BitLocker encryption on drive C, type:

```
manage-bde -changekey C: E:\
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde keypackage

09/02/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Generates a key package for a drive. The key package can be used in conjunction with the repair tool to repair corrupted drives.

Syntax

```
manage-bde -keypackage [<drive>] [-ID <keyprotectoryID>] [-path
<pathtoexternalkeydirectory>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-ID	Creates a key package using the key protector with the identifier specified by this ID value. **Tip:** Use the **manage-bde -protectors -get** command, along with the drive letter that you want to create a key package for, to get a list of available GUIDs to use as the ID value.
-path	Specifies the location to save the created key package.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To create a key package for drive C, based on the key protector identified by the GUID, and to save the key package to F:\Folder, type:

```
manage-bde -keypackage C: -id {84E151C1...7A62067A512} -path f:\Folder
```

Related links

- Command-Line Syntax Key
- manage-bde command

manage-bde upgrade

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Upgrades the BitLocker version.

Syntax

```
manage-bde -upgrade [<drive>] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To upgrade BitLocker encryption on drive C, type:

```
manage-bde –upgrade C:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

manage-bde wipefreespace

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Wipes the free space on the volume, removing any data fragments that may have existed in the space. Running this command on a volume encrypted using the **Used Space Only** encryption method provides the same level of protection as the **Full Volume Encryption** encryption method.

Syntax

```
manage-bde -wipefreespace|-w [<drive>] [-cancel] [-computername <name>] [{-?|/?}] [{-help|-h}]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Represents a drive letter followed by a colon.
-cancel	Cancels a wipe of free space that is in process.
-computername	Specifies that manage-bde.exe will be used to modify BitLocker protection on a different computer. You can also use **-cn** as an abbreviated version of this command.
`<name>`	Represents the name of the computer on which to modify BitLocker protection. Accepted values include the computer's NetBIOS name and the computer's IP address.
-? or /?	Displays brief Help at the command prompt.
-help or -h	Displays complete Help at the command prompt.

Examples

To wipe the free space on drive C, type either:

```
manage-bde -w C:
```

```
manage-bde -wipefreespace C:
```

To cancel the wipe of the free space on drive C, type either:

```
manage-bde -w -cancel C:
```

```
manage-bde -wipefreespace -cancel C:
```

Related links

- Command-Line Syntax Key
- manage-bde command

Feedback

Was this page helpful? Yes No

mapadmin

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The **mapadmin** command-line utility administers User Name Mapping on the local or remote computer running Microsoft Services for Network File System. If you are logged on with an account that does not have administrative credentials, you can specify a user name and password of an account that does.

Syntax

```
mapadmin [<computer>] [-u <user> [-p <password>]]
mapadmin [<computer>] [-u <user> [-p <password>]] {start | stop}
mapadmin [<computer>] [-u <user> [-p <password>]] config <option[...]>
mapadmin [<computer>] [-u <user> [-p <password>]] add -wu <windowsuser> -uu
<UNIXuser> [-setprimary]
mapadmin [<computer>] [-u <user> [-p <password>]] add -wg <windowsgroup> -ug
<UNIXgroup> [-setprimary]
mapadmin [<computer>] [-u <user> [-p <password>]] setprimary -wu
<Windowsuser> [-uu <UNIXuser>]
mapadmin [<computer>] [-u <user> [-p <password>]] setprimary -wg
<Windowsgroup> [-ug <UNIXgroup>]
mapadmin [<computer>] [-u <user> [-p <password>]] delete <option[...]>
mapadmin [<computer>] [-u <user> [-p <password>]] list <option[...]>
mapadmin [<computer>] [-u <user> [-p <password>]] backup <filename>
mapadmin [<computer>] [-u <user> [-p <password>]] restore <filename>
mapadmin [<computer>] [-u <user> [-p <password>]] adddomainmap -d
<Windowsdomain> {-y <<NISdomain>> | -f <path>}
mapadmin [<computer>] [-u <user> [-p <password>]] removedomainmap -d
<Windowsdomain> -y <<NISdomain>>
mapadmin [<computer>] [-u <user> [-p <password>]] removedomainmap -all
mapadmin [<computer>] [-u <user> [-p <password>]] listdomainmaps
```

Parameters

Expand table

Parameter	Description
`<computer>`	Specifies the remote computer running the User Name Mapping service that you want to administer. You can specify the computer using a Windows

Parameter	Description
	Internet Name Service (WINS) name or a Domain Name System (DNS) name, or by Internet Protocol (IP) address.
-u `<user>`	Specifies the user name of the user whose credentials are to be used. It might be necessary to add the domain name to the user name in the form *domain**username*.
-p `<password>`	Specifies the password of the user. If you specify the **-u** option but omit the **-p** option, you are prompted for the user's password.
`start | stop`	Starts or stops the User Name Mapping service.
config	Specifies general settings for User Name Mapping. The following options are available with this parameter: • **-r** `<dddd>:<hh>:<mm>`: Specifies the refresh interval for updating from the Windows and NIS databases in days, hours, and minutes. The minimum interval is 5 minutes. • **-i** `{yes | no}`: Turns simple mapping on (**yes**) or off (**no**). By default, mapping is turned on.
add	Creates a new mapping for a user or group. The following options are available with this parameter: • **-wu** `<name>`: Specifies the name of the Windows user for which a new mapping is being created. • **-uu** `<name>`: Specifies the name of the UNIX user for which a new mapping is being created. • **-wg** `<group>`: Specifies the name of the Windows group for which a new mapping is being created. • **-ug** `<group>`: Specifies the name of the UNIX group for which a new mapping is being created. • **-setprimary:** Specifies that the new mapping is the primary mapping.
setprimary	Specifies which mapping is the primary mapping for a UNIX user or group with multiple mappings. The following options are available with this parameter: • **-wu** `<name>`: Specifies the Windows user of the primary mapping. If more than one mapping for the user exists, use the **-uu** option to specify the primary mapping. • **-uu** `<name>`: Specifies the UNIX user of the primary mapping. • **-wg** `<group>`: Specifies the Windows group of the primary mapping. If more than one mapping for the group exists, use the **-ug** option to specify the primary mapping. • **-ug** `<group>`: Specifies the UNIX group of the primary mapping.
delete	Removes the mapping for a user or group. The following options are available for this parameter:

Parameter	Description
	• **-wu** `<user>`: Specifies the Windows user for which the mapping will be deleted, specified as `<windowsdomain>\<username>`. You must specify either the **-wu** or the **-uu** option, or both. If you specify both options, the particular mapping identified by the two options will be deleted. If you specify only the **-wu** option, all mappings for the specified user will be deleted. • **-uu** `<user>`: Specifies the UNIX user for whom the mapping will be deleted, specified as `<username>`. You must specify either the **-wu** or the **-uu** option, or both. If you specify both options, the particular mapping identified by the two options will be deleted. If you specify only the **-uu** option, all mappings for the specified user will be deleted. • **-wg** `<group>`: Specifies the Windows group for which the mapping will be deleted, specified as `<windowsdomain>\<username>`. You must specify either the **-wg** or the **-ug** option, or both. If you specify both options, the particular mapping identified by the two options will be deleted. If you specify only the **-wg** option, all mappings for the specified group will be deleted. • **-ug** `<group>`: Specifies the UNIX group for which the mapping will be deleted, specified as `<groupname>`. You must specify either the **-wg** or the **-ug** option, or both. If you specify both options, the particular mapping identified by the two options will be deleted. If you specify only the **-ug** option, all mappings for the specified group will be deleted.
list	Displays information about user and group mappings. The following options are available with this parameter: • **-all:** Lists both simple and advanced mappings for users and groups. • **-simple:** Lists all simple mapped users and groups. • **-advanced:** Lists all advanced mapped users and groups. Maps are listed in the order in which they are evaluated. Primary maps, marked with an asterisk (*), are listed first, followed by secondary maps, which are marked with a carat `(^)`. • **-wu** `<name>`: Lists the mapping for a specified Windows user. • **-wg** `<group>`: Lists the mapping for a Windows group. • **-uu** `<name>`: Lists the mapping for a UNIX user. • **-ug** `<group>`: Lists the mapping for a UNIX group.
backup	Saves User Name Mapping configuration and mapping data to the file specified by `<filename>`.
restore	Replaces configuration and mapping data with data from the file (specified by `<filename>`) that was created using the **backup** parameter.

Parameter	Description
adddomainmap	Adds a simple map between a Windows domain and an NIS domain or password and group files. The following options are available for this parameter: • **-d** `<windowsdomain>`: Specifies the Windows domain to be mapped. • **-y** `<NISdomain>`: Specifies the NIS domain to be mapped. You must use the **-n** `<NISserver>` parameter to specify the NIS server for the NIS domain specified by the **-y** option. • **-f** `<path>`: Specifies the fully-qualified path of directory containing the password and group files to be mapped. The files must be located on the computer being managed, and you can't use **mapadmin** to manage a remote computer to set up maps based on password and group files.
removedomainmap	Removes a simple map between a Windows domain and an NIS domain. The following options and argument are available for this parameter: • **-d** `<windowsdomain>`: Specifies the Windows domain of the map to be removed. • **-y** `<NISdomain>`: Specifies the NIS domain of the map to be removed. • **-all:** Specifies that all simple maps between Windows and NIS domains are to be removed. This will also remove any simple map between a Windows domain and password and group files.
listdomainmaps	Lists the Windows domains that are mapped to NIS domains or password and group files.

Remarks

- If you don't specify any paramters, the **mapadmin** command displays the current settings for User Name Mapping.
- For all options that specify a user or group name, the following formats can be used:
  - For Windows users, use the formats: `<domain>\<username>`, `\\<computer>\<username>`, `\<computer>\<username>`, or `<computer>\<username>`
  - For Windows groups, use the formats: `<domain>\<groupname>`, `\\<computer>\<groupname>`, `\<computer>\<groupname>`, or `<computer>\<groupname>`
  - For UNIX users, use the formats: `<NISdomain>\<username>`, `<username>@<NISdomain>`, `<username>@PCNFS`, or `PCNFS\<username>`

- For UNIX groups, use the formats: `<NISdomain>\<groupname>`, `<groupname>@<NISdomain>`, `<groupname>@PCNFS`, or `PCNFS\<groupname>`

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

md

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a directory or subdirectory. Command extensions, which are enabled by default, allow you to use a single **md** command to create intermediate directories in a specified path.

> **Note**
>
> This command is the same as the mkdir command.

Syntax

```
md [<drive>:]<path>
```

Parameters

Expand table

Parameter	Description
`<drive>`:	Specifies the drive on which you want to create the new directory.
`<path>`	Specifies the name and location of the new directory. The maximum length of any single path is determined by the file system. This is a required parameter.
/?	Displays help at the command prompt.

Examples

To create a directory named *Directory1* within the current directory, type:

```
md Directory1
```

To create the directory tree *Taxes**Property**Current* within the root directory, with command extensions enabled, type:

```
md \Taxes\Property\Current
```

To create the directory tree *Taxes**Property**Current* within the root directory as in the previous example, but with command extensions disabled, type the following sequence of commands:

```
md \Taxes
md \Taxes\Property
md \Taxes\Property\Current
```

Related links

- Command-Line Syntax Key
- mkdir command

Feedback

Was this page helpful? Yes No

mkdir

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a directory or subdirectory. Command extensions, which are enabled by default, allow you to use a single **mkdir** command to create intermediate directories in a specified path.

> **Note**
>
> This command is the same as the md command.

Syntax

```
mkdir [<drive>:]<path>
```

Parameters

Expand table

Parameter	Description
`<drive>:`	Specifies the drive on which you want to create the new directory.
`<path>`	Specifies the name and location of the new directory. The maximum length of any single path is determined by the file system. This is a required parameter.
/?	Displays help at the command prompt.

Examples

To create a directory named *Directory1* within the current directory, type:

```
mkdir Directory1
```

To create the directory tree *Taxes**Property**Current* within the root directory, with command extensions enabled, type:

```
mkdir \Taxes\Property\Current
```

To create the directory tree *Taxes**Property**Current* within the root directory as in the previous example, but with command extensions disabled, type the following sequence of commands:

```
mkdir \Taxes
mkdir \Taxes\Property
mkdir \Taxes\Property\Current
```

Related links

- Command-Line Syntax Key
- md command

Feedback

Was this page helpful? Yes No

mklink

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a directory or file symbolic or hard link.

Syntax

```
mklink [[/d] | [/h] | [/j]] <link> <target>
```

Parameters

Expand table

Parameter	Description
/d	Creates a directory symbolic link. By default, this command creates a file symbolic link.
/h	Creates a hard link instead of a symbolic link.
/j	Creates a Directory Junction.
`<link>`	Specifies the name of the symbolic link being created.
`<target>`	Specifies the path (relative or absolute) that the new symbolic link refers to.
/?	Displays help at the command prompt.

Examples

To create and remove a symbolic link named MyFolder from the root directory to the \Users\User1\Documents directory, and a hard link named Myfile.file to the example.file file located within the directory, type:

```
mklink /d \MyFolder \Users\User1\Documents
mklink /h \MyFile.file \User1\Documents\example.file
```

```
rd \MyFolder
del \MyFile.file
```

Related links

- Command-Line Syntax Key
- del command
- rd command
- New-Item in Windows PowerShell

Feedback

Was this page helpful? Yes No

mmc

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Using mmc command-line options, you can open a specific **mmc** console, open **mmc** in author mode, or specify that the 32-bit or 64-bit version of **mmc** is opened.

Syntax

```
mmc <path>\<filename>.msc [/a] [/64] [/32]
```

Parameters

Expand table

Parameter	Description
`<path>\<filename>.msc`	starts **mmc** and opens a saved console. You need to specify the complete path and file name for the saved console file. If you do not specify a console file, **mmc** opens a new console.
/a	Opens a saved console in author mode. Used to make changes to saved consoles.
/64	Opens the 64-bit version of **mmc** (mmc64). Use this option only if you are running a Microsoft 64-bit operating system and want to use a 64-bit snap-in.
/32	Opens the 32-bit version of **mmc** (mmc32). When running a Microsoft 64-bit operating system, you can run 32-bit snap-ins by opening mmc with this command-line option when you have 32-bit only snap-ins.

Remarks

- You can use environment variables to create command lines or shortcuts that don't depend on the explicit location of console files. For instance, if the path to a console file is in the system folder (for example, **mmc**

c:\winnt\system32\console_name.msc), you can use the expandable data string **%systemroot%** to specify the location (**mmc%systemroot%\system32\console_name.msc**). This may be useful if you're delegating tasks to people in your organization who are working on different computers.

- When consoles are opened using the **/a** option, they're opened in author mode, regardless of their default mode. This doesn't permanently change the default mode setting for files; when you omit this option, mmc opens console files according to their default mode settings.
- After you open **mmc** or a console file in author mode, you can open any existing console by clicking **Open** on the **Console** menu.
- You can use the command line to create shortcuts for opening **mmc** and saved consoles. A command-line command works with the **Run** command on the **Start** menu, in any command-prompt window, in shortcuts, or in any batch file or program that calls the command.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

mode

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays system status, changes system settings, or reconfigures ports or devices. If used without parameters, **mode** displays all the controllable attributes of the console and the available COM devices.

Serial port

Configures a serial communications port and sets the output handshake.

Syntax

```
mode com<m>[:] [baud=<b>] [parity=<p>] [data=<d>] [stop=<s>] [to={on|off}]
[xon={on|off}] [odsr={on|off}] [octs={on|off}] [dtr={on|off|hs}] [rts=
{on|off|hs|tg}] [idsr={on|off}]
```

Parameters

Expand table

Parameter	Description
`com<m>[:]`	Specifies the number of the async Prncnfg.vbshronous communications port.
`baud=<b>`	Specifies the transmission rate in bits per second. The valid values include: • **11** - 110 baud • **15** - 150 baud • **30** - 300 baud • **60** - 600 baud • **12** - 1200 baud • **24** - 2400 baud • **48** - 4800 baud • **96** - 9600 baud • **19** - 19,200 baud

Parameter	Description
`parity=<p>`	Specifies how the system uses the parity bit to check for transmission errors. The valid values include: • **n** - none • **e** - even (default value) • **o** - odd • **m** - mark • **s** - space Not all devices support using the **m** or **s** parameters.
`data=<d>`	Specifies the number of data bits in a character. Valid values range from **5** through **8**. The default value is **7**. Not all devices support the values **5** and **6**.
`stop=<s>`	Specifies the number of stop bits that define the end of a character: **1**, **1.5**, or **2**. If the baud rate is **110**, the default value is **2**. Otherwise, the default value is **1**. Not all devices support the value **1.5**.
`to={on | off}`	Specifies whether the device uses infinite time out processing. The default value is **off**. Turning this option **on** means that the device will never stop waiting to receive a response from a host or client computer.
`xon={on | off}`	Specifies whether the system allows the XON/XOFF protocol. This protocol provides flow control for serial communications, enhancing reliability, but reducing performance.
`odsr={on | off}`	Specifies whether the system turns on the Data Set Ready (DSR) output handshake.
`octs={on | off}`	Specifies whether the system turns on the Clear to Send (CTS) output handshake.
`dtr={on | off | hs}`	Specifies whether the system turns on the Data Terminal Ready (DTR) output handshake. Setting this value to **on** mode, provides a constant signal to show the terminal is ready to send data. Setting this value to **hs** mode provides a handshake signal between the two terminals.
`rts={on | off | hs | tg}`	Specifies whether the system turns on the Request to Send (RTS) output handshake. Setting this value to **on** mode, provides a constant signal to show the terminal is ready to send data. Setting this value to **hs** mode provides a handshake signal between the two terminals. Setting this value to **tg** mode provides a way to toggle between ready and not ready states.
`idsr={on | off}`	Specifies whether the system turns on the DSR sensitivity. You must turn this option on to use DSR handshaking.
/?	Displays help at the command prompt.

Device status

Displays the status of a specified device. If used without parameters, **mode** displays the status of all devices installed on your system.

Syntax

```
mode [<device>] [/status]
```

Parameters

Expand table

Parameter	Description
`<device>`	Specifies the name of the device for which you want to display the status. Standard names include, LPT1: through LPT3:, COM1: through COM9:, and CON.
/status	Requests the status of any redirected parallel printers. You can also use **/sta** as an abbreviated version of this command.
/?	Displays help at the command prompt.

Redirect printing

Redirects printer output. You must be a member of the Administrators group to redirect printing.

> **Note**
>
> To set up your system so that it sends parallel printer output to a serial printer, you must use the **mode** command twice. The first time, you must use **mode** to configure the serial port. The second time, you must use **mode** to redirect parallel printer output to the serial port you specified in the first **mode** command.

Syntax

```
mode LPT<n>[:]=COM<m>[:]
```

Parameters

Expand table

Parameter	Description
LPT`<n>`[:]	Specifies the number of the LPT to configure. Typically, this means providing a value from **LTP1: through LTP3:**, unless your system includes special parallel port support. This parameter is required.
COM`<m>`[:]	Specifies the COM port to configure. Typically, this means providing a value from **COM1: through COM9:**, unless your system has special hardware for additional COM ports. This parameter is required.
/?	Displays help at the command prompt.

Examples

To redirect a serial printer that operates at 4800 baud with even parity, and is connected to the COM1 port (the first serial connection on your computer), type:

```
mode com1 48,e,,,b
mode lpt1=com1
```

To redirect parallel printer output from LPT1 to COM1, and then to print a file using LPT1, type the following command before you print the file:

```
mode lpt1
```

This command prevents the redirection the file from LPT1 to COM1.

Select code page

Configures or queries the code page info for a selected device.

Syntax

```
mode <device> codepage select=<yyy>
mode <device> codepage [/status]
```

Parameters

Expand table

Parameter	Description
`<device>`	Specifies the device for which you want to select a code page. CON is the only valid name for a device. This parameter is required.
codepage	Specifies which code page to use with the specified device. You can also use **cp** as an abbreviated version of this command. This parameter is required.
select=`<yyy>`	Specifies the number of the code page to use with the device. The supported code pages, by country/region or language, include: • **437:** United States • **850:** Multilingual (Latin I) • **852:** Slavic (Latin II) • **855:** Cyrillic (Russian) • **857:** Turkish • **860:** Portuguese • **861:** Icelandic • **863:** Canadian-French • **865:** Nordic • **866:** Russian • **869:** Modern Greek This parameter is required.
/status	Displays the numbers of the current code pages selected for the specified device. You can also use **/sta** as an abbreviated version of this command. Regardless whether you specify **/status**, the **mode codepage** command will display the numbers of the code pages that are selected for the specified device.
/?	Displays help at the command prompt.

Display mode

Changes the size of the command prompt screen buffer

Syntax

```
mode con[:] [cols=<c>] [lines=<n>]
```

Parameters

Expand table

Parameter	Description
con[:]	Indicates that the change applies to the Command Prompt window. This parameter is required.
cols=`<c>`	Specifies the number of columns in the command prompt screen buffer. The default setting is 80 columns, but you can set this to any value. If you don't use the default, typical values are 40 and 135 columns. Using non-standard values can result in the command prompt app problems.
lines=`<n>`	Specifies the number of lines in the command prompt screen buffer. The default value is 25, but you can set this to any value. If you don't use the default, the other typical value is 50 lines.
/?	Displays help at the command prompt.

Typematic rate

Sets the keyboard typematic rate. The typematic rate is the speed at which Windows can repeat a character when you press the key on a keyboard.

> ⓘ Note
>
> Some keyboards don't recognize this command.

Syntax

```
mode con[:] [rate=<r> delay=<d>]
```

Parameters

Expand table

Parameter	Description
con[:]	Specifies the keyboard. This parameter is required.
rate=`<r>`	Specifies the rate at which a character is repeated on the screen when you hold down a key. The default value is 20 characters per second for IBM AT-compatible keyboards, and 21 for IBM PS/2-compatible keyboards, but you can use any value from 1 through 32. If you set this parameter, you must also set the **delay** parameter.
delay=`<d>`	Specifies the amount of time that will elapse after you press and hold down a key before the character output repeats. The default value is 2 (.50 seconds), but you can also use 1 (.25 seconds), 3 (.75 seconds), or 4 (1 second). If you set this parameter, you must also set the **rate** parameter.
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

more

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays one screen of output at a time.

> **Note**
>
> The **more** command, with different parameters, is also available from the Recovery Console.

Syntax

```
<command> | more [/c] [/p] [/s] [/t<n>] [+<n>]
more [[/c] [/p] [/s] [/t<n>] [+<n>]] < [<drive>:][<path>]<filename>
more [/c] [/p] [/s] [/t<n>] [+<n>] [<files>]
```

Parameters

Expand table

Parameter	Description
`<command>`	Specifies a command for which you want to display the output.
/c	Clears the screen before displaying a page.
/p	Expands form-feed characters.
/s	Displays multiple blank lines as a single blank line.
/t`<n>`	Displays tabs as the number of spaces specified by *n*.
+`<n>`	Displays the first file, beginning at the line specified by *n*.
`[<drive>:][<path>]<filename>`	Specifies the location and name of a file to display.

Parameter	Description
`<files>`	Specifies a list of files to display. Files must be separated using spaces.
/?	Displays help at the command prompt.

Remarks

- The following subcommands are accepted at the **more** prompt (`-- More --`), including:

  Expand table

Key	Action
SPACEBAR	Press the **SPACEBAR** to display the next screen.
ENTER	Press **ENTER** to display the file one line at a time.
f	Press **F** to display the next file listed on the command line.
q	Press **Q** to quit the **more** command.
=	Shows the line number.
p `<n>`	Press **P** to display the next *n* lines.
s `<n>`	Press **S** to skip the next *n* lines.
?	Press **?** to show the commands that are available at the **more** prompt.

- If you use the redirection character (`<`), you must also specify a file name as the source.
- If you use the pipe (`|`), you can use such commands as **dir**, **sort**, and **type**.

Examples

To view the first screen of information of a file named *Clients.new*, type one of the following commands:

```
more < clients.new
type clients.new | more
```

The **more** command displays the first screen of information from Clients.new, and you can press the SPACEBAR to see the next screen of information.

To clear the screen and remove all extra blank lines before displaying the file *Clients.new*, type one of the following commands:

```
more /c /s < clients.new
type clients.new | more /c /s
```

To display the current line number at the **more** prompt, type:

```
more =
```

The current line number is added to the **more** prompt, as `-- More [Line: 24] --`

To display a specific number of lines at the **more** prompt, type:

```
more p
```

The **more** prompt asks you for the number of lines to display, as follows: `-- More -- Lines:`. Type the number of lines to display, and then press ENTER. The screen changes to show only that number of lines.

To skip a specific number of lines at the **more** prompt, type:

```
more s
```

The **more** prompt asks you for the number of lines to skip, as follows: `-- More -- Lines:`. Type the number of lines to skip, and then press ENTER. The screen changes to show that those lines are skipped.

Related links

- Command-Line Syntax Key

- Windows Recovery Environment (WinRE)

Feedback

Was this page helpful? Yes No

mount

Article • 09/28/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

A command-line utility that mounts Network File System (NFS) network shares. When used without options or arguments, **mount** displays information about all mounted NFS file systems.

> **Note**
>
> This utility is available only if **Client for NFS** is installed.

Syntax

```
mount [-o <option>[...]] [-u:<username>] [-p:{<password> | *}] {\\
<computername>\<sharename> | <computername>:/<sharename>} {<devicename> | *}
```

Parameters

Expand table

Parameter	Description
-o rsize=`<buffersize>`	Sets the size in kilobytes of the read buffer. Acceptable values are 1, 2, 4, 8, 16, and 32; the default is 32 KB.
-o wsize=`<buffersize>`	Sets the size in kilobytes of the write buffer. Acceptable values are 1, 2, 4, 8, 16, and 32; the default is 32 KB.
-o timeout=`<seconds>`	Sets the time-out value in seconds for a remote procedure call (RPC). Acceptable values are 0.8, 0.9, and any integer in the range 1-60; the default is 0.8.
-o retry=`<number>`	Sets the number of retries for a soft mount. Acceptable values are integers in the range 1-10; the default is 1.
-o mtype=`{soft|hard}`	Sets the mount type for your NFS share. By default, Windows uses a soft mount. Soft mounts time out more easily when

Parameter	Description
	there are connection issues; however, to reduce I/O disruption during NFS server reboots, we recommend using a hard mount.
-o lang=`{euc-jp|euc-tw|euc-kr|shift-jis|Big5|Ksc5601|Gb2312-80|Ansi}`	Specifies the language encoding to configure on an NFS share. You can use only one language on the share. This value can include any of the following values: • **euc-jp:** Japanese • **euc-tw:** Chinese • **euc-kr:** Korean • **shift-jis:** Japanese • **Big5:** Chinese • **Ksc5601:** Korean • **Gb2312-80:** Simplified Chinese • **Ansi:** ANSI-encoded
-o fileaccess=`<mode>`	Specifies the default permission mode of new files created on the NFS share. Specify *mode* as a three-digit number in the form *ogw*, where *o*, *g*, and *w* are each a digit representing the access granted the file's owner, group, and the world, respectively. The digits must be in the range 0-7, including: • **0:** No access • **1:** x (execute access) • **2:** w (write access) • **3:** wx (write and execute access) • **4:** r (read access) • **5:** rx (read and execute access) • **6:** rw (read and write access) • **7:** rwx (read, write, and execute access)
-o anon	Mounts as an anonymous user.
-o nolock	Disables locking (default is **enabled**).
-o casesensitive	Forces file lookups on the server to be case sensitive.
-o sec=`{sys|krb5|krb5i|krb5p}`	Specifies the security mechanism for mounting an NFS share. This value can include any of the following: • **sec=sys:** No authentication or security checks are performed. Data transfers aren't encrypted. • **sec=krb5:** Specifies Kerberos authentication for mounting the NFS share. • **sec=krb5i:** Extends Kerberos security by adding data integrity checks between the client and server hasn't been tampered. • **sec=krb5p:** Enhances Kerberos security by enabling data privacy, which encrypts data exchanged between

Parameter	Description
	the client and server.
-u:`<username>`	Specifies the user name to use for mounting the share. If *username* isn't preceded by a backslash (\ ), it's treated as a UNIX user name.
-p:`<password>`	The password to use for mounting the share. If you use an asterisk (*), you're prompted for the password.
`<computername>`	Specifies the name of the NFS server.
`<sharename>`	Specifies the name of the file system.
`<devicename>`	Specifies the drive letter and name of the device. If you use an asterisk (*) this value represents the first available driver letter.

Examples

To mount the NFS share anonymously on network drive Z, type:

Windows Command Prompt

```
mount -o anon \\ServerIP\ShareName Z:
```

> **Note**
>
> In order for this parameter to function as intended, the NFS server must allow anonymous access.

To mount the NFS share and disable file locking for network drive Z, type:

Windows Command Prompt

```
mount -o nolock \\ServerIP\ShareName Z:
```

To mount the NFS share with permissions for the *ogw* set as rwx, rw, r, and to continuously reconnect to the NFS server if the connection is lost for network drive Z, type:

Windows Command Prompt

```
mount -o fileaccess=764 mtype=hard \\ServerIP\ShareName Z:
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

mountvol

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates, deletes, or lists a volume mount point. You can also link volumes without requiring a drive letter.

Syntax

```
mountvol [<drive>:]<path volumename>
mountvol [<drive>:]<path> /d
mountvol [<drive>:]<path> /l
mountvol [<drive>:]<path> /p
mountvol /r
mountvol [/n|/e]
mountvol <drive>: /s
```

Parameters

Expand table

Parameter	Description
`[<drive>:]<path>`	Specifies the existing NTFS directory where the mount point will reside.
`<volumename>`	Specifies the volume name that is the target of the mount point. The volume name uses the following syntax, where *GUID* is a globally unique identifier: `\\?\volume\{GUID}\`. The brackets `{ }` are required.
/d	Removes the volume mount point from the specified folder.
/l	Lists the mounted volume name for the specified folder.
/p	Removes the volume mount point from the specified directory, dismounts the basic volume, and takes the basic volume offline, making it unmountable. If other processes are using the volume, **mountvol** closes any open handles before dismounting the volume.
/r	Removes volume mount point directories and registry settings for volumes that are no longer in the system, preventing them from being automatically mounted

Parameter	Description
	and given their former volume mount point(s) when added back to the system.
/n	Disables automatic mounting of new basic volumes. New volumes are not mounted automatically when added to the system.
/e	Re-enables automatic mounting of new basic volumes.
/s	Mounts the EFI system partition on the specified drive.
/?	Displays help at the command prompt.

Remarks

- If you dismount your volume while using the **/p** parameter, the volume list will show the volume as not mounted until a volume mount point is created.
- If your volume has more than one mount point, use **/d** to remove the additional mount points before using **/p**. You can make the basic volume mountable again by assigning a volume mount point.
- If you need to expand your volume space without reformatting or replacing a hard drive, you can add a mount path to another volume. The benefit of using one volume with several mount paths is that you can access all local volumes by using a single drive letter (such as `C:`). You don't need to remember which volume corresponds to which drive letter—although you can still mount local volumes and assign them drive letters.

Examples

To create a mount point, type:

```
mountvol \sysmount \\?\volume\{2eca078d-5cbc-43d3-aff8-7e8511f60d0e}\
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

move

08/15/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Moves one or more files from one directory to another directory.

> **ⓘ Important**
>
> Moving encrypted files to a volume that doesn't support Encrypting File System (EFS) results in an error. You must first decrypt the files or move them to a volume that supports EFS.

Syntax

```
move [{/y|-y}] [<source>] [<target>]
```

Parameters

Expand table

Parameter	Description
/y	Stops prompting for confirmation that you want to overwrite an existing destination file. This parameter might be preset in the COPYCMD environment variable. You can override this preset by using the **/-y** parameter. The default is to prompt before overwriting files, unless the command is run from within a batch script.
/-y	Starts prompting for confirmation that you want to overwrite an existing destination file.
`<source>`	Specifies the path and name of the file or files to move. To move or rename a directory, the *source* should be the current directory path and name.
`<target>`	Specifies the path and name to move files to. To move or rename a directory, the *target* should be the desired directory path and name.
/?	Displays help at the command prompt.

Examples

To move all files with the .xls extension from the *\Data* directory to the *\Second_Q\Reports* directory, type:

```
move \data\*.xls \second_q\reports\
```

Related links

- Command-Line Syntax Key

mqbkup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Backs up MSMQ message files and registry settings to a storage device and restores previously-stored messages and settings.

Both the backup and the restore operations stop the local MSMQ service. If the MSMQ service was started beforehand, the utility will attempt to restart the MSMQ service at the end of the backup or the restore operation. If the service was already stopped before running the utility, no attempt to restart the service is made.

Before using the MSMQ Message Backup/Restore utility you must close all local applications that are using MSMQ.

Syntax

```
mqbkup {/b | /r} <folder path_to_storage_device>
```

Parameters

Expand table

Parameter	Description
/b	Specifies backup operation.
/r	Specifies restore operation.
`<folder path_to_storage_device>`	Specifies the path where the MSMQ message files and registry settings are stored.
/?	Displays help at the command prompt.

Remarks

- If a specified folder doesn't exist while performing either the backup or restore operation, the folder is automatically created by the utility.
- If you choose to specify an existing folder, it must be empty. If you specify a non-empty folder, the utility deletes every file and subfolder contained within it. In this case, you'll be prompted to give permission to delete existing files and subfolders. You can use the **/y** parameter to indicate that you agree beforehand to the deletion of all existing files and subfolders in the specified folder.
- The locations of folders used to store MSMQ message files are stored in the registry. Therefore, the utility restores MSMQ message files to the folders specified in the registry and not to the storage folders used before the restore operation.

Examples

To backup all MSMQ message files and registry settings, and to store them in the *msmqbkup* folder on your C: drive, type:

```
mqbkup /b c:\msmqbkup
```

To delete all existing files and subfolders in the *oldbkup* folder on your C: drive, and then to store MSMQ message files and registry settings in the folder, type:

```
mqbkup /b /y c:\oldbkup
```

To restore MSMQ messages and registry settings, type:

```
mqbkup /r c:\msmqbkup
```

Related links

- Command-Line Syntax Key
- MSMQ Powershell Reference

Feedback

Was this page helpful? Yes No

mqsvc

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Message Queuing technology enables apps running at different times to communicate across heterogeneous networks and systems that may be temporarily offline. Message Queuing provides guaranteed message delivery, efficient routing, security, and priority-based messaging. It can be used to implement solutions for both asynchronous and synchronous messaging scenarios. For more information about Message Queuing, see Message Queuing (MSMQ).

Syntax

```
mqsvc.exe
```

Related links

- Command-Line Syntax Key
- MSMQ Powershell Reference

Feedback

Was this page helpful? Yes No

mqtgsvc

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Monitors a queue for incoming messages and performs an action, in the form of an executable file or COM component, when the rules of a trigger are evaluated as true. For examples of how the Message Queuing Triggers service can be used, see Message Queuing Triggers.

Syntax

```
mqtgsvc.exe
```

Related links

- Command-Line Syntax Key
- MSMQ Powershell Reference

Feedback

Was this page helpful? Yes No

msdt

Article • 02/21/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Invokes a troubleshooting pack at the command line or as part of an automated script, and enables additional options without user input.

> **Note**
>
> The Microsoft Support Diagnostic Tool (MSDT) is deprecated. For more information, see Deprecated Windows features.

Syntax

```
msdt </id <name> | /path <name> | /cab < name>> <</parameter> [options] …
<parameter> [options]>>
```

Parameters

Expand table

Parameter	Description
/id `<packagename>`	Specifies which diagnostic package to run. For a list of available packages, see Available Troubleshooting packs.
/path `<directory|.diagpkg file|.diagcfg file>`	Specifies the full path to a diagnostic package. If you specify a directory, the directory must contain a diagnostic package. You cannot use the **/path** parameter in conjunction with the** /id**, **/dci**, or **/cab** parameters.
/dci `<passkey>`	Prepopulates the passkey field. This parameter is only used when a support provider has supplied a passkey.
/dt `<directory>`	Displays the troubleshooting history in the specified directory. Diagnostic results are stored in the user's **%LOCALAPPDATA%\Diagnostics** or **%LOCALAPPDATA%\ElevatedDiagnostics** directories.

Parameter	Description
/af `<answerfile>`	Specifies an answer file in XML format that contains responses to one or more diagnostic interactions.
/modal `<ownerHWND>`	Makes the troubleshooting pack modal to a window designated by the parent Console Window Handle (HWND), in decimal. This parameter is typically used by applications that launch a troubleshooting pack. For more information about obtaining Console Window Handles, see How to Obtain a Console Window Handle (HWND) .
/moreoptions `<true|false>`	Enables (true) or suppresses (false) the final troubleshooting screen that asks if the user wants to explore additional options. This parameter is typically used when the troubleshooting pack is launched by a troubleshooter that isn't part of the operating system.
/param `<parameters>`	Specifies a set of interaction responses at the command line, similar to an answer file. This parameter isn't typically used within the context of troubleshooting packs created with TSP Designer. For more information about developing custom parameters, see Windows Troubleshooting Platform.
/advanced	Expands the advanced link on the Welcome page by default when the troubleshooting pack is started.
/custom	Prompts the user to confirm each possible resolution before it's applied.

Return codes

Troubleshooting packs comprise a set of root causes, each of which describes a specific technical problem. After completing the troubleshooting pack tasks, each root cause returns a state of fixed, not fixed, detected (but not fixable), or not found. In addition to specific results reported in the troubleshooter user interface, the troubleshooting engine returns a code in the results describing, in general terms, whether or not the troubleshooter fixed the original problem. The codes are:

Expand table

Code	Description
-1	**Interruption:** The troubleshooter was closed before the troubleshooting tasks were completed.
0	**Fixed:** The troubleshooter identified and fixed at least one root cause, and no root causes remain in a not fixed state.
1	**Present, but not fixed:** The troubleshooter identified one or more root causes that remain

Code	Description
	in a not fixed state. This code is returned even if another root cause was fixed.
2	**Not found:** The troubleshooter didn't identify any root causes.

Related links

- Command-Line Syntax Key
- Available troubleshooting packs
- TroubleshootingPack PowerShell reference

Feedback

Was this page helpful? Yes No

msg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sends a message to a user on a Remote Desktop Session Host server.

> **Note**
>
> You must have Message special access permission to send a message.

Syntax

```
msg {<username> | <sessionname> | <sessionID>| @<filename> | *} [/server:<servername>] [/time:<seconds>] [/v] [/w] [<message>]
```

Parameters

Expand table

Parameter	Description
`<username>`	Specifies the name of the user that you want to receive the message. If you don't specify a user or a session, this command displays an error message. When specifying a session, it must be an active one.
`<sessionname>`	Specifies the name of the session that you want to receive the message. If you don't specify a user or a session, this command displays an error message. When specifying a session, it must be an active one.
`<sessionID>`	Specifies the numeric ID of the session whose user you want to receive a message.
`@<filename>`	Identifies a file containing a list of user names, session names, and session IDs that you want to receive the message.
*	Sends the message to all user names on the system.

Parameter	Description
/server:`<servername>`	Specifies the Remote Desktop Session Host server whose session or user you want to receive the message. If unspecified, **/server** uses the server to which you are currently logged on.
/time:`<seconds>`	Specifies the amount of time that the message you sent is displayed on the user's screen. After the time limit is reached, the message disappears. If no time limit is set, the message defaults to **60 seconds** and disappears.
/v	Displays information about the actions being performed.
/w	Waits for an acknowledgment from the user that the message has been received. Use this parameter with `/time:<*seconds*>` to avoid a possible long delay if the user does not immediately respond. Using this parameter with **/v** is also helpful.
`<message>`	Specifies the text of the message that you want to send. If no message is specified, you will be prompted to enter a message. To send a message that is contained in a file, type the less than (<) symbol followed by the file name.
/?	Displays help at the command prompt.

Examples

To send a message entitled, *Let's meet at 1PM today* to all sessions for *User1*, type:

```
msg User1 Let's meet at 1PM today
```

To send the same message to session *modeM02*, type:

```
msg modem02 Let's meet at 1PM today
```

To send the message to all sessions contained in the file *userlist*, type:

```
msg @userlist Let's meet at 1PM today
```

To send the message to all users who are logged on, type:

```
```

```
msg * Let's meet at 1PM today
```

To send the message to all users, with an acknowledgment time-out (for example, 10 seconds), type:

```
msg * /time:10 Let's meet at 1PM today
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

msiexec

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Provides the means to install, modify, and perform operations on Windows Installer from the command line.

Install options

Set the install type for launching an installation package.

Syntax

```
msiexec.exe [/i][/a][/j{u|m|/g|/t}][/x] <path_to_package>
```

Parameters

Expand table

Parameter	Description
/i	Specifies normal installation.
/a	Specifies administrative installation.
/ju	Advertise the product to the current user.
/jm	Advertise the product to all users.
/j/g	Specifies the language identifier used by the advertised package.
/j/t	Applies transform to the advertised package.
/x	Uninstalls the package.
`<path_to_package>`	Specifies the location and name of the installation package file.

Examples

To install a package named *example.msi* from the C: drive, using a normal installation process, type:

```
msiexec.exe /i "C:\example.msi"
```

Display options

You can configure what a user sees during the installation process, based on your target environment. For example, if you're distributing a package to all clients for manual installation, there should be a full UI. However, if you're deploying a package using Group Policy, which requires no user interaction, there should be no UI involved.

Syntax

```
msiexec.exe /i <path_to_package> [/quiet][/passive][/q{n|b|r|f}]
```

Parameters

Expand table

Parameter	Description
`<path_to_package>`	Specifies the location and name of the installation package file.
/quiet	Specifies quiet mode, which means there's no user interaction required.
/passive	Specifies unattended mode, which means the installation only shows a progress bar.
/qn	Specifies there's no UI during the installation process.
/qn+	Specifies there's no UI during the installation process, except for a final dialog box at the end.
/qb	Specifies there's a basic UI during the installation process.
/qb+	Specifies there's a basic UI during the installation process, including a final dialog box at the end.
/qr	Specifies a reduced UI experience during the installation process.

Parameter	Description
/qf	Specifies a full UI experience during the installation process.

Remarks

- The modal box isn't shown if the installation is cancelled by the user. You can use **qb+!** or **qb!+** to hide the **CANCEL** button.

Examples

To install package *C:\example.msi*, using a normal installation process and no UI, type:

```
msiexec.exe /i "C:\example.msi" /qn
```

Restart options

If your installation package overwrites files or attempts to change files that are in use, a reboot might be required before the installation completes.

Syntax

```
msiexec.exe /i <path_to_package> [/norestart][/promptrestart][/forcerestart]
```

Parameters

Expand table

Parameter	Description
`<path_to_package>`	Specifies the location and name of the installation package file.
/norestart	Stops the device from restarting after the installation completes.
/promptrestart	Prompts the user if a reboot is required.
/forcerestart	Restarts the device after the installation completes.

Examples

To install package *C:\example.msi*, using a normal installation process with no reboot at the end, type:

```
msiexec.exe /i "C:\example.msi" /norestart
```

Logging options

If you need to debug your installation package, you can set the parameters to create a log file with specific information.

Syntax

```
msiexec.exe [/i][/x] <path_to_package> [/L{i|w|e|a|r|u|c|m|o|p|v|x+|!|*}]
<path_to_log>
```

Parameters

Expand table

Parameter	Description
/i	Specifies normal installation.
/x	Uninstalls the package.
`<path_to_package>`	Specifies the location and name of the installation package file.
/li	Turns on logging and includes status messages in the output log file.
/lw	Turns on logging and includes non-fatal warnings in the output log file.
/le	Turns on logging and includes all error messages in the output log file.
/la	Turns on logging and includes information about when an action started in the output log file.
/lr	Turns on logging and includes action-specific records in the output log file.

Parameter	Description
/lu	Turns on logging and includes user request information in the output log file.
/lc	Turns on logging and includes the initial UI parameters in the output log file.
/lm	Turns on logging and includes out-of-memory or fatal exit information in the output log file.
/lo	Turns on logging and includes out-of-disk-space messages in the output log file.
/lp	Turns on logging and includes terminal properties in the output log file.
/lv	Turns on logging and includes verbose output in the output log file.
/lx	Turns on logging and includes extra debugging information in the output log file.
/l+	Turns on logging and appends the information to an existing log file.
/l!	Turns on logging and flushes each line to the log file.
/l*	Turns on logging and logs all information, except verbose information (**/lv**) or extra debugging information (**/lx**).
`<path_to_logfile>`	Specifies the location and name for the output log file.

Examples

To install package *C:\example.msi*, using a normal installation process with all logging information provided, including verbose output, and storing the output log file at *C:\package.log*, type:

```
msiexec.exe /i "C:\example.msi" /L*V "C:\package.log"
```

Update options

You can apply or remove updates using an installation package.

Syntax

```
```

```
msiexec.exe [/p][/update][/uninstall[/package<product_code_of_package>]]
<path_to_package>
```

Parameters

Expand table

Parameter	Description
/p	Installs a patch. If you're installing silently, you must also set the REINSTALLMODE property to *ecmus* and REINSTALL to *ALL*. Otherwise, the patch only updates the MSI cached on the target device.
/update	Install patches option. If you're applying multiple updates, you must separate them using a semi-colon (;).
/package	Installs or configures a product.

Examples

```
msiexec.exe /p "C:\MyPatch.msp"
msiexec.exe /p "C:\MyPatch.msp" /qb REINSTALLMODE="ecmus" REINSTALL="ALL"
msiexec.exe /update "C:\MyPatch.msp"
```

```
msiexec.exe /uninstall {1BCBF52C-CD1B-454D-AEF7-852F73967318} /package
{AAD3D77A-7476-469F-ADF4-04424124E91D}
```

Where the first GUID is the patch GUID, and the second one is the MSI product code to which the patch was applied.

Repair options

You can use this command to repair an installed package.

Syntax

```
```

```
msiexec.exe [/f{p|o|e|d|c|a|u|m|s|v}] <product_code>
```

Parameters

Expand table

Parameter	Description
/fp	Repairs the package if a file is missing.
/fo	Repairs the package if a file is missing, or if an older version is installed.
/fe	Repairs the package if file is missing, or if an equal or older version is installed.
/fd	Repairs the package if file is missing, or if a different version is installed.
/fc	Repairs the package if file is missing, or if checksum does not match the calculated value.
/fa	Forces all files to be reinstalled.
/fu	Repairs all the required user-specific registry entries.
/fm	Repairs all the required computer-specific registry entries.
/fs	Repairs all existing shortcuts.
/fv	Runs from source and re-caches the local package.

Examples

To force all files to be reinstalled based on the MSI product code to be repaired, *{AAD3D77A-7476-469F-ADF4-04424124E91D}*, type:

```
msiexec.exe /fa {AAD3D77A-7476-469F-ADF4-04424124E91D}
```

Set public properties

You can set public properties through this command. For information about the available properties and how to set them, see Public Properties.

Related links

- Command-Line Syntax Key
- Msiexec.exe Command-Line Options
- Standard Installer Command-Line Options

Feedback

Was this page helpful? Yes No

msinfo32

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Opens the System Information tool to display a comprehensive view of the hardware, system components, and software environment on the local computer.

Some System Information categories contain large amounts of data. You can use the **start /wait** command to optimize reporting performance for these categories. For more information, see System Information.

Syntax

```
msinfo32 [/pch] [/nfo <path>] [/report <path>] [/computer <computername>]
[/showcategories] [/category <categoryID>] [/categories {+<categoryID>(+
<categoryID>)|+all(-<categoryID>)}]
```

Parameters

Expand table

Parameter	Description
`<path>`	Specifies the file to be opened in the format *C:\Folder1\File1.xxx*, where *C* is the drive letter, *Folder1* is the folder, *File1* is the file name, and *xxx* is the file name extension. This file can be an **.nfo**, **.xml**, **.txt**, or **.cab** file.
`<computername>`	Specifies the name of the target or local computer. This can be a UNC name, an IP address, or a full computer name.
`<categoryID>`	Specifies the ID of the category item. You can obtain the category ID by using **/showcategories**.
/pch	Displays the System History view in the System Information tool.
/nfo	Saves the exported file as an **.nfo** file. If the file name that is specified in *path* does not end in an **.nfo** extension, the **.nfo** extension is automatically appended to the file name.

Parameter	Description
/report	Saves the file in *path* as a text file. The file name is saved exactly as it appears in *path*. The .txt extension is not appended to the file unless it is specified in path.
/computer	Starts the System Information tool for the specified remote computer. You must have the appropriate permissions to access the remote computer.
/showcategories	Starts the System Information tool with all available category IDs displayed, rather than displaying the friendly or localized names. For example, the Software Environment category is displayed as the **SWEnv** category.
/category	Starts System Information with the specified category selected. Use **/showcategories** to display a list of available category IDs.
/categories	Starts System Information with only the specified category or categories displayed. It also limits the output to the selected category or categories. Use **/showcategories** to display a list of available category IDs.
/?	Displays help at the command prompt.

Examples

To list the available category IDs, type:

```
msinfo32 /showcategories
```

To start the System Information tool with all available information displayed, except Loaded Modules, type:

```
msinfo32 /categories +all -loadedmodules
```

To display **System Summary** information and to create an .nfo file called *syssum.nfo*, which contains information in the **System Summary** category, type:

```
msinfo32 /nfo syssum.nfo /categories +systemsummary
```

To display resource conflict information and to create an .nfo file called *conflicts.nfo*, which contains information about resource conflicts, type:

```
msinfo32 /nfo conflicts.nfo /categories
+componentsproblemdevices+resourcesconflicts+resourcesforcedhardware
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

mstsc

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Creates connections to Remote Desktop Session Host servers or other remote computers and edits an existing Remote Desktop Connection (.rdp) configuration file.

Syntax

```
mstsc.exe [<connectionfile>] [/v:<server>[:<port>]] [/g:<gateway>] [/admin]
[/f[ullscreen]] [/w:<width> /h:<height>] [/public] [/span] [/multimon] [/l]
[/restrictedAdmin] [/remoteGuard] [/prompt] [/shadow:<sessionID> [/control]
[/noConsentPrompt]]
mstsc.exe /edit <connectionfile>
```

Parameters

Expand table

Parameter	Description
`<connectionfile>`	Specifies the name of an **.rdp** file for the connection.
/v:`<server>[:<port>]`	Specifies the remote computer and, optionally, the port number to which you want to connect.
/g:`<gateway>`	Specifies the RD Gateway server to use for the connection. The command only reads this parameter when you specify the endpoint PC with `/v`.
/admin	Connects you to a session for administering the server.
/f[ullscreen]	Starts Remote Desktop Connection in full-screen mode. `/f` and `/fullscreen` are equivalent.
/w:`<width>`	Specifies the width of the Remote Desktop window.
/h:`<height>`	Specifies the height of the Remote Desktop window.
/public	Runs Remote Desktop in public mode. In public mode, Remote Desktop doesn't cache passwords and bitmaps.

Parameter	Description
/span	Matches the Remote Desktop width and height to the local virtual desktop and spans the Remote Desktop session across multiple monitors. The monitors must have the same resolution and be aligned horizontally (side by side). This parameter differs from `/multimon`, which mirrors the client-side monitor layout instead of spanning a single desktop.
/multimon	Configures the Remote Desktop Services session monitor layout to be identical to the current client-side configuration.
/l	Enumerates the monitors attached to the local PC and the ID associated with each monitor. You can use the monitor IDs to populate the `selectedmonitors` RDP file setting.
/edit `<connectionfile>`	Opens the specified **.rdp** file for editing.
/restrictedAdmin	This mode won't send your credentials to the remote PC, which can protect you if you connect to a compromised device. Other PCs might not authenticate connections from the remote PC, which affects application functionality and compatibility. This mode implies the `/admin` parameter.
/remoteGuard	This mode prevents the command from sending your credentials to the remote PC, which can help protect your credentials if you connect to a compromised device. Unlike Restricted Administrator mode, Remote Guard also supports connections from the remote PC by redirecting all requests back to your device.
/prompt	Prompts you for your credentials when you connect to the remote PC.
/shadow: `<sessionID>`	Specifies the ID of the session to shadow.
/control	Allows control of the session when shadowing.
/noConsentPrompt	Allows shadowing without user consent.
/?	Shows the Remote Desktop Connection Usage dialog box.

Remarks

- Each user has a hidden Default.rdp file in the **Documents** folder.
- By default, Windows saves user-created **.rdp** files in the user's **Documents** folder, but you can save them anywhere.
- To span across monitors, the monitors must use the same resolution and must align horizontally (that is, side-by-side). There's currently no support for spanning multiple monitors vertically on the client system.

Examples

To connect to a session in full-screen mode, type:

```
mstsc /f
```

Use the equivalent `/fullscreen` parameter:

```
mstsc /v:computer1 /fullscreen
```

To span a session across horizontally aligned monitors of the same resolution, type:

```
mstsc /v:computer1 /span
```

To assign width/height, type:

```
mstsc /v:computer1 /w:1920 /h:1080
```

To open a file called `filename.rdp` for editing, type:

```
mstsc /edit filename.rdp
```

Related links

- Command-Line Syntax Key

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 09/08/2026

nbtstat

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays NetBIOS over TCP/IP (NetBT) protocol statistics, NetBIOS name tables for both the local computer and remote computers, and the NetBIOS name cache. This command also allows a refresh of the NetBIOS name cache and the names registered with Windows Internet Name Service (WINS). Used without parameters, this command displays Help information.

This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network Connections.

Syntax

```
nbtstat [/a <remotename>] [/A <IPaddress>] [/c] [/n] [/r] [/R] [/RR] [/s]
[/S] [<interval>]
```

Parameters

Expand table

Parameter	Description
/a `<remotename>`	Displays the NetBIOS name table of a remote computer, where *remotename* is the NetBIOS computer name of the remote computer. The NetBIOS name table is the list of NetBIOS names that corresponds to NetBIOS applications running on that computer.
/A `<IPaddress>`	Displays the NetBIOS name table of a remote computer, specified by the IP address (in dotted decimal notation) of the remote computer.
/c	Displays the contents of the NetBIOS name cache, the table of NetBIOS names and their resolved IP addresses.
/n	Displays the NetBIOS name table of the local computer. The status of **registered** indicates that the name is registered either by broadcast or with a WINS server.
/r	Displays NetBIOS name resolution statistics.

Parameter	Description
/R	Purges the contents of the NetBIOS name cache and then reloads the pre-tagged entries from the **Lmhosts** file.
/RR	Releases and then refreshes NetBIOS names for the local computer that is registered with WINS servers.
/s	Displays NetBIOS client and server sessions, attempting to convert the destination IP address to a name.
/S	Displays NetBIOS client and server sessions, listing the remote computers by destination IP address only.
`<interval>`	Displays selected statistics, pausing the number of seconds specified in *interval* between each display. Press CTRL+C to stop displaying statistics. If this parameter is omitted, **nbtstat** prints the current configuration information only once.
/?	Displays help at the command prompt.

Remarks

- The **nbtstat** command-line parameters are case-sensitive.
- The column headings generated by the **nbtstat** command, include:

Expand table

Heading	Description
Input	The number of bytes received.
Output	The number of bytes sent.
In/Out	Whether the connection is from the computer (outbound) or from another computer to the local computer (inbound).
Life	The remaining time that a name table cache entry will live before it is purged.
Local Name	The local NetBIOS name associated with the connection.
Remote Host	The name or IP address associated with the remote computer.
`<03>`	The last byte of a NetBIOS name converted to hexadecimal. Each NetBIOS name is 16 characters long. This last byte often has special significance because

Heading	Description
	the same name might be present several times on a computer, differing only in the last byte. For example, `<20>` is a space in ASCII text.
type	The type of name. A name can either be a unique name or a group name.
Status	Whether the NetBIOS service on the remote computer is running (registered) or a duplicate computer name has registered the same service (Conflict).
State	The state of NetBIOS connections.

- The possible NetBIOS connection states, include:

Expand table

State	Description
Connected	A session has been established.
listening	This endpoint is available for an inbound connection.
Idle	This endpoint has been opened but cannot receive connections.
Connecting	A session is in the connecting phase and the name-to-IP address mapping of the destination is being resolved.
Accepting	An inbound session is currently being accepted and will be connected shortly.
Reconnecting	A session is trying to reconnect (it failed to connect on the first attempt).
Outbound	A session is in the connecting phase and the TCP connection is currently being created.
Inbound	An inbound session is in the connecting phase.
Disconnecting	A session is in the process of disconnecting.
Disconnected	The local computer has issued a disconnect and it is waiting for confirmation from the remote system.

Examples

To display the NetBIOS name table of the remote computer with the NetBIOS computer name of *CORP07*, type:

```
```

```
nbtstat /a CORP07
```

To display the NetBIOS name table of the remote computer assigned the IP address of *10.0.0.99*, type:

```
nbtstat /A 10.0.0.99
```

To display the NetBIOS name table of the local computer, type:

```
nbtstat /n
```

To display the contents of the local computer NetBIOS name cache, type:

```
nbtstat /c
```

To purge the NetBIOS name cache and reload the pre-tagged entries in the local *Lmhosts* file, type:

```
nbtstat /R
```

To release the NetBIOS names registered with the WINS server and re-register them, type:

```
nbtstat /RR
```

To display NetBIOS session statistics by IP address every five seconds, type:

```
nbtstat /S 5
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

net print

Article • 03/21/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

> **Important**
>
> This command has been deprecated. However, you can perform many of the same tasks using the prnjobs command, Windows Management Instrumentation (WMI), PrintManagement in Powershell.

Displays information about a specified printer queue or a specified print job, or controls a specified print job.

Syntax

```
net print {\\<computername>\<sharename> | \\<computername> <jobnumber>
[/hold | /release | /delete]} [help]
```

Parameters

Expand table

Parameters	Description
`\\<computername>\<sharename>`	Specifies (by name) the computer and print queue about which you want to display information.
`\\<computername>`	Specifies (by name) the computer that hosts the print job you want to control. If you do not specify a computer, the local computer is assumed. Requires the `<jobnumber>` parameter.
`<jobnumber>`	Specifies the number of the print job you want to control. This number is assigned by the computer that hosts the print queue where the print job is sent. After a computer assigns a number to a print job, that number is not assigned to any other print jobs in any queue hosted by that computer. Required when using the `\\<computername>` parameter.

Parameters	Description
`[/hold | /release | /delete]`	Specifies the action to take with the print job. If you specify a job number, but don't specify any action, information about the print job is displayed. • **/hold** - Delays the job, allowing other print jobs to bypass it until it is released. • **/release** - Releases a print job that has been delayed. • **/delete** - Removes a print job from a print queue.
help	Displays help at the command prompt.

Remarks

- The `net print\\<computername>` command displays information about print jobs in a shared printer queue. The following is an example of a report for all print jobs in a queue for a shared printer named *LASER*:

```
printers at \\PRODUCTION
Name              Job #      Size      Status
------------------------------
LASER Queue       3 jobs               *printer active*
USER1          84        93844      printing
USER2          85        12555      Waiting
USER3          86        10222      Waiting
```

- The following is an example of a report for a print job:

```
Job #            35
Status           Waiting
Size             3096
remark
Submitting user  USER2
Notify           USER2
Job data type
Job parameters
additional info
```

Examples

To list the contents of the *Dotmatrix* print queue on the *Production* computer, type:

```
net print \\Production\Dotmatrix
```

To display information about job number *35* on the *Production* computer, type:

```
net print \\Production 35
```

To delay job number *263* on the *Production* computer, type:

```
net print \\Production 263 /hold
```

To release job number *263* on the *Production* computer, type:

```
net print \\Production 263 /release
```

Related links

- Command-Line Syntax Key
- print command reference
- prnjobs command
- Windows Management Instrumentation (WMI)
- PrintManagement in Powershell

Feedback

Was this page helpful? Yes No

net user

08/06/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `net user` command allows you to add, modify, or delete user accounts, and display detailed information about user accounts on a local computer or domain.

Syntax

```
net user [<UserName> {<Password> | *} [<Options>]] [/domain]
net user [<UserName> {<Password> | *} /add [<Options>] [/domain]]
net user [<UserName> [/delete] [/domain]]
net user username [/times:{times | all}]
net user username [/active: {yes | no}]
```

Parameters

Expand table

Parameter	Description
`<UserName>`	Specifies the name of the user account to add, delete, modify, or view. The name of the user account can have as many as 20 characters.
`<Password>`	Assigns or changes a password for the user's account. Type an asterisk (`*`) to produce a prompt for the password. The password isn't displayed when the user types it at the password prompt.
`<Options>`	Specifies a command-line option. Refer to the next table for descriptions of the command-line option syntax.
`/domain`	Performs the operation on the domain controller in the computer's primary domain.
`/add`	Adds a new user account. When combined with `/domain`, it specifies that the account should be created on the domain instead of the local machine.
`/delete`	Deletes the specified user account. When combined with `/domain`, it deletes the account from the domain rather than the local machine.

Parameter	Description
`/times:{<Day>[-<Day>][,<Day>[-<Day>]],<Time>[-<Time>][,<Time>[-<Time>]][;] | all}`	Specifies the days and times when users are permitted to sign into the computer. `<Time>` must be in one-hour increments. `<Day>` can be spelled out or abbreviated (`M, T, W, Th, F, Sa, Su`). Hours can be specified in either 12-hour (with AM/PM or A.M./P.M.) or 24-hour format. Use **all** to allow sign in at any time, or leave blank to prevent sign in entirely. Separate days and times with commas, and separate multiple day/time entries with semicolons (for example: **M,4AM-5PM;T,1PM-3PM**). Don't include spaces in the time designations.
`/active:{yes | no}`	Enables or disables the user account. If the user account isn't active, the user can't access resources on the computer. The default is **yes**.
?	Displays the command help information.

Expand table

Options parameter syntax	Description
`/comment:"<Text>"`	Provides a descriptive comment about the user's account. This comment can have as many as 48 characters. The text must be enclosed in double quotes.
`/countrycode:<NNN>`	Uses the operating system Country/Region codes to implement the specified language files for a user's Help and error messages. A value of 0 signifies the **default** Country/Region code.
`/expires:{{<MM/DD/YYYY> | <DD/MM/YYYY> | <mmm,dd,YYYY>} | never}`	Sets the expiration period for the specified user account. Expiration dates can be in [MM/DD/YYYY], [DD/MM/YYYY], or [mmm,dd,YYYY] formats, depending on the Country/Region code. `never` specifies that a user account doesn't have an expiration date. When you set an account to never expire, it means the account remains active indefinitely, unless manually disabled or deleted by an administrator. Accounts are set to never expire by default. The account expires at the beginning of the specified date. For the *month* value, you can use numbers (`1` for January), spell it out (`January`), or use a three-letter abbreviation (`Jan`, `Feb`, etc.). You can use two or four numbers for the year value. Use commas or slashes to separate parts of the date. Don't use spaces. If you omit `<YYYY>`, the next occurrence of the date is assumed. For example, the following entries are equivalent if entered between Jan. 10,

Options parameter syntax	Description
	2018, and Jan. 8, 2019: **Jan,9** **1/9/18** **January,9,2019** **1/9**
`/fullname:"<Name>"`	Specifies a user's full name rather than a user name. Enclose the name in quotation marks.
`/homedir:<Path>`	Sets the path for the user's home directory. The path must exist.
`/passwordchg:{yes | no}`	Specifies whether users can change their own password. The default is **yes**.
`/passwordreq:{yes | no}`	Specifies whether a user account must have a password. The default is **yes**.
`/profilepath:[<Path>]`	Sets a path for the user's sign in profile. This path points to a registry profile.
`/scriptpath:<Path>`	Specifies the relative path to the user's sign-in script. The path must not be absolute; use a relative path such as `\\domain\SysVol\domain\scripts`.
`/usercomment:"<Text>"`	Specifies that an administrator can add or change the "User comment" for the account. The text must be enclosed in double quotes.
`/workstations:{<ComputerName>[,...] | *}`	Lists as many as eight (8) workstations from which a user can sign into the network. Separate multiple entries in the list with commas. If `/workstations` has no list or if the list is an asterisk (`*`), users can sign in from any computer.

> ⓘ **Note**
>
> A password must satisfy the minimum password length value that is set with the **net accounts /minpwlen** command. A password can have as many as 127 characters.

Examples

To display a list of all user accounts for the local computer, run the following command:

Windows Command Prompt

```
net user
```

To display information about the user account **TommyH**, run the following command:

Windows Command Prompt

```
net user tommyh
```

To add a user account for **Jay Samson** (user name: **jays**), with sign in rights from 8 AM to 5 PM, Monday through Friday, with the mandatory password **Cyk4^g3B**, and use the user's full name, run the following command:

Windows Command Prompt

```
net user jays Cyk4^g3B /add /passwordreq:yes /times:monday-friday,8am-5pm
/fullname:"Jay Jamison"
```

To set the sign-in time from 8 AM to 5 PM for the username **MikeD** using 24-hour notation, run the following command:

Windows Command Prompt

```
net user miked /time:M-F,08:00-17:00
```

To set the sign-in time from 8 AM to 5 PM for **MikeD** using 12-hour notation, run the following command:

Windows Command Prompt

```
net user miked /time:M-F,8AM-5PM
```

To specify sign-in hours of 4 AM to 12 PM on Monday, 12 PM to 8 PM on Tuesday, and 8 AM to 5 PM Wednesday through Friday for username **AnibalS**, run the following command:

Windows Command Prompt

```
net user anibals /time:M,4AM-12PM;T,12PM-8PM;W-F,8AM-5PM
```

See also

Command-Line Syntax Key

netcfg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Installs the Windows Preinstallation Environment (WinPE), a lightweight version of Windows used to deploy workstations.

Syntax

```
netcfg [/v] [/e] [/winpe] [/l ] /c /i
```

Parameters

Expand table

Parameter	Description
/v	Runs in verbose (detailed) mode.
/e	Uses servicing environment variables during install and uninstall.
/winpe	Installs TCP/IP, NetBIOS, and Microsoft Client for Windows preinstallation environment (WinPE).
/l	Provides the location of the INF file.
/c	Provides the class of the component to be installed; **protocol**, **service**, or **client**.
/i `<comp-ID>`	Provides the component ID.
/s	Provides the type of components to show, including **\ta** for adapters or **n** for net components.
/b	Displays the binding paths, when followed by a string containing the name of the path.
/q `<comp-ID>`	Queries whether component ID is installed

Parameter	Description
/u `<comp-ID>`	Uninstalls the component ID.
/m	Outputs the binding map to NetworkBindingMap.txt in the current directory. Using with /v will also display the binding map to the console.
/d	Performs a cleanup on all networking devices. This will require a reboot.
/x	Performs a cleanup on networking devices, skipping those without physical object names. This will require a reboot.
/?	Displays help at the command prompt.

Examples

To install the protocol *example* using c:\oemdir\example.inf, type:

```
netcfg /l c:\oemdir\example.inf /c p /i example
```

To install the *MS_Server* service, type:

```
netcfg /c s /i MS_Server
```

To install TCP/IP, NetBIOS and Microsoft Client for Windows preinstallation environment, type:

```
netcfg /v /winpe
```

To display if component *MS_IPX* is installed, type:

```
netcfg /q MS_IPX
```

To uninstall component *MS_IPX*, type:

```
netcfg /u MS_IPX
```

To show all installed net components, type:

```
netcfg /s n
```

To display binding paths containing *MS_TCPIP*, type:

```
netcfg /b ms_tcpip
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

netdom

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

`Netdom` is a command-line tool that enables administrators to manage Active Directory (AD) domains and trust relationships from the command line. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). To learn more, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

You must run the `netdom` command from an elevated command prompt. The `netdom` allows you to performing the following actions:

- Add a device to a domain:
  - This process involves creating a computer account in the domain's directory. The computer and the domain exchange security credentials, allowing the domain to authenticate and authorize the computer.
  - You can also remove query, and move an existing computer account for a member workstation from one domain to another while maintaining the security descriptor on the computer account.
- Manage computer accounts for domain member workstations and member servers:
  - Computer accounts in a domain are similar to user accounts but represent computers. Managing these accounts involves making sure they're correctly configured and secure. It includes activities such as renaming the computer within the domain or updating its credentials to maintain its ability to communicate securely with the domain controllers.
- Establish one-way or two-way trust relationships between domains:
  - A one-way trust means that one domain trusts the users from another domain but not vice versa.
  - A two-way trust allows mutual access.
  - Trusts can be transitive, meaning they extend beyond two domains, or non-transitive, limited to the direct trust link. This is crucial for organizations with multiple domain environments to manage resources and user access efficiently.
- Verifies or resets the secure channel:

- A secure channel is a communication link established between a computer and a domain controller for exchanging authentication information. This channel needs to remain secure to protect sensitive data.
- If the secure channel becomes disrupted, a verification checks its status. If issues are detected, resetting the channel re-establishes this secure communication link by updating credentials and ensuring the computer can authenticate with the domain controller again.

- Manage trust relationships between domains:
  - Managing trust relationships involves routinely verifying that these relationships are still valid and functioning as intended. It also includes modifying or removing them per organization needs.
  - Proper management ensures that resources remain secure while accessible and helps to maintain an efficient cross-domain interaction.

Syntax

```
netdom [ ADD | COMPUTERNAME | HELP | JOIN | MOVE | MOVENT4BDC | QUERY | REMOVE |
RENAMECOMPUTER | RESET | RESETPWD | TRUST | VERIFY ]
```

Commands

Expand table

Command	Description
netdom add	Adds a workstation or server account to the domain.
netdom computername	Manages the primary and alternate names for a computer. This command can safely rename Active Directory domain controllers and member servers.
netdom join	Joins a workstation or member server to a domain. The act of joining a computer to a domain creates an account for the computer on the domain, if it doesn't already exist.
netdom move	Moves a workstation or member server to a new domain. The act of moving a computer to a new domain creates an account for the computer on the domain, if it doesn't already exist.

Command	Description
netdom movent4bdc	Renames a Windows NT 4.0 backup domain controller to reflect a domain name change. This can help Windows NT 4.0 domain renaming efforts.
netdom query	Queries the domain for information such as membership and trust.
netdom remove	Removes a workstation or server from the domain.
netdom renamecomputer	Renames a domain computer and its corresponding domain account. Use this command to rename domain workstations and member servers only. To rename domain controllers, use the `netdom computername` command.
netdom reset	Resets the secure connection between a workstation and a domain controller.
netdom resetpwd	Resets the computer account password for a domain controller.
netdom trust	Establishes, verifies, or resets a trust relationship between domains.
netdom verify	Verifies the secure connection between a workstation and a domain controller.

Remarks

- A trust relationship is a defined affiliation between domains that enables pass-through authentication.
- A one-way trust relationship between two domains means that one domain (the trusting domain) allows users who have accounts on the other domain (the trusted domain), access to its resources.
- The one-way trust relationship described is helpful in primary domain models, but it isn't the only type of trust relationship. When two one-way trusts are established between domains, it's known as a two-way trust. In two-way trusts, each domain treats the users from the trusted (and trusting) domain as its own users.
- If you specify the `/verbose` parameter, the output lists the success or failure of each transaction that is necessary to perform the operation.
- The `/reboot` parameter specifies that the computer specified by this operation is automatically rebooted after the completion of the operation.
- The default delay before the computer restarts is 20 seconds.

See also

Command-Line Syntax Key

netdom add

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom add` command adds a workstation or server account to a domain. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). To use `netdom add`, you must run the command from an elevated command prompt.

Syntax

```
netdom add machine [/Domain:domain] [/UserD:user] [/PasswordD:[password | *]]
                   [/Server:server] [/OU:ou path] [/DC] [/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer that you want to add.
`/domain:<Domain>`	Specifies the domain in which to create the account. If you don't specify this parameter, then `netdom add` uses the domain that the current computer belongs to.
`/userd:<User>`	Specifies the user account that makes the connection with the domain that you specify using the `/domain` parameter. If you don't specify this parameter, `netdom add` uses the current user account.
`/passwordd:<Password>` | `*`	Specifies the password of the user account that you specify using the `/userd` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.
`/server:<Server>`	Specifies the name of a domain controller that performs the *add* operation. This parameter can't be used with the `/ou` parameter.
`/ou:<OUPath>`	Specifies the organizational unit (OU) under which to create the account. You must use the full RFC 1779 distinguished name of the OU. If you don't

Parameter	Description
	specify this parameter, `netdom add` creates the account under the default OU for computer objects for that domain.
`/dc`	Specifies that a domain controller's machine account is to be created. This makes it possible for the computer accounts for new AD DS domain controllers. This parameter can't be used with the **/ou** parameter.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`help` | `/?`	Displays help at the command prompt.

Examples

To add the workstation **mywksta** to the domain **reskita** using the credentials of a domain administrator, run the following command:

Windows Command Prompt

```
netdom add mywksta /domain:reskita /userd:domainadmin /passwordd:*
```

To add a computer named **Srv01** to the domain **reskita.com** and specify the organizational unit (OU) where the computer account should be placed, run the following command:

Windows Command Prompt

```
netdom add Srv01 /domain:reskita /OU:"OU=Servers,DC=reskita,DC=com"
/userd:domainadmin /passwordd:*
```

> **Note**
>
> If you don't specify the `/ou` parameter, `netdom add` creates the account in the **Computers** container.

See also

Command-Line Syntax Key

netdom computername

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom computername` command manages the primary and alternate names for a computer. This command can safely rename Active Directory (AD) domain controllers (DC) and member servers. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). To use `netdom computername`, you must run the command from an elevated command prompt.

Syntax

```
netdom computername machine [/UserO:user] [/PasswordO:[password | *]]
                            [/UserD:user] [/PasswordD:[password | *]]
[/SecurePasswordPrompt]
                            /Add:<new-alternate-DNS-name> | /Remove:<alternate-
DNS-name>
                            /MakePrimary:<computer-dns-name>
                            /Enumerate[:{AlternateNames | PrimaryName | AllNames}]
                            /Verify
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer that you want to manage.
`/usero:<UserName>`	Specifies the user account that you want to use for the originating domain.
`/passwordo:<Password>` | `*`	Specifies the password that you want to use for the originating domain. If you specify `*`, you're prompted for the password.
`/userd:<UserName>`	Specifies the user account that you want to use for the destination domain.
`/passwordd:<Password>` | `*`	Specifies the password that you want to use for the destination

Parameter	Description
	domain. If you specify `*`, you're prompted for the password.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`/add:<NewAltDNSName>`	Creates a new alternate name. Specify a fully qualified domain name (FQDN), such as `comp1.example.com`.
`/remove:<AltDNSName>`	Deletes an existing alternate name. Specify an FQDN, such as `comp1.example.com`.
`/makeprimary:<ComputerDNSName>`	Makes an existing alternate name into the primary name. Specify an FQDN, such as `comp1.example.com`.
`/enumerate:ALTERNATENAMES` | `PRIMARYNAME` | `ALLNAMES`	Lists the primary name or any alternate names. Values: - `ALTERNATENAMES`: Lists alternate names only. - `PRIMARYNAME`: Lists the primary name only. - `ALLNAMES`: Lists the primary and any alternate names (default).
`/verify`	Checks if there's a DNS record and a service principal name (SPN) for each computer name.
`help` | `/?`	Displays help at the command prompt.

Examples

To give an alternate name for the DC **MyDC** in the **mydomain.com** domain, run the following command:

Windows Command Prompt

```
netdom computername mydc.mydomain.com /add:altDCname.mydomain.com
```

Before you can make an alternate name the primary name of a computer, the name must exist as an alternate name.

Once the alternate name is added, you can promote it to be the primary name. To rename the domain controller **MyDC** to **altDCname** in the **mydomain.com** domain, use the following command:

Windows Command Prompt

```
netdom computername mydc.mydomain.com /makeprimary:altdcname.mydomain.com
```

When you rename a member server, the process is similar. Use an existing alternate name as the new primary name.

See also

Command-Line Syntax Key

netdom join

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom join` command joins a workstation or member server to a domain. The act of joining a computer to a domain creates an account for the computer on the domain, if it doesn't already exist. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the Active Directory Domain Services Tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom join`, you must run the command from an elevated command prompt.

Syntax

```
netdom join machine /Domain:domain [/OU:ou path] [/UserD:user]
                    [/PasswordD:[password | *]]
                    [/UserO:user] [/PasswordO:[password | *]]
                    [/PasswordM:[password | *]]
                    [/ReadOnly]
                    [/REBoot[:Time in seconds]]
                    [/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer that you want to join to the domain.
`/domain:<Domain>`	Specifies the domain that you want to join the computer to. Using the Fully Qualified Domain Name (FQDN) format is recommended for clarity. If not specified, `netdom join` defaults to the domain to which the current computer belongs to.
`/ou:<OUPath>`	Specifies the organizational unit (OU) under which you want to create the account. You must specify the full RFC 1779 distinguished name of the OU. If not specified, `netdom join` creates the account under the default OU for computer objects for that domain.

Parameter	Description
`/userd:<Domain>\<User>`	Specifies the user account that makes the connection with the domain specified with the `/domain` parameter. If not specified, `netdom join` uses the current user account.
`/usero:<User>`	Specifies the user account that makes the connection with the computer you want to join to the domain. If not specified, `netdom join` uses the current user account.
`/passwordd:<Password>` | `*`	Specifies the password of the user account specified with the `/userd` parameter. If you specify `*`, you're prompted for the password.
`/passwordo:<Password>` | `*`	Specifies the password of the user account specified with the `/usero` parameter. If you specify `*`, you're prompted for the password.
`/passwordm:<Password>` | `*`	Specifies the password for the pre-created computer account identified by the machine name. If you specify `*` instead of a password, you're prompted for the password. This option must be used with the `/ReadOnly` parameter.
`/readonly`	Joins the domain using a pre-created computer account without writing to a domain controller. This doesn't require a writable domain controller. It must be used with the `/Domain` and `/PasswordM` parameters, and can't be used with the `/OU` parameter.
`/reboot:<seconds>`	Shuts down the computer and automatically reboots after the join operation completes. The *seconds* value is the number of seconds before automatic shutdown. The default is **20** seconds.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`help` | `/?`	Displays help at the command prompt.

Examples

To join the computer **mywksta** to the **devgroup.contoso.com** domain in the **Dsys/workstations** OU, run the following command:

Windows Command Prompt

```
netdom join mywksta /domain:devgroup.contoso.com
/OU:"OU=Workstations,OU=Dsys,DC=devgroup,DC=contoso,DC=com"
```

Besides adding the computer account to the domain, this command modifies the workstation to contain the appropriate shared secret to complete the *join* operation.

To join the computer **mywksta** to the domain **contoso.com**, run the following command:

Windows Command Prompt

```
netdom join mywksta /domain:contoso.com /userd:user01 /passwordd:*
```

This command prompts you to enter the password for the domain user "user01" securely.

See also

Command-Line Syntax Key

netdom move

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom move` command moves a workstation or member server to a new domain. The act of moving a computer to a new domain creates an account for the computer on the domain, if it doesn't already exist. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). To use `netdom move`, you must run the command from an elevated command prompt.

Syntax

```
netdom move machine /Domain:domain [/OU:ou path]
                    [/UserD:user] [/PasswordD:[password | *]]
                    [/UserO:user] [/PasswordO:[password | *]]
                    [/UserF:user] [/PasswordF:[password | *]]
                    [/Reboot[:Time in seconds]]
                    [/SecurePasswordPrompt]
```

Parameters

> **Note**
>
> When you move a computer to a new domain, `netdom move` doesn't delete its computer account in the former domain. However, if you supply credentials for the former domain, this command disables the old computer account.

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer that you want to move.
`/domain:<Domain>`	Specifies the domain to which you want to move the account. If you don't specify the parameter, then `netdom move` uses the domain to which the current computer belongs.

Parameter	Description
`/ou:<OUPath>`	Specifies the organizational unit (OU) under which to create the account. This must be the full RFC 1779 distinguished name of the OU. If you don't specify this parameter, `netdom move` creates the account under the default OU for computer objects for that domain.
`/userd:<User>`	Specifies the user account that makes the connection with the domain that you specify using the `/domain` parameter. If you don't specify this parameter, `netdom move` uses the current user account.
`/passwordd:<Password>` | `*`	Specifies the password of the user account that you specify using the `/userd` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.
`/usero:<User>`	Specifies the user account to make the connection with the computer that you want to move. If you don't specify this parameter, `netdom move` uses the current user account.
`/passwordo:<Password>` | `*`	Specifies the password of the user account that you specify using the `/usero` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.
`/userf:<User>`	Specifies the user account to make the connection with the computer's former domain (of which the computer was a member before the move). This parameter is used to disable the old computer account.
`/passwordf:<Password>` | `*`	Specifies the password of the user account that you specify using the `/userf` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.
`/reboot:<seconds>`	Shuts down the computer and automatically reboots after the move operation completes. The *seconds* value is the number of seconds before automatic shutdown. The default is **20** seconds.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`help` | `/?`	Displays help at the command prompt.

Examples

To move **mywksta** from its current domain into the **MyDomain** domain, run the following command:

```
Windows Command Prompt
```

```
netdom move mywksta /domain:mydomain /userd:mydomain\admin /passwordd:*
```

If the destination is an AD DS domain, this command updates the Security ID history for the workstation, retaining the security permissions that the computer account had previously.

To move **Srv01** to the domain **NewDomain** and specify the OU, run the following command:

Windows Command Prompt

```
netdom move svr01 /domain:newdomain
/ou:"OU=Workstations,OU=Resources,DC=NewDomain,DC=local"
/userd:NewDomain\DomainAdmin /passwordd:*
```

See also

Command-Line Syntax Key

netdom movent4bdc

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom movent4bdc` command renames a Windows NT 4.0 backup domain controller (DC). It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). To use `netdom movent4bdc`, you must run the command from an elevated command prompt.

This command is specific to legacy Windows Server operating systems.

Syntax

```
netdom movent4bdc machine [/Domain:domain] [/Reboot[:Time in seconds]]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the backup DC that you want to rename.
`/domain:<Domain>`	Specifies the new name of the domain.
`/reboot:<seconds>`	Shuts down the computer and automatically reboots after the move operation completes. The *seconds* value is the number of seconds before automatic shutdown. The default is **20** seconds.
`help` | `/?`	Displays help at the command prompt.

Examples

To rename **OldBDC** to **NewBDC** for the domain **reskita**, run the following command:

```
Windows Command Prompt
```

```
netdom movent4bdc oldbdc newbdc /domain:reskita
```

See also

Command-Line Syntax Key

netdom query

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom query` command retrieves information from a domain about its membership and trust relationships. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom query`, you must run the command from an elevated command prompt.

Syntax

```
netdom query [/Domain:domain] [/Server:server]
             [/UserD:user] [/PasswordD:[password | *]]
             [/Verify] [/Reset] [/Direct] [/SecurePasswordPrompt]
             WORKSTATION | SERVER | DC | OU | PDC | FSMO | TRUST
```

Parameters

Expand table

Parameter	Description
`/domain:<Domain>`	Specifies the domain to query for information. If you don't specify this parameter, then `netdom query` uses the domain to which the current computer belongs.
`/server:<Server>`	Specifies the name of the domain controller that performs the query.
`/userd:<User>`	Specifies the user account that makes the connection with the domain that you specify using the `/domain` parameter. If you don't specify this parameter, `netdom query` uses the current user account.
`/passwordd:<Password>` | `*`	Specifies the password of the user account that you specify using the `/userd` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.

Parameter	Description
`/verify`	Verifies the secure channel secrets for all listed memberships or trusts and displays the results. Only outbound trusts are verified. Domain administrator credentials are required for accurate verification.
`/reset`	Specifies resynchronization of the secure channel secrets for all enumerated memberships or trusts that are currently broken. Domain administrator credentials are required to reset all enumerated trusts or memberships.
`/direct`	Indicates that the query for trust relationships returns only direct trust relationships, rather than direct and indirect relationships. This parameter can't be used with the `/verify` parameter.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`workstation` | `server` | `dc` | `ou` | `pdc` | `fsmo` | `trust`	Specifies the type of list to generate. The following list shows the possible objects: - **Workstation**: Queries the domain for the list of workstations. - **Server**: Queries the domain for the list of servers. - **DC**: Queries the domain for the list of domain controllers. - **OU**: Queries the domain for the list of OUs under which the user that you specify can create a computer object. - **PDC**: Queries the domain for the current primary domain controller. - **FSMO**: Queries the domain for the current list of flexible single master operations (FSMO) role owners. - **Trust**: Queries the domain for the list of its trusts.
`help` | `/?`	Displays help at the command prompt.

Examples

To list all the *servers* in the domain **NorthAmerica**, run the following command:

Windows Command Prompt

```
netdom query /domain:NorthAmerica SERVER
```

To list all servers and verify the secure channel secret in the domain **NorthAmerica**, run the following command:

Windows Command Prompt

```
netdom query /domain:NorthAmerica SERVER /verify
```

To list all the direct trust relationships for the domain **NorthAmerica**, run the following command:

Windows Command Prompt

```
netdom query /domain:NorthAmerica /Direct TRUST
```

See also

Command-Line Syntax Key

netdom remove

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom remove` command removes a workstation or server from a domain. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom remove`, you must run the command from an elevated command prompt.

Syntax

```
netdom remove machine [/Domain:domain] [/UserD:user]
                      [/PasswordD:[password | *]]
                      [/UserO:user] [/PasswordO:[password | *]]
                      [/Reboot[:Time in seconds]] [/Force]
                      [/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer that you want to remove.
`/domain:<Domain>`	Specifies the domain from which you want to remove the computer. If not specified, `netdom remove` uses the domain that the current computer belongs to.
`/userd:<User>`	Specifies the user account that makes the connection with the domain specified using the `/domain` parameter. If not specified, uses the current user account.
`/passwordd:<Password>` | `*`	Specifies the password of the user account specified using the `/userd` parameter. If set to `*`, you're prompted for the password.
`/usero:<User>`	Specifies the user account to make the connection with the computer you

Parameter	Description
	want to remove. If not specified, uses the current user account.
`/passwordo:<Password>` | `*`	Specifies the password of the user account specified using the `/usero` parameter. If set to `*`, you're prompted for the password.
`/reboot:<seconds>`	Shuts down the computer and automatically reboots after the move operation completes. The *seconds* value is the number of seconds before automatic shutdown. The default is **20** seconds.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`help` | `/?`	Displays help at the command prompt.

Examples

To remove **mywksta** from the **MyDomain** domain, run the following command:

Windows Command Prompt

```
netdom remove mywksta /domain:MyDomain /userd:MyDomain\admin /passwordd:*
```

See also

Command-Line Syntax Key

netdom renamecomputer

10/22/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom renamecomputer` command renames a domain workstation or member server and its corresponding domain account. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom renamecomputer`, you must run the command from an elevated command prompt.

Syntax

```
netdom renamecomputer machine /NewName:new-name
                              [/UserD:user [/PasswordD:[password | *]]]
                              [/UserO:user [/PasswordO:[password | *]]]
                              [/Force] [/Reboot[:Time in seconds]]
                              [/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer that you want to rename.
`/newname:<NewComputerName>`	Specifies the new name of the computer.
`/userd:<User>`	Specifies the user account that you want to use for the destination domain.
`/passwordd:<Password>` | `*`	Specifies the password of the user account that you specify using the `/userd` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.
`/usero:<User>`	Specifies the user account that you want to use for the originating domain.

Parameter	Description
`/passwordo:<Password>` | `*`	Specifies the password of the user account that you specify using the `/usero` parameter. If you specify the value of this parameter as a wildcard character (`*`), this parameter prompts you for the password.
`/force`	When the `/force` parameter is specified, you aren't prompted for confirmation to rename a device.
`/reboot:<seconds>`	Shuts down the computer and automatically reboots after the move operation completes. The *seconds* value is the number of seconds before automatic shutdown. The default is **20** seconds.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`help` | `/?`	Displays help at the command prompt.

> **Note**
>
> Don't use the `netdom renamecomputer` command to rename an AD DS domain controller (DC). Doing so Might cause the DC to no longer function as a DC on the network. Instead, use the `netdom computername` command.

Examples

To rename the member server **MyServer** to **MyNewServer**, run the following command:

Windows Command Prompt

```
netdom renamecomputer MyServer /newname:MyNewServer
```

If you don't have the appropriate permissions, the `/userd` and `/passwordd` parameters must be specified.

See also

Command-Line Syntax Key

netdom reset

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom reset` command resets the secure connection between a workstation and a domain controller (DC). It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom reset`, you must run the command from an elevated command prompt.

Syntax

```
netdom reset machine [/Domain:domain] [/Server:server]
                     [/UserO:user] [/PasswordO:[password | *]]
[/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer whose secure connection you want to reset.
`/domain:<Domain>`	Specifies the domain with which to establish the secure connection. If not specified, `netdom reset` uses the domain to which the current computer belongs.
`/server:<Server>`	Specifies the DC to use to establish the secure connection.
`/usero:<User>`	Specifies the user account to use to make the secure connection with the computer that you want to reset. If not specified, `netdom reset` uses the current user account.
`/passwordo:<Password>` | `*`	Specifies the password of the user account specified using the `/usero` parameter. If you use the wildcard character (`*`), you're prompted for the password.

Parameter	Description
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`help` | `?`	Displays help at the command prompt.

Examples

To reset the secure channel secret that is maintained between **mywksta** and **devgroup.contoso.com**, run the following command:

Windows Command Prompt

```
netdom reset mywksta /domain:devgroup.contoso.com
```

To reset the secure channel between the primary DC for **NorthAmerica** and the backup DC **BackupDC**, run the following command:

Windows Command Prompt

```
netdom reset BackupDC /domain:NorthAmerica
```

Member servers frequently establish secure channel sessions with DCs that aren't local to their site. To force a secure channel session between member server **mywksta** and a specific DC, **MyOtherDC**, run the following command:

Windows Command Prompt

```
netdom reset mywksta /domain:MyDomain.contoso.com /Server:MyOtherDC
```

See also

Command-Line Syntax Key

netdom resetpwd

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom resetpwd` command resets the secure channel password of the machine account for a domain controller (DC). This command must be run on the DC whose machine account password you wish to reset. It doesn't support resetting passwords for remote machines or member servers. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT).

For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom resetpwd`, you must run the command from an elevated command prompt.

Syntax

```
netdom resetpwd /Server:domain-controller /UserD:user /PasswordD:[password | *]
                [/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`/server:<DC>`	Specifies the DC to use to set the computer account password.
`/userd:<User>]`	Specifies the user account to use to make the secure connection with the domain specified using the `/server` parameter. If omitted, it uses the current user account.
`/passwordd:<Password>` | `*`	Specifies the password for the user account when using the `/userd` parameter. If set to `*`, you're prompted for the password.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.

Parameter	Description
`help` | `/?`	Displays help at the command prompt.

Examples

To reset the secure channel password for a computer account associated with the DC **MyDC** and be securely prompted for the password, run the following command:

```
netdom resetpwd /server:MyDC /userd:admin /passwordd:*
```

See also

Command-Line Syntax Key

netdom trust

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom trust` command allows administrators to manage, establish, verify, or reset trust relationships between domains. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom trust`, you must run the command from an elevated command prompt.

> **Note**
>
> The `netdom trust` command can't be used to create a forest trust between two AD DS forests. To create a cross-forest trust between two AD DS forests, use the **Active Directory Domains and Trusts** snap-in to create and manage forest trusts. Scripting solution such as using PowerShell is also an option for managing these types of trusts if you need to automate the process.

Syntax

```
netdom trust trusting_domain_name /Domain:trusted_domain_name [/UserD:user]
             [/PasswordD:[password | *]] [/UserO:user] [/PasswordO:[password | *]]
             [/Verify] [/Reset] [/PasswordT:new_realm_trust_password]
             [/Add] [/Remove] [/Twoway] [/Realm] [/Kerberos]
             [/Transitive[:{yes | no}]]
             [/OneSide:{trusted | trusting}] [/Force] [/Quarantine[:{yes | no}]]
             [/NameSuffixes:trust_name [/ToggleSuffix:#]]
             [/EnableSIDHistory[:{yes | no}]] [/ForestTransitive[:{yes | no}]]
             [/CrossOrganization[:{yes | no}]] [/AddTLN:TopLevelName]
             [/AddTLNEX:TopLevelNameExclusion] [/RemoveTLN:TopLevelName]
             [/RemoveTLNEX:TopLevelNameExclusion] [/SecurePasswordPrompt]
             [/EnableTgtDelegation[:{yes | no}]] [/EnablePIMTrust[:{yes | no}]]
             [/AuthTargetValidation[:{yes | no}]] [/ChildDomain:childdomainname]
             [/InvokeTrustScanner]
```

Parameters

Expand table

Parameter	Description
`<TrustingDomainName>`	Specifies the name of the trusting domain.
`/domain:<TrustedDomainName>`	Specifies the name of the trusted domain or non-Windows realm. If not specified, the current domain to which the current computer belongs is used.
`/userd:<User>`	Specifies the user account to use for the connection with the domain specified using the `/domain` parameter. Defaults to the current user account if not specified.
`/passwordd:<Password>` | `*`	Specifies the password for the user account used with `/userd`. Use `*` to prompt for the password.
`/usero:<User>`	Specifies the user account to use for the connection with the trusting domain. Defaults to the current user account if not specified.
`/passwordo:<Password>` | `*`	Specifies the password for the user account used with `/usero`. Use `*` to prompt for the password.
`/verify`	Verifies the secure channel secrets for a specific trust relationship.
`/reset`	Resets the trust secret between trusted domains or between the domain controller (DC) and the workstation.
`/passwordt:<NewRealmTrustPassword>`	Sets a new trust password. This option is valid only with the `/add` or `/reset` parameters, and only if one of the specified domains is a non-Windows Kerberos realm. The trust password is configured on the Windows domain only, so credentials for the non-Windows domain aren't required.
`/add`	Creates a trust.
`/remove`	Removes a trust.
`/twoway`	Establishes a two-way trust relationship.
`/realm`	Creates the trust for a non-Windows Kerberos realm. Valid only with the `/add` parameter. The `/passwordt` parameter is required.
`/kerberos`	Uses the Kerberos protocol to verify authentication between a workstation and the specified domain. Requires credentials for both the source and target domains.
`/transitive:Yes` | `No`	Applies only to non-Windows Kerberos realm trusts. Use `yes` to make the trust transitive, or `no` to make it non-transitive. If not specified, displays the current transitivity setting.

Parameter	Description
`/oneside:trusted` | `trusting`	Specifies that the trust operation should be conducted on only one side of the trust relationship. Use `trusted` to apply the operation to the domain specified with the `/domain` parameter (the "trusted" domain), or use `trusting` to apply it to the "trusting" domain. This option is only valid with the `/add` and `/remove` parameters. When used with `/add`, the `/passwordt` parameter is also required. - **Trusted Domain**: This is the domain that is being trusted. In a trust relationship, the trusting domain allows users from the trusted domain to access its resources. The trusted domain's users are given certain permissions or access within the trusting domain. - **Trusting Domain**: This is the domain that trusts another domain (the trusted domain). It essentially means that the trusting domain is extending its trust to the users of the trusted domain, allowing them to access resources within the trusting domain.
`/force`	Removes both the trusted domain object and cross-reference object from the forest. The full DNS name must be specified for the domain. Valid with the `/remove` parameter and if specified, a child domain is removed.
`/quarantine:Yes` | `No`	Sets or clears the domain quarantine attribute. If not specified, displays the current state. `Yes` accepts only SIDs from the directly trusted domain. `No` accepts any SID (default). Specifying `/quarantine` without an option displays the current state.
`/namesuffixes:<TrustName>`	Lists the routed name suffixes for the specified trust. This parameter is valid only for a forest trust or a forest transitive non-Windows realm trust. Use `/usero` and `/passwordo` for authentication if needed. The `/domain` parameter isn't required for this operation.
`/togglesuffix:#`	Use this parameter with `/namesuffixes` to enable or disable a specific name suffix. Specify the number of the name entry as shown in the output of the preceding `/namesuffixes` command. You can't change the status of names that are in conflict until the conflicting name in the other trust is disabled. Always run `/namesuffixes` immediately before `/togglesuffix` because the order of name entries might change.
`/enablesidhistory:Yes` | `No`	Enables (`Yes`) or disables (`No`) migrated users in the trusted forest to use SID history to access resources. Valid only for outbound forest trusts. Only enable if you trust the administrators of the trusted forest. If an option isn't specified, the current state is displayed.
`/foresttransitive:Yes` | `No`	Marks the trust as forest transitive (yes) or not (no). Valid only for AD trusts and non-Windows realm trusts only on the root domain for a forest. If not specified, displays the current state.

Parameter	Description
`/selectiveauth:Yes` | `No`	Enables (`Yes`) or disables (`No`) selective authentication across the trust. Valid only on outbound forest and external trusts. If not specified, displays the current state.
`/addtln:<TopLevelName>`	Adds the specified top-level DNS name suffix to the forest trust info for the trust. Valid only for a forest transitive non-Windows realm trust and only on the root domain for a forest. Run `/namesuffixes` for a list of name suffixes.
`/addtlnex:<TopLevelNameExclusion>`	Adds the specified top-level name exclusion (DNS name suffix) to the forest trust info for the trust. Valid only for a forest transitive non-Windows realm trust and only on the root domain for a forest. Run `/namesuffixes` for a list of name suffixes.
`/removetln:<TopLevelName>`	Removes the specified top-level DNS name suffix from the forest trust info for the trust. Valid only for a forest transitive non-Windows realm trust and only on the root domain for a forest. Run `/namesuffixes` for a list of name suffixes.
`/removetlnex:<TopLevelNameExclusion>`	Removes the specified top-level name exclusion (DNS name suffix) from the forest trust info for the trust. Valid only for a forest transitive non-Windows realm trust and only on the root domain for a forest. Run `/namesuffixes` for a list of name suffixes.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.
`/enabletgtdelegation:Yes` | `No`	Enables (`Yes`) or disables (`No`) Kerberos full delegation on outbound forest trusts. When set to `No`, Kerberos full delegation is blocked, preventing services in the other forest from receiving forwarded Ticket Granting Tickets (TGTs). **Disabling** this option means that services in the other forest configured for "*Trust this computer/user for delegation to any service*" isn't able to use Kerberos full delegation with any account in this forest.
`/enablepimtrust:Yes` | `No`	Enables (`Yes`) or disables (`No`) Privileged Identity Management (PIM) trust behaviors for this trust. The trust must be marked as forest transitive before enabling this attribute. If `/enablepimtrust` is specified without `Yes` or `No`, the current state of this attribute is displayed.
`/authtargetvalidation:Yes` | `No`	Enables (`Yes`) or disables (`No`) authentication target validation for authentication requests on the specified trust. For forest trusts, you can limit this setting to a specific child domain using the `/childdomain` parameter.

Parameter	Description
	Disabling this validation might expose your environment to security risks from the remote forest and should only be done when necessary.
`/childdomain: <ChildDomainName>`	Use to target a child domain within a larger domain structure when performing trust-related operations to ensure that the trust operation applies directly to the child domain. This parameter is useful in scenarios where precise control over trust relationships is needed within complex domain environments.
`/invoketrustscanner`	Initiates a trust scan for the specified trusting domain. If the trusting domain is set to `*`, all trusts are scanned. This command must be executed locally on the primary DC. The trust scanner typically runs automatically. Use this command only for troubleshooting or support purposes.
`help` | `/?`	Displays help at the command prompt.

Examples

To set the domain **USA-Chicago** to trust the domain **NorthAmerica**, run the following command:

Windows Command Prompt

```
netdom trust USA-Chicago /domain:NorthAmerica /add /userd:NorthAmerica\admin
/passwordd:* /usero:USA-Chicago\admin /passwordo:*
```

To establish a two-way trust between the **engineering.contoso.com** domain and the **marketing.contoso.com** domain, run the following command:

Windows Command Prompt

```
netdom trust engineering.contoso.com /domain:marketing.contoso.com /add /twoway
/usero:admin@engineering.contoso.com /passwordo:*
/userd:admin@marketing.contoso.com /passwordd:*
```

To establish a one-way trust where the **NorthAmerica** domain trusts the non-Windows Kerberos realm **ATHENA**, run the following command:

Windows Command Prompt

```
netdom trust NorthAmerica /domain:ATHENA /add /passwordt:* /realm
```

> **Note**
>
> Verifying a specific trust relationship requires credentials unless the user has domain administrator privileges on both domains.

If you want to set the Kerberos realm **ATHENA** to trust the **NorthAmerica** domain, run the following command:

Windows Command Prompt

```
netdom trust NorthAmerica /domain:ATHENA /add /realm
```

To undo (remove) the trust that **USA-Chicago** has with **NorthAmerica**, run the following command:

Windows Command Prompt

```
netdom trust USA-Chicago /domain:NorthAmerica /remove
```

To reset the secure channel for the one-way trust between **NorthAmerica** and **USA-Chicago**, run the following command:

Windows Command Prompt

```
netdom trust USA-Chicago /domain:NorthAmerica /userd:NorthAmerica\admin
/passwordd:* /reset
```

To verify that the trust relationship between the **MyDomain** domain and the **devgroup.example.com** domain supports Kerberos authentication, run the following command:

Windows Command Prompt

```
netdom trust MyDomain /domain:devgroup.example.com /verify /kerberos
/userd:devgroup\admin /passwordd:* /usero:MyDomain\admin /passwordo:*
```

> **Note**
>
> You can't run this trust operation from a remote location. You must run the operation on the workstation that you want to test.

To enable or disable the first routed name suffix in the list generated by the previous command, run the following command:

Windows Command Prompt

```
netdom trust myTestDomain /domain:foresttrustpartnerdomain /namesuffixes
/togglesuffix:1
```

You can only add a DNS name suffix for a trust that is a forest transitive non-Windows realm trust. The same restriction applies to the parameters for managing name suffix routing within a forest trust:

- `/addtln`
- `/addtlnex`
- `/removetln`
- `/removetlnex`

See also

Command-Line Syntax Key

netdom verify

06/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016

The `netdom verify` command checks the secure channel between a specified computer and a domain controller to ensure trust and connectivity are intact. It's available if you have the Active Directory Domain Services (AD DS) server role installed. It's also available if you install the AD DS tools that are part of the Remote Server Administration Tools (RSAT). For more information, see How to Administer Microsoft Windows Client and Server Computers Locally and Remotely.

To use `netdom verify`, you must run the command from an elevated command prompt.

Syntax

```
netdom verify machine [/Domain:domain] [/UserO:user] [/PasswordO:[password | *]]
[/SecurePasswordPrompt]
```

Parameters

Expand table

Parameter	Description
`<machine>`	Specifies the name of the computer whose secure connection you want to verify.
`/domain:<Domain>`	Specifies the domain with which to verify the secure connection. If not specified, `netdom verify` uses the domain to which the current computer belongs.
`/usero:<User>`	Specifies the user account to use for verification. If not specified, the current user account is used.
`/passwordo:<Password>` | `*`	Specifies the password for the user account specified used with the `/usero` parameter. If set to `*`, you're prompted for the password.
`/securepasswordprompt`	Opens a secure credentials popup for entering credentials. This is useful when specifying smartcard credentials. This option is effective only when the password is entered as `*`.

Parameter	Description
`help` | `/?`	Displays help at the command prompt.

Examples

To verify that netdom maintains the secure channel secret between **mywksta** and **devgroup.contoso.com**, run the following command:

Windows Command Prompt

```
netdom verify mywksta /domain:devgroup.contoso.com
```

See also

Command-Line Syntax Key

Network shell (netsh)

08/12/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Network shell (netsh) is a versatile and comprehensive command-line utility for configuring, managing, and monitoring network components and server roles on Windows and Windows Server systems. With `netsh`, you can view and modify network settings, automate tasks, and troubleshoot network issues locally or remotely.

In many scenarios, the netsh commands offer equivalent functionality to the Microsoft Management Console (MMC) snap-ins for managing networking server roles and features. This allows administrators to perform configuration and management tasks either through the graphical MMC interface or by using command-line automation.

For example, you can configure Network Policy Server (NPS) by using either the NPS MMC snap-in or the netsh commands in the `netsh nps` context. In addition, there are netsh commands for network technologies, such as for IPv6, network bridge, and Remote Procedure Call (RPC), that aren't available in Windows Server as an MMC snap-in.

> **Important**
>
> It's recommended that you use Windows PowerShell to manage networking technologies in Windows and Windows Server rather than `netsh`.

Example syntax

```
netsh [-a <Aliasfile>] [-c <Context>] [-r <RemoteMachine>] [-u <DomainName>\
<Username>] [-p <Password> | *] [Command> | -f <ScriptFile>]
```

The following parameters are optional.

Expand table

Parameter	Description
-a	Specifies that you want to return to the netsh shell after running an alias file. An alias file is the text file that contains one or more netsh commands.

Parameter	Description
-c	Enters the specified netsh context. The context represents a specific set of available commands related to a particular feature or function.
-r	Specifies that you want the command to run on a remote computer using its computer name, a fully qualified domain name (FQDN), or IP address. The Remote Registry service must be running on the remote computer. If it's not running, Windows displays a "Network Path Not Found" error message. If you specify `-r` followed by another command, netsh runs the command on the remote computer and then returns to the command prompt. If you specify `-r` without another command, netsh opens in remote mode. The process is similar to using `set machine` at the netsh command shell. When you use `-r`, you set the target computer for the current instance of netsh only.
-u	Specifies that you want to run the netsh command under a specific user account using `DomainName\Username` format.
-p	Specifies that you want to provide a password for the user account. If `*` is used, this specifies you want to securely enter the password.
Command	This is the netsh command that you want to run.
-f	Specifies that you want to exit netsh after executing the commands from the script file you provide.

If your string value contains spaces between characters, you must enclose the string value in quotation marks. For example, `-r "contoso remote device"`. After you exit and reenter `netsh`, the target computer is reset as the local computer. You can run netsh commands on a remote computer by specifying a computer name stored in WINS, a UNC name, an internet name to be resolved by the DNS server, or an IP address.

> ⓘ **Note**
>
> To use certain netsh commands, such as the `netsh firewall` commands remotely on another computer by using the netsh `-r` parameter, the **Remote Registry** service must be running on the remote computer. If this service isn't running, you might receive a network communication error message.

Running netsh commands

Running `netsh` commands can be performed using an elevated command prompt or PowerShell window. You can change to the context that contains the command you want to

use depending on the networking components that are installed.

For example, if you type `netsh`, you're taken directly into the network shell. Once here, if you type `dhcp` and press **Enter**, you're taken into the context menu for `dhcpclient` commands.

Command contexts

Netsh interacts with other operating system components by using dynamic-link library (DLL) files. Each netsh helper DLL provides an extensive set of features called a *context*, which is a group of commands that's specific to a networking server role or feature. These contexts extend the functionality of netsh by providing configuration and monitoring support for one or more services, utilities, or protocols.

For example, dhcpmon.dll provides netsh with the context and set of commands necessary to configure and manage DHCP servers.

The `>` placement let's you know which context or subcontext you're in. You can obtain a list of netsh contexts by running `netsh ?` or `netsh help` and pressing **Enter** for any of the available commands.

Here's an example output:

```
The following commands are available:

Commands in this context:
..              - Goes up one context level.
?               - Displays a list of commands.
abort           - Discards changes made while in offline mode.
add             - Adds a configuration entry to a list of entries.
advfirewall     - Changes to the 'netsh advfirewall' context.
alias           - Adds an alias.
branchcache     - Changes to the 'netsh branchcache' context.
bridge          - Changes to the 'netsh bridge' context.
bye             - Exits the program.
commit          - Commits changes made while in offline mode.
delete          - Deletes a configuration entry from a list of entries.
dhcpclient      - Changes to the 'netsh dhcpclient' context.
dnsclient       - Changes to the 'netsh dnsclient' context.
dump            - Displays a configuration script.
exec            - Runs a script file.
exit            - Exits the program.
firewall        - Changes to the 'netsh firewall' context.
help            - Displays a list of commands.
http            - Changes to the 'netsh http' context.
interface       - Changes to the 'netsh interface' context.
ipsec           - Changes to the 'netsh ipsec' context.
```

```
ipsecdosprotection - Changes to the 'netsh ipsecdosprotection' context.
lan            - Changes to the 'netsh lan' context.
namespace      - Changes to the 'netsh namespace' context.
netio          - Changes to the 'netsh netio' context.
offline        - Sets the current mode to offline.
online         - Sets the current mode to online.
popd           - Pops a context from the stack.
pushd          - Pushes current context on stack.
quit           - Exits the program.
ras            - Changes to the 'netsh ras' context.
rpc            - Changes to the 'netsh rpc' context.
set            - Updates configuration settings.
show           - Displays information.
trace          - Changes to the 'netsh trace' context.
unalias        - Deletes an alias.
wfp            - Changes to the 'netsh wfp' context.
winhttp        - Changes to the 'netsh winhttp' context.
winsock        - Changes to the 'netsh winsock' context.

The following sub-contexts are available:
advfirewall branchcache bridge dhcpclient dnsclient firewall http interface ipsec
lan mbn namespace netio nlm ras rpc trace wcn wfp winhttp winsock wlan

To view help for a command, type the command, followed by a space, and then type
?.
```

Command subcontexts

The contexts might also include both commands and nested contexts known as *subcontexts*. For instance, the *interface* context contains subcontexts such as *ipv4* and *ipv6*, which allow you to manage specific network interface settings. For example:

```

```

```
netsh> interface help

The following commands are available:

Commands in this context:
6to4           - Changes to the 'netsh interface 6to4' context.
?              - Displays a list of commands.
dump           - Displays a configuration script.
fl48           - Changes to the 'netsh interface fl48' context.
fl68           - Changes to the 'netsh interface fl68' context.
help           - Displays a list of commands.
httpstunnel    - Changes to the 'netsh interface httpstunnel' context.
ipv4           - Changes to the 'netsh interface ipv4' context.
ipv6           - Changes to the 'netsh interface ipv6' context.
isatap         - Changes to the 'netsh interface isatap' context.
portproxy      - Changes to the 'netsh interface portproxy' context.
set            - Sets configuration information.
```

```
show            - Displays information.
tcp             - Changes to the 'netsh interface tcp' context.
teredo          - Changes to the 'netsh interface teredo' context.
udp             - Changes to the 'netsh interface udp' context.

The following sub-contexts are available:
 6to4 fl48 fl68 httpstunnel ipv4 ipv6 isatap portproxy tcp teredo udp

To view help for a command, type the command, followed by a space, and then
 type ?.
```

To perform tasks in another context without changing from your current context, type the context path of the command you want to use at the netsh shell. For example, to add a static IPv4 address for "NetLAN1" and specify both the subnet mask and gateway, type:

```
netsh interface ipv4 set address "NetLAN1" static <IP_Address> <Subnet_Mask>
<Gateway>
```

Formatting legend

You can use the following formatting legend to interpret and use correct command syntax when running commands interactively, in a batch file, or a script.

- Text in *italic* is information that you must supply when you type the command. For example, if a command has a parameter named *-UserName*, you must type the actual user name.
- Text in **bold** is information that you must type exactly as shown when you type the command.
- Text followed by an ellipsis **...** is a parameter that can be repeated several times in a command line.
- Text that's between brackets **[ ]** is an optional item.
- Text that's between braces **{ }** with choices separated by a pipe provides a set of choices from which you must select only one, such as `{enable|disable}`.
- Text that's formatted with the Courier font is code or program output.

String values for netsh commands

There are commands that contain parameters for which a string value is required. In the case where a string value contains spaces between characters, you're required to enclose the string value in double quotation marks.

When setting the `interface` parameter for IPv4 networks with a string value of **Wireless Network Connection** to DHCP, use double quotation marks around the string value. For example:

```
netsh interface ipv4> set address name="Wireless Network Connection" dhcp
```

Alternatively, for network interfaces, the index number can be used in place of the name associated to it. For example:

```
netsh interface ipv4> set address name=15 dhcp
```

To find the name or index of the connected interfaces in the interface subcontext, select your interface type and type `show interfaces` to view detailed information. For example:

```
netsh interface> ipv4 show interfaces
```

```
netsh interface> ipv6 show interfaces
```

If you aren't in the interface subcontext, you can run the *full* command for either interface. For example:

```
netsh interface ipv4 show interfaces
```

```
netsh interface ipv6 show interfaces
```

Using netsh in batch files

A batch file is a plain text file that contains a series of commands to be executed by the Windows Command Prompt (CMD). Batch files have a `.bat` or `.cmd` extension. They're used to

automate repetitive tasks, manage system configurations, and run sequences of commands without user intervention. Batch files are especially useful for tasks that involve multiple command-line instructions.

A batch script would be useful in these scenarios:

- **Bulk network configuration**: If you need to configure network settings for multiple computers, a batch file can streamline the process by automating the setup. Bulk configurations are helpful in enterprise environments or for IT professionals managing multiple machines.
- **Network reconfiguration**: Users switching between different network environments can use batch files to quickly change settings like IP addresses and DNS servers without manual intervention.
- **Backup and restore settings**: You can create a batch script to backup network settings before making changes, and another to restore them if needed.

Here's an example batch file. The lines beginning with "rem" are comments that let users know what action the commands perform:

batch

```
@echo off
echo Configuring network settings...

REM Set the IP address, subnet mask, and default gateway
netsh interface ip set address name="Ethernet" static 192.168.1.100 255.255.255.0
192.168.1.1

REM Set the DNS server address
netsh interface ip set dns name="Ethernet" static 8.8.8.8

REM Add a secondary DNS server
netsh interface ip add dns name="Ethernet" 8.8.4.4 index=2

REM Enable the firewall
netsh advfirewall set allprofiles state on

echo Network settings configured successfully.
pause
```

Here's a brief explanation of what this script does:

- It sets a static IP address, subnet mask, and default gateway for the network interface named "Ethernet".
- Configures the primary DNS server to be used.
- Adds a secondary (fallback) DNS server.

- Enables the Windows Firewall with Advanced Security for all profiles.

See also

- Command-Line Syntax Key

netsh add

07/24/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh add` command is used to install helper Dynamic Link Libraries (DLL) allowing for more specialized network configurations.

Syntax

```
netsh add helper [file]
```

Parameters

Expand table

Command	Description
helper	Adds a helper DLL file.

Example

To add a helper DLL located at **C:\dlls\HelperDLL.dll**, run the following command:

Windows Command Prompt

```
netsh add helper C:\dlls\HelperDLL.dll
```

netsh advfirewall

07/24/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh advfirewall` command configures and manages the Windows Firewall with Advanced Security utility. This tool allows administrators to configure firewall rules, security policies, and monitor various aspects of the firewall's behavior.

Syntax

The `netsh advfirewall` command provides various subcommands and contexts based on the specific operation you want to perform.

```
netsh advfirewall consec | dump | export | firewall | help | import | mainmode |
monitor | reset | set | show | ?
```

```
netsh advfirewall [consec | firewall | mainmode] <add> <delete> <dump> <help>
<set> <show> <?>
```

```
netsh advfirewall [monitor] <delete> <dump> <help> <show> <?>
```

```
netsh advfirewall export <path\filename>
netsh advfirewall import <path\filename>
netsh advfirewall reset [export <path\filename>]
```

```
netsh advfirewall set [allprofiles | currentprofile | domainprofile |
privateprofile | publicprofile] state <on> <off> <notconfigured>
netsh advfirewall set firewallpolicy <blockinbound> <blockinboundalways>
<allowinbound> <allowoutbound> <blockoutbound> <notconfigured>
netsh advfirewall set settings <localfirewallrules> <localconsecrules>
<inboundusernotification> <remotemanagement> <unicastresponsetomulticast>
netsh advfirewall set logging <allowedconnections> <droppedconnections> <filename>
```

```
<maxfilesize>
netsh advfirewall set global [statefulftp | statefulpptp] <enable> <disable>
<notconfigured>
netsh advfirewall set global ipsec [strongcrlcheck <value>] [saidletimemin
<value>] [defaultexemptions <value>] [ipsecthroughnat <value>] [authzcomputergrp
<value>] [authzusergrp <value>]
netsh advfirewall set global mainmode [mmkeylifetime <value>] [mmsecmethods
<value>] [mmforcedh <value>]
```

```
netsh advfirewall show [allprofiles | currentprofile | domainprofile |
privateprofile | publicprofile] <state> <firewallpolicy> <settings> <logging>
netsh advfirewall show global <ipsec> <statefulftp> <statefulpptp> <mainmode>
<categories>
netsh advfirewall show <store>
```

Parameters

Expand table

Command	Description
consec `<add>` `<delete>` `<help>` `<set>` `<show>` rule name=`<string>` consec `<dump>`	Switches to the `netsh advfirewall consec` context to configure connection security rules. • add - Adds a new connection security rule. • delete - Deletes all matching connection security rules. • dump - Displays a configuration script. • help - Displays a list of commands. • set - Sets new values for properties of an existing rule. • show - Displays a specified connection security rule.
dump	Displays a configuration script that can be used to restore the current configuration.
export `<path\filename>`	Exports the current firewall policy settings to a specified file.
firewall `<add>` `<delete>` `<help>` `<set>` `<show>` rule name=`<string>` firewall `<dump>`	Switches to the `netsh advfirewall firewall` context to configure firewall rules. • add - Adds a new inbound or outbound firewall rule. • delete - Deletes all matching firewall rules. • dump - Displays a configuration script. • help - Displays a list of commands. • set - Sets new values for properties of an existing rule. • show - Displays a specified firewall rule.

Command	Description
help *or* ?	Displays a list of commands and their descriptions in the current context.
import `<path\filename>`	Imports firewall policy settings from a specified file into the current policy store.
mainmode `<add>` `<delete>` `<help>` `<set>` `<show>` rule name=`<string>` mainmode `<dump>`	Switches to the `netsh advfirewall mainmode` context to configure main mode settings. • add - Adds a new mainmode rule. • delete - Deletes all matching mainmode rules. • dump - Displays a configuration script. • help - Displays a list of commands. • set - Sets new values for properties of an existing rule. • show - Displays a specified mainmode rule.
monitor delete `mmsa | qmsa <source destination | all>` monitor `<help>` monitor show `<consec>` `<currentprofile>` `<firewall>` `<mainmode>` `<mmsa>` `<qmsa>` monitor `<dump>`	Switches to the `netsh advfirewall monitor` context to display monitoring information. • delete - Deletes all matching security associations. The source destination is one or more IP addresses. `All` specifies that the operation should apply to all entries. • dump - Displays a configuration script. • help - Displays a list of commands. • show - Shows the runtime Firewall policy settings: - `consec` displays current consec state information. - `currentprofile` displays the currently active profiles. - `firewall` displays current firewall state information. - `mainmode` displays current mainmode state information. - `mmsa` displays the main mode SAs. - `qmsa` displays the quick mode SAs.
reset `<export path\filename>`	Resets the Windows Defender Firewall with Advanced Security policies to the default settings. The current active policy can be optionally exported to a specified file. In a Group Policy object, this command returns all settings to `notconfigured` and deletes all connection security and firewall rules.
set allprofiles | currentprofile | domainprofile | privateprofile | publicprofile | `parameter` `<value>`	Configures per-profile or global settings for the firewall. • allprofiles - Sets properties in all profiles. • currentprofile - Sets properties in the active profile. • domainprofile - Sets properties in the domain profile. • global - Sets the global properties. • privateprofile - Sets properties in the private profile. • publicprofile - Sets properties in the public profile. The `notconfigured` value is valid only for a Group Policy store.

Command	Description
	The available `parameter` and `<value>` are as follows: `state` - Configures the firewall state. • `<on>` - Enables the firewall. When the firewall is on, it actively filters network traffic based on the configured rules. • `<off>` - Disables the firewall. No filtering or protection is applied, and all network traffic is allowed. • `<notconfigured>` - sets the firewall state to its default configuration as per the system policy or administrative template settings. `firewallpolicy` - Configures the default *inbound* and *outbound* behavior. • `<blockinbound>` - Blocks inbound connections that don't match an inbound rule. • `<blockinboundalways>` - Blocks all inbound connections even if the connection matches a rule. • `<allowinbound>` - Allows inbound connections that don't match a rule. • `<allowoutbound>` - Allows outbound connections that don't match a rule. • `<blockoutbound>` - Blocks outbound connections that don't match a rule. • `<notconfigured>` - Returns the value to its unconfigured state. `settings` - Configures the firewall settings. • `localfirewallrules` - Merge local firewall rules with Group Policy rules. Valid when configuring a Group Policy store. • `localconsecrules` - Merge local connection security rules with Group Policy rules. Valid when configuring a Group Policy store. • `inboundusernotification` - Notify user when a program listens for inbound connections. • `remotemanagement` - Allow remote management of Windows Firewall. • `unicastresponsetomulticast` - Control stateful unicast response to multicast. • `<enable>` - Turns on the specified setting. • `<disable>` - Turns off the specified setting. • `<notconfigured>` - Restores the setting to the systems default configuration. `logging` - Configures logging settings. • `allowedconnections` - Determines whether successful inbound and outbound connections are recorded in the logs. `<on>` - Logs successful connections. `<off>` - Doesn't log connections. `<notconfigured>` - Reverts to the system's default behavior for logging successful connections.

Command	Description
	• `droppedconnections` - Specifies whether connections blocked by the firewall are logged. `<on>` - Logs all blocked connections. `<off>` - Don't log blocked connections. `<notconfigured>` - Reverts to the system's default behavior for logging blocked connections. • `filename` - Sets the name and location for storing the log file. `<string>` - The file path and file name designated for the log file, such as `C:\Logs\firewall.log`. `<notconfigured>` - Uses the default file path and name provided by the system. • `maxfilesize` - Sets the maximum size of the log file in kilobytes (KB). If the file exceeds this size, the oldest entries are overwritten. Acceptable file size is between 1 KB and 32,767 KB. `<notconfigured>` - Adopts the system's default maximum file size.
set global [statefulftp | statefulpptp] `<enable>` `<disable>` `<notconfigured>`	Configures the global settings including advanced IPsec options. The use of DES, MD5, and DHGroup1 isn't recommended. These cryptographic algorithms are provided for backward compatibility only. The mmsecmethods keyword default sets the policy to dhgroup2-aes128-sha1,dhgroup2-3des-sha1 `statefulftp` - This option controls the firewall's behavior regarding FTP (File Transfer Protocol) traffic. `statefulpptp` - This option manages the handling of PPTP (Point-to-Point Tunneling Protocol) traffic. `<enable>` - Activates stateful inspection for the selected protocol. This ensures that the firewall can dynamically adjust rules to correctly handle corresponding traffic. `<disable>` - Deactivates stateful inspection for the selected protocol. This might lead to issues with correctly handling protocol-specific traffic as the firewall doesn't dynamically adjust its rules. `<notconfigured>` - Resets the setting to the default behavior as determined by the system's policy or configuration. This allows the system to apply its default rules without user-specified changes.
set global ipsec `parameter` `<value>`	`strongcrlcheck` - Configures the enforcement level of Certificate Revocation List (CRL) checking. `<0>` - Disables CRL checking (default setting). `<1>` - Fails the connection if the certificate is revoked. `<2>` - Fails the connection on any CRL-related error. `<notconfigured>` - Resets the value to the not configured state.

Command	Description

	`saidletimemin` - Sets the idle time in minutes for a Security Association (SA) before it's considered inactive. Acceptable values range from `5` to `60` minutes.
	`<notconfigured>` - Defaults to `5` minutes.
	
	`defaultexemptions` - Manages default protocols exempted from IPsec protection. The default is to exempt IPv6 neighbordiscovery protocol and DHCP from IPsec. Acceptable values include:
	`<none>`
	`<neighbordiscovery>`
	`<icmp>`
	`<dhcp>`
	`<notconfigured>`
	
	`ipsecthroughnat` - Configures the conditions under which IPsec can establish security associations with devices behind a Network Address Translator (NAT). Acceptable values include:
	`<never>` - Associations aren't permitted (default).
	`<serverbehindnat>` - Only if the server is behind a NAT.
	`<serverandclientbehindnat>` - If both the server and client are behind a NAT.
	`<notconfigured>` - Defaults to `<never>`.
	
	`authzcomputergrp` - Specifies computer groups authorized to establish tunnel mode connections.
	`<none>` - No specific computer group is authorized to establish tunnel mode connections.
	`<SDDL string>` - Specifies a Security Descriptor Definition Language (SDDL) string that defines the group of computers authorized to establish connections.
	`<notconfigured>` - Specifies default settings.
	
	`authzusergrp` - Specifies user groups authorized to establish tunnel mode connections.
	`<none>` - No specific user group is authorized to establish tunnel mode connections.
	`<SDDL string>` - Defines the specific user groups authorized to establish connections.
	`<notconfigured>` - Specifies default settings.

set global mainmode `parameter` `<value>` | `<notconfigured>`	`mmkeylifetime` - Sets the duration for which the keys in Main Mode are valid. Key lifetime is defined by time in minutes [<num> min], number of sessions [<num> sess], or both. The minimum *lifetime* of the keys in minutes range from `1` minute and maximum of `2880` minutes. The minimum number of *sessions* for which the keys are valid range from `0` sessions to a maximum of `2,147,483,647` sessions.

Command	Description

	`mmsecmethods` - Configures the list of security proposals for the Main Mode IPsec negotiations, which includes key exchange, encryption, and integrity algorithms. You can use a comma-separated list of proposals in the format `keyexch:enc-integrity`. Each proposal consists of a set of algorithms from each category.
	• **keyexch**: Specifies the Diffie-Hellman (DH) or Elliptic Curve Diffie-Hellman groups used for the key exchange. Acceptable values are:
	`<dhgroup1>`
	`<dhgroup2>`
	`<dhgroup14>`
	`<dhgroup24>`
	`<ecdhp256>`
	`<ecdhp384>`
	• **enc**: Specifies the encryption algorithm used for securing the data. Acceptable values are:
	`<3des>`
	`<des>`
	`<aes128>`
	`<aes192>`
	`<aes256>`
	• **integrity**: Specifies the algorithm used to ensure the data integrity. Acceptable values are:
	`<md5>`
	`<sha1>`
	`<sha256>`
	`<sha384>`
	
	`mmforcedh` - Enforces the use of DH for key exchanges during IPsec main mode negotiations.
	`<yes>` - Forces the use of DH for key exchange.
	`<no>` - Doesn't force the use of DH for key exchange (default setting).

show allprofiles | currentprofile | domainprofile | privateprofile | publicprofile | `parameter`	Displays the current profile or global properties for the firewall. • allprofiles - Displays the properties for all profiles. • currentprofile - Displays the properties for the active profile. • domainprofile - Displays the properties in the domain profile. • global - Displays the global properties. • privateprofile - Displays the properties for the private profile. • publicprofile - Displays the properties for the public profile. • store - Displays the current policy store. If a parameter isn't specified for the profiles, all properties are displayed. The available parameters are as follows:

Command	Description
	`state`: Displays whether the Windows Defender Firewall with Advanced Security is currently enabled or disabled for each profile. `firewallpolicy`: Shows the default actions for inbound and outbound connections, indicating whether such connections are allowed or blocked by default for each profile. `settings`: Provides detailed properties of the firewall, including configuration settings for each profile. `logging`: Shows the current logging configuration, including log file location, size, and logging options for dropped packets and successful connections.
show global `parameter`	Displays the global settings for Windows Defender Firewall with Advanced Security. If no parameter is specified, all properties are displayed. The available parameters are as follows: `ipsec`: Displays the specific IPsec settings. `statefulftp`: Displays the configuration for Stateful FTP support. `statefulpptp`: Shows the configuration for Stateful PPTP support. `mainmode`: Displays the configuration for Main Mode, a phase in the IPsec negotiation process where both parties authenticate each other and establish a secure channel for further communication. `categories`: Displays the configuration related to Firewall Categories.

Remarks

Consec parameter remarks

add

```
- The rule name should be unique and can't be "all".
- When mode=tunnel, tunnel endpoints must be specified unless the action is
"noauthentication". If specific IP addresses are used, they must belong to the
same IP version.
  - For dynamic tunnels, tunnel endpoints can be set to "any".
  - Local tunnel endpoints aren't required for Client policies, for example,
"any".
  - Remote tunnel endpoints aren't required for Gateway policies, for example,
"any". Additionally, the action must be one of the following:
requireinrequireout, requireinclearout, or noauthentication.
- requireinclearout isn't valid when mode=Transport.
- At least one authentication must be specified.
```

```
- Auth1 and auth2 can be comma-separated lists of options.
- Computerpsk and computerntlm methods can't be specified together for auth1.
- Computercert can't be specified with user credentials for auth2.
- Certsigning options ecdsap256 and ecdsap384 are only supported on Windows
Vista SP1 and later.
- Qmsecmethods can be a list of proposals separated by a ",".
- For qmsecmethods,
integrity=md5|sha1|sha256|aesgmac128|aesgmac192|aesgmac256|aesgcm128|aesgcm192
|aesgcm256 and
encryption=3des|des|aes128|aes192|aes256|aesgcm128|aesgcm192|aesgcm256.
- If aesgcm128, aesgcm192, or aesgcm256 is specified, it must be used for both
ESP integrity and encryption.
- Aesgmac128, aesgmac192, aesgmac256, aesgcm128, aesgcm192, aesgcm256, sha256
are only supported on Windows Vista SP1 and later.
- Qmpfs=mainmode uses the main mode key exchange setting for PFS.
- The use of DES, MD5, and DHGroup1 isn't recommended. These cryptographic
algorithms are provided for backward compatibility only.
- The default value for certmapping and excludecaname is "no".
- The " characters within CA name must be replaced with \'
- For auth1ca and auth2ca, the CA name must be prefixed by "CN=".
- catype can be used to specify the Certification authority type
"catype=root/intermediate".
- authnoencap is supported on Windows 7 and later.
- authnoencap means that the computers will only use authentication, and won't
use any per packet encapsulation or encryption algorithms to protect
subsequent network packets exchanged as part of this connection.
- QMPFS and authnoencap can't be used together on the same rule.
- AuthNoEncap must be accompanied by at least one AH or ESP integrity suite.
- applyauthz can only be specified for tunnel mode rules.
- exemptipsecprotectedconnections can only be specified for tunnel mode rules.
By setting this flag to "Yes", ESP traffic is exempted from the tunnel. AH
only traffic won't be exempted from the tunnel.
- Valuemin, when specified, for a qmsecmethod should be between 5 to 2880
minutes. Valuekb, when specified, for a qmsecmethod should be between 20480 KB
to 2147483647 KB.
- Certhash specifies the thumbprint, or hash of the certificate.
- Followrenewal specifies whether to automatically follow renewal links in
certificates. Only applicable for certificate section (requires certhash).
- Certeku specifies the comma separated list of EKU OIDs to match in the
certificate.
- Certname specifies the string to match for certificate name (requires
certnametype).
- Certnametype specifies the certificate field for the certname to be matched
against (requires certname).
- Certcriteriatype specifies whether to take the action with the certificate
when selecting the local certificate, validating the peer certificate, or
both.
- Within a computercert authentication mapping, multiple certificates can be
referenced by separating each entry by using the "|" character.
```

Firewall parameter remarks

add

```
- Add a new inbound or outbound rule to the firewall policy.
- Rule name should be unique and can't be "all".
- If a remote computer or user group is specified, security must be:
authenticate, authenc, authdynenc, or authnoencap.
- Setting security to authdynenc allows systems to dynamically negotiate the
use of encryption for traffic that matches a given Windows Defender Firewall
rule. Encryption is negotiated based on existing connection security rule
properties. This option enables the ability of a machine to accept the first
TCP or UDP packet of an inbound IPsec connection as long as it's secured, but
not encrypted, using IPsec. Once the first packet is processed, the server
renegotiates the connection and upgrade it so that all subsequent
communications are fully encrypted.
- If action=bypass, the remote computer group must be specified when dir=in.
- If service=any, the rule applies only to services.
- ICMP type or code can be "any".
- Edge can only be specified for inbound rules.
- AuthEnc and authnoencap can't be used together.
- Authdynenc is valid only when dir=in.
- When authnoencap is set, the security=authenticate option becomes an
optional parameter.
```

Mainmode parameter remarks

add

```
- Add a new mainmode rule to the firewall policy.
- Rule name should be unique and can't be "all".
- Computerpsk and computerntlm methods can't be specified together for auth1.
- The use of DES, MD5, and DHGroup1 isn't recommended. These cryptographic
algorithms are provided for backward compatibility only.
- The minimum main mode keylifetime is mmkeylifetime=1min. The maximum main
mode mmkeylifetime=2880min. The minimum number of sessions=0 sessions. The
maximum=2,147,483,647 sessions.
- The mmsecmethods keyword default sets the policy to: dhgroup2-aes128-
sha1,dhgroup2-3des-sha1
- Certhash specifies the thumbprint, or hash of the certificate.
- Followrenewal specifies whether to automatically follow renewal links in
certificates. Only applicable for certificate section (requires certhash).
- Certeku specifies the comma separated list of EKU OIDs to match in the
certificate.
- Certname specifies the string to match for certificate name (requires
certnametype).
- Certnametype specifies the certificate field for the certname to be matched
```

```
against (requires certname).
- Certcriteriatype specifies whether to take the action with the certificate
when selecting the local certificate, validating the peer certificate, or
both.
```

Monitor parameter remarks

delete

```
- This command deletes the matching security association as specified by the
source destination pair.
- Source and destination are each a single IPv4 or IPv6 address.
```

Examples

> **Note**
>
> For detailed usage instructions and additional examples for all parameters, use the `help` or `?` commands. These commands provide syntax details and guidance for specific operations.

consec

To add a connection security rule requiring authentication, run the following command:

Windows Command Prompt

```
netsh advfirewall consec add rule name="RequireAuth" endpoint1=any endpoint2=any
action=requireinrequestout
```

To add a new connection security rule that uses Kerberos for authentication, run the following command:

Windows Command Prompt

```
netsh advfirewall consec add rule name="KerberosAuthRule" endpoint1=any
endpoint2=any action=requireinrequireout auth1=computerkerb
```

To delete a connection security rule by its name, run the following command:

Windows Command Prompt

```
netsh advfirewall consec delete rule name="RuleName"
```

dump

To save the current advanced firewall configuration to a text file, run the following command:

Windows Command Prompt

```
netsh advfirewall dump > "C:\path\to\firewall_dump.txt"
```

To display the current firewall configuration in CMD, run the following command:

Windows Command Prompt

```
netsh advfirewall dump
```

To include the advanced firewall state in the main dump for diagnostics, run the following command:

Windows Command Prompt

```
netsh advfirewall dump | findstr "State"
```

export

To export the current firewall policy for backup, run the following command:

Windows Command Prompt

```
netsh advfirewall export "C:\folder\firewall_backup.wfw"
```

firewall

To add a rule allowing inbound traffic on port 8080, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall add rule name="Allow8080" protocol=TCP dir=in
localport=8080 action=allow
```

To block outgoing traffic to a specific IP, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall add rule name="BlockOutIP" protocol=TCP dir=out
remoteip=192.168.1.100 action=block
```

To remove a specific firewall rule by name, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall delete rule name="MyRule"
```

help

To get help for all firewall commands, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall ?
```

To find out syntax details for adding a firewall rule, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall add rule ?
```

import

To import a saved firewall configuration from a file, run the following command:

Windows Command Prompt

```
netsh advfirewall import "C:\folder\firewall_config.wfw"
```

mainmode

To add a mainmode rule for security negotiation, run the following command:

Windows Command Prompt

```
netsh advfirewall mainmode add rule name="MyMainmodeRule" auth1=computerkerb
```

To show detailed settings of mainmode rules, run the following command:

Windows Command Prompt

```
netsh advfirewall mainmode show rule name=all
```

To remove a mainmode rule by specifying its name, run the following command:

Windows Command Prompt

```
netsh advfirewall mainmode delete rule name="MyMainmodeRule"
```

monitor

To monitor active connection security associations, run the following command:

Windows Command Prompt

```
netsh advfirewall monitor show mmsa
```

To display statistics and active connections, run the following command:

Windows Command Prompt

```
netsh advfirewall monitor show firewall
```

reset

To reset the firewall settings without additional confirmation, run the following command:

Windows Command Prompt

```
netsh advfirewall reset
```

set

To disable logging of dropped packets, run the following command:

Windows Command Prompt

```
netsh advfirewall set currentprofile logging droppedconnections disable
```

To enable notifications for blocked applications, run the following command:

Windows Command Prompt

```
netsh advfirewall set currentprofile settings allowunicastresponsetomulticast
enable
```

show

To display all active firewall rules, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall show rule name=all
```

To show detailed information on a specific rule by name, run the following command:

Windows Command Prompt

```
netsh advfirewall firewall show rule name="MyRuleName" verbose
```

netsh branchcache

07/25/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh branchcache` command is used in Windows to manage BranchCache, a feature that speeds up access to data in branch offices.

Syntax

```
netsh branchcache dump | exportkey | flush | help | importkey | reset | set | show
| smb | ?
```

```
netsh branchcache exportkey [outputfile=]<File Path> [passphrase]=<Pass Phrase>
netsh branchcache importkey [inputfile=]<File Path> [passphrase]=<Pass Phrase>
```

```
netsh branchcache set [cachesize | key | localcache | publicationcache |
publicationcachesize | service]
netsh branchcache set [cachesize | publicationcachesize] [size=]{DEFAULT|<number>}
[[percent=]{TRUE|FALSE}]
netsh branchcache set key [[passphrase=]<Pass Phrase>]
netsh branchcache set [localcache | publicationcache] [directory=]{DEFAULT|<File
Path>}
netsh branchcache set service [mode=]
{DISABLED|LOCAL|DISTRIBUTED|HOSTEDSERVER|HOSTEDCLIENT}
                              [[location]=<Host Name>]
                              [[clientauthentication]={DOMAIN|NONE}]
                              [[serveonbattery]={TRUE|FALSE}]
```

```
netsh branchcache show [hostedcache | localcache | publicationcache | status]
netsh branchcache show status [[detail=]{BASIC|ALL}]
```

```
netsh branchcache smb [dump | help | set | show]
```

```
netsh branchcache smb set [latency=]<number>
```

Parameters

Expand table

Parameter	Description
dump	Dumps the configuration details and settings related to BranchCache on your device.
exportykey `<FilePath>\<FileName>` `<Passphrase>`	Exports the key which the BranchCache service uses to protect content information. The key can later be imported on another device by using the `importkey` command. The output file is a directory path and name of the file where the key should be exported. The passphrase is required to import the key.
flush	Clears or deletes the local cache of content that BranchCache stores on your device.
help *or* ?	Displays help information for the given command.
importkey `<FilePath>\<FileName>` `<Passphrase>`	Imports a key for the BranchCache service to use to protect content information. The key must be previously exported using the `exportkey` command. If the service is currently running, this command restarts it in order to begin using the new key.
reset	Restarts the BranchCache service and clears both the content cache on the client side and any locally stored publication data. This command resets configuration settings to their default settings.
set [cachesize | publicationcachesize] [size=] {DEFAULT|`<number>`} [[percent=]{`TRUE`|`FALSE`}]	cachesize - Sets the size of the local cache. publicationcachesize - Sets the size of the local publication cache. size - Indicates the cache size as a percentage or an exact `<number>` of bytes. If set to `default`, the default configuration is restored. percent - Specifies whether the size value is interpreted as a percentage of the hard disk size or in the exact `<number>` of bytes. This parameter is ignored if the size

Parameter	Description
	value is set to `default`. `True` - Indicates the size is a percentage of the size of the hard disk. `False` - Indicates the size is an exact number of bytes (default option).
set key [[passphrase=]<Passphrase>]	key - Generates a new content information key. A passphrase is used to generate the key. If a passphrase isn't provided, a random key is generated. If two keys generated using the same passphrase, they're identical. A passphrase can be used to duplicate the same key on another device.
set [localcache | publicationcache] [directory=] {DEFAULT|<File Path>}	localcache - Sets the location of the local cache. publicationcache - Sets the location of the local publication cache. directory - Is the path to the directory in which the local cache or publication cache should be stored. When set to `default`, it restores the default cache location. This service is restarted, if necessary, in order to complete the change. Any existing cache files are moved to the new location.
set service [mode=]{DISABLED | LOCAL | DISTRIBUTED | HOSTEDSERVER | HOSTEDCLIENT} [[location]=<Host Name>] [[clientauthentication]={DOMAIN|NONE}] [[serveonbattery]={TRUE|FALSE}]	mode - Configures the BranchCache service mode. • `disabled` - Disables the BranchCache service. • `local` - Enables caching of data on the local device only. • `distributed` - Activates distributed caching, allowing data to be cached across multiple computers in a peer-to-peer configuration. • `hostedserver` - Sets the device as a Hosted Cache Server to serve cached data to clients. • `hostedclient` - Sets the device as a client of a Hosted Cache Server to receive and store cached data. location - Sets the location of the Hosted Cache Server, which is only necessary if the mode is set to `hostedclient` and is ignored in other modes. clientauthentication - Sets the method used by a Hosted Cache Server to authenticate connected clients. This parameter is only relevant when the mode is set to `hostedserver` and is ignored otherwise.

Parameter	Description
	• `domain` - Requires clients to be members of the same domain as the Hosted Cache Server for authentication. • `none` - No authentication method is implemented for clients. serveonbattery - Allows the local client to respond to peer requests for cached data while operating on battery power. This parameter is applicable only when the mode is set to `distributed` and is ignored otherwise. • `true` - Enables the servicer to provide cached data while on battery. • `false` - Disables the servicing of cached data requests when on battery power.
show [hostedcache | localcache | publicationcache | status]	`hostedcache` - Displays the configuration information status of the Hosted Cache server if the device is set up to function as one. `localcache` - Provides details about the local cache on the device, such as the size of the cache on the hard disk and cache location. `publicationcache` - Shows information about the publication cache settings and contents. `status` - Provides a summary of the current BranchCache configuration and operational status, including the mode BranchCache is set to.
show status [[detail=]{`BASIC`|`ALL`}]	detail - Specifies the level of detail for the displayed information. `basic` - Shows only essential information (default option). `all` - Displays comprehensive information available.
smb [dump] smb [set | show] latency [latency=]\`<number>`	Manages the BranchCache settings related to the Server Message Block (SMB) protocol. `dump` - Outputs a script that represents the current configuration settings. `set` - Modifies or assigns values to specific configuration parameters. `show` - Displays the current configuration parameters and their values. `latency` - Specifies the minimum link latency, in

Parameter	Description
	milliseconds, to the branch office before SMB utilizes BranchCache for caching.

Examples

dump

To dump the BranchCache configuration to a file, type the following command:

Windows Command Prompt

```
netsh branchcache dump > C:\BranchCacheConfig.txt
```

exportkey

To export the client key to a file, type the following command:

Windows Command Prompt

```
netsh branchcache exportkey C:\Keys\BranchCacheKey.bckey
```

To export a specific key with a password, type the following command:

Windows Command Prompt

```
netsh branchcache exportkey C:\Keys\BranchCacheKey.bckey mypassword
```

flush

To clear all cached content, type the following command:

Windows Command Prompt

```
netsh branchcache flush
```

help

To display help for a specific BranchCache command, such as exportkey, type the following command:

Windows Command Prompt

```
netsh branchcache exportkey help
```

> **Note**
>
> Both the `help` and `?` commands are used interchangeably.

importkey

To import a BranchCache key from a file, type the following command:

Windows Command Prompt

```
netsh branchcache importkey C:\Keys\BranchCacheKey.bckey
```

To import a key from a file with a password, type the following command:

Windows Command Prompt

```
netsh branchcache importkey C:\Keys\BranchCacheKey.bckey mypassword
```

reset

To reset BranchCache to its default state, type the following command:

Windows Command Prompt

```
netsh branchcache reset
```

set

To set the BranchCache mode to `distributed`, type the following command:

Windows Command Prompt

```
netsh branchcache set mode=distributed
```

To set the cache size to a specific percentage, such as `10%`, type the following command:

Windows Command Prompt

```
netsh branchcache set cachesize=10%
```

To configure a client to use a hosted cache server, type the following command:

Windows Command Prompt

```
netsh branchcache set service mode=HOSTEDCLIENT location=HOSTEDSERVERNAME
```

show

To show the current BranchCache status, type the following command:

Windows Command Prompt

```
netsh branchcache show status
```

To show the current BranchCache configuration, type the following command:

Windows Command Prompt

```
netsh branchcache show configuration
```

smb

To set the latency threshold to `80` milliseconds, type the following command:

Windows Command Prompt

```
netsh branchcache smb set latency=80
```

See also

- BranchCache
- BranchCache PowerShell module

netsh bridge

10/10/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh bridge` command configures network adapter bridge settings in Windows. By creating a network bridge, you can link two or more network segments, allowing devices on separate segments to communicate seamlessly as if they're part of a single network.

Syntax

```
netsh bridge [add | create | destroy | dump | help | list | remove | set | show |
?]
```

```
netsh bridge add <Adapter ID> to <Bridge GUID>
```

```
netsh bridge create [<Adapter ID #1> <Adapter ID #2>]
```

```
netsh bridge destroy <Bridge GUID>
```

```
netsh bridge list
```

```
netsh bridge remove <Adapter ID> from <Bridge GUID>
netsh bridge remove all from <Bridge GUID>
```

```
netsh bridge set adapter [id=]<integer> [[forcecompatmode=]enable|disable]
```

```
netsh bridge show adapter
```

Parameters

Expand table

Command	Description
bridge `add`	Adds a network adapter, specified by its Adapter ID, to an existing bridge identified by its Bridge GUID.
bridge `create`	Creates a new network bridge that includes two specified network adapters. The newly created bridge is assigned its own unique GUID, which can be retrieved using the `netsh bridge list` command.
bridge `destroy`	Removes all network adapters from the specified bridge, identified by its bridge GUID, and deletes the bridge.
bridge `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
bridge `list`	Displays all the created bridges that are uniquely identified by the bridge GUID.
bridge `remove`	Removes a network adapter, specified by its Adapter ID, from a bridge identified by its Bridge GUID. Using the `all` parameter removes all network adapters from the specified bridge and deletes the bridge.
bridge `set adapter`	Modifies the bridge configuration for the specified adapter. `id` - The identifier of the network adapter to configure. To view available adapter IDs, use the `netsh bridge show adapter` command. `forcecompatmode` - Specifies whether to enable or disable Layer 3 compatibility mode for the adapter: - `enable`: Turns on Layer 3 mode. - `disable`: Turns off Layer 3 mode.
bridge `show adapter`	Displays a list of all available network adapters, including their Adapter IDs, current settings, and status flags, indicating which adapters can participate in a network bridge.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To add a network adapter (Adapter ID 3) to an existing bridge, with Bridge GUID `{12345678-ABCD-EF01-2345-6789ABCDEF03}`, run the following command:

Windows Command Prompt

```
netsh bridge add adapter=3 bridge={12345678-ABCD-EF01-2345-6789ABCDEF03}
```

To create a new network bridge with two adapters (Adapter ID 1 and Adapter ID 2), run the following command:**

Windows Command Prompt

```
netsh bridge create adapter=1 adapter=2
```

To destroy a bridge and remove all its adapters (Bridge GUID `{12345678-ABCD-EF01-2345-6789ABCDEF02}`), run the following command:

Windows Command Prompt

```
netsh bridge destroy bridge={12345678-ABCD-EF01-2345-6789ABCDEF02}
```

To display all created network bridges and their GUIDs, run the following command:

Windows Command Prompt

```
netsh bridge list
```

To remove a network adapter (Adapter ID 2) from an existing bridge, with Bridge GUID `{12345678-ABCD-EF01-2345-6789ABCDEF02}`, run the following command:

Windows Command Prompt

```
netsh bridge remove adapter=2 bridge={12345678-ABCD-EF01-2345-6789ABCDEF02}
```

To display all available network adapters, run the following command:

Windows Command Prompt

```
netsh bridge show adapter
```

To modify an adapter (Adapter ID 1) in a bridge to enable Layer 3 compatibility mode, run the following command:

Windows Command Prompt

```
netsh bridge set adapter id=1 forcecompatmode=enable
```

See also

- Command-Line Syntax Key

netsh delete

07/25/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh delete` command is used to remove helper Dynamic Link Libraries (DLL) allowing for more specialized network configurations.

Syntax

```
netsh delete helper [file]
```

Parameters

Expand table

Command	Description
helper	Calls the helper DLL file.

Example

To remove a helper DLL located at **C:\dlls\HelperDLL.dll**, run the following command:

Windows Command Prompt

```
netsh delete helper C:\dlls\HelperDLL.dll
```

netsh dhcpclient

07/29/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh dhcpclient` command is used to configure and manage settings for the DHCP client, including enabling tracing and exporting diagnostic logs.

Syntax

```
netsh dhcpclient [list | help | ? | trace]
netsh dhcpclient list
netsh dhcpclient trace <enable> <disable>
netsh dhcpclient trace <dump>
```

Parameters

Expand table

Command	Description
list	Lists the available commands in the `netsh dhcpclient` context.
trace `<enable>` | `<disable>` `<dump>`	Controls tracing for the DHCP client, allowing you to enable or disable it. This feature is helpful for troubleshooting as it generates detailed logs of DHCP client activities. You can retrieve the last 100 traces to review recent DHCP interactions. `enable` - Enables tracing for the DHCP client. `disable` - Disables tracing for the DHCP client. `dump` - Exports the last 100 traces.
help or ?	Provides help for the list of commands available in the `netsh dhcpclient` context.

Example

To enable tracing for the DHCP client, run the following command:

Windows Command Prompt

```
netsh dhcpclient trace enable
```

To dump the last 100 traces for review in a specified path, run the following command:

Windows Command Prompt

```
netsh dhcpclient trace dump > C:\NetshTrace\dhcp_trace.txt
```

netsh dnsclient

07/28/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh dnsclient` command is used to configure and manage Domain Name System (DNS) client settings on a Windows computer. It allows administrators to add, delete, set, and display DNS server addresses, encryption settings, and global DNS client options. This command is useful for automating DNS configuration, troubleshooting name resolution issues, and managing advanced DNS features such as DNS-over-HTTPS (DoH) and DNS-over-TLS (DoT).

Syntax

```
netsh dnsclient [add | delete | dump | help | set | show | ?]
```

```
netsh dnsclient add dnsserver [name=]<string> [address=]<IP address> [[index=]
<integer>] [[validate=]yes|no]
netsh dnsclient add encryption [server=]<IP address> [dohtemplate=]<string>
[dothost=]<hostname>:<port> [[autoupgrade=]yes|no] [[udpfallback=]yes|no]
netsh dnsclient add global [[doh=]yes|no|auto] [[dot=]yes|no] [[ddr=]yes|no]
netsh dnsclient add interface [name=]<string> [[ddr=]yes|no]
[[ddrfallback=]yes|no]
```

```
netsh dnsclient delete dnsserver [name=]<string> [address=]<IP address> [[index=]
<integer>] [[validate=]yes|no]
netsh dnsclient delete encryption [server=]<IP address> [[protocol=]doh|dot]
netsh dnsclient delete global [global=](doh|dot|ddr|all)
netsh dnsclient delete interface [name=]<string> [setting=](ddr|all)
```

```
netsh dnsclient <dump>
```

```
netsh dnsclient set dnsserver [name=]<string> [source=]dhcp|static [[address=]<IP
address>|none] [[register=]none|primary|both] [[validate=]yes|no]
```

```
netsh dnsclient set encryption [server=]<IP address> [dohtemplate=]<string>
[dothost=]<hostname>:<port> [[autoupgrade=]yes|no] [[udpfallback=]yes|no]
netsh dnsclient set global [[doh=]yes|no|auto] [[dot=]yes|no] [[ddr=]yes|no]
netsh dnsclient set interface [name=]<string> [[ddr=]yes|no]
[[ddrfallback=]yes|no]
```

```
netsh dnsclient show encryption [[server=]<IP address>]
netsh dnsclient show [global | state]
```

Parameters

Expand table

Parameter	Description
add dnsserver `name=<string>` `address=<IP address>` `index=<integer>` `validate=<yes|no>`	Adds a new DNS server IP address to the statically configured list. By default, the DNS server is appended to the end of the list. If an index is specified, the server is inserted at that position, shifting other entries as needed. If DNS servers were previously assigned via DHCP, the new address replaces the existing list. When the validate option is set to `yes`, the newly added DNS server is validated. **name** - The name or index of the network interface to which the DNS server is added. **address** - The IP address of the DNS server to add. **index** - Sets the order of preference for the DNS server address. **validate** - Determines whether the DNS server address should be validated. Defaults to `yes`.
add encryption `server=<IP address>` `dohtemplate=<string>` `dothost=<hostname>:<port>` `autoupgrade=<yes|no>` `udpfallback=<yes|no>`	Adds a new encrypted DNS server to the list of statically configured secure resolvers. **server** - Specifies the IP address of the encrypted DNS server to add. **dohtemplate** - Defines the DoH template to use for the server. **dothost** - Sets the DoT host for the server. Format: `hostname:port`. Use `:` for the default configuration. **autoupgrade** - Indicates whether queries to this server should automatically upgrade to DoH or DoT. **udpfallback** - Determines if UDP/TCP should be used for this server if DoH queries fail.
add global `doh=<yes|no|auto>` `dot=<yes|no>` `ddr=<yes|no>`	Adds a global DNS configuration

Parameter	Description
	doh - Configures the global DoH settings. `No` disables DoH for the DNS client. `Yes` enables DoH usage according to interface, server, or other configurations. `Auto` forces all recognized DoH servers to use only DoH. **dot** - Configures the global DoT settings. `No` - disables DoT for the DNS client. `Yes` enables DoT usage according to interface, server, or other configurations. **ddr** - Configures the global Discovery of Designated Resolvers (DDR) settings. `No` disables DDR for the DNS client. `Yes` enables DDR usage according to interface, server, or other configurations.
add interface `name=<string>` `ddr=<yes|no>` `ddrfallback=<yes|no>`	Adds or configures DNS client settings for a specific network interface. **name** - The name or index of the network interface to configure. **ddr** - Enables or disables DDR for unencrypted DNS servers on the specified adapter. `Yes` performs DDR on all unencrypted servers configured for the adapter. `No` doesn't perform DDR by default on unencrypted servers for the adapter. **ddrfallback** - Specifies whether to allow fallback to unencrypted name resolution if secure resolution using DDR fails. `Yes` enables fallback to unencrypted name resolution. `No` disables fallback to unencrypted name resolution (default: disabled).
delete dnsserver `name=<string>` `address=<IP address>` `index=<integer>` `validate=<yes|no>`	Removes statically configured DNS server IP addresses from a specified interface. If the `validate` option is set to `yes`, the remaining DNS servers are validated after deletion. **name** - Specifies the name or index of the network interface from which DNS servers are deleted. **address** - Indicates which DNS server addresses to remove. You can specify a particular IP address to delete a single DNS server, or use `all` to remove all configured DNS server addresses from the interface. **validate** - Determines whether the DNS server settings should be validated after deletion. By default, validation is enabled (`yes`).
delete encryption `server=<IP address>` `protocol=<doh|dot>`	Deletes encrypted DNS server configurations from the list of statically configured secure resolvers. **server** - Specifies the IP address for the encrypted DNS server removal. **protocol** - Specifies the encryption protocol removal from the server.
delete global `global=<doh|dot|ddr|all>`	Restores the DNS settings to default values. **all** - Restores all settings to their default values.

Parameter	Description
	doh - Restores the default DoH global setting. **dot** - Restores the default DoT global setting. **ddr** - Restores the default DDR global setting.
delete interface `name=<string>` `setting=<ddr|all>`	Restores the interface DNS settings to default values. **name** - Specifies the name or index of the network interface whose DNS settings you want to restore to default. **setting** - Indicates which DNS setting on the interface should be reset. Use `all` to restore all settings to their default values, or `ddr` to restore only the DDR setting to its default.
dump	Outputs a configuration script containing the current DNS client settings.
set dnsserver `name=<string>` `source=<dhcp|static>` `address=<IP address>|none` `register=<none|primary|both>` `validate=<yes|no>`	Configures the DNS server settings to use either DHCP or a static list of DNS server IP addresses for the specified interface. The `address` option is available only when the source is set to `static`. If the `validate` option is set to `yes`, the newly configured DNS server is validated. **name** - Specifies the name or index of the network interface. **source** - Determines how DNS servers are configured for the interface. Use `dhcp` to obtain DNS servers automatically via DHCP, or `static` to specify DNS servers manually. **address** - Sets the DNS server IP address. Provide an IP address to add a DNS server, or use `none` to clear the list of DNS servers. **register** - Controls Dynamic DNS registration. Use `none` to disable registration, `primary` to register only under the primary DNS suffix, or `both` to register under both the primary and connection-specific suffixes. **validate** - Indicates whether the DNS server setting should be validated. Default value is `yes`.
set encryption `server=<IP address>` `dohtemplate=<string>` `dothost=<hostname>:<port>` `autoupgrade=<yes|no>` `udpfallback=<yes|no>`	Configures the encrypted DNS server settings in the list of statically defined secure resolvers. **server** - Specifies the IP address of the encrypted DNS server to modify. **dohtemplate** - Defines the DoH template to use for the server. **dothost** - Sets the DoT host for the server. Provide as `hostname:port`. Use `:` for the default configuration. **autoupgrade** - Indicates whether queries to this server should automatically upgrade to DoH or DoT. **udpfallback** - Determines if UDP/TCP should be used for this server if DoH queries fail.

Parameter	Description
set global `doh=<yes|no|auto>` `dot=<yes|no>` `ddr=<yes|no>`	Configures the global DNS configuration. **doh** - Is the global DoH setting. `No` forbids the use of DoH by the DNS client. `Yes` allows DoH use based on interface, server, or other configurations. `Auto` forces all well-known DoH servers to use only DoH. **dot** - Is the global DoT setting. `No` forbids the use of DoT by the DNS client. `Yes` allows DoT use based on interface, server, or other configurations. **ddr** - Is the global DDR setting. `No` disables the use of DDR by the DNS client. `Yes` enables DDR use based on interface, server, or other configurations.
set interface `name=<string>` `ddr=<yes|no>` `ddrfallback=<yes|no>`	Configures interface specific DNS settings. **name** - Specifies the name or index of the interface to be configured. **ddr** - Specifies the interface-specific DDR settings. `Yes` performs DDR on all unencrypted servers configured on the specified adapter. `No` doesn't perform DDR by default on unencrypted servers configured on the specified adapter. **ddrfallback** - Specifies if servers should fall back to unencrypted name resolution if secure resolution using DDR-retrieved settings fails, provided per-adapter DDR is enabled. Fallback is disabled by default. `Yes` enables fallback to unencrypted name resolution. `No` disables fallback to unencrypted name resolution.
show encryption `server=<IP address>`	Displays the encrypted DNS server settings from the list of statically configured secure resolvers. If no server is specified, information for all encrypted DNS servers is shown. **server** - Specifies the encrypted DNS server by IP address.
show `<global>`	Displays all configured DNS global settings.
show `<state>`	Displays the current configuration and status of the DNS client.
help or ?	Provides help for the list of commands available in the `netsh dnsclient` context.

Examples

To add a DNS configuration or setting (for example, a specific DNS server address), run the following command:

```
Windows Command Prompt
```

```
netsh dnsclient add dnsserver Ethernet=192.168.10.1
```

To delete a DNS configuration or setting (such as a specific DNS server address), run the following command:

Windows Command Prompt

```
netsh dnsclient delete dnsserver Ethernet=192.168.10.1
```

To set a static DNS server address for a specific network interface, run the following command:

Windows Command Prompt

```
netsh dnsclient set dnsserver name=Ethernet source=static address=192.168.10.1
register=primary validate=yes
```

To show globally configured DNS settings, run the following command:

Windows Command Prompt

```
netsh dnsclient show global
```

netsh dump

07/25/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh dump` command generates a script that saves the system's current network settings. By running this command, you produce a script that can be used to recreate the same settings later or on a different device.

Syntax

```
netsh dump
```

Parameters

Expand table

Command	Description
dump	Exports the network configuration details and settings to your device.

Example

To export your network configuration settings to a file named "netsh_config.txt" and save it in **C:\NetConfig**, run the following command:

Windows Command Prompt

```
netsh dump > "C:\NetConfig\netsh_config.txt"
```

netsh exec

07/25/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh exec` command executes a script file. This script file contains a series of `netsh` commands that are run sequentially when the script is executed. It automates network configuration tasks by performing complex operations without you needing to manually enter the entries.

Syntax

```
netsh exec <ScriptFile>
```

Parameters

Expand table

Command	Description
exec `<ScriptFile>`	Loads and runs the target script file.

Example

To load and run your script from the location **C:\NetshScripts\network_script.txt**, run the following command:

Windows Command Prompt

```
netsh exec "C:\NetshScripts\network_script.txt"
```

netsh http

08/19/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh http` command enables you to configure and manage HTTP settings for the HTTP Service in Windows. This service handles HTTP traffic for applications and system components, allowing you to control aspects such as SSL certificates, IP listen lists, cache parameters, and timeouts. Use this command to fine-tune HTTP behavior, enhance security, and troubleshoot network issues related to HTTP traffic.

Syntax

```
netsh http [add | delete | dump | flush | help | show | update | ?]
```

```
netsh http add cacheparam [type=](cacherangechunksize|maxcacheresponsesize)
                          [value=]<ulong>

netsh http add iplisten [ipaddress=]<IP address>

netsh http add setting [settingtype=]sslthrottle
                       [value=]<ulong>

netsh http add timeout [timeouttype=]idleconnectiontimeout|headerwaittimeout
                       [value=]<u-short>
```

```
netsh http add sslcert [hostnameport=<name:port> | ipport=<ipaddr:port> | ccs=
<port>] | json=[filepath]
                       [appid=<GUID>] [certhash=<string>] [certstorename=<string>]
[verifyclientcertrevocation=enable|disable]
                       [verifyrevocationwithcachedclientcertonly=enable|disable]
[usagecheck=enable|disable]
                       [revocationfreshnesstime=<u-int>] [urlretrievaltimeout=<u-
int>] [sslctlidentifier=<string>]
                       [sslctlstorename=<string>] [dsmapperusage=enable|disable]
[clientcertnegotiation=enable|disable]
                       [reject=enable|disable] [disablehttp2=enable|disable]
[disablequic=enable|disable]
                       [disablelegacytls=enable|disable]
[disabletls12=enable|disable] [disabletls13=enable|disable]
```

```
                        [disableocspstapling=enable|disable]
[enabletokenbinding=enable|disable] [logextendedevents=enable|disable]
                        [enablesessionticket=enable|disable]
[disablesessionid=enable|disable]
```

```
netsh http add urlacl [url=]<string> [user=]<string> [listen=]yes|no
[delegate=]yes|no [sddl=]<string>
```

```
netsh http delete cache [url=<string>] [recursive=yes|no]
netsh http delete iplisten [ipaddress=<IP address>]
netsh http delete setting [settingtype=]sslthrottle
netsh http delete sslcert ipport=<ipaddr:port> | hostnameport=<name:port> | ccs=
<port>
netsh http delete timeout [timeouttype=](idleconnectiontimeout|headerwaittimeout)
netsh http delete urlacl [url=]<string>
```

```
netsh http dump
```

```
netsh http flush logbuffer
```

```
netsh http show <cacheparam> <iplisten> <setting> <timeout>
netsh http show cachestate [url=<string>]
netsh http show servicestate [view=session|requestq|client] [verbose=yes|no]
netsh http show sslcert [ipport=<ipaddr:port> | hostnameport=<name:port> | ccs=
<port> | scopedccs=<name:port>] [json=enable]
netsh http show urlacl [url=<string>]
```

```
netsh http update sslcert [hostnameport=<name:port> | ipport=<ipaddr:port> | ccs=
<port>] | json=[filepath]
                          [appid=<GUID>] [certhash=<string>] [certstorename=
<string>] [verifyclientcertrevocation=enable|disable]

[verifyrevocationwithcachedclientcertonly=enable|disable]
[usagecheck=enable|disable]
                          [revocationfreshnesstime=<u-int>] [urlretrievaltimeout=
```

```
<u-int>] [sslctlidentifier=<string>]
                          [sslctlstorename=<string>]
[dsmapperusage=enable|disable] [clientcertnegotiation=enable|disable]
                          [reject=enable|disable] [disablehttp2=enable|disable]
[disablequic=enable|disable]
                          [disablelegacytls=enable|disable]
[disabletls12=enable|disable] [disabletls13=enable|disable]
                          [disableocspstapling=enable|disable]
[enabletokenbinding=enable|disable] [logextendedevents=enable|disable]
                          [enablesessionticket=enable|disable]
[disablesessionid=enable|disable]
```

```
netsh http show update sslpropertyex [hostnameport=<name:port> | ipport=
<ipaddr:port> | ccs=<port>] |
                                     [propertyid=0 receivewindow=<bytes>]
[propertyid=1 maxsettingsperframe=<value> maxsettingsperminute=<value>]
```

Parameters

Expand table

Command	Description
http `add cacheparam`	Configures and adds specific parameters to the HTTP Service cache. `type` - Specifies the type of parameter to be configured. Options are `cacherangechunksize` and `maxcacheresponsesize`. `value` - Denotes the parameter value, measured in bytes. If using a hexadecimal representation, prefix the value with `0x`.
http `add iplisten`	Adds a new IP address to the IP listen list for the HTTP Service. This list specifies the IP addresses to which the HTTP Service binds, excluding port numbers. The address '0.0.0.0' represents all IPv4 addresses, while '::' represents all IPv6 addresses. The format uses `ipaddress=IP_Address`.
http `add setting`	Adds a global configuration setting to the HTTP Service. `settingtype=sslthrottle` - Specifies the throttling limit for SSL connections. `value` - requires a non-negative integer value (ulong) to set the desired throttling level.
http `add sslcert`	Adds SSL certificate binding for a specified IP address or hostname and port, along with corresponding client certificate policies, to securely manage HTTPS connections for the HTTP Service.

Command	Description
	See Remarks for parameter description details.
http `add timeout`	Defines a global timeout setting for the HTTP Service. `timeouttype=` options include: • `idleconnectiontimeout` - Specifies the maximum duration that an idle connection can be maintained before being closed. • `headerwaittimeout` - Defines the maximum time to wait for the complete set of HTTP headers to be received. The `value` parameter requires a non-negative integer (u-short) representing the timeout duration in *seconds*.
http `add urlacl`	Allows non-administrator users and accounts to register the URL. The access permissions can be defined either by specifying an NT account name along with the `listen` and `delegate` parameters, or by using a Security Descriptor Definition Language (SDDL) string for the Discretionary Access Control List (DACL). `url=` - Specifies the fully qualified URL that you want to associate with the Access Control List (ACL). This must include the protocol (for example, `http://` or `https://`), domain, and any necessary port or path segments. `user=` - Identifies the user or user-group name that is granted access. This can be in the format of a single user name, a user group, or a security principal. `listen` - Determines whether the specified user has permission to register the URL. Options include: • `yes` - Grants the user permission to register the specified URL (default setting). • `no` - Prevents the user from registering the specified URL. `delegate` - Determines whether the specified user has permission to delegate the URL. Options include: • `yes` - Allows the user to delegate URL management to other users. • `no` - Prevents the user from delegating URL management to others (default setting). `sddl` - The SDDL string that provides a detailed description of the Discretionary Access Control List (DACL) associated with the URL ACL, specifying the permissions and access rights for users and groups.
http `delete cache`	Clears the entire URI cache or removes specific entries based on the specified URI. `url=` - Specifies the URL entry in the HTTP cache that you want to delete. `recursive` - Accepts a value of `yes` or `no`. When set to `yes`, it removes not only the specified URL entry but also all entries under that URL, effectively clearing the cache for the specified URL and any sub-URLs or associated resources. When set to `no`, only the specified URL is removed from the cache without affecting any subentries.

Command	Description
http `delete iplisten`	Removes an IP address from the IP listen list. `ipaddress=` - Specifies the IP address that you want to remove from the HTTP listener list.
http `delete setting`	Removes a global configuration setting, causing the HTTP Service to revert to its default behavior for that particular parameter. `settingtype=sslthrottle` - Removes the SSL throttle setting.
http `delete timeout`	Removes a global timeout setting, restoring the HTTP Service's timeout behavior to its default values. • `timeouttype=idleconnectiontimeout` - Specifies that you want to remove the setting for the idle connection timeout. The idle connection timeout defines how long an idle (inactive) TCP connection should be kept open before it's closed. Removing this setting reverts any custom idle timeout value to its default state. • `timeouttype=headerwaittimeout` - Specifies that you want to remove the setting for the header wait timeout. The header wait timeout determines how long the server waits for an HTTP request header to be received before timing out. Removing this setting reverts any custom header wait timeout value to its default state.
http `delete urlacl`	Deletes a reserved URL, freeing up the specified URL from being exclusively used by the HTTP Service. `url=` - Specifies the URL reservation that you want to delete.
http `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
http `flush logbuffer`	Clears the internal buffers for the log files.
http `show cacheparam` | `iplisten` | `setting` | `timeout`	• Displays the cache parameters of the HTTP Service measured in bytes (`cacheparam`). • Lists all IP addresses currently included in the IP listen list (`iplisten`). • Displays the current configuration settings of the HTTP Service in seconds (`setting`). • Displays the timeout values set for the HTTP Service in seconds (`timeout`).
http `show cachestate`	Enumerates all resources and their properties stored in the HTTP response cache, or it provides detailed information about a specific resource and its associated properties. `url=` - Specifies a particular URL for which you want to view the cache state.

Command	Description
http `show servicestate`	Displays a current snapshot of the HTTP Service. `view=` - Specifies which aspect of the HTTP service state you wish to view. You can choose one of the following options: • `session` - Displays information about active sessions, including connection and session details. • `requestq` - Shows the state of the request queue, providing insights into the number of requests being processed and queued. • `client` - Provides information about client connections, such as the number of connected clients and connection details. `verbose=` - Determines the level of detail included in the output. • `yes` - Provides detailed information, offering a more comprehensive view of the selected service state. • `no` - Presents a summary with less detail, useful for a quick overview of the service state.
http `show sslcert`	Lists SSL server certificate bindings along with their associated client certificate policies. If no specific binding is provided, all available bindings are displayed. `scopedccs=name:port` - Specifies a scoped central certificate store (CCS) entry by providing a hostname and port. Similar to the `hostnameport` parameter, it focuses the output on certificates related to a specific scoped CCS configuration. Available subparameters are: • `ipport=ipaddr:port` • `hostnameport=name:port` • `ccs=port` • `json=enable` See Remarks for parameter description details.
http `show urlacl`	Displays the DACLs for a specified reserved URL or for all reserved URLs, offering insights into their access permissions. `url=` - Specifies the URL reservation for which you want to view details.
http `update sslcert`	Updates an existing SSL server certificate binding and associated client certificate policies for a specified IP address or hostname and port. See Remarks for parameter description details.
http `update sslpropertyex`	Modifies the parameters of an existing SSL server certificate binding and its related client certificate policies for a given IP address or hostname and port. • `propertyid=0 receivewindow=<bytes>` - Updates the receive window size for SSL connections. `propertyid=0` identifies this specific property, and `receivewindow=`

Command	Description
	`<bytes>` sets the size of the receive window in bytes. • `propertyid=1 maxsettingsperframe=<value> maxsettingsperminute=<value>` - Updates the HTTP/2 settings related to SSL connections. Here, `propertyid=1` specifies this particular property, while `maxsettingsperframe=<value>` sets the maximum number of settings frames allowed per frame, and `maxsettingsperminute=<value>` sets the maximum number of settings frames permitted per minute. Other available subparameters are: • `hostnameport=name:port` • `ipport=ipaddr:port` • `ccs=port` See Remarks for parameter description details.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Remarks

- `hostnameport=name:port`: Specifies the binding using a hostname and port number. This option binds the SSL certificate to a specific hostname (such as `www.example.com`) on a specified port (typically port 443 for HTTPS).
- `ipport=ipaddr:port`: Specifies the binding using an IP address and port number. This binds the SSL certificate to a particular IP address (such as `192.168.1.1`) and a port.
- `ccs=port`: Refers to a Central Certificate Store (CCS) port. In this context, CCS allows for centralized management of SSL certificates across multiple servers, using a specific port for SSL bindings.
- `json=filepath`: Uses a JSON file for configuring the SSL binding. This option allows for complex or bulk configuration settings to be specified in JSON format that the HTTP Service can parse and apply.
- `appid=GUID`: The unique identifier (GUID) for the application associated with the certificate binding.
- `certhash=<string>`: The certificate hash (often represented as a thumbprint) of the SSL certificate to be bound.
- `certstorename=<string>`: The name of the certificate store where the SSL certificate is located (for example, `MY` for the personal store).

- `verifyclientcertrevocation=`: Enables or disables verification of client certificate revocation status. Options are `enable` or `disable`.
- `verifyrevocationwithcachedclientcertonly=`: Determines whether only cached client certificates are used for revocation checking. Options are `enable` or `disable`.
- `usagecheck=`: Checks the certificate usage policies. Options are `enable` or `disable`. Default is **enabled**.
- `revocationfreshnesstime=`: Sets the freshness time for revocation data in *seconds*. If the value is set to `0`, then the new CRL updates only if the previous one expires.
- `urlretrievaltimeout=`: Defines the timeout for retrieving revocation URLs in *milliseconds*.
- `sslctlidentifier=<string>`: The identifier for the SSL certificate trust list (CTL) to be used.
- `sslctlstorename=<string>`: Specifies the name of the certificate store under the `LOCAL_MACHINE` registry hive where the SSL CTL is stored.
- `dsmapperusage=`: Enables or disables the use of the Active Directory DSMapper for client certificate mapping. Options are `enable` or `disable`.
- `clientcertnegotiation=`: Controls negotiation for client certificates during the SSL handshake. Options are `enable` or `disable`. Default is **disabled**.
- `reject=`: Specifies the rejection of SSL connections based on certain criteria. Options are `enable` or `disable`.
- `disablehttp2=`: Enables or disables HTTP/2 protocol support. Options are `enable` or `disable`.
- `disablequic=`: Enables or disables QUIC protocol support. Options are `enable` or `disable`.
- `disablelegacytls=`: Enables or disables the use of legacy TLS protocols. Options are `enable` or `disable`.
- `disabletls12=` | `disabletls13=`: Enables or disables specific versions of the TLS protocol (TLS 1.2 or TLS 1.3). Options are `enable` or `disable`.
- `disableocspstapling=`: Enables or disables OCSP stapling, which is used for checking certificate revocation status. Options are `enable` or `disable`.
- `enabletokenbinding=`: Enables or disables token binding for enhanced security. Options are `enable` or `disable`.

- `logextendedevents=`: Enables or disables logging of extended events for diagnostic purposes. Options are `enable` or `disable`.
- `enablesessionticket=`: Controls whether SSL session tickets are enabled for session resumption. Options are `enable` or `disable`.
- `disablesessionid=`: Controls whether SSL session IDs are disabled for session resumption. Options are `enable` or `disable`.

Examples

To reserve the URL `http://example.com:8080/` for the user `DOMAIN\UserName` with listening permissions enabled, run the following command:

Windows Command Prompt

```
netsh http add urlacl url=http://example.com:8080/ user=DOMAIN\UserName listen=yes
```

To remove a specific IP address from the list of addresses HTTP.sys listens on, run the following command:

Windows Command Prompt

```
netsh http delete iplisten ipaddress=192.168.1.100
```

To display the cache state for a specific URL, run the following command:

Windows Command Prompt

```
netsh http show cachestate url=http://example.com
```

To remove a URL reservation, run the following command:

Windows Command Prompt

```
netsh http delete urlacl url=http://+:80/MyApp
```

To display details about a specific URL reservation, run the following command:

Windows Command Prompt

```
netsh http show urlacl url=http://+:80/MyApp
```

To view SSL certificate details for a specific IP and port, run the following command:

Windows Command Prompt

```
netsh http show sslcert ipport=192.168.1.100:443
```

To update HTTP/2 settings regarding SSL connections on a specific IP and port, allowing `100` frames at `1000` minutes, run the following command:

Windows Command Prompt

```
netsh http update sslpropertyex ipport=192.168.1.100:443 propertyid=1
maxsettingsperframe=100 maxsettingsperminute=1000
```

See also

- Command-Line Syntax Key

netsh interface

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh interface` command provides a comprehensive set of tools for configuring and managing network interfaces in Windows. Using this command, you can enable or disable network adapters, assign static or dynamic IP addresses, configure DNS and WINS servers, rename interfaces, and monitor interface statistics. The interface context is useful for troubleshooting connectivity issues, optimizing network settings, and automating network configuration tasks across physical and virtual adapters.

Additionally, `netsh interface` supports advanced features such as configuring port proxies, managing IPv4 and IPv6 settings, and controlling tunneling protocols like 6to4, ISATAP, and Teredo. You can also use it to adjust TCP and UDP parameters, set routing policies, and manage global network behaviors. These capabilities make `netsh interface` a powerful tool for both admins and power users who need granular control over network configuration and diagnostics.

Syntax

```
netsh interface [6to4 | clat | dump | fl48 | fl68 | help | httpstunnel | ipv4 | ipv6
| isatap | portproxy | set | show | tcp | teredo | udp | ?]
```

```
netsh interface 6to4 dump
netsh interface 6to4 set interface [name=]<string> [routing=]
(enabled|disabled|default)
netsh interface 6to4 set relay [name=]<string>|default
[state=]enabled|disabled|automatic|default [interval=]<integer>
netsh interface 6to4 set routing [routing=]enabled|disabled|automatic|default
[sitelocals=]enabled|disabled|default
netsh interface 6to4 set state [state=]enabled|disabled|default
[undoonstop=]enabled|disabled|default
netsh interface 6to4 show <interface> <relay> <routing> <state>
```

```
netsh interface clat dump
netsh interface clat set global [permit=]disabled|enabled
```

```
[pref64fromdns=]disabled|enabled [pref64fromra=]disabled|enabled
netsh interface clat show global
```

```
netsh interface fl48 dump
netsh interface fl48 show virtualinterfaces
```

```
netsh interface fl68 dump
netsh interface fl68 show virtualinterfaces
```

```
netsh interface httpstunnel add interface [type=]server|client [url=]<string>
[state=]enabled|disabled|default
                                        [authmode=]none|certificates
netsh interface httpstunnel delete interface
netsh interface httpstunnel dump
netsh interface httpstunnel reset
netsh interface httpstunnel set interface [url=]<string>
[state=]enabled|disabled|default [authmode=]none|certificates
netsh interface httpstunnel show interfaces [store=]active|persistent
netsh interface httpstunnel show statistics [interface=]<string>
```

```
netsh interface ipv4 add address [name=]<string> [address=]<IPv4 address>[/<integer>
[mask=]<IPv4 mask>]
                                [type=]unicast|anycast [gateway=]<IPv4 address>
[gwmetric=]<integer>
                                [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                                [subinterface=]<string> [store=]active|persistent
[skipassource=]true|false
netsh interface ipv4 add dnsservers [name=]<string> [address=]<IPv4 address>
[index=]<integer> [validate=]yes|no
netsh interface ipv4 add excludedportrange [protocol=]tcp|udp [startport=]<integer>
[numberofports=]<integer>
                                          [store=]active|persistent
netsh interface ipv4 add neighbors [interface=]<string> [address=]<IPv4 address>
[neighbor=]<string>
                                  [subinterface=]<string> [store=]active|persistent
netsh interface ipv4 add route [prefix=]<IPv4 address>/<integer> [interface=]
<string> [nexthop=]<IPv4 address>
                               [siteprefixlength=]<integer> [metric=]<integer>
[publish=]no|age|yes
                               [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                               [store=]active|persistent
```

```
netsh interface ipv4 add winsservers [name=]<string> [address=]<IP address> [index=]
<integer>
netsh interface ipv4 delete address [name=]<string> [address=]<IPv4 address>
[gateway=]<IPv4 address>|all
                                    [store=]active|persistent
netsh interface ipv4 delete arpcache [name=]<string> [address=]<IPv4 address>
[subinterface=]<string>
                                     [store=]active|persistent
netsh interface ipv4 delete destinationcache [interface=]<string> [address=]<IP
address>
netsh interface ipv4 delete dnsservers [name=]<string> [address=]<IPv4 address>
[validate=]yes|no
netsh interface ipv4 delete excludedportrange [protocol=]tcp|udp [startport=]
<integer> [numberofports=]<integer>
                                               [store=]active|persistent
netsh interface ipv4 delete neighbors [name=]<string> [address=]<IPv4 address>
[subinterface=]<string>
                                      [store=]active|persistent
netsh interface ipv4 delete route [prefix=]<IPv4 address>/<integer> [interface=]
<string> [nexthop=]<IPv4 address>
                                   [store=]active|persistent
netsh interface ipv4 delete winsservers [name=]<string> [address=]<IP address>|all
netsh interface ipv4 dump
netsh interface ipv4 install
netsh interface ipv4 reset
netsh interface ipv4 set address [name=]<string> [source=]dhcp|static [address=]
<IPv4 address>[/<integer>]
                                  [mask=]<IPv4 mask> [gateway=]<IPv4 address>|none
[gwmetric=]<integer>
                                  [type=]unicast|anycast [subinterface=]<string>
[store=]active|persistent
netsh interface ipv4 set compartment [compartment=]<integer> [defaultcurhoplimit=]
<integer>
                                     [store=]active|persistent
netsh interface ipv4 set dnsservers [name=]<string> [source=]dhcp|static [address=]
<IP address>|none
                                    [register=]none|primary|both [validate=]yes|no
netsh interface ipv4 set dynamicportrange [protocol=]tcp|udp [startport=]<integer>
[numberofports=]<integer>
                                          [store=]active|persistent
netsh interface ipv4 set global [defaultcurhoplimit=]<integer> [neighborcachelimit=]
<integer>
                                [routecachelimit=]<integer> [reassemblylimit=]
<integer>
                                [icmpredirects=]enabled|disabled
[sourceroutingbehavior=]drop|forward|dontforward
                                [taskoffload=]enabled|disabled
[dhcpmediasense=]enabled|disabled
                                [mediasenseeventlog=]enabled|disabled
[mldlevel=]none|sendonly|all
                                [mldversion=]version1|version2|version3
[multicastforwarding=]enabled|disabled
                                [groupforwardedfragments=]enabled|disabled
[randomizeidentifiers=]enabled|disabled
                                [store=]active|persistent
```

```
[addressmaskreply=]enabled|disabled [minmtu=]<integer>
                                [flowlabel=]enabled|disabled
[loopbacklargemtu=]enabled|disabled
                                [loopbackworkercount=]enabled|disabled
[loopbackexecutionmode=]inline|adaptive|worker
                                [sourcebasedecmp=]enabled|disabled
[reassemblyoutoforderlimit=]<integer>
                                [slaacprivacylevel=]<integer> [slaacmaxdadattempts=]
<integer>
                                [routepolicies=]enabled|disabled
[recursivereassembly=]enabled|disabled
                                [icmpjumbograms=]enabled|disabled
netsh interface ipv4 set interface [interface=]<string>
[[forwarding=]enabled|disabled]
                                   [advertise=]enabled|disabled [mtu=]<integer>
[siteprefixlength=]<integer>
                                   [nud=]enabled|disabled [basereachabletime=]
<integer>
                                   [retransmittime=]<integer> [dadtransmits=]
<integer>
                                   [routerdiscovery=]enabled|disabled|dhcp
[managedaddress=]enabled|disabled
                                   [otherstateful=]enabled|disabled
[weakhostsend=]enabled|disabled
                                   [weakhostreceive=]enabled|disabled
[ignoredefaultroutes=]enabled|disabled
                                   [advertisedrouterlifetime=]<0-65535>
[advertisedefaultroute=]enabled|disabled
                                   [currenthoplimit=]<0-255>
[store=]active|persistent
                                   [forcearpndwolpattern=]enabled|disabled
[enabledirectedmacwolpattern=]enabled|disabled
                                   [ecncapability=]ecndisabled|ect1|ect0|application
[rabaseddnsconfig=]enabled|disabled
                                   [dhcpstaticipcoexistence=]enabled|disabled
netsh interface ipv4 set neighbors [interface=]<string> [address=]<IPv4 address>
[neighbor=]<string> [store=]active|persistent
netsh interface ipv4 set route [prefix=]<IPv4 address>/<integer> [interface=]
<string> [nexthop=]<IPv4 address>
                                [siteprefixlength=]<integer> [metric=]<integer>
[publish=]no|age|yes
                                [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                                [store=]active|persistent
netsh interface ipv4 set subinterface [interface=]<string> [mtu=]<integer>
[subinterface=]<string> [store=]active|persistent
netsh interface ipv4 set winsservers [name=]<string> [source=]dhcp|static [address=]
<IP address>|none
netsh interface ipv4 show addresses [name=]<string>
netsh interface ipv4 show compartments [compartment=]<integer>
[level=]normal|verbose [store=]active|persistent
netsh interface ipv4 show config [name=]<string>
netsh interface ipv4 show destinationcache [interface=]<string> [address=]<IP
address> [level=]normal|verbose
netsh interface ipv4 show dnsservers [name=]string
```

```
netsh interface ipv4 show dynamicportrange [protocol=]tcp|udp
[store=]active|persistent
netsh interface ipv4 show excludedportrange [protocol=]tcp|udp
[store=]active|persistent
netsh interface ipv4 show global [store=]active|persistent
netsh interface ipv4 show icmpstats [rr=]<integer>
netsh interface ipv4 show interfaces [interface=]<string> [rr=]<integer>
[level=]normal|verbose [store=]active|persistent
netsh interface ipv4 show ipaddresses [interface=]<string> [level=]normal|verbose
[store=]active|persistent
netsh interface ipv4 show ipnettomedia [rr=]<integer>
netsh interface ipv4 show ipstats [rr=]<integer>
netsh interface ipv4 show joins [interface=]<string> [level=]normal|verbose
netsh interface ipv4 show neighbors [interface=]<string> [address=]<IP address>
[subinterface=]<string>
                                    [level=]normal|verbose [store=]active|persistent
netsh interface ipv4 show offload [name=]<string>
netsh interface ipv4 show route [level=]normal|verbose [store=]active|persistent
netsh interface ipv4 show subinterfaces [interface=]<string> [subinterface=]<string>
[level=]normal|verbose
                                          [store=]active|persistent
netsh interface ipv4 show tcpconnections [localaddress=]<IPv4 address> [localport=]
<integer> [remoteaddress=]<IPv4 address>
                                           [remoteport=]<integer> [rr=]<integer>
netsh interface ipv4 show tcpstats [rr=]<integer>
netsh interface ipv4 show udpconnections [localaddress=]<IPv4 address> [localport=]
<integer> [rr=]<integer>
netsh interface ipv4 show udpstats [rr=]<integer>
netsh interface ipv4 show winsservers [name=]string
netsh interface ipv4 uninstall
```

```
netsh interface ipv6 6to4 dump
netsh interface ipv6 6to4 set interface [name=]<string>
[routing=]enabled|disabled|default
netsh interface ipv6 6to4 set relay [name=]<string>|default
[state=]enabled|disabled|automatic|default [interval=]<integer>
netsh interface ipv6 6to4 set routing [routing=]enabled|disabled|automatic|default
[sitelocals=]enabled|disabled|default
netsh interface ipv6 6to4 set state [state=]enabled|disabled|default
[undoonstop=]enabled|disabled|default
netsh interface ipv6 6to4 show <interface> <relay> <routing> <state>
netsh interface ipv6 add address [interface=]<string> [address=]<IPv6 address>
[/<integer>] [type=]unicast|anycast
                                  [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                                  [store=]active|persistent [skipassource=]true|false
netsh interface ipv6 add dnsservers [name=]<string> [address=]<IPv6 address>
[index=]<integer> [validate=]yes|no
netsh interface ipv6 add excludedportrange [protocol=]tcp|udp [startport=]<integer>
[numberofports=]<integer>
                                             [store=]active|persistent
netsh interface ipv6 add neighbors [interface=]<string> [address=]<IPv6 address>
```

```
[neighbor=]<string> [subinterface=]<string>
                                    [store=]active|persistent
netsh interface ipv6 add potentialrouter [interface=]<string> [address=]<IPv6
address>
netsh interface ipv6 add prefixpolicy [prefix=]<IPv6 address>/<integer>
[precedence=]<integer> [label=]<integer>
                                      [store=]active|persistent
netsh interface ipv6 add route [prefix=]<IPv6 address>/<integer> [interface=]
<string> [nexthop=]<IPv6 address>
                                [siteprefixlength=]<integer> [metric=]<integer>
[publish=]no|age|yes
                                [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                                [store=]active|persistent
netsh interface ipv6 add v6v4tunnel [interface=]<string> [localaddress=]<IPv4
address> [remoteaddress=]<IPv4 address>
netsh interface ipv6 delete address [interface=]<string> [address=]<IPv6 address>
[store=]active|persistent
netsh interface ipv6 delete destinationcache [interface=]<string> [address=]<IP
address>
netsh interface ipv6 delete dnsservers [name=]<string> [address=]<IPv6 address>|all
[validate=]yes|no
netsh interface ipv6 delete excludedportrange [protocol=]tcp|udp [startport=]
<integer> [numberofports=]<integer>
                                                [store=]active|persistent
netsh interface ipv6 delete interface [interface=]<string>
netsh interface ipv6 delete neighbors [interface=]<string> [address=]<IPv6 address>
[subinterface=]<string>
                                      [store=]active|persistent
netsh interface ipv6 delete potentialrouter [interface=]<string> [address=]<IPv6
address>
netsh interface ipv6 delete prefixpolicy [prefix=]<IPv6 address>/<integer>
[store=]active|persistent
netsh interface ipv6 delete route [prefix=]<IPv6 address>/<integer> [interface=]
<string> [nexthop=]<IPv6 address>
                                    [store=]active|persistent
netsh interface ipv6 dump
netsh interface ipv6 isatap dump
netsh interface ipv6 isatap set router [name=]<string>|default
[state=]enabled|disabled|default [interval=]<integer>
netsh interface ipv6 isatap set state [state=]enabled|disabled|default
netsh interface ipv6 isatap show <router> <state>
netsh interface ipv6 reset
netsh interface ipv6 set address [interface=]<string> [address=]<IPv6 address>
[type=]unicast|anycast
                                  [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                                  [store=]active|persistent
netsh interface ipv6 set compartment [compartment=]<integer> [defaultcurhoplimit=]
<integer> [store=]active|persistent
netsh interface ipv6 set dnsservers [name=]<string> [source=]dhcp|static [address=]
<IP address>|none
                                    [register=]none|primary|both [validate=]yes|no
netsh interface ipv6 set dynamicportrange [protocol=]tcp|udp [startport=]<integer>
[numberofports=]<integer>
```

```
                                        [store=]active|persistent
netsh interface ipv6 set global [defaultcurhoplimit=]<integer> [neighborcachelimit=]
<integer>
                                [routecachelimit=]<integer> [reassemblylimit=]
<integer>
                                [icmpredirects=]enabled|disabled
[sourceroutingbehavior=]drop|forward|dontforward
                                [taskoffload=]enabled|disabled
[dhcpmediasense=]enabled|disabled
                                [mediasenseeventlog=]enabled|disabled
[mldlevel=]none|sendonly|all
                                [mldversion=]version1|version2|version3
[multicastforwarding=]enabled|disabled
                                [groupforwardedfragments=]enabled|disabled
[randomizeidentifiers=]enabled|disabled
                                [store=]active|persistent
[addressmaskreply=]enabled|disabled [minmtu=]<integer>
                                [flowlabel=]enabled|disabled
[loopbacklargemtu=]enabled|disabled
                                [loopbackworkercount=]enabled|disabled
[loopbackexecutionmode=]inline|adaptive|worker
                                [sourcebasedecmp=]enabled|disabled
[reassemblyoutoforderlimit=]<integer>
                                [slaacprivacylevel=]<integer> [slaacmaxdadattempts=]
<integer>
                                [routepolicies=]enabled|disabled
[recursivereassembly=]enabled|disabled
                                [icmpjumbograms=]enabled|disabled
netsh interface ipv6 set interface [interface=]<string>
[[forwarding=]enabled|disabled]
                                   [advertise=]enabled|disabled [mtu=]<integer>
[siteprefixlength=]<integer>
                                   [nud=]enabled|disabled [basereachabletime=]
<integer>
                                   [retransmittime=]<integer> [dadtransmits=]
<integer>
                                   [routerdiscovery=]enabled|disabled|dhcp
[managedaddress=]enabled|disabled
                                   [otherstateful=]enabled|disabled
[weakhostsend=]enabled|disabled
                                   [weakhostreceive=]enabled|disabled
[ignoredefaultroutes=]enabled|disabled
                                   [advertisedrouterlifetime=]<0-65535>
[advertisedefaultroute=]enabled|disabled
                                   [currenthoplimit=]<0-255>
[store=]active|persistent
                                   [forcearpndwolpattern=]enabled|disabled
[enabledirectedmacwolpattern=]enabled|disabled
                                   [ecncapability=]ecndisabled|ect1|ect0|application
[rabaseddnsconfig=]enabled|disabled
                                   [dhcpstaticipcoexistence=]enabled|disabled
netsh interface ipv6 set neighbors [interface=]<string> [address=]<IPv6 address>
[neighbor=]<string> [store=]active|persistent
netsh interface ipv6 set prefixpolicy [prefix=]<IPv6 address>/<integer>
[precedence=]<integer> [label=]<integer>
```

```
                                        [store=]active|persistent
netsh interface ipv6 set privacy [prefix=]<IPv6 address>/<integer> [precedence=]
<integer> [label=]<integer>
                                    [store=]active|persistent
netsh interface ipv6 set route [prefix=]<IPv6 address>/<integer> [interface=]
<string> [nexthop=]<IPv6 address>
                                   [siteprefixlength=]<integer> [metric=]<integer>
[publish=]no|age|yes
                                   [validlifetime=]<integer>|infinite
[preferredlifetime=]<integer>|infinite
                                   [store=]active|persistent
netsh interface ipv6 set slaacsecretkey [key=]<string>
netsh interface ipv6 set subinterface [interface=]<string> [mtu=]<integer>
[subinterface=]<string> [store=]active|persistent
netsh interface ipv6 set teredo
[type]=disabled|client|enterpriseclient|natawareclient|server|default
                                    [servername=]<hostname>|<IPv4 address>|default
[refreshinterval=]<integer>|default
                                    [clientport=]<integer>|default [servervirtualip=]
<IPv4 address>|default
netsh interface ipv6 show addresses [interface=]<string> [level=]normal|verbose
[store=]active|persistent
netsh interface ipv6 show compartments [compartment=]<integer>
[level=]normal|verbose [store=]active|persistent
netsh interface ipv6 show destinationcache [interface=]<string> [address=]<IP
address> [level=]normal|verbose
netsh interface ipv6 show dnsservers [name=]string
netsh interface ipv6 show dynamicportrange [protocol=]tcp|udp
[store=]active|persistent
netsh interface ipv6 show excludedportrange [protocol=]tcp|udp
[store=]active|persistent
netsh interface ipv6 show global [store=]active|persistent
netsh interface ipv6 show interfaces [interface=]<string> [rr=]<integer>
[level=]normal|verbose [store=]active|persistent
netsh interface ipv6 show ipstats [rr=]<integer>
netsh interface ipv6 show joins [interface=]<string> [level=]normal|verbose
netsh interface ipv6 show neighbors [interface=]<string> [address=]<IP address>
[subinterface=]<string>
                                      [level=]normal|verbose [store=]active|persistent
netsh interface ipv6 show offload [name=]<string>
netsh interface ipv6 show potentialrouters [interface=]<string>
netsh interface ipv6 show prefixpolicies [store=]active|persistent
netsh interface ipv6 show privacy [store=]active|persistent
netsh interface ipv6 show route [level=]normal|verbose [store=]active|persistent
netsh interface ipv6 show siteprefixes
netsh interface ipv6 show slaacsecretkey [store=]active|persistent
netsh interface ipv6 show subinterfaces [interface=]<string> [subinterface=]<string>
[level=]normal|verbose
                                         [store=]active|persistent
netsh interface ipv6 show tcpstats [rr=]<integer>
netsh interface ipv6 show teredo
netsh interface ipv6 show tfofallback
netsh interface ipv6 show udpstats [rr=]<integer>
```

```
netsh interface isatap dump
netsh interface isatap set router [name=]<string>|default
[state=]enabled|disabled|default [interval=]<integer>
netsh interface isatap set state [state=]enabled|disabled|default
netsh interface isatap show <router> <state>
```

```
netsh interface portproxy add v4tov4 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv4 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv4 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy add v4tov6 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv6 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv4 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy add v6tov4 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv4 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv6 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy add v6tov6 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv6 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv6 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy deleted v4tov4 [listenport=]<integer>|<servicename>
[listenaddress=]<IPv4 address>|<hostname>
                                        [protocol=]tcp
netsh interface portproxy deleted v4tov6 [listenport=]<integer>|<servicename>
[listenaddress=]<IPv4 address>|<hostname>
                                        [protocol=]tcp
netsh interface portproxy deleted v6tov4 [listenport=]<integer>|<servicename>
[listenaddress=]<IPv6 address>|<hostname>
                                        [protocol=]tcp
netsh interface portproxy deleted v6tov6 [listenport=]<integer>|<servicename>
[listenaddress=]<IPv6 address>|<hostname>
                                        [protocol=]tcp
netsh interface portproxy dump
netsh interface portproxy reset
netsh interface portproxy set v4tov4 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv4 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv4 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy set v4tov6 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv6 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv4 address>|<hostname>
                                    [protocol=]tcp
```

```
netsh interface portproxy set v6tov4 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv4 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv6 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy set v6tov6 [listenport=]<integer>|<servicename>
[connectaddress=]<IPv6 address>|<hostname>
                                    [connectport=]<integer>|<servicename>
[listenaddress=]<IPv6 address>|<hostname>
                                    [protocol=]tcp
netsh interface portproxy show all
netsh interface portproxy show <v4tov4> <v4tov6> <v6tov4> <v6tov6>
```

```
netsh interface set interface [name=]IfName [admin=]ENABLED|DISABLED
[connect=]CONNECTED|DISCONNECTED [newname=]NewName
```

```
netsh interface show interface [name=]string
```

```
netsh interface tcp add supplementalport
[template=]internet|datacenter|automatic|custom|compat [localport=]*|<integer>
                                        [remoteport=]*|<integer>
netsh interface tcp add supplementalsubnet
[template=]internet|datacenter|automatic|custom|compat [family=]ipv4|ipv6
                                          [address=]<IPv4 or IPv6 address>/<prefix
length>
netsh interface tcp delete supplementalport [localport=]*|<number> [remoteport=]*|
<number>
netsh interface tcp delete supplementalsubnet [family=]ipv4|ipv6 [address=]<IPv4 or
IPv6 address>/<prefix length>
netsh interface tcp dump
netsh interface tcp reload
netsh interface tcp rundown
netsh interface tcp set global [rss=]disabled|enabled|default

[autotuninglevel=]disabled|highlyrestricted|restricted|normal|experimental
                                [ecncapability=]disabled|enabled|default
[timestamps=]disabled|enabled|default
                                [initialrto=]<300-3000>
[rsc=]disabled|enabled|default
                                [nonsackrttresiliency=]disabled|enabled|default
[maxsynretransmissions=]<2-8>
                                [fastopen=]disabled|enabled|default
[fastopenfallback=]disabled|enabled|default
                                [hystart=]disabled|enabled|default
[prr=]disabled|enabled|default
```

```
[pacingprofile=]off|initialwindow|slowstart|always|default
netsh interface tcp set heuristics [wsh=]disabled|enabled|default
[forcews=]disabled|enabled|default
netsh interface tcp set security [mpp=]disabled|enabled|default [startport=]
<integer> [numberofports=]<integer>
                                  [profiles=]disabled|enabled|default
netsh interface tcp set supplemental
[template=]automatic|datacenter|internet|compat|custom [minrto=]<integer>
                                      [icw=]<integer>
[congestionprovider=]none|ctcp|dctcp|cubic|bbr2|default
                                      [enablecwndrestart=]disabled|enabled|default
[delayedacktimeout=]<integer>
                                      [delayedackfrequency=]<integer>
[rack=]disabled|enabled|default
                                      [taillossprobe=]disabled|enabled|default
netsh interface tcp show global [store=]active|persistent
netsh interface tcp show heuristics [heuristics=]wsh|forcews
netsh interface tcp show rscstats [interface=]<string>
netsh interface tcp show security [store=]active|persistent
netsh interface tcp show supplemental
[template=]automatic|datacenter|internet|compat|custom
netsh interface tcp show supplementalports [level=]normal|verbose
netsh interface tcp show supplementalsubnets [level=]normal|verbose
```

```
netsh interface teredo dump
netsh interface teredo set state
[type]=disabled|client|enterpriseclient|natawareclient|server|default
                                  [servername=]<hostname>|<IPv4 address>|default
[refreshinterval=]<integer>|default
                                  [clientport=]<integer>|default [servervirtualip=]
<IPv4 address>|default
netsh interface teredo show state
```

```
netsh interface udp dump
netsh interface udp reset
netsh interface udp set global [uro=]disabled|enabled|default
[uso=]disabled|enabled|default
netsh interface udp show global [store=]active|persistent
```

Parameters

interface 6to4

Expand table

Command	Description
interface `6to4 set interface`	Configures settings for a specified 6to4 interface in the legacy IPv4-based context. `name` - Specifies the interface name or index. `routing` - Sets whether the interface acts as a router: - `enabled`: Turns on routing for this interface. - `disabled`: Turns off routing for this interface. - `default`: Uses the system's default routing setting.
interface `6to4 set relay`	Configures the relay server for 6to4 tunneling in the legacy IPv4-based context. `name` - Specifies the name of the 6to4 relay or use `default` for the system default. `state` - Sets the state of relay name resolution: - `enabled`: Forces the relay to be active and handle 6to4 traffic. - `disabled`: Turns off the relay. It doesn't process any 6to4 traffic. - `automatic`: Lets Windows decide when to enable or disable the relay based on network conditions. - `default`: Resets the relay state to its default configuration. `interval` - Sets the relay name resolution interval in minutes.
interface `6to4 set routing`	Configures routing options for 6to4 tunneling in the legacy IPv4-based context. `routing` - Controls how 6to4 routing is handled: - `enabled`: Forces 6to4 routing to be active. - `disabled`: Turns off 6to4 routing. - `automatic`: Lets Windows decide when to enable/disable routing based on network conditions. - `default`: Resets routing to the system default. `sitelocals` - Manages the use of site-local addresses in 6to4 routing: - `enabled`: Allows site-local addresses. - `disabled`: Blocks site-local addresses. - `default`: Resets to the system default for site-local address handling.
interface `6to4 set state`	Configures the 6to4 service in the legacy IPv4-based context `state` - Specifies the operational mode for the 6to4 service: - `disabled`: Disables the 6to4 service; no 6to4 interfaces are created. - `enabled`: Enables the 6to4 service and configures 6to4 addresses for each public IPv4 address. - `automatic`: Enables the 6to4 service to automatically configure itself for each available public IPv4 address. - `default`: Restores the default 6to4 behavior, equivalent to `disabled`. `undoonstop` - Determines whether 6to4 is disabled when the service stops.
interface `6to4 show`	Displays the 6to4 interface configuration and status in the legacy IPv4-based context.

Command	Description
	`interface` - Displays the configuration details of the 6to4 interface. `relay` - Displays information about the 6to4 relay server. `routing` - Displays the current routing state for 6to4. `state` - Displays the current operational state of the 6to4 service.

interface clat

Expand table

Command	Description
interface `clat set global`	Configures global Customer-side Translator (CLAT) settings, including enabling or disabling CLAT and setting DNS or RA preferences. If both `pref64fromdns` and `pref64fromra` are disabled, CLAT can't be activated because this combination is invalid. `permit` - Specifies if CLAT is allowed on noncellular interfaces. `pref64fromdns` - Determines whether DNS is used as the source for prefix information. If `pref64fromra` is also enabled, DNS is used only as a fallback. `pref64fromra` - Determines whether router advertisements (RA) are used as the source for prefix information.
interface `clat show global`	Displays the current global CLAT configuration settings.

interface dump

Expand table

Command	Description
interface `dump` interface `6to4 dump` interface `clat dump` interface `fl48 dump` interface `fl68 dump` interface `httpstunnel dump` interface `ipv4 dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.

Command	Description
interface `ipv6 6to4 dump` interface `ipv6 dump` interface `ipv6 isatap dump` interface `isatap dump` interface `portproxy dump` interface `tcp dump` interface `teredo dump` interface `udp dump`	

interface fl48

Expand table

Command	Description
interface `fl48 show virtualinterfaces`	Displays a list of all FL48 virtual interfaces.

interface fl68

Expand table

Command	Description
interface `fl68 show virtualinterfaces`	Displays a list of all FL68 virtual interfaces.

interface httpstunnel

Expand table

Command	Description
interface `httpstunnel add interface`	Creates and configures an IPHTTPS client or server interface, allowing you to specify connection properties and authentication settings.

Command	Description
	`type` - Specifies the interface type: - `client`: Configures a client interface. - `server`: Configures a server interface (valid only on Windows Server). `url` - For a server interface, the URL to listen for HTTP(S) requests. For a client interface, the URL to send HTTP(S) requests. `state` - Sets the operational state: - `default`: Server interface is always configured. Client interface is configured on demand (default for client). - `enabled`: Interface is always configured (default for server). - `disabled`: Disables the interface. `authmode` - Specifies authentication mode: - `none`: No client authentication during SSL handshake (default). - `certificates`: Uses client certificates for authentication (server interface only, valid only for Windows Server).
interface `httpstunnel delete interface`	Removes an existing IPHTTPS client or server interface configuration.
interface `httpstunnel reset`	Removes all IPHTTPS client and server interface configurations from the local system.
interface `httpstunnel set interface`	Modifies the configuration of an existing IPHTTPS client or server interface. `url` - For a server interface, the URL to listen for HTTP(S) requests. For a client interface, the URL to send HTTP(S) requests. `state` - Sets the operational state: - `default`: Server interface is always configured. Client interface is configured on demand (default for client). - `enabled`: Interface is always configured (default for server). - `disabled`: Disables the interface. `authmode` - Specifies authentication mode: - `none`: No client authentication during SSL handshake (default). - `certificates`: Uses client certificates for authentication (server interface only, valid only for Windows Server).
interface `httpstunnel show interfaces`	Displays details for all network interfaces, or for a specified interface if provided. `store` - Specifies which interfaces to display: - `active`: Shows only active interfaces. - `persistent`: Shows all interfaces (default).

Command	Description
interface `httpstunnel show statistics`	Shows IPHTTPS interface statistics. `interface` - Specifies the interface name for which statistics are displayed.

interface ipv4

Expand table

Command	Description
interface `ipv4 add address`	Assigns a static IPv4 address and default gateway to a specified interface. If the interface is currently configured for DHCP, DHCP is disabled when this command is executed. `name` - Specifies the interface name or index. `address` - The IPv4 address to assign. Optionally, include a subnet prefix length (example: `192.168.1.10/24`). `mask` - Specifies the subnet mask for the IP address. `type` - Specifies the address type. Use `unicast` to assign a unicast address (default), or `anycast` to assign an anycast address. `gateway` - The IPv4 address of the default gateway. `gwmetric` - The metric value for the default gateway. `validlifetime` - Duration of the address or route that remains valid. Default is `infinite`. `preferredlifetime` - Duration of the address or route that's preferred. Default is `infinite`. `subinterface` - Specifies the LUID of the subinterface for the default gateway. Required only for interfaces with multiple subinterfaces. `store` - Specifies the address persistence. Use `active` for a temporary address removed on reboot, or `persistent` to retain the address after reboot (default). `skipassource` - If `true`, the address isn't used as the source for outgoing packets unless explicitly specified. Default is `false`.
interface `ipv4 add dnsservers`	Adds a new DNS server IP address to the static list for the specified interface. By default, the new DNS server is appended to the end of the list. If an index is provided, the server is inserted at that position, shifting others as needed. If the interface previously used DHCP for DNS, the static address replaces the DHCP-assigned list. When the `validate` option is set to `yes`, the new DNS server is verified before being added. `name` - Specifies the name or index of the interface where the DNS server is added. `address` - The IPv4 address of the DNS server to add. `index` - Sets the preference order for the DNS server address.

Command	Description
	`validate` - Indicates whether to validate the DNS server setting using `yes` (default) or `no`.
interface `ipv4 add excludedportrange`	Excludes a specified range of consecutive ports from being used by the system. `protocol` - Specifies the protocol to exclude. Use `tcp` to exclude TCP ports or `udp` to exclude UDP ports. `startport` - The starting port number for the exclusion range. `numberofports` - The total number of consecutive ports to exclude, beginning with `startport`. `store` - Determines the persistence of the exclusion. Use `active` for a temporary exclusion until the next reboot, or `persistent` to retain the exclusion after reboot (default).
interface `ipv4 add neighbors`	Adds a neighbor entry to the specified network interface. This is typically used to associate an IP address with a link-layer (MAC) address. `interface` - The name or index of the network interface where the neighbor entry will be added. `address` - The IPv4 address of the neighbor. `neighbor` - The link-layer (MAC) address to associate with the specified IP address. `subinterface` - The LUID of the subinterface, required only for interfaces with multiple subinterfaces. `store`: Determines whether the entry is temporary (`active`, removed on next reboot) or permanent (`persistent`, retained after reboot; this is the default).
interface `ipv4 add route`	Adds a network route for a specified IP prefix, allowing traffic to be directed through a chosen interface and gateway. `prefix` - The IP prefix for which to add a route. `interface` - The name or index of the interface to use for the route. `nexthop` - The gateway address, if the prefix isn't on-link. `siteprefixlength` - The prefix length for the entire site, if on-link. `metric` - The metric value for the route. `publish` - Controls advertisement in Route Advertisements. Use `no` (default) to not advertise, `age` to advertise with a finite lifetime, or `yes` to advertise with an infinite lifetime. `validlifetime` - The duration for which the route remains valid. Default is `infinite`. `preferredlifetime` - The duration for which the route is preferred. Default is equal to `validlifetime`. `store` - Determines persistence. Use `active` for a temporary change (until next boot), or `persistent` for a change that remains after reboot (default).
interface `ipv4 add winsservers`	Adds a new WINS server IP address to the static list for the specified interface. By default, the server is appended to the end of the list. If an index is provided, the

Command	Description
	server is inserted at that position, shifting other entries as needed. If the interface previously received WINS servers via DHCP, the new address replaces the DHCP-assigned list. `name` - The name or index of the interface to which the WINS server is added. `address` - The IP address of the WINS server to add. `index` - Sets the preference order for the specified WINS server address.
interface `ipv4 delete address`	Removes a static IPv4 address or default gateway from the specified interface. `name` - The interface name or index. `address` - The static IPv4 address to delete. `gateway` - Specify either a particular default gateway IP address to delete, or use `all` to delete all default gateway IP addresses on the interface. `store` - Use `active` to make the deletion temporary (until next boot), or `persistent` to make it permanent (default).
interface `ipv4 delete arpcache`	Deletes entries from the Address Resolution Protocol (ARP) cache. Flushing the ARP cache removes all ARP entries. If an interface is specified, only that interface's cache is flushed. If an address is specified, only the matching ARP entry is deleted. `name` - Specifies the interface name or index. `address` - The IPv4 address of the neighbor to remove. `subinterface` - The LUID of the subinterface. `store` - Determines which store to delete from. By default, entries in both stores are deleted: - `active`: Entry may reappear on next boot. - `persistent`: Entry is removed permanently.
interface `ipv4 delete destinationcache`	Flushes the destination cache, which stores routing information for IPv4 destinations. If an interface is specified, only that interface's cache is flushed. If an address is also specified, only that destination cache entry is deleted. `interface` - The name or index of the interface. `address` - The destination address.
interface `ipv4 delete dnsservers`	Deletes statically configured DNS server IP addresses for a specific interface. If `validate` is set to `yes`, the remaining DNS servers are validated. `name` - The name or index of the interface where DNS servers are deleted. `address` - The IP address of the DNS server to delete, or `all` to delete all configured DNS server addresses. `validate` - Specifies whether to validate the DNS server setting after deletion, `yes` (default) or `no`.
interface `ipv4 delete excludedportrange`	Deletes a previously set exclusion for a range of TCP or UDP ports. The `startport` and `numberofports` must match the previously created exclusion.

Command	Description
	`protocol` - `tcp` to delete a TCP exclusion, or `udp` for UDP. `startport` - The starting port number of the exclusion. `numberofports` - The number of consecutive ports to delete, starting from `startport`. `store` - `active` for a temporary deletion (until next boot), or `persistent` for a permanent deletion (default).
interface `ipv4 delete neighbors`	Deletes entries from the IPv4 neighbor cache. By default, entries in both stores are deleted. `name` - The interface name or index. `address` - The neighbor's IP address. `subinterface` - The subinterface LUID. `store` - `active` for a temporary deletion (entry may reappear on next boot), or `persistent` for a permanent deletion.
interface `ipv4 delete route`	Deletes an IPv4 route from the routing table. `prefix` - The IP prefix of the route to delete (for example, `192.168.1.0/24`). `interface` - The name or index of the interface associated with the route. `nexthop` - The gateway address for the route, if the prefix isn't on-link. `store` - Specifies whether the deletion is temporary (`active`, until next boot) or permanent (`persistent`). By default, entries in both stores are deleted.
interface `ipv4 delete winsservers`	Deletes statically configured WINS server IP addresses for a specific interface. `name` - The name or index of the interface where WINS servers are deleted. `address` - The IP address of the WINS server to delete, or use `all` to delete all configured WINS server addresses for the interface.
interface `ipv4 install`	Installs the IP protocol. A system restart is required for the changes to take effect.
interface `ipv4 reset`	Removes all user-configured settings. You must restart your computer for the default settings to take effect.
interface `ipv4 set address`	Enables or disables DHCP for IP address configuration. Removes any previous static IP addresses and default gateways, and can add a new static IP address and default gateway. `name` - Interface name or index. `source` - `dhcp` enables DHCP for configuring IP addresses for the specified interface. `static` disables DHCP and must be specified if an address or gateway is being configured. `address` - IPv4 address to add or modify, optionally followed by the subnet prefix length. `mask` - The IP subnet mask for the specified IP address. `gateway` - Either a specific IPv4 address for the default gateway or `none` (default,

Command	Description
	no gateway). `gwmetric` - The metric value for the default gateway. Set only if gateway is specified. `type` - `unicast` (default) marks the address as a unicast address while `anycast` marks it as anycast. `subinterface` - The LUID of the subinterface on which the default gateway exists. Needed only for interfaces with multiple subinterfaces. `store` - Determines whether the configuration change is temporary (`active`, lasts until the next reboot) or permanent (`persistent`, remains after reboot; default).
interface `ipv4 set compartment`	Modifies compartment settings for a network interface. `compartment` - Specifies the compartment ID to configure. `defaultcurhoplimit` - Sets the default hop limit for outgoing packets in the compartment. `store` - Determines persistence: - `active`: Applies until next reboot. - `persistent`: Retains the setting after reboot.
interface `ipv4 set dnsservers`	Sets DNS server configuration to either DHCP or static mode. When `source` is `static`, the `address` option is available for configuring a static list of DNS server IP addresses for the specified interface. If `validate` is `yes`, the newly set DNS server is validated. `name` - Name or index of the interface. `source` - `dhcp` sets DHCP as the source for configuring DNS servers. `static` sets the source to local static configuration. `address` - IP address for a DNS server or `none` to clear the list. `register` - `none` disables Dynamic DNS registration, `primary` registers under the primary DNS suffix only, `both` registers under both the primary and connection-specific suffixes. `validate` - Specifies whether validation of the DNS server setting is performed, `yes` (default) or `no`.
interface `ipv4 set dynamicportrange`	Modifies the range of ports used for dynamic port assignment (wildcard port assignment). `protocol` - Determines the protocol type: - `tcp` modifies the dynamic port range for TCP. - `udp` for UDP. `startport` - Starting port for dynamic port assignment. `numberofports` - Number of ports available for dynamic port assignment. `store` - Determines whether the configuration change is temporary (`active`, lasts until the next reboot) or permanent (`persistent`, remains after reboot (default)).
interface `ipv4 set global`	Configures the global IPv4 stack settings.

Command	Description
	`addressmaskreply` - Enables (`enabled`) or disables (`disabled`) responding to ICMP address mask packets. `defaultcurhoplimit` - Sets the default hop limit for outgoing packets. `dhcpmediasense` - Enables (`enabled`) or disables (`disabled`) media sense (network connection detection). `flowlabel` - Enables (`enabled`) or disables (`disabled`) flow label support. `groupforwardedfragments` - Enables (`enabled`) or disables (`disabled`) grouping of fragments before forwarding. `icmpjumbograms` - Enables (`enabled`) or disables (`disabled`) ICMP jumbograms (IPv6 only). `icmpredirects` - Enables (`enabled`) or disables (`disabled`) updating the path cache in response to ICMP redirect packets. `loopbackexecutionmode` - Sets execution mode for loopback processing: - `inline`: Loopback packet processing happens directly in the context of the thread that sends the packet. This is the simplest and fastest mode, but it can block the sending thread if processing takes time. - `adaptive`: The system dynamically chooses between inline and worker modes based on current load and performance. - `worker`: Loopback packet processing is offloaded to a separate worker thread. This prevents blocking the sending thread, which can improve responsiveness under heavy load, but may introduce a slight delay. `loopbacklargemtu` - Enables (`enabled`) or disables (`disabled`) large IP MTU for the loopback path. `loopbackworkercount` - Number of worker threads (1-16) for loopback processing in adaptive mode. `mediasenseeventlog` - Enables (`enabled`) or disables (`disabled`) media sense (network connection detection) event logging. `minmtu` - Sets the minimum IP MTU to avoid fragmentation. For IPv4, range is 352-576. For IPv6, it's fixed at 1280. `mldlevel` - Sets multicast support level: - `none`: Multicast packets can't be sent or received. - `sendonly`: Multicast packets can be sent but not received. - `all`: Multicast packets can be sent and received. `mldversion` - The maximum Multicast Listener Discovery (MLD) version supported by the host. `multicastforwarding` - Enables (`enabled`) or disables (`disabled`) forwarding of multicast packets. `neighborcachelimit` - The maximum number of neighbor cache entries no longer referenced. `randomizeidentifiers` - Enables (`enabled`) or disables (`disabled`) randomization of interface identifiers. `reassemblylimit` - The maximum size of the reassembly buffer. `reassemblyoutoforderlimit` - The maximum number of out-of-order fragments for reassembly. `recursivereassembly` - Enables (`enabled`) or disables (`disabled`) recursive reassembly of fragments (IPv6 only). `routecachelimit` - The maximum number of route cache entries.

Command	Description
	`routepolicies` - Enables (`enabled`) or disables (`disabled`) route policies. `slaacmaxdadattempts` - The number of duplicate address detection attempts for SLAAC addresses (IPv6 only). `slaacprivacylevel` - The privacy level for SLAAC address suffixes (IPv6 only): - `0`: Use EUI-64. - `1`: Use randomized EUI-64. - `2`: Use randomized EUI-64, prefix, and secret key. `sourcebasedecmp` - Enables (`enabled`) or disables (`disabled`) source-based Equal-Cost Multi-Path (ECMP). `sourceroutingbehavior` - Controls behavior for source-routed packets. Per RFC 5095, forwarding of source-routed IPv6 packets is removed. `forward` is equivalent to `dontforward`. `store` - Specifies persistence: - `active`: Setting lasts until next boot. - `persistent`: Setting is retained after reboot (default). `taskoffload` - Enables (`enabled`) or disables (`disabled`) task offload.
interface `ipv4 set interface`	Modifies the configuration of an IPv4 network interface. `advertise` - Enables (`enabled`) or disables (`disabled`) whether the interface sends RA. By default, advertisements are disabled. `advertisedefaultroute` - Determines whether the interface is advertised as a default router. `advertisedrouterlifetime` - Sets the router lifetime (in seconds) advertised in RA with a value of **0** to **65535**. Default is 1,800 seconds. `basereachabletime` - Sets the base time (in milliseconds) that a neighbor is considered reachable. `currenthoplimit` - Sets the hop limit for outbound IPv4 traffic. Valid range is **0** to **255**. `dadtransmits` - Sets the number of Duplicate Address Detection (DAD) attempts. A value of **0** disables DAD. `dhcpstaticipcoexistence` - Enables (`enabled`) or disables (`disabled`) whether DHCP and static IP addresses are allowed to coexist on the interface. `ecncapability` - Controls Explicit Congestion Notification (ECN) marking for outgoing packets: - `ecndisabled`: Disables ECN marking. Outgoing packets won't use ECN bits, so congestion is signaled only by packet loss. - `ect0`: Sets ECN bits to ECT(0), indicating the packet is ECN-capable. Routers can mark congestion without dropping packets. - `ect1`: Sets ECN bits to ECT(1), another ECN-capable codepoint (rarely used; functionally similar to ECT(0)). - `application`: Leaves ECN marking up to the application. The application decides whether and how to set ECN bits. `enabledirectedmacwolpattern` - Enables (`enabled`) or disables (`disabled`) directed MAC packet patterns to wake the computer. `forcearpndwolpattern` - Enables (`enabled`) or disables (`disabled`) address resolution and neighbor discovery packet patterns to wake the computer from

Command	Description
	sleep. `forwarding` - Enables (`enabled`) or disables (`disabled`) packet forwarding on this interface. When enabled, the interface can route packets to other interfaces. `ignoredefaultroutes` - When set to `enabled`, default routes on this interface are ignored. Default is `disabled`. `interface` - Specifies the name or index of the network interface to configure. `managedaddress` - Enables (`enabled`) or disables (`disabled`) managed address configuration, allowing addresses to be assigned via DHCPv6. This setting may be overridden if router discovery is enabled and advertising is disabled. `mtu` - Sets the Maximum Transmission Unit (MTU) for the interface. If not specified, the link's default MTU is used. `nud` - Enables (`enabled`) or disables (`disabled`) Neighbor Unreachability Detection (NUD), which helps determine if a neighbor is reachable. `otherstateful` - Enables (`enabled`) or disables (`disabled`) other stateful configuration options, such as obtaining additional settings via DHCPv6. This may also be overridden by router discovery settings. `rabaseddnsconfig` - Enables (`enabled`) or disables (`disabled`) DNS configuration via RA as defined in RFC 6106. `retransmittime` - Specifies the time (in milliseconds) between retransmissions of neighbor solicitation messages. `routerdiscovery` - Enables, disables, or allows DHCP to control router discovery on the interface. `siteprefixlength` - Defines the default length of the global IPv6 prefix for the site. `store` - Specifies whether the configuration change is temporary (`active`, until next reboot) or persistent (`persistent`, remains after reboot). `weakhostreceive` - Enables weak host receive behavior, allowing the interface to receive packets for IP addresses not assigned to it. `weakhostsend` - Enables weak host send behavior, allowing the interface to send packets even if the source IP doesn't belong to the interface.
interface `ipv4 set neighbors`	Configures a static neighbor entry for an IPv4 interface. `interface` - The name or index of the network interface to configure. `address` - IPv4 address of the neighbor. `neighbor` - Link-layer (MAC) address to associate with the specified IPv4 address. `store` - Determines persistence: - `active` Entry is removed on next boot. - `persistent`: Entry remains after reboot (default).
interface `ipv4 set route`	Modifies parameters for an IPv6 route. When `publish` is set to `no` or `age`, the route is deleted after the valid lifetime expires. If `publish` is `age`, RAs include the remaining valid lifetime. If `publish` is set to `yes`, the route is never deleted and RAs always show the specified valid lifetime. `prefix` - The IPv6 prefix of the route to modify. `interface` - The name or index of the interface for the route.

Command	Description
	`nexthop` - The gateway IPv6 address, if the prefix isn't on-link. `siteprefixlength` - The prefix length for the entire site if on-link. `metric` - The metric value for the route. `publish` - Controls advertisement in RA: - `no`: Not advertised (default). - `age`: Advertised with finite lifetime. - `yes`: Advertised with infinite lifetime. `validlifetime` - The duration value of the route to remain valid (default is infinite). `preferredlifetime` - The duration of the route that's preferred (default is infinite). `store` - Determines persistence: - `active` Entry is removed on next boot. - `persistent`: Entry remains after reboot (default). Time values can be expressed in days, hours, minutes, and seconds (`1d2h3m4s`).
interface `ipv4 set subinterface`	Modifies the subinterface configuration parameters. `interface` - The name or index of the interface to configure. `mtu` - Sets the MTU for the subinterface. Defaults to the link's natural MTU if not specified. `subinterface` - The LUID of the subinterface. Required only for interfaces with multiple subinterfaces. `store` - Specifies persistence: - `active`: Setting lasts until next boot. `persistent`: Setting is retained after reboot.
interface `ipv4 set winsservers`	Configures the WINS server settings for a specified interface. Use `source=dhcp` to configure WINS servers via DHCP. Use `source=static` to specify a static list of WINS server IP addresses for the interface. The `address` option is available only when `source=static`. `name` - Specifies the interface name or index. `source` - Sets the source for WINS server configuration: - `dhcp`: Uses DHCP to configure WINS servers. - `static`: - Uses a static list of WINS server IP addresses. `address` - Specifies the WINS server IP address. `none` - Clears the WINS servers list.
interface `ipv4 show addresses`	Displays the IP address configuration for all interfaces or a specific interface. `name` - Specifies the interface name or index.
interface `ipv4 show compartments`	Displays information about all compartments, or details for a specified compartment. `compartment` - Specifies the compartment ID to display information for.

Command	Description
	`level` - Sets the output detail level. Use `normal` to show one line per compartment (default if no compartment is specified), or `verbose` to show detailed information (default if a compartment is specified). `store` - Chooses which compartments to display: - `active`: Shows current compartments (default). - `persistent`: Shows compartments saved for future use.
interface `ipv4 show config`	Displays the IP address and TCP/IP configuration details for one or more interfaces, including assigned DNS and WINS servers. `name` - Specifies the interface name or index.
interface `ipv4 show destinationcache`	Shows destination cache entries. If `interface` is specified, it displays the cache for that interface only. If `address` is also specified, shows only the matching destination cache entry. `interface` - Specifies the interface name or index. `address` - Sets the destination address. `level` - Chooses output detail: - `normal`: Displays one line per entry (default if no address is given). - `verbose`: Shows extra information for each entry (default if an address is given).
interface `ipv4 show dnsservers`	Displays the DNS server settings for one or more specified network interfaces. `name` - Specifies the interface name or index.
interface `ipv4 show dynamicportrange`	Shows the configuration parameters for dynamic port ranges. `protocol` - Specifies the protocol type: - `tcp`: Displays the dynamic port range for TCP. - `udp`: Displays the range for UDP. `store` - Selects which configuration to show: - `active`: Displays current stack settings (default). - `persistent`: Displays saved settings.
interface `ipv4 show excludedportrange`	Shows all excluded port ranges. `protocol` - Specifies the protocol type: - `tcp`: Displays the dynamic port range for TCP. - `udp`: Displays the range for UDP. `store` - Selects which configuration to show: - `active`: Displays current stack settings (default). - `persistent`: Displays saved settings.
interface `ipv4 show global`	Shows the global configuration parameters. `store` - Selects which configuration to show:

Command	Description
	- `active`: Displays current stack settings (default). - `persistent`: Displays saved settings.
interface `ipv4 show icmpstats`	Displays the ICMP statistics information. `rr` - Specifies the interval, in seconds, at which the interface ICMP statistics are refreshed.
interface `ipv4 show interfaces`	Shows information about all interfaces, or about a specified interface if provided. `interface` - Specifies the name or index of the network interface. `rr` - Sets the refresh interval, in seconds, for updating interface statistics. `level` - Sets the detail level for output. - `normal`: Shows one line per interface (default if no interface is specified). - `verbose`: Displays detailed information (default if an interface is specified). `store` - Chooses which interfaces to display: - `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv4 show ipaddresses`	Shows information about all IPv4 addresses, or about a specified interface if provided. `interface` - Specifies the name or index of the network interface. `level` - Sets the detail level for output. - `normal`: Shows one line per interface (default if no interface is specified). - `verbose`: Displays detailed information (default if an interface is specified). `store` - Chooses which interfaces to display: - `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv4 show ipnettomedia`	Displays the IP network to physical media mappings information. `rr` - Specifies the refresh interval, in seconds, for updating interface IP net-to-media mappings.
interface `ipv4 show ipstats`	Displays the IP statistics information. `rr` - Specifies the refresh interval, in seconds, for updating IP statistics.
interface `ipv4 show joins`	Shows IP multicast groups joined on one or more interfaces. If no interface is specified, multicast groups for all interfaces are displayed. `interface` - Specifies the name or index of the network interface. `level` - Sets the output detail level: - `normal`: Shows one line per interface (default if no interface is specified). - `verbose`: Displays detailed information (default if an interface is specified).

Command	Description
interface `ipv4 show neighbors`	Displays neighbor cache entries for IPv4 interfaces. If you specify an interface, only its cache is shown. If you also specify a subinterface, only entries for that subinterface are displayed. If you provide an address, only the entry for that specific neighbor is shown. `interface` - The name or index of the network interface. `address` - The IPv4 address of the neighbor. `subinterface` - The LUID of the subinterface. `level` - Specifies the output detail: - `normal`: Shows one line per subinterface (default if no subinterface is given). - `verbose`: Shows extra details (default if a subinterface is given). `store` - Chooses which interfaces to display: - `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv4 show offload`	Displays the offload information of a specific interface or interfaces. `name` - Specifies the interface name or index.
interface `ipv4 show route`	Shows entries in the route table. `level` - Specifies the detail of route information to display: - `normal`: Shows standard routes (default). - `verbose`: Includes loopback routes. `store` - Selects which route table to display. Use `active` for current stack information (default), or `persistent` for saved configuration.
interface `ipv4 show subinterfaces`	Shows information about all subinterfaces, or about all subinterfaces on a specified interface if provided. `interface` - Specifies the interface name or index. `subinterface` - Specifies the LUID of the subinterface, used only for interfaces with multiple subinterfaces. `level` - Sets the output detail: - `normal`: Shows one line per subinterface (default if no subinterface is specified). - `verbose`: Displays detailed information for each subinterface (default if a subinterface is specified). `store` - Selects which subinterfaces to display: - `active`: Shows currently active subinterfaces (default). - `persistent`: Shows persistent subinterfaces.
interface `ipv4 show tcpconnections`	Displays the status of TCP connections. `localaddress` - Specifies the local IP address for the connection. `localport` - Specifies the local port number for the connection. `remoteaddress` - Specifies the remote IP address for the connection.

Command	Description
	`remoteport` - Specifies the remote port number for the connection. `rr` - Sets the refresh rate, in seconds, for displaying TCP connection information.
interface `ipv4 show tcpstats`	Displays the TCP connection statistics. `rr` - Specifies the refresh interval, in seconds, for updating TCP statistics information.
interface `ipv4 show udpconnections`	Displays the UDP connection status information. `localaddress` - Specifies the local IP address of the UDP endpoint. `localport` - Specifies the local port number of the UDP endpoint. `rr` - Sets the refresh interval, in seconds, for displaying UDP connection information.
interface `ipv4 show udpstats`	Displays the UDP connection statistics. `rr` - Sets the refresh interval, in seconds, for displaying UDP statistics information.
interface `ipv4 show winsservers`	Displays the WINS server configuration for a specific interface or interfaces. `name` - Specifies the interface name or index.
interface `ipv4 uninstall`	Removes the IPv4 protocol from the system. A system restart is required for the changes to take effect.

interface ipv6

Expand table

Command	Description
interface `ipv6 6to4 set interface`	Configures settings for a specified 6to4 interface within the IPv6 context. `name` - Specifies the interface name or index. `routing` - Sets whether the interface acts as a router: - `enabled`: Turns on routing for this interface. - `disabled`: Turns off routing for this interface. - `default`: Uses the system's default routing setting.
interface `ipv6 6to4 set relay`	Configures the 6to4 relay server for IPv6 tunneling. `name` - Specifies the name of the 6to4 relay or use `default` for the system default. `state` - Sets the state of relay name resolution: - `enabled`: Forces the relay to be active and handle 6to4 traffic. - `disabled`: Turns off the relay. It doesn't process any 6to4 traffic.

Command	Description
	- `automatic`: Lets Windows decide when to enable or disable the relay based on network conditions. - `default`: Resets the relay state to its default configuration. `interval` - Sets the relay name resolution interval in minutes.
interface `ipv6 6to4 set routing`	Configures routing options for 6to4 tunneling in the IPv6 configuration context. `routing` - Controls how 6to4 routing is handled: - `enabled`: Forces 6to4 routing to be active. - `disabled`: Turns off 6to4 routing. - `automatic`: Lets Windows decide when to enable/disable routing based on network conditions. - `default`: Resets routing to the system default. `sitelocals` - Manages the use of site-local addresses in 6to4 routing: - `enabled`: Allows site-local addresses. - `disabled`: Blocks site-local addresses. - `default`: Resets to the system default for site-local address handling.
interface `ipv6 6to4 set state`	Configures the 6to4 service in the IPv6 configuration context. `state` - Specifies the operational mode for the 6to4 service: - `disabled`: Disables the 6to4 service; no 6to4 interfaces are created. - `enabled`: Enables the 6to4 service and configures 6to4 addresses for each public IPv4 address. - `automatic`: Enables the 6to4 service to automatically configure itself for each available public IPv4 address. - `default`: Restores the default 6to4 behavior, equivalent to `disabled`. `undoonstop` - Determines whether 6to4 is disabled when the service stops.
interface `ipv6 6to4 show`	Displays the 6to4 interface configuration and status in the IPv6 configuration context. `interface` - Displays the configuration details of the 6to4 interface. `relay` - Displays information about the 6to4 relay server. `routing` - Displays the current routing state for 6to4. `state` - Displays the current operational state of the 6to4 service.
interface `ipv6 add address`	Assigns a static IPv4 address and default gateway to a specified interface. If the interface is currently configured for DHCP, DHCP is disabled when this command is executed. `interface` - Specifies the interface name or index. `address` - The IPv6 address to assign. Optionally, include a subnet prefix length (example: `192.168.1.10/24`). `type` - Specifies the address type. Use `unicast` to assign a unicast address (default), or `anycast` to assign an anycast address. `validlifetime` - Duration of the address or route that remains valid. Default is

Command	Description
	`infinite`. `store` - Specifies the address persistence. Use `active` for a temporary address removed on reboot, or `persistent` to retain the address after reboot (default). `skipassource` - If `true`, the address isn't used as the source for outgoing packets unless explicitly specified. Default is `false`.
interface `ipv6 add dnsservers`	Adds a new DNS server IPv6 address to the static list for the specified interface. By default, the new DNS server is appended to the end of the list. If an index is provided, the server is inserted at that position, shifting others as needed. If the interface previously used DHCP for DNS, the static address replaces the DHCP-assigned list. When the `validate` option is set to `yes`, the new DNS server is verified before being added. `name` - Specifies the name or index of the interface where the DNS server is added. `address` - The IPv6 address of the DNS server to add. `index` - Sets the preference order for the DNS server address. `validate` - Indicates whether to validate the DNS server setting using `yes` (default) or `no`.
interface `ipv6 add excludedportrange`	Excludes a specified range of consecutive ports from being used by the system for IPv6 traffic. `protocol` - Specifies the protocol to exclude. Use `tcp` to exclude TCP ports or `udp` to exclude UDP ports. `startport` - The starting port number for the exclusion range. `numberofports` - The total number of consecutive ports to exclude, beginning with `startport`. `store` - Determines the persistence of the exclusion. Use `active` for a temporary exclusion until the next reboot, or `persistent` to retain the exclusion after reboot (default).
interface `ipv6 add neighbors`	Adds a neighbor entry to the specified IPv6 network interface. This is typically used to associate an IP address with a link-layer (MAC) address. `interface` - The name or index of the network interface where the neighbor entry is added. `address` - The IPv6 address of the neighbor. `neighbor` - The link-layer (MAC) address to associate with the specified IP address. `subinterface` - The LUID of the subinterface, required only for interfaces with multiple subinterfaces. `store`: Determines whether the entry is temporary (`active`, removed on next reboot) or permanent (`persistent`, retained after reboot; this is the default).
interface `ipv6 add potentialrouter`	Adds a potential router to the specified interface.

Command	Description
	`interface` - Name or index of the interface. `address` - The IPv6 address of the potential router.
interface `ipv6 add prefixpolicy`	Adds a policy that influences how source and destination IPv6 addresses are selected for a specified prefix. `prefix` - The IPv6 prefix to which the policy applies. `precedence` - The precedence value used to sort destination addresses. `label` - The label used to match source and destination address prefixes. `store` - Determines persistence: - `active`: The policy is temporary and removed on reboot. - `persistent`: The policy remains after reboot (default).
interface `ipv6 add route`	Adds a network route for a specified IPv6 prefix. `prefix` - The IP prefix for which to add a route. `interface` - The name or index of the interface to use for the route. `nexthop` - The gateway address, if the prefix isn't on-link. `siteprefixlength` - The prefix length for the entire site, if on-link. `metric` - The metric value for the route. `publish` - Controls advertisement in Route Advertisements. Use `no` (default) to not advertise, `age` to advertise with a finite lifetime, or `yes` to advertise with an infinite lifetime. `validlifetime` - The duration for which the route remains valid. Default is `infinite`. `preferredlifetime` - The duration for which the route is preferred. Default is equal to `validlifetime`. `store` - Determines persistence. Use `active` for a temporary change (until next boot), or `persistent` for a change that remains after reboot (default).
interface `ipv6 add v6v4tunnel`	Creates an IPv6-over-IPv4 tunnel interface. <rb> `interface` - The friendly name for the tunnel interface. `localaddress` - The IPv4 address of the local tunnel endpoint. `remoteaddress` - The IPv4 address of the remote tunnel endpoint.
interface `ipv6 delete address`	Removes a static IPv6 address or default gateway from the specified interface. `interface` - The interface name or index. `address` - The static IPv6 address to delete. `store` - Use `active` to make the deletion temporary (until next boot), or `persistent` to make it permanent (default).
interface `ipv6 delete destinationcache`	Flushes the destination cache, which stores routing information for IPv6 destinations. If an interface is specified, only that interface's cache is flushed. If an address is also specified, only that destination cache entry is deleted.

Command	Description
	`interface` - The name or index of the interface. `address` - The destination address.
interface `ipv6 delete dnsservers`	Deletes statically configured DNS server IP addresses for a specific interface. If `validate` is set to `yes`, the remaining DNS servers are validated. `name` - The name or index of the interface where DNS servers are deleted. `address` - The IPv6 address of the DNS server to delete, or `all` to delete all configured DNS server addresses. `validate` - Specifies whether to validate the DNS server setting after deletion, `yes` (default) or `no`.
interface `ipv6 delete excludedportrange`	Deletes a previously set exclusion for a range of TCP or UDP ports. The `startport` and `numberofports` must match the previously created exclusion. `protocol` - `tcp` to delete a TCP exclusion, or `udp` for UDP. `startport` - The starting port number of the exclusion. `numberofports` - The number of consecutive ports to delete, starting from `startport`. `store` - `active` for a temporary deletion (until next boot), or `persistent` for a permanent deletion (default).
interface `ipv6 delete interface`	Removes a specified interface from the IPv6 stack. Only point-to-point interfaces that were manually configured can be deleted. `interface` - Name or index of the interface to delete.
interface `ipv6 delete neighbors`	Deletes entries from the IPv6 neighbor cache. By default, entries in both stores are deleted. `interface` - The interface name or index. `address` - The neighbor's IP address. `subinterface` - The subinterface LUID. `store` - `active` for a temporary deletion (entry may reappear on next boot), or `persistent` for a permanent deletion.
interface `ipv6 delete potentialrouter`	Removes a potential router from the specified interface. `interface` - Name or index of the interface. `address` - The IPv6 address of the potential router.
interface `ipv6 delete prefixpolicy`	Removes a policy that influences how source and destination IPv6 addresses are selected for a specified prefix. `prefix` - The IPv6 prefix to which the policy applies. `store` - Determines persistence:

Command	Description
	- `active`: The policy is temporary and removed on reboot. - `persistent`: The policy remains after reboot (default).
interface `ipv6 delete route`	Deletes an IPv6 route from the routing table. `prefix` - The IP prefix of the route to delete (for example, `192.168.1.0/24`). `interface` - The name or index of the interface associated with the route. `nexthop` - The gateway address for the route, if the prefix isn't on-link. `store` - Specifies whether the deletion is temporary (`active`, until next boot) or permanent (`persistent`). By default, entries in both stores are deleted.
interface `ipv6 isatap set router`	Configures the ISATAP router settings. `name` - Specifies the ISATAP router name or use `default` for the system default. `state` - Sets the router name resolution state: - `enabled`: Enables router name resolution for ISATAP. - `disabled`: Disables router name resolution for ISATAP. - `default`: Restores the default router name resolution state. `interval` - Sets the router name resolution interval in minutes.
interface `ipv6 isatap set state`	Sets the operational state of the ISATAP service. `state` - Specifies the ISATAP service state: - `disabled`: Disables the ISATAP service. No ISATAP interfaces are created. - `enabled`: Enables the ISATAP service and configures a link-local address on each ISATAP interface. Additional addresses may be assigned by the ISATAP server. - `default`: Attempts to contact the ISATAP server. If unreachable, no addresses are configured on the ISATAP interface.
interface `ipv6 isatap show`	Displays the following information: `router` - Displays information about the configured ISATAP router. `state` - Displays the current operational state of the ISATAP service.
interface `ipv6 reset`	Removes all user-configured IPv6 settings and restores defaults after a system restart.
interface `ipv6 set address`	Modifies the IPv6 address configuration for a specified interface. `interface` - Interface name or index. `address` - The IPv6 address to add or modify. `type` - Specifies the address type. `unicast` (default) marks the address as a unicast address while `anycast` marks it as anycast. `validlifetime` - The duration value of the route to remain valid (default is infinite). `preferredlifetime` - The duration of the route that's preferred (default is infinite).

Command	Description
	`store` - Determines whether the configuration change is temporary (`active`, lasts until the next reboot) or permanent (`persistent`, remains after reboot; default).
interface `ipv6 set compartment`	Updates compartment settings for a network interface. `compartment` - Specifies the compartment ID to configure. `defaultcurhoplimit` - Sets the default hop limit for outgoing packets in the compartment. `store` - Determines persistence: - `active`: Applies until next reboot. - `persistent`: Retains the setting after reboot.
interface `ipv6 set dnsservers`	Sets DNS server configuration to either DHCP or static mode. When `source` is `static`, the `address` option is available for configuring a static list of DNS server IP addresses for the specified interface. If `validate` is `yes`, the newly set DNS server is validated. `name` - Name or index of the interface. `source` - `dhcp` sets DHCP as the source for configuring DNS servers. `static` sets the source to local static configuration. `address` - IP address for a DNS server or `none` to clear the list. `register` - `none` disables Dynamic DNS registration, `primary` registers under the primary DNS suffix only, `both` registers under both the primary and connection-specific suffixes. `validate` - Specifies whether validation of the DNS server setting is performed, `yes` (default) or `no`.
interface `ipv6 set dynamicportrange`	Modifies the range of ports used for dynamic port assignment (wildcard port assignment). `protocol` - Determines the protocol type: - `tcp` modifies the dynamic port range for TCP. - `udp` for UDP. `startport` - Starting port for dynamic port assignment. `numberofports` - Number of ports available for dynamic port assignment. `store` - Determines whether the configuration change is temporary (`active`, lasts until the next reboot) or permanent (`persistent`, remains after reboot (default)).
interface `ipv6 set global`	Configures the global IPv6 stack settings. `addressmaskreply` - Enables (`enabled`) or disables (`disabled`) responding to ICMP address mask packets. `defaultcurhoplimit` - Sets the default hop limit for outgoing packets. `dhcpmediasense` - Enables (`enabled`) or disables (`disabled`) media sense (network connection detection). `flowlabel` - Enables (`enabled`) or disables (`disabled`) flow label support. `groupforwardedfragments` - Enables (`enabled`) or disables (`disabled`) grouping of

Command	Description
	fragments before forwarding. `icmpjumbograms` - Enables (`enabled`) or disables (`disabled`) ICMP jumbograms (IPv6 only). `icmpredirects` - Enables (`enabled`) or disables (`disabled`) updating the path cache in response to ICMP redirect packets. `loopbackexecutionmode` - Sets execution mode for loopback processing: - `inline`: Loopback packet processing happens directly in the context of the thread that sends the packet. This is the simplest and fastest mode, but it can block the sending thread if processing takes time. - `adaptive`: The system dynamically chooses between inline and worker modes based on current load and performance. - `worker`: Loopback packet processing is offloaded to a separate worker thread. This prevents blocking the sending thread, which can improve responsiveness under heavy load, but may introduce a slight delay. `loopbacklargemtu` - Enables (`enabled`) or disables (`disabled`) large IP MTU for the loopback path. `loopbackworkercount` - Number of worker threads (1-16) for loopback processing in adaptive mode. `mediasenseeventlog` - Enables (`enabled`) or disables (`disabled`) media sense (network connection detection) event logging. `minmtu` - Sets the minimum IP MTU to avoid fragmentation. For IPv4, range is 352-576. For IPv6, it's fixed at 1280. `mldlevel` - Sets multicast support level: - `none`: Multicast packets can't be sent or received. - `sendonly`: Multicast packets can be sent but not received. - `all`: Multicast packets can be sent and received. `mldversion` - The maximum Multicast Listener Discovery (MLD) version supported by the host. `multicastforwarding` - Enables (`enabled`) or disables (`disabled`) forwarding of multicast packets. `neighborcachelimit` - The maximum number of neighbor cache entries no longer referenced. `randomizeidentifiers` - Enables (`enabled`) or disables (`disabled`) randomization of interface identifiers. `reassemblylimit` - The maximum size of the reassembly buffer. `reassemblyoutoforderlimit` - The maximum number of out-of-order fragments for reassembly. `recursivereassembly` - Enables (`enabled`) or disables (`disabled`) recursive reassembly of fragments (IPv6 only). `routecachelimit` - The maximum number of route cache entries. `routepolicies` - Enables (`enabled`) or disables (`disabled`) route policies. `slaacmaxdadattempts` - The number of duplicate address detection attempts for SLAAC addresses (IPv6 only). `slaacprivacylevel` - The privacy level for SLAAC address suffixes (IPv6 only): - `0`: Use EUI-64. - `1`: Use randomized EUI-64. - `2`: Use randomized EUI-64, prefix, and secret key.

Command	Description
	`sourcebasedecmp` - Enables (`enabled`) or disables (`disabled`) source-based Equal-Cost Multi-Path (ECMP). `sourceroutingbehavior` - Controls behavior for source-routed packets. Per RFC 5095, forwarding of source-routed IPv6 packets is removed. `forward` is equivalent to `dontforward`. `store` - Specifies persistence: - `active`: Setting lasts until next boot. - `persistent`: Setting is retained after reboot (default). `taskoffload` - Enables (`enabled`) or disables (`disabled`) task offload.
interface `ipv6 set interface`	Modifies the configuration of an IPv6 network interface. `advertise` - Enables (`enabled`) or disables (`disabled`) whether the interface sends RA. By default, advertisements are disabled. `advertisedefaultroute` - Determines whether the interface is advertised as a default router. `advertisedrouterlifetime` - Sets the router lifetime (in seconds) advertised in RA with a value of **0** to **65535**. Default is 1,800 seconds. `basereachabletime` - Sets the base time (in milliseconds) that a neighbor is considered reachable. `currenthoplimit` - Sets the hop limit for outbound IPv6 traffic. Valid range is **0** to **255**. `dadtransmits` - Sets the number of Duplicate Address Detection (DAD) attempts. A value of **0** disables DAD. `dhcpstaticipcoexistence` - Enables (`enabled`) or disables (`disabled`) whether DHCP and static IP addresses are allowed to coexist on the interface. `ecncapability` - Controls Explicit Congestion Notification (ECN) marking for outgoing packets: - `ecndisabled`: Disables ECN marking. Outgoing packets won't use ECN bits, so congestion is signaled only by packet loss. - `ect0`: Sets ECN bits to ECT(0), indicating the packet is ECN-capable. Routers can mark congestion without dropping packets. - `ect1`: Sets ECN bits to ECT(1), another ECN-capable codepoint (rarely used; functionally similar to ECT(0)). - `application`: Leaves ECN marking up to the application. The application decides whether and how to set ECN bits. `enabledirectedmacwolpattern` - Enables (`enabled`) or disables (`disabled`) directed MAC packet patterns to wake the computer. `forcearpndwolpattern` - Enables (`enabled`) or disables (`disabled`) address resolution and neighbor discovery packet patterns to wake the computer from sleep. `forwarding` - Enables (`enabled`) or disables (`disabled`) packet forwarding on this interface. When enabled, the interface can route packets to other interfaces. `ignoredefaultroutes` - When set to `enabled`, default routes on this interface are ignored. Default is `disabled`. `interface` - Specifies the name or index of the network interface to configure. `managedaddress` - Enables (`enabled`) or disables (`disabled`) managed address

Command	Description
	configuration, allowing addresses to be assigned via DHCPv6. This setting may be overridden if router discovery is enabled and advertising is disabled. `mtu` - Sets the Maximum Transmission Unit (MTU) for the interface. If not specified, the link's default MTU is used. `nud` - Enables (`enabled`) or disables (`disabled`) Neighbor Unreachability Detection (NUD), which helps determine if a neighbor is reachable. `otherstateful` - Enables (`enabled`) or disables (`disabled`) other stateful configuration options, such as obtaining additional settings via DHCPv6. This may also be overridden by router discovery settings. `rabaseddnsconfig` - Enables (`enabled`) or disables (`disabled`) DNS configuration via RA as defined in RFC 6106. `retransmittime` - Specifies the time (in milliseconds) between retransmissions of neighbor solicitation messages. `routerdiscovery` - Enables, disables, or allows DHCP to control router discovery on the interface. `siteprefixlength` - Defines the default length of the global IPv6 prefix for the site. `store` - Specifies whether the configuration change is temporary (`active`, until next reboot) or persistent (`persistent`, remains after reboot). `weakhostreceive` - Enables weak host receive behavior, allowing the interface to receive packets for IP addresses not assigned to it. `weakhostsend` - Enables weak host send behavior, allowing the interface to send packets even if the source IP doesn't belong to the interface.
interface `ipv6 set neighbors`	Configures a static neighbor entry for an IPv6 interface. `interface` - The name or index of the network interface to configure. `address` - IPv6 address of the neighbor. `neighbor` - Link-layer (MAC) address to associate with the specified IPv6 address. `store` - Determines persistence: - `active` Entry is removed on next boot. - `persistent`: Entry remains after reboot (default).
interface `ipv6 set prefixpolicy`	Modifies the source and destination address selection policy for a given IPv6 prefix. `prefix` - The IPv6 prefix for which to add or modify a policy. `precedence` - The precedence value used for address selection ordering. `label` - The label value for matching during address selection. `store` - Determines persistence: - `active` Entry is removed on next boot. - `persistent`: Entry remains after reboot (default).
interface `ipv6 set privacy`	Modifies parameters related to temporary IPv6 address generation. `state` - Enables (`enabled`) or disables (`disabled`) temporary address generation. `maxdadattempts` - The number of duplicate address detection attempts (default is

Command	Description
	5). `maxvalidlifetime` - The maximum valid lifetime for a temporary address (default is seven days). `maxpreferredlifetime` - The maximum preferred lifetime for a temporary address (default is one day). `regeneratetime` - The time before deprecating a temporary address when a new address is generated (default is 5 seconds). `maxrandomtime` - The maximum random delay at startup (default is 10 minutes). `store` - Determines persistence: - `active` Entry is removed on next boot. - `persistent`: Entry remains after reboot (default). Time values can be expressed in days, hours, minutes, and seconds (`1d2h3m4s`).
interface `ipv6 set route`	Modifies parameters for an IPv6 route. When `publish` is set to `no` or `age`, the route is deleted after the valid lifetime expires. If `publish` is `age`, RAs include the remaining valid lifetime. If `publish` is set to `yes`, the route is never deleted and RAs always show the specified valid lifetime. `prefix` - The IPv6 prefix of the route to modify. `interface` - The name or index of the interface for the route. `nexthop` - The gateway IPv6 address, if the prefix isn't on-link. `siteprefixlength` - The prefix length for the entire site if on-link. `metric` - The metric value for the route. `publish` - Controls advertisement in RA: - `no`: Not advertised (default). - `age`: Advertised with finite lifetime. - `yes`: Advertised with infinite lifetime. `validlifetime` - The duration value of the route to remain valid (default is infinite). `preferredlifetime` - The duration of the route that's preferred (default is infinite). `store` - Determines persistence: - `active` Entry is removed on next boot. - `persistent`: Entry remains after reboot (default). Time values can be expressed in days, hours, minutes, and seconds (`1d2h3m4s`).
interface `ipv6 set slaacsecretkey`	Sets the RFC 7217 secret key for IPv6 SLAAC privacy. The key must be a 32-character hexadecimal string. This secret is used when the global SLAAC privacy level is set to 2 (`netsh interface ipv6 set global slaacprivacylevel=2`). `key` - A 32-character hexadecimal string used as the secret key for IPv6 SLAAC privacy.
interface `ipv6 set subinterface`	Modifies the subinterface configuration parameters. `interface` - The name or index of the interface to configure.

Command	Description
	`mtu` - Sets the MTU for the subinterface. Defaults to the link's natural MTU if not specified. `subinterface` - The LUID of the subinterface. Required only for interfaces with multiple subinterfaces. `store` - Specifies persistence: - `active`: Setting lasts until next boot. `persistent`: Setting is retained after reboot.
interface `ipv6 set teredo`	Configures the Teredo operational state. Use `default` for any parameter to restore its system default. The `type=server` option is available only on Windows Server editions. `type` - Specifies the Teredo mode: - `disabled`: Turns off the Teredo service. - `client`: Enables the Teredo service on the client. - `enterpriseclient`: Skips managed network detection. - `natawareclient`: Enables Teredo client detection and NAT type. - `server`: Enables the server (Windows Server only). - `default`: Sets to the system default (client). `servername` - Sets the Teredo server name or IPv4 address. `refreshinterval` - Defines the client refresh interval in seconds. `clientport` - Sets the UDP port for the Teredo client. If omitted, the system chooses the port. `servervirtualip` - Specifies the IPv4 address for the server's virtual IP (not used in client mode).
interface `ipv6 show addresses`	Shows all IPv6 addresses, or only those for a specified interface if provided. `interface` - Specifies the interface name or index to display addresses for. `level` - Sets the output detail: - `normal`: Displays one line per interface (default if no interface is specified). -`verbose`: Displays detailed information (default if an interface is specified). `store` - Chooses which addresses to show: - `active`: Displays addresses currently in use (default). -`persistent`: Displays addresses saved for future use.
interface `ipv6 show compartments`	Displays information about all compartments, or details for a specified compartment. `compartment` - Specifies the compartment ID to display information for. `level` - Sets the output detail level. Use `normal` to show one line per compartment (default if no compartment is specified), or `verbose` to show detailed information (default if a compartment is specified). `store` - Chooses which compartments to display: - `active`: Shows current compartments (default). - `persistent`: Shows compartments saved for future use.

Command	Description
interface `ipv6 show destinationcache`	Shows destination cache entries. If `interface` is specified, displays the cache for that interface only. If `address` is also specified, shows only the matching destination cache entry. `interface` - Specifies the interface name or index. `address` - Sets the destination address. `level` - Chooses output detail: - `normal`: Displays one line per entry (default if no address is given). - `verbose`: Shows extra information for each entry (default if an address is given).
interface `ipv6 show dnsservers`	Displays the DNS server settings for one or more specified network interfaces. `name` - Specifies the interface name or index.
interface `ipv6 show dynamicportrange`	Shows the configuration parameters for dynamic port ranges. `protocol` - Specifies the protocol type: - `tcp`: Displays the dynamic port range for TCP. - `udp`: Displays the range for UDP. `store` - Selects which configuration to show: - `active`: Displays current stack settings (default). - `persistent`: Displays saved settings.
interface `ipv6 show excludedportrange`	Shows all excluded port ranges. `protocol` - Specifies the protocol type: - `tcp`: Displays the dynamic port range for TCP. - `udp`: Displays the range for UDP. `store` - Selects which configuration to show: - `active`: Displays current stack settings (default). - `persistent`: Displays saved settings.
interface `ipv6 show global`	Shows the global configuration parameters. `store` - Selects which configuration to show: - `active`: Displays current stack settings (default). - `persistent`: Displays saved settings.
interface `ipv6 show interfaces`	Shows information about all interfaces, or about a specified interface if provided. `interface` - Specifies the name or index of the network interface. `rr` - Sets the refresh interval, in seconds, for updating interface statistics. `level` - Sets the detail level for output. - `normal`: Shows one line per interface (default if no interface is specified). - `verbose`: Displays detailed information (default if an interface is specified). `store` - Chooses which interfaces to display:

Command	Description
	- `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv6 show ipstats`	Displays the IPv6 statistics information. `rr` - Specifies the refresh interval, in seconds, for updating IP statistics.
interface `ipv6 show joins`	Shows IPv6 multicast groups joined on one or more interfaces. If no interface is specified, multicast groups for all interfaces are displayed. `interface` - Specifies the name or index of the network interface. `level` - Sets the output detail level: - `normal`: Shows one line per interface (default if no interface is specified). - `verbose`: Displays detailed information (default if an interface is specified).
interface `ipv6 show neighbors`	Displays neighbor cache entries for IPv6 interfaces. If you specify an interface, only its cache is shown. If you also specify a subinterface, only entries for that subinterface are displayed. If you provide an address, only the entry for that specific neighbor is shown. `interface` - The name or index of the network interface. `address` - The IPv6 address of the neighbor. `subinterface` - The LUID of the subinterface. `level` - Specifies the output detail: - `normal`: Shows one line per subinterface (default if no subinterface is given). - `verbose`: Shows extra details (default if a subinterface is given). `store` - Chooses which interfaces to display: - `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv6 show offload`	Displays the offload information of a specific interface or interfaces. `name` - Specifies the interface name or index.
interface `ipv6 show potentialrouters`	Lists all potential routers detected by the system, or only those associated with a specified interface if provided. `interface` - The name or index of the network interface.
interface `ipv6 show prefixpolicies`	Displays the prefix policy table, which determines how source and destination IPv6 addresses are selected for network connections. `store` - Specifies which set of prefix policies you want to view: - `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv6 show privacy`	Displays the current IPv6 privacy settings.

Command	Description
	`store` - Specifies which configuration store to use when displaying privacy settings: - `active`: Shows currently active interfaces (default). - `persistent`: Shows interfaces saved for future use.
interface `ipv6 show route`	Shows entries in the route table. `level` - Specifies the detail of route information to display: - `normal`: Shows standard routes (default). - `verbose`: Includes loopback routes. `store` - Selects which route table to display. Use `active` for current stack information (default), or `persistent` for saved configuration.
interface `ipv6 show siteprefixes`	Displays the current IPv6 site prefix table.
interface `ipv6 show slaacsecretkey`	Displays the RFC 7217 secret key used for IPv6 SLAAC privacy when the privacy level is set to 2. `store` - Specifies which key to display: - `active`: Shows the key currently in use by the stack (default). - `persistent`: Shows the key saved for future use.
interface `ipv6 show subinterfaces`	Shows information about all subinterfaces, or about all subinterfaces on a specified interface if provided. `interface` - Specifies the interface name or index. `subinterface` - Specifies the LUID of the subinterface, used only for interfaces with multiple subinterfaces. `level` - Sets the output detail: - `normal`: Shows one line per subinterface (default if no subinterface is specified). - `verbose`: Displays detailed information for each subinterface (default if a subinterface is specified). `store` - Selects which subinterfaces to display: - `active`: Shows currently active subinterfaces (default). - `persistent`: Shows persistent subinterfaces.
interface `ipv6 show tcpstats`	Displays the TCP connection statistics. `rr` - Specifies the refresh interval, in seconds, for updating TCP statistics information.
interface `ipv6 show teredo`	Displays the current configuration and operational status of the Teredo tunneling protocol.
interface `ipv6 show tfofallback`	Displays the current status and statistics for TCP Fast Open (TFO) fallback on IPv6 interfaces. This helps determine if TFO is being used or if connections are falling back to standard TCP.

Command	Description
interface `ipv6 show udpstats`	Displays the UDP connection statistics. `rr` - Sets the refresh interval, in seconds, for displaying UDP statistics information.

interface isatap

Expand table

Command	Description
interface `isatap set router`	Configures the ISATAP router settings. `name` - Specifies the ISATAP router name or use `default` for the system default. `state` - Sets the router name resolution state: - `enabled`: Enables router name resolution, allowing the system to resolve the ISATAP router name to an IPv6 address. - `disabled`: Disables router name resolution, preventing the system from resolving the ISATAP router name. - `default`: Restores the router name resolution state to the system default setting. `interval` - Sets the router name resolution interval in minutes.
interface `isatap set state`	Configures the operational state of the ISATAP service. `state` - Specifies the ISATAP service mode: - `enabled`: Enables the ISATAP service and configures a link-local address on each ISATAP interface. Additional addresses may be assigned by the ISATAP server. - `disabled`: Disables the ISATAP service. No ISATAP interfaces are created. - `default`: Attempts to contact the ISATAP server. If unreachable, no addresses are configured on the ISATAP interface.
interface `isatap show router`	Displays information about the configured ISATAP router.
interface `isatap show state`	Displays the current operational state of the ISATAP service.

interface portproxy

Expand table

Command	Description
interface `portproxy add`	Adds a proxy rule to listen on an IPv4 address and port, forwarding incoming TCP connections to another IPv4 address and port.

Command	Description
`v4tov4`	`listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv4 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv4 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv4 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy add v4tov6`	Adds a proxy rule to listen on an IPv4 address and port, forwarding incoming TCP connections to an IPv6 address and port. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv6 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv6 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv4 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy add v6tov4`	Adds a proxy rule to listen on an IPv6 address and port, forwarding incoming TCP connections to an IPv4 address and port. `listenport` - The IPv6 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv4 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv4 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv6 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy add v6tov6`	Adds a proxy rule to listen on an IPv6 address and port, forwarding incoming TCP connections to another IPv6 address and port. `connectaddress` - The IPv6 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv6 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv6 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy delete v4tov4`	Deletes a proxy rule that listens on an IPv4 address and port, forwarding TCP connections to another IPv4 address and port. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the

Command	Description
	proxy listens for incoming connections. `listenaddress` - The local IPv6 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy delete v4tov6`	Deletes a proxy rule that listens on an IPv4 address and port, forwarding TCP connections to an IPv6 address and port. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `listenaddress` - The local IPv6 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy delete v6tov4`	Deletes a proxy rule that listens on an IPv6 address and port, forwarding TCP connections to an IPv4 address and port. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `listenaddress` - The local IPv6 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy delete v6tov6`	Deletes a proxy rule that listens on an IPv6 address and port, forwarding TCP connections to another IPv6 address and port. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `listenaddress` - The local IPv6 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy reset`	Resets all port proxy rules and restores the port proxy configuration to its default state.
interface `portproxy set v4tov4`	Updates an existing proxy rule that listens on an IPv4 address and port, forwarding TCP connections to another IPv4 address and port. Use this to change the destination or listening parameters for a v4-to-v4 proxy. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv4 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv4 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv4 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy set v4tov6`	Updates an existing proxy rule that listens on an IPv4 address and port, forwarding TCP connections to an IPv6 address and port. Use this to modify the forwarding details for a v4-to-v6 proxy.

Command	Description
	`listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv4 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv4 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv4 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy set v6tov4`	Updates an existing proxy rule that listens on an IPv6 address and port, forwarding TCP connections to an IPv4 address and port. Use this to change the destination or listening parameters for a v6-to-v4 proxy. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv4 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv4 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv4 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy set v6tov6`	Updates an existing proxy rule that listens on an IPv6 address and port, forwarding TCP connections to another IPv6 address and port. Use this to modify the forwarding details for a v6-to-v6 proxy. `listenport` - The IPv4 TCP port number (1-65535) or service name on which the proxy listens for incoming connections. `connectaddress` - The IPv4 address, or hostname, to which incoming connections are forwarded. `connectport` - The IPv4 TCP port number (1-65535) or service name on the destination address. `listenaddress` - The local IPv4 address or hostname to listen on. `protocol` - The protocol to use. Only `tcp` is supported.
interface `portproxy show all`	Displays all configured port proxy rules.
interface `portproxy show v4tov4`	Displays the configuration details for proxy rules forwarding IPv4 connections to IPv4.
interface `portproxy show v4tov6`	Displays the configuration details for proxy rules forwarding IPv4 connections to IPv6.

Command	Description
interface `portproxy show v6tov4`	Displays the configuration details for proxy rules forwarding IPv6 connections to IPv4.
interface `portproxy show v6tov6`	Displays the configuration details for proxy rules forwarding IPv6 connections to IPv6.

> **Note**
>
> See the following remarks for the `portproxy` command usage:
>
> - All parameters except `protocol` are required.
> - If you omit `listenaddress`, it defaults to `0.0.0.0`.
> - Port numbers *must* be valid integers in the range 1-65535.
> - Service names (for example, `listenport=MyService`) are resolved using the system's services file, which also applies to `connectport`.

interface set

Expand table

Command	Description
interface `set interface`	Changes the configuration of a network interface. `name` - The current name of the network interface you want to modify. `admin` - Controls whether the interface is `enabled` or `disabled`. `connect` - Sets the connection state, `connected` or `disconnected`, for non-LAN interfaces. `newname` - Assigns a new name to the interface (LAN only). You must specify at least one subparameter other than `name`. If you use `connect=CONNECTED`, the interface is enabled even if you also specify `admin=DISABLED`.

interface show

Expand table

Command	Description
interface `show interface`	Displays the parameters and status of network interfaces. `name` - Specifies the name of a particular interface to display its current settings. If omitted, information for all interfaces is shown.

interface tcp

Expand table

Command	Description
interface `tcp add supplementalport`	Adds a TCP template for a specific source and destination port pair. `template` - Specifies the TCP template to apply: - `internet`: Configures the port for typical internet-facing scenarios. - `datacenter`: Optimizes the port for use in a datacenter environment. - `automatic`: Lets the system choose the most appropriate template based on detected environment or other heuristics. - `custom`: Allows you to specify your own settings instead of using a predefined template. - `compat`: Uses compatibility settings, typically for legacy systems or applications that require older networking behaviors. `localport` - Sets the source port. Use `*` for all ports or specify a port number. `remoteport` - Sets the destination port. Use `*` for all ports or specify a port number.
interface `tcp add supplementalsubnet`	Adds a TCP template filter for traffic destined to a specific subnet. `template` - Specifies the TCP template to apply: - `internet`: Configures the port for typical internet-facing scenarios. - `datacenter`: Optimizes the port for use in a datacenter environment. - `automatic`: Lets the system choose the most appropriate template based on detected environment or other heuristics. - `custom`: Allows you to specify your own settings instead of using a predefined template. - `compat`: Uses compatibility settings, typically for legacy systems or applications that require older networking behaviors. `family` - Sets the address family, either `ipv4` or `ipv6`. `address` - Specifies the subnet address in IPv4 or IPv6 notation, optionally followed by a prefix length.
interface `tcp delete supplementalport`	Removes the specified port entry from the TCP supplemental filter table. `localport` - Specifies the source port to remove. Use `*` to match all ports or provide a specific port number.

Command	Description
	`remoteport` - Specifies the destination port to remove. Use `*` to match all ports or provide a specific port number.
interface `tcp delete supplementalsubnet`	Removes a TCP template filter for a specific subnet. `family` - Specifies the address family, either `ipv4` or `ipv6`. `address` - Specifies the subnet address in IPv4 or IPv6 notation, optionally followed by a prefix length.
interface `tcp reload`	Reloads persisted TCP configuration state from disk. This is an experimental command and **isn't** intended for production use. **Don't use**.
interface `tcp rundown`	Forces a connection rundown on all active TCP trace sessions.
interface `tcp set global`	Configures system-wide TCP settings that apply to all network connections. `rss` - Sets receive-side scaling (RSS): - `disabled`: Disables RSS. - `enabled`: Enables RSS. - `default`: Restores the system default. `autotuninglevel` - Sets how the TCP receive window automatically adjusts: - `disabled`: Keeps the receive window fixed at its default size. - `highlyrestricted`: Allows limited growth of the receive window, conservatively. - `restricted`: Allows some growth, but with restrictions in certain cases. - `normal`: Lets the receive window expand for most network conditions. - `experimental`: Enables maximum growth for testing or unusual scenarios. `ecncapability` - Turns on ECN (Explicit Congestion Notification) (`enabled`) or off (`disabled`). Use `default` to reset to the system setting. `timestamps` - Controls the use of TCP timestamps: - `disabled`: Disables timestamps. - `enabled`: Negotiates timestamps for outgoing connections and enables them for incoming if the peer supports it. - `allowed`: Enables timestamps for incoming connections if the peer supports it (default). `initialrto` - Sets the initial timeout (in milliseconds) for TCP SYN retransmissions. Default is `3000`. `rsc` - Enables (`enabled`) or disables (`disabled`) receive segment coalescing. Use `default` to restore the system setting. `nonsackrttresiliency` - Enables (`enabled`) or disables (`disabled`) round-trip time resiliency for clients that don't support SACK. Default is `disabled`. `maxsynretransmissions` - Sets how many times TCP retransmits SYN packets when trying to connect. Default is `2`. `fastopen` - Enables (`enabled`) or disables (`disabled`) TCP Fast Open. Use `default` to restore the system setting. `fastopenfallback` - Enables (`enabled`) or disables (`disabled`) fallback behavior for TCP Fast Open. Use `default` to restore the system setting.

Command	Description
	`hystart` - Enables (`enabled`) or disables (`disabled`) the HyStart algorithm for slow start. Use `default` to restore the system setting. `prr` - Enables (`enabled`) or disables (`disabled`) the Proportional Rate Reduction algorithm. Use `default` to restore the system setting. `pacingprofile` - Controls when TCP packet pacing is used: - `off`: Disables pacing. - `initialwindow`: Paces only the initial congestion window. - `slowstart`: Paces during slow start. - `always`: Pacing is always enabled. - `default`: Same as `off`.
interface `tcp set heuristics`	Configures TCP heuristics for all connections. `wsh` - This setting isn't used and has no effect. `forcews` - Controls force window scaling on retransmit: - `disabled`: Disables forced window scaling. - `enabled`: Enables forced window scaling. - `default`: Restores the window scaling to the system default (enabled).
interface `tcp set security`	Sets the TCP security parameters, including memory pressure protection (MPP). Use `mpp` to enable or disable MPP globally or for a specific port range. If no port range is specified, the setting applies globally. Port range settings are effective only when MPP is enabled system-wide. `mpp` - Controls memory pressure protection: - `disabled`: Disables MPP. - `enabled`: Enables MPP. - `default`: Restores to the system default. `startport` - Sets the starting port number (1-65535) for the port range. `numberofports` - Specifies how many ports to include, starting from `startport`. `profiles` - Controls profile usage for memory pressure protection: - `disabled`: Disables profile usage. - `enabled`: Enables profile usage. - `default`: Restores the system default.
interface `tcp set supplemental`	Configures TCP supplemental template settings for specific port or subnet filters. Use this command to fine-tune TCP behavior for different network scenarios. If only the `template` parameter is specified, the default global template is modified. If additional parameters are included, the specified template is updated with those values. When setting `minrto`, ensure `delayedacktimeout` is less than `minrto`. For best results, enable both `rack` and `taillossprobe` together to improve loss recovery. `template` - Specifies the TCP template to apply: - `internet`: Configures the port for typical internet-facing scenarios. - `datacenter`: Optimizes the port for use in a datacenter environment. - `automatic`: Lets the system choose the most appropriate template based on

Command	Description
	detected environment or other heuristics. - `custom`: Allows you to specify your own settings instead of using a predefined template. - `compat`: Uses compatibility settings, typically for legacy systems or applications that require older networking behaviors. `minrto` - Sets the minimum retransmission timeout in milliseconds (20-300 ms). `icw` - Sets the initial congestion window size in multiples of the maximum segment size (2-64 MSS). `congestionprovider` - Selects the congestion control algorithm: - `none`: No specific algorithm is set. - `ctcp`: Compound TCP improves throughput on high-speed networks. - `dctcp`: Data Center TCP, optimized for data center environments. - `cubic`: Default in many Linux systems, good for high-bandwidth, high-latency networks. - `bbr2`: Bottleneck Bandwidth and Round-trip propagation time, aims for high throughput and low latency. - `default`: Uses the system's default algorithm. `enablecwndrestart` - Enables (`enabled`) or disables (`disabled`) restarting the congestion window after idle periods. Use `default` to use the system's default behavior. `delayedacktimeout` - Sets the delayed acknowledgment timeout in milliseconds (10-600 ms). `delayedackfrequency` - Sets how many segments are received before sending an ACK (1-255). `rack` - Enables (`enabled`) or disables (`disabled`) Recent Acknowledgment (RACK) time-based loss recovery. Use `default` to use the system's default behavior. `taillossprobe` - Enables (`enabled`) or disables (`disabled`) Tail Loss Probe for faster recovery from packet loss at the end of a transfer. Use `default` to use the system's default behavior.
interface `tcp show global`	Displays global TCP parameters such as autotuning and congestion control. `store` - Selects which configuration to show: - `active`: Displays the current in-memory configuration (default). - `persistent`: Displays settings that are saved and persist after reboot.
interface `tcp show heuristics`	Displays TCP heuristics, which are algorithms that adjust TCP behavior based on network conditions. `wsh` - Displays Windows Scaling heuristics. `forcews` - Displays forced window scaling settings.
interface `tcp show rscstats`	Displays Receive Segment Coalescing (RSC) statistics for a specified network interface. `interface` - The name or index of the network interface.

Command	Description
interface `tcp show security`	Displays TCP security settings such as SYN attack protection. `store` - Selects the level of security to show: - `active`: Displays the current in-memory configuration (default). - `persistent`: Displays settings that are saved and persist after reboot.
interface `tcp show supplemental`	Displays the current TCP supplemental template settings. If no parameters are specified, the default global TCP template is shown. If a `template=` parameter is provided, details for that specific template are displayed. `template` - Specifies the TCP template to display: - `automatic`: Uses system-selected settings based on environment. - `datacenter`: Optimized for datacenter workloads. - `internet`: Optimized for typical internet scenarios. - `compat`: Provides legacy compatibility settings. - `custom`: Shows user-defined custom settings.
interface `tcp show supplementalports`	Displays which ports are using supplemental TCP settings. `level` - Sets the output detail: - `normal`: Displays basic information for supplemental ports. - `verbose`: Displays detailed information for supplemental ports.
interface `tcp show supplementalsubnets`	Displays which subnets are using supplemental TCP settings. `level` - Sets the output detail: - `normal`: Displays basic information for supplemental subnets. - `verbose`: Displays detailed information for supplemental subnets.

interface teredo

Expand table

Command	Description
interface `teredo set state`	Configures the Teredo operational state. Use `default` for any parameter to restore its system default. The `type=server` option is available only on Windows Server editions. `type` - Specifies the Teredo mode: - `disabled`: Turns off the Teredo service. - `client`: Enables the Teredo service on the client. - `enterpriseclient`: Skips managed network detection. - `natawareclient`: Enables Teredo client detection and NAT type. - `server`: Enables the server (Windows Server only).

Command	Description
	- `default`: Sets to the system default (client). `servername` - Sets the Teredo server name or IPv4 address. `refreshinterval` - Defines the client refresh interval in seconds. `clientport` - Sets the UDP port for the Teredo client. If omitted, the system chooses the port. `servervirtualip` - Specifies the IPv4 address for the server's virtual IP (not used in client mode).
interface `teredo show state`	Displays the current configuration and operational status of the Teredo tunneling protocol.

interface udp

Expand table

Command	Description
interface `udp reset`	Removes all user-configured UDP settings and restores all UDP parameters to their default values.
interface `udp set global`	Configures the global UDP offload settings for all network connections. `uro` - Enables (`enabled`), disables (`disabled`), or restores (`default`) UDP Receive Offload (URO), which allows the network adapter to offload UDP *receive* processing from the CPU. `uso` - Enables (`enabled`), disables (`disabled`), or restores (`default`) UDP Segmentation Offload (USO), which allows the network adapter to offload UDP *send* processing from the CPU.
interface `udp show global`	Displays the global UDP offload settings for all network connections. `store` - Specifies which configuration to display: - `active`: Shows the current in-memory settings (default). - `persistent`: Shows settings that persist after reboot.

interface help

Expand table

Command	Description
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To enable routing on the 6to4 interface named "MyInterface", run the following command:

Windows Command Prompt

```
netsh interface 6to4 set interface name=MyInterface routing=enabled
```

To enable CLAT and allow IPv6 prefix discovery from DNS and RAs, run the following command:

Windows Command Prompt

```
netsh interface clat set global permit=enabled pref64fromdns=enabled
pref64fromra=enabled
```

To add a new HTTPS tunnel interface as a client with the URL `https://tunnel.example.com`, set its state to enabled and use certificate authentication, run the following command:

Windows Command Prompt

```
netsh interface httpstunnel add interface type=client
url="https://tunnel.example.com" state=enabled authmode=certificates
```

To modify an existing HTTPS tunnel interface with the URL `https://tunnel.example.com`, set its state to disabled and authentication mode to none, run the following command:

Windows Command Prompt

```
netsh interface httpstunnel set interface url="https://tunnel.example.com"
state=disabled authmode=none
```

To add the IPv4 address 192.168.1.10 with subnet mask 255.255.255.0 as a unicast address on the interface "Ethernet", set the default gateway to 192.168.1.1 and make the configuration persistent, run the following command:

Windows Command Prompt

```
netsh interface ipv4 add address name="Ethernet" address=192.168.1.10
mask=255.255.255.0 type=unicast gateway=192.168.1.1 store=persistent
```

To delete an excluded TCP port range starting at port 5000 with a total of 20 ports from the active configuration, run the following command:

Windows Command Prompt

```
netsh interface ipv4 delete excludedportrange protocol=tcp startport=5000
numberofports=20 store=active
```

To set the IPv4 address source to DHCP on the interface named "Wi-Fi", run the following command:

Windows Command Prompt

```
netsh interface ipv4 set address name="Wi-Fi" source=dhcp
```

To set a static IPv4 address of 10.0.0.2 with subnet mask 255.255.255.0 and gateway 10.0.0.1 on the interface "LAN1", and save the settings persistently, run the following command:

Windows Command Prompt

```
netsh interface ipv4 set address name="LAN1" source=static address=10.0.0.2
mask=255.255.255.0 gateway=10.0.0.1 store=persistent
```

To enable 6to4 routing and allow site-local addresses, run the following command:

Windows Command Prompt

```
netsh interface ipv6 6to4 set routing routing=enabled sitelocals=enabled
```

To add a prefix policy for the IPv6 prefix 2001:db8::/32 with precedence 40 and label "2", and store it persistently, run the following command:

Windows Command Prompt

```
netsh interface ipv6 add prefixpolicy prefix=2001:db8::/32 precedence=40 label=2
store=persistent
```

To delete the neighbor entry for address fe80::1 on the interface named "Ethernet" from the active configuration, run the following command:

Windows Command Prompt

```
netsh interface ipv6 delete neighbors interface="Ethernet" address=fe80::1
store=active
```

To enable ISATAP state, run the following command:

Windows Command Prompt

```
netsh interface ipv6 isatap set state state=enabled
```

To set the DNS IPv6 server address to 2001:4860:4860::8888 on the network interface "Wi-Fi", with static configuration and enable DNS registration with both primary and alternate suffixes, run the following command:

Windows Command Prompt

```
netsh interface ipv6 set dnsservers name="Wi-Fi" source=static
address=2001:4860:4860::8888 register=both validate=yes
```

To set the IPv6 interface "Ethernet" to disable forwarding, enable advertisement, set MTU to 1400, and make the configuration persistent, run the following command:

Windows Command Prompt

```
netsh interface ipv6 set interface interface="Ethernet" forwarding=disabled
advertise=enabled mtu=1400 store=persistent
```

To configure Teredo as a client and set the server name to `teredo.example.com`, run the following command:

Windows Command Prompt

```
netsh interface ipv6 set teredo type=client servername=teredo.example.com
```

To display IPv6 interface information with verbose details for the interface named "Ethernet", run the following command:

Windows Command Prompt

```
netsh interface ipv6 show interfaces interface="Ethernet" level=verbose
```

To set the ISATAP router to `isatap.example.com` and enable the state with a router discovery interval of 30 seconds, run the following command:

Windows Command Prompt

```
netsh interface isatap set router name=isatap.example.com state=enabled interval=30
```

To add a TCP port proxy that listens on port 8080 at address 127.0.0.1 and forwards connections to 192.168.1.10 on port 80, run the following command:

Windows Command Prompt

```
netsh interface portproxy add v4tov4 listenport=8080 listenaddress=127.0.0.1
connectaddress=192.168.1.10 connectport=80 protocol=tcp
```

To delete an existing v6tov4 TCP port proxy listening on address "2001:db8::1" at port 5000, run the following command:

Windows Command Prompt

```
netsh interface portproxy deleted v6tov4 listenport=5000 listenaddress=2001:db8::1
protocol=tcp
```

To disable the interface named "Wi-Fi", run the following command:

Windows Command Prompt

```
netsh interface set interface name="Wi-Fi" admin=DISABLED
```

To rename the interface "Ethernet" to "WiredLAN", run the following command:

Windows Command Prompt

```
netsh interface set interface name="Ethernet" newname="WiredLAN"
```

To add a supplemental TCP port configuration for the `Internet` template, for local port 443 and *any* remote port, run the following command:

Windows Command Prompt

```
netsh interface tcp add supplementalport template=internet localport=443
remoteport=*
```

To delete a supplemental subnet for IPv6 address 2001:db8::/64, run the following command:

Windows Command Prompt

```
netsh interface tcp delete supplementalsubnet family=ipv6 address=2001:db8::/64
```

To set the global TCP settings to enable RSS, use normal autotuning level, enable ECN capability, and set maximum SYN retransmissions to 5, run the following command:

Windows Command Prompt

```
netsh interface tcp set global rss=enabled autotuninglevel=normal
ecncapability=enabled maxsynretransmissions=5
```

To enable TCP heuristics for Windows Scaling Hint and Force Window Scaling, run the following command:

Windows Command Prompt

```
netsh interface tcp set heuristics wsh=enabled forcews=enabled
```

To configure Teredo state to `natawareclient` with server name `teredo.example.com`, refresh interval of 60, and client port 3544, run the following command:

Windows Command Prompt

```
netsh interface teredo set state type=natawareclient servername=teredo.example.com
refreshinterval=60 clientport=3544
```

To enable URO and USO globally, run the following command:

Windows Command Prompt

```
netsh interface udp set global uro=enabled uso=enabled
```

See also

- Command-Line Syntax Key

Last updated on 10/31/2025

netsh ipsec

08/04/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh ipsec` command is used to configure and manage Internet Protocol Security (IPsec) policies and settings. IPsec is a comprehensive protocol suite that secures network communications by authenticating and encrypting each IP packet within a communication session, ensuring the confidentiality, integrity, and authenticity of the data being transmitted.

Syntax

```
netsh ipsec [dump | dynamic | help | static | ?]
```

```
netsh ipsec dynamic add mmpolicy [name=]<string> [qmpermm=]<integer> [mmlifetime=]
<integer> [softsaexpirationtime=]<integer> [mmsecmethods=](sec#1 sec#2 ... sec#n)
netsh ipsec dynamic add qmpolicy [name=]<string> [soft=](yes|no) [pfsgroup=]
(GRP1|GRP2|GRP3|GRPMM|NOPFS) [qmsecmethods=](neg#1 neg#2 ... neg#n)
netsh ipsec dynamic add rule [srcaddr=](ipv4|ipv6|ipv4-ipv4|ipv6-ipv6|dns|server)
[dstaddr=](ipv4|ipv6|ipv4-ipv4|ipv6-ipv6|dns|server)
                              [mmpolicy=]<string> [qmpolicy=]<string> [protocol=]
(ANY|ICMP|TCP|UDP|RAW|<integer>) [srcport=]<port> [dstport=]<port>
                              [mirrored=](yes|no) [conntype=](lan|dialup|all)
[actioninbound=](permit | block | negotiate) [actionoutbound=]
(permit|block|negotiate)
                              [srcmask=](mask|prefix) [dstmask=](mask|prefix)
[tunneldstaddress=](ip|dns) [kerberos=](yes|no) [psk=]<preshared key>
                              [rootca=]"<certificate> certmap:(yes|no)
excludecaname:(yes|no)"
```

```
netsh ipsec dynamic delete <all> [[mmpolicy|qmpolicy [name=]<string>|all]
netsh ipsec dynamic delete rule [srcaddr=](ipv4|ipv6|ipv4-ipv4|ipv6-
ipv6|dns|server) [dstaddr=](ipv4|ipv6|ipv4-ipv4|ipv6-ipv6|dns|server)
                                 [protocol=](ANY|ICMP|TCP|UDP|RAW|<integer>)
[srcport=]<port> [dstport=]<port> [mirrored=](yes|no)
                                 [conntype=](lan|dialup|all) [srcmask=]
(mask|prefix) [dstmask=](mask|prefix) [tunneldstaddress=](ip|dns)
```

```
netsh ipsec <dump>
```

```
netsh ipsec static add filter [filterlist=]<string> [srcaddr=](ipv4|ipv6|ipv4-
ipv4|ipv6-ipv6|dns|server)
                              [dstaddr=](ipv4|ipv6|ipv4-ipv4|ipv6-ipv6|dns|server)
                              [description=]<string> [protocol=]
(ANY|ICMP|TCP|UDP|RAW|<integer>)
                              [mirrored=](yes|no) [srcmask=](mask|prefix)
[dstmask=](mask|prefix) [srcport=]<port> [dstport=]<port>
```

```
netsh ipsec static add filteraction [name=]<string> [description=]<string>
[qmpfs=](yes|no) [inpass=](yes|no)
                                    [soft=](yes|no) [action=]
(permit|block|negotiate) [qmsecmethods=](neg#1 neg#2 ... neg#n)
```

```
netsh ipsec static add filterlist [name=]<string> [description=]<string>
```

```
netsh ipsec static add policy [name=]<string> [description=]<string> [mmpfs=]
(yes|no) [qmpermm=]<integer> [mmlifetime=]<integer>
                              [activatedefaultrule=](yes|no) [pollinginterval=]
<integer> [assign=](yes|no) [mmsecmethods=](sec#1 sec#2 ... sec#n)
```

```
netsh ipsec static add rule [name=]<string> [policy=]<string> [filterlist=]
<string> [filteraction=]<string> [tunnel=](ip|dns)
                            [conntype=](lan|dialup|all) [activate=](yes|no)
[description=]<string> [kerberos=](yes|no)
                            [psk=]<preshared key> [rootca=]"<certificate> certmap:
(yes|no) excludecaname:(yes|no)"
```

```
netsh ipsec static delete filter [filterlist=]<string> [srcaddr=](ipv4|ipv6|ipv4-
ipv4|ipv6-ipv6|dns|server)
                                 [dstaddr=](ipv4|ipv6|ipv4-ipv4|ipv6-
ipv6|dns|server) [protocol=](ANY|ICMP|TCP|UDP|RAW|<integer>)
```

```
                                    [mirrored=](yes|no) [srcmask=](mask|prefix)
[dstmask=](mask|prefix) [srcport=]<port> [dstport=]<port>
```

```
netsh ipsec static delete [filteraction | filterlist | policy] [name=]<string>
netsh ipsec static delete all
```

```
netsh ipsec static delete rule [name=]<string> [id=]<integer> [all] [policy=]
<string>
```

```
netsh ipsec static [exportpolicy | importpolicy] [file=]<string>
```

```
netsh ipsec static set batch [mode=](enable|disable)
```

```
netsh ipsec static set defaultrule [policy=]<string> [qmpfs=](yes|no) [activate=]
(yes|no) [qmsecmethods=](neg#1 neg#2 ... neg#n)
                                    [kerberos=](yes|no) [psk=]<preshared key>
[rootca=]"<certificate> certmap:(yes|no) excludecaname:(yes|no)"
```

```
netsh ipsec static set filteraction [name=]<string> [guid=]<guid> [newname=]
<string> [description=]<string> [qmpfs=](yes|no)
                                     [inpass=](yes|no) [soft=](yes|no) [action=]
(permit|block|negotiate) [qmsecmethods=](neg#1 neg#2 ... neg#n)
```

```
netsh ipsec static set filterlist [name=]<string> [guid=]<guid> [newname=]<string>
[description=]<string>
```

```
netsh ipsec static set policy [name=]<string> [guid=]<guid> [newname=]<string>
[description=]<string> [mmpfs=](yes|no) [qmpermm=]<integer>
```

```
                              [mmlifetime=]<integer> [activatedefaultrule=]
(yes|no) [pollinginterval=]<integer> [assign=](yes|no)
                              [gponame=]<string> [mmsecmethods=](sec#1 sec#2 ...
sec#n)
```

```
netsh ipsec static set rule [name=]<string> [id=]<integer> [policy=]<string>
[newname=]<string> [description=]<string> [filterlist=]<string>
                             [filteraction=]<string> [tunnel=](ip|dns) [conntype=]
(lan|dialup|all) [activate=](yes|no) [kerberos=](yes|no)
                             [psk=]<preshared key> [rootca=]"<certificate> certmap:
(yes|no) excludecaname:(yes|no)"
```

```
netsh ipsec static set store [location=](local|domain) [domain=]<string>
```

```
netsh ipsec static show all [format=](list|table) [wide=](yes|no)
```

```
netsh ipsec static show filteraction [name=]<string> [rule=]<string> [all]
[level=](verbose|normal)
                                     [format=](list|table) [wide=](yes|no)
```

```
netsh ipsec static show filterlist [name=]<string> [rule=]<string> [all] [level=]
(verbose|normal)
                                   [format=](list|table) [resolvedns=](yes|no)
[wide=](yes|no)
```

```
netsh ipsec static show gpoassignedpolicy [name=]<string>
```

```
netsh ipsec static show policy [name=]<string> [all] [level=](verbose|normal)
[format=](list|table) [wide=](yes|no)
```

```
netsh ipsec static show rule [name=]<string> [id=]<integer> [all] [default]
[policy=]<string> [type=](tunnel|tranport)
                              [level=](verbose|normal) [format=](list|table)
[wide=](yes|no)
```

```
netsh ipsec static show <store>
```

Parameters

Expand table

Command	Description
ipsec `dynamic add mmpolicy`	Adds a Main Mode policy to the Security Policy Database (SPD). `softsaexpirationtime` - Specifies the time, in *seconds*, before a soft expiration of the security association. Other subparameters: • `name` • `qmpermm` • `mmlifetime` • `mmsecmethods` See Remarks for description details.
ipsec `dynamic add qmpolicy`	Adds a Quick Mode policy to the SPD. `pfsgroup` - Specifies the Perfect Forward Secrecy (PFS) group for key exchange. Options are: • `GRP1`, `GRP2`, `GRP3` - Specifies different Diffie-Hellman groups (security levels). • `GRPMM` - Specifies the Main Mode group. • `NOPFS` - Specifies no PFS (default setting). Other subparameters: • `name` • `soft` See Remarks for description details.
ipsec `dynamic add rule`	Adds a new IPsec rule to the dynamic policy configuration. Unlike static policies, dynamic policies are typically temporary and can be used for

Command	Description
	testing, troubleshooting, or temporarily modifying security configurations without changing the permanent static policy settings. `actioninbound` - Specifies the action to be taken on inbound traffic. `actionoutbound` - Specifies the action to be taken on outbound traffic. Options are: • `permit`: Allows the matching traffic to pass through without any IPsec protection. • `block`: Prevents the matching traffic from passing through. • `negotiate`: Requires that the matching traffic performs IPsec negotiation to establish a secure connection. Other subparameters: • `srcaddr` • `dstaddr` • `mmpolicy` • `qmpolicy` • `protocol` • `srcport` • `dstport` • `mirrored` • `conntype` • `srcmask` • `dstmask` • `tunneldstaddress` • `kerberos` • `psk` • `rootca` See Remarks for description details.
ipsec `dynamic delete all`	Removes *all* dynamic IPsec rules from the current policy configuration. Other subparameters: • `mmpolicy` • `qmpolicy` • `name` • `all` See Remarks for description details.
ipsec `dynamic delete rule`	Removes a specific IPsec rule from the dynamic policy configuration. When you execute this command, you're instructing the system to delete a rule that was dynamically added to the IPsec policy engine. This action affects how network traffic is treated in terms of security policies.

Command	Description
	Other subparameters: • `srcaddr` • `dstaddr` • `protocol` • `srcport` • `dstport` • `mirrored` • `conntype` • `srcmask` • `dstmask` • `tunneldstaddress` See Remarks for description details.
ipsec `dump`	Exports the current IPsec configuration settings to a text file.
ipsec `static add filter`	Adds a specific traffic filter to an existing IPsec policy. This filter defines which network traffic (based on criteria like IP addresses, protocols, and ports) is subject to the security rules and actions specified by the IPsec policy. Other subparameters: • `filterlist` • `srcaddr` • `dstaddr` • `description` • `protocol` • `mirrored` • `srcmask` • `dstmask` • `srcport` • `dstport` See Remarks for description details.
ipsec `static add filteraction`	Adds a filter action within an IPsec policy. A filter action specifies the actions that should be taken on traffic that matches the criteria of an IPsec filter. These actions determine how matching traffic is handled, such as whether it should be allowed, blocked, or secured. Other subparameters: • `name` • `description` • `qmpfs` • `inpass` • `soft`

Command	Description
	• `action` • `qmsecmethods` See Remarks for description details.
ipsec `static add filterlist`	A new filter list in an IPsec policy configuration in Windows. A filter list is essentially a collection of one or more filters, each defining specific criteria for network traffic, such as source and destination IP addresses, ports, and protocols. Other subparameters: • `name` • `description` See Remarks for description details.
ipsec `static add policy`	Adds a new IPsec policy. This command doesn't apply any specific rules or filters by itself, rather, it serves as the starting point for defining how you want to manage or secure network traffic on the device. Other subparameters: • `name` • `description` • `mmpfs` • `qmpermm` • `mmlifetime` • `activatedefaultrule` • `pollinginterval` • `assign` • `mmsecmethods` See Remarks for description details.
ipsec `static add rule`	Adds a rule to an existing IPsec policy. Once you've created an IPsec policy, you'll typically need to define specific rules that dictate how traffic matching certain criteria should be handled. Other subparameters: • `name` • `policy` • `filterlist` • `filteraction` • `tunnel` • `conntype` • `activate` • `description` • `kerberos`

Command	Description
	• `psk` • `rootca` See Remarks for description details.
ipsec `static delete filter`	Removes a specified filter from the IPsec configuration. Filters in IPsec are used to define criteria for matching network traffic, such as specifying particular IP addresses, protocols, or port numbers to which certain rules or actions should apply. Other subparameters: • `filterlist` • `srcaddr` • `dstaddr` • `protocol` • `mirrored` • `srcmask` • `dstmask` • `srcport` • `dstport` See Remarks for description details.
ipsec `static delete all` netsh `ipsec static delete [filteraction | filterlist | policy]`	Removes `all` IPsec policies, rules, filters, filter actions, and associated components from the system. This command can also remove *specific* components or from the IPsec configuration. Other subparameters: • `name` See Remarks for description details.
ipsec `static delete rule`	Removes a specific rule from an existing IPsec policy. Other subparameters: • `name` • `id` • `all` • `policy` See Remarks for description details.
ipsec `static [exportpolicy | importpolicy] file=<string>`	This command is used to manage the export and import of IPsec policies, allowing you to save or load IPsec configurations to or from a file. This can be useful for backing up configurations or applying the same configuration to multiple systems.

Command	Description
	• `exportpolicy` - Exports the current IPsec policy configuration to a specified file. • `importpolicy` - Imports an IPsec policy configuration from a specified file. • `file` - Specifies the path and name of the file you wish to use for exporting or importing the policy.
ipsec `static set batch mode=(enable|disable)`	Executes multiple IPsec commands in a group, reducing the chances of policy inconsistencies during changes. When you enable batch mode, you can enter multiple IPsec commands, and they're queued up. Once you complete entering the commands and exit batch mode, the policy changes are applied all at once. When you **enable** batch mode, it allows multiple IPsec policy commands to be processed as a batch, which means they can be applied together to minimize the time during which policy changes might be inconsistent or incomplete. When you **disable** batch mode, each command is executed independently as you enter them.
ipsec `static set defaultrule`	Configures the default IPsec rule that is applied in the absence of more specific rules. Other subparameters: • `policy` • `qmpfs` • `activate` • `qmsecmethods` • `kerberos` • `psk` • `rootca` See Remarks for description details.
ipsec `static set filteraction`	Modifies the action associated with a specific IPsec filter. Filter actions determine how traffic is handled, such as permitting or blocking it. Other subparameters: • `name` • `guid` • `newname` • `description` • `qmpfs` • `inpass` • `soft` • `action` • `qmsecmethods`

Command	Description
	See Remarks for description details.
ipsec `static set filterlist`	Updates an existing filter list, which consists of filters defining which traffic is affected by IPsec policies. Other subparameters: • `name` • `guid` • `newname` • `description` See Remarks for description details.
ipsec `static set policy`	Alters the rules and settings for secure communication, including filter lists and actions set in the IPsec policy. `gponame` - Specifies the name of the Group Policy Object (GPO) where the IPsec policy is set. Other subparameters: • `name` • `guid` • `newname` • `description` • `mmpfs` • `qmpermm` • `mmlifetime` • `activatedefaultrule` • `pollinginterval` • `assign` • `mmsecmethods` See Remarks for description details.
ipsec `static set rule`	Modifies a specific IPsec rule within a policy. Other subparameters: • `name` • `id` • `policy` • `newname` • `description` • `filterlist` • `filteraction` • `tunnel` • `conntype`

Command	Description
	• `activate` • `kerberos` • `psk` • `rootca` See Remarks for description details.
ipsec `static set store`	Specifies which configuration store, either the local policy store or the Active Directory-based store, that the IPsec commands apply to when you configure or view IPsec settings. `location` - Specifies where the IPsec policies are stored. Options are: `local`: Indicates that the policies are stored on the local computer. `domain`: Indicates that the policies are stored in a domain-level policy, which is managed through Active Directory (AD). `domain=` : Designates the domain name when the `location` is set to `domain`. Replace `<string>` with the actual domain name you're using. This is applicable only if the location is specified as `domain`, and it determines which domain's policies are being accessed or altered.
ipsec `static show all`	Displays all *currently* configured IPsec policies, rules, filter lists, and actions. Other subparameters: • `format` • `wide` See Remarks for description details.
ipsec `static show filteraction`	Displays details about the filter actions configured in the system. `resolvedns` - Specifies whether to resolve and display the Domain Name System (DNS) names for the IP addresses in the filter list. Options are: `yes`: The command attempts to resolve the IP addresses to their corresponding DNS names. `no`: The command displays the raw IP addresses without attempting DNS resolution. Other subparameters: • `name` • `rule` • `all` • `level` • `format` • `wide` See Remarks for description details.

Command	Description
ipsec `static show filterlist`	Displays the filter lists defined in the system, which helps identify how traffic is matched against policy rules. Other subparameters: • `name` • `rule` • `all` • `level` • `format` • `wide` See Remarks for description details.
ipsec `static show gpoassignedpolicy`	Displays the IPsec policies assigned through GPO. Other subparameters: • `name` See Remarks for description details.
ipsec `static show policy`	Displays detailed information about *specific* IPsec policies, including rules, filters, and actions. Other subparameters: • `name` • `all` • `level` • `format` • `wide` See Remarks for description details.
ipsec `static show rule`	Displays information for a specified IPsec rule. Other subparameters: • `name` • `id` • `all` • `policy` • `type` • `level` • `format` • `wide` See Remarks for description details.

Command	Description
ipsec `static show store`	Displays the current store context that is being used for IPsec commands. It shows whether the commands are targeting the local policy store or the Active Directory-based store.

Remarks

- **name**: This is the *name* you set for the mmpolicy, qmpolicy, filteraction, filterlist, policy, rule, or gpoassignedpolicy.
- **qmpermm**: Specifies the *number* of Quick Mode sessions allowed per Main Mode session in Internet Key Exchange (IKE) .
- **mmlifetime**: Specifies the duration, in *minutes*, before the Main Mode security association is renegotiated in IKE.
- **mmsecmethods**: Specifies one or more security methods, separated by spaces, using the format "*ConfAlg-HashAlg-GroupNum*".
  - **ConfAlg**: Encryption algorithm, such as `DES` or `3DES`.
  - **HashAlg**: Hash algorithm, such as `MD5` or `SHA1`.
  - **GroupNum**: Diffie-Hellman group number, where `1` = Low, `2` = Medium, `3` = DH2048.

  The use of DES and MD5 isn't recommended. These cryptographic algorithms are provided for backward compatibility only.
- **soft**: Specifies whether to permit unsecured communication with computers that don't support IPsec using `yes` or `no` values.
- **qmsecmethods**: Specifies the IPsec security offer using one of the following formats:
  - *ESP[ConfAlg,AuthAlg]:k/s*
  - *AH[HashAlg]:k/s*
  - *AH[HashAlg]+ESP[ConfAlg,AuthAlg]:k/s*

    Where:
    - **ConfAlg**: Encryption algorithm (`DES`, `3DES`, or `None`)
    - **AuthAlg**: Authentication algorithm (`MD5`, `SHA1`, or `None`)

    - **HashAlg**: Hash algorithm (`MD5` or `SHA1`)
    - **k**: Lifetime in kilobytes
    - **s**: Lifetime in seconds
- **srcaddr** / **dstaddr**: Specifies the source address (starting point of the network traffic) and destination address (endpoint of the network traffic) respectively. Options are:
    - `ipv4`: Designates an IPv4 address.
    - `ipv6`: Designates an IPv6 address.
    - `ipv4-ipv4`: Represents an IPv4 range.
    - `ipv6-ipv6`: Represents an IPv6 range.
    - `dns`: Uses a DNS name to specify the source *or* the destination.
    - `server`: Refers to a server address.
- **mmpolicy**: Refers to the Main Mode policy to be used for the rule. Main Mode policies define the security settings for establishing secure, authenticated communications.
- **qmpolicy**: Indicates the Quick Mode policy to be used, which defines the security settings for maintaining secured traffic after the initial connection is established.
- **protocol**: This parameter allows you to match the rule to target specific IP protocols. Options are:
    - `ANY`: Allows all protocols.
    - `ICMP`: Also represented by the protocol *number* `1`.
    - `TCP`: Also represented by the protocol *number* `6`.
    - `UDP`: Also represented by the protocol *number* `17`.
    - `RAW`: Represents raw socket access, which allows applications to have direct access to lower-level protocols. This can be useful for applications that need to construct their own packet headers or work outside the standard transport protocols.
- **srcport**: Defines the source *port number* for the traffic. This parameter specifies the port from which traffic originates.
- **dstport**: Defines the destination *port number* for the traffic. This parameter specifies the port to which traffic is sent.

- **mirrored**: Indicates whether the rule should automatically be applied in both directions. Values are `yes` or `no`.
- **conntype**: Specifies the type of connection. This determines the network interface to which the rule applies. Options include:
  - `lan`: Specifies that the rule addition or removal applies only to a Local Area Network (LAN) or wireless connection.
  - `dialup`: Specifies that the rule addition or removal applies only to dial-up (phone line) connection.
  - `all`: Specifies that the rule addition or removal applies to all types of network connections.
- **srcmask**: Defines the *source* address mask or prefix to be used. This parameter allows for subnet or prefix-based matching of source addresses.
- **dstmask**: Defines the *destination* address mask or prefix to be used. This parameter allows for subnet or prefix-based matching of destination addresses.
- **tunneldstaddress**: Specifies the tunnel destination address, which can be an `ip` address or `dns` name. This is used when setting up tunnel mode IPsec rules.
- **kerberos**: Indicates whether Kerberos authentication is used. Options are `yes` or `no`.
- **psk**: Refers to the pre-shared key used in authentication. Specifying this parameter is necessary if the authentication method you're configuring relies on it. If you're using a different type of authentication, such as certificates, then the `psk` parameter isn't used.
- **rootca**: Provides options for root certificate authorities. The `certmap` option determines whether certificates are mapped (`yes` or `no`), and `excludecaname` determines if CA names are excluded (`yes` or `no`).
- **description**: Specifies a *description* you give to a filter, filteraction, filterlist, policy, or rule.
- **qmpfs**: Specifies whether Quick Mode Perfect Forward Secrecy (PFS) is required. PFS ensures that session keys aren't compromised even if the server's private key is compromised. Options are `yes` or `no`.
- **inpass**: Specifies whether inbound traffic should be permitted to pass through unprotected. Options are `yes` or `no`.
- **guid**: Specifies the GUID corresponding to a filteraction, filterlist, or policy.

- **newname**: Specifies a new name that you want to assign to an existing filteraction, filterlist, policy, or rule.
- **mmpfs**: Specifies whether Main Mode Perfect Forward Secrecy (PFS) should be used. PFS ensures that if one session's keys are compromised, it doesn't affect the security of subsequent sessions. Options are `yes` or `no`.
- **id**: Specifies the unique ID of the rule you wish to delete, configure, or display.
- **all**: Specifies one of the following actions:
  - Deletes all dynamic or static IPsec configurations or rules.
  - Displays all static, filteraction, filterlist, policies, or rules for the IPsec settings.
- **policy**: Specifies a particular IPsec policy by its name. The `<string>` value should be replaced with the actual name of the IPsec policy you wish to target.
- **pollinginterval**: Specifies the interval, in *seconds*, which the system checks for changes in the IPsec policy.
- **assign**: Specifies whether the newly created or modified IPsec policy should be set as the active policy. Options are `yes` or `no`.
- **action**: This parameter specifies the type of action that should be taken when a packet matches a particular IPsec filter. Options are:
  - `permit`: Allows the matching traffic to pass through without any IPsec protection.
  - `block`: Prevents the matching traffic from passing through.
  - `negotiate`: Requires that the matching traffic performs IPsec negotiation to establish a secure connection.
- **filterlist**: Specifies the *name* (the `<string>` value) of the filter list you created or wish to reference.
- **filteraction**: Specifies the *name* (the `<string>` value) of the action you created within the IPsec policy. Options are:
  - `permit`: Allows the matching traffic to pass through without any IPsec protection.
  - `block`: Prevents the matching traffic from passing through.
  - `negotiate`: Requires that the matching traffic performs IPsec negotiation to establish a secure connection.

- **activate**: Specifies to activate (`yes`) or deactivate (`no`) a specific rule within a filter list, or the default rule for IPsec policies. The default rule typically applies to traffic that doesn't match any specific filters or rules.
- **activatedefaultrule**: Specifies whether the default rule within that policy should be active (`yes`) or inactive (`no`) from the start. This command is also used to modify (set) an existing IPsec policy to change the activation state of the default rule.
- **tunnel**: Specifies an addition or modification of how traffic is tunneled through a rule. Options are:
  - `ip`: This option specifies that the rule uses an IP address for the tunnel. You typically provide the tunnel's start and end IP addresses if you're configuring an IPsec tunnel mode.
  - `dns`: This option specifies that the rule uses DNS names to define the endpoints of the tunnel. This is useful when you want to set up tunnels using domain names rather than static IP addresses.
- **wide**: Shows detailed information in a more verbose format. Options are `yes` or `no`.
- **level**: Specifies the amount of detail to dump in an output. Options are:
  - `verbose`: Provides a detailed, in-depth view, which can be useful for troubleshooting or comprehensive analysis.
  - `normal`: Displays standard information, offering a more general output.
- **format**: Specifies the presentation style of the output. Options are:
  - `list`: Displays the information in a list format, which is often easier to read for detailed inspection of individual entries.
  - `table`: Displays the information in a table format, which can be more compact and provide a quick comparison of multiple entries at a glance.

Examples

To add a Main Mode IPsec policy named "MyMM-Policy" with 100 Quick Mode sessions, a 4800-minute lifetime, 1800-second soft expiration, and 3DES/SHA1 security, run the following command:

```
Windows Command Prompt
```

```
netsh ipsec dynamic add mmpolicy name="MyMM-Policy" qmpermm=100 mmlifetime=4800
softsaexpirationtime=1800 mmsecmethods="3DES-SHA1"
```

To add a Quick Mode IPsec policy named "MyQM-Policy" with unsecured communication fallback enabled, set to group 2 PFS, and two security methods for ESP: 3DES/SHA1 and AES128/SHA256, run the following command:

Windows Command Prompt

```
netsh ipsec dynamic add qmpolicy name="MyQM-Policy" soft=yes pfsgroup=GRP2
qmsecmethods="ESP:3DES-SHA1 ESP:AES128-SHA256"
```

To create a security rule that permits TCP traffic on port 80 between the source IP `192.168.1.10` and destination IP `192.168.1.20` over a LAN connection, using the 'MainModePolicy' and 'QuickModePolicy', run the following command:

Windows Command Prompt

```
netsh ipsec dynamic add rule srcaddr=192.168.1.10 dstaddr=192.168.1.20
mmpolicy="MainModePolicy" qmpolicy="QuickModePolicy" protocol=TCP srcport=80
dstport=80 mirrored=no conntype=lan actioninbound=permit actionoutbound=permit
```

To delete all IPsec dynamic policies, including both main mode and quick mode policies, run the following command:

Windows Command Prompt

```
netsh ipsec dynamic delete all
```

To delete a specific quick mode policy named "QuickModePolicy", run the following command:

Windows Command Prompt

```
netsh ipsec dynamic delete qmpolicy name="QuickModePolicy"
```

To delete a specific IPsec rule that permits TCP traffic on port 80 between the source IP `192.168.1.10` and destination IP `192.168.1.20` over a LAN connection, run the following command:

Windows Command Prompt

```
netsh ipsec dynamic delete rule srcaddr=192.168.1.10 dstaddr=192.168.1.20
```

```
protocol=TCP srcport=80 dstport=80 conntype=lan
```

To export the current IPsec configuration to a script file that can be used to reconfigure the system or another system with the same settings, run the following command:

Windows Command Prompt

```
netsh ipsec dump > C:\Users\Admin\MyFolder\ipsec_settings.txt
```

To add an IPsec filter that permits TCP traffic on port `80` from source IP `192.168.1.10` to destination IP `192.168.1.20` with mirrored configuration, run the following command:

Windows Command Prompt

```
netsh ipsec static add filter filterlist=MyWebTraffic srcaddr=192.168.1.10
dstaddr=192.168.1.20 protocol=TCP srcport=80 dstport=80 description="Allow web
traffic" mirrored=yes
```

To create a filter action that allows traffic to pass without engaging in IPsec negotiation or encryption, run the following command:

Windows Command Prompt

```
netsh ipsec static add filteraction name=AllowTraffic description="Permit traffic
without negotiation" qmpfs=no inpass=yes soft=no action=permit
```

To create a new IPsec filter list for grouping related IPsec filters, run the following command:

Windows Command Prompt

```
netsh ipsec static add filterlist name=WebTrafficFilters description="Filters for
managing web traffic"
```

To create an IPsec policy named "SecurePolicy" that uses specific encryption methods and settings to protect sensitive network traffic, run the following command:

Windows Command Prompt

```
netsh ipsec static add policy name=SecurePolicy description="Policy for securing
sensitive traffic" mmpfs=yes qmpermm=50 mmlifetime=28800 activatedefaultrule=no
pollinginterval=600 assign=yes mmsecmethods="3DES-SHA1 DHGroup2;AES128-SHA256
DHGroup14"
```

To add an IPsec rule named "WebServerRule" that implements security measures for LAN connections to a web server by using specified filter lists and actions with authentication, run the following command:

Windows Command Prompt

```
netsh ipsec static add rule name=WebServerRule policy=SecurePolicy
filterlist=WebTrafficFilters filteraction=AllowTraffic tunnel=no conntype=lan
activate=yes description="Secures web server traffic with specified filters and
Kerberos authentication" kerberos=yes
```

To delete a specific IPsec filter that manages TCP traffic between source IP `192.168.1.10` and destination IP `192.168.1.20` within the filter list "MyFilterList," run the following command:

Windows Command Prompt

```
netsh ipsec static delete filter filterlist=MyFilterList srcaddr=192.168.1.10
dstaddr=192.168.1.20 protocol=TCP mirrored=yes srcport=80 dstport=80
```

To delete an IPsec rule named "WebServerRule" from the policy "SecurePolicy," run the following command:

Windows Command Prompt

```
netsh ipsec static delete rule name=WebServerRule policy=SecurePolicy
```

To export all IPsec policies to a file named "MyPolicy.txt" for backup or transfer to another system, run the following command:

Windows Command Prompt

```
netsh ipsec static exportpolicy file=C:\Policies\MyPolicy.txt
```

To enable batch mode for IPsec commands, which allows multiple IPsec operations to be executed as a single batch for efficiency, run the following command:

Windows Command Prompt

```
netsh ipsec static set batch mode=enable
```

To set a default IPsec rule in the "SecurePolicy" policy with specific security methods and authentication settings, run the following command:

Windows Command Prompt

```
netsh ipsec static set defaultrule policy=SecurePolicy qmpfs=yes activate=yes
qmsecmethods="3DES-SHA1;AES128-SHA256" kerberos=yes psk=yourPreSharedKey
```

To modify an existing IPsec filter action named "OldActionName" by renaming it to "NewActionName" and setting specific security behaviors, run the following command:

Windows Command Prompt

```
netsh ipsec static set filteraction name=OldActionName newname=NewActionName
description="Updated filter action" qmpfs=yes inpass=no soft=no action=negotiate
qmsecmethods="AES256-SHA1;AES128-SHA256"
```

To modify an existing IPsec filter list named "OldListName" by renaming it to "NewListName" and updating its description, run the following command:

Windows Command Prompt

```
netsh ipsec static set filterlist name=OldListName newname=NewListName
description="Updated filter list for enhanced security"
```

To update an existing IPsec policy named "OldPolicyName" by renaming it to "NewPolicyName" with reduced reauthentication frequency, and enhanced security settings, run the following command:

Windows Command Prompt

```
netsh ipsec static set policy name=OldPolicyName newname=NewPolicyName
description="Updated policy for enhanced security measures" mmpfs=yes qmpermm=100
mmlifetime=36000 activatedefaultrule=yes pollinginterval=1200 assign=no
mmsecmethods="AES256-SHA1 DHGroup14;AES128-SHA256 DHGroup2"
```

To update an existing IPsec rule named "OldRuleName" by renaming it to "NewRuleName" within the policy "SecurePolicy," tunneling the LAN traffic by IP and incorporating specific security configurations, run the following command:

Windows Command Prompt

```
netsh ipsec static set rule name=OldRuleName policy=SecurePolicy
newname=NewRuleName description="Updated rule for enhanced server protection"
filterlist=WebTrafficFilters filteraction=NegotiateSecurity tunnel=ip conntype=lan
activate=yes kerberos=yes psk=mySecureKey
```

To store an IPsec policy in a domain location, run the following command:

```
```

```
netsh ipsec static set store location=domain domain=MyDomain.com
```

To display all IPsec policies in a list format with concise details, run the following command:

```
netsh ipsec static show all format=list wide=no
```

To display detailed information about a specific IPsec filter action with verbose output in a table format, run the following command:

```
netsh ipsec static show filteraction name="ExampleFilterAction" level=verbose
format=table wide=yes
```

To display information about all IPsec filter lists with verbose output in a list format where DNS names aren't resolved to IPs, run the following command:

```
netsh ipsec static show filterlist all level=verbose format=list resolvedns=no
wide=yes
```

To display the IPsec policy assigned to a GPO with the name "MyGPO," run the following command:

```
netsh ipsec static show gpoassignedpolicy name=MyGPO
```

To display detailed information about a specific IPsec policy called "MyPolicy," in a verbose, list format and without limiting the output width, run the following command:

```
netsh ipsec static show policy name=MyPolicy all level=verbose format=list
wide=yes
```

To display detailed information about the IPsec rule named "MyRule" with an ID of `123` associated with the policy "MyPolicy," in verbose mode and formatted as a table without width restriction, run the following command:

```
netsh ipsec static show rule name=MyRule id=123 all policy=MyPolicy type=tunnel
level=verbose format=table wide=yes
```

See also

- Command-Line Syntax Key

netsh lan

07/28/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh lan` command allows you to manage and configure local area network (LAN) settings on your computer.

Syntax

```
netsh lan add | delete | dump | export | help | reconnect | set | show | ?
```

```
netsh lan add profile [filename=]<string> [interface=]<string>
netsh lan delete profile [interface=]<string>
netsh lan export profile [folder=]<string> [interface=]<string>
```

```
netsh lan reconnect [interface=]<string>
```

```
netsh lan set autoconfig [enabled=] <yes|no> [interface=] <string>
netsh lan set blockperiod [value=]<0-60>
netsh lan set eapuserdata [filename=]<string> [allusers]<yes|no> [interface=]
<string>
```

```
netsh lan set profileparameter [[interface=]<string>]
                               [authMode=]
<machineOrUser|machineOnly|userOnly|guest>
                               [ssoMode=]<preLogon|postLogon|none> [maxDelay=]<1-
120>
                               [allowDialog=]<yes|no> [userVLAN=]<yes|no>
                               [oneXEnabled=]<yes|no> [oneXEnforced=]<yes|no>
```

```
netsh lan set trace [mode=]<yes|no|persistent>
```

```
netsh lan show interfaces
netsh lan show profiles [interface=]<string>
netsh lan show [settings | tracing]
```

Parameters

Expand table

Command	Description
add profile `filename=<string>` `interface=<string>`	Adds a new profile configuration entry to a table for LAN settings. **filename** - The name of the profile XML file. **interface** - The name of the network interface the profile is added to.
delete profile `interface=<string>`	Removes a profile configuration entry from a table for LAN settings. **interface** - The name of the network interface.
dump	Displays a script of the current LAN configuration. Can also be exported to a text file.
export profile `folder=<string>` `interface=<string>`	Exports the LAN profiles as XML files for backup or restore. **folder** - The name of the folder where the profile XML files is saved to. **interface** - The name of the network interface.
reconnect `interface=<string>`	Re-establishes a connection on a network interface. If the string value contains spaces, it must be wrapped in double quotes (" "). **interface** - The name of the network interface to reconnect to.
set allowexplicitcreds `allow=<yes|no>`	Allows or disallows the use of shared user credentials. The `allow` parameter required.
set autoconfig `enabled=<yes|no>` `interface=<string>`	**enabled** - specifies whether autoconfiguration is set to on or off. **interface** - The name of the network interface. Both parameters are required.
set blockperiod `value=<0-60>`	**value** - Sets the duration, in minutes, where automatic connection attempts to a network are paused. The blocked state is cleared when

Command	Description
	you manually attempt to connect to a LAN, when a session changes, or when the network media reconnects.
set eapuserdata `filename=<string>` `allusers=<yes|no>` `interface=<string>`	**filename** - Specifies the name of the user data XML file. **allusers** - Specifies whether userdata should be saved for all users or the current user. **interface** - The name of the network interface. Both `filename` and `allusers` parameters are required. If the interface name isn't specified, the machine userdata is set.
set profileparameter `interface=<string>` `authMode=<machineOrUser` | `machineOnly` | `userOnly` | `guest>` `ssoMode=<preLogon` | `postLogon` | `none>` `maxDelay=<1-120>` `allowDialog=<yes|no>` `userVLAN=<yes|no>` `oneXEnabled=<yes|no>` `oneXEnforced=<yes|no>`	Configures or modifies LAN profile settings. **interface** - The name of the network interface. **authMode** - The method type used for authenticating credentials. **ssoMode** - The type of single sign-on (SSO) used (if any). **maxDelay** - The timeout value to establish an SSO connection. **allowDialog** - Enables or disables a dialog for prelogon. Prelogon is relevant in scenarios where network authentication is needed before a user logs into the system, such as with 802.1X authentication. **userVLAN** - Specifies whether the network connection can switch to a different Virtual Local Area Network (VLAN) after a user authenticates. **oneXEnabled** - Specifies whether 802.1X authentication is enabled on the network interface. **oneXEnforced** - Specifies if 802.1X authentication is enforced. When enforced, the network interface requires successful 802.1X authentication before allowing network access.
set tracing `mode=<yes|no|persistent>`	Specifies if you want to enable or disable logging trace files for the LAN connection for diagnostic purposes. If set to `persistent`, tracing is enabled even after a network adapter is disabled, re-enabled, or a system is rebooted.
show `interfaces` | `profiles` | `settings` | `tracing`	Displays the current LAN settings and information. `interfaces` - Displays the local LAN interfaces. `profiles` - Displays all local configured LAN profiles. `settings` - Displays the local global LAN settings. `tracing` - Displays if LAN tracing is enabled or disabled.
help or ?	Displays a list of commands and their descriptions in the current context.

Remarks

- **add profile**: The `interface` parameter is optional. The filename parameter is required and specifies the XML file that contains the profile data.
- **delete profile**: The `interface` parameter is optional. If you specify an interface, the profile is deleted only from that interface. If no interface is specified, the profile is deleted from the machine profile.
- **export profile**: The `interface` parameter is optional. If an interface name is specified, the profile for that interface is exported as "InterfaceName.xml". If no interface is specified, all profiles are exported, and the machine profile is saved as "MachineProfile.xml".

  The folder name is required and must point to an existing folder accessible from the local computer. You can use an absolute path or a relative path based on the current working directory. The value "." refers to the current directory, while ".." refers to its parent directory. Universal Naming Convention (UNC) paths aren't supported.
- **reconnect**: The `interface` parameter is optional. If an interface name is specified, only that interface reconnects. If no interface is specified, all interfaces reconnect.
- **set profileparameter**: Modifies the profile on specified interface. If the interface is specified, then only profile on that interface is modified. At least one parameter besides `interface` must be specified.

Examples

To add a new LAN profile from an XML file to a specific interface, run the following command:

Windows Command Prompt

```
netsh lan add profile filename="C:\Profiles\MyLANProfile.xml" interface="Ethernet"
```

To re-establish a connection on a network interface, run the following command:

Windows Command Prompt

```
netsh lan reconnect interface="Ethernet"
```

To enable autoconfiguration on a network interface, run the following command:

Windows Command Prompt

```
netsh lan set autoconfig enabled=yes interface="Ethernet"
```

To configure a LAN profile to use machine-only authentication mode, run the following command:

Windows Command Prompt

```
netsh lan set profileparameter interface="Ethernet" authMode=machineOnly
```

To set the block period for automatic connection attempts to 30 minutes, run the following command:

Windows Command Prompt

```
netsh lan set blockperiod value=30
```

To save Extensible Authentication Protocol (EAP) user data to an XML file for all users on a specific interface, run the following command:

Windows Command Prompt

```
netsh lan set eapuserdata filename="C:\UserData\EAPConfig.xml" allusers=yes
interface="Ethernet"
```

To display all the local configured LAN profiles, run the following command:

Windows Command Prompt

```
netsh lan show profiles
```

netsh mbn

10/10/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh mbn` command is used to configure, monitor, and troubleshoot Mobile Broadband (MBN) interfaces, including cellular modems and SIM-based network connections. It provides a range of options for managing profiles, connections, radio states, and advanced settings for mobile broadband devices on Windows.

The **WWAN AutoConfig** service must be running on your device for certain commands and parameters to function.

Syntax

```
netsh mbn [add | connect | delete | disconnect | dump | help | set | show | test |
?]
```

```
netsh mbn add dmprofile [interface=]<string> [name=]<string>
netsh mbn add profile [interface=]<string> [name=]<string>
```

```
netsh mbn connect [interface=]<string> [connmode=]tmp|name [name=]<string>
```

```
netsh mbn delete dmprofile [interface=]<string> [name=]<string>
netsh mbn delete profile [interface=]<string> [name=]<string>
```

```
netsh mbn disconnect [interface=]<string>
```

```
netsh mbn set acstate [interface=]<string>
[state=]autooff|autoon|manualoff|manualon
netsh mbn set dataenablement [interface=]<string> [profileset=]internet|mms|all
[mode=]yes|no
netsh mbn set dataroamcontrol [interface=]<string> [profileset=]internet|mms|all
[state=]none|partner|all
netsh mbn set enterpriseapnparams [interface=]<string>
[allowusercontrol=]yes|no|nc [allowuserview=]yes|no|nc
                                  [profileaction=]add|delete|modify|nc
netsh mbn set highestconncategory [interface=]<string>
[highestcc=]admim|user|operator|device
netsh mbn set powerstate [interface=]<string> [state=]on|off
netsh mbn set profileparameter [name=]<string> [[interface=]<string>]
[cost]=default|unrestricted|fixed|variable
netsh mbn set slotmapping [interface=]<string> [slotindex=]<integer>
netsh mbn set tracing [mode=]yes|no
```

```
netsh mbn show acstate [interface=]<string>
               basestationinfo [interface=]<string>
               capability [interface=]<string>
               connection [interface=]<string>
               d3cold [interface=]<string>
               dataenablement [interface=]<string>
               dataroamcontrol [interface=]<string>
               dmprofiles [name=]<string> [interface=]<string>
               enterpriseapnparams [interface=]<string>
               highestconncategory [interface=]<string>
               homeprovider [interface=]<string>
               interfaces
               McmInfo [interface=]<string>
               netlteattachinfo [interface=]<string>
               pin [interface=]<string>
               pinlist [interface=]<string>
               preferredproviders [interface=]<string>
               profiles [name=]<string> [interface=]<string> [purpose=]<string>
               profilestate [interface=]<string> [name=]<string>
               provisionedcontexts [interface=]<string> [name=]<string>
               purpose
               radio [interface=]<string>
               readyinfo [interface=]<string> [slotindex=]<integer>
               signal [interface=]<string>
               slotmapping [interface=]<string>
               slotstatus [interface=]<string>
               smsconfig [interface=]<string>
               tracing
               UICCCardAdditionalInfoWithEFHPLMNwAct [interface=]<string>
               visibleproviders [interface=]<string>
```

```
netsh mbn test [feature=<feature area>] [testPath=<path>] [taefPath=<path>]
[param=<test input params>]
```

Parameters

Expand table

Command	Description
mbn `add`	`dmprofile` - Adds a Mobile Broadband Device Management Config Profile for the given interface for all users: `interface` - The interface name. `name` - The XML profile name. `profile` - Adds a Mobile Broadband Network Profile for the given interface for all users: `interface` - The interface name. `name` - The XML profile name. The `interface` and `name` parameters are required.
mbn `connect`	Connects to a Mobile Broadband Network using a specified profile. The device must be registered before running this command. The `interface` parameter is required. `connmode` - Determines how connection parameters are provided: - `tmp`: Use a profile XML file for a temporary connection. - `name`: Use a profile that was previously stored with the `netsh mbn add profile` command. `name` - If `connmode` is set to `tmp`, specify the profile XML file name. If `connmode` is set to `name`, specify the stored profile name.
mbn `delete`	`dmprofile` - Removes a Mobile Broadband Device Management Config Profile for the given interface. `interface` - The interface name. `name` - The XML profile name. `profile` - Removes a Mobile Broadband Network Profile from the given interface: `interface` - The interface name. `name` - The XML profile name. The `interface` and `name` parameters are required.
mbn `disconnect`	Disconnects from the Mobile Broadband Network on the specified interface: `interface` - The interface name. The `interface` parameter is required.

Command	Description
mbn `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
mbn `set`	Configures one of the following settings: `acstate` - Configures the auto-connect behavior for Mobile Broadband data on the interface. The parameters `interface` and `state` are required. `state` - Configures one of the following values: - `autooff`: Turns auto connect token off. - `autoon`: Turns auto-connect token on. - `manualoff`: Turns off the manual connect token. - `manualon`: Turns on the manual connect token. `dataenablement` - Enables or disables Mobile Broadband data for a specified profile set and interface. The parameters `interface`, `profileset`, and `mode` are required. `profileset` - Is the name of the profile set: - `internet`, `mms`, or `all` - `mode`: Enables (`yes`) or disables (`no`) the target profile set. `dataroamcontrol` - Adjusts data roaming settings for Mobile Broadband on the selected profile set and interface. The parameters `interface`, `profileset`, and `mode` are required. `profileset` - Is the name of the profile set: - `internet`, `mms`, or `all` - `state`: Applies changes to the home carrier only (`none`), home and partner carriers only (`partner`), or home, partner, and roaming carriers (`all`). `enterpriseapnparams` - Modifies enterprise Access Point Name (APN) parameters for Mobile Broadband on the specified interface. The parameters `interface`, `allowusercontrol`, `allowuserview`, and `profileaction` are required. - `allowusercontrol`: Allows (`yes`) or disallows (`no`) end user control of enterpriseAPN. Use `nc` to leave at default state. - `allowuserview`: Allows (`yes`) or disallows (`no`) end user view of enterpriseAPN. Use `nc` to leave at default state. - `profileaction`: Adds (`add`), deletes (`delete`) or modifies (`modify`) an enterpriseAPN profile. Use `nc` to leave at default state. `highestconncategory` - Specifies the highest allowed connection category for Mobile Broadband on the given interface. The parameters `interface` and `highestcc` are required. - Sets one of the provisioned profile types: - `admin` - `user` - `operator` - `device` `powerstate` - Switches the Mobile Broadband radio on or off for the specified interface.

Command	Description
	The parameters `interface` and `state` are required. `state`: Enables (`yes`) or disables (`no`) the radio transceiver. `profileparameter` - Updates parameters within a Mobile Broadband Network Profile. The `name` parameter and at least one other parameter, such as `interface` or `cost`, must be specified. - `name`: The profile name to be modified. - `interface`: The interface name. - `cost`: The cost associated with the profile. `slotmapping` - Assigns modem slot mapping for Mobile Broadband on the specified interface. The parameters `interface` and `slotindex` are required. - `interface`: The interface name. - `slotindex`: The interface slot index number to be set. `tracing` - Turns tracing on or off for Mobile Broadband operations. The `mode` parameter is required. - `mode`: Enables (`yes`) or (`disables`) tracing for the Mobile Broadband.
mbn `show`	Displays on of the following information: `acstate interface=` - Displays the auto connect state for Mobile Broadband data on the specified interface. `basestationinfo interface=` - Displays base station information for the specified interface. `capability interface=` - Displays capability details for the specified interface. `connection interface=` - Displays current connection details for the specified interface. `d3cold interface=` - Indicates whether the modem supports D3Cold power state for the specified interface. `dataenablement interface=` - Displays the data enablement status for the specified interface. `dataroamcontrol interface=` - Displays the data roaming control status for the specified interface. `dmprofiles` - Lists Direct Management Config profiles configured on the system. Use the `name` and `interface` parameters to filter for specific profiles on certain interfaces. `enterpriseapnparams interface=` - Displays enterprise APN parameters for the specified interface. `highestconncategory interface=` - Displays the highest connection category for the specified interface.

Command	Description
	`homeprovider interface=` - Displays home provider details for the specified interface. `interfaces` - Lists all Mobile Broadband interfaces on the system. `McmInfo interface=` - Displays MCM-related information for the specified interface. `netlteattachinfo interface=` - Displays LTE attach information for the specified interface. `pin interface=` - Displays PIN information for the specified interface. `pinlist interface=` - Displays the list of PINs for the specified interface. `preferredproviders interface=` - Displays the list of preferred providers for the specified interface. `profiles` - Lists all configured profiles on the system. Use the `name` and `interface` parameters to filter for specific profiles on certain interfaces. If `purpose` is provided, only profiles with the matching purpose GUID are displayed. The following strings are acceptable: - `GUID` - `internet` - `supl` - `mms` - `ims` - `allhost` `profilestate` - Displays the state of a profile for the specified interface. The `interface` and `name` parameters are required. `provisionedcontexts interface=` - Displays provisioned context information for the specified interface. `purpose` - Displays available purpose group GUIDs. `radio interface=` - Displays radio state information for the specified interface. `readyinfo` - Displays readiness information for the specified interface. The `interface` parameter is required while `slotindex` is optional. `signal interface=` - Displays signal strength and quality for the specified interface. `slotmapping interface=` - Displays modem slot mapping for the specified interface. `slotstatus interface=` - Displays modem slot status for the specified interface. `smsconfig interface=` - Displays SMS configuration details for the specified interface.

Command	Description
	`tracing` - Indicates whether tracing is enabled or disabled. `UICCCardAdditionalInfoWithEFHPLMNwAct interface=` - Displays EFInfo in UICC for the specified interface. `visibleproviders interface=` - Displays the list of visible providers for the specified interface.
mbn `test`	Performs mobile broadband testing operations. `feature` (required) - Is the feature area: - `connectivity` - `power` - `radio` - `esim` - `esimhlk` - `sms` - `dssa` - `lte` - `bringup` - `nr` `testpath` - Is the path containing the test binaries. `taefpath` - Is the path containing the TAEF binaries. `param` - Tests other fields using colon separated values: - `connectivity`, `radio`, or `bringup`: The `AccessString` or `UserName` (if applicable), and `Password` (if applicable). - `esim`: The `ActivationCode`. The `testPath` and `taefPath` parameters are optional on amd64 devices where HLK Server is installed.
mbn `help` or `?`	Displays a list of commands and their descriptions in the current context.

Remarks

- Values given to the `interface` parameter must be wrapped in double quotes (" ") for interface names containing a space.
- If you're in the `mbn` context, you can run `show interfaces` to display all interface names.

Examples

To add a Device Management profile for the mobile broadband interface named "Cellular", with the profile name "EnterpriseDM", run the following command:

Windows Command Prompt

```
netsh mbn add dmprofile interface=Cellular name=EnterpriseDM
```

To connect the mobile broadband interface "Cellular" using a temporary profile XML file named "temp_profile.xml", run the following command:

Windows Command Prompt

```
netsh mbn connect interface=Cellular connmode=tmp name=temp_profile.xml
```

To disconnect the mobile broadband interface "Cellular", run the following command:

Windows Command Prompt

```
netsh mbn disconnect interface=Cellular
```

To turn on auto-connect for the "Cellular" interface, run the following command:

Windows Command Prompt

```
netsh mbn set acstate interface=Cellular state=autoon
```

To add an enterprise APN profile for the "Cellular" interface, allowing user control, and view, run the following command:

Windows Command Prompt

```
netsh mbn set enterpriseapnparams interface=Cellular allowusercontrol=yes
allowuserview=yes profileaction=add
```

To display the tracing status for Mobile Broadband commands, run the following command:

Windows Command Prompt

```
netsh mbn show tracing
```

To perform a connectivity test with the AccessString "internet", run the following command:

Windows Command Prompt

```
netsh mbn test feature=connectivity param="AccessString=internet"
```

To perform an eSIM test that selects the DSSA category, run the following command:

Windows Command Prompt

```
netsh mbn test feature=esim param="/select:\"(@Category='DSSA')\""
```

See also

- Command-Line Syntax Key

netsh namespace

07/29/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh namespace` command retrieves and displays namespace policy information from a Windows system.

Syntax

```
netsh namespace [dump] [show] [help] [?]
netsh namespace show <effectivepolicy> <policy> [namespace=] <string>
```

Parameters

Expand table

Parameter	Description
`dump`	Outputs a configuration script containing the current namespace policy settings. This script can be saved and used to restore settings if needed.
`show`	Displays information about namespace policies. • `effectivepolicy` - Displays the effective namespace policies in place. You can specify a namespace to view only the relevant policy entries for that namespace. • `policy` - Displays the entries in the namespace policy table. You can specify a namespace to restrict the output to policies specific to that namespace.
`help` or `?`	Provides help for the list of commands available in the `netsh namespace` context.

Examples

To retrieve the policies for a specific namespace, run the following command:

Windows Command Prompt

```
netsh namespace show policy namespace=[YourNamespace]
```

See also

- Command-Line Syntax Key

netsh netio

08/12/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh netio` command is used to manage network binding filters, allowing administrators to control which clients can bind to specific network providers.

Syntax

```
netsh netio [add | delete | dump | help | show | ?]
```

```
netsh netio add bindingfilter [npi=]<string> [client=]<string> [provider=]<string>
                              [type=](block|singleclient) [store=]
(active|persistent)
```

```
netsh netio delete bindingfilter [npi=]<string> [client=]<string> [provider=]
<string>
                                 [store=](active|persistent)
```

```
netsh netio dump
```

```
netsh netio show bindingfilter [store=](active|persistent)
```

Parameters

Expand table

Command	Description
netio `add bindingfilter`	Adds a new binding filter. • **npi** - Specifies the NPI (Network Programming Interface) GUID or name. • **client** - Specifies the client GUID or name. • **provider** - Specifies the provider GUID or name. • **type** - Determines the filter type. `block` prevents the specified client from binding to the provider. `singleclient` allows only the specified client to bind to the provider (default). • **store** - Specifies filter persistence. `active` indicates the filter is temporary and will disappear on the next boot. `persistent` indicates the filter remains in effect across reboots (default).
netio `delete bindingfilter`	Removes an existing binding filter. • **npi** - Specifies the NPI GUID or name. • **client** - Specifies the client GUID or name. • **provider** - Specifies the provider GUID or name. • **store** - Specifies filter persistence. `active` indicates the filter removal is temporary will reappear after the system is rebooted. `persistent` indicates the filter removal is permanent and remains in effect across system reboots (default).
netio `dump`	Creates a script containing the current netio configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
netio `show bindingfilter`	Displays all binding filters. **store** - Specifies the filter persistence. `active` displays the filters in the stack (default). `persistent` displays all persistent filters.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To block a specified client from binding to a provider persistently, you can run the following command:

Windows Command Prompt

```
netsh netio add bindingfilter npi="NPI_Name" client="Client_Name"
provider="Provider_Name" type=block store=persistent
```

To permanently remove a binding filter, you can run the following command:

Windows Command Prompt

```
netsh netio delete bindingfilter npi="NPI_Name" client="Client_Name"
provider="Provider_Name" store=persistent
```

To display all active binding filters, you can run the following command:

Windows Command Prompt

```
netsh netio show bindingfilter store=active
```

See also

- Command-Line Syntax Key

netsh nlm

10/10/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh nlm` command allows admins to manage the Network Location Manager (NLM) settings and behavior by manually overriding how Windows classifies the network connection quality.

Syntax

```
netsh nlm [ dump | enum | help | query | set | show | ?]
```

```
netsh nlm enum <connection> <network>
```

```
netsh nlm query all
```

```
netsh nlm set connectionquality [<guid> [good|bad|ignore]]
```

```
netsh nlm show <connectivity> <cost>
```

Parameters

Expand table

Command	Description
nlm `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be

Command	Description
	replicated on another system.
nlm `enum`	`connection` - Enumerates the current INetworkConnection objects. `network` - Enumerates the current INetwork objects.
nlm `query all`	Queries all network states from all supported services and writes to a file. A file path and name must be specified to save results.
nlm `set connectionquality`	Manually sets the connection quality of a network interface, overriding the connection quality classification by the Network classifier. `guid` - Is the interface GUID. `good` - Forces Windows to consider the specified network interface's connection quality as stable and reliable. `bad` - Forces Windows to consider the specified network interface's connection quality as unreliable where certain features might disable themselves, or services might limit their operations. `ignore` - Forces Windows to not override its own assessment. Windows determines the connection quality by itself, based on its normal detection algorithms (reverting automatic classification).
nlm `show`	`connectivity` - Displays the current connectivity state from the INetworkListManager. `cost` - Displays the current cost from the INetworkCostManager.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To enumerate the current network connections, run the following command:

Windows Command Prompt

```
netsh nlm enum connection
```

To query all network states and save the results to a file, run the following command:

Windows Command Prompt

```
netsh nlm query all <Filepath\Filename>
```

To manually set the connection quality of a network interface to "good", run the following command:

Windows Command Prompt

```
netsh nlm set connectionquality <My_GUID> good
```

To display the current connectivity state, run the following command:

Windows Command Prompt

```
netsh nlm show connectivity
```

See also

- Command-Line Syntax Key

netsh ras

10/03/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh ras` command manages Remote Access Service (RAS) settings. RAS provides network connectivity for remote users, such as VPN or dial-up connections. By using this command, admins can configure, monitor, and troubleshoot remote access features, including user authentication, connection profiles, and server settings.

Syntax

```
netsh ras [aaaa | add | delete | diagnostics | dump | help | ip | ipv6 | set |
show | ?]
```

```
netsh ras aaaa add acctserver [name=]server [secret=]secret [init-
score=]InitialScore [port=]port
                              [timeout=]timeout [messages=]ENABLED|DISABLED
netsh ras aaaa add authserver [name=]server [secret=]secret [init-
score=]InitialScore [port=]port
                              [timeout=]timeout [signature=]ENABLED|DISABLED
```

```
netsh ras aaaa delete acctserver [name=]server
netsh ras aaaa delete authserver [name=]server
```

```
netsh ras aaaa set accounting [provider=]WINDOWS|RADIUS|NONE
netsh ras aaaa set acctserver [name=]server [secret=]secret [init-
score=]InitialScore [port=]port
                              [timeout=]timeout [messages=]ENABLED|DISABLED
netsh ras aaaa set authentication [name=]server [secret=]secret [init-
score=]InitialScore [port=]port
                                  [timeout=]timeout [messages=]ENABLED|DISABLED
netsh ras aaaa set authserver [name=]server [secret=]secret [init-
score=]InitialScore [port=]port
                              [timeout=]timeout [signature=]ENABLED|DISABLED
netsh ras aaaa set ipsecpolicy [psk=]ENABLED|DISABLED [secret=]<Pre-shared key>
```

```
netsh ras aaaa show <accounting>
netsh ras aaaa show acctserver [name=]server
netsh ras aaaa show <authentication>
netsh ras aaaa show authserver [name=]server
netsh ras aaaa show <ipsecpolicy>
```

```
netsh ras add authtype [type=]PAP|MD5CHAP|MSCHAPv2|EAP|CERT
netsh ras add link [type=]SWC|LCP
netsh ras add multilink [type=]MULTI|BACP
netsh ras add registeredserver [domain=]domain [server=]server
```

```
netsh ras delete authtype [type=]PAP|MD5CHAP|MSCHAPv2|EAP|CERT
netsh ras delete link [type=]SWC|LCP
netsh ras delete multilink [type=]MULTI|BACP
netsh ras delete registeredserver [domain=]domain [server=]server
```

```
netsh ras diagnostics <dump>
netsh ras diagnostics set cmtracing [state=]ENABLED|DISABLED
netsh ras diagnostics set loglevel [events=]error|warn|all|none
netsh ras diagnostics set modemtracing [state=]ENABLED|DISABLED
netsh ras diagnostics set rastracing [component=]component
[state=]ENABLED|DISABLED
netsh ras diagnostics set securityeventlog [state=]ENABLED|DISABLED
netsh ras diagnostics set tracefacilities [state=]ENABLED|DISABLED|CLEAR
netsh ras diagnostics show <all> <cmtracing> <configuration> <installation>
                           <loglevel> <logs> <modemtracing> <rastracing>
                           <securityeventlog> <tracefascilities>
```

```
netsh ras dump
```

```
netsh ras ip add range [from=] [to=]
netsh ras ip delete pool
netsh ras ip delete range [from=] [to=]
netsh ras ip dump
netsh ras ip set access [mode=]ALL|SERVERONLY
```

```
netsh ras ip set addrassign [method=]AUTO|POOL
netsh ras ip set addrreq [mode=]ALLOW|DENY
netsh ras ip set broadcastnameresolution [mode=]ENABLED|DISABLED
netsh ras ip set negotiation [mode=]ALLOW|DENY
netsh ras ip set preferredadapter [name=]<interface_name>
```

```
netsh ras ipv6 dump
netsh ras ipv6 set access [mode=]ALL|SERVERONLY
netsh ras ipv6 set negotiation [mode=]ALLOW|DENY
netsh ras ipv6 set prefix [prefix=]<IPv6 prefix>
netsh ras ipv6 set routeradvertise [mode=]ENABLED|DISABLED
netsh ras ipv6 show <config>
```

```
netsh ras set authmode [mode=]STANDARD|NODCC|BYPASS
netsh ras set client [name=]<str_clientname> [state=]disconnect|resetstats
netsh ras set conf [confstate=]ENABLED|DISABLED
netsh ras set ikev2connection [idletimeout=]<idle_timeout> [nwoutagetime=]
<nw_outage_time>
netsh ras set ikev2saexpiry [saexpirytime=]<sa_expiry_time> [sadatasizelimit=]
<sa_datasize_limit>
netsh ras set portstatus [name=]<port_name>
netsh ras set sstp-ssl-cert [name=]certname [hash=]<hash>
netsh ras set type [ipv4rtrtype=]lanonly|lananddd|none
[ipv6rtrtype=]lanonly|lananddd|none [rastype=]IPv4|IPv6|BOTH|NONE
netsh ras set user [name=]<username> [dialin=]PERMIT|DENY|POLICY
[cbpolicy=]NONE|CALLER|ADMIN [cbnumber=]<callback number>
netsh ras set wanports [device=]<devicename> [rasinonly=]enabled|disabled
[ddinout=]enabled|disabled [ddoutonly=]enabled|disabled [phone=]<phonenumber>
[maxports=]<numports>
```

```
netsh ras show activeservers
netsh ras show authmode
netsh ras show authtype
netsh ras show client [name=]<str_clientname>
netsh ras show conf
netsh ras show ikev2connection
netsh ras show ikev2saexpiry
netsh ras show link
netsh ras show multilink
netsh ras show portstatus [name=]<port_name> [state=]<enum>
netsh ras show registeredserver
netsh ras show sstp-ssl-cert
netsh ras show status
netsh ras show type
```

```
netsh ras show user [name=]<username> [mode=]PERMIT|REPORT
netsh ras show wanports [device=]<devicename>
```

Parameters

Expand table

Command	Description
ras `aaaa add`	Inserts one or more entries into the AAAA table. `acctserver` - Adds a RADIUS accounting server. `authserver` - Adds a RADIUS authentication server. See Remarks for subparameters.
ras `aaaa delete`	Removes one or more entries from the AAAA table. `acctserver` - Removes a RADIUS accounting server. `authserver` - Removes a RADIUS authentication server. See Remarks for subparameters.
ras `aaaa set accounting`	Specifies which service handles accounting for remote access connections. Options are: `windows` - Uses Windows security for accounting operations. `radius` - Sends accounting data to a configured RADIUS server. `none` - Disables accounting. No records are kept for connection activities.
ras `aaaa set acctserver`	Configures an IP address or name of a RADIUS server to use for accounting. See Remarks for subparameters.
ras `aaaa set authentication`	Specifies which service is used to authenticate remote access connections. Options are: `windows` - Uses Windows security for authentication. `radius` - Uses a configured RADIUS server for authentication.
ras `aaaa set authserver`	Configures an IP address or name of a RADIUS server to pass authentication requests. See Remarks for subparameters.
ras `aaaa set ipsecpolicy`	Sets the IPsec policy for L2TP connections. `psk` determines if a custom IPsec policy using a preshared key is enabled for L2TP connections. Options are:

Command	Description
	`enabled` - Allows the use of a custom IPsec policy with a specified preshared key. `disabled` - Uses the default Certificate Authority-based IPsec policy. `secret` - Specifies the preshared key to be used when `psk` is set to `enabled`.
ras `aaaa show accounting`	Displays the current accounting provider.
ras `aaaa show acctserver`	Displays detailed information about an accounting server. If no server name is provided using `name=`, all configured accounting servers are displayed.
ras `aaaa show authentication`	Displays the authentication provider.
ras `aaaa show authserver`	Displays detailed information about an authentication server. If no server name is provided using `name=`, all configured accounting servers are displayed.
ras `aaaa show ipsecpolicy`	Displays the IPsec policy for L2TP connection. The **Routing and Remote Access** service must be running to obtain results.
ras `add authtype`	Specifies the authentication types that the Remote Access server attempts to negotiate with connecting clients. The server negotiates authentication methods in order from most secure to least secure. Once both the client and server agree on an authentication type, Point-to-Point Protocol (PPP) negotiation proceeds according to the relevant RFC standards. See Remarks for authentication subparameters.
ras `add link`	Adds to the list of link properties PPP negotiates. See Remarks for link subparameters.
ras `add multilink`	Adds to the list of multilink types PPP negotiates. See Remarks for multilink subparameters.
ras `add registeredserver`	Registers the specified Windows computer as a Remote Access server within the Active Directory (AD) of the designated domain. See Remarks for `server` and `domain` subparameters.
ras `delete authtype`	Removes an authentication type from the list of types that the Remote Access server negotiates. See Remarks for authentication subparameters.
ras `delete link`	Removes from the list of link properties PPP negotiates.

Command	Description
	See Remarks for link subparameters.
ras `delete multilink`	Removes from the list of multilink PPP types that are negotiated. See Remarks for multilink subparameters.
ras `delete registeredserver`	Unregisters the specified Windows computer from the list of Remote Access servers registered in the AD of the designated domain. See Remarks for `server` and `domain` subparameters.
ras `diagnostics set`	Performs one of the following actions: `cmtracing` - Enables or disables Connection Manager tracing, which logs events related to connection establishment and management. `loglevel` - Sets the global log level for Routing and Remote Access Service (RRAS). Options include `error`, `warn`, `all`, or `none`. `modemtracing` - Enables or disables detailed tracing of modem settings and messages during network connections. `rastracing` - Enables or disables extended tracing for a specified RRAS component to assist with troubleshooting. To trace all components, use `state=*`. The following list describes the available components: - rasman - rasppp - rasauth - rasip - rasl2tp - raspptp - rassstp - rasgre - raseap - rasqec - rasmobility - rascustom `securityeventlog` - Enables or disables logging of security events. Security Event logs can be viewed using Event Viewer. `tracefacilities` - Enables, disables, or clears extended tracing for all RRAS components. See Remarks for `state` subparameters.
ras `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to

Command	Description
ras `diagnostics dump` ras `ip dump` ras `ipv6 dump`	be replicated on another system.
ras `ip add range` ras `ip delete range`	Adds or removes a range of the static IP address pool that the Remote Access server using `from` and `to` as the range in the `X.X.X.X` format.
ras `ip delete pool`	Deletes all ranges from the static IP address pool.
ras `ip set access`	Sets what type of access level to grant. See Remarks for `mode` subparameters.
ras `ip set addrassign`	Sets the method by which the Remote Access server assigns IP addresses to client devices. Options are: `auto` - Automatically assigns IP addresses to clients using DHCP. If a DHCP server is unavailable, the server assigns a random private IP address. `pool` - Assigns IP addresses from the Remote Access server's configured address pool.
ras `ip set addrreq`	Specifies whether dial-in clients are permitted to request their own IP addresses. See Remarks for `mode` subparameters.
ras `ip set broadcastnameresolution`	Specifies whether to enable or disable broadcast name resolution using NetBIOS over TCP/IP. See Remarks for `mode` subparameters.
ras `ip set negotiation`	Specifies whether the Remote Access server allows IPv4 to be configured for any client connections it accepts. See Remarks for `mode` subparameters.
ras `ip set preferredadapter`	Specifies the preferred adapter for Routing and Remote Access Service. See Remarks for `name` subparameters.
ras `ipv6 set access`	Sets what type of access level to grant for IPv6. See Remarks for `mode` subparameters.
ras `ipv6 set negotiation`	Specifies whether the Remote Access server allows IPv6 to be configured for any client connections it accepts.

Command	Description
	See Remarks for `mode` subparameters.
ras `ipv6 set prefix`	Sets the static IPv6 prefix that the Remote Access server uses to advertise to client devices in the format `prefix=X:X:X:X::`.
ras `ipv6 set routeradvertise`	Specifies the router advertisement mode on the server. See Remarks for `mode` subparameters.
ras `ipv6 show config`	Displays the current Remote Access IPv6 configuration.
ras `set authmode`	Sets which authentication mode dial-in clients should use. See Remarks for `mode` subparameters.
ras `set client`	Disconnects or resets the statistics of the username attached to the client device. See Remarks for `name` and `state` subparameters.
ras `set conf`	Configures whether the RAS is `enabled` (on) or `disabled` (off) using `confstate=`. If set to `disabled`, all configurations are also removed.
ras `set ikev2connection`	Sets the idle timeout and network outage time values for IKEv2 client connections. `idletimeout` - Sets the idle timeout in minutes for IKEv2 client connections. If the client is inactive for this period, the connection is disconnected. Valid range is 5 minutes to 2,879 minutes (less than 48 hours). `nwoutagetime` - Sets the network outage time in minutes for IKEv2 client connections. If the network is unavailable for this duration, the connection is terminated. Minimum value is 2 minutes.
ras `set ikev2saexpiry`	Sets the IKEv2 Security Association (SA) expiration controls. `saexpirytime` - Sets the SA expiry time in minutes for IKEv2 client connections. The minimum value is 5 minutes and the maximum is 2879 minutes (less than 48 hours). `sadatasizelimit` - Sets the SA data size limit in megabytes for IKEv2 connections. The minimum value is 1 MB.
ras `set portstatus`	Resets the statistics information of RAS ports. If a name isn't specified, all active ports statistics are resets. See Remarks for `name` subparameters.
ras `set sstp-ssl-cert`	Configures the current Secure Socket Tunneling Protocol (SSTP) certificate.

Command	Description
	`name` - Specifies the certificate name to use for SSTP. `hash` - Specifies the SHA-1 hash of the certificate to use for SSTP.
ras `set type`	Configures the router and RAS roles on the device. Options are: `ipv4rtrtype` - Sets the IPv4 router mode: - `lanonly`: The computer acts as a LAN-only router and doesn't support Demand Dial or VPN connections. - `lananddd`: The computer acts as both a LAN and Demand Dial router, supporting VPN connections. - `none`: The computer isn't enabled as an IPv4 router. `ipv6rtrtype` - Sets the IPv6 router mode: - `lanonly`: The computer acts as a LAN-only router and doesn't support Demand Dial or VPN connections. - `lananddd`: The computer acts as both a LAN and Demand Dial router, supporting VPN connections. - `none`: The computer isn't enabled as an IPv6 router. `rastype` - Sets the RAS configuration: - `ipv4`: Configured as a Remote Access Server for IPv4. - `ipv6`: Configured as a Remote Access Server for IPv6. - `both`: Configured as a Remote Access Server for both IPv4 and IPv6. - `none`: Not configured as a RAS.
ras `set user`	Configures user Remote Access properties. Options are: `username` - The user's logon name. `dialin` - Determines if the user is allowed to dial in: - `PERMIT`: Allows the user to dial in. - `DENY`: Prevents the user from dialing in. - `POLICY`: Uses remote access policies to determine dial-in permission. The `POLICY` option isn't available for users in mixed-mode domains. For these users, `POLICY` is treated as `DENY`. `cbpolicy` - Specifies the user's callback policy. Callback allows the server to call the user back to save call costs: - `NONE`: Callback isn't allowed. - `CALLER`: The user can specify a callback number at call time. - `ADMIN`: The server always calls the user at the specified callback number. `cbnumber` - The callback number used when `cbpolicy` is set to `ADMIN`.
ras `set wanports`	Configures options for RAS WAN ports. `device` - Specifies the device name of the WAN port.

Command	Description
	`rasinonly` - Enables (`enabled`) or disables (`disabled`) inbound remote access connections. `ddinout` - Enables (`enabled`) or disables (`disabled`) demand dial inbound and outbound routing connections. `ddoutonly` - Enables (`enabled`) or disables (`disabled`) demand dial outbound routing connections. `phone` - Sets the phone number for the device. Accepts a string value such as a phone number, VPI/VCI, or IP address. `maxports` - Specifies the maximum number of ports available for the device.
ras `show`	Displays information based on one of the following parameters: `activeservers` - Displays advertisements from Remote Access servers. `authmode` - Shows the current authentication mode. `authtype` - Lists enabled authentication types. `client` - Displays connected Remote Access clients and their status. To display a specific client, add `name=`. `conf` - Shows the server's configuration state. `ikev2connection` - Displays idle timeout and network outage settings for IKEv2 connections. `ikev2saexpiry` - Shows IKEv2 SA expiration controls. `link` - Lists PPP link properties to be negotiated. `multilink` - Lists PPP multilink types to be negotiated. `portstatus` - Displays the current status of RAS ports. To display a specific port and its state, use `name=` and `state=` subparameters. `registeredserver` - Indicates if a computer is registered as a Remote Access server in AD. `sstp-ssl-cert` - Shows the current SSTP certificate configuration. `status` - Displays Routing and Remote Access Server status. `type` - Shows router and RAS roles configured on the computer. `user` - Displays Remote Access properties for users. To display specific user properties, use `name=` and `mode=` subparameters. `wanports` - Shows configuration options for RAS WAN ports.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Remarks

`name` - Specify the DNS name or IP address of the RADIUS server or client device, username, network interface name, or network port name.

`secret` - Defines the shared secret used for authentication between the client and the RADIUS server.

`init-score` - Sets the initial priority score for the server. Higher scores indicate higher priority.

`port` - Indicates the port number used to send authentication or accounting requests to the RADIUS server.

`timeout` - Determines the time (in seconds) to wait for a response before marking the RADIUS server as unavailable.

`messages` - Controls whether accounting on/off messages are sent to the RADIUS server. Options are:

- `Enabled`
- `Disabled`

`server` - Specifies the DNS name or computer name to register as a Remote Access server. If omitted, the command registers the computer from which it's run.

`domain` - Specifies the AD domain where the server should be registered. If omitted, the computer's primary domain is used by default.

`state` - For diagnostics, sets tracing and logging options. Options are: `Enable`, `Disable`, and `Clear`. For client devices, `disconnect` removes the specified client and `resetstats` resets the statistical counters for the specified client, such as packets received, error counts, session counters, and others without disconnecting the client. For the RAS ports on your device, it displays which state the specified port is set to, such as:

- nonoperational (nonoperational ports)
- disconnected (the port is disconnected or not active)
- callingback (if the port is a callback port)
- listening (the port is listening for incoming calls)
- authenticating (the connected user is being authenticated)
- connected (authenticated and connected)
- initializing (initializing port for use)

`mode` - Is a multi use parameter where it can perform the following actions:

- Grant access levels for a device with options being `all` where clients are able to reach the remote access server and any networks to which it's connected, and `serveronly` where clients only reach the remote access server.

- You can `allow` or `deny` IP negotiation for client connections and if dial-in clients can request their own IP addresses.
- You can `enable` or `disable` broadcast name resolution where RAS clients resolve NetBIOS names on the local subnet when other resolution methods are unavailable.
- You can `enable` or `disable` router advertisement. A default route is a route with a zero-length prefix (`::/0`). To advertise itself as a default router, a computer acting as an IPv6 router must be configured with a default route (`::/0`) that's published.
- You can determine whether dial-in clients should be authenticated. Options are:
  - `STANDARD` - All devices must authenticate.
  - `NODCC` - Direct-connect devices aren't authenticated.
  - `BYPASS` - Authentication isn't required for any devices.
- You can filter user Remote Access information using `report` to show all users and `permit` to show only users whose dial-in permission is set to PERMIT.

`type` - This is either the type of authentication the Remote Access server uses to negotiate, link property PPP, or multilink PPP. Options are:

- **Authentication**
  - `PAP` - Password Authentication Protocol. Transmits credentials in clear text and is the least secure authentication method.
  - `MD5CHAP` - Challenge Handshake Authentication Protocol using MD5 hashing. Credentials are protected by a challenge-response mechanism with MD5 encryption.
  - `MSCHAPv2` - Microsoft Challenge Handshake Authentication Protocol version 2. Provides enhanced security over earlier versions, including mutual authentication and stronger encryption.
  - `EAP` - Extensible Authentication Protocol. Supports multiple authentication methods, such as smart cards, certificates, and token devices, allowing flexible and extensible authentication.
  - `CERT` - Certificate-based authentication for IKEv2 connections. Uses digital certificates to verify identity and establish secure connections.
- **Link PPP**
  - `SWC` - Enables software-based compression using Microsoft Point-to-Point Compression (MPPC), which reduces the size of transmitted data to optimize bandwidth usage.

- `LCP` - Activates Link Control Protocol (LCP) extensions from the PPP protocol suite, allowing negotiation of link parameters and management of the PPP connection.

- **Multilink PPP**
  - `MULTI` - Enables multilink PPP sessions, allowing multiple physical links to be combined into a single logical connection for increased bandwidth and redundancy.
  - `BACP` - Activates Bandwidth Allocation Control Protocol, which dynamically manages bandwidth allocation across multiple links in a multilink PPP session to optimize performance.

Examples

To add a new RADIUS accounting server called "AcctSrv1" with a shared secret of `P@ssw0rd`, an initial score of 50, listening on UDP port 1813, a 30-second timeout, and accounting messages enabled, run the following command:

Windows Command Prompt

```
netsh ras aaaa add acctserver name=AcctSrv1 secret=P@ssw0rd init-score=50
port=1813 timeout=30 messages=ENABLED
```

To remove an existing RADIUS authentication server named "AuthSrv1", run the following command:

Windows Command Prompt

```
netsh ras aaaa delete authserver name=AuthSrv1
```

To enable MS-CHAPv2 authentication for incoming RAS connections, run the following command:

Windows Command Prompt

```
netsh ras add authtype type=MSCHAPv2
```

To remove the LCP link-control protocol from RAS, run the following command:

Windows Command Prompt

```
netsh ras delete link type=LCP
```

To log only warning-level RAS diagnostic events, run the following command:

Windows Command Prompt

```
netsh ras diagnostics set loglevel events=warn
```

To specify "Ethernet 2" as the preferred network adapter for RAS IP connections, run the following command:

Windows Command Prompt

```
netsh ras ip set preferredadapter name="Ethernet 2"
```

To set the IPv6 prefix 2001:0db8:85a3::/64 that RAS hands out to clients, run the following command:

Windows Command Prompt

```
netsh ras ipv6 set prefix prefix=2001:0db8:85a3::/64
```

To set the IKEv2 SA to expire after 1 hour (3,600 seconds) or once 1 GB (1,073,741,824 bytes) of data is transferred, run the following command:

Windows Command Prompt

```
netsh ras set ikev2saexpiry saexpirytime=3600 sadatasizelimit=1073741824
```

To display the complete RAS configuration, run the following command:

Windows Command Prompt

```
netsh ras show conf
```

See also

- Command-Line Syntax Key

netsh rpc

08/20/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh rpc` command enables administrators to configure, manage, and troubleshoot Remote Procedure Call (RPC) service settings, including selective binding, firewall filtering, and access control for RPC traffic.

Syntax

```
netsh rpc [add | delete | dump | filter | help | reset | show | ?]
```

```
netsh rpc add <subnet>
```

```
netsh rpc delete <subnet>
```

```
netsh rpc dump
```

```
netsh rpc filter add condition [field=]<string> [matchtype=]<string> [data=]
<string>
netsh rpc filter add filter
netsh rpc filter add rule [layer=]<string> [actiontype=]<string> [[filterkey=]
<string>]
                          [persistence=]volatile [audit=]enable
[auditparameters=]enable
```

```
netsh rpc reset
```

```
netsh rpc show
```

Parameters

Expand table

Command	Description
rpc `add`	Adds one or more subnets (IP address) to an "add" list managed by the RPC service.
rpc `delete`	Adds one or more subnets (IP address) to a "delete" list, implying these subnets are to be removed from existing configurations.
rpc `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
rpc `filter add condition`	Creates precise firewall rules to control RPC traffic based on specific conditions. `field=<string>` - Specifies the aspect of the RPC communication where the condition applies. For `um` (User Mode), acceptable values are: • `if_uuid` - Interface UUID • `if_version` - Interface version • `if_flag` - Interface flag • `dcom_app_id` - DCOM Application ID • `image_name` - The executable image involved in the RPC call • `protocol` - Protocol used (like TCP) • `auth_type` - Type of authentication used • `auth_level` - Level of authentication enforced • `sec_encrypt_alg` - Security encryption algorithm • `sec_key_size` - Size of the security key • `remote_user_token` - Token for remote user authentication • `local_addr_v4` - Local IPv4 address • `local_addr_v6` - Local IPv6 address • `remote_addr_v4` - Remote IPv4 address • `remote_addr_v6` - Remote IPv6 address • `local_port` - Local communication port • `pipe` - Named pipe used in the connection • `opnum` - Operation number For `epmap` (Endpoint Mapper), acceptable values are: • Similar values as user mode but without `opnum`. For `ep_add` (Endpoint Address), acceptable values are: • `process_with_if_uuid` - Process associated with the interface UUID • `protocol`: Protocol used

Command	Description
	• `ep_value` - Endpoint value • `ep_flags` - Flags related to the endpoint. For `proxy_conn` (Proxy Connection), acceptable values are: `server_name`: Name of the server involved • `server_port` - Port on the server • `proxy_auth_type` - Type of proxy authentication • `client_token` - Token for client authentication • `client_cert_key_name` - Name of the client's certificate key • `client_cert_oid` - OID of the client's certificate For `proxy_if` (Proxy Interface), acceptable values are: • Similar to `proxy_conn`, with some overlapping fields. `matchtype=<string>` - Determines the type of comparison or match conducted on the field's value. Acceptable values are: • `equal` - Exact match • `greater` - Greater than a specified value • `less` - Less than a specified value • `greater_or_equal` - Greater than or equal to a specified value • `less_or_equal` - Less than or equal to a specified value • `range` - Within a specified range • `all_set` - Bitwise matching commonly used for flags or settings that use bit fields • `any_set` - Bitwise matching commonly used for flags or settings that use bit fields • `none_set` - Bitwise matching commonly used for flags or settings that use bit fields `data=<string>` - The actual value used in the condition that corresponds to the specified field.
rpc `filter add filter`	Adds an RPC firewall filter. Before you can add a filter, you must first define at least one rule and one or more conditions.
rpc `filter add rule`	Adds an RPC firewall filter rule. `layer=<string>` - Specifies the layer where the filter is applied. Acceptable values are: • `um` - User Mode • `epmap` - Endpoint Mapper • `ep_add` - Endpoint Address • `proxy_conn` - Proxy Connection • `proxy_if` - Proxy Interface `actiontype=<string>` - Defines the action to take when the filter matches. Acceptable values are: • `block` - Blocks the RPC traffic • `permit` - Allows the RPC traffic. • `continue` - Continues to the next filter for evaluation.

Command	Description
	`filterkey=<string>` - Specifies a unique UUID that identifies the RPC firewall filter. Useful for managing or referencing specific filters. `persistence` - Determines whether the filter remains after a reboot. By default, filters are persistent. `audit` - Enables auditing for this filter. Only filters with the "permit" action type can have auditing enabled. Auditing isn't supported at the `ep_add` layer. `auditparameters` - Enables auditing of RPC call parameter buffers for this filter. Requires `audit` to be enabled. Auditing of parameters is only supported at the `um` layer.
rpc `reset`	Resets the selective binding configuration so that the system listens on all network interfaces (no subnets are restricted).
rpc `show`	Displays the current selective binding configuration, listing each subnet and its binding state on the system.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To add the specified subnet `192.168.1.0` to the RPC "add list", run the following command:

Windows Command Prompt

```
netsh rpc add 192.168.1.0
```

To add a filter condition that applies to RPC traffic where the local IPv4 address is exactly `192.168.1.10`, run the following command:

Windows Command Prompt

```
netsh rpc filter add condition field=local_addr_v4 matchtype=equal
data=192.168.1.10
```

To apply a rule to the Endpoint Mapper layer permitting traffic for a specific UUID, along with persistence after a reboot and logged for auditing, run the following command:

Windows Command Prompt

```
netsh rpc filter add rule layer=epmap actiontype=permit filterkey={Your-UUID}
persistence=yes audit=yes
```

See also

- Command-Line Syntax Key

netsh set

07/29/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh set` command allows administrators to specify the target machine for subsequent network configuration tasks, enabling operations on either local or remote systems. If a machine name isn't specified, the local machine is used. A username and password can't be used to connect to the local machine.

Syntax

```
netsh set machine [name=]<string> [user=][[DomainName\]UserName] [pwd=][Password | *]
```

Parameters

Expand table

Parameter	Description
`<name>`	Specifies the device name or network address of the machine on which you want to perform actions.
`<user>`	Provides the username credentials required to authenticate and connect to the remote machine. If your network uses a domain, specify the username in the format `DomainName\UserName`.
`<pwd>`	Supplies the password associated with the specified user account. If you enter an asterisk (`*`) instead of a password, you're prompted to input the password interactively for added security.

Examples

To specify a target machine using its IP address in a domain environment where the password is prompted, run the following command:

Windows Command Prompt

```
netsh set machine name=192.168.1.10 user=DOMAIN\Admin pwd=*
```

See also

- Command-Line Syntax Key

netsh show

07/29/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh show` command displays information about defined command aliases and available dynamic link library (DLL) modules.

Syntax

```
netsh show <alias> <helper>
```

Parameters

Expand table

Parameter	Description
`alias`	Lists all the defined aliases in the `netsh` context. Aliases are shortcuts that can be created for longer commands or sequences.
`helper`	Lists all the top-level DLL helpers available.

Examples

To see the aliases used within your environment, run the following command:

Windows Command Prompt

```
netsh show alias
```

To see a list of the DLL helper files used within your environment, run the following command:

Windows Command Prompt

```
netsh show helper
```

See also

- Command-Line Syntax Key

netsh trace

10/06/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh trace` command allows you to capture detailed network traffic and trace events, providing valuable data for diagnosing and troubleshooting network problems on Windows systems. It supports a range of options for filtering, exporting, and analyzing trace data.

Syntax

```
netsh trace [convert | correlate | diagnose | dump | export | help | merge |
postreset | show | start | ?]
```

```
netsh trace convert [input=]tracefilename.etl [output=]filename
[dump=]CSV|XML|EVTX|TXT|No [report=]yes|no
                    [overwrite=]yes|no [metadata=]yes|no [tmfpath=]pathname
[manpath=]pathname
```

```
netsh trace correlate [input=]tracefilename.etl [output=]newtracefilename.etl
[filter=]Activity_ID
                      [overwrite=]yes|no [retaincorrelationevents=]yes|no
[retainpii=]yes|no
                      [retainglobalevents=]yes|no
```

```
netsh trace diagnose [scenario=]<scenarioname> [namedAttribute=]<attributeValue>
                     [saveSessionTrace=]<yes|no> [report=]<yes|no> [capture=]
<yes|no>
```

```
netsh trace dump
```

```
netsh trace export [scenario=]<scenarioname> [file=]<filename>
```

```
netsh trace merge
```

```
netsh trace postreset
```

```
netsh trace show <capturefilterhelp>
                 <globalkeywordsandlevels>
                 <helperclass> [name=]<helperclassname>
                 <interfaces>
                 <provider> [name=]<providerIdOrName>
                 <providerfilerhelp>
                 <providers>
                 <scenario> [name=]<scenarioname>
                 <scenarios>
                 <status>
```

```
netsh trace start [sessionname]=<sessionname> [scenario=]<scenario1,scenario2>
[globalKeywords=]keywords
                  [globalLevel=]level [[capture=]yes|no]
[capturetype=]physical|vmswitch|both
                  [report=]yes|no|disabled [persistent=]yes|no
[traceFile=]path\filename
                  [maxSize=]filemaxsize [fileMode=]single|circular|append
[overwrite=]yes|no
                  [correlation=]yes|no|disabled [capturefilters]
[provider=]providerIdOrName
                  [keywords=]keywordMaskOrSet [level=]level [bufferSize=]
<bufferSize>
                  [provider=]provider2IdOrName [providerFilter=]yes|no
[keywords=]keyword2MaskOrSet
                  [perfMerge=]yes|no [level=]level2 ...
```

Parameters

> ⓘ **Note**

Certain commands and parameters might only function fully in previous versions of Windows and Windows Server, which are no longer in support.

Expand table

Command	Description
trace `convert`	Converts a trace file to an HTML report. `input`: Specifies the input Event Trace Log (ETL) file. `output`: Sets the output file name. If not specified, defaults to the input file name. `dump`: Chooses the output format: `TXT` (default), `CSV`, `XML`, `EVTX`, or `No` (no dump). `report`: Generates an HTML report if set to `yes`. Default is `no`. `overwrite`: Overwrites existing files if set to `yes`. Default is `no`. `metadata`: Includes metadata traces in the output if set to `yes`. Default is `no`. `tmfpath`: Provides the path to the Trace Message Format (TMF) files for decoding the Windows Software Trace Preprocessor (WPP) traces. `manpath`: Provides the path to manifest files for decoding Event Traces for Windows (ETW) traces.
trace `correlate`	Normalizes or filters a trace file, producing a new output file. You can filter by Activity ID (a GUID in the format `{xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx}`) to include only related events. The command can also retain correlation events, personal data (such as packet capture events), and global events. `input`: Specifies the input ETL trace file. `output`: Sets the output file name and path. `filter`: Outputs only events related to the specified Activity GUID. Default is none. `overwrite`: Overwrites existing files if set to `yes`. Default is `no`. `retaincorrelationevents`: Retains correlation events if set to `yes`. Default is `no`. `retainpii`: Retains events containing personal data if set to `yes` (default). `retainglobalevents`: Retains global events if set to `yes` (default).
trace `diagnose`	Starts a diagnostic session. `scenario`: Specifies a diagnostic scenario name (example, `InternetClient`). `namedAttribute`: Supply the additional details or options for the scenario, which is dependent on the scenario chosen. `saveSessionTrace`: If set to `yes`, saves the trace data from the session for later analysis. If `no`, the trace isn't saved. This option defaults to `yes` if `report=yes`. `report`: If set to `yes`, generates a diagnostic report after running the scenario. If `no` (default), no report is created. `capture`: If set to `yes`, captures network traffic during the diagnosis. If `no` (default), traffic isn't captured.
trace `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another

Command	Description
	system.
trace `export`	Exports a scenario to a Windows Performance Recorder (WPR) profile. `scenario`: Specifies the name of the tracing scenario to export. `file`: Sets the output file name and path for the exported WPR profile.
trace `merge`	Merges multiple trace files into one and enriches the output with symbol and PerfTrack metadata for improved analysis.
trace `postreset`	Restores the global tracing settings to default after a network stack reset.
trace `show`	Displays information for the following parameters: `capturefilterhelp`: Lists all supported capture filters and provides usage examples. `globalkeywordsandlevels`: Displays available global keywords and tracing levels for configuration. `helperclass`: Provides information about helper classes used in tracing. `interfaces`: Lists all network interfaces available for tracing. `provider`: Displays detailed information about a specific trace provider. `providerfilterhelp`: Lists supported provider filters and explains their usage. `providers`: Shows all trace providers available on the system. `scenario`: Displays details about a specific tracing scenario. `scenarios`: Lists all available tracing scenarios. `status`: Shows the current tracing configuration and session status.
trace `start`	Starts a trace session. `sessionname`: Specifies the name of the trace session. `scenario`: Specifies a predefined scenario (`LAN`, `InternetClient`, etc.) `globalKeywords`: Filters events for all providers using ETW keywords. `globalLevel`: Sets the verbosity for global events: - `1` (Critical): Only critical events. - `2` (Error): Critical and error events. - `3` (Warning): Critical, error, and warning events. - `4` (Informational): Adds informational events. - `5` (Verbose): Captures all events, including detailed debug info. `capture`: Whether to capture network packets (`yes` or `no`). `capturetype`: Specifies the packet capture type (`physical`, `vmswitch`, or `both`). `report`: Generate an HTML report after tracing (`yes`, `no`, or `disabled`). `persistent`: Keeps the session running after a system reboot (`yes` or `no`). `traceFile`: Full path and filename for the trace output. `maxSize`: Maximum size of the trace file in megabytes. `fileMode`: Specifies the method of file writing mode: - `single`: All data is written to one file up to the specified file size in megabytes. - `circular`: Overwrites oldest data when file reaches max size (acts as a ring buffer).

Command	Description
	- `append`: Adds new trace data to the end of an existing file. `overwrite`: Overwrites existing trace file (`yes` or `no`). `correlation`: Enables correlation for events (`yes`, `no`, or `disabled`). `capturefilters`: Filters for captured packets (IP, TCP port, etc.). `provider`: Specifies the ETW provider by GUID or name. `keywords`: *Provider*-specific keyword filtering. `level`: *Provider*-specific verbosity. `bufferSize`: Size of the trace buffer in megabytes. `providerFilter`: Enables *provider*-specific filtering (`yes` or `no`). `perfMerge`: Merges performance data (`yes` or `no`). Some parameters are scenario-specific and might be ignored if not relevant. Defaults are used if omitted. Using `capture=yes` can generate large files quickly. `fileMode=circular` overwrites old data when max size is reached.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To convert an ETL log file to XML, run the following command:

Windows Command Prompt

```
netsh trace convert input="C:\Logs\mytrace.etl" output="C:\Logs\mytrace.xml"
```

To analyze and correlate the data between multiple ETL files into a single correlated file, run the following command:

Windows Command Prompt

```
netsh trace correlate input="C:\Logs\trace1.etl,C:\Logs\trace2.etl"
output="C:\Logs\correlated_trace.etl"
```

To export an ETL to a text file (`.txt`), run the following command:

Windows Command Prompt

```
netsh trace export input="C:\Logs\mytrace.etl" output="C:\Logs\mytrace.txt"
```

To merge multiple ETL files into a single trace file, run the following command:

Windows Command Prompt

```
netsh trace merge input="C:\Logs\trace1.etl,C:\Logs\trace2.etl"
output="C:\Logs\merged_trace.etl"
```

To list all network interfaces available for tracing, run the following command:

Windows Command Prompt

```
netsh trace show interfaces
```

To start a new trace session and save the log to a specific file path, run the following command:**

Windows Command Prompt

```
netsh trace start capture=yes tracefile="C:\Logs\networktrace.etl"
```

See also

- Command-Line Syntax Key

netsh wcn

07/29/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh wcn` command allows administrators to configure and manage Wireless Configuration Network (WCN) settings on local or remote Windows devices. It provides commands to query, enroll, and export wireless configuration details.

Syntax

```
netsh wcn [dump] [enroll] [query] [help] [?]
netsh wcn enroll [enrolleeUUID=] <string> [password=] <string> [profile=] <string>
[interface=] <string> [type=] (auto | ethernet | 802dot11)
netsh wcn query [SSID=] <string> [interface=] <string>
```

Parameters

Expand table

Parameter	Description
`dump`	Exports the current WCN settings and configuration as a script for documentation or replication on another system.
`enroll`	Initiates the process to connect to a wireless network using WCN protocols. • `enrolleeUUID` - Specifies the Wi-Fi Protected Setup (WPS) Universally Unique Identifier (UUID) of the device you wish to enroll. This is a unique identifier provided by the device. • `password` - The WPS password or PIN associated with the device. This is used during the enrollment process to authenticate the device to the network. • `profile` - The name of the network profile to be used. This profile contains the configuration settings for the network connection. • `interface` - Specifies the friendly name of the network interface on which the enrollment process is performed. If omitted, the default network interface is used. This parameter is optional. • `type` - Determines the transport method to be used for the enrollment. This can specify different transport technologies, such as: **auto** (automatically selects the available network interfaces), **ethernet** (a wired Ethernet connection), **802dot11** (a wireless 802.11 connection). This parameter is optional.

Parameter	Description
`query`	Retrieves and displays detailed information about a WCN device, including network settings, security details, and device-specific information. • `ssid` - Specifies the SSID (Service Set Identifier) of the 802.11 wireless network you wish to query. • `interface` - Specifies the friendly name of the network interface you want to use for the query. This parameter is optional.
`help` or `?`	Provides a list of commands available in the `netsh wcn` context.

Examples

To initiate the enrollment process for a device with a specific UUID using a wireless 802.11 connection, run the following command:

Windows Command Prompt

```
netsh wcn enroll enrolleeUUID=your-device-uuid password=your-password
profile=your-ssid type=802dot11
```

To enroll a device using a wired Ethernet connection on a specified network interface, run the following command:

Windows Command Prompt

```
netsh wcn enroll enrolleeUUID=your-device-uuid password=your-password
profile=your-ssid interface=your-interface-name type=ethernet
```

To query a wireless network with a specific SSID using the default network interface, run the following command:

Windows Command Prompt

```
netsh wcn query SSID=your-network-ssid
```

To query detailed information about a wireless network with a specific SSID using a specified network interface, run the following command:

Windows Command Prompt

```
netsh wcn query SSID=your-network-ssid interface=your-interface-name
```

Remarks

The `enroll` parameter starts WPS enrollment with the device identified by the enrollee UUID. The specified network profile is applied using the chosen interface, or the default if none is given.

The device password is usually a 4-digit or 8-digit PIN. Refer to your device's documentation for its UUID and PIN. If no transport type is set, or "auto" is used, the system selects the best available option.

See also

- Command-Line Syntax Key

netsh wfp

08/11/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh wfp` command is used to manage and troubleshoot the Windows Filtering Platform (WFP). It enables you to capture, display, and analyze filtering rules, network events, and diagnostics related to network traffic filtering. This tool is useful for administrators who need to investigate network policies, monitor traffic, and resolve issues with WFP configuration.

Syntax

```
netsh wfp [capture | dump | help | set | show | ?]
```

```
netsh wfp capture start cab=<on|off> traceonly=<on|off> keywords=
<none|bcast|mcast|bcast+mcast> file=<path>
netsh wfp capture status
netsh wfp capture stop
```

```
netsh wfp dump
```

```
netsh wfp set options netevents=<on|off> keywords=<none|bcast|mcast|bcast+mcast>
txnwatchdog=<milliseconds>
```

```
netsh wfp show appid file=<path>
               boottimepolicy file=<path|->
               filters file=<path|-> protocol=<ipprotocol> localaddr=<ipaddress>
remoteaddr=<ipaddress> localport=<port>
                       appid=<path> userid=<user> dir=<in|out> verbose=<on|off>
               ikeevents file=<path|-> remoteaddr=<ipaddress>
               netevents file=<path|-> protocol=<ipprotocol> localaddr=<ipaddress>
remoteaddr=<ipaddress> localport=<port>
                         remoteport=<port> appid=<path> userid=<user> timewindow=
```

```
 <seconds>
                options options=<netevents|keywords|txnwatchdog>
                security type=
 <callout|engine|filter|ikesadb|ipsecsadb|layer|netevents|provider|providercontext|
 sublayer>
                          guid=<guid>
                state file=<path|->
                sysports file=<path|->
```

Parameters

Expand table

Command	Description
netsh `wfp capture start`	Starts a capture session for network events processed by WFP. **cab** - Specifies whether to compress the output files into a single `.cab` archive. • `on` (default): Combines all files into one `.cab` file. • `off`: Leaves files uncompressed, useful for manual troubleshooting. **traceonly** - Determines if only event tracing data is collected. • `on`: Captures only event tracing, resulting in smaller output files. • `off` (default): Captures all relevant data. **keywords** - Sets the network event keywords for the capture session. Options include: • `NONE`: Excludes both to minimize file size for extended captures. • `BCAST`: Includes broadcast events. • `MCAST`: Includes multicast events. • `BCAST+MCAST` Includes both broadcast & multicast events (default). **file** - Specifies the output file name. The default is `wfpdiag.cab`. If the `cab` parameter is set to `on`, the file name shouldn't include the extension since the `.cab` is automatically added to the output file.
netsh `wfp capture status`	Displays the status of the current capture session, indicating whether a session is active and providing details about the ongoing capture process.
netsh `wfp capture stop`	Stops an active capture session that was started with the `capture start` command. Use this command to end data collection and finalize the output files for analysis.
netsh `wfp dump`	Generates a script reflecting the current WFP configuration. You can save this script to a file and later use it to restore any modified settings.
netsh `wfp set options`	Configures global options for the Windows Filtering Platform. Only one option parameter can be modified per command execution. **netevents** - Enables or disables the collection of network events.

Command	Description
	• `on`: Network events are captured. • `off`: Network event collection is disabled. **keywords** - Specifies which types of network events to capture. Options include: • `NONE`: No broadcast or multicast events are captured (default). • `BCAST`: Captures broadcast events only. • `MCAST`: Captures multicast events only. • `BCAST+MCAST`: Captures both broadcast and multicast events. **txnwatchdog** - Sets the timeout (in milliseconds) for transaction watchdog events. Provide an integer value to specify the timeout. If set to zero or not specified, an event is triggered for every transaction.
netsh `wfp show appid`	Displays the device path of the executable using `file=<path>` format.
netsh `wfp show boottimepolicy`	Displays the boot-time policy and filters using `file=<path>` as the output file name. By default, the output is saved as `btpol.xml`. If you set this parameter to a dash (`file=-`), the output is displayed directly in the console instead of being saved to a file.
netsh `wfp show filters`	Displays filters matching the specified traffic parameters. • **file** - Specifies the output file name in the `file=<path>` format. The default is `filters.xml`. If set to a dash (`file=-`), the output is displayed in the console instead of being saved to a file. • **protocol** - The IP protocol number, for example, use `6` for TCP or `17` for UDP. To learn more, see IANA Protocol Numbers. • **localaddr** / **remoteaddr** - Specifies the local or remote IP addresses respectively. Both must be either IPv4 or IPv6, and if both are specified, they must belong to the same address family. • **localport** / **remoteport** - Specifies the local or remote port numbers respectively. • **appid** - The application path on the local host that is sending or receiving traffic. Accepts either an NT path (`\device\harddiskvolume1\windows\system32\ftp.exe`) or a DOS path (`C:\Windows\System32\ftp.exe`). The specified path must exist. • **userid** - The user associated with the traffic on the local host. Can be a security identifier (SID) such as `S-1-5-18` or a user name such as `NT AUTHORITY\SYSTEM`. • **dir** - The direction of the connection. By default, both inbound and outbound filters are shown. Use `IN` to display only inbound filters or `OUT` for outbound filters. • **verbose** - Controls the level of detail in the output. Set to `ON` to display all matching filters, or `OFF` (default) to suppress filters unlikely to affect connectivity.
netsh `wfp show ikeevents`	Displays recent Internet Key Exchange (IKE) epoch events matching the specified parameters. • **file** - Specifies the output file name for the results in the format `file=<path>`. The default is `netevents.xml`. If you set this parameter to a dash (`file=-`), the output is displayed directly in the console instead of being saved to a file.

Command	Description
	• **remoteaddr** - The remote IP address to filter events by. Accepts either an IPv4 or IPv6 address.
netsh `wfp show netevents`	Displays recent network events matching the specified traffic parameters. • **file** - Specifies the output file name in the `file=<path>` format. The default is `filters.xml`. If set to a dash (`file=-`), the output is displayed in the console instead of being saved to a file. • **protocol** - The IP protocol number, for example, use `6` for TCP or `17` for UDP. • **localaddr** / **remoteaddr** - Specifies the local or remote IP addresses respectively. Both must be either IPv4 or IPv6, and if both are specified, they must belong to the same address family. • **localport** / **remoteport** - Specifies the local or remote port numbers respectively. • **appid** - The application path on the local host that is sending or receiving traffic. Accepts either an NT path (`\device\harddiskvolume1\windows\system32\ftp.exe`) or a DOS path (`C:\Windows\System32\ftp.exe`). The specified path must exist. • **userid** - The user associated with the traffic on the local host. Can be a security identifier (SID) such as `S-1-5-18` or a user name such as `NT AUTHORITY\SYSTEM`. • **timewindow** - Limits the displayed network events to a specified time frame in seconds.
netsh `wfp show options`	Specifies which global WFP option to display in `optionsfor=` format. Options are: • `NETEVENTS`: Shows whether network event diagnostics are enabled or disabled. • `KEYWORDS`: Displays the types of network events (such as broadcast or multicast) that are being captured for diagnostics. • `TXNWATCHDOG`: Displays the current timeout value (in milliseconds) for the transaction watchdog event.
netsh `wfp show security`	Displays the specified security descriptor. **type** - Specifies the WFP object type whose security descriptor you want to display in `type=` format. Options include: • `CALLOUT` • `ENGINE` • `FILTER` • `IKESADB` • `IPSECSADB` • `LAYER` • `NETEVENTS` • `PROVIDER` • `PROVIDERCONTEXT` • `SUBLAYER` **guid** - For object types that support individual security descriptors, this parameter specifies the object's GUID in `guid=<guid>` format. If omitted, the command uses

Command	Description
	`IID_NULL` to retrieve the security descriptor for the entire type container. The following object types support per-object security descriptors: • `callout` • `filter` • `layer` • `provider` • `providercontext` • `sublayer`
netsh `wfp show state`	Displays the current state of WFP and IPsec. **file** - Specifies the output file name in `file=` format. By default, the output is saved as `wfpstate.xml`. If you set this parameter to a dash (`file=-`), the output is displayed directly in the console instead of being saved to a file.
netsh `wfp show sysports`	Displays system ports used by the TCP/IP Stack and RPC subsystem. **file** - Specifies the output file name in `file=` format. By default, the output is saved as `sysports.xml`. If you set this parameter to a dash (`file=-`), the output is displayed directly in the console instead of being saved to a file.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To start a capture session for network events and compress the output files into a single `.cab` archive, run the following command:

Windows Command Prompt

```
netsh wfp capture start cab=on
```

To start a capture session that collects only event tracing data and includes both broadcast and multicast events, run the following command:

Windows Command Prompt

```
netsh wfp capture start traceonly=on keywords=BCAST+MCAST
```

To enable the collection of network events and capture broadcast events only, run the following command:

Windows Command Prompt

```
netsh wfp set options netevents=on keywords=BCAST
```

To set the transaction watchdog timeout to 5,000 milliseconds, run the following command:

Windows Command Prompt

```
netsh wfp set options txnwatchdog=5000
```

To display the application path for a specified file, run the following command:

```
netsh wfp show appid file="C:\folder\app.exe"
```

To display whether network event diagnostics are enabled or disabled, run the following command:

Windows Command Prompt

```
netsh wfp show options optionsfor=NETEVENTS
```

To display the security descriptor for a specific WFP object type, for example, a filter with a known GUID, run the following command:

Windows Command Prompt

```
netsh wfp show security type=FILTER guid=YourGUID
```

See also

- Command-Line Syntax Key

netsh winhttp

08/20/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh winhttp` command configures and manages the WinHTTP (Windows HTTP Services) settings. WinHTTP is a Microsoft API used by applications to make HTTP requests which allows administrators to manage network communications for applications that use WinHTTP.

Syntax

```
netsh winhttp [dump | help | import | reset | set| show | ?]
```

```
netsh winhttp dump
```

```
netsh winhttp import proxy [source=]ie
```

```
netsh winhttp reset <autoproxy> <proxy> <tracing>
```

```
netsh winhttp set advproxy <settings-scope> <settings-file> <settings>
netsh winhttp set proxy [proxy-server=]<server name> [bypass-list=]<hosts list>
netsh winhttp set tracing
```

```
netsh winhttp show <advproxy> <proxy> <tracing>
```

Parameters

Expand table

Command	Description
http `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
winhttp `import proxy source=ie`	Import proxy settings into the WinHTTP configuration from the Internet Explorer (IE) settings. There isn't any direct built-in support for importing settings from other browsers using this command.
winhttp `reset` `<autoproxy>` | `<proxy>` | `<tracing>`	Provides options to reset various components of the WinHTTP configuration, such as: • `autoproxy` - Resets the WinHTTP proxy autodiscovery service. This is used to clear any settings or cache related to the automatic detection of proxy servers. • `proxy` - Resets the WinHTTP proxy setting to `DIRECT`, effectively removing any configured proxy server and allowing direct internet access without a proxy. • `tracing` - For this operation, the system advises using the `netsh trace stop` command, which is intended to stop and reset any tracing activities.
winhttp `set advproxy`	Configures advanced proxy settings for WinHTTP, providing detailed configurations like bypass lists or multiple proxy settings. `setting-scope` - Specifies the scope of the settings. Acceptable values are: • `user` - Applies the settings to the current user. • `machine` - Applies the settings system-wide. `settings-file` - Specifies a path to a JSON file containing the proxy configuration settings. This file should contain a JSON object with the relevant properties. `settings` - Specifies the proxy settings directly in JSON format. The JSON object must include the following properties: • `Proxy` - A string value representing the proxy server. • `ProxyBypass` - A string value listing hosts that should bypass the proxy. • `AutoconfigUrl` - A string value for the URL of an automatic configuration script. • `AutoDetect` - A boolean value indicating whether to automatically detect proxy settings (true or false).
winhttp `set proxy`	Configures the basic proxy settings for WinHTTP by specifying a proxy server and an optional list of hosts that should bypass the proxy. • `proxy-server=` - Specifies the proxy server to be used for both HTTP and HTTPS protocols. You can designate a single server, and optionally, a port number (for example, `proxy.example.com:8080`). • `bypass-list=` - Provides a list of hostnames or IP addresses that should not use the proxy server and instead be accessed directly. The list should be separated by

Command	Description
	semicolons. Use an empty string (`""`) to bypass the proxy for all short-name hosts (non-fully qualified domain names).
winhttp `set tracing`	Related to starting network tracing activities for troubleshooting or monitoring purposes. It suggests using the following to initiate tracing, `netsh trace start scenario=InternetClient`, which starts tracing network traffic activities specifically for internet client scenarios.
winhttp `show <advproxy>` | `<proxy>` | `<tracing>`	Provides options to display various WinHTTP configurations, such as: • `advproxy` - Displays the current advanced proxy settings for WinHTTP. This includes any detailed proxy configurations that have been set. • `proxy` - Displays the current basic proxy settings for WinHTTP. This reveals the proxy server in use and any bypass lists that have been configured. • `tracing` - Displays tracing information. It's advised to use the `netsh trace show` command.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To import proxy settings from Internet Explorer into WinHTTP, run the following command:

Windows Command Prompt

```
netsh winhttp import proxy source=ie
```

To reset the WinHTTP proxy settings to their default state, run the following command:

Windows Command Prompt

```
netsh winhttp reset proxy
```

To set advanced proxy settings for WinHTTP at the machine level using a JSON configuration string, run the following command:

Windows Command Prompt

```
netsh winhttp set advproxy setting-scope=machine settings="
{\"Proxy\":\"proxy.example.com:8080\",\"ProxyBypass\":\"*.example.com\",\"Autoconf
igUrl\":\"http://config.example.com/proxy.pac\",\"AutoDetect\":true}"
```

To display the current WinHTTP proxy settings, run the following command:

Windows Command Prompt

```
netsh winhttp show proxy
```

See also

- Command-Line Syntax Key

netsh winsock

08/12/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh winsock` command manages the Windows Sockets (Winsock) API, which is responsible for network communication between applications and network services.

Syntax

```
netsh winsock [audit | dump | help | remove | reset | set | show | ?]
```

```
netsh winsock audit trail
netsh winsock dump
netsh winsock remove provider <catalog id>
netsh winsock reset
netsh winsock set autotuning <on|off>
netsh winsock show <autotuning> <catalog>
```

Parameters

Expand table

Command	Description
winsock `audit trail`	Displays the audit trail of Layered Service Providers (LSP) that are installed and uninstalled.
winsock `dump`	Creates a script containing the current Winsock configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
winsock `remove provider <catalog id>`	Removes a specific Winsock LSP from the system using its catalog ID. This ID can be found when you run `winsock show catalog`. The Namespace Providers are presented as a GUID rather than a catalog ID.
winsock `reset`	Resets the Winsock catalog to a clean state, removing any custom LSPs to resolve network problems caused by corrupted Winsock settings. It doesn't affect Winsock Name Space Provider entries.

Command	Description
winsock `set autotuning`	This command toggles Winsock send autotuning, which enables dynamic send buffering to improve overall network throughput. Options are `on` or `off`.
winsock `show`	`autotuning` - Displays whether Winsock send autotuning is currently enabled or disabled. `catalog` - Displays the contents of the Winsock catalog, listing all registered Winsock LSPs and namespace providers.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To view the audit trail of LSPs, run the following command:

Windows Command Prompt

```
netsh winsock audit trail
```

To remove a specific Winsock LSP using its catalog ID, run the following command:

Windows Command Prompt

```
netsh winsock remove provider <catalog_id>
```

To enable Winsock send autotuning for better throughput, run the following command:

Windows Command Prompt

```
netsh winsock set autotuning on
```

To display the contents of the Winsock catalog, run the following command:

Windows Command Prompt

```
netsh winsock show catalog
```

See also

- Command-Line Syntax Key

netsh wlan

10/09/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `netsh wlan` command enables administrators to configure, manage, and troubleshoot wireless networking on Windows. It offers subcommands for connecting to Wi-Fi networks, managing wireless profiles, filtering SSIDs, exporting settings, and diagnosing connectivity problems.

Syntax

```
netsh wlan [add | connect | delete | disconnect | dump | export | help | IHV |
refresh | reportissues | set | show | start | stop | ?]
```

```
netsh wlan add filter [permission=]allow|block|denyall [ssid=]<string>
[networktype=]infrastructure|adhoc
netsh wlan add profile [filename=]<string> [interface=]<string> [user=]all|current
```

```
netsh wlan connect [name=]<string> [ssid=]<string> [interface=]<string>
```

```
netsh wlan delete filter [permission=]allow|block|denyall [ssid=]<string>
[networktype=]infrastructure|adhoc
netsh wlan delete profile [name=]<string> [[interface=]<string>]
```

```
netsh wlan disconnect [interface=]<string>
```

```
netsh wlan dump
```

```
netsh wlan export <hostednetworkprofile>
netsh wlan export profile [name=]<string> [folder=]<string> [[interface=]<string>]
[key=<string>]
```

```
netsh wlan IHV <startlogging> <stoplogging>
```

```
netsh wlan refresh hostednetwork [data=]key
```

```
netsh wlan reportissues
```

```
netsh wlan set allowexplicitcreds [allow=]yes|no
netsh wlan set autoconfig [enabled=]yes|no [interface=]<string>
netsh wlan set blockednetworks [display=]show|hide
netsh wlan set blockperiod [value=]0-60
netsh wlan set createalluserprofile [enabled=]yes|no
netsh wlan set hostednetwork [mode=]allow|disallow [ssid=]<ssid> [key=]
<passphrase> [keyUsage=]persistent|temporary
netsh wlan set profileorder [name=]<string> [interface=]<string> [priority=]
<integer>
netsh wlan set profileparameter [name=]<string> [interface=]<string> [SSIDname=]
<string> [ConnectionType=]ESS|IBSS
                                [autoSwitch=yes|no] [ConnectionMode=auto|manual]
[nonBroadcast=yes|no]
                                [Randomization=]yes|no|daily
[authentication=]open|shared|WPA|WPA2|WPAPSK|WPA2PSK
                                [encryption=]none|WEP|TKIP|AES
[keyType=]networkKey|passphrase [keyIndex=]1-4
                                [keyMaterial=<string>] [PMKCacheMode=]yes|no
[PMKCacheSize=]1-255 [PMKCacheTTL=300-86400]
                                [preAuthMode=]yes|no [preAuthThrottle=]1-16
[FIPS=]yes|no [useOneX=]yes|no

[authMode=]machineOrUser|machineOnly|userOnly|guest
[ssoMode=]preLogon|postLogon|none
                                [maxDelay=]1-120 [allowDialog=]yes|no
[userVLAN=]yes|no [heldPeriod=]1-3600
                                [AuthPeriod=1-3600] [StartPeriod=]1-3600
[maxStart=]1-100 [maxAuthFailures=]1-100
                                [cacheUserData=]yes|no
```

```
[cost=]default|unrestricted|fixed|variable
netsh wlan set profiletype [name=]<string> [profiletype=]all|current [interface=]
<string>
netsh wlan set randomization [enabled=]yes|no [interface=]<string>
netsh wlan set tracing [mode=]yes|no|persistent
```

```
netsh wlan show all
netsh wlan show allowexplicitcreds
netsh wlan show autoconfig
netsh wlan show blockednetworks
netsh wlan show createalluserprofile
netsh wlan show drivers
netsh wlan show filters
netsh wlan show hostednetwork
netsh wlan show interfaces
netsh wlan show networks
netsh wlan show onlyUseGPProfilesforAllowedNetworks
netsh wlan show profiles
netsh wlan show randomization
netsh wlan show settings
netsh wlan show tracing
netsh wlan show wirelesscapabilities
netsh wlan show wlanreport
```

```
netsh wlan start hostednetwork
```

```
netsh wlan stop hostednetwork
```

Parameters

Expand table

Command	Description
wlan `add`	`filter` - Adds a wireless network into either the wireless network *allowed* or *disallowed* list. The parameter `ssid` is required if the permission is set to `allow` or `block`. If `denyall` is set, `ssid` isn't required. `permission` - Sets the permission type of the filter. - `allow`

Command	Description
	- `block` - `denyall` `ssid` - The SSID of the wireless network. `networktype` - The network type of the wireless network. - `infrastructure` - `adhoc` `profile` - Adds a WLAN profile to the specified network interface: - `filename` (required): Specifies the XML file containing the profile data. - `interface` (optional): The name of the wireless interface, as listed by the `show interface` command. If provided, the profile is added only to the specified interface; otherwise, it's added to all wireless interfaces. - `user` (optional): Determines whether the profile applies to all (`all`) users or only the current (`current`) user. By default, the profile is applied to all users.
wlan `connect`	Connects to a wireless network using the specified profile. If more than one network interface is available, you must specify which interface to use, otherwise the interface parameter can be omitted. The profile name is required, while the Service Set Identifier (SSID) is optional. If the profile contains only one SSID, it's used automatically. If multiple SSIDs exist in the profile, you must specify the SSID. When you specify an interface, wildcard names aren't supported. If the chosen interface is already connected to a wireless network, the command disconnects from the current network before attempting to connect to the new one. If the current and target networks are the same, the command returns success without making changes. `ssid`: Specifies the SSID of the wireless network. `name`: Specifies the profile name to use for the connection attempt. `interface`: Specifies the interface name from which the connection is attempted.
wlan `delete`	`filter` - Removes a wireless network from the wireless network *allowed* or *disallowed* list. The parameter `ssid` is required if the permission is set to `allow` or `block`. If `denyall` is set, `ssid` isn't required. `permission` - Sets the permission type of the filter. - `allow` - `block` - `denyall` `ssid` - The SSID of the wireless network. `networktype` - The network type of the wireless network. - `infrastructure` - `adhoc` `profile` - Removes a WLAN profile from the specified network interface:

Command	Description
	- `name` (required): Specifies the name of the profile to remove. The profile name can have wildcard characters and multiple profiles with the matching name is removed. - `interface` (optional): The name of the wireless interface, as listed by the `show interface` command. If provided, the profile is removed only from the specified interface; otherwise, it's removed from all wireless interfaces.
wlan `disconnect`	Disconnects the device from the wireless network on the specified interface. The `interface=` parameter is required if there are two or more available interfaces on the device. Wildcard characters can be used to specify multiple interfaces.
wlan `dump`	Creates a script containing the current context configuration. The script can be saved to a file and used to restore settings if they're altered or need to be replicated on another system.
wlan `export`	Exports selected WLAN profiles to XML files in the specified folder. Each exported profile is saved as "Interface Name-Profile Name.xml". `hostednetworkprofile` - Exports the hosted network profile to an XML file. `profile` - Exports the specified profiles to an XML file. Options are: - `name`: Specifies the profile name to export. - `folder`: Specifies the destination folder for the exported profile XML files. - `interface`: Specifies the network interface associated with the profile. - `key`: To display the security key in plain text, use `key=clear`. The `folder`, `name`, and `interface` parameters are optional. - If `name` is specified, only that profile is exported; otherwise, all profiles are exported. - If `folder` is provided, it must be an existing, accessible local folder (absolute or relative path). "." refers to the current directory, and ".." to its parent. UNC paths aren't supported. By default, profiles are saved in the current directory. - If `interface` is specified, only the profile on that interface is exported; otherwise, all profiles with the given name are exported. - If the `key` parameter is set to plain text and the caller is a local administrator, the XML file includes the key in plain text. Otherwise, the key is encrypted in the output file.
wlan `IHV`	`startlogging` - Starts verbose logging for Wi-Fi Independent Hardware Vendor (IHV) events. `stoplogging` - Stops verbose logging for Wi-Fi IHV events.
wlan `refresh`	Refreshes the settings of the hosted wireless network. `data` - Instructs the WLAN service to update the hosted network with a new security key.

Command	Description
wlan `reportissues`	Generates a WLAN smart trace report to help diagnosing and reporting wireless network issues.
wlan `set allowexplicitcreds`	`allow` - Enables (`yes`) or disables (`no`) the use of explicit credentials for WLAN connections. This parameter is required.
wlan `set autoconfig`	`enabled` - Turns WLAN auto-configuration to on (`yes`) or off (`no`). `interface` - Specifies the wireless network interface to configure. If `enabled=no` is set, then Windows doesn't connect to any wireless network automatically from the specified interface.
wlan `set blockednetworks`	`display` - Shows (`show`) or hides (`hide`) blocked wireless networks in the network list. If `display=hide` is set, blocked networks aren't displayed in the list of available wireless networks.
wlan `set blockperiod`	`value` - Sets the period (in minutes, `0` to `60`) to block a wireless network after a failed connection attempt. The blocked status is cleared when you manually attempt to connect, when the user session changes, or when the network interface reconnects.
wlan `set createalluserprofile`	`enabled` - Allows (`yes`) or disallows (`no`) creation of all-user wireless profiles. This parameter is required.
wlan `set hostednetwork`	`mode` - Enables (`allow`) or disables (`disallow`) the hosted network feature. `ssid` - Sets the SSID (network name) for the hosted network. `key` - Specifies the passphrase for the hosted network. The user security key should be a string with 8 to 63 ASCII characters. `keyUsage` - Determines if the key is `persistent` or `temporary`. If `keyUsage` is set to `persistent`, the security key is saved and reused whenever the hosted network is started in the future. If set to `temporary`, the key is used only for the current or next hosted network session and is deleted when the hosted network is stopped. If `keyUsage` isn't specified, the key defaults to `persistent`.
wlan `set profileorder`	Sets the preference order of a wireless network profile for a specific interface. All three parameters (`name`, `interface`, and `priority`) are required. Only *User* profiles can have their order changed. Group Policy profiles are read-only and always take precedence over *User* profiles. The `priority` parameter determines the new position of the profile in the preferred *User* profile list. For example, setting `priority=1` moves the profile to the top of the *User* profile list. `name` - Specifies the name of the wireless profile. `interface` - Specifies the wireless network interface to apply the profile order. `priority` - Sets the priority (integer value) of the profile.
wlan `profileparameter`	Modifies the specified profile. If the interface is specified, then only the profile on that interface is modified. The `name` parameter is required. At least one other parameter besides the profile name and interface must also be specified.

Command	Description
	`name`: Specifies the name of the wireless profile. `interface`: Specifies the wireless network interface. `SSIDname`: Specifies the SSID for the profile. `ConnectionType` - Specifies whether the network is infrastructure (`ESS`) or ad-hoc (`IBSS`). `ConnectionMode` - Sets the connection mode to automatic (`auto`) or manual (`manual`). Must be set to `manual` for ad-hoc (`IBSS`) networks. `autoSwitch` - Determines if the device automatically switches to a more preferred network when available (`yes` or `no`). `randomization` - Enables (`yes`) or disables (`no`) MAC address randomization for privacy. If `daily` is set, MAC address randomization refreshes every day. `nonBroadcast` - Indicates whether to connect to a hidden (non-broadcast) network (`yes` or `no`). `authentication` - Sets the authentication method used for the connection: - `open`: No authentication (password) required. - `shared`: Shared WEP key authentication. - `WPA` - `WPA2` - `WPAPSK` - `WPA2PSK` `encryption` - Specifies the encryption protocol for wireless traffic (`WEP`, `TKIP`, `AES`, or `none`). `keyType` - Defines if the shared key is a `networkkey` or a `passphrase`. `keyIndex` - Selects the key index (`1` to `4`) used for encrypting WEP wireless traffic. The value represents one of four stored WEP keys. `keyMaterial` - Specifies the network key or passphrase. `PMKCacheMode` - Enables (`yes`) or disables (`no`) PMK caching (only for WPA2 networks). `PMKCacheSize` - Sets the number of entries (`1` to `255`) in the PMK cache on the client. `PMKCacheTTL` - Specifies how long in seconds (`300` to `86400`) a PMK cache entry is retained. `preAuthMode` - Enables (`yes`) or disables (`no`) pre-authentication (only for WPA2 networks). `preAuthThrottle` - Sets the number of pre-authentication attempts on neighboring access points from `1` to `16`. `FIPS` - Turns on (`yes`) FIPS mode or off (`no`) for enhanced security compliance. `useOneX` - Enables (`yes`) or disables (`no`) 802.1X authentication. `authMode` - Selects the type of credentials used for authentication (`machineoruser`, `machineonly`, `useronly`, or `guest`). `ssoMode` - Specifies the single sign-on mode to use, if any (`prelogon`, `postlogon`, or `none`). `maxDelay` - Sets the maximum time in seconds allowed to establish a single sign-on connection (`1` to `120`). `allowDialog` - Allows (`allow`) or prevents (`no`) dialog prompts during pre-logon.

Command	Description
	`userVLAN` - Indicates if the network switches to a different VLAN after user authentication (`yes` or `no`). `heldPeriod` - Sets the interval in seconds (`1` to `3600`) between authentication attempts. `AuthPeriod` - Maximum time in seconds (`1` to `3600`) the client waits for a response from the authenticator. `StartPeriod` - Time in seconds (`1` to `3600`) to wait before sending an EAPOL-Start message. `maxStart` - Maximum number of EAPOL-Start messages sent during authentication from `1` to `100`. `maxAuthFailures` - Maximum number of authentication failures allowed for a set of credentials from `1` to `100`. `cacheUserData` - Enables (`yes`) or disables (`no`) caching of user credentials for future use. `cost` - Sets the cost category for the network profile: - `default`: Treats Wi-Fi as non-metered unless otherwise specified. - `unrestricted`: Treats the network as unlimited. - `fixed`: Marks the network as metered with a fixed data limit. - `variable`: Marks the network as metered, but assumes the cost or data availability might change (roaming charges or dynamic pricing).
wlan `set profiletype`	Changes the profile type for the specified profile. If the interface is specified, then only the profile on that interface is changed. The `name` and `profiletype` parameters are required. `name` - The profile name to be changed. `profiletype` - Specifies the desired profile type using `all` or `current` for a single user. `interface` - Specifies the name of the interface on which the profile is set.
wlan `set randomization`	`enabled` - Enables (`yes`) or disables (`no`) MAC address randomization. `interface` - Specifies the wireless network interface. This command fails if the interface doesn't support randomization or if randomization is blocked by the system admin. Both parameters are required.
wlan `set tracing`	`mode` - Enables (`yes`) or disables (`no`) tracing, with the option to make tracing `persistent`. If enabled, trace logs for wireless LAN are collected and saved to the trace files. Under persistent mode, tracing is still in effect after system reboot. If mode is set to disable, then both persistent and non-persistent tracing stops.
wlan `show`	Displays the following information: `all` - Provides detailed information about all wireless devices and networks on the system. `allowexplicitcreds` - Displays the current configuration for using explicit user

Command	Description
	credentials with wireless connections. `autoconfig` - Whether automatic wireless network configuration is enabled or disabled. `blockednetworks` - Displays the current visibility settings for blocked wireless networks. `createalluserprofile` - If users are permitted to create wireless profiles available to all users. `drivers` - Lists properties and details of installed wireless LAN drivers. `filters` - Displays the current lists of allowed and blocked wireless networks. `hostednetwork` - Displays the status and properties of the hosted wireless network. `interfaces` - Lists all wireless LAN interfaces detected on the system. `networks` - Displays all visible wireless networks in range. `onlyUseGPProfilesforAllowedNetworks` - Indicates whether only Group Policy-configured profiles are used for allowed networks. `profiles` - Lists all configured wireless network profiles. `randomization` - Indicates the status of MAC address randomization for wireless connections. `settings` - Displays global configuration settings for wireless LAN. `tracing` - Whether wireless LAN tracing is currently active. `wirelesscapabilities` - Lists the wireless features and capabilities supported by the system. `wlanreport` - Generates a report summarizing recent wireless network sessions and activity.
wlan `start hostednetwork`	Starts the WLAN service on the hosted network.
wlan `stop hostednetwork`	Stops the WLAN service on the hosted network.
`help` *or* `?`	Displays a list of commands and their descriptions in the current context.

Examples

To add a Wi-Fi profile filter that only allows connection to a specific network SSID, run the following command:

Windows Command Prompt

```
netsh wlan add filter permission=allow ssid="MyNetwork" networktype=infrastructure
```

To connect to a Wi-Fi network named "CorpWiFi" using the interface "Wi-Fi", run the following command:

Windows Command Prompt

```
netsh wlan connect name=CorpWiFi interface="Wi-Fi"
```

To disconnect from a Wi-Fi network on the interface "Wi-Fi", run the following command:

Windows Command Prompt

```
netsh wlan disconnect interface="Wi-Fi"
```

To export the Wi-Fi profile named "CorpWiFi" to the folder "C:\WiFiProfiles" including the security key, run the following command:

Windows Command Prompt

```
netsh wlan export profile name="CorpWiFi" folder="C:\WiFiProfiles" key=clear
```

To start logging IHV events related to WLAN, run the following command:

Windows Command Prompt

```
netsh wlan IHV startlogging
```

To refresh the hosted network and update its key or password, run the following command:

Windows Command Prompt

```
netsh wlan refresh hostednetwork data=MyKey
```

To set the SSID name to "CorpWiFi_5G" and the connection mode to auto for the Wi-Fi profile "CorpWiFi" on the interface "Wi-Fi", run the following command:

Windows Command Prompt

```
netsh wlan set profileparameter name="CorpWiFi" interface="Wi-Fi"
SSIDname="CorpWiFi_5G" ConnectionMode=auto
```

To set the profile type for all profiles on the Wi-Fi profile "CorpWiFi" on the interface "Wi-Fi", run the following command:

Windows Command Prompt

```
netsh wlan set profiletype name="CorpWiFi" profiletype=all interface="Wi-Fi"
```

To enable WLAN tracing in persistent mode, run the following command:

Windows Command Prompt

```
netsh wlan set tracing mode=persistent
```

See also

- Command-Line Syntax Key

netstat

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Displays active TCP connections, ports on which the computer is listening, Ethernet statistics, the IP routing table, IPv4 statistics (for the IP, ICMP, TCP, and UDP protocols), and IPv6 statistics (for the IPv6, ICMPv6, TCP over IPv6, and UDP over IPv6 protocols). Used without parameters, this command displays active TCP connections.

> **Important**
>
> This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network Connections.

Syntax

```
netstat [-a] [-b] [-e] [-n] [-o] [-p <Protocol>] [-q] [-r] [-s] [<interval>]
```

Parameters

Expand table

Parameter	Description
-a	Displays all active TCP connections and the TCP and UDP ports on which the computer is listening.
-b	Displays the executable involved in creating each connection or listening port. In some cases well-known executables host multiple independent components, and in these cases the sequence of components involved in creating the connection or listening port is displayed. In this case the executable name is in [] at the bottom, on top is the component it called, and so forth until TCP/IP was reached. Note that this option can be time-consuming and will fail unless you have sufficient permissions.
-e	Displays Ethernet statistics, such as the number of bytes and packets sent and received. This parameter can be combined with **-s**.
-n	Displays active TCP connections, however, addresses and port numbers are expressed

Parameter	Description
	numerically and no attempt is made to determine names.
-o	Displays active TCP connections and includes the process ID (PID) for each connection. You can find the application based on the PID on the Processes tab in Windows Task Manager. This parameter can be combined with **-a**, **-n**, and **-p**.
-p `<Protocol>`	Shows connections for the protocol specified by *Protocol*. In this case, the *Protocol* can be tcp, udp, tcpv6, or udpv6. If this parameter is used with **-s** to display statistics by protocol, *Protocol* can be tcp, udp, icmp, ip, tcpv6, udpv6, icmpv6, or ipv6.
-q	Displays all connections, listening ports, and bound nonlistening TCP ports. Bound nonlistening ports may or may not be associated with an active connection.
-s	Displays statistics by protocol. By default, statistics are shown for the TCP, UDP, ICMP, and IP protocols. If the IPv6 protocol is installed, statistics are shown for the TCP over IPv6, UDP over IPv6, ICMPv6, and IPv6 protocols. The **-p** parameter can be used to specify a set of protocols.
-r	Displays the contents of the IP routing table. This is equivalent to the route print command.
`<interval>`	Redisplays the selected information every *interval* seconds. Press CTRL+C to stop the redisplay. If this parameter is omitted, this command prints the selected information only once.
/?	Displays help at the command prompt.

Remarks

- The **netstat** command provides statistics for the following:

Expand table

Parameter	Description
Proto	The name of the protocol (TCP or UDP).
Local address	The IP address of the local computer and the port number being used. The name of the local computer that corresponds to the IP address and the name of the port is shown unless the **-n** parameter is specified. If the port is not yet established, the port number is shown as an asterisk (*).
Foreign address	The IP address and port number of the remote computer to which the socket is connected. The names that corresponds to the IP address and the port are shown unless the **-n** parameter is specified. If the port is not yet established, the port number is shown as an asterisk (*).
State	Indicates the state of a TCP connection, including: ◦ CLOSE_WAIT ◦ CLOSED

Parameter	Description
	◦ ESTABLISHED ◦ FIN_WAIT_1 ◦ FIN_WAIT_2 ◦ LAST_ACK ◦ LISTEN ◦ SYN_RECEIVED ◦ SYN_SEND ◦ TIMED_WAIT

Examples

To display both the Ethernet statistics and the statistics for all protocols, type:

```
netstat -e -s
```

To display the statistics for only the TCP and UDP protocols, type:

```
netstat -s -p tcp udp
```

To display active TCP connections and the process IDs every 5 seconds, type:

```
netstat -o 5
```

To display active TCP connections and the process IDs using numerical form, type:

```
netstat -n -o
```

To display all connections, listening ports, and bound nonlistening ports in numerical form along with the process ID and executable for each, type:

```
netstat -anobq
```

Related links

- Command-Line Syntax Key

Last updated on 09/08/2026

nfsadmin

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

A command-line utility that administers Server for NFS or Client for NFS on the local or remote computer running Microsoft Services for Network File System (NFS). Used without parameters, nfsadmin server displays the current Server for NFS configuration settings and nfsadmin client displays the current Client for NFS configuration settings.

Syntax

```
nfsadmin server [computername] [-u Username [-p Password]] -l
nfsadmin server [computername] [-u Username [-p Password]] -r {client | all}
nfsadmin server [computername] [-u Username [-p Password]] {start | stop}
nfsadmin server [computername] [-u Username [-p Password]] config
option[...]
nfsadmin server [computername] [-u Username [-p Password]] creategroup
<name>
nfsadmin server [computername] [-u Username [-p Password]] listgroups
nfsadmin server [computername] [-u Username [-p Password]] deletegroup
<name>
nfsadmin server [computername] [-u Username [-p Password]] renamegroup
<oldname> <newname>
nfsadmin server [computername] [-u Username [-p Password]] addmembers
<hostname>[...]
nfsadmin server [computername] [-u Username [-p Password]] listmembers
nfsadmin server [computername] [-u Username [-p Password]] deletemembers
<hostname><groupname>[...]
nfsadmin client [computername] [-u Username [-p Password]] {start | stop}
nfsadmin client [computername] [-u Username [-p Password]] config
option[...]
```

General Parameters

Expand table

Parameter	Description
computername	Specifies the remote computer you want to administer. You can specify the computer using a Windows Internet Name Service (WINS) name or a Domain

Parameter	Description
	Name System (DNS) name, or by Internet Protocol (IP) address.
-u Username	Specifies the user name of the user whose credentials are to be used. It might be necessary to add the domain name to the user name in the form *domain\username*.
-p Password	Specifies the password of the user specified using the **-u** option. If you specify the **-u** option but omit the **-p** option, you are prompted for the user's password.

Server for NFS-related parameters

Expand table

Parameter	Description
-l	Lists all locks held by clients.
-r `{client|all}`	Releases the locks held by a client or, if all is specified, by all clients.
start	Starts the Server for NFS service.
stop	Stops the Server for NFS service.
config	Specifies general settings for Server for NFS. You must supply at least one of the following options with the **config** command argument: • **mapsvr=**`<server>` - Sets server as the User Name Mapping server for Server for NFS. Although this option continues to be supported for compatibility with previous versions, you should use the sfuadmin utility instead. • **auditlocation=**`{eventlog|file|both|none}` - Specifies whether events will be audited and where the events will be recorded. One of the following arguments is required: ◦ **eventlog** - Specifies that audited events will be recorded only in the Event Viewer application log. ◦ **file** - Specifies that audited events will be recorded only in the file specified by `config fname`. ◦ **both** - Specifies that audited events will be recorded in the Event Viewer application log as well as the file specified by `config fname`. ◦ **none** - Specifies that events aren't audited. • **fname=**`<file>` - Sets the file specified by file as the audit file. The default is **%sfudir%\log\nfssvr.log**. • **fsize=**`<size>` - Sets size as the maximum size in megabytes of the audit file. The default maximum size is **7 MB**. • `audit=[+|-]mount [+|-]read [+|-]write [+|-]create [+|-]delete [+|-]locking [+|-]all` - Specifies the events to be logged. To start logging an event, type a plus sign (+) before the event name; to stop

Parameter	Description
	logging an event, type a minus sign (-) before the event name. If the sign is omitted, the + sign is assumed. Don't use **all** with any other event name. • **lockperiod=** `<seconds>` - Specifies the number of seconds that Server for NFS will wait to reclaim locks after a connection to Server for NFS has been lost and then reestablished or after the Server for NFS service has been restarted. • **portmapprotocol=** `{TCP|UDP|TCP+UDP}` - Specifies which transport protocols Portmap supports. The default setting is **TCP+UDP**. • **mountprotocol=** `{TCP|UDP|TCP+UDP}` - Specifies which transport protocols mount supports. The default setting is **TCP+UDP**. • **nfsprotocol=** `{TCP|UDP|TCP+UDP}` - Specifies which transport protocols Network File System (NFS) supports. The default setting is **TCP+UDP** • **nlmprotocol=** `{TCP|UDP|TCP+UDP}` - Specifies which transport protocols Network Lock Manager (NLM) supports. The default setting is **TCP+UDP**. • **nsmprotocol=** `{TCP|UDP|TCP+UDP}` - Specifies which transport protocols Network Status Manager (NSM) supports. The default setting is **TCP+UDP**. • **enableV3=** `{yes|no}` - Specifies whether NFS version 3 protocols will be supported. The default setting is **yes**. • **renewauth=** `{yes|no}` - Specifies whether client connections will be required to be reauthenticated after the period specified by config renewauthinterval. The default setting is **no**. • **renewauthinterval=** `<seconds>` - Specifies the number of seconds that elapse before a client is forced to be reauthenticated if `config renewauth` is set to **yes**. The default value is **600 seconds**. • **dircache=** `<size>` - Specifies the size in kilobytes of the directory cache. The number specified as size must be a multiple of 4 between 4 and 128. The default directory cache size is **128 KB**. • **translationfile=** `<file>` - Specifies a file containing mapping information for replacing characters in the names of files when moving them from Windows-based to UNIX-based file systems. If file is not specified, then file name character translation is disabled. If the value of **translationfile** is changed, you must restart the server for the change to take effect. • **dotfileshidden=** `{yes|no}` - Specifies whether files with names beginning with a period (.) are marked as hidden in the Windows file system, and consequently hidden from NFS clients. The default setting is **no**. • **casesensitivelookups=** `{yes|no}` - Specifies whether directory lookups are case sensitive (require exact matching of character case). You must also disable Windows kernel case-insensitivity to support case-sensitive file names. To support case-sensitivity, change the **DWord** value of the registry key, `HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\kernel`, to **0**.

Parameter	Description
	• **ntfscase=**`{lower|upper|preserve}` - Specifies whether the case of characters in the names of files in the NTFS file system will be returned in lowercase, uppercase, or in the form stored in the directory. The default setting is **preserve**. This setting can't be changed if **casesensitivelookups** is set to **yes**.
creategroup `<name>`	Creates a new client group, giving it the specified name.
listgroups	Displays the names of all client groups.
deletegroup `<name>`	Removes the client group specified by name.
renamegroup `<oldname>` `<newname>`	Changes the name of the client group specified by *oldname* to *newname*.
addmembers `<hostname>[...]`	Adds a *host* to the client group specified by *name*.
listmembers `<name>`	Lists the host computers in the client group specified by *name*.
deletemembers `<hostname>` `<groupname>[...]`	Removes the client specified by *host* from the client group specified by *group*.

Client for NFS-related parameters

Expand table

Parameter	Description
start	Starts the Client for NFS service.
stop	Stops the Client for NFS service.
config	Specifies general settings for Client for NFS. You must supply at least one of the following options with the **config** command argument: • **fileaccess=**`<mode>` - Specifies the default permission mode for files created on Network File System (NFS) servers. The **mode** argument consists of a three digit number, from 0 to 7 (inclusive), which represent the default permissions granted the user, group, and others. The digits translate to UNIX-style permissions as follows: *0=none*, *1=x (execute)*, *2=w (write only)*, *3=wx (write and execute)*, *4=r (read only)*, *5=rx (read and execute)*, *6=rw (read and write)*,

Parameter	Description
	and *7=rwx (read, write, and execute)*. For example, `fileaccess=750` gives read, write, and execute permissions to the owner, read and execute permissions to the group, and no access permission to others. • **mapsvr=** `<server>` - Sets server as the User Name Mapping server for Client for NFS. Although this option continues to be supported for compatibility with previous versions, you should use the sfuadmin utility instead. • **mtype=** `{hard|soft}` - Specifies the default mount type. For a hard mount, Client for NFS continues to retry a failed RPC until it succeeds. For a soft mount, Client for NFS returns failure to the calling application after retrying the call the number of times specified by the retry option. • **retry=** `<number>` - Specifies the number of times to try to make a connection for a soft mount. This value must be from 1 to 10, inclusive. The default is **1**. • **timeout=** `<seconds>` - Specifies the number of seconds to wait for a connection (remote procedure call). This value must be *0.8*, *0.9*, or an integer from *1 to 60*, inclusive. The default is **0.8**. • **protocol=** `{TCP|UDP|TCP+UDP}` - Specifies which transport protocols the client supports. The default setting is **TCP+UDP**. • **rsize=** `<size>` -Specifies the size, in kilobytes, of the read buffer. This value can be *0.5, 1, 2, 4, 8, 16,* or *32*. The default is **32**. • **wsize=** `<size>` - Specifies the size, in kilobytes, of the write buffer. This value can be *0.5, 1, 2, 4, 8, 16,* or *32*. The default is **32**. • **perf=default** - Restores the following performance settings to default values, *mtype*, *retry*, *timeout*, *rsize*, or *wsize*.

Examples

To stop Server for NFS or Client for NFS, type:

```
nfsadmin server stop
nfsadmin client stop
```

To start Server for NFS or Client for NFS, type:

```
nfsadmin server start
nfsadmin client start
```

To set Server for NFS to not be case-sensitive, type:

```
nfsadmin server config casesensitive=no
```

To set Client for NFS to be case-sensitive, type:

```
nfsadmin client config casesensitive=yes
```

To display all the current Server for NFS or Client for NFS options, type:

```
nfsadmin server config
nfsadmin client config
```

Related links

- Command-Line Syntax Key
- NFS cmdlets reference

Feedback

Was this page helpful? Yes No

nfsshare

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Controls Network File System (NFS) shares. Used without parameters, this command displays all Network File System (NFS) shares exported by Server for NFS.

Syntax

```
nfsshare <sharename>=<drive:path> [-o <option=value>...]
nfsshare {<sharename> | <drive>:<path> | * } /delete
```

Parameters

Expand table

Parameter	Description
-o anon=`{yes|no}`	Specifies whether anonymous (unmapped) users can access the share directory.
-o rw=`[<host>[:<host>]...]`	Provides read-write access to the shared directory by the hosts or client groups specified by *host*. You must separate host and group names with a colon (:). If *host* isn't specified, all hosts and client groups (except those specified with the **ro** option) get read-write access. If neither the **ro** nor the **rw** option is set, all clients have read-write access to the shared directory.
-o ro=`[<host>[:<host>]...]`	Provides read-only access to the shared directory by the hosts or client groups specified by *host*. You must separate host and group names with a colon (:). If *host* isn't specified, all clients (except those specified with the **rw** option) get read-only access. If the **ro** option is set for one or more clients, but the **rw** option isn't set, only the clients specified with the **ro** option have access to the shared directory.

Parameter	Description
-o encoding=`{euc-jp|euc-tw|euc-kr|shift-jis|Big5|Ksc5601|Gb2312-80|Ansi)`	Specifies the language encoding to configure on an NFS share. You can use only one language on the share. This value can include any of the following values: • **euc-jp:** Japanese • **euc-tw:** Chinese • **euc-kr:** Korean • **shift-jis:** Japanese • **Big5:** Chinese • **Ksc5601:** Korean • **Gb2312-80:** Simplified Chinese • **Ansi:** ANSI-encoded
-o anongid=`<gid>`	Specifies that anonymous (unmapped) users access the share directory using *gid* as their group identifier (GID). The default is **-2**. The anonymous GID is used when reporting the owner of a file owned by an unmapped user, even if anonymous access is disabled.
-o anonuid=`<uid>`	Specifies that anonymous (unmapped) users access the share directory using *uid* as their user identifier (UID). The default is **-2**. The anonymous UID is used when reporting the owner of a file owned by an unmapped user, even if anonymous access is disabled.
-o root=`[<host>[:<host>]...]`	Provides root access to the shared directory by the hosts or client groups specified by *host*. You must separate host and group names with a colon (:). If *host* isn't specified, all clients get root access. If the **root** option isn't set, no clients have root access to the shared directory.
/delete	If *sharename* or `<drive>:<path>` is specified, this parameter deletes the specified share. If a wildcard (*) is specified, this parameter deletes all NFS shares.
/?	Displays help at the command prompt.

Remarks

- If *sharename* as the only parameter, this command lists the properties of the NFS share identified by *sharename*.
- If *sharename* and `<drive>:<path>` are used, this command exports the folder identified by `<drive>:<path>` as *sharename*. If you use the **/delete** option, the specified folder stops being available to NFS clients.

Related links

- Command-Line Syntax Key
- Services for Network File System Command Reference
- NFS cmdlets reference

Feedback

Was this page helpful? Yes No

nfsstat

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

A command-line utility that displays statistical info about the Network File System (NFS) and Remote Procedure Call (RPC) calls. Used without parameters, this command displays all of the statistical data without resetting anything.

Syntax

```
nfsstat [-c][-s][-n][-r][-z][-m]
```

Parameters

Expand table

Parameter	Description
-c	Displays only the client-side NFS and RPC and NFS calls sent and rejected by the client. To display NFS or RPC information only, combine this flag with the **-n** or **-r** parameter.
-s	Displays only the server-side NFS and RPC and NFS calls sent and rejected by the server. To display NFS or RPC information only, combine this flag with the **-n** or **-r** parameter.
-m	Displays information about mount flags set by mount options, mount flags internal to the system, and other mount information.
-n	Displays NFS information for both the client and server. To display only the NFS client or server information, combine this flag with the **-c** or **-s** parameter.
-r	Displays RPC information for both the client and server. To display only the RPC client or server information, combine this flag with the **-c** or **-s** parameter.
-z	Resets the call statistics. This flag is only available to the root user and can be combined with any of the other parameters to reset particular sets of statistics after displaying them.

Examples

To display information about the number of RPC and NFS calls sent and rejected by the client, type:

```
nfsstat -c
```

To display and print the client NFS call-related information, type:

```
nfsstat -cn
```

To display RPC call-related information for both the client and server, type:

```
nfsstat -r
```

To display information about the number of RPC and NFS calls received and rejected by the server, type:

```
nfsstat -s
```

To reset all call-related information to zero on the client and server, type:

```
nfsstat -z
```

Related links

- Command-Line Syntax Key
- Services for Network File System Command Reference
- NFS cmdlets reference

Feedback

Was this page helpful? Yes No

nlbmgr

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configure and manage your Network Load Balancing clusters and all cluster hosts from a single computer, using the Network Load Balancing Manager. You can also use this command to replicate the cluster configuration to other hosts.

You can start the Network Load Balancing Manager from the command-line using the command **nlbmgr.exe**, which is installed in the **systemroot\System32** folder.

Syntax

```
nlbmgr [/noping][/hostlist <filename>][/autorefresh <interval>][/help | /?]
```

Parameters

Expand table

Parameter	Description
/noping	Prevents the Network Load Balancing Manager from pinging the hosts prior to trying to contact them through Windows Management Instrumentation (WMI). Use this option if you have disabled Internet Control Message Protocol (ICMP) on all available network adapters. If the Network Load Balancing Manager attempts to contact a host that isn't available, you'll experience a delay when using this option.
/hostlist `<filename>`	Loads the hosts specified in filename into the Network Load Balancing Manager.
/autorefresh `<interval>`	Causes the Network Load Balancing Manager to refresh its host and cluster information every `<interval>` seconds. If no interval is specified, the information is refreshed every 60 seconds.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- NetworkLoadBalancingClusters cmdlets reference

Feedback

Was this page helpful? Yes No

nslookup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information that you can use to diagnose Domain Name System (DNS) infrastructure. Before using this tool, you should be familiar with how DNS works. The nslookup command-line tool is available only if you have installed the TCP/IP protocol.

Syntax

```
nslookup [exit | finger | help | ls | lserver | root | server | set | view]
[options]
```

Parameters

Expand table

Parameter	Description
nslookup exit	Exits the nslookup command-line tool.
nslookup finger	Connects with the finger server on the current computer.
nslookup help	Displays a short summary of subcommands.
nslookup ls	Lists information for a DNS domain.
nslookup lserver	Changes the default server to the specified DNS domain.
nslookup root	Changes the default server to the server for the root of the DNS domain name space.
nslookup server	Changes the default server to the specified DNS domain.
nslookup set	Changes configuration settings that affect how lookups function.
nslookup set	Prints the current values of the configuration settings.

Parameter	Description
all	
nslookup set class	Changes the query class. The class specifies the protocol group of the information.
nslookup set d2	Turns exhaustive Debugging mode on or off. All fields of every packet are printed.
nslookup set debug	Turns Debugging mode on or off.
nslookup set domain	Changes the default DNS domain name to the name specified.
nslookup set port	Changes the default TCP/UDP DNS name server port to the value specified.
nslookup set querytype	Changes the resource record type for the query.
nslookup set recurse	Tells the DNS name server to query other servers if it doesn't have the information.
nslookup set retry	Sets the number of retries.
nslookup set root	Changes the name of the root server used for queries.
nslookup set search	Appends the DNS domain names in the DNS domain search list to the request until an answer is received. This applies when the set and the lookup request contain at least one period, but do not end with a trailing period.
nslookup set srchlist	Changes the default DNS domain name and search list.
nslookup set timeout	Changes the initial number of seconds to wait for a reply to a request.
nslookup set type	Changes the resource record type for the query.
nslookup set vc	Specifies to use or not use a virtual circuit when sending requests to the server.
nslookup view	Sorts and lists the output of the previous **ls** subcommand or commands.

Remarks

- The nslookup command-line tool has two modes: interactive and noninteractive.
  - If you need to look up only a single piece of data, or you're using nslookup in scripts, command lines, or PowerShell, use the noninteractive mode. In noninteractive mode, also called command mode, the first command line parameter is the name or IP address of the computer that you want to look up. The second parameter is the name or IP address of a DNS name server. If you omit the second argument, nslookup uses the default DNS name server.
  - If you need to look up more than one piece of data or set several configurations, you can use interactive mode. To enter interactive mode, type a hyphen (-) instead of the first parameter in the nslookup command line. Enter the name or IP address of a DNS name server for the second parameter. If you omit the second argument, nslookup uses the default DNS name server. You can also invoke interactive mode by simply entering `nslookup` at the command prompt, and then entering names or IP addresses to search for in the interactive command line.
- Once you enter `nslookup -` or `nslookup` alone, the command prompt changes to the interactive prompt `>`. While in interactive mode, you can:
  - Enter names or IP addresses, `set` variables, and other options on separate lines.
  - Interrupt interactive commands at any time by pressing CTRL+B.
  - Exit, by entering `exit`.
  - Treat a built-in command as a computer name by preceding it with the escape character (`\`). An unrecognized command is interpreted as a computer name.
- If the computer to find is an IP address and the query is for an **A** or **PTR** resource record type, the name of the computer is returned.
- If the computer to find is a name and doesn't have a trailing period, the default DNS domain name is appended to the name. This behavior depends on the state of the following **set** subcommands: **domain**, **srchlist**, **defname**, and **search**.
- If the lookup request fails, the command-line tool provides one of the following error messages:

Expand table

Error message	Description
timed out	The server didn't respond to a request after a certain amount of time and a certain number of retries. You can set the time-out period with

Error message	Description
	the nslookup set timeout command. You can set the number of retries with the nslookup set retry command.
No response from server	No DNS name server is running on the server computer.
No records	The DNS name server doesn't have resource records of the current query type for the computer, although the computer name is valid. The query type is specified with the nslookup set querytype command.
Nonexistent domain	The computer or DNS domain name doesn't exist.
Connection refused or Network is unreachable	The connection to the DNS name server or finger server couldn't be made. This error commonly occurs with the **ls** and **finger** requests.
Server failure	The DNS name server found an internal inconsistency in its database and couldn't return a valid answer.
Refused	The DNS name server refused to service the request.
format error	The DNS name server found that the request packet wasn't in the proper format. It may indicate an error in **nslookup**.

Examples

In nslookup noninteractive mode, you specify parameters and options in the Windows command line or script. In interactive mode, you specify arguments and options on separate lines at the interactive command prompt.

Noninteractive mode

In nslookup noninteractive mode, the first parameter is the computer to find, and the second parameter is the DNS name server to use. If you don't specify a second parameter, nslookup uses the default DNS name server. The following examples use `nslookup` in noninteractive mode.

- The following example looks up the IP addresses for the domain name `mydomain.com` on the DNS name server at `1.1.1.1`:

  Windows Command Prompt

  ```
  nslookup mydomain.com 1.1.1.1
  ```

- The following example looks up the domain name for the IP address `4.4.4.4` on the default DNS name server:

  Windows Command Prompt

  ```
  nslookup 4.4.4.4
  ```

- To specify options, you can use `nslookup -<option>`. For example, the following command turns on the nslookup `debug` option to get more information about packets sent.

  Windows Command Prompt

  ```
  nslookup -debug mydomain.com
  ```

- To return certain types of records or information, use the `-type=<resourcerecordtype>` option. For example, the following command returns only IPv6 record types:

  Windows Command Prompt

  ```
  nslookup -type=AAAA mydomain.com
  ```

- You can combine options and resource record type queries in command lines. The following example enables debug output, retrieves both IPv6 and IPv4 addresses, doesn't attempt to use the search domain, uses recursive lookup, and uses the 1.1.1.1 DNS lookup server:

  Windows Command Prompt

  ```
  nslookup -debug -type=A+AAAA -nosearch -recurse mydomain.com 1.1.1.1
  ```

Interactive mode

To use interactive mode, enter `-` instead of the first parameter of a nslookup command line, or simply enter `nslookup`. The command prompt then changes to the interactive prompt `>`. The following examples show interactive mode commands.

- The following command places nslookup in interactive mode and sets `1.1.1.1` as the default DNS lookup server:

  Windows Command Prompt

```
nslookup - 1.1.1.1
```

- The following command at the interactive prompt returns nslookup option and parameter settings for the current server:

```
Windows Command Prompt
set all
```

- The following command at the interactive prompt returns the IP addresses for `mydomain.com`:

```
Windows Command Prompt
mydomain.com
```

- The following command at the interactive prompt changes the default DNS name server to `4.4.4.4`:

```
Windows Command Prompt
server 4.4.4.4
```

- The following command at the interactive prompt sets the query resource record type to `HINFO`:

```
Windows Command Prompt
set type=HINFO
```

- The following command at the interactive prompt exits interactive mode and returns to the Windows command prompt:

```
Windows Command Prompt
exit
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup /exit

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Exits the nslookup command-line tool.

Syntax

```
nslookup /exit
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup /finger

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Connects with the finger server on the current device.

Syntax

```
finger [<username>] [{[>] <filename> | [>>] <filename>}]
```

Parameters

Expand table

Parameter	Description
`<username>`	Specifies the name of the user to look up.
`<filename>`	Specifies a file name in which to save the output. You can use the greater than (`>`) and double greater than (`>>`) characters to redirect the output in the usual manner.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup help

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the subcommand help text.

Syntax

```
help
```

```
?
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup ls

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists DNS domain information.

Syntax

```
ls [<option>] <DNSdomain> [{[>] <filename>|[>>] <filename>}]
```

Parameters

Expand table

Parameter	Description
`<option>`	The valid options include: • **-t:** Lists all records of the specified type. For more information, see nslookup set querytype. • **-a:** Lists aliases of computers in the DNS domain. This parameter is the same as **-t CNAME** • **-d:** Lists all records for the DNS domain. This parameter is the same as **-t ANY** • **-h:** Lists CPU and operating system information for the DNS domain. This parameter is the same as **-t HINFO** • **-s:** Lists well-known services of computers in the DNS domain. This parameter is the same as **-t WKS**.
`<DNSdomain>`	Specifies the DNS domain for which you want information.
`<filename>`	Specifies a file name to use for the saved output. You can use the greater than (`>`) and double greater than (`>>`) characters to redirect the output in the usual manner.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Remarks

- The default output of this command includes computer names and their associated IP addresses.
- If your output is directed to a file, hash marks are added for every 50 records received from the server.

Related links

- Command-Line Syntax Key
- nslookup set querytype

Feedback

Was this page helpful? Yes No

nslookup lserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the initial server to the specified Domain Name System (DNS) domain.

This command uses the initial server to look up the information about the specified DSN domain. If you want to lookup information using the current default server, use the nslookup server command.

Syntax

```
lserver <DNSdomain>
```

Parameters

Expand table

Parameter	Description
`<DNSdomain>`	Specifies the DNS domain for the initial server.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup server

Feedback

Was this page helpful? Yes No

nslookup root

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the default server to the server for the root of the Domain Name System (DNS) domain name space. Currently, the ns.nic.ddn.mil name server is used. You can change the name of the root server using the nslookup set root command.

> **Note**
>
> This command is the same as `lserver ns.nic.ddn.mil`.

Syntax

```
root
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup set root

Feedback

Was this page helpful? Yes No

nslookup server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the default server to the specified Domain Name System (DNS) domain.

This command uses the current default server to look up the information about the specified DNS domain. If you want to lookup information using the initial server, use the nslookup lserver command.

Syntax

```
server <DNSdomain>
```

Parameters

Expand table

Parameter	Description
`<DNSdomain>`	Specifies the DNS domain for the default server.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup lserver

Feedback

Was this page helpful? Yes No

nslookup set

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes configuration settings that affect how lookups function.

Syntax

```
set all [class | d2 | debug | domain | port | querytype | recurse | retry |
root | search | srchlist | timeout | type | vc] [options]
```

Parameters

Expand table

Parameter	Description
nslookup set all	Lists all current settings.
nslookup set class	Changes the query class, which specifies the protocol group of the information.
nslookup set d2	Turns the verbose debugging mode on or off.
nslookup set debug	Turns off debugging mode completely.
nslookup set domain	Changes the default Domain Name System (DNS) domain name to the specified name.
nslookup set port	Changes the default TCP/UDP Domain Name System (DNS) name server port to the specified value.
nslookup set querytype	Changes the resource record type for the query.
nslookup set recurse	Tells the Domain Name System (DNS) name server to query other servers if it doesn't find any information.
nslookup set retry	Sets the number of retries.

Parameter	Description
nslookup set root	Changes the name of the root server used for queries.
nslookup set search	Appends the Domain Name System (DNS) domain names in the DNS domain search list to the request until an answer is received.
nslookup set srchlist	Changes the default Domain Name System (DNS) domain name and search list.
nslookup set timeout	Changes the initial number of seconds to wait for a reply to a lookup request.
nslookup set type	Changes the resource record type for the query.
nslookup set vc	Specifies whether to use a virtual circuit when sending requests to the server.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set all

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Outputs the current configuration setting values, including the default server and computer (the host).

Syntax

```
set all
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set class

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the query class. The class specifies the protocol group of the information.

Syntax

```
set class=<class>
```

Parameters

Expand table

Parameter	Description
`<class>`	The valid values include: • **IN:** Specifies the Internet class. This is the default value. • **CHAOS:** Specifies the Chaos class. • **HESIOD:** Specifies the MIT Athena Hesiod class. • **ANY:** Specifies to use any of the previously listed values.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set d2

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Turns the verbose debugging mode on or off. All fields of every packet are printed.

Syntax

```
set [no]d2
```

Parameters

Expand table

Parameter	Description
nod2	Turns off the verbose debugging mode. This is the default value.
d2	Turns on the verbose debugging mode.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set debug

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Turns debugging mode on or off.

Syntax

```
set [no]debug
```

Parameters

Expand table

Parameter	Description
nodebug	Turns off debugging mode. This is the default value.
debug	Turns on debugging mode. By turning debugging mode on, you can view more information about the packet sent to the server and the resulting answer.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set domain

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the default Domain Name System (DNS) domain name to the specified name.

Syntax

```
set domain=<domainname>
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Specifies a new name for the default DNS domain name. The default value is the name of the host.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Remarks

- The default DNS domain name is appended to a lookup request depending on the state of the **defname** and **search** options.
- The DNS domain search list contains the parents of the default DNS domain if it has at least two components in its name. For example, if the default DNS domain is mfg.widgets.com, the search list is named both mfg.widgets.com and widgets.com.
- Use the nslookup set srchlist command to specify a different list and the nslookup set all command to display the list.

Related links

- Command-Line Syntax Key
- nslookup set srchlist
- nslookup set all

Feedback

Was this page helpful? Yes No

nslookup set port

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the default TCP/UDP Domain Name System (DNS) name server port to the specified value.

Syntax

```
set port=<port>
```

Parameters

Expand table

Parameter	Description
`<port>`	Specifies the new value for the default TCP/UDP DNS name server port. The default port is **53**.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set querytype

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the resource record type for the query. For information about resource record types, see Request for Comment (Rfc) 1035.

> **Note**
>
> This command is the same as the nslookup set type command.

Syntax

```
set querytype=<resourcerecordtype>
```

Parameters

Expand table

Parameter	Description
`<resourcerecordtype>`	Specifies a DNS resource record type. The default resource record type is **A**, but you can use any of the following values: • **A:** Specifies a computer's IP address. • **ANY:** Specifies a computer's IP address. • **CNAME:** Specifies a canonical name for an alias. • **GID** Specifies a group identifier of a group name. • **HINFO:** Specifies a computer's CPU and type of operating system. • **MB:** Specifies a mailbox domain name. • **MG:** Specifies a mail group member. • **MINFO:** Specifies mailbox or mail list information. • **MR:** Specifies the mail rename domain name. • **MX:** Specifies the mail exchanger. • **NS:** Specifies a DNS name server for the named zone. • **PTR:** Specifies a computer name if the query is an IP address; otherwise, specifies the pointer to other information. • **SOA:** Specifies the start-of-authority for a DNS zone.

Parameter	Description
	• **TXT:** Specifies the text information. • **UID:** Specifies the user identifier. • **UINFO:** Specifies the user information. • **WKS:** Describes a well-known service.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup set type

Feedback

Was this page helpful? Yes No

nslookup set recurse

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Tells the Domain Name System (DNS) name server to query other servers if it can't find the information on the specified server.

Syntax

```
set [no]recurse
```

Parameters

Expand table

Parameter	Description
norecurse	Stops the Domain Name System (DNS) name server from querying other servers if it can't find the information on the specified server.
recurse	Tells the Domain Name System (DNS) name server to query other servers if it can't find the information on the specified server. This is the default value.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup set retry

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

This command sets the number of times a request is resent to a server for information, before giving up.

> **Note**
>
> To change the length of time before the request times out, use the nslookup set timeout command.

Syntax

```
set retry=<number>
```

Parameters

Expand table

Parameter	Description
`<number>`	Specifies the new value for the number of retries. The default number of retries is **1**.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup set timeout

Last updated on 10/30/2025

nslookup set root

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the name of the root server used for queries.

> **Note**
>
> This command supports the nslookup root command.

Syntax

```
set root=<rootserver>
```

Parameters

Expand table

Parameter	Description
`<rootserver>`	Specifies the new name for the root server. The default value is **ns.nic.ddn.mil**.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup root

Feedback

Was this page helpful? Yes No

nslookup set search

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Appends the Domain Name System (DNS) domain names in the DNS domain search list to the request until an answer is received. This applies when the set and the lookup request contain at least one period, but do not end with a trailing period.

Syntax

```
set [no]search
```

Parameters

Expand table

Parameter	Description
nosearch	Stops appending the Domain Name System (DNS) domain names in the DNS domain search list for the request.
search	Appends the Domain Name System (DNS) domain names in the DNS domain search list for the request until an answer is received. This is the default value.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful?

Yes No

nslookup set srchlist

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the default Domain Name System (DNS) domain name and search list. This command overrides the default DNS domain name and search list of the nslookup set domain command.

Syntax

```
set srchlist=<domainname>[/...]
```

Parameters

Expand table

Parameter	Description
`<domainname>`	Specifies new names for the default DNS domain and search list. The default domain name value is based on the host name. You can specify a maximum of six names separated by slashes (/).
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Remarks

- Use the nslookup set all command to display the list.

Examples

To set the DNS domain to *mfg.widgets.com* and the search list to the three names:

```
set srchlist=mfg.widgets.com/mrp2.widgets.com/widgets.com
```

Related links

- Command-Line Syntax Key
- nslookup set domain
- nslookup set all

Feedback

Was this page helpful? Yes No

nslookup set timeout

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Changes the number of seconds to wait for a reply to a lookup request. Use the nslookup set retry command to determine the number of times to try to send the request.

Syntax

```
set timeout=<number>
```

Parameters

Expand table

Parameter	Description
`<number>`	Specifies the number of seconds to wait for a reply. The default number of seconds to wait is **2**.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Examples

To set the timeout for getting a response to 5 seconds:

```
set timeout=5
```

Related links

- Command-Line Syntax Key
- nslookup set retry

Last updated on 10/30/2025

nslookup set type

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the resource record type for the query. For information about resource record types, see Request for Comment (Rfc) 1035.

> **Note**
>
> This command is the same as the nslookup set querytype command.

Syntax

```
set type=<resourcerecordtype>
```

Parameters

Expand table

Parameter	Description
`<resourcerecordtype>`	Specifies a DNS resource record type. The default resource record type is **A**, but you can use any of the following values: • **A:** Specifies a computer's IP address. • **ANY:** Specifies a computer's IP address. • **CNAME:** Specifies a canonical name for an alias. • **GID** Specifies a group identifier of a group name. • **HINFO:** Specifies a computer's CPU and type of operating system. • **MB:** Specifies a mailbox domain name. • **MG:** Specifies a mail group member. • **MINFO:** Specifies mailbox or mail list information. • **MR:** Specifies the mail rename domain name. • **MX:** Specifies the mail exchanger. • **NS:** Specifies a DNS name server for the named zone. • **PTR:** Specifies a computer name if the query is an IP address; otherwise, specifies the pointer to other information. • **SOA:** Specifies the start-of-authority for a DNS zone.

Parameter	Description
	• **TXT:** Specifies the text information. • **UID:** Specifies the user identifier. • **UINFO:** Specifies the user information. • **WKS:** Describes a well-known service.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup set type

Feedback

Was this page helpful? Yes No

nslookup set vc

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Specifies whether to use a virtual circuit when sending requests to the server.

Syntax

```
set [no]vc
```

Parameters

Expand table

Parameter	Description
novc	Specifies to never use a virtual circuit when sending requests to the server. This is the default value.
vc	Specifies to always use a virtual circuit when sending requests to the server.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

nslookup view

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sorts and lists the output of the previous **ls** commands or subcommands.

Syntax

```
view <filename>
```

Parameters

Expand table

Parameter	Description
`<filename>`	Specifies the name of the file containing output from the previous **ls** commands or subcommands.
/?	Displays help at the command prompt.
/help	Displays help at the command prompt.

Related links

- Command-Line Syntax Key
- nslookup ls

Feedback

Was this page helpful? Yes No

ntbackup

Article • 05/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The `ntbackup` command backs up and restores your computer and files from a command prompt. This command is replaced with the wbadmin command.

> **Important**
>
> The `wbadmin` command can't recover backups created by using the `ntbackup` commands. The Windows NT Backup - Restore utility is needed to recover from legacy backups.

Related links

- Command-Line Syntax Key
- wbadmin

Feedback

Was this page helpful? Yes No

ntcmdprompt

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Runs the command interpreter **Cmd.exe**, rather than **Command.com**, after running a Terminate and Stay Resident (TSR) or after starting the command prompt from within an MS-DOS application.

Syntax

```
ntcmdprompt
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- When **Command.com** is running, some features of **Cmd.exe**, such as the **doskey** display of command history, aren't available. If you would prefer to run the **Cmd.exe** command interpreter after you've started a Terminate and Stay Resident (TSR) or started the command prompt from within an application based on MS-DOS, you can use the **ntcmdprompt** command. However, keep in mind that the TSR may not be available for use when you are running **Cmd.exe**. You can include the **ntcmdprompt** command in your **Config.nt** file or the equivalent custom startup file in an application's program information file (Pif).

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

ntfrsutl

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Dumps the internal tables, thread, and memory information for the NT File Replication Service (NTFRS) from both the local and remote servers. The recovery setting for NTFRS in Service Control Manager (SCM) can be critical to locating and keeping important log events on the computer. This tool provides a convenient method of reviewing those settings.

Syntax

```
ntfrsutl[idtable|configtable|inlog|outlog][<computer>]
ntfrsutl[memory|threads|stage][<computer>]
ntfrsutl ds[<computer>]
ntfrsutl [sets][<computer>]
ntfrsutl [version][<computer>]
ntfrsutl poll[/quickly[=[<n>]]][/slowly[=[<n>]]][/now][<computer>]
```

Parameters

Expand table

Parameter	Description
idtable	Specifies the ID table.
configtable	Specifies the FRS configuration table.
inlog	Specifies the inbound log.
outlog	Specifies the outbound log.
`<computer>`	Specifies the computer.
memory	Specifies the memory usage.
threads	Specifies the memory usage.
stage	Specifies the memory usage.

Parameter	Description
ds	Lists the NTFRS service's view of the DS.
sets	Specifies the active replica sets.
version	Specifies the API and NTFRS service versions.
poll	Specifies the current polling intervals. • `/quickly` - Polls quickly until it retrieves a stable configuration. • `/quickly=` - Polls quickly every default number of minutes. • `/quickly=<n>` - Polls quickly every *n* minutes. • `/slowly` - Polls slowly until it retrieves a stable configuration. • `/slowly=` - Polls slowly every default number of minutes. • `/slowly=<n>` - Polls slowly every *n* minutes. • `/now` - Polls now.
/?	Displays help at the command prompt.

Examples

To determine the polling interval for file replication, type:

```
C:\Program Files\SupportTools>ntfrsutl poll wrkstn-1
```

To determine the current NTFRS application program interface (API) version, type:

```
C:\Program Files\SupportTools>ntfrsutl version
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

openfiles

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables an administrator to query, display, or disconnect files and directories that have been opened on a system. This command also enables or disables the system **Maintain Objects List** global flag.

openfiles /disconnect

Enables an administrator to disconnect files and folders that have been opened remotely through a shared folder.

Syntax

```
openfiles /disconnect [/s <system> [/u [<domain>\]<username> [/p
[<password>]]]] {[/id <openfileID>] | [/a <accessedby>] | [/o {read | write
| read/write}]} [/op <openfile>]
```

Parameters

Expand table

Parameter	Description
/s `<system>`	Specifies the remote system to connect to (by name or IP address). Don't use backslashes. If you don't use the **/s** option, the command is run on the local computer by default. This parameter applies to all files and folders that are specified in the command.
/u `[<domain>\]` `<username>`	Runs the command using the permissions of the specified user account. If you don't use the **/u** option, system permissions are used by default.
/p `[<password>]`	Specifies the password of the user account that is specified in the **/u** option. If you don't use the **/p** option, a password prompt appears when the command is run.

Parameter	Description
/id `<openfileID>`	Disconnects open files by the specified file ID. You can use the wildcard character (*) with this parameter. Note: You can use the **openfiles /query** command to find the file ID.
/a `<accessedby>`	Disconnects all open files associated with the user name specified in the *accessedby* parameter. You can use the wildcard character (*) with this parameter.
/o `{read | write | read/write}`	Disconnects all open files with the specified open mode value. Valid values are **Read**, **Write**, or **Read/Write**. You can use the wildcard character (*) with this parameter.
/op `<openfile>`	Disconnects all open file connections that are created by a specific open file name. You can use the wildcard character (*) with this parameter.
/?	Displays help at the command prompt.

Examples

To disconnect all open files with the *file ID 26843578*, type:

```
openfiles /disconnect /id 26843578
```

To disconnect all open files and directories accessed by the user *hiropln*, type:

```
openfiles /disconnect /a hiropln
```

To disconnect all open files and directories with *read/write mode*, type:

```
openfiles /disconnect /o read/write
```

To disconnect the directory with the open file name *C:\testshare*, regardless of who is accessing it, type:

```
```

```
openfiles /disconnect /a * /op c:\testshare\
```

To disconnect all open files on the remote computer *srvmain* that are being accessed by the user *hiropln*, regardless of their ID, type:

```
openfiles /disconnect /s srvmain /u maindom\hiropln /id *
```

openfiles /query

Queries and displays all open files.

Syntax

```
openfiles /query [/s <system> [/u [<domain>\]<username> [/p [<password>]]]]
[/fo {TABLE | LIST | CSV}] [/nh] [/v]
```

Parameters

Expand table

Parameter	Description
/s `<system>`	Specifies the remote system to connect to (by name or IP address). Don't use backslashes. If you don't use the **/s** option, the command is run on the local computer by default. This parameter applies to all files and folders that are specified in the command.
/u `[<domain>\]` `<username>`	Runs the command using the permissions of the specified user account. If you don't use the **/u** option, system permissions are used by default.
/p `[<password>]`	Specifies the password of the user account that is specified in the **/u** option. If you don't use the **/p** option, a password prompt appears when the command is run.
[/fo `{TABLE |` `LIST | CSV}`]	Displays the output in the specified format. Valid values include: • **TABLE** - Displays output in a table. • **LIST** - Displays output in a list. • **CSV** - Displays output in Comma Separated Values (CSV) format.

Parameter	Description
/nh	Suppresses column headers in the output. Valid only when the **/fo** parameter is set to **TABLE** or **CSV**.
/v	Specifies that detailed (verbose) information be displayed in the output.
/?	Displays help at the command prompt.

Examples

To query and display all open files, type:

```
openfiles /query
```

To query and display all open files in table format without headers, type:

```
openfiles /query /fo table /nh
```

To query and display all open files in list format with detailed information, type:

```
openfiles /query /fo list /v
```

To query and display all open files on the remote system *srvmain* by using the credentials for the user *hiropln* on the *maindom* domain, type:

```
openfiles /query /s srvmain /u maindom\hiropln /p p@ssW23
```

> **Note**
>
> In this example, the password is supplied on the command line. To prevent displaying the password, leave out the **/p** option. You'll be prompted for the password, which won't be echoed to the screen.

openfiles /local

Enables or disables the system **Maintain Objects List** global flag. If used without parameters, **openfiles /local** displays the current status of the **Maintain Objects List** global flag.

> **Note**
>
> Changes made by using the **on** or **off** option don't take effect until you restart the system. Enabling the **Maintain Objects List** global flag might slow down your system.

Syntax

```
openfiles /local [on | off]
```

Parameters

Expand table

Parameter	Description
`[on | off]`	Enables or disables the system **Maintain Objects List** global flag, which tracks local file handles.
/?	Displays help at the command prompt.

Examples

To check the current status of the **Maintain Objects List** global flag, type:

```
openfiles /local
```

By default, the **Maintain Objects List** global flag is disabled, and the following message appears, `INFO: The system global flag 'maintain objects list' is currently disabled.`

To enable the **Maintain Objects List** global flag, type:

```
openfiles /local on
```

The following message appears when the global flag is enabled, `SUCCESS: The system global flag 'maintain objects list' is enabled. This will take effect after the system is restarted.`

To disable the **Maintain Objects List** global flag, type:

```
openfiles /local off
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pagefileconfig

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

> **Important**
>
> This command has been deprecated.

Enables an administrator to display and configure a system's paging file Virtual Memory settings. For descriptions and usage information, see pagefileconfig.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

path

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the command path in the PATH environment variable, specifying the set of directories used to search for executable (.exe) files. If used without parameters, this command displays the current command path.

Syntax

```
path [[<drive>:]<path>[;...][;%PATH%]]
path ;
```

Parameters

Expand table

Parameter	Description
`[<drive>:]<path>`	Specifies the drive and directory to set in the command path. The current directory is always searched before the directories specified in the command path.
;	Separates directories in the command path. If used without other parameters, ; clears the existing command paths from the PATH environment variable and directs Cmd.exe to search only in the current directory.
`%PATH%`	Appends the command path to the existing set of directories listed in the PATH environment variable. If you include this parameter, Cmd.exe replaces it with the command path values found in the PATH environment variable, eliminating the need to manually enter these values at the command prompt.
/?	Displays help at the command prompt.

Remarks

- The Windows operating system searches using default file name extensions in the following order of precedence: .exe, .com, .bat, and .cmd. Which means if you're

looking for a batch file named, acct.bat, but have an app named acct.exe in the same directory, you must include the .bat extension at the command prompt.

- If two or more files in the command path have the same file name and extension, this command first searches for the specified file name in the current directory. Then, it searches the directories in the command path in the order that they're listed in the PATH environment variable.

- If you place the **path** command in your Autoexec.nt file, the Windows operating system automatically appends the specified MS-DOS subsystem search path every time you log on to your computer. Cmd.exe does not use the Autoexec.nt file. When started from a shortcut, Cmd.exe inherits the environment variables set in My Computer/Properties/Advanced/Environment.

Examples

To search the paths *c:\user\taxes*, *b:\user\invest*, and *b:\bin* for external commands, type:

```
path c:\user\taxes;b:\user\invest;b:\bin
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pathping

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Provides information about network latency and network loss at intermediate hops between a source and destination. This command sends multiple echo Request messages to each router between a source and destination, over a period of time, and then computes results based on the packets returned from each router. Because this command displays the degree of packet loss at any given router or link, you can determine which routers or subnets might be having network problems. Used without parameters, this command displays help.

> **Note**
>
> This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network Connections.
>
> Additionally, this command identifies which routers are on the path, same as using the tracert command. However, this command also sends pings periodically to all of the routers over a specified time period and computes statistics based on the number returned from each.

Syntax

```
pathping [/n] [/h <maximumhops>] [/g <hostlist>] [/p <Period>] [/q
<numqueries> [/w <timeout>] [/i <IPaddress>] [/4 <IPv4>] [/6 <IPv6>]
[<targetname>]
```

Parameters

Expand table

Parameter	Description
/n	Prevents **pathping** from attempting to resolve the IP addresses of intermediate

Parameter	Description
	routers to their names. This might expedite the display of **pathping** results.
/h `<maximumhops>`	Specifies the maximum number of hops in the path to search for the target (destination). The default is 30 hops.
/g `<hostlist>`	Specifies that the echo Request messages use the **Loose Source Route** option in the IP header with the set of intermediate destinations specified in *hostlist*. With loose source routing, successive intermediate destinations can be separated by one or multiple routers. The maximum number of addresses or names in the host list is **9**. The *hostlist* is a series of IP addresses (in dotted decimal notation) separated by spaces.
/p `<period>`	Specifies the number of milliseconds to wait between consecutive pings. The default is 250 milliseconds (1/4 second). This parameter sends individual pings to each intermediate hop. Because of this, the interval between two pings sent to the same hop is *period* multiplied by the number of hops.
/q `<numqueries>`	Specifies the number of echo Request messages sent to each router in the path. The default is 100 queries.
/w `<timeout>`	Specifies the number of milliseconds to wait for each reply. The default is 3000 milliseconds (3 seconds). This parameter sends multiple pings in parallel. Because of this, the amount of time specified in the *timeout* parameter isn't bounded by the amount of time specified in the *period* parameter for waiting between pings.
/i `<IPaddress>`	Specifies the source address.
/4 `<IPv4>`	Specifies that pathping uses IPv4 only.
/6 `<IPv6>`	Specifies that pathping uses IPv6 only.
`<targetname>`	Specifies the destination, which is identified either by IP address or host name.
/?	Displays help at the command prompt.

Remarks

- All parameters are case-sensitive.
- To avoid network congestion and to minimize the effects of burst losses, pings should be sent at a sufficiently slow pace.

Example of the pathping command output

```
```

```
D:\>pathping /n contoso1
Tracing route to contoso1 [10.54.1.196]
over a maximum of 30 hops:
  0  172.16.87.35
  1  172.16.87.218
  2  192.168.52.1
  3  192.168.80.1
  4  10.54.247.14
  5  10.54.1.196
computing statistics for 125 seconds...
            Source to Here   This Node/Link
Hop  RTT    Lost/Sent = Pct  Lost/Sent = Pct  address
  0                                           172.16.87.35
                                 0/ 100 =  0%   |
  1   41ms     0/ 100 =  0%     0/ 100 =  0%  172.16.87.218
                               13/ 100 = 13%   |
  2   22ms    16/ 100 = 16%     3/ 100 =  3%  192.168.52.1
                                 0/ 100 =  0%   |
  3   24ms    13/ 100 = 13%     0/ 100 =  0%  192.168.80.1
                                 0/ 100 =  0%   |
  4   21ms    14/ 100 = 14%     1/ 100 =  1%  10.54.247.14
                                 0/ 100 =  0%   |
  5   24ms    13/ 100 = 13%     0/ 100 =  0%  10.54.1.196
Trace complete.
```

When **pathping** is run, the first results list the path. Next, a busy message is displayed for approximately 90 seconds (the time varies by hop count). During this time, information is gathered from all routers previously listed and from the links between them. At the end of this period, the test results are displayed.

In the above sample report, the **This Node/Link**, **Lost/Sent** = **Pct** and **address** columns show that the link between *172.16.87.218* and *192.168.52.1* is dropping 13% of the packets. The routers at hops 2 and 4 are also dropping packets addressed to them, but this loss doesn't affect their ability to forward traffic that isn't addressed to them.

The loss rates displayed for the links, identified as a vertical bar (|) in the **address** column, indicate link congestion that is causing the loss of packets that are being forwarded on the path. The loss rates displayed for routers (identified by their IP addresses) indicate that these routers might be overloaded.

Related links

- Command-Line Syntax Key
- tracert command

Feedback

Was this page helpful? Yes No

pause

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Suspends the processing of a batch program, displaying the prompt, `Press any key to continue . . .`

Syntax

```
pause
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- If you press CTRL+C to stop a batch program, the following message appears, `Terminate batch job (Y/N)?`. If you press **Y** (for yes) in response to this message, the batch program ends and control returns to the operating system.
- You can insert the **pause** command before a section of the batch file that you might not want to process. When **pause** suspends processing of the batch program, you can press CTRL+C and then press **Y** to stop the batch program.

Examples

To create a batch program that prompts the user to change disks in one of the drives, type:

```
@echo off
:Begin
copy a:*.*
echo Put a new disk into Drive A
pause
goto begin
```

In this example, all the files on the disk in Drive A are copied to the current directory. After the message prompts you to put a new disk in Drive A, the **pause** command suspends processing so that you can change disks and then press any key to resume processing. This batch program runs in an endless loop—the **goto begin** command sends the command interpreter to the Begin label of the batch file.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pbadmin

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

> **Important**
>
> This command has been deprecated.

Administers phone books. Used without parameters, pbadmin starts Phone Book Administrator. For descriptions and usage information, see pbadmin.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pentnt

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

> **Important**
>
> This command has been deprecated.

Detects floating point division error (if present) in the Pentium chip, disables floating point hardware, and turns on floating point emulation. For descriptions and usage information, see pentnt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

perfmon

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Start Windows Reliability and Performance Monitor in a specific standalone mode.

Syntax

```
perfmon </res|report|rel|sys>
```

Parameters

Expand table

Parameter	Description
/res	Starts the Resource View.
/report	Starts the System Diagnostics Data Collector Set and displays a report of the results.
/rel	Starts the Reliability Monitor.
/sys	Starts the Performance Monitor.

Related links

- Command-Line Syntax Key
- Windows Performance Monitor

Feedback

Was this page helpful? Yes No

ping

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Verifies IP-level connectivity to another TCP/IP computer by sending Internet Control Message Protocol (ICMP) echo Request messages. The receipt of the corresponding echo Reply messages is displayed, along with round-trip times. ping is the primary TCP/IP command used to troubleshoot connectivity, reachability, and name resolution. Used without parameters, this command displays Help content.

You can also use this command to test both the computer name and the IP address of the computer. If pinging the IP address is successful, but pinging the computer name isn't, you might have a name resolution problem. In this case, make sure the computer name you're specifying can be resolved through the local Hosts file, by using Domain Name System (DNS) queries, or through NetBIOS name resolution techniques.

> **Note**
>
> This command is available only if the Internet Protocol (TCP/IP) is installed as a component in the properties of a network adapter in Network Connections.

Syntax

```
ping [/t] [/a] [/n <count>] [/l <size>] [/f] [/I <TTL>] [/v <TOS>] [/r
<count>] [/s <count>] [{/j <hostlist> | /k <hostlist>}] [/w <timeout>] [/R]
[/S <Srcaddr>] [/4] [/6] <targetname>
```

Parameters

Expand table

Parameter	Description
/t	Specifies ping continue sending echo Request messages to the destination until interrupted. To interrupt and display statistics, press CTRL+ENTER. To interrupt and quit this command, press CTRL+C.

Parameter	Description
/a	Specifies reverse name resolution be performed on the destination IP address. If this operation is successful, ping displays the corresponding host name.
/n `<count>`	Specifies the number of echo Request messages be sent. The default is 4.
/l `<size>`	Specifies the length, in bytes, of the **Data** field in the echo Request messages. The default is 32. The maximum size is 65,500.
/f	Specifies that echo Request messages are sent with the **Do not Fragment** flag in the IP header set to 1 (available on IPv4 only). The echo Request message can't be fragmented by routers in the path to the destination. This parameter is useful for troubleshooting path Maximum Transmission Unit (PMTU) problems.
/i `<TTL>`	Specifies the value of the Time To Live (TTL) field in the IP header for echo Request messages sent. The default is the default TTL value for the host. The maximum *TTL* is 255.
/v `<TOS>`	Specifies the value of the Type Of Service (TOS) field in the IP header for echo Request messages sent (available on IPv4 only). The default is 0. *TOS* is specified as a decimal value from 0 through 255.
/r `<count>`	Specifies the **Record Route** option in the IP header is used to record the path taken by the echo Request message and corresponding echo Reply message (available on IPv4 only). Each hop in the path uses an entry in the **Record Route** option. If possible, specify a *count* equal to or greater than the number of hops between the source and destination. The *count* must be a minimum of 1 and a maximum of 9.
/s `<count>`	Specifies that the **Internet timestamp** option in the IP header is used to record the time of arrival for the echo Request message and corresponding echo Reply message for each hop. The *count* must be a minimum of 1 and a maximum of 4. This parameter is required for link-local destination addresses.
/j `<hostlist>`	Specifies the echo Request messages use the **Loose Source Route** option in the IP header with the set of intermediate destinations specified in *hostlist* (available on IPv4 only). With loose source routing, successive intermediate destinations can be separated by one or multiple routers. The maximum number of addresses or names in the host list is 9. The host list is a series of IP addresses (in dotted decimal notation) separated by spaces.
/k `<hostlist>`	Specifies the echo Request messages use the **Strict Source Route** option in the IP header with the set of intermediate destinations specified in *hostlist* (available on IPv4 only). With strict source routing, the next intermediate destination must be directly reachable (it must be a neighbor on an interface of the router). The maximum number of addresses or names in the host list is 9. The host list is a series of IP addresses (in dotted decimal notation) separated by spaces.
/w `<timeout>`	Specifies the amount of time, in milliseconds, to wait for the echo Reply message corresponding to a given echo Request message. If the echo Reply message isn't

Parameter	Description
	received within the time-out, the "Request timed out" error message is displayed. The default time-out is 4000 (4 seconds).
/R	Specifies the round-trip path is traced (available on IPv6 only).
/S `<Srcaddr>`	Specifies the source address to use (available on IPv6 only).
/4	Specifies IPv4 used to ping. This parameter isn't required to identify the target host with an IPv4 address. It's only required to identify the target host by name.
/6	Specifies IPv6 used to ping. This parameter isn't required to identify the target host with an IPv6 address. It's only required to identify the target host by name.
`<targetname>`	Specifies the host name or IP address of the destination.
/?	Displays help at the command prompt.

Example of the ping command output

```
C:\>ping example.microsoft.com
    pinging example.microsoft.com [192.168.239.132] with 32 bytes of data:
    Reply from 192.168.239.132: bytes=32 time=101ms TTL=124
    Reply from 192.168.239.132: bytes=32 time=100ms TTL=124
    Reply from 192.168.239.132: bytes=32 time=120ms TTL=124
    Reply from 192.168.239.132: bytes=32 time=120ms TTL=124
```

Examples

To ping the destination 10.0.99.221 and resolve 10.0.99.221 to its host name, type:

```
ping /a 10.0.99.221
```

To ping the destination 10.0.99.221 with 10 echo Request messages, each of which has a Data field of 1000 bytes, type:

```
ping /n 10 /l 1000 10.0.99.221
```

To ping the destination 10.0.99.221 and record the route for 4 hops, type:

```
ping /r 4 10.0.99.221
```

To ping the destination 10.0.99.221 and specify the loose source route of 10.12.0.1-10.29.3.1-10.1.44.1, type:

```
ping /j 10.12.0.1 10.29.3.1 10.1.44.1 10.0.99.221
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pktmon

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Packet Monitor (Pktmon) is an in-box, cross-component network diagnostics tool for Windows. It can be used for advanced packet capture and event collection, drop detection, filtering, and counting. Pktmon is especially helpful in virtualization scenarios such as container networking and SDN, because it provides visibility within the networking stack.

Syntax

```
pktmon { filter | list | start | stop | status | unload | counters | reset |
etl2txt | etl2pcap | hex2pkt | help } [options]
```

Commands

Expand table

Command	Description
pktmon filter	Manage packet filters.
pktmon list	List packet processing components.
pktmon start	Start packet capture and event collection.
pktmon stop	Stop data collection.
pktmon status	Query current status.
pktmon unload	Unload PktMon driver.
pktmon counters	Display current packet counters.
pktmon reset	Reset packet counters to zero.
pktmon etl2txt	Convert log file to text format.
pktmon etl2pcap	Convert log file to pcapng format.

Command	Description
pktmon hex2pkt	Decode packet in hexadecimal format.
pktmon help	Show help text for specific command.

Related links

- Packet Monitor overview
- Pktmon support for Microsoft Network Monitor (Netmon)

Feedback

Was this page helpful? Yes No

pktmon counters

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Pktmon counters allows you to query and display current packet counters from monitored components to confirm the presence of expected traffic and get a high-level view of how the traffic flowed in the machine.

Syntax

```
pktmon counters [--type { all | flow | drop }] [--include-hidden] [--zero]
[--drop-reason] [--live] [--refresh-rate <n>] [--json]
```

Parameters

Expand table

Parameter	Description
-t, --type	Select which types of counters to show. Supported values are **all** counters (default), **flow** (flows only), or **drop** (drops only).
-z, --zero	Show counters that are zero in both directions.
-i, --include-hidden	Show counters from components that are hidden by default.
-r, --drop-reason	Show the most recent drop reason for each drop counter.
--live	Automatically refresh the counters. Press **Ctrl+C** to stop.
--refresh-rate <n>	Number of times to refresh the counters per second, from 1 to 30. Default is 10.
--json	Output the counters in JSON format. Implies -i and -r.

Related links

- Pktmon
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon etl2pcap

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Convert pktmon log file to pcapng format. Dropped packets are not included by default. These logs can be analyzed using Wireshark (or any pcapng analyzer).

Syntax

```
pktmon etl2pcap <file> [--out <name>] [--drop-only] [--component-id <id>]
```

Where `<file>` is the ETL file to convert.

Parameters

Expand table

Parameter	Description
-o, --out <name>	Name of the formatted pcapng file.
-d, --drop-only	Convert dropped packets only.
-c, --component-id <id>	Filter packets by a specific component ID.

Output filtering

All information about the packet drop reports and packet flow through the networking stack is lost in pcapng format output. Log contents should be carefully prefiltered to show the complete conversion. For example:

- Pcapng format doesn't distinguish between a flowing packet and a dropped packet. To separate all the packets in the capture from dropped packets, generate two pcapng files; one that contains all the packets (`pktmon etl2pcap log.etl --out log-capture.etl`), and another that contains only dropped packets (`pktmon`

`etl2pcap log.etl --drop-only --out log-drop.etl`). This way you're able to analyze the dropped packets in a separate log.

- Pcapng format doesn't distinguish between different networking components where a packet was captured. For such multilayered scenarios, specify the desired component ID in the pcapng output `pktmon etl2pcap log.etl --component-id 5`. Repeat this command for each set of component IDs that you're interested in.

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon etl2txt

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Convert ETL log file to text format.

Syntax

```
pktmon etl2txt <file> [--out <name>] [--stats-only] [--timestamp-only] [--
metadata]
                      [--tmfpath <path>] [--brief] [--verbose <n>] [--hex]
                      [--no-ethernet] [--vxlan <port>]
```

Where `<file>` is the ETL file to convert.

Parameters

Expand table

Parameter	Description
-o, --out <name>	Name of the formatted text file.
-s, --stats-only	Display log file statistical information.
-t, --timestamp-only	Use timestamp only prefix for events and packets.
-m, --metadata	Print event metadata, such as logging level and keywords.
-p, --tmfpath <path>	Path to TMF files for decoding WPP traces. Multiple paths should be separated by semicolons. All WPP traces are skipped when this option is not specified.

Network packet formatting options

Expand table

Parameter	Description
-b, --brief	Use abbreviated packet format.
-v, --verbose <n>	Verbosity level from 1 to 3.
-x, --hex	Include hexadecimal format.
-e, --no-ethernet	Don't print ethernet header.
-l, --vxlan <port>	Custom VXLAN port.

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon filter

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Pktmon filter allows you to list, add, or remove packet filters.

Syntax

```
pktmon filter { list | add | remove } [OPTIONS | help]
```

Parameters

Expand table

Parameter	Description
pktmon filter list	Display active packet filters.
pktmon filter add	Add a filter to control which packets are reported.
pktmon filter remove	Remove all packet filters.

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon filter add

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Pktmon filter add allows you to add a filter to control which packets are reported. For a packet to be reported, it must match all conditions specified in at least one filter. Up to 32 filters can be active at once.

Syntax

```
pktmon filter add <name> [-m <mac> [mac2]] [-v <vlan>] [-d { IPv4 | IPv6 |
number }]
                         [-t { TCP [flags...] | UDP | ICMP | ICMPv6 | number
}]
                         [-i <ip> [ip2]] [-p <port> [port2]] [-b] [-e
[port]]
```

You may provide an optional name or description of the filter.

> **Note**
>
> When two MACs (-m), IPs (-i), or ports (-p) are specified, the filter matches packets that contain both. It will not distinguish between source or destination for this purpose.

Parameters

You can supply parameters for Ethernet frame, IP header, TCP/UDP header, cluster heartbeat, and encapsulation.

Expand table

Parameter	Description
-m, --mac[-address]	Match source or destination MAC address. See note above.

Parameter	Description
-v, --vlan	Match by VLAN ID (VID) in the 802.1Q header.
-d, --data-link[-protocol], --ethertype	Match by data link (layer 2) protocol. Can be IPv4, IPv6, ARP, or a protocol number.
-t, --transport[-protocol], --ip-protocol	Match by transport (layer 4) protocol. Can be TCP, UDP, ICMP, ICMPv6, or a protocol number. To further filter TCP packets, an optional list of TCP flags to match can be provided. Supported flags are FIN, SYN, RST, PSH, ACK, URG, ECE, and CWR.
-i, --ip[-address]	Match source or destination IP address. See note above. To match by subnet, use CIDR notation with the prefix length.
-p, --port	Match source or destination port number. See note above.
-b, --heartbeat	Match RCP heartbeat messages over UDP port 3343.
-e, --encap	Apply above filtering parameters to both inner and outer encapsulation headers. Supported encapsulation methods are VXLAN, GRE, NVGRE, and IP-in-IP. Custom VXLAN port is optional, and defaults to 4789.

Examples

The following set of filters will capture any ICMP traffic from or to the IP address 10.0.0.10 along with any traffic on port 53.

PowerShell

```
C:\Test> pktmon filter add -i 10.0.0.10 -t icmp
C:\Test> pktmon filter add -p 53
```

The following filter will capture all the SYN packets sent or received by the IP address 10.0.0.10:

PowerShell

```
C:\Test> pktmon filter add -i 10.0.0.10 -t tcp syn
```

The following filter called **MyPing** pings 10.10.10.10 using the ICMP protocol:

PowerShell

```
C:\Test> pktmon filter add MyPing -i 10.10.10.10 -t ICMP
```

The following filter called **MySmbSyb** captures TCP synchronized SMB traffic:

PowerShell

```
C:\Test> pktmon filter add MySmbSyn -i 10.10.10.10 -t TCP SYN -p 445
```

The following filter called **MySubnet** captures traffic on the subnet mask 255.255.255.0, or /24 in CIDR notation:

PowerShell

```
C:\Test> pktmon filter add MySubnet -i 10.10.10.0/24
```

Other references

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon hex2pkt

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Decode packet in hexadecimal format.

Syntax

```
pktmon hex2pkt [--type { Ethernet | IP | HTTP }]
```

Parameters

Expand table

Parameter	Description
-t, --type	Packet type to decode. Options are Ethernet, IP, and HTTP. Default is Ethernet.

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon list

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists all active networking components that can be monitored, allowing you to examine the networking stack layout. The command shows networking components (drivers) arranged by adapters bindings.

Syntax

```
pktmon list [--all] [--include-hidden] [--json]
```

Parameters

Expand table

Parameter	Description
-a, --all	Show all component types. Only network adapters are displayed by default.
-i, --include-hidden	Show components that are hidden by default.
--json	Output the list in JSON format. Implies -i and -a.

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon reset
- Pktmon start
- Pktmon status

- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon reset

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Reset counters to zero.

Syntax

```
pktmon reset [-counters]
```

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon start
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon start

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts packet capture and event collection.

Syntax

```
pktmon start [--capture [--counters-only] [--comp <selector>] [--type
<type>] [--pkt-size <bytes>] [--flags <mask>]]
             [--trace --provider <name> [--keywords <k>] [--level <n>] ...]
             [--file-name <name>] [--file-size <size>] [--log-mode <mode>]
```

Packet capture parameters

Use **-c** or **--capture** to enable packet capture and packet counters, along with the following optional parameters.

Expand table

Parameter	Description
-o, --counters-only	Collect packet counters only. No packet logging.
--comp	Select components to capture packets on. Can be all components (all), NICs only (nics), or a list of component IDs. Defaults to all.
--type	Select which packets to capture. Can be all, flow, or drop. Default is all.
--pkt-size <bytes>	Number of bytes to log from each packet. To always log the entire packet, set this to 0. Default is 128 bytes.
--flags <mask>	Hexadecimal bitmask that controls information logged during packet capture. Default is 0x012. Packet capture flags, below.

Packet capture flags

The following flags apply to the **--flags** parameter (see above).

Expand table

Flag	Description
0x001	Internal Packet Monitor errors.
0x002	Information about components, counters, and filters. This information is added to the end of the log file.
0x004	Source and destination information for the first packet in NET_BUFFER_LIST group.
0x008	Select packet metadata from NDIS_NET_BUFFER_LIST_INFO enumeration.
0x010	Raw packet, truncated to the size specified in the [--pkt-size] parameter.

Event collection parameters

Use **-t** or **--trace** to enable event collection, along with the following optional parameters.

Expand table

Parameter	Description
-p, --provider <name>	Event provider name or GUID. For multiple providers, use this parameter more than once.
-k, --keywords <k>	Hexadecimal bitmask that controls which events are logged for the corresponding provider. Default is 0xFFFFFFFF.
-l, --level <n>	Logging level for the corresponding provider. Default is 4 (info level).

Logging parameters

Use the following parameters for logging:

Expand table

Parameter	Description
-f, --file-name <name>	Log file name. Default is PktMon.etl.
-s, --file-size <size>	Maximum log file size in megabytes. Default is 512 MB.
-m, --log-mode	Sets the logging mode (see below). Default is circular.

Logging modes

The following modes apply to the **-m** or **--log-mode** parameter (see above).

Expand table

Mode	Description
circular	New events overwrite the oldest ones when the log is full.
multi-file	A new log file is created each time the log is full. Log files are sequentially numbered: PktMon1.etl, PktMon2.etl, etc. No limited on the number of captured events.
real-time	Display events and packets on screen at real time. No log file is created. Press **Ctrl+C** to stop monitoring.
memory	Like circular, but the entire log is stored in memory. It is written to a file when pktmon is stopped. Memory buffer size is specified in [--file-size] parameter.

Examples

Example 1: Packet capture

PowerShell

```
C:\Test> pktmon start --capture
```

Example 2: Packet counters only

PowerShell

```
C:\Test> pktmon start --capture --counters-only
```

Example 3: Event logging

PowerShell

```
C:\Test> pktmon start --trace -p Microsoft-Windows-TCPIP -p Microsoft-
Windows-NDIS
```

Example 4: Packet capture with event logging

PowerShell

```
C:\Test> pktmon start --capture --trace -p Microsoft-Windows-TCPIP -k 0xFF -
```

```
1 4
```

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon status
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon status

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Query current Packet Monitor status.

Syntax

```
pktmon status [--buffer-info]
```

Parameters

Expand table

Parameter	Description
-b, --buffer-info	Display ETW buffer information.

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon unload
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pktmon unload

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Stop the PktMon driver service and unload PktMon.sys. Effectively equivalent to 'sc.exe stop PktMon'. Measurement (if active) will immediately stop, and any state will be deleted (counters, filters, etc.).

Syntax

```
pktmon unload
```

Related links

- Pktmon
- Pktmon counters
- Pktmon etl2pcap
- Pktmon etl2txt
- Pktmon filter
- Pktmon filter add
- Pktmon hex2pkt
- Pktmon list
- Pktmon reset
- Pktmon start
- Pktmon status
- Packet Monitor overview

Feedback

Was this page helpful? Yes No

pnpunattend

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Audits a computer for device drivers, and perform unattended driver installations, or search for drivers without installing and, optionally, report the results to the command line. Use this command to specify the installation of specific drivers for specific hardware devices.

Prerequisites

Preliminary preparation is required for older versions of the Windows operating system. Prior to using this command, you must complete the following tasks:

1. Create a directory for the drivers you want to install. For example, create a folder at **C:\Drivers\Video** for video adapter drivers.

2. Download and extract the driver package for your device. Copy the contents of the subfolder that contains the INF file for your version of the operating system and any subfolders to the video folder that you created. For example, copy the video driver files to **C:\Drivers\Video**.

3. Add a system environment path variable to the folder you created in step 1.For example, **C:\Drivers\Video**.

4. Create the following registry key, and then for the **DriverPaths** key you create, set the **Value Data** to **1**.

5. For Windows® 7 navigate the registry path: **HKEY_LOCAL_Machine\Software\Microsoft\Windows NT\CurrentVersion**, and then create the keys: **UnattendSettings\PnPUnattend\DriverPaths**

Syntax

```
PnPUnattend.exe auditsystem [/help] [/?] [/h] [/s] [/l]
```

Parameters

Expand table

Parameter	Description
auditsystem	Specifies online driver install. Required, except when this command is run with either the **/help** or **/?** parameters.
/s	Optional. Specifies to search for drivers without installing.
/l	Optional. Specifies to display the log information for this command in the command prompt.
`/? | /help`	Optional. Displays help for this command at the command prompt.

Examples

To command shows how to use the **PNPUnattend.exe** to audit a computer for possible driver updates, and then report the findings to the command prompt, type:

```
pnpunattend auditsystem /s /l
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pnputil

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Pnputil.exe is a command line utility that you can use to manage the driver store. You can use this command to add driver packages, remove driver packages, and list driver packages that are in the store.

Syntax

```
pnputil.exe [-f | -i] [ -? | -a | -d | -e ] <INF name>
```

Parameters

Expand table

Parameter	Description
-a	Specifies to add the identified INF file.
-d	Specifies to delete the identified INF file.
-e	Specifies to enumerate all third-party INF files.
-f	Specifies to force the deletion of the identified INF file. Can't be used in conjunction with the **–i** parameter.
-i	Specifies to install the identified INF file. Can't be used in conjunction with the **-f** parameter.
/?	Displays help at the command prompt.

Examples

To add an INF file, named USBCAM.INF, type:

```
```

```
pnputil.exe -a a:\usbcam\USBCAM.INF
```

To add all INF files, located in c:\drivers, type:

```
pnputil.exe -a c:\drivers\*.inf
```

To add and install the USBCAM.INF driver, type:

```
pnputil.exe -i -a a:\usbcam\USBCAM.INF
```

To enumerate all third-party drivers, type:

```
pnputil.exe –e
```

To delete the INF file and driver named oem0.inf, type:

```
pnputil.exe -d oem0.inf
```

Related links

- Command-Line Syntax Key
- popd command

Feedback

Was this page helpful? Yes No

popd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

The **popd** command changes the current directory to the directory that was most recently stored by the **pushd** command.

Every time you use the **pushd** command, a single directory is stored for your use. However, you can store multiple directories by using the **pushd** command multiple times. The directories are stored sequentially in a virtual stack, so if you use the **pushd** command once, the directory in which you use the command is placed at the bottom of the stack. If you use the command again, the second directory is placed on top of the first one. The process repeats every time you use the **pushd** command.

If you use the **popd** command, the directory on the top of the stack is removed and the current directory is changed to that directory. If you use the **popd** command again, the next directory on the stack is removed. If command extensions are enabled, the **popd** command removes any drive-letter assignments created by the **pushd** command.

Syntax

```
popd
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Examples

To change the current directory from the one in which the batch program was run, and then to change it back, type:

```
@echo off
rem This batch file deletes all .txt files in a specified directory
pushd %1
del *.txt
popd
cls
echo All text files deleted in the %1 directory
```

Related links

- Command-Line Syntax Key
- pushd

Feedback

Was this page helpful? Yes No

PowerShell

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Windows PowerShell is a task-based command-line shell and scripting language designed especially for system administration. Built on the .NET Framework, Windows PowerShell helps IT professionals and power users control and automate the administration of the Windows operating system and applications that run on Windows.

Using PowerShell.exe

The **PowerShell.exe** command-line tool starts a Windows PowerShell session in a Command Prompt window. When you use **PowerShell.exe**, you can use its optional parameters to customize the session. For example, you can start a session that uses a particular execution policy or one that excludes a Windows PowerShell profile. Otherwise, the session is the same as any session that is started in the Windows PowerShell console.

- To start a Windows PowerShell session in a Command Prompt window, type `PowerShell`. A **PS** prefix is added to the command prompt to indicate that you are in a Windows PowerShell session.
- To start a session with a particular execution policy, use the **ExecutionPolicy** parameter, and type:

  PowerShell

  ```
  PowerShell.exe -ExecutionPolicy Restricted
  ```

- To start a Windows PowerShell session without your Windows PowerShell profiles, use the **NoProfile** parameter, and type:

  PowerShell

  ```
  PowerShell.exe -NoProfile
  ```

- To start a session , use the **ExecutionPolicy** parameter, and type:

  PowerShell

```
PowerShell.exe -ExecutionPolicy Restricted
```

- To see the PowerShell.exe help file, type:

```PowerShell
PowerShell.exe -help
PowerShell.exe -?
PowerShell.exe /?
```

- To end a Windows PowerShell session in a Command Prompt window, type `exit`. The typical command prompt returns.

Remarks

- For a complete list of the **PowerShell.exe** command-line parameters, see about_PowerShell.Exe.
- For information about other ways to start Windows PowerShell, see Starting Windows PowerShell.
- Windows PowerShell runs on the Server Core installation option of Windows Server operating systems. However, features that require a graphic user interface, such as the Windows PowerShell Integrated Scripting Environment (ISE), and the Out-GridView and Show-Command cmdlets, don't run on Server Core installations.

Related links

- about_PowerShell.Exe
- about_PowerShell_Ise.exe
- Windows PowerShell

Feedback

Was this page helpful? Yes No

PowerShell_ise

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Windows PowerShell Integrated Scripting Environment (ISE) is a graphical host application that enables you to read, write, run, debug, and test scripts and modules in a graphic-assisted environment. Key features such as IntelliSense, Show-Command, snippets, tab completion, syntax-coloring, visual debugging, and context-sensitive Help provide a rich scripting experience.

Using PowerShell.exe

The **PowerShell_ISE.exe** tool starts a Windows PowerShell ISE session. When you use **PowerShell_ISE.exe**, you can use its optional parameters to open files in Windows PowerShell ISE or to start a Windows PowerShell ISE session with no profile or with a multithreaded apartment.

- To start a Windows PowerShell ISE session in a Command Prompt window, in Windows PowerShell, or at the **Start** menu, type:

  ```
  PowerShell
  ```

  ```
  PowerShell_Ise.exe
  ```

- To open a script (.ps1), script module (.psm1), module manifest (.psd1), XML file, or any other supported file in Windows PowerShell ISE, type:

  ```
  PowerShell
  ```

  ```
  PowerShell_Ise.exe <filepath>
  ```

  In Windows PowerShell 3.0, you can use the optional **File** parameter as follows:

  ```
  PowerShell
  ```

  ```
  PowerShell_Ise.exe -file <filepath>
  ```

- To start a Windows PowerShell ISE session without your Windows PowerShell profiles, use the **NoProfile** parameter. (The **NoProfile** parameter is introduced in

Windows PowerShell 3.0.), type:

```PowerShell
PowerShell_Ise.exe -NoProfile
```

- To see the PowerShell_ISE.exe help file, type:

```PowerShell
PowerShell_Ise.exe -help
PowerShell_Ise.exe -?
PowerShell_Ise.exe /?
```

Remarks

- For a complete list of the **PowerShell_ISE.exe** command-line parameters, see about_PowerShell_Ise.Exe.
- For information about other ways to start Windows PowerShell, see Starting Windows PowerShell.
- Windows PowerShell runs on the Server Core installation option of Windows Server operating systems. However, because Windows PowerShell ISE requires a graphic user interface, it does not run on Server Core installations.

Related links

- about_PowerShell_Ise.exe

Feedback

Was this page helpful? Yes No

print

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sends a text file to a printer. A file can print in the background if you send it to a printer connected to a serial or parallel port on the local computer.

> **Note**
>
> You can perform many configuration tasks from the command prompt by using the Mode command, including configuring a printer connected to a parallel or a serial port, displaying printer status, or preparing a printer for code page switching.

Syntax

```
print [/d:<printername>] [<drive>:][<path>]<filename>[ ...]
```

Parameters

Expand table

Parameter	Description
/d: `<printername>`	Specifies the printer that you want to print the job. To print to a locally connected printer, specify the port on your computer where the printer is connected. Valid values for parallel ports are **LPT1**, **LPT2**, and **LPT3**. Valid values for serial ports are **COM1**, **COM2**, **COM3**, and **COM4**. You can also specify a network printer by using its queue name (`\\server_name\printer_name`). If you don't specify a printer, the print job is sent to **LPT1** by default.
`<drive>`:	Specifies the logical or physical drive where the file you want to print is located. This parameter isn't required if the file you want to print is located on the current drive.
`<path>`	Specifies the location of the file you want to print. This parameter isn't required if the file you want to print is located in the current directory.

Parameter	Description
`<filename>[ ...]`	Required. Specifies the file you want to print. You can include multiple files in one command.
/?	Displays help at the command prompt.

Examples

To send the **report.txt** file, located in the current directory, to a printer connected to **lpt2** on the local computer, type:

```
print /d:lpt2 report.txt
```

To send the **report.txt** file, located in the **c:\accounting** directory, to the **printer1** print queue on the **/d:\copyroom** server, type:

```
print /d:\\copyroom\printer1 c:\accounting\report.txt
```

Related links

- Command-Line Syntax Key
- Print Command Reference
- Mode command

Feedback

Was this page helpful? Yes No

prncnfg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Configures or displays configuration information about a printer. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the prncnfg file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\prncnfg`.

Syntax

```
cscript prncnfg {-g | -t | -x | -?} [-S <Servername>] [-P <Printername>] [-z
<newprintername>] [-u <Username>] [-w <password>] [-r <portname>] [-l
<location>] [-h <sharename>] [-m <comment>] [-f <separatorfilename>] [-y
<datatype>] [-st <starttime>] [-ut <untiltime>] [-i <defaultpriority>] [-o
<priority>] [<+|->shared] [<+|->direct] [<+|->hidden] [<+|->published] [<+|-
>rawonly] [<+|->queued] [<+|->enablebidi] [<+|->keepprintedjobs] [<+|-
>workoffline] [<+|->enabledevq] [<+|->docompletefirst]
```

Parameters

Expand table

Parameter	Description
-g	Displays configuration information about a printer.
-t	Configures a printer.
-x	Renames a printer.
-S `<Servername>`	Specifies the name of the remote computer that hosts the printer that you want to manage. If you don't specify a computer, the local computer is used.
-P `<Printername>`	Specifies the name of the printer that you want to manage. Required.
-z `<newprintername>`	Specifies the new printer name. Requires the **-x** and **-P** parameters.

Parameter	Description
-u `<Username>` -w `<password>`	Specifies an account with permissions to connect to the computer that hosts the printer that you want to manage. All members of the target computer's local Administrators group have these permissions, but the permissions can also be granted to other users. If you don't specify an account, you must be logged on under an account with these permissions for the command to work.
-r `<portname>`	Specifies the port where the printer is connected. If this is a parallel or a serial port, then use the ID of the port (for example, LPT1 or COM1). If this is a TCP/IP port, use the port name that was specified when the port was added.
-l `<location>`	Specifies the printer location, such as **Copyroom**. If the location contains spaces, use quotation marks around the text, such as "**Copy Room**".
-h `<sharename>`	Specifies the printer's share name.
-m `<comment>`	Specifies the printer's comment string.
-f `<separatorfilename>`	Specifies a file that contains the text that appears on the separator page.
-y `<datatype>`	Specifies the data types that the printer can accept.
-st `<starttime>`	Configures the printer for limited availability. Specifies the time of day the printer is available. If you send a document to a printer when it is unavailable, the document is held (spooled) until the printer becomes available. You must specify time as a 24-hour clock. For example, to specify 11:00 P.M., type **2300**.
-ut `<endtime>`	Configures the printer for limited availability. Specifies the time of day the printer is no longer available. If you send a document to a printer when it is unavailable, the document is held (spooled) until the printer becomes available. You must specify time as a 24-hour clock. For example, to specify 11:00 P.M., type **2300**.
-o `<priority>`	Specifies a priority that the spooler uses to route print jobs into the print queue. A print queue with a higher priority receives all its jobs before any queue with a lower priority.
-i `<defaultpriority>`	Specifies the default priority assigned to each print job.
`{+|-}`shared	Specifies whether this printer is shared on the network.
`{+|-}`direct	Specifies whether the document should be sent directly to the printer without being spooled.
`{+|-}`published	Specifies whether this printer should be published in active directory. If you publish the printer, other users can search for it based on its

Parameter	Description
	location and capabilities (such as color printing and stapling).
{+|-}hidden	Reserved function.
{+|-}rawonly	Specifies whether only raw data print jobs can be spooled in this queue.
{+|-}}queued	Specifies that the printer should not begin to print until after the last page of the document is spooled. The printing program is unavailable until the document has finished printing. However, using this parameter ensures that the whole document is available to the printer.
{+|-}keepprintedjobs	Specifies whether the spooler should retain documents after they are printed. Enabling this option allows a user to resubmit a document to the printer from the print queue instead of from the printing program.
{+|-}workoffline	Specifies whether a user is able to send print jobs to the print queue if the computer is not connected to the network.
{+|-}enabledevq	Specifies whether print jobs that don't match the printer setup (for example, PostScript files spooled to non-PostScript printers) should be held in the queue rather than being printed.
{+|-}docompletefirst	Specifies whether the spooler should send print jobs with a lower priority that have completed spooling before sending print jobs with a higher priority that have not completed spooling. If this option is enabled and no documents have completed spooling, the spooler will send larger documents before smaller ones. You should enable this option if you want to maximize printer efficiency at the cost of job priority. If this option is disabled, the spooler always sends higher priority jobs to their respective queues first.
{+|-}enablebidi	Specifies whether the printer sends status information to the spooler.
/?	Displays help at the command prompt.

Examples

To display configuration information for the printer named *colorprinter_2* with a print queue hosted by the remote computer named *HRServer*, type:

```
cscript prncnfg -g -S HRServer -P colorprinter_2
```

To configure a printer named *colorprinter_2* so that the spooler in the remote computer named *HRServer* keeps print jobs after they have been printed, type:

```
cscript prncnfg -t -S HRServer -P colorprinter_2 +keepprintedjobs
```

To change the name of a printer on the remote computer named *HRServer* from *colorprinter_2* to *colorprinter 3*, type:

```
cscript prncnfg -x -S HRServer -P colorprinter_2 -z "colorprinter 3"
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

prndrvr

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds, deletes, and lists printer drivers. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the prndrvr file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\prndrvr`.

Used without parameters, **prndrvr** displays command-line help.

Syntax

```
cscript prndrvr {-a | -d | -l | -x | -?} [-m <model>] [-v {0|1|2|3}] [-e <environment>] [-s <Servername>] [-u <Username>] [-w <password>] [-h <path>] [-i <inf file>]
```

Parameters

Expand table

Parameter	Description
-a	Installs a driver.
-d	Deletes a driver.
-l	Lists all printer drivers installed on the server specified by the **-s** parameter. If you don't specify a server, Windows lists the printer drivers installed on the local computer.
-x	Deletes all printer drivers and additional printer drivers not in use by a logical printer on the server specified by the **-s** parameter. If you don't specify a server to remove from the list, Windows deletes all unused printer drivers on the local computer.
-m `<model_name>`	Specifies (by name) the driver you want to install. Drivers are often named for the model of printer they support. See the printer documentation for more

Parameter	Description
	information.
`-v {0|1|2|3}`	Specifies the version of the driver you want to install. See the description of the **-e**parameter for information on which versions are available for which environment. If you don't specify a version, the version of the driver appropriate for the version of Windows running on the computer where you are installing the driver is installed.
-e `<environment>`	Specifies the environment for the driver you want to install. If you don't specify an environment, the environment of the computer where you are installing the driver is used. The supported environment parameters are: **Windows NT x86**, **Windows x64** or **Windows IA64**.
-s `<Servername>`	Specifies the name of the remote computer that hosts the printer that you want to manage. If you don't specify a computer, the local computer is used.
-u `<Username>` -w `<password>`	Specifies an account with permissions to connect to the computer that hosts the printer that you want to manage. All members of the target computer's local Administrators group have these permissions, but the permissions can also be granted to other users. If you don't specify an account, you must be logged on under an account with these permissions for the command to work.
-h `<path>`	Specifies the path to the driver file. If you don't specify a path, the path to the location where Windows was installed is used.
-i `<filename.inf>`	Specifies the complete path and file name for the driver you want to install. If you don't specify a file name, the script uses one of the inbox printer .inf files in the inf subdirectory of the Windows directory. if the driver path is not specified, the script searches for driver files in the driver.cab file.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "Computer Name").
- The **-x** parameter deletes all additional printer drivers (drivers installed for use on clients running alternate versions of Windows), even if the primary driver is in use. If the fax component is installed, this option also deletes fax drivers. The primary fax driver is deleted if it is not in use (that is, if there is no queue using it). If the primary fax driver is deleted, the only way to re-enable fax is to reinstall the fax component.

Examples

To list all drivers on the local \printServer1 server, type:

```
cscript prndrvr -l -s
```

To add a version 3 Windows x64 printer driver for the Laser printer model 1 model of printer using the c:\temp\Laserprinter1.inf driver information file for a driver stored in the c:\temp folder, type:

```
cscript prndrvr -a -m Laser printer model 1 -v 3 -e Windows x64 -i
c:\temp\Laserprinter1.inf -h c:\temp
```

To delete a version 3 Windows x64 printer driver for Laser printer model 1, type:

```
cscript prndrvr -a -m Laser printer model 1 -v 3 -e Windows x64
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

prnjobs

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Pauses, resumes, cancels, and lists print jobs. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the prnjobs file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\prnjobs.vbs`.

Syntax

```
cscript prnjobs {-z | -m | -x | -l | -?} [-s <Servername>] [-p
<Printername>] [-j <JobID>] [-u <Username>] [-w <password>]
```

Parameters

Expand table

Parameter	Description
-z	Pauses the print job specified by the **-j** parameter.
-m	Resumes the print job specified by the **-j** parameter.
-x	Cancels the print job specified by the **-j** parameter.
-l	Lists all the print jobs in a print queue.
-s `<Servername>`	Specifies the name of the remote computer that hosts the printer that you want to manage. If you don't specify a computer, the local computer is used.
-p `<Printername>`	Required. Specifies the name of the printer that you want to manage.
-j `<JobID>`	Specifies (by ID number) the print job you want to cancel.
-u `<Username>` -w `<password>`	Specifies an account with permissions to connect to the computer that hosts the printer that you want to manage. All members of the target computer's local Administrators group have these permissions, but the permissions can

Parameter	Description
	also be granted to other users. If you don't specify an account, you must be logged on under an account with these permissions for the command to work.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "Computer Name").

Examples

To pause a print job with a job ID of 27 sent to the remote computer named HRServer for printing on the printer named colorprinter, type:

```
cscript prnjobs.vbs -z -s HRServer -p colorprinter -j 27
```

To list all current print jobs in the queue for the local printer named colorprinter_2, type:

```
cscript prnjobs.vbs -l -p colorprinter_2
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

prnmngr

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds, deletes, and lists printers or printer connections, in addition to setting and displaying the default printer. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the prnmngr file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\prnmngr`.

Syntax

```
cscript prnmngr {-a | -d | -x | -g | -t | -l | -?}[c] [-s <Servername>] [-p
<Printername>] [-m <printermodel>] [-r <portname>] [-u <Username>]
[-w <password>]
```

Parameters

Expand table

Parameter	Description
-a	Adds a local printer connection.
-d	Deletes a printer connection.
-x	Deletes all printers from the server specified by the **-s** parameter. If you don't specify a server, Windows deletes all printers on the local computer.
-g	Displays the default printer.
-t	Sets the default printer to the printer specified by the **-p** parameter.
-l	Lists all printers installed on the server specified by the **-s** parameter. If you don't specify a server, Windows lists the printers installed on the local computer.

Parameter	Description
c	Specifies that the parameter applies to printer connections. Can be used with the **-a** and **-x** parameters.
-s `<Servername>`	Specifies the name of the remote computer that hosts the printer that you want to manage. If you don't specify a computer, the local computer is used.
-p `<Printername>`	Specifies the name of the printer that you want to manage.
-m `<Modelname>`	Specifies (by name) the driver you want to install. Drivers are often named for the model of printer they support. See the printer documentation for more information.
-r `<portname>`	Specifies the port where the printer is connected. If this is a parallel or a serial port, use the ID of the port (for example, LPT1: or COM1:). If this is a TCP/IP port, use the port name that was specified when the port was added.
-u `<Username>` -w `<password>`	Specifies an account with permissions to connect to the computer that hosts the printer that you want to manage. All members of the target computer's local Administrators group have these permissions, but the permissions can also be granted to other users. If you don't specify an account, you must be logged on under an account with these permissions for the command to work.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "Computer Name").

Examples

To add a printer named colorprinter_2 that is connected to LPT1 on the local computer and requires a printer driver called color printer Driver1, type:

```
cscript prnmngr -a -p colorprinter_2 -m "color printer Driver1" -r lpt1:
```

To delete the printer named colorprinter_2 from the remote computer named HRServer, type:

```
cscript prnmngr -d -s HRServer -p colorprinter_2
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

prnport

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates, deletes, and lists standard TCP/IP printer ports, in addition to displaying and changing port configuration. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the prnport file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\prnport`.

Syntax

```
cscript prnport {-a | -d | -l | -g | -t | -?} [-r <portname>] [-s
<Servername>] [-u <Username>] [-w <password>] [-o {raw | lpr}] [-h
<Hostaddress>] [-q <Queuename>] [-n <portnumber>] -m{e | d} [-i <SNMPindex>]
[-y <communityname>] -2{e | -d}
```

Parameters

Expand table

Parameter	Description
-a	Creates a standard TCP/IP printer port.
-d	Deletes a standard TCP/IP printer port.
-l	Lists all standard TCP/IP printer ports on the computer specified by the **-s** parameter.
-g	Displays the configuration of a standard TCP/IP printer port.
-t	Configures the port settings for a standard TCP/IP printer port.
-r `<portname>`	Specifies the port to which the printer is connected.
-s `<Servername>`	Specifies the name of the remote computer that hosts the printer that you want to manage. If you don't specify a computer, the local computer is used.

Parameter	Description
-u `<Username>` -w `<password>`	Specifies an account with permissions to connect to the computer that hosts the printer that you want to manage. All members of the target computer's local Administrators group have these permissions, but the permissions can also be granted to other users. If you don't specify an account, you must be logged on under an account with these permissions for the command to work.
-o `{raw|lpr}`	Specifies which protocol the port uses: TCP raw or TCP lpr. The TCP raw protocol is a higher performance protocol on Windows than the lpr protocol. If you use TCP raw, you can optionally specify the port number by using the **-n** parameter. The default port number is 9100.
-h `<Hostaddress>`	Specifies (by IP address) the printer for which you want to configure the port.
-q `<Queuename>`	Specifies the queue name for a TCP raw port.
-n `<portnumber>`	Specifies the port number for a TCP raw port. The default port number is 9100.
-m`{e|d}`	Specifies whether SNMP is enabled. The parameter **e** enables SNMP. The parameter **d** disables SNMP.
-i `<SNMPindex`	Specifies the SNMP index, if SNMP is enabled. For more information, see **Rfc 1759** at the Rfc editor website.
-y `<communityname>`	Specifies the SNMP community name, if SNMP is enabled.
-2`{e|-d}`	Specifies whether double spools (also known as respooling) are enabled for TCP lpr ports. Double spools are necessary because TCP lpr must include an accurate byte count in the control file that is sent to the printer, but the protocol cannot get the count from the local print provider. Therefore, when a file is spooled to a TCP lpr print queue, it is also spooled as a temporary file in the system32 directory. TCP lpr determines the size of the temporary file and sends the size to the server running LPD. The parameter **e** enables double spools. The parameter **d** disables double spools.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "Computer Name").

Examples

To display all standard TCP/IP printing ports on the server \Server1, type:

```
cscript prnport -l -s Server1
```

To delete the standard TCP/IP printing port on the server \Server1 that connects to a network printer at 10.2.3.4, type:

```
cscript prnport -d -s Server1 -r IP_10.2.3.4
```

To add a standard TCP/IP printing port on the server \Server1 that connects to a network printer at 10.2.3.4 and uses the TCP raw protocol on port 9100, type:

```
cscript prnport -a -s Server1 -r IP_10.2.3.4 -h 10.2.3.4 -o raw -n 9100
```

To enable SNMP, specify the "public" community name and set the SNMP index to 1 on a network printer at 10.2.3.4 shared by the server \Server1, type:

```
cscript prnport -t -s Server1 -r IP_10.2.3.4 -me -y public -i 1 -n 9100
```

To add a standard TCP/IP printing port on the local computer that connects to a network printer at 10.2.3.4 and automatically get the device settings from the printer, type:

```
cscript prnport -a -r IP_10.2.3.4 -h 10.2.3.4
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

prnqctl

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Prints a test page, pauses or resumes a printer, and clears a printer queue. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the prnqctl file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\prnqctl`.

Syntax

```
cscript Prnqctl {-z | -m | -e | -x | -?} [-s <Servername>] [-p
<Printername>] [-u <Username>] [-w <password>]
```

Parameters

Expand table

Parameter	Description
-z	Pauses printing on the printer specified by the **-p** parameter.
-m	Resumes printing on the printer specified by the **-p** parameter.
-e	Prints a test page on the printer specified by the **-p** parameter.
-x	Cancels all print jobs on the printer specified by the **-p** parameter.
-s `<Servername>`	Specifies the name of the remote computer that hosts the printer that you want to manage. If you don't specify a computer, the local computer is used.
-p `<Printername>`	Required. Specifies the name of the printer that you want to manage.
-u `<Username>` -w `<password>`	Specifies an account with permissions to connect to the computer that hosts the printer that you want to manage. All members of the target computer's local Administrators group have these permissions, but the permissions can

Parameter	Description
	also be granted to other users. If you don't specify an account, you must be logged on under an account with these permissions for the command to work.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "Computer Name").

Examples

To print a test page on the Laserprinter1 printer shared by the \Server1 computer, type:

```
cscript prnqctl -e -s Server1 -p Laserprinter1
```

To pause printing on the Laserprinter1 printer on the local computer, type:

```
cscript prnqctl -z -p Laserprinter1
```

To cancel all print jobs on the Laserprinter1 printer on the local computer, type:

```
cscript prnqctl -x -p Laserprinter1
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

prompt

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the Cmd.exe command prompt, including displaying any text you want, such as the name of the current directory, the time and date, or the Microsoft Windows version number. If used without parameters, this command resets the command prompt to the default setting, which is the current drive letter and directory followed by the greater than symbol (>).

Syntax

```
prompt [<text>]
```

Parameters

Expand table

Parameter	Description
`<text>`	Specifies the text and information that you want to include in the command prompt.
/?	Displays help at the command prompt.

Remarks

- The character combinations you can include instead of, or in addition to, one or more character strings in the *text* parameter:

Expand table

Character	Description
$q	= (Equal sign)
$$	$ (Dollar sign)

Character	Description
$t	Current time
$d	Current date
$p	Current drive and path
$v	Windows version number
$n	Current drive
$g	> (Greater than sign)
$l	< (Less than sign)
$b	| (Pipe symbol)
$_	ENTER-LINEFEED
$e	ANSI escape code (code 27)
$h	Backspace (to delete a character that has been written to the command line)
$a	& (Ampersand)
$c	((Left parenthesis)
$f	) (Right parenthesis)
$s	Space

- When command extensions are enabled the **prompt** command supports the following formatting characters:

Expand table

Character	Description
$+	Zero or more plus sign (+) characters, depending on the depth of the **pushd** directory stack (one character for each level pushed).
$m	The remote name associated with the current drive letter or the empty string if current drive is not a network drive.

- If you include the **$p** character in the text parameter, your disk is read after you enter each command (to determine the current drive and path). This can take extra time, especially for floppy disk drives.

Examples

To set a two-line command prompt with the current time and date on the first line and the greater than sign on the next line, type:

```
prompt $d$s$s$t$_$g
```

The prompt is changed as follows, where the date and time are current:

```
Fri 06/01/2007  13:53:28.91
```

To set the command prompt to display as an arrow (`-->`), type:

```
prompt --$g
```

To manually change the command prompt to the default setting (the current drive and path followed by the greater than sign), type:

```
prompt $p$g
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

pubprn

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Publishes a printer to the Active Directory Domain Services. This command is a Visual Basic script located in the `%WINdir%\System32\printing_Admin_Scripts\<language>` directory. To use this command at a command prompt, type **cscript** followed by the full path to the pubprn file, or change directories to the appropriate folder. For example: `cscript %WINdir%\System32\printing_Admin_Scripts\en-US\pubprn`.

Syntax

```
cscript pubprn {<servername> | <UNCprinterpath>} LDAP://CN=<container>,DC=
<container>
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the name of the Windows server that hosts the printer that you want to publish. If you don't specify a computer, the local computer is used.
`<UNCprinterpath>`	The Universal Naming Convention (UNC) path to the shared printer that you want to publish.
`LDAP://CN=<Container>,DC=<Container>`	Specifies the path to the container in Active Directory Domain Services where you want to publish the printer.
/?	Displays help at the command prompt.

Remarks

- If the information that you supply contains spaces, use quotation marks around the text (for example, "Computer Name").

Examples

To publish all printers on the \Server1 computer to the MyContainer container in the MyDomain.company.com domain, type:

```
cscript pubprn Server1 LDAP://CN=MyContainer,DC=MyDomain,DC=company,DC=Com
```

To publish the Laserprinter1 printer on the \\Server1 server to the MyContainer container in the MyDomain.company.com domain, type:

```
cscript pubprn \\Server1\Laserprinter1
LDAP://CN=MyContainer,DC=MyDomain,DC=company,DC=Com
```

Related links

- Command-Line Syntax Key
- Print Command Reference

Feedback

Was this page helpful? Yes No

pushd

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Stores the current directory for use by the **popd** command, and then changes to the specified directory.

Every time you use the **pushd** command, a single directory is stored for your use. However, you can store multiple directories by using the **pushd** command multiple times. The directories are stored sequentially in a virtual stack, so if you use the **pushd** command once, the directory in which you use the command is placed at the bottom of the stack. If you use the command again, the second directory is placed on top of the first one. The process repeats every time you use the **pushd** command.

If you use the **popd** command, the directory on the top of the stack is removed and the current directory is changed to that directory. If you use the **popd** command again, the next directory on the stack is removed. If command extensions are enabled, the **popd** command removes any drive-letter assignment created by the **pushd** command.

Syntax

```
pushd [<path>]
```

Parameters

Expand table

Parameter	Description
`<path>`	Specifies the directory to make the current directory. This command supports relative paths.
/?	Displays help at the command prompt.

Remarks

- If command extensions are enabled, the **pushd** command accepts either a network path or a local drive letter and path.
- If you specify a network path, the **pushd** command temporarily assigns the highest unused drive letter (starting with Z:) to the specified network resource. The command then changes the current drive and directory to the specified directory on the newly assigned drive. If you use the **popd** command with command extensions enabled, the **popd** command removes the drive-letter assignment created by **pushd**.

Examples

To change the current directory from the one in which the batch program was run, and then to change it back:

```
@echo off
rem This batch file deletes all .txt files in a specified directory
pushd %1
del *.txt
popd
cls
echo All text files deleted in the %1 directory
```

Related links

- Command-Line Syntax Key
- popd command

Feedback

Was this page helpful? Yes No

pushprinterconnections

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Reads Deployed Printer Connection settings from Group Policy and deploys/removes printer connections as needed.

> **Important**
>
> This utility is for use in machine startup or user logon scripts, and shouldn't be run from the command line.

Syntax

```
pushprinterconnections <-log> <-?>
```

Parameters

Expand table

Parameter	Description
<-log>	Writes a per user debug log file to *%temp*, or writes a per machine debug log to *%windir%\temp*.
<-?>	Displays Help at the command prompt.

Related links

- Command-Line Syntax Key
- Print Command Reference
- Deploy Printers by Using Group Policy

Feedback

Was this page helpful? Yes No

pwlauncher

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables or disables the Windows To Go Startup Options (pwlauncher). The **pwlauncher** command-line tool allows you to configure the computer to boot into a Windows To Go workspace automatically (assuming one is present), without requiring you to enter your firmware or change your startup options.

Windows To Go Startup Options allow a user to configure their computer to boot from USB from within Windows-without ever entering their firmware, as long as their firmware supports booting from USB. Enabling a system to always boot from USB first has implications that you should consider. For example, a USB device that includes malware could be booted inadvertently to compromise the system, or multiple USB drives could be plugged in to cause a boot conflict. For this reason, the default configuration has the Windows To Go Startup Options disabled by default. In addition, administrator privileges are required to configure Windows To Go Startup Options. If you enable the Windows To Go startup options using the pwlauncher command-line tool or the **Change Windows To Go Startup Options** app the computer will attempt to boot from any USB device that is inserted into the computer before it is started.

Syntax

```
pwlauncher {/enable | /disable}
```

Parameters

Expand table

Parameter	Description
/enable	Enables Windows To Go startup options, so the computer will automatically boot from a USB device when present.
/disable	Disables Windows To Go startup options, so the computer can't be booted from a USB device unless configured manually in the firmware.

Parameter	Description
/?	Displays help at the command prompt.

Examples

To enable boot from USB:

```
pwlauncher /enable
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

qappsrv

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays a list of all Remote Desktop Session Host servers on the network. To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

> **Note**
>
> This command is the same as the **query termserver command**.

Related links

- Command-Line Syntax Key
- query termserver command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

qprocess

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about processes that are running on a Remote Desktop Session Host server. To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

> **Note**
>
> This command is the same as the **query process command**.

Related links

- Command-Line Syntax Key
- query process command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

query commands

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about processes, sessions, and Remote Desktop Session Host servers. To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
query process
query session
query termserver
query user
```

Parameters

Expand table

Parameter	Description
query process	Displays information about processes running on an Remote Desktop Session Host server.
query session	Displays information about sessions on a Remote Desktop Session Host server.
query termserver	Displays a list of all Remote Desktop Session Host servers on the network.
query user	Displays information about user sessions on a Remote Desktop Session Host server.

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

query process

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about processes that are running on a Remote Desktop Session Host server. You can use this command to find out which programs a specific user is running, and also which users are running a specific program. This command returns the following information:

- User who owns the process
- Session that owns the process
- ID of the session
- Name of the process
- ID of the process

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
query process [*|<processID>|<username>|<sessionname>|/id:<nn>|
<programname>] [/server:<servername>]
```

Parameters

Expand table

Parameter	Description
*	Lists the processes for all sessions.

Parameter	Description
`<processID>`	Specifies the numeric ID identifying the process that you want to query.
`<username>`	Specifies the name of the user whose processes you want to list.
`<sessionname>`	Specifies the name of the active session whose processes you want to list.
/id:`<nn>`	Specifies the ID of the session whose processes you want to list.
`<programname>`	Specifies the name of the program whose processes you want to query. The .exe extension is required.
/server: `<servername>`	Specifies the Remote Desktop Session Host server whose processes you want to list. If unspecified, the server where you are currently logged on is used.
/?	Displays help at the command prompt.

Remarks

- Administrators have full access to all **query process** functions.
- If you don't specify the *<username>*, *<sessionname>*, */id:*`<nn>`, *<programname>*, or * parameters, this query displays only the processes that belong to the current user.
- When **query process** returns information, a greater than `(>)` symbol is displayed before each process that belongs to the current session.

Examples

To display information about the processes being used by all sessions, type:

```
query process *
```

To display information about the processes being used by *session ID 2*, type:

```
query process /ID:2
```

Related links

- Command-Line Syntax Key
- query command
- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

query session

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about sessions on a Remote Desktop Session Host server. The list includes information not only about active sessions but also about other sessions that the server runs.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
query session [<sessionname> | <username> | <sessionID>] [/server:
<servername>] [/mode] [/flow] [/connect] [/counter]
```

Parameters

Expand table

Parameter	Description
`<sessionname>`	Specifies the name of the session that you want to query.
`<username>`	Specifies the name of the user whose sessions you want to query.
`<sessionID>`	Specifies the ID of the session that you want to query.
/server: `<servername>`	Identifies the rd Session Host server to query. The default is the current server.
/mode	Displays current line settings.
/flow	Displays current flow-control settings.
/connect	Displays current connect settings.

Parameter	Description
/counter	Displays current counters information, including the total number of sessions created, disconnected, and reconnected.
/?	Displays help at the command prompt.

Remarks

- A user can always query the session to which the user is currently logged on. To query other sessions, the user must have special access permission.
- If you don't specify a session using the *<username>*, *<sessionname>*, or *sessionID* parameters, this query will display information about all active sessions in the system.
- When **query session** returns information, a greater than `(>)` symbol is displayed before the current session. For example:

```
C:\>query session
    SESSIONNAME     USERNAME        ID STATE    TYPE    DEVICE
    console         Administrator1  0 active    wdcon
    >rdp-tcp#1      User1           1 active    wdtshare
    rdp-tcp                         2 listen    wdtshare
                                    4 idle
                                    5 idle
```

  Where:
  - **SESSIONNAME** specifies the name assigned to the session.
  - **USERNAME** indicates the user name of the user connected to the session.
  - **STATE** provides information about the current state of the session.
  - **TYPE** indicates the session type.
  - **DEVICE**, which isn't present for the console or network-connected sessions, is the device name assigned to the session.
  - Any sessions in which the initial state is configured as DISABLED won't show up in the **query session** list until they're enabled.

Examples

To display information about all active sessions on server *Server2*, type:

```
query session /server:Server2
```

To display information about active session *modeM02*, type:

```
query session modeM02
```

Related links

- Command-Line Syntax Key
- query command
- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

query termserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays a list of all Remote Desktop Session Host servers on the network. This command searches the network for all attached Remote Desktop Session Host servers and returns the following information:

- Name of the server
- Network (and node address if the /address option is used)

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
query termserver [<servername>] [/domain:<domain>] [/address] [/continue]
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the name that identifies the Remote Desktop Session Host server.
/domain: `<domain>`	Specifies the domain to query for terminal servers. You don't need to specify a domain if you are querying the domain in which you are currently working.
/address	Displays the network and node addresses for each server.
/continue	Prevents pausing after each screen of information is displayed.
/?	Displays help at the command prompt.

Examples

To display information about all Remote Desktop Session Host servers on the network, type:

```
query termserver
```

To display information about the Remote Desktop Session Host server named *Server3*, type:

```
query termserver Server3
```

To display information about all Remote Desktop Session Host servers in domain *CONTOSO*, type:

```
query termserver /domain:CONTOSO
```

To display the network and node address for the Remote Desktop Session Host server named *Server3*, type:

```
query termserver Server3 /address
```

Related links

- Command-Line Syntax Key
- query command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

query user

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about user sessions on a Remote Desktop Session Host server. You can use this command to find out if a specific user is logged on to a specific Remote Desktop Session Host server. This command returns the following information:

- Name of the user
- Name of the session on the Remote Desktop Session Host server
- Session ID
- State of the session (active or disconnected)
- Idle time (the number of minutes since the last keystroke or mouse movement at the session)
- Date and time the user logged on

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
query user [<username> | <sessionname> | <sessionID>] [/server:<servername>]
```

Parameters

Expand table

Parameter	Description
`<username>`	Specifies the logon name of the user that you want to query.

Parameter	Description
`<sessionname>`	Specifies the name of the session that you want to query.
`<sessionID>`	Specifies the ID of the session that you want to query.
/server: `<servername>`	Specifies the Remote Desktop Session Host server that you want to query. Otherwise, the current Remote Desktop Session Host server is used. This parameter is only required if you're using this command from a remote server.
/?	Displays help at the command prompt.

Remarks

- To use this command, you must have Full Control permission or special access permission.
- If you don't specify a user using the *<username>*, *<sessionname>*, or *sessionID* parameters, a list of all users who are logged on to the server is returned. Alternatively, you can also use the **query session** command to display a list of all sessions on a server.
- When **query user** returns information, a greater than `(>)` symbol is displayed before the current session.

Examples

To display information about all users logged on the system, type:

```
query user
```

To display information about the user *USER1* on server *Server1*, type:

```
query user USER1 /server:Server1
```

Related links

- Command-Line Syntax Key

- query command
- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

quser

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about user sessions on a Remote Desktop Session Host server. You can use this command to find out if a specific user is logged on to a specific Remote Desktop Session Host server. This command returns the following information:

- Name of the user
- Name of the session on the Remote Desktop Session Host server
- Session ID
- State of the session (active or disconnected)
- Idle time (the number of minutes since the last keystroke or mouse movement at the session)
- Date and time the user logged on

> **Note**
>
> This command is the same as the query user command. To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
quser [<username> | <sessionname> | <sessionID>] [/server:<servername>]
```

Parameters

Expand table

Parameter	Description
`<username>`	Specifies the logon name of the user that you want to query.
`<sessionname>`	Specifies the name of the session that you want to query.
`<sessionID>`	Specifies the ID of the session that you want to query.
/server: `<servername>`	Specifies the Remote Desktop Session Host server that you want to query. Otherwise, the current Remote Desktop Session Host server is used. This parameter is only required if you're using this command from a remote server.
/?	Displays help at the command prompt.

Remarks

- To use this command, you must have Full Control permission or special access permission.
- If you don't specify a user using the <*username*>, <*sessionname*>, or *sessionID* parameters, a list of all users who are logged on to the server is returned. Alternatively, you can also use the **query session** command to display a list of all sessions on a server.
- When **quser** returns information, a greater than `(>)` symbol is displayed before the current session.

Examples

To display information about all users logged on the system, type:

```
quser
```

To display information about the user *USER1* on server *Server1*, type:

```
quser USER1 /server:Server1
```

Related links

- Command-Line Syntax Key
- query user command
- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

qwinsta

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays information about sessions on a Remote Desktop Session Host server. The list includes information not only about active sessions but also about other sessions that the server runs.

> **Note**
>
> This command is the same as the query session command. To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
qwinsta [<sessionname> | <username> | <sessionID>] [/server:<servername>]
[/mode] [/flow] [/connect] [/counter]
```

Parameters

Expand table

Parameter	Description
`<sessionname>`	Specifies the name of the session that you want to query.
`<username>`	Specifies the name of the user whose sessions you want to query.
`<sessionID>`	Specifies the ID of the session that you want to query.
/server: `<servername>`	Identifies the rd Session Host server to query. The default is the current server.
/mode	Displays current line settings.
/flow	Displays current flow-control settings.

Parameter	Description
/connect	Displays current connect settings.
/counter	Displays current counters information, including the total number of sessions created, disconnected, and reconnected.
/?	Displays help at the command prompt.

Remarks

- A user can always query the session to which the user is currently logged on. To query other sessions, the user must have special access permission.
- If you don't specify a session using the <*username*>, <*sessionname*>, or *sessionID* parameters, this query will display information about all active sessions in the system.
- When **qwinsta** returns information, a greater than `(>)` symbol is displayed before the current session. For example:

```
C:\>qwinsta
    SESSIONNAME     USERNAME        ID STATE    TYPE    DEVICE
    console         Administrator1  0 active    wdcon
    >rdp-tcp#1      User1           1 active    wdtshare
    rdp-tcp                         2 listen    wdtshare
                                    4 idle
                                    5 idle
```

  Where:
  - **SESSIONNAME** specifies the name assigned to the session.
  - **USERNAME** indicates the user name of the user connected to the session.
  - **STATE** provides information about the current state of the session.
  - **TYPE** indicates the session type.
  - **DEVICE**, which isn't present for the console or network-connected sessions, is the device name assigned to the session.
  - Any sessions in which the initial state is configured as DISABLED won't show up in the **qwinsta** list until they're enabled.

Examples

To display information about all active sessions on server *Server2*, type:

```
qwinsta /server:Server2
```

To display information about active session *modeM02*, type:

```
qwinsta modeM02
```

Related links

- Command-Line Syntax Key
- query session command
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

rd

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a directory.

The **rd** command can also run from the Windows Recovery Console, using different parameters. For more information, see Windows Recovery Environment (WinRE).

> **Note**
>
> This command is the same as the rmdir command.

Syntax

```
rd [<drive>:]<path> [/s [/q]]
```

Parameters

Expand table

Parameter	Description
`[<drive>:]<path>`	Specifies the location and the name of the directory that you want to delete. *Path* is required. If you include a backslash () at the beginning of the specified *path*, then the *path* starts at the root directory (regardless of the current directory).
/s	Deletes a directory tree (the specified directory and all its subdirectories, including all files).
/q	Specifies quiet mode. Does not prompt for confirmation when deleting a directory tree. The **/q** parameter works only if **/s** is also specified. **CAUTION:** When you run in quiet mode, the entire directory tree is deleted without confirmation. Make sure that important files are moved or backed up before using the **/q** command-line option.
/?	Displays help at the command prompt.

Remarks

- You can't delete a directory that contains files, including hidden or system files. If you attempt to do so, the following message appears:

  `The directory is not empty`

  Use the **dir /a** command to list all files (including hidden and system files). Then use the **attrib** command with **-h** to remove hidden file attributes, **-s** to remove system file attributes, or **-h -s** to remove both hidden and system file attributes. After the hidden and file attributes have been removed, you can delete the files.

- You can't use the **rd** command to delete the current directory. If you attempt to delete the current directory, the following error message appears:

  `The process can't access the file because it is being used by another process.`

  If you receive this error message, you must change to a different directory (not a subdirectory of the current directory), and then try again.

Examples

To change to the parent directory so you can safely remove the desired directory, type:

```
cd ..
```

To remove a directory named *test* (and all its subdirectories and files) from the current directory, type:

```
rd /s test
```

To run the previous example in quiet mode, type:

```
rd /s /q test
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rdpsign

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables you to digitally sign a Remote Desktop Protocol (.rdp) file.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
rdpsign /sha1 <hash> [/q | /v |] [/l] <file_name.rdp>
```

Parameters

Expand table

Parameter	Description
/sha1 `<hash>`	Specifies the thumbprint, which is the Secure Hash Algorithm 1 (SHA1) hash of the signing certificate that is included in the certificate store. Used in Windows Server 2012 R2 and older.
/sha256 `<hash>`	Specifies the thumbprint, which is the Secure Hash Algorithm 256 (SHA256) hash of the signing certificate that is included in the certificate store. Replaces /sha1 in Windows Server 2016 and newer.
/q	Quiet mode. No output when the command succeeds and minimal output if the command fails.
/v	verbose mode. Displays all warnings, messages, and status.
/l	Tests the signing and output results without actually replacing any of the input files.
`<file_name.rdp>`	The name of the .rdp file. You must specify the .rdp file (or files) to sign by

Parameter	Description
	using the full file name. Wildcard characters are not accepted.
/?	Displays help at the command prompt.

Remarks

- The SHA1 or SHA256 certificate thumbprint should represent a trusted .rdp file publisher. To obtain the certificate thumbprint, open the **Certificates** snap-in, double-click the certificate that you want to use (either in the local computer's certificates store or in your personal certificates store), click the **details** tab, and then in the **Field** list, click **Thumbprint**.

  > ⓘ **Note**
  >
  > When you copy the thumbprint for use with the rdpsign.exe tool, you must remove any spaces.

- The signed output files overwrite the input files.
- If multiple files are specified, and if any of the .rdp files can't be read or written to, the tool continues to the next file.

Examples

To sign an .rdp file named *file1.rdp*, navigate to the folder where you saved the .rdp file, and then type:

```
rdpsign /sha1 hash file1.rdp
```

> ⓘ **Note**
>
> The *hash* value represents the SHA1 certificate thumbprint, without any spaces.

To test whether digital signing will succeed for an .rdp file without actually signing the file, type:

```
```

```
rdpsign /sha1 hash /l file1.rdp
```

To sign multiple .rdp files that are named, *file1.rdp*, *file2.rdp*, and *file3.rdp*, type (including the spaces between file names):

```
rdpsign /sha1 hash file1.rdp file2.rdp file3.rdp
```

See Also

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

recover (DiskPart)

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Refreshes the state of all disks in a disk group, attempt to recover disks in an invalid disk group, and resynchronizes mirrored volumes and RAID-5 volumes that have stale data. This command operates on disks that are failed or failing. It also operates on volumes that are failed, failing, or in failed redundancy state.

This command operates on groups of dynamic disks. If this command is used on a group with a basic disk, it won't return an error, but no action will be taken.

> **Note**
>
> A disk that is part of a disk group must be selected for this operation to succeed. Use the select disk command to select a disk and shift the focus to it.

Syntax

```
recover [noerr]
```

Parameters

Expand table

Parameter	Description
noerr	For scripting only. When an error is encountered, DiskPart continues to process commands as if the error did not occur. Without this parameter, an error causes DiskPart to exit with an error code.

Examples

To recover the disk group that contains the disk with focus, type:

```
recover
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil` tool is a command-line utility designed to manage and repair volumes formatted with the Resilient File System (ReFS) in Windows. This utility provides a wide range of functionalities to ensure the integrity, performance, and optimization of ReFS volumes. It's an essential tool for maintaining robust file system operations and addressing various issues that might arise.

Commands

Expand table

Command	Description
compression	Enables and manages volume compression to save storage space.
dedup	Identifies and consolidates duplicate clusters to optimize storage usage.
fixboot	Repairs boot sectors to ensure proper system startup.
iometrics	Tracks and reports input/output metrics for volume performance analysis.
leak	Detects and resolves memory or resource leaks in the file system.
salvage	Performs recovery operations on a corrupt volume to retrieve data.
streamsnapshot	Manages snapshots of data streams for backup or recovery purposes.
triage	Analyzes and addresses file system corruptions.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil compression

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil compression` command is used to query or set the compression parameters on a specified Resilient File System (ReFS) volume. You can check the current compression status, apply compression using different formats, or decompress a compressed volume.

Syntax

```
refsutil compression <drive> <[/q]> | <[/c] [/f <format>] [/e <engine>] [/cs <size>]>
```

Parameters

Expand table

Parameters	Description
`<drive>`	Specifies the drive letter of the volume to be compressed in the format `E:`.
`/q`	Queries the volume's current compression parameters.
`/c`	Compresses the volume with the specified compression parameters.
`/f <format>`	Specifies the compression format to use. Valid options include: • `LZ4`: A fast compression algorithm. • `ZSTD`: A compression algorithm that offers a good balance between compression ratio and speed. • `NONE`: Used to *decompress* a previously compressed volume. When using `NONE`, omit the engine and compression chunk size parameters.
`/e <engine>`	Defines the compression level to use, which varies by format: • `LZ4`: Levels `1`, `3-12`. The default is `1`. Levels `3-12` use the LZ4HC algorithm, which provides higher compression ratios at the cost of slower compression speeds.

Parameters	Description
	• `ZSTD`: Levels `1-22`. The default is `3`. Higher levels provide better compression ratios but reduce compression speed and might require more memory (especially levels `20` and greater). Decompression speed remains consistent regardless of the chosen compression level. Omitting the `/e` parameter or using the value `0` sets the default level for any given compression format.
`/cs <size>`	Specifies the compression chunk size in bytes. This value must be to the power of two (2^*n*), at least equal to the volume's cluster size, and no greater than `64MB`. Larger chunk sizes can enhance compression ratios but might negatively impact read performance for smaller data amounts. Beyond `1MB`, the benefits diminish significantly, so larger sizes aren't recommended. If omitted or set to `0`, the volume's cluster size is used.

> ⓘ **Note**
>
> For the `/e` parameter, the default compression values are subject to change.

Examples

To query the F: volume compression parameters, run the command:

```
refsutil compression F: /q
```

To compress the F: volume using the `LZ4` format with level 12 compression and a chunk size of 524,288 bytes (512 KB), run the command:

```
refsutil compression F: /c /f LZ4 /e 12 /cs 524288
```

To compress the F: volume using the `ZSTD` format with level 15 compression and a chunk size of 131,072 bytes (128 KB), run the command:

```
refsutil compression F: /c /f ZSTD /e 15 /cs 131072
```

To compress the F: volume using the `ZSTD` format with default compression parameters, run the command:

```
refsutil compression F: /c /f ZSTD
```

To decompress the F: volume, run the command:

```
refsutil compression F: /c /f NONE
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil dedup

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil compression` command manages data deduplication on a specified Resilient File System (ReFS) volume. You can either deduplicate the volume or scan it to see how much space can be saved by deduplicating equivalent clusters. You can also control CPU usage and choose between memory-mapped file I/O or asynchronous reads.

Syntax

```
refsutil dedup <drive> [/d] [/s] [/cpu <percentage>] [/mm]
```

Parameters

Expand table

Parameter	Description
`drive`	Specifies the volume path in the `E:` format or a mount point.
`/d`	Deduplicates the volume. This parameter can't be used with the `/s` parameter.
`/s`	Scans the volume to determine how much space can be saved by deduplicating equivalent clusters. This parameter can't be used with the `/d` parameter.
`/cpu percentage`	Specifies the maximum percentage of CPU to use. Acceptable values are between 1-100.
`/mm`	Uses memory-mapped file I/O to read files for deduplication. This parameter must be used with either the `/d` or `/s` parameters.

Examples

To scan the D: volume and see how much space you can save by deduplicating equivalent clusters, run the command:

```
refsutil dedup D: /s
```

To deduplicate the D: volume using 50% of CPU resources, run the command:

```
refsutil dedup D: /d /cpu 50
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil fixboot

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil fixboot` command repairs or updates the boot sector of a specified ReFS (Resilient File System) volume. It ensures the boot sector is correctly configured based on the provided drive, version, and cluster size.

> **⚠ Warning**
>
> This operation modifies the boot sector(s) of a volume and may lead to complete data loss. Proceed with caution.

Syntax

```
refsutil fixboot <drive> <MajorVersion> <MinorVersion> <ClusterSize> [-f] [-w <Dir>] [-smr <SMRBandSizeInMB>] [-x]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Specifies the drive letter of the volume to be fixed in the format `E:`.
`<MajorVersion>`	Specifies the major version of the ReFS volume.
`<MinorVersion>`	Specifies the minor version of the ReFS volume.
`<ClusterSize>`	Specifies the cluster size of the volume, either 4096 or 65536 bytes.
`-f`	Forces the operation to ignore any valid boot sector found and fix the boot sector based on the provided parameters.
`-w <Dir>`	Writes the changes to one or more boot sectors if needed and backs up one or more current boot sectors into the specified backup directory `<DIR>`.

Parameter	Description
`-smr <SMRBandSizeInMB>`	Specifies that the volume is in SMR (Shingled Magnetic Recording) format. The cluster size must be 65536 bytes. `SMRBandSizeinMB` can be either 128 MB or 256 MB.
`-x`	Forces the volume to dismount if necessary and invalidates all open handles to the volume.

Examples

To fix the boot sector of a ReFS volume on drive D: with a major version of 3, minor version of 3, and a cluster size of `4096` bytes, run this command:

```
refsutil fixboot D: 3 3 4096
```

To fix the boot sector of the ReFS volume on drive D: with a major version of 2, minor version of 1, and a cluster size of `65536` bytes, run the following command. This command forces the operation to ignore any valid boot sector found and writes the necessary changes to the boot sector, then backing up the boot sector into the directory `C:\Backup`:

```
refsutil fixboot D: 2 1 65536 -f -w C:\Backup
```

To fix the boot sector of the ReFS volume on drive D: with a major version of 2, minor version of 0, and a cluster size of `65536` bytes, run the following command. The command specifies that the volume is in SMR format with a band size of 128 MB:

```
refsutil fixboot D: 2 0 65536 -smr 128
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil iometrics

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil iometrics` command allows you to query or set volume heat parameters for tracking input/output (IO) metrics on a ReFS (Resilient File System) volume. This tool helps monitor and manage the performance and usage patterns on the volume.

Syntax

```
refsutil iometrics <drive> <[/q]> | <[/s] [/st <seconds>] [/sp <seconds>]
[/sg <number>]>
```

Parameters

Expand table

Parameters	Description
`<drive>`	Specifies the drive letter of the volume to be compressed in the format `E:`.
`/q`	Queries the current volume activity tracking settings.
`/s`	Configures the volume activity tracking settings.
`/st <seconds>`	Specifies how long in seconds the volume should track IO activity.
`/sp <seconds>`	Specifies how often in seconds the IO metrics should be tracked.
`/sg <number>`	Specifies the number of periods to track per volume tracking unit.

ReFS enforces values for `/st`, `/sp`, and `/sg` to be within suitable ranges and adjusts them to the filesystem minimums if needed. These values can be omitted, in which case ReFS uses default settings. The current minimum values are:

- `/st`: 3,600 seconds (1 hour)
- `/sp`: 10 seconds
- `/sg`: 5 periods

Examples

To query the volume activity tracking settings on the F: drive, run the command:

```
refsutil iometrics F: /q
```

To set the volume activity tracking settings on drive F: with default values, run the command:

```
refsutil iometrics F: /s
```

To set the volume activity tracking settings on drive F: to track IO activity for 604,800 seconds or 7 (seven) days, run the command:

```
refsutil iometrics F: /s /st 604800
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil leak

Article • 04/03/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil leak` command assists administrators in detecting and diagnosing storage leaks, which occur when storage space is consumed inefficiently or unnecessarily. When the specified Resilient File System (ReFS) volume is scanned, this command identifies the location and size of leaks, helping to optimize storage usage and maintain system performance.

Syntax

```
refsutil leak <drive> [/a] [/x] [/v] [/d] [/q] [/t <count>] [/s <file>]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Specifies the drive letter in the format `E:` or the path to the volume mount point.
`/a`	If leak detection discovers corruption, this parameter attempts to fix it and rerun the leak detection. With this parameter enabled, leak detection restarts once for each corrupted directory found on the volume.
`/x`	Operates with an exclusive lock on the volume. Without this parameter, leak detection creates a snapshot of the volume for processing.
`/v`	Enables verbose output, providing more detailed information.
`/d`	Only performs a diagnosis. Leaks are reported to standard output (stdout) but isn't fixed.
`/q`	Queries the required space for the leak analysis operation.
`/t <count>`	Specifies how many CPU threads to use for leak detection. Utilizing more threads speeds up the process on a multi-core CPU. If set to `0`, it runs the detection in a single thread (not recommended). The default is `4` threads.

Parameter	Description
`/s <file>`	Uses the specified `<file>` as a scratch buffer for the operation. The `<file>` must be able to grow to the size reported by the `/q` parameter. If this parameter is omitted, a scratch buffer is created in the TEMP directory.

> **Note**
>
> Using the `/d` and `/a` parameters together will triage all found directory and file corruptions, but doesn't fix leaks.

Examples

To perform a leak analysis and generate a detailed report using F:\Scratch.tmp as the temporary file using `6` CPU threads, run this command:

```
refsutil leak D: /a /t 6 /s F:\Scratch.tmp
```

To perform a leak analysis and display detailed output using `5` CPU threads, run this command:

```
refsutil leak D: /a /d /v /t 5
```

To check the amount of space needed for a leak analysis on the D: drive, run this command:

```
refsutil leak D: /q
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil salvage

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil salvage` command attempts to diagnose severely damaged Resilient File System (ReFS) volumes. It works by identifying files that remain intact despite the damage. Once these recoverable files are identified, the tool copies them to a different, specified volume.

The `refsutil salvage` tool operates in two phases, Scan Phase and Copy Phase:

- **Scan Phase**: Involves diagnosing the damaged ReFS volume and identifying recoverable files. In manual mode, you're able to pause and resume this phase. Progress and logs are stored in the working directory, allowing you to continue the scan from where it was left off.
- **Copy Phase**: Involves copying the identified recoverable files to a specified target volume. This phase can be run independently in manual mode. Logs and progress for this phase are also saved in the working directory.

In automatic mode, both the Scan Phase and Copy Phase run sequentially without user intervention. All progress and logs are saved in the working directory, ensuring that the process can be monitored and managed efficiently.

You typically don't need to use the `refsutil salvage` tool unless the volume appears as RAW, indicating corruption. If the volume is in a read-only state, the data is still accessible and can be retrieved without this tool.

> **Note**
>
> The `refsutil salvage` command is only supported in the following operating systems:
>
> - Windows 10 Pro for Workstations or later
> - Windows Server 2019 or later

Syntax

```
```

```
refsutil salvage -QA | -FA | -C <source volume> <working directory> <target
directory> <options>
refsutil salvage -D | -QS | -FS | -IC <source volume> <working directory>
<options>
refsutil salvage -SL <source volume> <working directory> <target directory>
<file list> <options>
```

Parameters

Expand table

Parameter	Description
`<source volume>`	Specifies the ReFS volume to process in the format `E:` or the path to the volume mount point.
`<working directory>`	Specifies the location to store temporary information and logs. It must **not** be located on the `<source volume>`.
`<target directory>`	Specifies the location where identified files are copied to. It must **not** be located on the `<source volume>`.
`<file list>`	Specifies a file containing a list of specific files to be salvaged from the source volume to the target directory. This file must be generated during the Scan Phase using the `foundfiles.<volume signature>.txt` file.
`<options>`	• `-m`: Recovers all possible files, including deleted ones. • `-v`: Specifies to use verbose mode for detailed logging. • `-sv`: Skips the ReFS volume version check and assume the highest volume version that this tool can handle. • `-hl`: Assumes that the volume supports hard links. • `-x`: Forces the volume to dismount first, if necessary. All opened handles to the volume are then invalid.

> ⚠ **Warning**
>
> The `-m` parameter causes the process to take longer to run and can also lead to unexpected results. Similarly, the `-sv` parameter can also lead to unexpected results.

Remarks

In repairing ReFS volumes, two modes are featured, Automatic Mode and Manual Mode. Automatic Mode scans and repairs the volume automatically without user intervention, while Manual Mode offers advanced users detailed control over the repair process, allowing them to review and decide on specific actions.

Expand table

Automatic mode	Manual mode	Description
Quick Automatic		`refsutil salvage -QA` This mode performs a Quick Scan Phase followed by a Copy Phase. It operates faster by assuming that critical structures of the volume remain intact, eliminating the need for a full-volume scan. As a result, it minimizes the recovery of outdated files, directories, or volumes.
Full Automatic		`refsutil salvage -FA` This mode performs a Full Scan Phase followed by a Copy Phase. This mode might take a long time as it scans the entire volume for any recoverable files, directories, or volumes.
	Diagnose Phase	`refsutil salvage -D` Attempts to verify if the `<source volume>` is a ReFS volume and assesses its mountability. If the volume isn't mountable, the tool identifies and provides the reasons. This phase operates independently.
	Quick Scan Phase	`refsutil salvage -QS` Quickly scans the `<source volume>` for recoverable files. This mode operates faster by assuming that certain critical structures of the volume remain intact, eliminating the need for a full-volume scan. As a result, it minimizes the recovery of outdated files, directories, or volumes. Discovered files are logged in the `foundfiles.<volume signature>.txt` file located in the `<working directory>`. If the Scan Phase was interrupted previously, running the command with the `-QS` flag resumes the scan from the last checkpoint.
	Full Scan Phase	`refsutil salvage -FS` Scans the entire `<source volume>` for recoverable files. This mode might take a significant amount of time as it performs a comprehensive scan of the volume. Discovered files are logged in the `foundfiles.<volume signature>.txt` file located in the `<working`

Automatic mode	Manual mode	Description
		`directory>`. If the Scan Phase was interrupted previously, running the command with the `-FS` flag resumes the scan from the last checkpoint.

	Copy Phase	`refsutil salvage -C`
		
		• Copies all files listed in the `foundfiles.<volume signature>.txt` file to the `<target directory>`. If the Scan Phase was interrupted prematurely, the `foundfiles.<volume signature>.txt` file might not be created, and no files are copied to the `<target directory>`.
		
		`refsutil salvage -SL`
		
		• Copies all files listed in `<file list>` from the `<source volume>` to the `<target directory>`. The `<file list>` must be generated during the Scan Phase, although the scan doesn't need to be completed. To create a `<file list>`, copy the `foundfiles.<volume signature>.txt` file to a new file, remove entries for files that shouldn't be restored, and retain entries for files that should be restored. The Select-String cmdlet can help filter the `foundfiles.<volume signature>.txt` to include only specific paths, extensions, or file names.
		
		`refsutil salvage -IC`
		
		• Advanced users can recover files using an interactive console. This mode requires files generated during one of the Scan Phases.

Examples

To perform an automatic quick analysis scan, which quickly checks the volume for issues and attempts repairs without user input, run this command:

```
refsutil salvage -QA E: C:\Temp
```

To perform a full analysis scan automatically, which thoroughly checks the volume for issues and attempts repairs without user input, run this command:

```
```

```
refsutil salvage -FA E: C:\Temp
```

To perform a comprehensive scan with user control, allowing you to review and decide on specific actions during the repair process, run this command:

```
refsutil salvage -C E: C:\Temp D:\Recovered
```

To perform a quick scan that provides a detailed report of the issues found, enabling verbose logging for more information, run this command:

```
refsutil salvage -QS E: C:\Temp -v
```

To skip the version check during a full scan, which assumes the highest volume version that the tool can handle (useful if you know the tool is compatible with your volume), run this command:

```
refsutil salvage -FS E: C:\Temp -sv
```

To salvage specific files listed in a file list, created during the Scan Phase, and to force dismount the volume if necessary, run this command:

```
refsutil salvage -SL E: C:\Temp D:\Recovered FileList.txt -x
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil streamsnapshot

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil streamsnapshot` command allows you to manage snapshots of data streams within files in the Resilient File System (ReFS). You can create, delete, list, or compare snapshots of data streams at specified points in time. The `file.dat` file contains the data streams for which you want to create, delete, list, or compare snapshots. Essentially, it's the target file upon which this utility performs its operations.

Syntax

```
refsutil streamsnapshot [/c <name>] [/l] [/d <name>] [/q <name>]
<file[:stream]>
```

Parameters

Expand table

Parameter	Description
`/c <name>`	Creates a snapshot named `<name>` of the stream specified by `file[:stream]` at the current point in time. This option is mutually exclusive with `/d`, `/l`, and `/q`.
`/d <name>`	Deletes the snapshot named `<name>` from the specified file `<file>`. This parameter ignores the `[:stream]` portion of the file path and is mutually exclusive with `/c`, `/l`, and `/q`.
`/l <name>`	Lists all snapshots for the stream specified by `file[:stream]` that match the given `<name>`. Wildcards are allowed in `<name>`. This parameter is mutually exclusive with `/c`, `/d`, and `/q`.
`/q <name>`	Lists all modifications between the snapshot `<name>` and the current state of the stream specified by `[:stream]`. Both must reference streams in the same snapshot chain, with `<name>` being older than `[:stream]`. This parameter is mutually exclusive with `/c`, `/d`, and `/l`.

Note

If `[:stream]` isn't provided, this utility acts on the default unnamed data stream, `$DATA`.

Examples

To create a snapshot named "snapshot_jan_2024" of the stream at the current point in time, run the following command:

```
refsutil streamsnapshot /c "snapshot_jan_2024" C:\file.dat
```

To delete a snapshot named "snapshot_jan_2024" from the file, run the following command:

```
refsutil streamsnapshot /d "snapshot_jan_2024" C:\file.dat
```

To list all snapshots for the stream in the file matching any pattern using the `*` wildcard, run the following command:

```
refsutil streamsnapshot /l "*" C:\file.dat
```

To list all modifications between the snapshot "snapshot_jan_2024" and the stream "snapshot_feb_2024", run the following command:

```
refsutil streamsnapshot /q "snapshot_jan_2024" C:\file.dat:snapshot_feb_2024
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

refsutil triage

Article • 04/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `refsutil triage` command is used to scrub directory contents or global tables on a ReFS (Resilient File System) volume, ensuring data integrity and consistency.

Syntax

```
refsutil triage <drive> [/s <id>] [/g] [/v]
```

Parameters

Expand table

Parameter	Description
`<drive>`	Specifies the target drive, either as a drive letter in the format `E:` or as a path to the volume's mount point.
`/s <id>`	Scrubs the directory specified by `<id>`, where `<id>` is the file identifier for the directory. This option scrubs all files within the specified directory. This parameter can't be used with the `/g` parameter.
`/g`	Scrubs the global tables of the volume. This parameter can't be used with the `/s <id>` parameter.
`/v`	Displays verbose output, providing detailed information during the scrubbing operation.

Examples

To perform a directory scrub on the directory with file ID `15100` on the D: drive, run the following command:

```
```

```
refsutil triage D: /s 15100
```

To perform a directory scrub on the directory with file ID `16040` on the D: drive with verbose output, run the following command:

```
refsutil triage D: /s 16040 /v
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg commands

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Performs operations on registry subkey information and values in registry entries.

Some operations enable you to view or configure registry entries on local or remote computers, while others allow you to configure only local computers. Using **reg** to configure the registry of remote computers limits the parameters that you can use in some operations. Check the syntax and parameters for each operation to verify that they can be used on remote computers.

> **⊗ Caution**
>
> Don't edit the registry directly unless you have no alternative. The registry editor bypasses standard safeguards, allowing settings that can degrade performance, damage your system, or even require you to reinstall Windows. You can safely alter most registry settings by using the programs in Control Panel or Microsoft Management Console (MMC). If you must edit the registry directly, back it up first.

Syntax

```
reg add
reg compare
reg copy
reg delete
reg export
reg import
reg load
reg query
reg restore
reg save
reg unload
```

Parameters

Expand table

Parameter	Description
reg add	Adds a new subkey or entry to the registry.
reg compare	Compares specified registry subkeys or entries.
reg copy	Copies a registry entry to a specified location on the local or remote computer.
reg delete	Deletes a subkey or entries from the registry.
reg export	Copies the specified subkeys, entries, and values of the local computer into a file for transfer to other servers.
reg import	Copies the contents of a file that contains exported registry subkeys, entries, and values into the registry of the local computer.
reg load	Writes saved subkeys and entries into a different subkey in the registry.
reg query	Returns a list of the next tier of subkeys and entries that are located under a specified subkey in the registry.
reg restore	Writes saved subkeys and entries back to the registry.
reg save	Saves a copy of specified subkeys, entries, and values of the registry in a specified file.
reg unload	Removes a section of the registry that was loaded using the **reg load** operation.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg add

Article • 01/04/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a new subkey or entry to the registry.

Syntax

```
reg add <keyname> [/v valuename | /ve] [/t datatype] [/s separator] [/d data] [/f] [/reg:32 | /reg:64]
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey or entry to be added. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: • **HKCR** (HKEY_CLASSES_ROOT) • **HKCU** (HKEY_CURRENT_USER) • **HKLM** (HKEY_LOCAL_MACHINE) • **HKU** (HKEY_USERS) • **HKCC** (HKEY_CURRENT_CONFIG) If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
/v `<Valuename>`	Specifies the name of the add registry entry.
/ve	Specifies that the added registry entry has a null value.
/t `<Type>`	Specifies the type for the registry entry. *Type* must be one of the following: • REG_SZ

Parameter	Description
	• REG_MULTI_SZ • REG_DWORD_BIG_ENDIAN • REG_DWORD • REG_BINARY • REG_DWORD_LITTLE_ENDIAN • REG_LINK • REG_FULL_RESOURCE_DESCRIPTOR • REG_EXPAND_SZ
/s `<Separator>`	Specifies the character to be used to separate multiple instances of data when the **REG_MULTI_SZ** data type is specified and more than one entry is listed. If not specified, the default separator is **\0**.
/d `<Data>`	Specifies the data for the new registry entry.
/f	Adds the registry entry without prompting for confirmation.
/reg:32	Specifies the key should be accessed using the 32-bit registry view.
/reg:64	Specifies the key should be accessed using the 64-bit registry view.
/?	Displays help at the command prompt.

Remarks

- Subtrees can't be added with this operation. This version of **reg** doesn't ask for confirmation when adding a subkey.
- The return values for the **reg add** operation are:

Expand table

Value	Description
0	Success
1	Failure

- For the **REG_EXPAND_SZ** key type, use the caret symbol (^) with % inside the /d parameter.

Examples

To add the key *HKLM\Software\MyCo* on remote computer *ABC*, type:

```
reg add \\ABC\HKLM\Software\MyCo
```

To add a registry entry to *HKLM\Software\MyCo* with a value named *Data*, the type *REG_BINARY*, and data of *fe340ead*, type:

```
reg add HKLM\Software\MyCo /v Data /t REG_BINARY /d fe340ead
```

To add a multi-valued registry entry to *HKLM\Software\MyCo* with a value named *MRU*, the type *REG_MULTI_SZ*, and data of *fax\0mail\0*, type:

```
reg add HKLM\Software\MyCo /v MRU /t REG_MULTI_SZ /d fax\0mail\0
```

To add an expanded registry entry to *HKLM\Software\MyCo* with a value named *Path*, the type *REG_EXPAND_SZ*, and data of *%systemroot%*, type:

```
reg add HKLM\Software\MyCo /v Path /t REG_EXPAND_SZ /d ^%systemroot^%
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg compare

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Compares specified registry subkeys or entries.

Syntax

```
reg compare <keyname1> <keyname2> [{/v Valuename | /ve}] [{/oa | /od | /os | on}] [/s]
```

Parameters

Expand table

Parameter	Description
`<keyname1>`	Specifies the full path of the subkey or entry to be added. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
`<keyname2>`	Specifies the full path of the second subkey to be compared. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. Specifying only the computer name in *keyname2* causes the operation to use the path to the subkey specified in *keyname1*. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
/v `<Valuename>`	Specifies the value name to compare under the subkey.
/ve	Specifies that only entries that have a value name of null should be compared.

Parameter	Description
/oa	Specifies that all differences and matches are displayed. By default, only the differences are listed.
/od	Specifies that only differences are displayed. This is the default behavior.
/os	Specifies that only matches are displayed. By default, only the differences are listed.
/on	Specifies that nothing is displayed. By default, only the differences are listed.
/s	Compares all subkeys and entries recursively.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg compare** operation are:

Expand table

Value	Description
0	The comparison is successful and the result is identical.
1	The comparison failed.
2	The comparison was successful and differences were found.

- The symbols displayed in the results, include:

Expand table

Symbol	Description
=	*KeyName1* data is equal to *KeyName2* data.
<	*KeyName1* data is less than *KeyName2* data.
>	*KeyName1* data is greater than *KeyName2* data.

Examples

To compare all values under the key **MyApp** with all values under the key **SaveMyApp**, type:

```
reg compare HKLM\Software\MyCo\MyApp HKLM\Software\MyCo\SaveMyApp
```

To compare the value for the Version under the key **MyCo** and the value for the Version under the key **MyCo1**, type:

```
reg compare HKLM\Software\MyCo HKLM\Software\MyCo1 /v Version
```

To compare all subkeys and values under HKLM\Software\MyCo on the computer named ZODIAC, with all subkeys and values under HKLM\Software\MyCo on the local computer, type:

```
reg compare \\ZODIAC\HKLM\Software\MyCo \\. /s
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg copy

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies a registry entry to a specified location on the local or remote computer.

Syntax

```
reg copy <keyname1> <keyname2> [/s] [/f]
```

Parameters

Expand table

Parameter	Description
`<keyname1>`	Specifies the full path of the subkey or entry to be added. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
`<keyname2>`	Specifies the full path of the second subkey to be compared. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
/s	Copies all subkeys and entries under the specified subkey.
/f	Copies the subkey without prompting for confirmation.
/?	Displays help at the command prompt.

Remarks

- This command doesn't ask for confirmation when copying a subkey.
- The return values for the **reg compare** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To copy all subkeys and values under the key MyApp to the key SaveMyApp, type:

```
reg copy HKLM\Software\MyCo\MyApp HKLM\Software\MyCo\SaveMyApp /s
```

To copy all values under the key MyCo on the computer named ZODIAC to the key MyCo1 on the current computer, type:

```
reg copy \\ZODIAC\HKLM\Software\MyCo HKLM\Software\MyCo1
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg delete

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a subkey or entries from the registry.

Syntax

```
reg delete <keyname> [{/v valuename | /ve | /va}] [/f]
```

Parameters

Expand table

Parameter	Description
`<keyname1>`	Specifies the full path of the subkey or entry to be deleted. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
/v `<valuename>`	Deletes a specific entry under the subkey. If no entry is specified, then all entries and subkeys under the subkey will be deleted.
/ve	Specifies that only entries that have no value will be deleted.
/va	Deletes all entries within the specified key. Subkey entries that reside within the specified key are not deleted.
/f	Deletes the existing registry subkey or entry without asking for confirmation.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg delete** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To delete the registry key Timeout and its all subkeys and values, type:

```
reg delete HKLM\Software\MyCo\MyApp\Timeout
```

To delete the registry value MTU under HKLM\Software\MyCo on the computer named ZODIAC, type:

```
reg delete \\ZODIAC\HKLM\Software\MyCo /v MTU
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg export

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies the specified subkeys, entries, and values of the local computer into a file for transfer to other servers.

Syntax

```
reg export <keyname> <filename> [/y]
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey. The export operation only works with the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If the registry key name contains a space, enclose the key name in quotes.
`<filename>`	Specifies the name and path of the file to be created during the operation. The file must have a .reg extension.
/y	Overwrites any existing file with the name *filename* without prompting for confirmation.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg export** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To export the contents of all subkeys and values of the key MyApp to the file AppBkUp.reg, type:

```
reg export HKLM\Software\MyCo\MyApp AppBkUp.reg
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg import

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies the contents of a file that contains exported registry subkeys, entries, and values into the registry of the local computer.

Syntax

```
reg import <filename>
```

Parameters

Expand table

Parameter	Description
`<filename>`	Specifies the name and path of the file that has content to be copied into the registry of the local computer. This file must be created in advance by using **reg export**.
`/reg:32`	Specifies the key should be accessed using the 32-bit registry view.
`/reg:64`	Specifies the key should be accessed using the 64-bit registry view.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg import** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To import registry entries from the file named AppBkUp.reg, type:

```
reg import AppBkUp.reg
```

Related links

- Command-Line Syntax Key
- reg export command

Feedback

Was this page helpful? Yes No

reg load

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Writes saved subkeys and entries into a different subkey in the registry. This command is intended for use with temporary files that are used for troubleshooting or editing registry entries.

Syntax

```
reg load <keyname> <filename>
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey to be loaded. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
`<filename>`	Specifies the name and path of the file to be loaded. This file must be created in advance by using the **reg save** command, and must have a .hiv extension.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg load** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To load the file named TempHive.hiv to the key HKLM\TempHive, type:

```
reg load HKLM\TempHive TempHive.hiv
```

Related links

- Command-Line Syntax Key
- reg save command

Feedback

Was this page helpful? Yes No

reg query

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Returns a list of the next tier of subkeys and entries that are located under a specified subkey in the registry.

Syntax

```
reg query <keyname> [{/v <valuename> | /ve}] [/s] [/se <separator>] [/f
<data>] [{/k | /d}] [/c] [/e] [/t <Type>] [/z] [/reg:32] [/reg:64]
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
/v `<valuename>`	Specifies the registry value name that is to be queried. If omitted, all value names for *keyname* are returned. *Valuename* for this parameter is optional if the **/f** option is also used.
/ve	Runs a query for value names that are empty.
/s	Specifies to query all subkeys and value names recursively.
/se `<separator>`	Specifies the single value separator to search for in the value name type **REG_MULTI_SZ**. If *separator* isn't specified, **\0** is used.
/f `<data>`	Specifies the data or pattern to search for. Use double quotes if a string contains spaces. If not specified, a wildcard (*) is used as the search pattern.

Parameter	Description
/k	Specifies to search in key names only. Must be used with /f.
/d	Specifies to search in data only.
/c	Specifies that the query is case sensitive. By default, queries are not case sensitive.
/e	Specifies to return only exact matches. By default, all the matches are returned.
/t `<Type>`	Specifies registry types to search. Valid types are: **REG_SZ**, **REG_MULTI_SZ**, **REG_EXPAND_SZ**, **REG_DWORD**, **REG_BINARY**, **REG_NONE**. If not specified, all types are searched.
/z	Specifies to include the numeric equivalent for the registry type in search results.
/reg:32	Specifies the key should be accessed using the 32-bit registry view.
/reg:64	Specifies the key should be accessed using the 64-bit registry view.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg query** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To display the value of the name value Version in the HKLM\Software\Microsoft\ResKit key, type:

```
reg query HKLM\Software\Microsoft\ResKit /v Version
```

To display all subkeys and values under the key HKLM\Software\Microsoft\ResKit\Nt\Setup on a remote computer named ABC, type:

```
reg query \\ABC\HKLM\Software\Microsoft\ResKit\Nt\Setup /s
```

To display all the subkeys and values of the type REG_MULTI_SZ using # as the separator, type:

```

reg query HKLM\Software\Microsoft\ResKit\Nt\Setup /se #
```

To display the key, value, and data for exact and case sensitive matches of SYSTEM under the HKLM root of data type REG_SZ, type:

```

reg query HKLM /f SYSTEM /t REG_SZ /c /e
```

To display the key, value, and data that match **0F** in the data under the HKCU root key of data type REG_BINARY, type:

```

reg query HKCU /f 0F /d /t REG_BINARY
```

To display the keys, values, and data that match **asp.net** under the key HKLM\SOFTWARE\Microsoft and all subkeys, type:

```

reg query HKLM\SOFTWARE\Microsoft /s /f asp.net
```

To display only the keys that match **asp.net** under the key HKLM\SOFTWARE\Microsoft and all subkeys, type:

```

reg query HKLM\SOFTWARE\Microsoft /s /f asp.net /k
```

To display the value and data for value names of null (default) under HKLM\SOFTWARE, type:

```
```

```
reg query HKLM\SOFTWARE /ve
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

reg restore

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Writes saved subkeys and entries back to the registry.

Syntax

```
reg restore <keyname> <filename>
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey to be restored. The restore operation only works with the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If the registry key name contains a space, enclose the key name in quotes.
`<filename>`	Specifies the name and path of the file with content to be written into the registry. This file must be created in advance by using the **reg save** command, and must have a .hiv extension.
/?	Displays help at the command prompt.

Remarks

- Before editing any registry entries, you must save the parent subkey using the **reg save** command. If the edit fails, you can then restore the original subkey using the **reg restore** operation.
- The return values for the **reg restore** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To restore the file named NTRKBkUp.hiv into the key HKLM\Software\Microsoft\ResKit, and overwrite the existing contents of the key, type:

```
reg restore HKLM\Software\Microsoft\ResKit NTRKBkUp.hiv
```

Related links

- Command-Line Syntax Key
- reg save command

Feedback

Was this page helpful? Yes No

reg save

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Saves a copy of specified subkeys, entries, and values of the registry in a specified file.

Syntax

```
reg save <keyname> <filename> [/y]
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
`<filename>`	Specifies the name and path of the created file. If no path is specified, the current path is used.
/y	Overwrites an existing file with the name *filename* without prompting for confirmation.
/?	Displays help at the command prompt.

Remarks

- Before editing any registry entries, you must save the parent subkey using the **reg save** command. If the edit fails, you can then restore the original subkey using the **reg restore** operation.

- The return values for the **reg save** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To save the hive MyApp into the current folder as a file named AppBkUp.hiv, type:

```
reg save HKLM\Software\MyCo\MyApp AppBkUp.hiv
```

Related links

- Command-Line Syntax Key
- reg restore command

Feedback

Was this page helpful? Yes No

reg unload

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a section of the registry that was loaded using the **reg load** operation.

Syntax

```
reg unload <keyname>
```

Parameters

Expand table

Parameter	Description
`<keyname>`	Specifies the full path of the subkey. To specify a remote computer, include the computer name (in the format `\\<computername>\`) as part of the *keyname*. Omitting `\\<computername>\` causes the operation to default to the local computer. The *keyname* must include a valid root key. Valid root keys for the local computer are: **HKLM**, **HKCU**, **HKCR**, **HKU**, and **HKCC**. If a remote computer is specified, valid root keys are: **HKLM** and **HKU**. If the registry key name contains a space, enclose the key name in quotes.
/?	Displays help at the command prompt.

Remarks

- The return values for the **reg unload** operation are:

Expand table

Value	Description
0	Success
1	Failure

Examples

To unload the hive TempHive in the file HKLM, type:

```
reg unload HKLM\TempHive
```

> **⊗ Caution**
>
> Don't edit the registry directly unless you have no alternative. The registry editor bypasses standard safeguards, allowing settings that can degrade performance, damage your system, or even require you to reinstall Windows. You can safely alter most registry settings by using the programs in Control Panel or Microsoft Management Console (MMC). If you must edit the registry directly, back it up first.

Related links

- Command-Line Syntax Key
- reg load command

Feedback

Was this page helpful? Yes No

regini

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Modifies the registry from the command line or a script, and applies changes that were preset in one or more text files. You can create, modify, or delete registry keys, in addition to modifying the permissions on the registry keys.

For details on the format and content of the text script file that regini.exe uses to make changes to the registry, see How to change registry values or permissions from a command line or a script.

Syntax

```
regini [-m \\machinename | -h hivefile hiveroot][-i n] [-o outputwidth][-b]
textfiles...
```

Parameters

Expand table

Parameter	Description
-m `<\\computername>`	Specifies the remote computer name with a registry that is to be modified. Use the format **\ComputerName**.
-h `<hivefile hiveroot>`	Specifies the local registry hive to modify. You must specify the name of the hive file and the root of the hive in the format **hivefile hiveroot**.
-i `<n>`	Specifies the level of indentation to use to indicate the tree structure of registry keys in the command output. The **regdmp.exe** tool (which gets a registry key's current permissions in binary format) uses indentation in multiples of four, so the default value is **4**.
-o `<outputwidth>`	Specifies the width of the command output, in characters. If the output will appear in the command window, the default value is the width of the window. If the output is directed to a file, the default value is **240** characters.

Parameter	Description
-b	Specifies that **regini.exe** output is backward compatible with previous versions of **regini.exe**.
textfiles	Specifies the name of one or more text files that contain registry data. Any number of ANSI or Unicode text files can be listed.

Remarks

The following guidelines apply primarily to the content of the text files that contain registry data that you apply by using **regini.exe**.

- Use the semicolon as an end-of-line comment character. It must be the first non-blank character in a line.
- Use the backslash to indicate continuation of a line. The command will ignore all characters from the backslash up to (but not including) the first non-blank character of the next line. If you include more than one space before the backslash, it is replaced by a single space.
- Use hard-tab characters to control indentation. This indentation indicates the tree structure of the registry keys; however, these characters are converted to a single space regardless of their position.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

regsvr32

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Registers .dll files as command components in the registry.

Syntax

```
regsvr32 [/u] [/s] [/n] [/i[:cmdline]] <Dllname>
```

Parameters

Expand table

Parameter	Description
/u	Unregisters server.
/s	Prevents displaying messages.
/n	Prevents calling **DllRegisterServer**. This parameter requires you to also use the **/i** parameter.
/i: `<cmdline>`	Passes an optional command-line string (*cmdline*) to **DllInstall**. If you use this parameter with the **/u** parameter, it calls **DllUninstall**.
`<Dllname>`	The name of the .dll file that will be registered.
/?	Displays help at the command prompt.

Examples

To register the .dll for the Active Directory Schema, type:

```
regsvr32 schmmgmt.dll
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

relog

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Extracts performance counters from performance counter logs into other formats, such as text-TSV (for tab-delimited text), text-CSV (for comma-delimited text), binary-BIN, or SQL.

> **Note**
>
> For more information about incorporating **relog** into your Windows Management Instrumentation (WMI) scripts, see the Scripting blog.

Syntax

```
relog [<filename> [<filename> ...]] [/a] [/c <path> [<path> ...]] [/cf
<filename>] [/f  {bin|csv|tsv|SQL}] [/t <value>] [/o
{outputfile|DSN!CounterLog}] [/b <M/D/YYYY> [[<HH>:] <MM>:] <SS>] [/e
<M/D/YYYY> [[<HH>:] <MM>:] <SS>] [/config {<filename>|i}] [/q]
```

Parameters

Expand table

Parameter	Description
`filename [filename ...]`	Specifies the pathname of an existing performance counter log. You can specify multiple input files.
-a	Appends output file instead of overwriting. This option does not apply to SQL format where the default is always to append.
-c `path [path ...]`	Specifies the performance counter path to log. To specify multiple counter paths, separate them with a space and enclose the counter paths in quotation marks (for example, `"path1 path2"`).
-cf filename	Specifies the pathname of the text file that lists the performance counters to be included in a relog file. Use this option to list counter

Parameter	Description
	paths in an input file, one per line. Default setting is all counters in the original log file are relogged.
-f `{bin | csv | tsv | SQL}`	Specifies the pathname of the output file format. The default format is **bin**. For a SQL database, the output file specifies the `DSN!CounterLog`. You can specify the database location by using the ODBC manager to configure the DSN (Database System Name).
-t value	Specifies sample intervals in *n* records. Includes every nth data point in the relog file. Default is every data point.
-o `{Outputfile | SQL:DSN!Counter_Log}`	Specifies the pathname of the output file or SQL database where the counters will be written. **Note:** For the 64-bit and 32-bit versions of relog.exe, you must define a DSN in the ODBC Data Source (64-bit and 32-bit respectively) on the system. Use the "SQL Server" ODBC driver to define a DSN.
-b `<M/D/YYYY> [[<HH>:]<MM>:]<SS>]`	Specifies the beginning time to copy the first record from the input file. Date and time must be in this exact format M/D/YYYYHH:MM:SS.
-e `<M/D/YYYY> [[<HH>:]<MM>:]<SS>]`	Specifies the end time to copy the last record from the input file. Date and time must be in this exact format M/D/YYYYHH:MM:SS.
-config `{filename | i}`	Specifies the pathname of the settings file that contains command-line parameters. If you're using a configuration file, you can use **-i** as a placeholder for a list of input files that can be placed on the command line. If you're using the command line, don't use **-i**. You can also use wildcards, such as `*.blg` to specify several input file names at once.
-q	Displays the performance counters and time ranges of log files specified in the input file.
-y	Bypasses prompting by answering "yes" to all questions.
/?	Displays help at the command prompt.

Remarks

- The general format for counter paths is as follows: `[\<computer>] \<object> [<parent>\<instance#index>] \<counter>]` where the parent, instance, index, and counter components of the format may contain either a valid name or a wildcard character. The computer, parent, instance, and index components aren't necessary for all counters.
- You determine the counter paths to use based on the counter itself. For example, the **LogicalDisk** object has an instance `<index>`, so you must provide the `<#index>`

or a wildcard. Therefore, you could use the following format: `\LogicalDisk(*/*#*)\\*`.

- In comparison, the **Process** object doesn't require an instance `<index>`. Therefore, you can use the following format: `\Process(*)\ID Process`.
- If a wildcard character is specified in the **Parent** name, all instances of the specified object that match the specified instance and counter fields will be returned.
- If a wildcard character is specified in the **Instance** name, all instances of the specified object and parent object will be returned if all instance names corresponding to the specified index match the wildcard character.
- If a wildcard character is specified in the **Counter** name, all counters of the specified object are returned.
- Partial counter path string matches (for example, pro*) aren't supported.
- Counter files are text files that list one or more of the performance counters in the existing log. Copy the full counter name from the log or the **/q** output in `<computer>\<object>\<instance>\<counter>` format. List one counter path on each line.
- When run, the **relog** command copies specified counters from every record in the input file, converting the format if necessary. Wildcard paths are allowed in the counter file.
- Use the **/t** parameter to specify that input files are inserted into output files at intervals of every `nth` record. By default, data is relogged from every record.
- You can specify that your output logs include records from before the beginning time (that is, **/b**) to provide data for counters that require computation values of the formatted value. The output file will have the last records from input files with timestamps less than the **/e** (that is, end time) parameter.
- The contents of the setting file used with the **/config** option should have the following format: `<commandoption>\<value>`, where `<commandoption>` is a command line option and `<value>` specifies its value.

##Q# Examples

To resample existing trace logs at fixed intervals of 30, list counter paths, output files, and formats, type:

```
```

```
relog c:\perflogs\daily_trace_log.blg /cf counter_file.txt /o
c:\perflogs\reduced_log.csv /t 30 /f csv
```

To resample existing trace logs at fixed intervals of 30, list counter paths, and output file, type:

```
relog c:\perflogs\daily_trace_log.blg /cf counter_file.txt /o
c:\perflogs\reduced_log.blg /t 30
```

To resample existing trace logs into a database, type:

```
relog "c:\perflogs\daily_trace_log.blg" -f sql -o
"SQL:sql2016x64odbc!counter_log"
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

ren

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Renames files or directories.

> **Note**
>
> This command is the same as the rename command.

Syntax

```
ren [<drive>:][<path>]<filename1> <filename2>
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]<filename1>`	Specifies the location and name of the file or set of files you want to rename. *Filename1* can include wildcard characters (* and ?).
`<filename2>`	Specifies the new name for the file. You can use wildcard characters to specify new names for multiple files.
/?	Displays help at the command prompt.

Remarks

- You can't specify a new drive or path when renaming files. You also can't use this command to rename files across drives or to move files to a different directory.
- Characters represented by wildcard characters in *filename2* will be identical to the corresponding characters in *filename1*.

- *Filename2* must be a unique file name. If *filename2* matches an existing file name, the following message appears: `Duplicate file name or file not found`.

Examples

To change all the .txt file name extensions in the current directory to .doc extensions, type:

```
ren *.txt *.doc
```

To change the name of a directory from *Chap10* to *Part10*, type:

```
ren chap10 part10
```

Related links

- Command-Line Syntax Key
- rename command

Feedback

Was this page helpful? Yes No

rename

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Renames one or more files or directories. Use wildcard characters (`*` and `?`) to rename multiple files in a single command.

> **Note**
>
> This command is the same as the ren command.

Syntax

syntax

```
rename [<drive>:][<path>]<filename1> <filename2>
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>] <filename1>`	Specifies the location and name of the file or set of files you want to rename. *Filename1* can include wildcard characters (* and ?).
`<filename2>`	Specifies the new name for the file. Use wildcard characters to specify new names for multiple files.
`/?`	Shows help at the command prompt.

Remarks

- You can't specify a new drive or path when renaming files. You also can't use this command to rename files across drives or to move files to a different directory.
- Wildcard characters in *filename2* represent the corresponding characters in *filename1*.

- *Filename2* must be a unique file name. If *filename2* matches an existing file name, the following message appears: `Duplicate file name or file not found`.

Examples

To change all the .txt file name extensions in the current directory to .doc extensions, type:

Windows Command Prompt

```
rename *.txt *.doc
```

When you use wildcards in *filename2*, the command replaces each wildcard with the characters in the *same position* in the original name. A **?** keeps the single character in that position, and a * keeps all remaining characters from that position to the end of the name or extension. Because the match is positional, the results can be surprising.

To replace a fixed-length prefix in a set of file names while keeping the rest of each name, type:

Windows Command Prompt

```
rename img*.jpg pic*.jpg
```

Because `img` and `pic` are both three characters, each * starts at the same position, so only the prefix changes:

Expand table

Original file name	New file name
`img001.jpg`	`pic001.jpg`
`img_final.jpg`	`pic_final.jpg`

> **Note**
>
> The replacement prefix must be the same length as the text it replaces. For example, `rename img*.jpg photo*.jpg` renames `img001.jpg` to `photo1.jpg`, because `photo` is five characters and overwrites the first five characters (`img00`) of the original name.

To change a single character at a known position and keep the rest of the name, type:

Windows Command Prompt

```
rename ??-*.log ??_*.log
```

This command matches names whose third character is a hyphen and replaces only that hyphen with an underscore:

Expand table

Original file name	New file name
`ab-server.log`	`ab_server.log`
`db-audit.log`	`db_audit.log`

To append an extension without removing the current one, type:

Windows Command Prompt

```
rename *.log *.log.bak
```

The * keeps the original name and extension, and the command appends `.bak` to the end:

Expand table

Original file name	New file name
`server.log`	`server.log.bak`
`app.debug.log`	`app.debug.log.bak`

To change the name of a directory from *Chap10* to *Part10*, type:

Windows Command Prompt

```
rename chap10 part10
```

Related links

- Command-Line Syntax Key
- ren command

ⓘ**Note:** The author created this article with assistance from AI. Learn more

Last updated on 09/08/2026

repair-bde

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Attempts to reconstruct critical parts of a severely damaged drive and salvage recoverable data if the drive was encrypted by using BitLocker and if it has a valid recovery password or recovery key for decryption.

> **ⓘ Important**
>
> If the BitLocker metadata data on the drive is corrupt, you must be able to supply a backup key package in addition to the recovery password or recovery key. If you used the default key back up setting for Active Directory Domain Services, your key package is backed up there. You can use the **BitLocker: Use BitLocker Recovery Password Viewer** to obtain the key package from AD DS.
>
> Using the key package and either the recovery password or recovery key, you can decrypt portions of a BitLocker-protected drive, even if the disk is corrupted. Each key package works only for a drive with the corresponding drive identifier.

Syntax

```
repair-bde <inputvolume> <outputvolumeorimage> [-rk] [-rp] [-pw] [-kp] [-lf] [-f] [{-?|/?}]
```

> **⚠ Warning**
>
> The contents of the output volume will be **completely deleted and overwritten** by the decrypted contents from the damaged BitLocker drive. If you want to save any existing data on the selected target drive, move the existing data to other reliable backup media first, before running the `repair-bde` command.

Parameters

Expand table

Parameter	Description
`<inputvolume>`	Identifies the drive letter of the BitLocker-encrypted drive that you want to repair. The drive letter must include a colon; for example: **C:**. If the path to a key package isn't specified, this command searches the drive for a key package. In the event that the hard drive is damaged, this command might not be able to find the package and will prompt you to provide the path.
`<outputvolumeorimage>`	Identifies the drive on which to store the content of the repaired drive. All information on the output drive will be overwritten.
-rk	Identifies the location of the recovery key that should be used to unlock the volume. This command can also be specified as **-recoverykey**.
-rp	Identifies the numerical recovery password that should be used to unlock the volume. This command can also be specified as **-recoverypassword**.
-pw	Identifies the password that should be used to unlock the volume. This command can also be specified as **-password**
-kp	Identifies the recovery key package that can be used to unlock the volume. This command can also be specified as **-keypackage**.
-lf	Specifies the path to the file that will store Repair-bde error, warning, and information messages. This command may also be specified as **-logfile**.
-f	Forces a volume to be dismounted even if it cannot be locked. This command can also be specified as **-force**.
-? or /?	Displays Help at the command prompt.

Limitations

The following limitations exist for the this command:

- This command can't repair a drive that failed during the encryption or decryption process.
- This command assumes that if the drive has any encryption, then the drive has been fully encrypted.

Examples

To attempt to repair drive C:, to write the content from drive C: to drive D: using the recovery key file (RecoveryKey.bek) stored on drive F:, and to write the results of this attempt to the log file (log.txt) on drive Z:, type:

```
repair-bde C: D: -rk F:\RecoveryKey.bek –lf Z:\log.txt
```

To attempt to repair drive C: and to write the content from drive C: to drive D: using the 48-digit recovery password specified, type:

```
repair-bde C: D: -rp 111111-222222-333333-444444-555555-666666-777777-888888
```

> **Note**
>
> The recovery password should be typed in eight blocks of six digits with a hyphen separating each block.

To force drive C: to dismount, attempt to repair drive C:, and then to write the content from drive C: to drive D: using the recovery key package and recovery key file (RecoveryKey.bek) stored on drive F:, type:

```
repair-bde C: D: -kp F:\RecoveryKeyPackage -rk F:\RecoveryKey.bek -f
```

To attempt to repair drive C: and to write the content from drive C: to drive D:, where you must type a password to unlock drive C: (when prompted), type:

```
repair-bde C: D: -pw
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

replace

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Replace existing files in a directory. If used with the **/a** option, this command adds new files to a directory instead of replacing existing files.

Syntax

```
replace [<drive1>:][<path1>]<filename> [<drive2>:][<path2>] [/a] [/p] [/r]
[/w]
replace [<drive1>:][<path1>]<filename> [<drive2>:][<path2>] [/p] [/r] [/s]
[/w] [/u]
```

Parameters

Expand table

Parameter	Description
`[<drive1>:][<path1>]<filename>`	Specifies the location and name of the source file or set of files. The *filename* option is required, and can include wildcard characters (* and **?**).
`[<drive2>:][<path2>]`	Specifies the location of the destination file. You can't specify a file name for files you replace. If you don't specify a drive or path, this command uses the current drive and directory as the destination.
/a	Adds new files to the destination directory instead of replacing existing files. You can't use this command-line option with the **/s** or **/u** command-line option.
/p	Prompts you for confirmation before replacing a destination file or adding a source file.
/r	Replaces Read-only and unprotected files. If you attempt to replace a Read-only file, but you don't specify **/r**, an error results and stops the replacement operation.

Parameter	Description
/w	Waits for you to insert a disk before the search for source files begins. If you don't specify **/w**, this command begins replacing or adding files immediately after you press ENTER.
/s	Searches all subdirectories in the destination directory and replaces matching files. You can't use **/s** with the **/a** command-line option. The command doesn't search subdirectories that are specified in *Path1*.
/u	Replaces only those files on the destination directory that are older than those in the source directory. You can't use **/u** with the **/a** command-line option.
/?	Displays help at the command prompt.

Remarks

- As this command adds or replaces files, the file names appear on the screen. After this command is done, a summary line is displayed in one of the following formats:

```
nnn files added
nnn files replaced
no file added
no file replaced
```

- If you're using floppy disks and you need to switch disks while running this command, you can specify the **/w** command-line option so that this command waits for you to switch the disks.
- You can't use this command to update hidden files or system files.
- The following table shows each exit code and a brief description of its meaning:

Expand table

Exit code	Description
0	This command successfully replaced or added the files.
1	This command encountered an incorrect version of MS-DOS.
2	This command couldn't find the source files.
3	This command couldn't find the source or destination path.

Exit code	Description
5	The user doesn't have access to the files that you want to replace.
8	There is insufficient system memory to carry out the command.
11	The user used the wrong syntax on the command line.

> **Note**
>
> You can use the ERRORLEVEL parameter on the **if** command line in a batch program to process exit codes that are returned by this command.

Examples

To update all the versions of a file named *Phones.cli* (which appear in multiple directories on drive C:), with the latest version of the *Phones.cli* file from a floppy disk in drive A:, type:

```
replace a:\phones.cli c:\ /s
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rexec

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Runs a specified command on a remote host. The remote host must be running a rexecd service (or daemon) for rexec to connect to.

> **Important**
>
> The **rexec** command has been deprecated, and isn't guaranteed to be supported in Windows.

Feedback

Was this page helpful? Yes No

risetup

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates an operating system image by pulling the files from the original Windows Server 2003 CD or from a customized distribution folder containing these files.

> **Important**
>
> The **rexec** command has been deprecated, and isn't guaranteed to be supported in Windows.

Feedback

Was this page helpful? Yes No

rmdir

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a directory.

The **rmdir** command can also run from the Windows Recovery Console, using different parameters. For more information, see Windows Recovery Environment (WinRE).

> **Note**
>
> This command is the same as the rd command.

Syntax

```
rmdir [<drive>:]<path> [/s [/q]]
```

Parameters

Expand table

Parameter	Description
`[<drive>:]<path>`	Specifies the location and the name of the directory that you want to delete. *Path* is required. If you include a backslash () at the beginning of the specified *path*, then the *path* starts at the root directory (regardless of the current directory).
/s	Deletes a directory tree (the specified directory and all its subdirectories, including all files).
/q	Specifies quiet mode. Does not prompt for confirmation when deleting a directory tree. The **/q** parameter works only if **/s** is also specified. **CAUTION:** When you run in quiet mode, the entire directory tree is deleted without confirmation. Make sure that important files are moved or backed up before using the **/q** command-line option.
/?	Displays help at the command prompt.

Remarks

- You can't delete a directory that contains files, including hidden or system files. If you attempt to do so, the following message appears:

  `The directory is not empty`

  Use the **dir /a** command to list all files (including hidden and system files). Then use the **attrib** command with **-h** to remove hidden file attributes, **-s** to remove system file attributes, or **-h -s** to remove both hidden and system file attributes. After the hidden and file attributes have been removed, you can delete the files.

- You can't use the **rmdir** command to delete the current directory. If you attempt to delete the current directory, the following error message appears:

  `The process can't access the file because it is being used by another process.`

  If you receive this error message, you must change to a different directory (not a subdirectory of the current directory), and then try again.

Examples

To change to the parent directory so you can safely remove the desired directory, type:

```

cd ..
```

To remove a directory named *test* (and all its subdirectories and files) from the current directory, type:

```

rmdir /s test
```

To run the previous example in quiet mode, type:

```

rmdir /s /q test
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

robocopy

Article • 03/17/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies file data from one location to another.

Syntax

Windows Command Prompt

```
robocopy <source> <destination> [<file>[ ...]] [<options>]
```

For example, to copy a file named *yearly-report.mov* from *c:\reports* to a file share *\\marketing\videos* while enabling multi-threading for higher performance (with the **/mt** parameter) and the ability to restart the transfer in case it's interrupted (with the **/z** parameter), type:

Windows Command Prompt

```
robocopy c:\reports "\\marketing\videos" yearly-report.mov /mt /z
```

> **Important**
>
> If any data is copied from the *root* of a device, the destination directory adopts the "hidden" and "system" attributes during the copy process.

Parameters

Expand table

Parameter	Description
`<source>`	Specifies the path to the source directory.
`<destination>`	Specifies the path to the destination directory.
`<file>`	Specifies the file or files to be copied. Wildcard characters (* or ?) are supported. If you don't specify this parameter, `*.*` is used as the default value.

Parameter	Description
`<options>`	Specifies the options to use with the **robocopy** command, including **copy**, **file**, **retry**, **logging**, and **job** options.

Copy options

Expand table

Option	Description
/s	Copies subdirectories. This option automatically excludes empty directories.
/e	Copies subdirectories. This option automatically includes empty directories.
/lev:`<n>`	Copies only the top *n* levels of the source directory tree.
/z	Copies files in restartable mode. In restartable mode, should a file copy be interrupted, robocopy can pick up where it left off rather than recopying the entire file.
/b	Copies files in backup mode. In backup mode, robocopy overrides file and folder permission settings (ACLs), which might otherwise block access.
/zb	Copies files in restartable mode. If file access is denied, switches to backup mode.
/j	Copies using unbuffered I/O (recommended for large files).
/efsraw	Copies all encrypted files in EFS RAW mode.
/copy: `<copyflags>`	Specifies which file properties to copy. The valid values for this option are: • **D** - Data • **A** - Attributes • **T** - Time stamps • **X** - Skip alt data streams • **S** - NTFS access control list (ACL) • **O** - Owner information • **U** - Auditing information The default value for the **/COPY** option is **DAT** (data, attributes, and time stamps). The **X** flag is ignored if either **/B** or **/ZB** is used.
/dcopy: `<copyflags>`	Specifies what to copy in directories. The valid values for this option are: • **D** - Data • **A** - Attributes • **T** - Time stamps • **E** - Extended attribute • **X** - Skip alt data streams

Option	Description
	The default value for this option is **DA** (data and attributes).
/sec	Copies files with security (equivalent to **/copy:DATS**).
/copyall	Copies all file information (equivalent to **/copy:DATSOU**).
/nocopy	Copies no file information (useful with **/purge**).
/secfix	Fixes file security on all files, even skipped ones.
/timfix	Fixes file times on all files, even skipped ones.
/purge	Deletes destination files and directories that no longer exist in the source. Using this option with the **/e** option and a destination directory, allows the destination directory security settings to not be overwritten.
/mir	Mirrors a directory tree (equivalent to **/e** plus **/purge**). Using this option with the **/e** option and a destination directory, overwrites the destination directory security settings.
/mov	Moves files, and deletes them from the source after they're copied.
/move	Moves files and directories, and deletes them from the source after they're copied.
/a+: [RASHCNET]	Adds the specified attributes to copied files. The valid values for this option are: • **R** - Read only • **A** - Archive • **S** - System • **H** - Hidden • **C** - Compressed • **N** - Not content indexed • **E** - Encrypted • **T** - Temporary
/a-: [RASHCNETO]	Removes the specified attributes from copied files. The valid values for this option are: • **R** - Read only • **A** - Archive • **S** - System • **H** - Hidden • **C** - Compressed • **N** - Not content indexed • **E** - Encrypted • **T** - Temporary • **O** - Offline
/create	Creates a directory tree and zero-length files only.

Option	Description
/fat	Creates destination files by using 8.3 character-length FAT file names only.
/256	Turns off support for paths longer than 256 characters.
/mon:`<n>`	Monitors the source and runs again when more than *n* changes are detected.
/mot:`<m>`	Monitors the source and runs again in *m* minutes if changes are detected.
/rh:hhmm-hhmm	Specifies run times when new copies can be started.
/pf	Checks run times on a per file (not per-pass) basis.
/ipg:`<n>`	Specifies the inter-packet gap to free bandwidth on slow lines.
/sj	Copies junctions (soft-links) to the destination path instead of link targets.
/sl	Don't follow symbolic links and instead create a copy of the link.
/mt:`<n>`	Creates multi-threaded copies with *n* threads. *n* must be an integer between 1 and 128. The default value for *n* is 8. For better performance, redirect your output using **/log** option. The **/mt** parameter can't be used with the **/ipg** and **/efsraw** parameters.
/nodcopy	Copies no directory info (the default **/dcopy:DA** is done).
/nooffload	Copies files without using the Windows Copy Offload mechanism.
/compress	Requests network compression during file transfer, if applicable.
/sparse:`<y|n>`	Enables or disables retaining the sparse state of files during copy process. If no option is selected, it defaults to **yes** (enabled).
/noclone	Doesn't attempt block cloning as an optimization.

> **ⓘ Important**
>
> When using the **/secfix** copy option, specify the type of security information you want to copy, using one of these additional copy options:
>
> - **/copyall**
> - **/copy:o**
> - **/copy:s**
> - **/copy:u**
> - **/sec**

> **Note**
>
> The **/mt** parameter was introduced in Windows Server 2008 R2 and its functionality applies to current versions of Windows Server.

Copy file throttling options

Expand table

Option	Description
/iomaxsize:`<n>`[kmg]	The requested max i/o size per read/write cycle in *n* **k**ilobytes, **m**egabytes, or **g**igabytes.
/iorate:`<n>`[kmg]	The requested i/o rate in *n* **k**ilobytes **m**egabytes, or **g**igabytes per second.
/threshold:`<n>`[kmg]	The file size threshold for throttling in *n* **k**ilobytes, **m**egabytes, or **g**igabytes (see Remarks).

These throttling options are used to specify the maximum I/O bandwidth that Robocopy allows to be used in bytes per second. If not specifying in bytes per second, whole numbers can be used if **k**, **m**, or **g** are specified. The minimum I/O bandwidth that is throttled is **524288** bytes even if a lesser value is specified.

File selection options

Expand table

Option	Description
/a	Copies only files for which the **Archive** attribute is set.
/m	Copies only files for which the **Archive** attribute is set, and resets the **Archive** attribute.
/ia:`[RASHCNETO]`	Includes only files for which any of the specified attributes are set. The valid values for this option are: • **R** - Read only • **A** - Archive • **S** - System • **H** - Hidden • **C** - Compressed • **N** - Not content indexed • **E** - Encrypted • **T** - Temporary

Option	Description
	• **O** - Offline
/xa:`[RASHCNETO]`	Excludes files for which any of the specified attributes are set. The valid values for this option are: • **R** - Read only • **A** - Archive • **S** - System • **H** - Hidden • **C** - Compressed • **N** - Not content indexed • **E** - Encrypted • **T** - Temporary • **O** - Offline
/xf `<filename>[ ...]`	Excludes files that match the specified names or paths. Wildcard characters (***** and **?**) are supported.
/xd `<directory>[ ...]`	Excludes directories that match the specified names and paths.
/xc	Excludes existing files with the same timestamp, but different file sizes.
/xn	Source directory files newer than the destination are excluded from the copy.
/xo	Source directory files older than the destination are excluded from the copy.
/xx	Excludes extra files and directories present in the destination but not the source. Excluding extra files doesn't delete files from the destination.
/xl	Excludes "lonely" files and directories present in the source but not the destination. Excluding lonely files prevents any new files from being added to the destination.
/im	Include modified files (differing change times).
/is	Includes the same files. Same files are identical in name, size, times, and all attributes.
/it	Includes "tweaked" files. Tweaked files have the same name, size, and times, but different attributes.
/max:`<n>`	Specifies the maximum file size (to exclude files bigger than *n* bytes).
/min:`<n>`	Specifies the minimum file size (to exclude files smaller than *n* bytes).
/maxage:`<n>`	Specifies the maximum file age to exclude files older than *n* days or a date based on when the files were last *modified*.

Option	Description
/minage:<n>	Specifies the minimum file age to exclude files newer than *n* days or a date based on when the files were last *modified*.
/maxlad:<n>	Specifies the maximum last access date (excludes files unused since *n*).
/minlad:<n>	Specifies the minimum last access date (excludes files used since *n*) If *n* is less than 1900, *n* specifies the number of days. Otherwise, *n* specifies a date in the format YYYYMMDD.
/xj	Excludes junction points, which are normally included by default.
/fft	Assumes FAT file times (two-second precision).
/dst	Compensates for one-hour DST time differences.
/xjd	Excludes junction points for directories.
/xjf	Excludes junction points for files.

Retry options

Expand table

Option	Description
/r:<n>	Specifies the number of retries on failed copies. The default value of *n* is 1,000,000 (one million retries).
/w:<n>	Specifies the wait time between retries, in seconds. The default value of *n* is 30 (wait time 30 seconds).
/reg	Saves the values specified in the **/r** and **/w** options as default settings in the registry.
/tbd	Specifies that the system waits for share names to be defined (retry error 67).
/lfsm	Operate in low free space mode that enables copy, pause, and resume (see Remarks).
/lfsm:<n> [kmg]	Specifies the floor size in *n* **k**ilobytes, **m**egabytes, or **g**igabytes.

Logging options

Expand table

Option	Description
/l	Specifies that files are to be listed only (and not copied, deleted, or time stamped).
/x	Reports all extra files, not just the ones that are selected.
/v	Produces verbose output, and shows all skipped files.
/ts	Includes source file time stamps in the output.
/fp	Includes the full path names of the files in the output.
/bytes	Prints sizes as bytes.
/ns	Specifies that file sizes aren't to be logged.
/nc	Specifies that file classes aren't to be logged.
/nfl	Specifies that file names aren't to be logged.
/ndl	Specifies that directory names aren't to be logged.
/np	Specifies to not display the progress of the copying operation (the number of files or directories copied so far).
/eta	Shows the estimated time of arrival (ETA) of the copied files.
/log:`<logfile>`	Writes the status output to the log file (overwrites the existing log file).
/log+:`<logfile>`	Writes the status output to the log file (appends the output to the existing log file).
/unilog:`<logfile>`	Writes the status output to the log file as unicode text (overwrites the existing log file).
/unilog+:`<logfile>`	Writes the status output to the log file as Unicode text (appends the output to the existing log file).
/tee	Writes the status output to the console window, and to the log file.
/njh	Specifies that there's no job header.
/njs	Specifies that there's no job summary.
/unicode	Displays the status output as unicode text.

Job options

Expand table

Option	Description
/job: `<jobname>`	Specifies that parameters are to be derived from the named job file. To run `/job:jobname`, you must first run the `/save:jobname` parameter to create the job file.
/save: `<jobname>`	Specifies that parameters are to be saved to the named job file. This must be ran before running `/job:jobname`. All copy, retry, and logging options must be specified before this parameter.
/quit	Quits after processing command line (to view parameters).
/nosd	Indicates that no source directory is specified.
/nodd	Indicates that no destination directory is specified.
/if	Includes the specified files.

Remarks

- Using **/PURGE** or **/MIR** on the root directory of the volume formerly caused robocopy to apply the requested operation on files inside the System Volume Information directory as well. This is no longer the case as if either is specified, robocopy skips any files or directories with that name in the top-level source and destination directories of the copy session.
- Modified files classification applies only when both source and destination filesystems support change timestamps, such as NTFS, and the source and destination files have different change times but are otherwise the same. These files aren't copied by default. Specify **/IM** to include them.
- The **/DCOPY:E** flag requests that extended attribute copying should be attempted for directories. Robocopy continues the copy operation even if a directory's EAs couldn't be copied. This flag isn't included in **/COPYALL**.
- If either **/IoMaxSize** or **/IoRate** are specified, robocopy enables copy file throttling to reduce system load. Both can be adjusted to optimal values and copy parameters, but the system and robocopy are allowed to adjust them to allowed values as necessary.
- If **/Threshold** is used, it specifies a minimum file size for engaging throttling. Files below that size aren't throttled. Values for all three parameters can be followed by an optional suffix character such as [KMG] (kilobytes, megabytes, gigabytes).
- Using **/LFSM** requests robocopy to operate in 'low free space mode'. In this mode, robocopy pauses whenever a file copy would cause the destination volume's free

space to go below a 'floor' value. This value can be explicitly specified using **/LFSM:*n***[KMG] flag.

- If **/LFSM** is specified with no explicit floor value, the floor is set to 10% of the destination volume's size. Low free space mode is incompatible with **/MT** and **/EFSRAW**.

Exit (return) codes

Expand table

Value	Description
0	No files were copied. No failure was encountered. No files were mismatched. The files already exist in the destination directory; therefore, the copy operation was skipped.
1	All files were copied successfully.
2	There are some additional files in the destination directory that aren't present in the source directory. No files were copied.
3	Some files were copied. Additional files were present. No failure was encountered.
5	Some files were copied. Some files were mismatched. No failure was encountered.
6	Additional files and mismatched files exist. No files were copied and no failures were encountered meaning that the files already exist in the destination directory.
7	Files were copied, a file mismatch was present, and additional files were present.
8	Several files didn't copy.

> **Note**
>
> Any value equal to or greater than **8** indicates that there was at least one failure during the copy operation.

Examples

It's highly recommended when running the `robocopy` command to create a log file that can be viewed once the process completes verifying its integrity. In the following examples, each one uses the `/LOG:` parameter. To append any log information to the same log file, use the `/LOG+:` parameter instead.

To copy all files and subdirectories, including empty directories, from the "Records" folder to the "Backup" folder on drive "D", type:

Windows Command Prompt

```
robocopy C:\Users\Admin\Records D:\Backup /E /ZB /LOG:C:\Logs\Backup.log
```

To mirror the contents of the "Records" folder to the "Backup" folder on drive "D", delete any files in the destination that don't exist in the source with 2 retries and waiting 5 seconds between each retry, type:

Windows Command Prompt

```
robocopy C:\Users\Admin\Records D:\Backup /MIR /R:2 /W:5
/LOG:C:\Logs\Backup.log
```

To copy all files and subdirectories that aren't empty from the "Records" folder to the "Backup" folder on drive "D", retaining the file data, attributes, and timestamps with 16 multi-threaded copy operation, type:

Windows Command Prompt

```
robocopy C:\Users\Admin\Records D:\Backup /S /E /COPY:DAT /MT:16
/LOG:C:\Logs\Backup.log
```

To move files and subdirectories, excluding empty directories, from the "Records" folder to the "Backup" folder on drive "D", and exclude files older than 7 days, type:

Windows Command Prompt

```
robocopy C:\Users\Admin\Records D:\Backup /S /MAXAGE:7 /MOV
/LOG:C:\Logs\Backup.log
```

To copy all files and subdirectories, including empty directories, from the "Records" folder to the "Backup" folder on drive "D" showing the estimated time for each file and delete any files and directories in the destination that don't exist from the source, type:

Windows Command Prompt

```
robocopy C:\Users\Admin\Records D:\Backup /ETA /PURGE
/LOG:C:\Logs\Backup.log
```

To copy all files and subdirectories from the folder named "Records" on the "C" drive to a folder named "Backup" on the "D" drive while limiting the I/O rate to 1 megabyte per

second during the copy operation, type:

Windows Command Prompt

```
robocopy C:\Records D:\Backup /iorate:1m
```

To skip copying files from a source folder to a destination folder when the files already exist in the destination folder, regardless of whether they're newer, older, or modified, type:

Windows Command Prompt

```
robocopy C:\Source C:\Destination /XC /XN /XO
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

route

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays and modifies the entries in the local IP routing table. If used without parameters, **route** displays help at the command prompt.

> **Important**
>
> This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network Connections.

Syntax

Windows Command Prompt

```
route [/f] [/p] [<command> [<destination>] [mask <netmask>] [<gateway>] [metric
<metric>]] [if <interface>]]
```

Parameters

Expand table

Parameter	Description
/f	Clears the routing table of all entries that are not host routes (routes with a netmask of 255.255.255.255), the loopback network route (routes with a destination of 127.0.0.0 and a netmask of 255.0.0.0), or a multicast route (routes with a destination of 224.0.0.0 and a netmask of 240.0.0.0). If this is used in conjunction with one of the commands (such as add, change, or delete), the table is cleared prior to running the command.
/p	When used with the add command, the specified route is added to the registry and is used to initialize the IP routing table whenever the TCP/IP protocol is started. By default, added routes are not preserved when the TCP/IP protocol is started. When used with the print command, the list of persistent routes is displayed. This parameter is ignored for all other commands. Persistent routes are stored in the registry location **HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters\PersistentRoutes**.
`<command>`	Specifies the command you want to run. The valid commands include: • **add** - Adds a route. • **change** - Modifies an existing route. • **delete:** - Deletes a route or routes. • **print** - Prints a route or routes.

Parameter	Description
`<destination>`	Specifies the network destination of the route. The destination can be an IP network address (where the host bits of the network address are set to 0), an IP address for a host route, or 0.0.0.0 for the default route.
`<mask>`	Specifies the next parameter for the 'netmask' value.
`<netmask>`	Specifies the network destination subnet mask. Defaults to 255.255.255.255 if not specified.
`<gateway>`	Specifies the forwarding or next hop IP address over which the set of addresses defined by the network destination and subnet mask are reachable. For locally attached subnet routes, the gateway address is the IP address assigned to the interface that is attached to the subnet. For remote routes, available across one or more routers, the gateway address is a directly reachable IP address that is assigned to a neighboring router.
metric `<metric>`	Specifies an integer cost metric (ranging from 1 to 9999) for the route, which is used when choosing among multiple routes in the routing table that most closely match the destination address of a packet being forwarded. The route with the lowest metric is chosen. The metric can reflect the number of hops, the speed of the path, path reliability, path throughput, or administrative properties.
if `<interface>`	Specifies the interface index for the interface over which the destination is reachable. For a list of interfaces and their corresponding interface indexes, use the display of the route print command. You can use either decimal or hexadecimal values for the interface index. For hexadecimal values, precede the hexadecimal number with 0x. When the if parameter is omitted, the interface is determined from the gateway address.
/?	Displays help at the command prompt.

Remarks

- Large values in the **metric** column of the routing table are the result of allowing TCP/IP to automatically determine the metric for routes in the routing table based on the configuration of IP address, subnet mask, and default gateway for each LAN interface. Automatic determination of the interface metric, enabled by default, determines the speed of each interface and adjusts the metrics of routes for each interface so that the fastest interface creates the routes with the lowest metric. To remove the large metrics, disable the automatic determination of the interface metric from the advanced properties of the TCP/IP protocol for each LAN connection.

- Names can be used for *destination* if an appropriate entry exists in the local *Networks* file stored in the `systemroot\System32\Drivers\\` folder. Names can be used for the *gateway* as long as they can be resolved to an IP address through standard host name resolution techniques such as Domain Name System (DNS) queries, use of the local Hosts file stored in the `systemroot\system32\drivers\\` folder, and NetBIOS name resolution.

- if the command is **print** or **delete**, the *gateway* parameter can be omitted and wildcards can be used for the destination and gateway. The *destination* value can be a wildcard value specified by an asterisk `(*)`. If the destination specified contains an asterisk `(*)` or a

question mark (?), it's treated as a wildcard, and only matching destination routes are printed or deleted. The asterisk matches any string, and the question mark matches any single character. For example, `10.\*.1, 192.168.\*`, `127.\*`, and `\*224\*` are all valid uses of the asterisk wildcard.

- Using an unsupported combination of a destination and subnet mask (netmask) value displays a "Route: bad gateway address netmask" error message. This error message appears when the destination contains one or more bits set to 1 in bit locations where the corresponding subnet mask bit is set to 0. To test this condition, express the destination and subnet mask using binary notation. The subnet mask in binary notation consists of a series of 1 bits, representing the network address portion of the destination, and a series of 0 bits, representing the host address portion of the destination. Check to determine whether there are bits in the destination that are set to 1 for the portion of the destination that is the host address (as defined by the subnet mask).

Examples

To display the entire contents of the IP routing table, type:

Windows Command Prompt

```
route print
```

To display the routes in the IP routing table that begin with 10, type:

Windows Command Prompt

```
route print 10.*
```

To add a default route with the default gateway address of 192.168.12.1, type:

Windows Command Prompt

```
route add 0.0.0.0 mask 0.0.0.0 192.168.12.1
```

To add a route to the destination 10.41.0.0 with the subnet mask of 255.255.0.0 and the next hop address of 10.27.0.1, type:

Windows Command Prompt

```
route add 10.41.0.0 mask 255.255.0.0 10.27.0.1
```

To add a persistent route to the destination 10.41.0.0 with the subnet mask of 255.255.0.0 and the next hop address of 10.27.0.1, type:

Windows Command Prompt

```
route /p add 10.41.0.0 mask 255.255.0.0 10.27.0.1
```

To add a route to the destination 10.41.0.0 with the subnet mask of 255.255.0.0, the next hop address of 10.27.0.1, and the cost metric of 7, type:

Windows Command Prompt

```
route add 10.41.0.0 mask 255.255.0.0 10.27.0.1 metric 7
```

To add a route to the destination 10.41.0.0 with the subnet mask of 255.255.0.0, the next hop address of 10.27.0.1, and using the interface index 0x3, type:

Windows Command Prompt

```
route add 10.41.0.0 mask 255.255.0.0 10.27.0.1 if 0x3
```

To delete the route to the destination 10.41.0.0 with the subnet mask of 255.255.0.0, type:

Windows Command Prompt

```
route delete 10.41.0.0 mask 255.255.0.0
```

To delete all routes in the IP routing table that begin with 10, type:

Windows Command Prompt

```
route delete 10.*
```

To change the next hop address of the route with the destination of 10.41.0.0 and the subnet mask of 255.255.0.0 from 10.27.0.1 to 10.27.0.25, type:

Windows Command Prompt

```
route change 10.41.0.0 mask 255.255.0.0 10.27.0.25
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rpcinfo

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists programs on remote computers. The **rpcinfo** command-line utility makes a remote procedure call (RPC) to an RPC server and reports what it finds.

Syntax

```
rpcinfo [/p [<node>]] [/b <program version>] [/t <node program> [<version>]]
[/u <node program> [<version>]]
```

Parameters

Expand table

Parameter	Description
/p `[<node>]`	lists all programs registered with the port mapper on the specified host. If you do not specify a node (computer) name, the program queries the port mapper on the local host.
/b `<program version>`	Requests a response from all network nodes that have the specified program and version registered with the port mapper. You must specify both a program name or number and a version number.
/t `<node program> [\<version>]`	Uses the TCP transport protocol to call the specified program. You must specify both a node (computer) name and a program name. If you do not specify a version, the program calls all versions.
/u `<node program> [\<version>]`	Uses the UDP transport protocol to call the specified program. You must specify both a node (computer) name and a program name. If you do not specify a version, the program calls all versions.
/?	Displays help at the command prompt.

Examples

To list all programs registered with the port mapper, type:

```
rpcinfo /p [<node>]
```

To request a response from network nodes that have a specified program, type:

```
rpcinfo /b <program version>
```

To use Transmission Control Protocol (TCP) to call a program, type:

```
rpcinfo /t <node program> [<version>]
```

Use User Datagram Protocol (UDP) to call a program:

```
rpcinfo /u <node program> [<version>]
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rpcping

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Confirms the RPC connectivity between the computer running Microsoft Exchange Server and any of the supported Microsoft Exchange Client workstations on the network. This utility can be used to check if the Microsoft Exchange Server services are responding to RPC requests from the client workstations via the network.

Syntax

```
rpcping [/t <protseq>] [/s <server_addr>] [/e <endpoint>
        |/f <interface UUID>[,majorver]] [/O <interface object UUID]
        [/i <#_iterations>] [/u <security_package_id>] [/a <authn_level>]
        [/N <server_princ_name>] [/I <auth_identity>] [/C <capabilities>]
        [/T <identity_tracking>] [/M <impersonation_type>]
        [/S <server_sid>] [/P <proxy_auth_identity>] [/F <RPCHTTP_flags>]
        [/H <RPC/HTTP_authn_schemes>] [/o <binding_options>]
        [/B <server_certificate_subject>] [/b] [/E] [/q] [/c]
        [/A <http_proxy_auth_identity>] [/U <HTTP_proxy_authn_schemes>]
        [/r <report_results_interval>] [/v <verbose_level>] [/d]
```

Parameters

Expand table

Parameter	Description
/t `<protseq>`	Specifies the protocol sequence to use. Can be one of the standard RPC protocol sequences: ncacn_ip_tcp, ncacn_np, or ncacn_http. If not specified, default is ncacn_ip_tcp.
/s `<server_addr>`	Specifies the server address. If not specified, the local machine will be pinged.
/e `<endpoint>`	Specifies the endpoint to ping. If none is specified, the endpoint mapper on the target machine will be pinged. This option is mutually exclusive with the interface (**/f**) option.

Parameter	Description
/o `<binding_options>`	Specifies the binding options for the RPC ping.
/f `<interface UUID>[,Majorver]`	Specifies the interface to ping. This option is mutually exclusive with the endpoint option. The interface is specified as a UUID. if the *majorver* is not specified, version 1 of the interface will be sought. When interface is specified, **rpcping** will query the endpoint mapper on the target machine to retrieve the endpoint for the specified interface. The endpoint mapper will be queried using the options specified in the command line.
/O `<object UUID>`	Specifies the object UUID if the interface registered one.
/i `<#_iterations>`	Specifies the number of calls to make. The default is 1. This option is useful for measuring connection latency if multiple iterations are specified.
/u `<security_package_id>`	Specifies the security package (security provider) RPC will use to make the call. The security package is identified as a number or a name. If a number is used it is the same number as in the RpcBindingSetAuthInfoEx API. If you specify this option, you must specify an authentication level other than *none*. There's no default for this option. If it isn't specified, RPC won't use security for the ping. The list below shows the names and numbers. Names are not case sensitive: • Negotiate / 9 or one of nego, snego or negotiate • NTLM / 10 or NTLM • SChannel / 14 or SChannel • Kerberos / 16 or Kerberos • Kernel / 20 or Kernel
/a `<authn_level>`	Specifies the authentication level to use. If this option is specified, the security package ID (**/u**) must also be specified. If this option isn't specified, RPC won't use security for the ping. There's no default for this option. Possible values are: • connect • call • pkt • integrity • privacy
/N `<server_princ_name>`	Specifies a server principal name. This field can be used only when authentication level and security package are selected.

Parameter	Description
/l `<auth_identity>`	Allows you to specify alternative identity to connect to the server. The identity is in the form user,domain,password. If the user name, domain, or password have special characters that can be interpreted by the shell, enclose the identity in double quotes. You can specify `\*` instead of the password and RPC will prompt you to enter the password without echoing it on the screen. If this field is not specified, the identity of the logged on user will be used. This field can be used only when authentication level and security package are selected.
/C `<capabilities>`	Specifies a hexadecimal bitmask of flags. This field can be used only when authentication level and security package are selected.
/T `<identity_tracking>`	Specifies static or dynamic. If not specified, dynamic is the default. This field can be used only when authentication level and security package are selected.
/M `<impersonation_type>`	Specifies anonymous, identify, impersonate or delegate. Default is impersonate. This field can be used only when authentication level and security package are selected.
/S `<server_sid>`	Specifies the expected SID of the server. This field can be used only when authentication level and security package are selected.
/P `<proxy_auth_identity>`	Specifies the identity to authenticate with to the RPC/HTTP proxy. Has the same format as for the **/l** option. You must specify security package (**/u**), authentication level (**/a**), and authentication schemes (**/H**) in order to use this option.
/F `<RPCHTTP_flags>`	Specifies the flags to pass for RPC/HTTP front end authentication. The flags may be specified as numbers or names The currently recognized flags are: • Use SSL / 1 or ssl or use_ssl • Use first auth scheme / 2 or first or use_first You must specify security package (**/u**) and authentication level (**/a**) to use this option.
/H `<RPC/HTTP_authn_schemes>`	Specifies the authentication schemes to use for RPC/HTTP front end authentication. This option is a list of numerical values or names separated by comma. Example: Basic,NTLM. Recognized values are (names are not case sensitive): • Basic / 1 or Basic

Parameter	Description
	• NTLM / 2 or NTLM • Certificate / 65536 or Cert You must specify security package (**/u**) and authentication level (**/a**) in order to use this option.
/B `<server_certificate_subject>`	Specifies the server certificate subject. You must use SSL for this option to work. You must specify security package (**/u**) and authentication level (**/a**) in order to use this option.
/b	Retrieves the server certificate subject from the certificate sent by the server and prints it to a screen or a log file. Valid only when the Proxy echo only option (/E) and the use SSL options are specified. You must specify security package (**/u**) and authentication level (**/a**) in order to use this option.
/R	Specifies the HTTP proxy. If *none*, the RPC proxy is used. The value *default* means to use the IE settings in your client machine. Any other value will be treated as the explicit HTTP proxy. If you do not specify this flag, the default value is assumed, that is, the IE settings are checked. This flag is valid only when the **/E** (echo Only) flag is enabled.
/E	Restricts the ping to the RPC/HTTP proxy only. The ping does not reach the server. Useful when trying to establish whether the RPC/HTTP proxy is reachable. To specify an HTTP proxy, use the /R flag. If an HTTP proxy is specified in the /o flag, this option will be ignored. You must specify security package (**/u**) and authentication level (**/a**) in order to use this option.
/q	Specifies quiet mode. Does not issue any prompts except for passwords. Assumes *Y* response to all queries. Use this option with care.
/c	Use smart card certificate. rpcping will prompt user to choose smart card.
/A	Specifies the identity with which to authenticate to the HTTP proxy. Has the same format as for the /I option. You must specify authentication schemes (/U), security package (**/u**) and authentication level (**/a**) in order to use this option.
/U	Specifies the authentication schemes to use for HTTP proxy authentication. This option is a list of numerical values or names separated by comma. Example: Basic,NTLM. Recognized values are (names are not case sensitive):

Parameter	Description
	• Basic / 1 or Basic • NTLM / 2 or NTLM You must specify security package (**/u**) and authentication level (**/a**) in order to use this option.
/r	If multiple iterations are specified, this option will make **rpcping** display current execution statistics periodically instead after the last call. The report interval is given in seconds. Default is 15.
/v	Tells **rpcping** how verbose to make the output. Default value is 1. 2 and 3 provide more output from **rpcping**.
/d	Launches RPC network diagnostic UI.
/p	Specifies to prompt for credentials if authentication fails.
/?	Displays help at the command prompt.

Examples

To find out if the Exchange server you connect through RPC/HTTP is accessible, type:

```
rpcping /t ncacn_http /s exchange_server /o RpcProxy=front_end_proxy /P
username,domain,* /H Basic /u NTLM /a connect /F 3
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rsh

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Runs commands on remote computers running the RSH service or daemon.

> **ⓘ Important**
>
> The **rexec** command has been deprecated, and isn't guaranteed to be supported in Windows.

After installing the subsystem for UNIX-based Applications, you can then open a C Shell or Korn Shell command window and run **rsh**. For more information, type **man rsh** at the C Shell or Korn Shell prompt.

Feedback

Was this page helpful? Yes No

rundll32

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Loads and runs 32-bit dynamic-link libraries (DLLs). There are no configurable settings for Rundll32. Help information is provided for a specific DLL you run with the **rundll32** command.

You must run the **rundll32** command from an elevated command prompt. To open an elevated command prompt, click **Start**, right-click **Command Prompt**, and then click **Run as administrator**.

Syntax

```
rundll32 <DLLname>
```

Parameters

Expand table

Parameter	Description
Rundll32 printui.dll,PrintUIEntry	Displays the printer user interface.

Remarks

Rundll32 can only call functions from a DLL explicitly written to be called by Rundll32.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

rundll32 printui.dll,PrintUIEntry

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Automates many printer configuration tasks. printui.dll is the executable file that contains the functions used by the printer configuration dialog boxes. These functions can also be called from within a script or a command-line batch file, or they can be run interactively from the command prompt.

Syntax

```
rundll32 printui.dll PrintUIEntry [baseparameter] [modificationparameter1]
[modificationparameter2] [modificationparameterN]
```

You can also use the following alternate syntaxes, although the examples in this topic use the previous syntax:

```
rundll32 printui.dll,PrintUIEntry [baseparameter] [modificationparameter1]
[modificationparameter2] [ModificationParameterN]
```

```
rundll32 printui PrintUIEntry [baseparameter] [modificationparameter1]
[modificationparameter2] [modificationparameterN]
```

```
rundll32 printui,PrintUIEntry [baseparameter] [modificationparameter1]
[modificationparameter2] [modificationparameterN]
```

Parameters

There are two types of parameters: base parameters and modification parameters. Base parameters specify the function that the command is to perform. Only one of these

parameters can appear in a given command line. Then, you can modify the base parameter by using one or more of the modification parameters if they are applicable to the base parameter (not all modification parameters are supported by all base parameters).

Expand table

Base Parameters	Description
/dl	Deletes the local printer.
/dn	Deletes a network printer connection.
/dd	Deletes a printer driver.
/e	Displays the printing preferences for a given printer.
/ga	adds a per computer printer connection (the connection is available to any user on that computer when they log on).
/ge	Displays per computer printer connections on a computer.
/gd	Deletes a per computer printer connection (the connection is deleted the next time a user logs on).
/ia	Installs a printer driver by using an .inf file.
/id	Installs a printer driver by using the add printer Driver Wizard.
/if	Installs a printer by using an .inf file.
/ii	Installs a printer by using the add printer wizard with an .inf file.
/il	Installs a printer by using the add printer wizard.
/in	Connects to a remote network printer.
/ip	Installs a printer by using the Network printer Installation Wizard (available from the user interface from print Management).
/k	prints a test page on a printer.
/o	Displays the queue for a printer.
/p	Displays the properties of a printer. When you use this parameter, you must also specify a value for the modification parameter **/n[name]**.
/s	Displays the properties of a print server. If you want to view the local print server, you do not need to use a modification parameter. However, if you want to view a remote print server, you must specify the **/c[name]** modification parameter.

Base Parameters	Description
/Ss	Specifies what type of information for a printer will be stored. If none of the values for **/Ss** are specified, the default behavior is as if all of them were specified. Use this base parameter with the following values placed at the end of the command line: • **2**: Stores the information contained in the printer s printER_INFO_2 structure. This structure contains the basic information about the printer such as its name, server name, port name, and share name. • **7**: Used to store the directory service information contained in the printER_INFO_7 structure. • **c**: Stores the color profile information for a printer. • **d**: Stores printer specific data such as the printer s hardware ID. • **s**: Stores the printer s security descriptor. • **g**: Stores the information in the printer s global DEVmode structure. • **m**: Stores the minimal settings for the printer. This is equivalent to specifying **2 d**, and **g**. • **u**: Stores the information in the printer s per user DEVmode structure.
/Sr	Specifies what information about a printer is restored and how conflicts in settings are handled. Use with the following values placed at the end of the command line: • **2**: Restores the information contained in the printer s printER_INFO_2 structure. This structure contains the basic information about the printer such as its name, server name, port name, and share name. • **7**: Restores the directory service information contained in the printER_INFO_7 structure. • **c**: Restores the color profile information for a printer. • **d**: Restores printer specific data, such as the printer s hardware ID. • **s**: Restores the printer s security descriptor. • **g**: Restores the information in the printer s global DEVmode structure. • **m**: Restores the minimal settings for the printer. This is equivalent to specifying **2**, **d**, and **g**. • **u** Restores the information in the printe s per user DEVmode structure. • **r**: If the printer name stored in the file is different from the name of the printer being restored to, then use the current printer name. This cannot be specified with **f**. If neither **r** nor **f** is specified and the names do not match, restoration of the settings fails. • **f**: If the printer name stored in the file is different from the name of the printer being restored to, then use the printer name in the file. This cannot be specified with **r**. If neither **f** nor **r** is specified and the names do not match, restoration of the settings fails. • **p**: If the port name in the file being restored from does not match the current port name of the printer being restored to, the printer s current port name is used. • **h**: If the printer being restored to could not be shared using the resource share name in the saved settings file, then attempt to share the printer with either the current share name or a new generated share name if neither **H**

Base Parameters	Description
	nor **h** is specified and the printer being restored to cannot be shared with the saved share name, then restoration fails. • **h**: If the printer being restored to cannot be shared with the saved share name, then do not share the printer. If neither **H** nor **h** is specified and the printer being restored to cannot be shared with the saved share name, then restoration fails. • **i**: If the driver in the saved settings file does not match the driver for the printer being restored to, then the restoration fails.
/Xg	Retrieves the settings for a printer.
/Xs	Sets the settings for a printer.
/y	Sets the printer being installed as the default printer.
/?	Displays the in-product help for the command and its associated parameters.
@[file]	Specifies a command-line argument file and directly inserts the text in that file into the command line.

Expand table

Modification Parameters	Description
/a[file]	Specifies the binary file name.
/b[name]	Specifies the base printer name.
/c[name]	Specifies the computer name if the action to be performed is on a remote computer.
/f[file]	Species the Universal Naming Convention (UNC) path and name of the .inf file name or the output file name, depending on the task that you are performing. Use **/F[file]** to specify a dependent .inf file.
/F[file]	Specifies the UNC path and name of a .inf file that the .inf file specified with **/f[file]** depends on.
/h[architecture]	Specifies the driver architecture. Use one of the following: **x86**, **x64**, or **Itanium**.
/j[provider]	Specifies the print provider name.
/l[path]	Specifies the UNC path where the printer driver files that you are using are located.

Modification Parameters	Description
/m[model]	Specifies the driver model name. (This value can be specified in the .inf file.)
/n[name]	Specifies the printer name.
/q	Runs the command with no notifications to the user.
/r[port]	Specifies the port name.
/u	Specifies to use the existing printer driver if it is already installed.
/t[#]	Specifies the zero-based index page to start on.
/v[version]	Specifies the driver version. If you do not also specify a value for **/K**, you must specify one of the following values: **type 2 - Kernel mode** or **type 3 - User mode**.
/w	prompts the user for a driver if the driver is not found in the .inf file that is specified by **/f**.
/Y	Specifies that printer names should not be automatically generated.
/z	Specifies to not automatically share the printer being installed.
/K	changes the meaning of the parameter **/h[architecture]** to accept **2** in place of **x86**, **3** in place of **x64**, or **4** in place of **Itanium**. It also changes the value of the parameter **/v[version]** to accept **2** in the place of **type 2 - Kernel mode** and **3** in place of **type 3 - User mode**.
/Z	Shares the printer that is being installed. Only use with the **/if** parameter.
/Mw[message]	Displays a warning message to the user before committing the changes specified in the command line.
/Mq[message]	Displays a confirmation message to the user before committing the changes specified in the command line.
/W[flags]	Specifies any parameters or options for the add printer wizard, the add printer Driver Wizard, and the Network printer Installation Wizard. **r**: Enables the wizards to be restarted from the last page.
/G[flags]	Specifies global parameters and options that you want to use. **w**: Suppresses setup driver warnings to the user.

Remarks

- The **PrintUIEntry** keyword is case sensitive, and you must enter the syntax for this command with the exact capitalization shown in the examples in this topic.

- For more examples, at a command prompt type: **rundll32 printui.dll,PrintUIEntry /?**

Examples

To add a new remote printer, printer1, for a computer, Client1, which is visible for the user account where this command is run, type:

```
rundll32 printui.dll PrintUIEntry /in /n\\client1\printer1
```

To add a printer using the add printer wizard and using an .inf file, InfFile.inf, located on drive c: at Infpath, type:

```
rundll32 printui.dll PrintUIEntry /ii /f c:\Infpath\InfFile.inf
```

To delete an existing printer, printer1, on a computer, Client1, type:

```
rundll32 printui.dll PrintUIEntry /dn /n\\client1\printer1
```

To add a per computer printer connection, printer2, for all users of a computer, Client2, type (the connection will be applied when a user logs on):

```
rundll32 printui.dll PrintUIEntry /ga /n\\client2\printer2
```

To delete a per computer printer connection, printer2, for all users of a computer, Client2, type (the connection will be deleted when a user logs on):

```
rundll32 printui.dll PrintUIEntry /gd /n\\client2\printer2
```

To view the properties of the print server, printServer1, type:

```
```

```
rundll32 printui.dll PrintUIEntry /s /t1 /c\\printserver1
```

To view the properties of a printer, printer3, type:

```
rundll32 printui.dll PrintUIEntry /p /n\\printer3
```

Related links

- rundll32
- print Command Reference

Feedback

Was this page helpful? Yes No

rwinsta

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Resets (deletes) a session on a Remote Desktop Session Host server.

> **Note**
>
> This command is the same as the reset session command.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Related links

- reset session
- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

sc.exe config

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Modifies the value of a service's entries in the registry and in the Service Control Manager database.

Syntax

```
sc.exe [<servername>] config [<servicename>] [type= {own | share | kernel | filesys | rec | adapt | interact type= {own | share}}] [start= {boot | system | auto | demand | disabled | delayed-auto}] [error= {normal | severe | critical | ignore}] [binpath= <binarypathname>] [group= <loadordergroup>] [tag= {yes | no}] [depend= <dependencies>] [obj= {<accountname> | <objectname>}] [displayname= <displayname>] [password= <password>]
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the name of the remote server on which the service is located. The name must use the Universal Naming Convention (UNC) format (for example, \myserver). To run SC.exe locally, don't use this parameter.
`<servicename>`	Specifies the service name returned by the **getkeyname** operation.
`type= {own | share | kernel | filesys | rec | adapt | interact type= {own | share}}`	Specifies the service type. The options include: • **own** - Specifies a service that runs in its own process. It doesn't share an executable file with other services. This is the default value. • **share** - Specifies a service that runs as a shared process. It shares an executable file with other services. • **kernel** - Specifies a driver. • **filesys** - Specifies a file system driver. • **rec** - Specifies a file system-recognized driver that identifies file systems used on the computer. • **adapt** - Specifies an adapter driver that identifies hardware devices such as keyboards, mice, and disk drives.

Parameter	Description
	• **interact** - Specifies a service that can interact with the desktop, receiving input from users. Interactive services must be run under the LocalSystem account. This type must be used in conjunction with **type= own** or **type= shared** (for example, **type= interact type= own**). Using **type= interact** by itself will generate an error.
`start= {boot | system | auto | demand | disabled | delayed-auto}`	Specifies the start type for the service. The options include: • **boot** - Specifies a device driver that is loaded by the boot loader. • **system** - Specifies a device driver that is started during kernel initialization. • **auto** - Specifies a service that automatically starts each time the computer is restarted and runs even if no one logs on to the computer. • **demand** - Specifies a service that must be started manually. This is the default value if **start=** is not specified. • **disabled** - Specifies a service that cannot be started. To start a disabled service, change the start type to some other value. • **delayed-auto** - Specifies a service that starts automatically a short time after other auto services are started.
`error= {normal | severe | critical | ignore}`	Specifies the severity of the error if the service fails to start when the computer is started. The options include: • **normal** - Specifies that the error is logged and a message box is displayed, informing the user that a service has failed to start. Startup will continue. This is the default setting. • **severe** - Specifies that the error is logged (if possible). The computer attempts to restart with the last-known good configuration. This could result in the computer being able to restart, but the service may still be unable to run. • **critical** - Specifies that the error is logged (if possible). The computer attempts to restart with the last-known good configuration. If the last-known good configuration fails, startup also fails, and the boot process halts with a Stop error. • **ignore** - Specifies that the error is logged and startup continues. No notification is given to the user beyond recording the error in the Event Log.
`binpath= <binarypathname>`	Specifies a path to the service binary file. There is no default for **binpath=**, and this string must be supplied. Additionally, **ntsd -d** can be specified in front of the string for debugging. For more information, see Debugging using CDB and NTSD.
`group= <loadordergroup>`	Specifies the name of the group of which this service is a member. The list of groups is stored in the registry, in the

Parameter	Description
	HKLM\System\CurrentControlSet\Control\ServiceGroupOrder subkey. The default value is null.
`tag= {yes | no}`	Specifies whether or not to obtain a TagID from the CreateService call. Tags are used only for boot-start and system-start drivers.
`depend= <dependencies>`	Specifies the names of services or groups that must start before this service. The names are separated by forward slashes (/).
`obj= {<accountname> | <objectname>}`	Specifies a name of an account in which a service will run, or specifies a name of the Windows driver object in which the driver will run. The default setting is **LocalSystem**.
`displayname= <displayname>`	Specifies a descriptive name for identifying the service in user interface programs. For example, the subkey name of one particular service is **wuauserv**, which has a more friendly display name of Automatic Updates.
`password= <password>`	Specifies a password. This is required if an account other than the LocalSystem account is used.
/?	Displays help at the command prompt.

Remarks

- Each command-line option (parameter) must include the equal sign as part of the option name.
- A space is required between an option and its value (for example, **type= own**. If the space is omitted, the operation fails.

Examples

To specify a binary path for the *NewService* service, type:

```
sc.exe config NewService binpath= c:\windows\system32\NewServ.exe
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

sc.exe create

Article • 09/29/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a subkey and entries for a service in the registry and in the Service Control Manager database.

Syntax

```
sc.exe [<servername>] create [<servicename>] [type= {own | share | kernel | filesys | rec | interact type= {own | share}}] [start= {boot | system | auto | demand | disabled | delayed-auto}] [error= {normal | severe | critical | ignore}] [binpath= <binarypathname>] [group= <loadordergroup>] [tag= {yes | no}] [depend= <dependencies>] [obj= {<accountname> | <objectname>}] [displayname= <displayname>] [password= <password>]
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the name of the remote server on which the service is located. The name must use the Universal Naming Convention (UNC) format (for example, \myserver). To run SC.exe locally, don't use this parameter.
`<servicename>`	Specifies the service name returned by the **getkeyname** operation.
`type= {own | share | kernel | filesys | rec | interact type= {own | share}}`	Specifies the service type. The options include: • **own** - Specifies a service that runs in its own process. It doesn't share an executable file with other services. This is the default value. • **share** - Specifies a service that runs as a shared process. It shares an executable file with other services. • **kernel** - Specifies a driver. • **filesys** - Specifies a file system driver. • **rec** - Specifies a file system-recognized driver that identifies file systems used on the computer. • **interact** - Specifies a service that can interact with the desktop, receiving input from users. Interactive services must be run under the LocalSystem account. This type must be used in conjunction with

Parameter	Description
	type= own or **type= share** (for example, **type= interact type= own**). Using **type= interact** by itself will generate an error.
`start= {boot | system | auto | demand | disabled | delayed-auto}`	Specifies the start type for the service. The options include: • **boot** - Specifies a device driver that is loaded by the boot loader. • **system** - Specifies a device driver that is started during kernel initialization. • **auto** - Specifies a service that automatically starts each time the computer is restarted and runs even if no one logs on to the computer. • **demand** - Specifies a service that must be started manually. This is the default value if **start=** is not specified. • **disabled** - Specifies a service that cannot be started. To start a disabled service, change the start type to some other value. • **delayed-auto** - Specifies a service that starts automatically a short time after other auto services are started.
`error= {normal | severe | critical | ignore}`	Specifies the severity of the error if the service fails to start when the computer is started. The options include: • **normal** - Specifies that the error is logged and a message box is displayed, informing the user that a service has failed to start. Startup will continue. This is the default setting. • **severe** - Specifies that the error is logged (if possible). The computer attempts to restart with the last-known good configuration. This could result in the computer being able to restart, but the service may still be unable to run. • **critical** - Specifies that the error is logged (if possible). The computer attempts to restart with the last-known good configuration. If the last-known good configuration fails, startup also fails, and the boot process halts with a Stop error. • **ignore** - Specifies that the error is logged and startup continues. No notification is given to the user beyond recording the error in the Event Log.
`binpath= <binarypathname>`	Specifies a path to the service binary file. There is no default for **binpath=**, and this string must be supplied.
`group= <loadordergroup>`	Specifies the name of the group of which this service is a member. The list of groups is stored in the registry, in the **HKLM\System\CurrentControlSet\Control\ServiceGroupOrder** subkey. The default value is null.
`tag= {yes | no}`	Specifies whether or not to obtain a TagID from the CreateService call. Tags are used only for boot-start and system-start drivers.

Parameter	Description
`depend= <dependencies>`	Specifies the names of services or groups that must start before this service. The names are separated by forward slashes (/).
`obj= {<accountname> | <objectname>}`	Specifies a name of an account in which a service will run, or specifies a name of the Windows driver object in which the driver will run. The default setting is **LocalSystem**.
`displayname= <displayname>`	Specifies a friendly name for identifying the service in user interface programs. For example, the subkey name of one particular service is **wuauserv**, which has a more friendly display name of Automatic Updates.
`password= <password>`	Specifies a password. This is required if an account other than the LocalSystem account is used.
/?	Displays help at the command prompt.

Remarks

- Each command-line option (parameter) must include the equal sign as part of the option name.
- A space is required between an option and its value (for example, **type= own**. If the space is omitted, the operation fails.

Examples

To create and register a new binary path for the *NewService* service, type:

```
sc.exe \\myserver create NewService binpath= c:\windows\system32\NewServ.exe
```

```
sc.exe create NewService binpath= c:\windows\system32\NewServ.exe type=
share start= auto depend= +TDI NetBIOS
```

To learn more about the `sc.exe` command, see SC commands.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

sc delete

08/15/2025

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Deletes a service subkey from the registry. If the service is running or if another process has an open handle to the service, the service is marked for deletion.

> **Note**
>
> We don't recommend you to use this command to delete built-in operating system services such as DHCP, DNS, or Internet Information Services. To install, remove, or reconfigure operating system roles, services and components, see Install or Uninstall Roles, Role Services, or Features

Syntax

```
sc <server> delete [service name]
```

Parameters

Expand table

Parameter	Description
`<server>`	Specifies the name of the remote server on which the service is located. The name must use the Universal Naming Convention (UNC) format (for example, \\MyServer). To run `sc` locally, don't use this parameter.
`<service name>`	Specifies the service name returned by the **getkeyname** operation.
/?	Displays help at the command prompt.

Examples

To delete the service subkey **NewServ** from the registry on the local computer, type:

```
sc delete NewServ
```

Related links

- Command-Line Syntax Key

Sc.exe query

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Obtains and displays information about the specified service, driver, type of service, or type of driver.

Syntax

```
sc.exe [<servername>] query [<servicename>] [type= {driver | service | all}]
[type= {own | share | interact | kernel | filesys | rec | adapt}] [state=
{active | inactive | all}] [bufsize= <Buffersize>] [ri= <Resumeindex>]
[group= <groupname>]
```

Parameters

Expand table

Parameter	Description
`<servername>`	Specifies the name of the remote server on which the service is located. The name must use the Universal Naming Convention (UNC) format (for example, \myserver). To run SC.exe locally, don't use this parameter.
`<servicename>`	Specifies the service name returned by the **getkeyname** operation. This **query** parameter isn't used in conjunction with other **query** parameters (other than *servername*).
`type= {driver | service | all}`	Specifies what to enumerate. The options include: • **driver** - Specifies that only drivers are enumerated. • **service** - Specifies only services are enumerated. This is the default value. • **all** - Specifies that both drivers and services are enumerated.
`type= {own | share | interact | kernel | filesys | rec | adapt}`	Specifies the type of services or type of drivers to be enumerated. The options include: • **own** - Specifies a service that runs in its own process. It doesn't share an executable file with other services. This is the default value.

Parameter	Description
	• **share** - Specifies a service that runs as a shared process. It shares an executable file with other services. • **kernel** - Specifies a driver. • **filesys** - Specifies a file system driver. • **rec** - Specifies a file system-recognized driver that identifies file systems used on the computer. • **interact** - Specifies a service that can interact with the desktop, receiving input from users. Interactive services must be run under the LocalSystem account. This type must be used in conjunction with **type= own** or **type= shared** (for example, **type= interact type= own**). Using **type= interact** by itself will generate an error.
`state= {active | inactive | all}`	Specifies the started state of the service to be enumerated. The options include: • **active** - Specifies all active services. This is the default value. • **inactive** - Specifies all paused or stopped services. • **all** - Specifies all services.
`bufsize= <Buffersize>`	Specifies the size (in bytes) of the enumeration buffer. The default buffer size is 1,024 bytes. You should increase the size of the buffer when the display resulting from a query goes over 1,024 bytes.
`ri= <Resumeindex>`	Specifies the index number at which enumeration is to begin or resume. The default value is 0 (zero). If more information is returned than what the default buffer can display, use this parameter with the `bufsize=` parameter.
`group= <Groupname>`	Specifies the service group to be enumerated. By default, all groups are enumerated. By default, all groups are enumerated (**group= **).
/?	Displays help at the command prompt.

Remarks

- Each command-line option (parameter) must include the equal sign as part of the option name.
- A space is required between an option and its value (for example, **type= own**. If the space is omitted, the operation fails.
- The **query** operation displays the following information about a service: SERVICE_NAME (service's registry subkey name), TYPE, STATE (as well as states which are not available), WIN32_EXIT_B, SERVICE_EXIT_B, CHECKPOINT, and WAIT_HINT.

- The **type=** parameter can be used twice in some cases. The first appearance of the **type=** parameter specifies whether to query services, drivers, or both (**all**). The second appearance of the **type=** parameter specifies a type from the **create** operation to further narrow a query's scope.
- When the display results from a **query** command exceed the size of the enumeration buffer, a message similar to the following is displayed:

```
Enum: more data, need 1822 bytes start resume at index 79

To display the remaining **query** information, rerun **query**,
setting **bufsize=** to be the number of bytes and setting **ri=** to
the specified index. For example, the remaining output would be
displayed by typing the following at the command prompt:

sc.exe query bufsize= 1822 ri= 79
```

Examples

To display information for active services only, type either of the following commands:

```
sc.exe query
sc.exe query type= service
```

To display information for active services, and to specify a buffer size of 2,000 bytes, type:

```
sc.exe query type= all bufsize= 2000
```

To display information for the *wuauserv* service, type:

```
sc.exe query wuauserv
```

To display information for all services (active and inactive), type:

```
sc.exe query state= all
```

To display information for all services (active and inactive), beginning at line 56, type:

```

sc.exe query state= all ri= 56
```

To display information for interactive services, type:

```

sc.exe query type= service type= interact
```

To display information for drivers only, type:

```

sc.exe query type= driver
```

To display information for drivers in the *Network Driver Interface Specification (NDIS) group*, type:

```

sc.exe query type= driver group= NDIS
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

schtasks commands

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Schedules commands and programs to run periodically or at a specific time, adds and removes tasks from the schedule, starts and stops tasks on demand, and displays and changes scheduled tasks.

> ⓘ **Note**
>
> The **schtasks.exe** tool performs the same operations as **Scheduled Tasks** in **Control Panel**. You can use these tools together and interchangeably.

Required permissions

- To schedule, view, and change all tasks on the local computer, you must be a member of the Administrators group.
- To schedule, view, and change all tasks on the remote computer, you must be a member of the Administrators group on the remote computer, or you must use the **/u** parameter to provide the credentials of an Administrator of the remote computer.
- You can use the **/u** parameter in a **/create** or **/change** operation if the local and remote computers are in the same domain, or if the local computer is in a domain that the remote computer domain trusts. Otherwise, the remote computer can't authenticate the user account specified, and it can't verify that the account is a member of the Administrators group.
- The task you plan to run must have the appropriate permission; these permissions vary by task. By default, tasks run with the permissions of the current user of the local computer, or with the permissions of the user specified by the **/u** parameter, if one is included. To run a task with permissions of a different user account or with system permissions, use the **/ru** parameter.

Syntax

```
schtasks /change
schtasks /create
schtasks /delete
schtasks /end
schtasks /query
schtasks /run
```

Parameters

Expand table

Parameter	Description
schtasks change	Changes one or more of the following properties of a task: • The program that the task runs (/tr) • The user account under which the task runs (/ru) • The password for the user account (/rp) • Adds the interactive-only property to the task (/it)
schtasks create	Schedules a new task.
schtasks delete	Deletes a scheduled task.
schtasks end	Stops a program started by a task.
schtasks query	Displays tasks scheduled to run on the computer.
schtasks run	Starts a scheduled task immediately. The **run** operation ignores the schedule, but uses the program file location, user account, and password saved in the task to run the task immediately.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

schtasks change

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes one or more of the following properties of a task:

- The program that the task runs (**/tr**)
- The user account under which the task runs (**/ru**)
- The password for the user account (**/rp**)
- Adds the interactive-only property to the task (**/it**)

Required permissions

- To schedule, view, and change all tasks on the local computer, you must be a member of the Administrators group.
- To schedule, view, and change all tasks on the remote computer, you must be a member of the Administrators group on the remote computer, or you must use the **/u** parameter to provide the credentials of an Administrator of the remote computer.
- You can use the **/u** parameter in a **/create** or **/change** operation if the local and remote computers are in the same domain, or if the local computer is in a domain that the remote computer domain trusts. Otherwise, the remote computer can't authenticate the user account specified, and it can't verify that the account is a member of the Administrators group.
- The task you plan to run must have the appropriate permission; these permissions vary by task. By default, tasks run with the permissions of the current user of the local computer, or with the permissions of the user specified by the **/u** parameter, if one is included. To run a task with permissions of a different user account or with system permissions, use the **/ru** parameter.

Syntax

```
```

```
schtasks /change /tn <Taskname> [/s <computer> [/u [<domain>\]<user> [/p
<password>]]] [/ru <username>] [/rp <password>] [/tr <Taskrun>] [/st
<Starttime>] [/ri <interval>] [/rl <level>] [{/et <Endtime> | /du
<duration>} [/k]] [/sd <Startdate>] [/ed <Enddate>] [/{ENABLE | DISABLE}]
[/it] [/z]
```

Parameters

Expand table

Parameter	Description
/tn `<Taskname>`	Identifies the task to be changed. Enter the task name (**Note**: Task names that have a space in its name are required to be wrapped in double quotes).
/s `<computer>`	Specifies the name or IP address of a remote computer (with or without backslashes). The default is the local computer.
/u `[<domain>]`	Runs this command with the permissions of the specified user account. By default, the command runs with the permissions of the current user of the local computer. The specified user account must be a member of the Administrators group on the remote computer. The **/u** and **/p** parameters are valid only when you use **/s**.
/p `<password>`	Specifies the password of the user account specified in the **/u** parameter. If you use the **/u** parameter without the **/p** parameter or the password argument, schtasks will prompt you for a password. The **/u** and **/p** parameters are valid only when you use **/s**.
/ru `<username>`	Changes the user name under which the scheduled task has to run. For the system account, valid values are "", "*NT AUTHORITY\SYSTEM*", or "*SYSTEM*".
/rp `<password>`	Specifies a new password for the existing user account, or the user account specified by the **/ru** parameter. This parameter is ignored with used with the local System account.
/tr `<Taskrun>`	Changes the program that the task runs. Enter the fully qualified path and file name of an executable file, script file, or batch file. If you don't add the path, **schtasks** assumes that the file is in the `<systemroot>\System32` directory. The specified program replaces the original program run by the task.
/st `<Starttime>`	Specifies the start time for the task, using the 24-hour time format, HH:mm. For example, a value of 14:30 is equivalent to the 12-hour time of 2:30 PM.
/ri `<interval>`	Specifies the repetition interval for the scheduled task, in minutes. Valid range is 1 - 599940 (599940 minutes = 9999 hours). If either the **/et** or **/du** parameters are specified, the default is **10 minutes**.
/rl `<level>`	Specifies the Run Level for the job. Acceptable values are LIMITED (scheduled tasks will be ran with the least level of privileges, such as Standard User accounts) and

Parameter	Description
	HIGHEST (scheduled tasks will be ran with the highest level of privileges, such as Superuser accounts). (**Note**: LIMITED is the default value).
/et `<Endtime>`	Specifies the end time for the task, using the 24-hour time format, HH:mm. For example, a value of 14:30 is equivalent to the 12-hour time of 2:30 PM.
/du `<duration>`	A value that specifies the duration to run the task. The time format is HH:mm (24-hour time). For example, a value of 14:30 is equivalent to the 12-hour time of 2:30 PM.
/k	Stops the program that the task runs at the time specified by **/et** or **/du**. Without **/k**, schtasks doesn't start the program again after it reaches the time specified by **/et** or **/du** nor does it stop the program if it's still running. This parameter is optional and valid only with a MINUTE or HOURLY schedule.
/sd `<Startdate>`	Specifies the first date on which the task should be run. The date format is MM/DD/YYYY.
/ed `<Enddate>`	Specifies the last date on which the task should be run. The format is MM/DD/YYYY.
/ENABLE	Specifies to enable the scheduled task.
/DISABLE	Specifies to disable the scheduled task.
/it	Specifies to run the scheduled task only when the run as user (the user account under which the task runs) is logged on to the computer. This parameter has no effect on tasks that run with system permissions or tasks that already have the interactive-only property set. You can't use a change command to remove the interactive-only property from a task. By default, run as user is the current user of the local computer when the task is scheduled or the account specified by the **/u** parameter, if one is used. However, if the command includes the **/ru** parameter, then the run as user is the account specified by the **/ru** parameter.
/z	Specifies to delete the task upon the completion of its schedule.
/?	Displays help at the command prompt.

Remarks

- The **/tn** and **/s** parameters identify the task. The **/tr**, **/ru**, and **/rp** parameters specify properties of the task that you can change.
- The **/ru** and **/rp** parameters specify the permissions under which the task runs. The **/u** and **/p** parameters specify the permissions used to change the task.

- To change tasks on a remote computer, the user must be logged on to the local computer with an account that is a member of the Administrators group on the remote computer.
- To run a **/change** command with the permissions of a different user (**/u**, **/p**), the local computer must be in the same domain as the remote computer or must be in a domain that the remote computer domain trusts.
- The System account doesn't have interactive logon rights. Users don't see, and can't interact with, programs run with system permissions. To identify tasks with the **/it** property, use a verbose query (**/query /v**). In a verbose query display of a task with **/it**, the Logon Mode field has a value of Interactive only.

Examples

To change the program that the Virus Check task runs from *VirusCheck.exe* to *VirusCheck2.exe*, type:

```
schtasks /change /tn Virus Check /tr C:\VirusCheck2.exe
```

This command uses the **/tn** parameter to identify the task and the **/tr** parameter to specify the new program for the task. (You can't change the task name.)

To change the password of the user account for the *RemindMe* task on the remote computer, *Svr01*, type:

```
schtasks /change /tn RemindMe /s Svr01 /rp p@ssWord3
```

This procedure is required whenever the password for a user account expires or changes. If the password saved in a task is no longer valid, then the task doesn't run. The command uses the **/tn** parameter to identify the task and the **/s** parameter to specify the remote computer. It uses the **/rp** parameter to specify the new password, *p@ssWord3*.

To change the ChkNews task, which starts Notepad.exe every morning at 9:00 A.M., to start Internet Explorer instead, type:

```
```

```
schtasks /change /tn ChkNews /tr c:\program files\Internet
Explorer\iexplore.exe /ru DomainX\Admin01
```

The command uses the **/tn** parameter to identify the task. It uses the **/tr** parameter to change the program that the task runs and the /ru parameter to change the user account under which the task runs. The **/ru** and **/rp** parameters, which provide the password for the user account, is not used. You must provide a password for the account, but you can use the **/ru** and **/rp** parameter and type the password in clear text, or wait for SchTasks.exe to prompt you for a password, and then enter the password in obscured text.

To change the SecurityScript task so that it runs with permissions of the System account, type:

```
schtasks /change /tn SecurityScript /ru
```

The command uses the **/ru** parameter to indicate the System account. Because tasks run with System account permissions do not require a password, SchTasks.exe does not prompt for one.

To add the interactive-only property to MyApp, an existing task, type:

```
schtasks /change /tn MyApp /it
```

This property assures that the task runs only when the run as user, that is, the user account under which the task runs, is logged on to the computer. The command uses the **/tn** parameter to identify the task and the **/it** parameter to add the interactive-only property to the task. Because the task already runs with the permissions of my user account, you don't need to change the **/ru** parameter for the task.

Related links

- Command-Line Syntax Key
- schtasks create command
- schtasks delete command

- schtasks end command
- schtasks query command
- schtasks run command

Feedback

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schtasks create

Article • 10/10/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Schedules a task.

Syntax

```
schtasks /create /sc <scheduletype> /tn <taskname> /tr <taskrun> [/s
<computer> [/u [<domain>\]<user> [/p <password>]]] [/ru {[<domain>\]<user> |
system}] [/rp <password>] [/mo <modifier>] [/d <day>[,<day>...] | *] [/m
<month>[,<month>...]] [/i <idletime>] [/st <starttime>] [/ri <interval>]
[{/et <endtime> | /du <duration>} [/k]] [/sd <startdate>] [/ed <enddate>]
[/it] [/np] [/z] [/xml <xmlfile>] [/v1] [/f] [/rl <level>] [/delay
<delaytime>] [/hresult]
```

Parameters

Expand table

Parameter	Description
/sc `<scheduletype>`	Specifies the schedule type. The valid values include: • **MINUTE** - Specifies the number of minutes before the task should run. • **HOURLY** - Specifies the number of hours before the task should run. • **DAILY** - Specifies the number of days before the task should run. • **WEEKLY** Specifies the number of weeks before the task should run. • **MONTHLY** - Specifies the number of months before the task should run. • **ONCE** - Specifies that that task runs once at a specified date and time. • **ONSTART** - Specifies that the task runs every time the system starts. You can specify a start date, or run the task the next time the system starts. • **ONLOGON** - Specifies that the task runs whenever a user (any user) logs on. You can specify a date, or run the task the next time the user logs on. • **ONIDLE** - Specifies that the task runs whenever the system is idle for a specified period of time. You can specify a date, or run the task the

Parameter	Description
	next time the system is idle. • **ONEVENT** - Specifies that the task runs based on an event that matches information from the system event log including the EventID.
/tn `<taskname>`	Specifies a name for the task. Each task on the system must have a unique name and must conform to the rules for file names, not exceeding 238 characters. Use quotation marks to enclose names that include spaces. To store your scheduled task in a different folder, run **/tn** `<folder name\task name>`.
/tr `<Taskrun>`	Specifies the program or command that the task runs. Type the fully qualified path and file name of an executable file, script file, or batch file. The path name must not exceed 262 characters. If you don't add the path, **schtasks** assumes that the file is in the `<systemroot>\System32` directory.
/s `<computer>`	Specifies the name or IP address of a remote computer (with or without backslashes). The default is the local computer.
/u `[<domain>]`	Runs this command with the permissions of the specified user account. The default is the permissions of the current user of the local computer. The **/u** and **/p** parameters are valid only when you use **/s**. The permissions of the specified account are used to schedule the task and to run the task. To run the task with the permissions of a different user, use the **/ru** parameter. The user account must be a member of the Administrators group on the remote computer. Also, the local computer must be in the same domain as the remote computer, or must be in a domain that is trusted by the remote computer domain.
/p `<password>`	Specifies the password of the user account specified in the **/u** parameter. If you use the **/u** parameter without the **/p** parameter or the password argument, schtasks will prompt you for a password. The **/u** and **/p** parameters are valid only when you use **/s**.
/ru {[`<domain>`]`<user>` | `system`}	Runs the task with permissions of the specified user account. By default, the task runs with the permissions of the current user of the local computer, or with the permission of the user specified by the **/u** parameter, if one is included. The **/ru** parameter is valid when scheduling tasks on local or remote computers. The valid options include: • **Domain** - Specifies an alternate user account. • **System** - Specifies the local System account, a highly privileged account used by the operating system and system services.
/rp `<password>`	Specifies the password for the existing user account, or the user account specified by the **/ru** parameter. If you don't use this parameter when specifying a user account, SchTasks.exe will prompt you for the password next time you sign in. Don't use the **/rp** parameter for tasks that run with

Parameter	Description
	System account credentials (**/ru System**). The System account doesn't have a password and SchTasks.exe doesn't prompt for one.
/mo `<modifiers>`	Specifies how often the task runs within its schedule type. The valid options include: • **MINUTE** - Specifies that the task runs every <n> minutes. You can use any value between 1 - 1439 minutes. By default, this is 1 minute. • **HOURLY** - Specifies that the task runs every <n> hours. You can use any value between 1 - 23 hours. By default, this is 1 hour. • **DAILY** - Specifies that the task runs every <n> days. You can use any value between 1 - 365 days. By default, this is 1 day. • **WEEKLY** - Specifies that the task runs every <n> weeks. You can use any value between 1 - 52 weeks. By default, this is 1 week. • **MONTHLY** - Specifies that the task runs every <n> months. You can use any of the following values: ◦ A number between 1 - 12 months ◦ **LASTDAY** - To run the task on the last day of the month ◦ **FIRST, SECOND, THIRD, or FOURTH along with the `/d <day>` parameter** - Specifies the particular week and day to run the task. For example, on the third Wednesday of the month. • **ONCE** - Specifies that the task runs once. • **ONSTART** - Specifies that the task runs at startup. • **ONLOGON** - Specifies that the task runs when the user specified by the **/ru** parameter logs on. • **ONIDLE** - Specifies that the task runs after the system is idle for the number of minutes specified by the **/i** parameter
/d DAY[,DAY...]	Specifies how often the task runs within its schedule type. The valid options include: • **WEEKLY** - Specifies that the task runs weekly by providing a value between 1-52 weeks. Optionally, you can also add a specific day of the week by adding a value of MON - SUN or a range of [MON - SUN...]). • **MONTHLY** - Specifies that the task runs weekly each month by providing a value of FIRST, SECOND, THIRD, FOURTH, LAST. Optionally, you can also add a specific day of the week by adding a value of MON - SUN or by providing a number between 1 - 12 months. If you use this option, you can also add a specific day of the month, by providing a number between 1-31. **NOTE:** The date value of 1 - 31 is valid only without the **/mo** parameter, or if the **/mo** parameter is monthly (1 - 12). The default is day 1 (the first day of the month).
/m MONTH[,MONTH...]	Specifies a month or months of the year during which the scheduled task should run. The valid options include JAN - DEC and `*` (every month). The **/m** parameter is valid only with a MONTHLY schedule. It's required when

Parameter	Description
	the LASTDAY modifier is used. Otherwise, it's optional and the default value is `*` (every month).
/i `<Idletime>`	Specifies how many minutes the computer is idle before the task starts. A valid value is a whole number from 1 to 999. This parameter is valid only with an ONIDLE schedule, and then it's required.
/st `<Starttime>`	Specifies the start time for the task, using the 24-hour time format, HH:mm. The default value is the current time on the local computer. The **/st** parameter is valid with MINUTE, HOURLY, DAILY, WEEKLY, MONTHLY, and ONCE schedules. It's required for a ONCE schedule.
/ri `<interval>`	Specifies the repetition interval for the scheduled task, in minutes. This isn't applicable for schedule types: MINUTE, HOURLY, ONSTART, ONLOGON,ONIDLE, and ONEVENT. Valid range is 1 - 599940 (599940 minutes = 9999 hours). If either the **/et** or **/du** parameters are specified, the default is **10 minutes**.
/et `<endtime>`	Specifies the time of day that a minute or hourly task schedule ends in <HH:MM> 24-hour format. After the specified end time, schtasks does not start the task again until the start time recurs. By default, task schedules have no end time. This parameter is optional and valid only with a MINUTE or HOURLY schedule.
/du `<duration>`	Specifies a maximum length of time for a minute or hourly schedule in <HHHH:MM> 24-hour format. After the specified time elapses, schtasks does not start the task again until the start time recurs. By default, task schedules have no maximum duration. This parameter is optional and valid only with a MINUTE or HOURLY schedule.
/k	Stops the program that the task runs at the time specified by **/et** or **/du**. Without **/k**, schtasks doesn't start the program again after it reaches the time specified by **/et** or **/du** nor does it stop the program if it's still running. This parameter is optional and valid only with a MINUTE or HOURLY schedule.

Parameter	Description
/sd `<Startdate>`	Specifies the date on which the task schedule starts. The default value is the current date on the local computer. The format for **Startdate** varies with the locale selected for the local computer in **Regional and Language Options**. Only one format is valid for each locale. The valid date formats include (be sure to choose the format most similar to the format selected for **Short date** in **Regional and Language Options** on the local computer): • `<MM>//` - Specifies to use month-first formats, such as English (United States) and Spanish (Panama). • `<DD>//` - Specifies to use day-first formats, such as Bulgarian and Dutch (Netherlands). • `<YYYY>//` - Specifies to use for year-first formats, such as Swedish and French (Canada).
/ed `<Enddate>`	Specifies the date on which the schedule ends. This parameter is optional. It isn't valid in a ONCE, ONSTART, ONLOGON, ONIDLE, or ONEVENT schedule. By default, schedules have no ending date. The default value is the current date on the local computer. The format for **Enddate** varies with the locale selected for the local computer in **Regional and Language Options**. Only one format is valid for each locale. The valid date formats include (be sure to choose the format most similar to the format selected for **Short date** in **Regional and Language Options** on the local computer): • `<MM>//` - Specifies to use month-first formats, such as English (United States) and Spanish (Panama). • `<DD>//` - Specifies to use day-first formats, such as Bulgarian and Dutch (Netherlands). • `<YYYY>//` - Specifies to use for year-first formats, such as Swedish and French (Canada).
/ec `<channelname>`	Specifies the event channel name triggered by the ONEVENT schedule type that matches a system event log criteria.
/it	Specifies to run the scheduled task only when the run as user (the user account under which the task runs) is logged on to the computer. This parameter has no effect on tasks that run with system permissions or tasks that already have the interactive-only property set. You can't use a change command to remove the interactive-only property from a task. By default, run as user is the current user of the local computer when the task is scheduled or the account specified by the **/u** parameter, if one is used. However, if the command includes the **/ru** parameter, then the run as user is the account specified by the **/ru** parameter.
/np	No password is stored. The task runs non-interactively as the given user. Only local resources are available.
/z	Specifies to delete the task upon the completion of its schedule.

Parameter	Description
/xml `<xmlfile>`	Creates a task specified in the XML file. Can be combined with the **/ru** and **/rp** parameters, or with the **/rp** parameter by itself if the XML file already contains the user account information.
/v1	Creates a task visible to pre-Vista operating systems. This is not compatible with the **/XML** parameter.
/f	Specifies to create the task and suppress warnings if the specified task already exists.
/rl `<level>`	Specifies the Run Level for the job. Acceptable values are **LIMITED** (scheduled tasks will be ran with the least level of privileges, such as Standard User accounts) and **HIGHEST** (scheduled tasks will be ran with the highest level of privileges, such as Superuser accounts). The default value is **Limited**.
/delay `<delaytime>`	Specifies the wait time to delay running the task after it's triggered in mmmm:ss format. This is only valid for the ONSTART, ONLOGON, and ONEVENT schedule types.
/hresult	Specifies the process exit code to be in HRESULT format.
/?	Displays help at the command prompt.

To schedule a task to run every `<n>` minutes

In a minute schedule, the **/sc minute** parameter is required. The **/mo** (modifier) parameter is optional and specifies the number of minutes between each run of the task. The default value for **/mo** is *1* (every minute). The **/et** (end time) and **/du** (duration) parameters are optional and can be used with or without the **/k** (end task) parameter.

Examples

- To schedule a security script, *Sec.vbs*, to run every 20 minutes, type:

  ```
  schtasks /create /sc minute /mo 20 /tn "Security Script" /tr
  \\central\data\scripts\sec.vbs
  ```

  Because this example doesn't include a starting date or time, the task starts 20 minutes after the command completes, and runs every 20 minutes thereafter whenever the system is running. Notice that the security script source file is

located on a remote computer, but that the task is scheduled and executes on the local computer.

- To schedule a security script, *Sec.vbs*, to run on the local computer every 100 minutes between 5:00 P.M. and 7:59 A.M. each day, type:

```
schtasks /create /tn "Security Script" /tr sec.vbs /sc minute /mo 100
/st 17:00 /et 08:00 /k
```

  This example uses the **/sc** parameter to specify a minute schedule and the **/mo** parameter to specify an interval of 100 minutes. It uses the **/st** and **/et** parameters to specify the start time and end time of each day's schedule. It also uses the **/k** parameter to stop the script if it's still running at 7:59 A.M. Without **/k**, schtasks wouldn't start the script after 7:59 A.M., but if the instance started at 6:20 A.M. was still running, it wouldn't stop it.

To schedule a task to run every `<n>` hours

In an hourly schedule, the **/sc hourly** parameter is required. The **/mo** (modifier) parameter is optional and specifies the number of hours between each run of the task. The default value for **/mo** is *1* (every hour). The **/k** (end task) parameter is optional and can be used with either **/et** (end at the specified time) or **/du** (end after the specified interval).

Examples

- To schedule the MyApp program to run every five hours, beginning on the first day of March 2002, type:

```
schtasks /create /sc hourly /mo 5 /sd 03/01/2002 /tn MyApp /tr
c:\apps\myapp.exe
```

  In this example, the local computer uses the **English (Zimbabwe)** option in **Regional and Language Options**, so the format for the start date is MM/DD/YYYY (03/01/2002).

- To schedule the MyApp program to run hourly, beginning at five minutes past midnight, type:

```
schtasks /create /sc hourly /st 00:05 /tn MyApp /tr c:\apps\myapp.exe
```

- To schedule the MyApp program to run every 3 hours, for 10 hours total, type:

```
schtasks /create /tn MyApp /tr MyApp.exe /sc hourly /mo 3 /st 00:00 /du
0010:00
```

In this example, the task runs at 12:00 A.M., 3:00 A.M., 6:00 A.M., and 9:00 A.M. Because the duration is 10 hours, the task isn't run again at 12:00 P.M. Instead, it starts again at 12:00 A.M. the next day. Also, because the program runs for just a few minutes, the **/k** parameter, which stops the program if it's still running when the duration expires, isn't necessary.

To schedule a task to run every `<n>` days

In a daily schedule, the **/sc daily** parameter is required. The **/mo** (modifier) parameter is optional and specifies the number of days between each run of the task. The default value for **/mo** is *1* (every day).

Examples

- To schedule the MyApp program to run once a day, every day, at 8:00 A.M. until December 31, 2021, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc daily /st 08:00
/ed 31/12/2021
```

In this example, the local computer system is set to the **English (United Kingdom)** option in **Regional and Language Options**, so the format for the end date is DD/MM/YYYY (31/12/2021). Additionally, because this example doesn't include the **/mo** parameter, the default interval of *1* is used to run the command every day.

- To schedule the MyApp program to run every twelve days at 1:00 P.M. (13:00) beginning on December 31, 2021, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc daily /mo 12 /sd
12/31/2002 /st 13:00
```

In this example, the system is set to the **English (Zimbabwe)** option in **Regional and Language Options**, so the format for the end date is MM/DD/YYYY (12/31/2021).

- To schedule a security script, *Sec.vbs*, to run every 70 days, type:

```
schtasks /create /tn "Security Script" /tr sec.vbs /sc daily /mo 70 /it
```

In this example, the **/it** parameter is used to specify that the task runs only when the user under whose account the task runs is logged onto the computer. Because the task runs with the permissions of a specific user account, this task only runs when that user is logged on.

> **Note**
>
> To identify tasks with the interactive-only (**/it**) property, use a verbose query (**/query /v**). In a verbose query display of a task with /it, the **Logon Mode** field has a value of Interactive only.

To schedule a task to run every `<n>` weeks

In a weekly schedule, the **/sc weekly** parameter is required. The **/mo** (modifier) parameter is optional and specifies the number of weeks between each run of the task. The default value for **/mo** is *1* (every week).

Weekly schedules also have an optional **/d** parameter to schedule the task to run on specified days of the week, or on all days (). The default is *MON (Monday)*. The every day () option is equivalent to scheduling a daily task.

Examples

- To schedule the MyApp program to run on a remote computer every six weeks, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc weekly /mo 6 /s
Server16 /u Admin01
```

Because this example leaves out the **/d** parameter, the task runs on Mondays. This example also uses the **/s** parameter to specify the remote computer and the **/u** parameter to run the command with the permissions of the user's Administrator account. Additionally, because the **/p** parameter is left out, SchTasks.exe prompts the user for the Administrator account password, and because the command is run remotely, all paths in the command, including the path to MyApp.exe, refer to paths on the remote computer.

- To schedule a task to run every other Friday, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc weekly /mo 2 /d
FRI
```

This example uses the **/mo** parameter to specify the two-week interval and the **/d** parameter to specify the day of the week. To schedule a task that runs every Friday, leave out the **/mo** parameter or set it to *1*.

To schedule a task to run every `<n>` months

In this schedule type, the **/sc monthly** parameter is required. The **/mo** (modifier) parameter, which specifies the number of months between each run of the task, is optional and the default is *1* (every month). This schedule type also has an optional **/d** parameter to schedule the task to run on a specified date of the month. The default is *1* (the first day of the month).

Examples

- To schedule the MyApp program to run on the first day of every month, type:

```
schtasks /create /tn MyApp /tr myapp.exe /sc monthly
```

The default value for both the **/mo** (modifier) parameter and the **/d** (day) parameter is *1*, so you don't need to use either of those parameters for this example.

- To schedule the MyApp program to run every three months, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /mo 3
  ```

  This example uses the **/mo** parameter to specify an interval of 3 months.

- To schedule the MyApp program to run every other month on the 21st day of the month at midnight for a year, from July 2, 2002 to June 30, 2003, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /mo 2 /d
  21 /st 00:00 /sd 2002/07/01 /ed 2003/06/30
  ```

  This example uses the **/mo** parameter to specify the monthly interval (every two months), the **/d** parameter to specify the date, the **/st** parameter to specify the time, and the **/sd** and **/ed** parameters to specify the start date and end date, respectively. Also in this example, the local computer is set to the **English (South Africa)** option in **Regional and Language Options**, so the dates are specified in the local format, YYYY/MM/DD.

To schedule a task to run on a specific day of the week

The day of the week schedule is a variation of the weekly schedule. In a weekly schedule, the **/sc weekly** parameter is required. The **/mo** (modifier) parameter is optional and specifies the number of weeks between each run of the task. The default value for **/mo** is *1* (every week). The **/d** parameter, which is optional, schedules the task to run on specified days of the week, or on all days (*). The default is *MON (Monday)*. The every day option `(/d *)` is equivalent to scheduling a daily task.

Examples

- To schedule the MyApp program to run every week on Wednesday, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc weekly /d WED
  ```

This example uses the **/d** parameter to specify the day of the week. Because the command leaves out the **/mo** parameter, the task runs every week.

- To schedule a task to run on Monday and Friday of every eighth week, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc weekly /mo 8 /d
MON,FRI
```

This example uses the **/d** parameter to specify the days and the **/mo** parameter to specify the eight-week interval.

To schedule a task to run on a specific week of the month

In this schedule type, the **/sc monthly** parameter, the **/mo** (modifier) parameter, and the **/d** (day) parameter are required. The **/mo** (modifier) parameter specifies the week on which the task runs. The **/d** parameter specifies the day of the week. You can specify only one day of the week for this schedule type. This schedule also has an optional **/m** (month) parameter that lets you schedule the task for particular months or every month (*). The default for the **/m** parameter is every month (*).

Examples

- To schedule the MyApp program to run on the second Sunday of every month, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /mo SECOND
/d SUN
```

This example uses the **/mo** parameter to specify the second week of the month and the **/d** parameter to specify the day.

- To schedule the MyApp program to run on the first Monday in March and September, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /mo FIRST
```

```
/d MON /m MAR,SEP
```

This example uses the **/mo** parameter to specify the first week of the month and the **/d** parameter to specify the day. It uses the **/m** parameter to specify the month, separating the month arguments with a comma.

To schedule a task to run on a specific day each month

In this schedule type, the **/sc monthly** parameter and the **/d** (day) parameter are required. The **/d** parameter specifies a date of the month (1 - 31), not a day of the week, and you can specify only one day in the schedule. The **/m** (month) parameter is optional, with the default being every month *()*, while the **/mo** (modifier) parameter isn't valid with this schedule type.

Schtasks.exe won't let you schedule a task for a date that's not in a month specified by the **/m** parameter. For example, trying to schedule the 31st day of February. However, if you don't use the **/m** parameter, and schedule a task for a date that doesn't appear in every month, then the task won't run in the shorter months. To schedule a task for the last day of the month, use the last day schedule type.

Examples

- To schedule the MyApp program to run on the first day of every month, type:

```

schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly
```

Because the default modifier is *none* (no modifier), this command uses the default day of *1*, and the default month of *every month*, without requiring any additional parameters.

- To schedule the MyApp program to run on May 15 and June 15 at 3:00 P.M. (15:00), type:

```

schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /d 15 /m
MAY,JUN /st 15:00
```

This example uses the **/d** parameter to specify the date and the **/m** parameter to specify the months. It also uses the **/st** parameter to specify the start time.

To schedule a task to run on the last day of a month

In the last day schedule type, the **/sc monthly** parameter, the **/mo LASTDAY** (modifier) parameter, and the **/m** (month) parameter are required. The **/d** (day) parameter isn't valid.

Examples

- To schedule the MyApp program to run on the last day of every month, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /mo
  lastday /m *
  ```

  This example uses the **/mo** parameter to specify the last day and the **/m** parameter with the wildcard character (*) to indicate that the program runs every month.

- To schedule the MyApp program to run on the last day of February and the last day of March at 6:00 P.M., type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /mo
  lastday /m FEB,MAR /st 18:00
  ```

  This example uses the **/mo** parameter to specify the last day, the **/m** parameter to specify the months, and the **/st** parameter to specify the start time.

To schedule to run once

In the run-once schedule type, the **/sc once** parameter is required. The **/st** parameter, which specifies the time that the task runs, is required. The **/sd** parameter, which specifies the date that the task runs, is optional, while the **/mo** (modifier) and **/ed** (end date) parameters aren't valid.

Schtasks won't let you schedule a task to run once if the date and time specified are in the past, based on the time of the local computer. To schedule a task that runs once on a remote computer in a different time zone, you must schedule it before that date and time occurs on the local computer.

Example

- To schedule the MyApp program to run at midnight on January 1, 2003, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc once /sd
  01/01/2003 /st 00:00
  ```

  This example uses the **/sc** parameter to specify the schedule type and the **/sd** and **/st** parameters to specify the date and time. Also in this example, the local computer uses the **English (United States)** option in **Regional and Language Options**, the format for the start date is MM/DD/YYYY.

To schedule a task to run every time the system starts

In the on-start schedule type, the **/sc onstart** parameter is required. The **/sd** (start date) parameter is optional and the default is the current date.

Example

- To schedule the MyApp program to run every time the system starts, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc onstart
  ```

  In this example, the local computer uses the **English (United States)** option in **Regional and Language Options**, the format for the start date is MM/DD/YYYY.

To schedule a task to run when a user logs on

The on logon schedule type schedules a task that runs whenever any user logs on to the computer. In the on logon schedule type, the **/sc onlogon** parameter is required. The

/sd (start date) parameter is optional and the default is the current date.

Example

- To schedule a task that runs when a user logs on to a remote computer, type:

  ```
  schtasks /create /tn "Start Web Site" /tr c:\myiis\webstart.bat /sc
  onlogon /s Server23
  ```

  This example schedules a batch file to run every time a user (any user) logs on to the remote computer. It uses the **/s** parameter to specify the remote computer. Because the command is remote, all paths in the command, including the path to the batch file, refer to a path on the remote computer.

To schedule a task to run when the system is idle

The on idle schedule type schedules a task that runs whenever there is no user activity during the time specified by the **/i** parameter. In the on idle schedule type, the **/sc onidle** parameter and the **/i** parameter are required. The **/sd** (start date) is optional and the default is the current date.

Example

- To schedule the MyApp program to run whenever the computer is idle, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc onidle /i 10
  ```

  This example uses the required **/i** parameter to specify that the computer must remain idle for ten minutes before the task starts.

To schedule a task to run now

Schtasks doesn't have a Run Now option, but you can simulate that option by creating a task that runs once and starts in a few minutes.

Example

- To schedule a task to run once, on November 13, 2020 at 2:18 P.M. local time, type:

```
schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc once /st 14:18 /sd
11/13/2002
```

  In this example, the local computer uses the **English (United States)** option in **Regional and Language Options**, so the format for the start date is MM/DD/YYYY.

To schedule a task that runs with different permissions

You can schedule tasks of all types to run with permissions of an alternate account on both the local and a remote computer. In addition to the parameters required for the particular schedule type, the **/ru** parameter is required and the **/rp** parameter is optional.

Examples

- To run the MyApp program on the local computer, type:

```
schtasks /create /tn MyApp /tr myapp.exe /sc weekly /d TUE /ru Admin06
```

  This example uses the **/ru** parameter to specify that the task should run with the permissions of the user's Administrator account (*Admin06*). Also in this example, the task is scheduled to run every Tuesday, but you can use any schedule type for a task run with alternate permissions.

  In response, SchTasks.exe prompts for the run as password for the *Admin06* account, and then displays a success message:

```
Please enter the run as password for Admin06: ********
SUCCESS: The scheduled task MyApp has successfully been created.
```

- To run the MyApp program on the *Marketing* computer every four days, type:

```
schtasks /create /tn MyApp /tr myapp.exe /sc daily /mo 4 /s Marketing
/u Marketing\Admin01 /ru Reskits\User01
```

This example uses the **/sc** parameter to specify a daily schedule, and the **/mo** parameter to specify an interval of four days. Additionally, this example uses the **/s** parameter to provide the name of the remote computer and the **/u** parameter to specify an account with permission to schedule a task on the remote computer (*Admin01 on the Marketing computer*). Finally, this example uses the **/ru** parameter to specify that the task should run with the permissions of the user's non-Administrator account (*User01* in the *Reskits* domain). Without the **/ru** parameter, the task would run with the permissions of the account specified by **/u**.

When running this example, Schtasks first requests the password of the user named by the **/u** parameter (to run the command) and then requests the password of the user named by the **/ru** parameter (to run the task). After authenticating the passwords, schtasks displays a message indicating that the task is scheduled:

```
Type the password for Marketing\Admin01:********
Please enter the run as password for Reskits\User01: ********
SUCCESS: The scheduled task MyApp has successfully been created.
```

- To run schedule the *AdminCheck.exe* program to run on the Public computer every Friday at 4:00 A.M.,, but only if the administrator of the computer is logged on, type:

```
schtasks /create /tn "Check Admin" /tr AdminCheck.exe /sc weekly /d FRI
/st 04:00 /s Public /u Domain3\Admin06 /ru Public\Admin01 /it
```

This example uses the **/sc** parameter to specify a weekly schedule, the **/d** parameter to specify the day, and the **/st** parameter to specify the start time. It also uses the **/s** parameter to provide the name of the remote computer, the **/u** parameter to specify an account with permission to schedule a task on the remote computer, the **/ru** parameter to configure the task to run with the permissions of the administrator of the Public computer (*Public\Admin01*), and the **/it** parameter to indicate that the task runs only when the *Public\Admin01* account is logged on.

> **Note**
>
> To identify tasks with the interactive-only (**/it**) property, use a verbose query (`/query /v`). In a verbose query display of a task with **/it**, the **Logon Mode** field has a value of **Interactive only**.

To schedule a task that runs with system permissions

Tasks of all types can run with permissions of the **System** account on both the local and a remote computer. In addition to the parameters required for the particular schedule type, the **/ru system** (or **/ru**) parameter is required, while the **/rp** parameter isn't valid.

> **Important**
>
> The **System** account doesn't have interactive logon rights. Users can't see or interact with programs or tasks run with system permissions. The **/ru** parameter determines the permissions under which the task runs, not the permissions used to schedule the task. Only Administrators can schedule tasks, regardless of the value of the **/ru** parameter.
>
> To identify tasks that run with system permissions, use a verbose query (`/query /v`). In a verbose query display of a system-run task, the **Run As User** field has a value of **NT AUTHORITY\SYSTEM** and the **Logon Mode** field has a value of **Background only**.

Examples

- To schedule the MyApp program to run on the local computer with permissions of the **System** account, type:

  ```
  schtasks /create /tn MyApp /tr c:\apps\myapp.exe /sc monthly /d 15 /ru System
  ```

  In this example, the task is scheduled to run on the fifteenth day of every month, but you can use any schedule type for a task run with system permissions. Additionally, this example uses the **/ru System** parameter to specify the system

security context. Because system tasks don't use a password, the **/rp** parameter is left out.

In response, SchTasks.exe displays an informational message and a success message, without prompting for a password:

```
INFO: The task will be created under user name (NT AUTHORITY\SYSTEM).
SUCCESS: The Scheduled task MyApp has successfully been created.
```

- To schedule the MyApp program to run on the *Finance01* computer every morning at 4:00 A.M., using system permissions, type:

```
schtasks /create /tn MyApp /tr myapp.exe /sc daily /st 04:00 /s
Finance01 /u Admin01 /ru System
```

This example uses the **/tn** parameter to name the task and the **/tr** parameter to specify the remote copy of the MyApp program, the **/sc** parameter to specify a daily schedule, but leaves out the **/mo** parameter because *1* (every day) is the default. This example also uses the **/st** parameter to specify the start time, which is also the time the task will run each day, the **/s** parameter to provide the name of the remote computer, the **/u** parameter to specify an account with permission to schedule a task on the remote computer, and the **/ru** parameter to specify that the task should run under the System account. Without the **/ru** parameter, the task would run using the permissions of the account specified by the **/u** parameter.

Schtasks.exe requests the password of the user named by the **/u** parameter and, after authenticating the password, displays a message indicating that the task is created and that it will run with permissions of the **System** account:

```
Type the password for Admin01:**********

INFO: The Schedule Task MyApp will be created under user name (NT
AUTHORITY\
SYSTEM).
SUCCESS: The scheduled task MyApp has successfully been created.
```

To schedule a task that runs more than one program

Each task runs only one program. However, you can create a batch file that runs multiple programs and then schedule a task to run the batch file.

1. Using a text editor, such as Notepad, create a batch file that includes the name and fully qualified path to the .exe file required to start the Event Viewer (Eventvwr.exe) and System Monitor (Perfmon.exe) programs.

   ```
   C:\Windows\System32\Eventvwr.exe
   C:\Windows\System32\Perfmon.exe
   ```

2. Save the file as *MyApps.bat*, open schtasks.exe, and then create a task to run *MyApps.bat* by typing:

   ```
   schtasks /create /tn Monitor /tr C:\MyApps.bat /sc onlogon /ru
   Reskit\Administrator
   ```

   This command creates the Monitor task, which runs whenever anyone logs on. It uses the **/tn** parameter to name the task, the **/tr** parameter to run MyApps.bat, the **/sc** parameter to indicate the OnLogon schedule type and the **/ru** parameter to run the task with the permissions of the user's Administrator account.

   As a result of this command, whenever a user logs on to the computer, the task starts both Event Viewer and System Monitor.

To schedule a task that runs on a remote computer

To schedule a task to run on a remote computer, you must add the task to the remote computer's schedule. Tasks of all types can be scheduled on a remote computer, but the following conditions must be met:

- You must have permission to schedule the task. As such, you must be logged on to the local computer with an account that is a member of the Administrators group on the remote computer, or you must use the **/u** parameter to provide the credentials of an Administrator of the remote computer.

- You can use the **/u** parameter only when the local and remote computers are in the same domain or the local computer is in a domain that the remote computer domain trusts. Otherwise, the remote computer cannot authenticate the user account specified and it cannot verify that the account is a member of the Administrators group.

- The task must have sufficient permission to run on the remote computer. The permissions required vary with the task. By default, the task runs with the permission of the current user of the local computer or, if the **/u** parameter is used, the task runs with the permission of the account specified by the **/u** parameter. However, you can use the **/ru** parameter to run the task with permissions of a different user account or with system permissions.

Examples

- To schedule the MyApp program (as an administrator) to run on the *SRV01* remote computer every ten days starting immediately,type:

  ```
  schtasks /create /s SRV01 /tn MyApp /tr c:\program
  files\corpapps\myapp.exe /sc daily /mo 10
  ```

  This example uses the **/s** parameter to provide the name of the remote computer. Because the local current user is an Administrator of the remote computer, the **/u** parameter, which provides alternate permissions for scheduling the task, isn't necessary.

  > ⓘ **Note**
  >
  > When scheduling tasks on a remote computer, all parameters refer to the remote computer. Therefore, the file specified by the **/tr** parameter refers to the copy of MyApp.exe on the remote computer.

- To schedule the MyApp program (as a user) to run on the *SRV06* remote computer every three hours, type:

  ```
  schtasks /create /s SRV06 /tn MyApp /tr c:\program
  files\corpapps\myapp.exe /sc hourly /mo 3 /u reskits\admin01 /p
  R43253@4$ /ru SRV06\user03 /rp MyFav!!Pswd
  ```

Because Administrator permissions are required to schedule a task, the command uses the **/u** and **/p** parameters to provide the credentials of the user's Administrator account (*Admin01* in the *Reskits* domain). By default, these permissions are also used to run the task. However, because the task does not need Administrator permissions to run, the command includes the **/u** and **/rp** parameters to override the default and run the task with permission of the user's non-Administrator account on the remote computer.

- To schedule the MyApp program (as a user) to run on the *SRV02* remote computer on the last day of every month.

```
schtasks /create /s SRV02 /tn MyApp /tr c:\program
files\corpapps\myapp.exe /sc monthly /mo LASTDAY /m * /u
reskits\admin01
```

Because the local current user (*user03*) isn't an Administrator of the remote computer, the command uses the **/u** parameter to provide the credentials of the user's Administrator account (*Admin01* in the *Reskits* domain). The Administrator account permissions will be used to schedule the task and to run the task.

Because the command did not include the **/p** (password) parameter, schtasks prompts for the password. Then it displays a success message and, in this case, a warning:

```
Type the password for reskits\admin01:********

SUCCESS: The scheduled task MyApp has successfully been created.
WARNING: The scheduled task MyApp has been created, but may not run
because the account information could not be set.
```

This warning indicates that the remote domain could not authenticate the account specified by the **/u** parameter. In this case, the remote domain could not authenticate the user account because the local computer isn't a member of a domain that the remote computer domain trusts. When this occurs, the task job appears in the list of scheduled tasks, but the task is actually empty and it won't run.

The following display from a verbose query exposes the problem with the task. In the display, note that the value of **Next Run Time** is **Never** and that the value of **Run As User** is **Could not be retrieved from the task scheduler database**.

Had this computer been a member of the same domain or a trusted domain, the task would have been successfully scheduled and would have run as specified.

```
HostName: SRV44
TaskName: MyApp
Next Run Time: Never
Status:
Logon mode: Interactive/Background
Last Run Time: Never
Last Result: 0
Creator: user03
Schedule: At 3:52 PM on day 31 of every month, start
starting 12/14/2001
Task To Run: c:\program files\corpapps\myapp.exe
Start In: myapp.exe
Comment: N/A
Scheduled Task State: Disabled
Scheduled Type: Monthly
Start Time: 3:52:00 PM
Start Date: 12/14/2001
End Date: N/A
Days: 31
Months: JAN,FEB,MAR,APR,MAY,JUN,JUL,AUG,SEP,OCT,NO
V,DEC
Run As User: Could not be retrieved from the task sched
uler database
Delete Task If Not Rescheduled: Enabled
Stop Task If Runs X Hours and X Mins: 72:0
Repeat: Every: Disabled
Repeat: Until: Time: Disabled
Repeat: Until: Duration: Disabled
Repeat: Stop If Still Running: Disabled
Idle Time: Disabled
Power Management: Disabled
```

Remarks

- To run the **/create** command with the permissions of a different user, use the **/u** parameter. The **/u** parameter is valid only for scheduling tasks on remote computers.
- To view more `schtasks /create` examples, type `schtasks /create /?` at a command prompt.
- To schedule a task that runs with permissions of a different user, use the **/ru** parameter. The **/ru** parameter is valid for tasks on local and remote computers.

- To use the **/u** parameter, the local computer must be in the same domain as the remote computer or it must be in a domain that the remote computer domain trusts. Otherwise, either the task isn't created, or the task job is empty and the task doesn't run.
- Schtasks always prompts for a password unless you provide one, even when you schedule a task on the local computer using the current user account. This is normal behavior for schtasks.
- Schtasks doesn't verify program file locations or user account passwords. If you don't enter the correct file location or the correct password for the user account, the task is created, but it won't run. Also, if the password for an account changes or expires, and you don't change the password saved in the task, then the task won't run.
- The **System** account doesn't have interactive logon rights. Users don't see and can't interact with programs run with system permissions.
- Each task runs only one program. However, you can create a batch file that starts multiple tasks, and then schedule a task that runs the batch file.
- You can test a task as soon as you create it. Use the run operation to test the task and then check the SchedLgU.txt file (SystemRoot\SchedLgU.txt) for errors.

Related links

- Command-Line Syntax Key
- schtasks change command
- schtasks delete command
- schtasks end command
- schtasks query command
- schtasks run command

Feedback

Was this page helpful? Yes No

schtasks delete

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes a scheduled task from the schedule. This command doesn't delete the program that the task runs or interrupt a running program.

Syntax

```
schtasks /delete /tn {<taskname> | *} [/f] [/s <computer> [/u [<domain>\]
<user> [/p <password>]]]
```

Parameters

Expand table

Parameter	Description
/tn `{<taskname> | *}`	Identifies the task to be deleted. If you use the `*`, this command deletes all tasks scheduled for the computer, not just the tasks scheduled by the current user.
/f	Suppresses the confirmation message. The task is deleted without warning.
/s `<computer>`	Specifies the name or IP address of a remote computer (with or without backslashes). The default is the local computer.
/u `[<domain>]`	Runs this command with the permissions of the specified user account. By default, the command runs with the permissions of the current user of the local computer. The specified user account must be a member of the Administrators group on the remote computer. The **/u** and **/p** parameters are valid only when you use **/s**.
/p `<password>`	Specifies the password of the user account specified in the **/u** parameter. If you use the **/u** parameter without the **/p** parameter or the password argument, schtasks will prompt you for a password. The **/u** and **/p** parameters are valid only when you use **/s**.
/?	Displays help at the command prompt.

Examples

To delete the *Start Mail* task from the schedule of a remote computer.

```
schtasks /delete /tn Start Mail /s Svr16
```

This command uses the **/s** parameter to identify the remote computer.

To delete all tasks from the schedule of the local computer, including tasks scheduled by other users.

```
schtasks /delete /tn * /f
```

This command uses the **/tn** * parameter to represent all tasks on the computer and the **/f** parameter to suppress the confirmation message.

Related links

- Command-Line Syntax Key
- schtasks change command
- schtasks create command
- schtasks end command
- schtasks query command
- schtasks run command

Feedback

Was this page helpful? Yes No

schtasks end

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Stops only the instances of a program started by a scheduled task. To stop other processes, you must use the TaskKill command.

Syntax

```
schtasks /end /tn <taskname> [/s <computer> [/u [<domain>\]<user> [/p <password>]]]
```

Parameters

Expand table

Parameter	Description
/tn `<taskname>`	Identifies the task that started the program. This parameter is required.
/s `<computer>`	Specifies the name or IP address of a remote computer (with or without backslashes). The default is the local computer.
/u `[<domain>]`	Runs this command with the permissions of the specified user account. By default, the command runs with the permissions of the current user of the local computer. The specified user account must be a member of the Administrators group on the remote computer. The **/u** and **/p** parameters are valid only when you use **/s**.
/p `<password>`	Specifies the password of the user account specified in the **/u** parameter. If you use the **/u** parameter without the **/p** parameter or the password argument, schtasks will prompt you for a password. The **/u** and **/p** parameters are valid only when you use **/s**.
/?	Displays help at the command prompt.

Examples

To stop the instance of Notepad.exe started by the *My Notepad* task, type:

```
schtasks /end /tn "My Notepad"
```

To stop the instance of Internet Explorer started by the *InternetOn* task on the remote computer, *Svr01*,type:

```
schtasks /end /tn InternetOn /s Svr01
```

Related links

- Command-Line Syntax Key
- schtasks change command
- schtasks create command
- schtasks delete command
- schtasks query command
- schtasks run command

Feedback

Was this page helpful? Yes No

schtasks query

Article • 02/16/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists all the tasks scheduled to run on the computer.

Syntax

```
schtasks [/query] [/fo {TABLE | LIST | CSV}] [/nh] [/v] [/s <computer> [/u
[<domain>\]<user> [/p <password>]]] [/tn <taskname>] [/xml] [/hresult]
```

Parameters

Expand table

Parameter	Description
/query	Optionally, specifies the name of the operation. Using this query without any parameters performs a query of all tasks on the system.
/fo `<format>`	Specifies the output format. The valid values are *TABLE*, *LIST*, or *CSV*.
/nh	Removes column headings from the table display. This parameter is valid with the *TABLE* or *CSV* output formats.
/v	Adds the advanced properties of the task to the display. This parameter is valid with the *LIST* or *CSV* output formats.
/s `<computer>`	Specifies the name or IP address of a remote computer (with or without backslashes). The default is the local computer.
/u `[<domain>]`	Runs this command with the permissions of the specified user account. By default, the command runs with the permissions of the current user of the local computer. The specified user account must be a member of the Administrators group on the remote computer. The **/u** and **/p** parameters are valid only when you use **/s**.
/p `<password>`	Specifies the password of the user account specified in the **/u** parameter. If you use the **/u** parameter without the **/p** parameter or the password argument, schtasks will prompt you for a password. The **/u** and **/p** parameters are valid only when you use **/s**.

Parameter	Description
/tn `<taskname>`	Queries a specific task by path and name. If **/tn** is used, `<taskname>` cannot be blank. Assumes from root directory "\" folder by default (thus \ is omittable if the task is located in the root directory). Does not accept wildcards. Use quotation marks to enclose names that include spaces.
/xml	Outputs all task definitions on the system to XML format. If **/tn** is specified, outputs the specific task name to XML format. The full path and name must be used.
/hresult	Specifies the process exit code to be in HRESULT format.
/?	Displays help at the command prompt.

Examples

To list all tasks scheduled for the local computer, type:

```
schtasks
schtasks /query
```

These commands produce the same result and can be used interchangeably.

To request a detailed display of the tasks on the local computer,type:

```
schtasks /query /fo LIST /v
```

This command uses the **/v** parameter to request a detailed (verbose) display and the **/fo LIST** parameter to format the display as a list for easy reading. You can use this command to verify that a task you created has the intended recurrence pattern.

To request a list of tasks scheduled for a remote computer and to add the tasks to a comma-separated log file on the local computer, type:

```
schtasks /query /s Reskit16 /fo csv /nh >> \\svr01\data\tasklogs\p0102.csv
```

You can use this command format to collect and track tasks that are scheduled for multiple computers. This command uses the **/s** parameter to identify the remote computer, *Reskit16*, the **/fo** parameter to specify the format and the **/nh** parameter to

suppress the column headings. The >> append symbol redirects the output to the task log, *p0102.csv*, on the local computer, *Svr01*. Because the command runs on the remote computer, the local computer path must be fully qualified.

To get information for the *Office Subscription Maintenance* task using the **/tn** parameter, which can also be viewed via Task Scheduler in the location *\Microsoft\Office* with the root directory "****" seen under the **Task Scheduler Library** folder, type:

```
schtasks /query /tn "\Microsoft\Office\Office Subscription Maintenance"
```

Related links

- Command-Line Syntax Key
- schtasks change command
- schtasks create command
- schtasks delete command
- schtasks end command
- schtasks run command

Feedback

Was this page helpful? Yes No

schtasks run

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Starts a scheduled task immediately. The run operation ignores the schedule, but uses the program file location, user account, and password saved in the task to run the task immediately. Running a task does not affect the task schedule and does not change the next run time scheduled for the task.

Syntax

```
schtasks /run /tn <taskname> [/s <computer> [/u [<domain>\]<user> [/p <password>]]]
```

Parameters

Expand table

Parameter	Description
/tn `<taskname>`	Identifies the task to start. This parameter is required.
/s `<computer>`	Specifies the name or IP address of a remote computer (with or without backslashes). The default is the local computer.
/u `[<domain>]`	Runs this command with the permissions of the specified user account. By default, the command runs with the permissions of the current user of the local computer. The specified user account must be a member of the Administrators group on the remote computer. The **/u** and **/p** parameters are valid only when you use **/s**.
/p `<password>`	Specifies the password of the user account specified in the **/u** parameter. If you use the **/u** parameter without the **/p** parameter or the password argument, schtasks will prompt you for a password. The **/u** and **/p** parameters are valid only when you use **/s**.
/?	Displays help at the command prompt.

Remarks

- Use this operation to test your tasks. If a task doesn't run, check the Task Scheduler Service transaction log, `<Systemroot>\SchedLgU.txt` for errors.
- To run a task remotely, the task must be scheduled on the remote computer. When you run the task, it runs only on the remote computer. To verify that a task is running on a remote computer, use Task Manager or the Task Scheduler Service transaction log, `<Systemroot>\SchedLgU.txt`.

Examples

To start the *Security Script* task, type:

```
schtasks /run /tn "Security Script"
```

To start the *Update* task on a remote computer, Svr01, type:

```
schtasks /run /tn Update /s Svr01
```

Related links

- Command-Line Syntax Key
- schtasks change command
- schtasks create command
- schtasks delete command
- schtasks end command
- schtasks query command

Feedback

Was this page helpful? Yes No

scwcmd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

The Scwcmd.exe command-line tool included with the Security Configuration Wizard (SCW) can be used to perform the following tasks:

- Analyze one or many servers with an SCW-generated policy.
- Configure one or many servers with an SCW-generated policy.
- Register a Security Configuration Database extension with SCW.
- Rollback SCW policies.
- Transform an SCW-generated policy into native files that are supported by Group Policy.
- View analysis results in HTML format.

> **Note**
>
> If you use **scwcmd** to configure, analyze, or roll back a policy on a remote server, SCW must be installed on the remote server.

Syntax

```
scwcmd analyze
scwcmd configure
scwcmd register
scwcmd rollback
scwcmd transform
scwcmd view
```

Parameters

Expand table

Parameter	Description
scwcmd analyze	Determines whether a computer is in compliance with a policy.
scwcmd configure	Applies an SCW-generated security policy to a computer.
scwcmd register	Extends or customizes the SCW Security Configuration Database by registering a Security Configuration Database file that contains role, task, service, or port definitions.
scwcmd rollback	Applies the most recent rollback policy available, and then deletes that rollback policy.
scwcmd transform	Transforms a security policy file generated by using SCW into a new Group Policy object (GPO) in Active Directory Domain Services.
scwcmd view	Renders an .xml file by using a specified .xsl transform.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

scwcmd analyze

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Determines whether a computer is in compliance with a policy. Results are returned in an .xml file.

This command also accepts a list of computer names as input. To view the results in your browser, use **scwcmd view** and specify `%windir%\security\msscw\TransformFiles\scwanalysis.xsl` as the .xsl transform.

Syntax

```
scwcmd analyze [[[/m:<computername> | /ou:<OuName>] /p:<policy>] | /i:
<computerlist>] [/o:<resultdir>] [/u:<username>] [/pw:<password>] [/t:
<threads>] [/l] [/e]
```

Parameters

Expand table

Parameter	Description
/m: `<computername>`	Specifies the NetBIOS name, DNS name, or IP address of the computer to analyze. If the **/m** parameter is specified, then the **/p** parameter must also be specified.
/ou: `<OuName>`	Specifies the fully qualified domain name (FQDN) of an organizational unit (OU) in Active Directory Domain Services. If the **/ou** parameter is specified, then the **/p** parameter must also be specified. All computers in the OU will be analyzed against the given policy.
/p: `<policy>`	Specifies the path and file name of the .xml policy file to be used to perform the analysis.
/i: `<computerlist>`	Specifies the path and file name of an .xml file that contains a list of computers along with their expected policy files. All computers in the .xml file will be analyzed against their corresponding policy files. A sample .xml file is `%windir%\security\SampleMachineList.xml`.

Parameter	Description
/o:`<resultdir>`	Specifies the path and directory where the analysis result files should be saved. The default is the current directory.
/u:`<username>`	Specifies an alternate user credential to use when performing the analysis on a remote computer. The default is the logged on user.
/pw:`<password>`	Specifies an alternate user credential to use when performing the analysis on a remote computer. The default is the password of the logged on user.
/t:`<threads>`	Specifies the number of simultaneous outstanding analysis operations that should be maintained during the analysis. The value range is 1-1000, with a default value of 40.
/l	Causes the analysis process to be logged. One log file will be generated for each computer being analyzed. The log files will be stored in the same directory as the result files. Use the **/o** option to specify the directory for the result files.
/e	Log an event to the Application Event log if a mismatch is found.
/?	Displays help at the command prompt.

Examples

To analyze a security policy against the file *webpolicy.xml*, type:

```
scwcmd analyze /p:webpolicy.xml
```

To analyze a security policy on the computer named *webserver* against the file *webpolicy.xml* by using the credentials of the *webadmin* account, type:

```
scwcmd analyze /m:webserver /p:webpolicy.xml /u:webadmin
```

To analyze a security policy against the file *webpolicy.xml*, with a *maximum of 100 threads*, and output the results to a file named results in the *resultserver* share, type:

```
scwcmd analyze /i:webpolicy.xml /t:100 /o:\\resultserver\results
```

To analyze a security policy for the *WebServers OU* against the file *webpolicy.xml* by using the *DomainAdmin* credentials, type:

```
scwcmd analyze /ou:OU=WebServers,DC=Marketing,DC=ABCCompany,DC=com
/p:webpolicy.xml /u:DomainAdmin
```

Related links

- Command-Line Syntax Key
- scwcmd configure command
- scwcmd register command
- scwcmd rollback command
- scwcmd transform command
- scwcmd view command

Feedback

Was this page helpful? Yes No

scwcmd configure

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Applies a Security Configuration Wizard (SCW)-generated security policy to a computer. This command-line tool also accepts a list of computer names as input.

Syntax

```
scwcmd configure [[[/m:<computername> | /ou:<OuName>] /p:<policy>] | /i:
<computerlist>] [/u:<username>] [/pw:<password>] [/t:<threads>]
```

Parameters

Expand table

Parameter	Description
/m: `<computername>`	Specifies the NetBIOS name, DNS name, or IP address of the computer to configure. If the **/m** parameter is specified, then the **/p** parameter must also be specified.
/ou: `<OuName>`	Specifies the fully qualified domain name (FQDN) of an organizational unit (OU) in Active Directory Domain Services. If the **/ou** parameter is specified, then the **/p** parameter must also be specified. All computers in the OU will be configured against the given policy.
/p: `<policy>`	Specifies the path and file name of the .xml policy file to be used to perform the configuration.
/i: `<computerlist>`	Specifies the path and file name of an .xml file that contains a list of computers along with their expected policy files. All computers in the .xml file will be analyzed against their corresponding policy files. A sample .xml file is `%windir%\security\SampleMachineList.xml`.
/u: `<username>`	Specifies an alternate user credential to use when performing the configuration on a remote computer. The default is the logged on user.
/pw: `<password>`	Specifies an alternate user credential to use when performing the configuration on a remote computer. The default is the password of the logged on user.

Parameter	Description
/t:`<threads>`	Specifies the number of simultaneous outstanding configuration operations that should be maintained during the analysis. The value range is 1-1000, with a default value of 40.
/l	Causes the analysis process to be logged. One log file will be generated for each computer being analyzed. The log files will be stored in the same directory as the result files. Use the **/o** option to specify the directory for the result files.
/e	Log an event to the Application Event log if a mismatch is found.
/?	Displays help at the command prompt.

Examples

To configure a security policy against the file *webpolicy.xml*, type:

```
scwcmd configure /p:webpolicy.xml
```

To configure a security policy for the computer at *172.16.0.0* against the file *webpolicy.xml* by using the credentials of the *webadmin* account, type:

```
scwcmd configure /m:172.16.0.0 /p:webpolicy.xml /u:webadmin
```

To configure a security policy on all computers on the list *campusmachines.xml* with a *maximum of 100 threads*, type:

```
scwcmd configure /i:campusmachines.xml /t:100
```

To configure a security policy for the *WebServers OU* against the file *webpolicy.xml* by using the *DomainAdmin* credentials, type:

```
scwcmd configure /ou:OU=WebServers,DC=Marketing,DC=ABCCompany,DC=com
/p:webpolicy.xml /u:DomainAdmin
```

Related links

- Command-Line Syntax Key
- scwcmd analyze command
- scwcmd register command
- scwcmd rollback command
- scwcmd transform command
- scwcmd view command

Feedback

Was this page helpful? Yes No

scwcmd register

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Extends or customizes the Security Configuration Wizard (SCW) Security Configuration Database by registering a Security Configuration Database file that contains role, task, service, or port definitions.

Syntax

```
scwcmd register /kbname:<MyApp> [/kbfile:<kb.xml>] [/kb:<path>] [/d]
```

Parameters

Expand table

Parameter	Description
/kbname: `<MyApp>`	Specifies the name under which the Security Configuration Database extension will be registered. This parameter must be specified.
/kbfile: `<kb.xml>`	Specifies the path and file name of the Security Configuration Database file used to extend or customize the base Security Configuration Database. To validate that the Security Configuration Database file is compliant with the SCW schema, use the `%windir%\security\KBRegistrationInfo.xsd` schema definition file. This option must be provided unless the **/d** parameter is specified.
/kb: `<path>`	Specifies the path to the directory that contains the SCW Security Configuration Database files to be updated. If this option is not specified, `%windir%\security\msscw\kbs` is used.
/d	Unregisters a Security Configuration Database extension from the Security Configuration Database. The extension to unregister is specified by the **/kbname** parameter. (The **/kbfile** parameter shouldn't be specified.) The Security Configuration Database to unregister the extension from is specified by the **/kb** parameter.
/?	Displays help at the command prompt.

Examples

To register the Security Configuration Database file named *SCWKBForMyApp.xml* under the name *MyApp* in the location `\\kbserver\kb`, type:

```
scwcmd register /kbfile:d:\SCWKBForMyApp.xml /kbname:MyApp /kb:\\kbserver\kb
```

To unregister the Security Configuration Database *MyApp*, located at `\\kbserver\kb`, type:

```
scwcmd register /d /kbname:MyApp /kb:\\kbserver\kb
```

Related links

- Command-Line Syntax Key
- scwcmd analyze command
- scwcmd configure command
- scwcmd rollback command
- scwcmd transform command
- scwcmd view command

Feedback

Was this page helpful? Yes No

scwcmd rollback

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Applies the most recent rollback policy available, and then deletes that rollback policy.

Syntax

```
scwcmd rollback /m:<computername> [/u:<username>] [/pw:<password>]
```

Parameters

Expand table

Parameter	Description
/m:`<computername>`	Specifies the NetBIOS name, DNS name, or IP address of a computer where the rollback operation should be performed.
/u:`<username>`	Specifies an alternate user account to use when performing a remote rollback. The default is the logged on user.
/pw:`<password>`	Specifies an alternate user credential to use when performing a remote rollback. The default is the logged on user.
/?	Displays help at the command prompt.

Examples

To roll back the security policy on a computer at IP address *172.16.0.0*, type:

```
scwcmd rollback /m:172.16.0.0
```

Related links

- Command-Line Syntax Key
- scwcmd analyze command
- scwcmd configure command
- scwcmd register command
- scwcmd transform command
- scwcmd view command

Feedback

Was this page helpful? Yes No

scwcmd transform

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Transforms a security policy file generated by using the Security Configuration Wizard (SCW) into a new Group Policy Object (GPO) in Active Directory Domain Services. The transform operation does not change any settings on the server where it is performed. After the transform operation has completed, an administrator must link the GPO to the desired OUs to deploy the policy to servers.

> **Important**
>
> Domain administrator credentials are needed to complete the transform operation.
>
> Internet Information Services (IIS) security policy settings can't be deployed by using Group Policy.
>
> Firewall policies that list approved apps shouldn't be deployed to servers unless the Windows Firewall service started automatically when the server was last started.

Syntax

```
scwcmd transform /p:<policyfile.xml> /g:<GPOdisplayname>
```

Parameters

Expand table

Parameter	Description
/p: `<policyfile.xml>`	Specifies the path and file name of the .xml policy file that should be applied. This parameter must be specified.
/g: `<GPOdisplayname>`	Specifies the display name of the GPO. This parameter must be specified.

Parameter	Description
/?	Displays help at the command prompt.

Examples

To create a GPO named *FileServerSecurity* from a file named *FileServerPolicy.xml*, type:

```
scwcmd transform /p:FileServerPolicy.xml /g:FileServerSecurity
```

Related links

- Command-Line Syntax Key
- scwcmd analyze command
- scwcmd configure command
- scwcmd register command
- scwcmd rollback command
- scwcmd view command

Feedback

Was this page helpful? Yes No

scwcmd view

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Renders an .xml file by using a specified .xsl transform. This command can be useful for displaying Security Configuration Wizard (SCW) .xml files by using different views.

Syntax

```
scwcmd view /x:<Xmlfile.xml> [/s:<Xslfile.xsl>]
```

Parameters

Expand table

Parameter	Description
/x: `<Xmlfile.xml>`	Specifies the .xml file to be viewed. This parameter must be specified.
/s: `<Xslfile.xsl>`	Specifies the .xsl transform to apply to the .xml file as part of the rendering process. This parameter is optional for SCW .xml files. When the **view** command is used to render a SCW .xml file, it will automatically try to load the correct default transform for the specified .xml file. If an .xsl transform is specified, the transform must be written under the assumption that the .xml file is in the same directory as the .xsl transform.
/?	Displays help at the command prompt.

Example

To view *Policyfile.xml* by using the *Policyview.xsl* transform, type:

```
scwcmd view /x:C:\policies\Policyfile.xml /s:C:\viewers\Policyview.xsl
```

Related links

- Command-Line Syntax Key
- scwcmd analyze command
- scwcmd configure command
- scwcmd register command
- scwcmd rollback command
- scwcmd transform command

Feedback

Was this page helpful? Yes No

secedit commands

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Configures and analyzes system security by comparing your current security configuration against specified security templates.

> **Note**
>
> The Microsoft Management Console (MMC) and the Security Configuration and Analysis snap-in are not available on Server Core.

Syntax

```
secedit /analyze
secedit /configure
secedit /export
secedit /generaterollback
secedit /import
secedit /validate
```

Parameters

Expand table

Parameter	Description
secedit /analyze	Allows you to analyze current systems settings against baseline settings that are stored in a database. The analysis results are stored in a separate area of the database and can be viewed in the Security Configuration and Analysis snap-in.
secedit /configure	Allows you to configure a system with security settings stored in a database.
secedit /export	Allows you to export security settings stored in a database.

Parameter	Description
secedit /generaterollback	Allows you to generate a rollback template with respect to a configuration template.
secedit /import	Allows you to import a security template into a database so that the settings specified in the template can be applied to a system or analyzed against a system.
secedit /validate	Allows you to validate the syntax of a security template.

Remarks

- If there is no filepath specified, all filenames will default to the current directory.
- Your analysis results are stored in a separate area of the database and can be viewed in the Security Configuration and Analysis snap-in to the MMC.
- If your security templates are created by using the Security Template snap-in, and if you run the Security Configuration and Analysis snap-in against those templates, the following files are created:

Expand table

File	Description
scesrv.log	◦ **Location:** `%windir%\security\logs` ◦ **Created by:** Operating system ◦ **File type:** Text ◦ **Refresh rate:** Overwritten when `secedit analyze`, `secedit configure`, `secedit export` or `secedit import` is run. ◦ **Content:** Contains the results of the analysis grouped by policy type.
user-selected name.sdb	◦ **Location:** `%windir%\<user account>\Documents\Security\Database` ◦ **Created by:** Running the Security Configuration and Analysis snap-in ◦ **File type:** Proprietary ◦ **Refresh rate:** Updated whenever a new security template is created. ◦ **Content:** Local security policies and user-created security templates.
user-selected name.log	◦ **Location:** User-defined, but defaults to `%windir%\<user account>\Documents\Security\Logs` ◦ **Created by:** Running the `secedit analyze` or `secedit configure` commands, or by using the Security Configuration and Analysis snap-in. ◦ **File type:** Text ◦ **Refresh rate:** Overwritten when `secedit analyze` or `secedit configure` is run, or by using the Security Configuration and Analysis snap-in.

File	Description
	◦ **Content:** Log file name, date and time, and the results of the analysis or investigation.
user-selected name.inf	◦ **Location:** `%windir%\*<user account>\Documents\Security\Templates` ◦ **Created by:** Running the Security Template snap-in. ◦ **File type:** Text ◦ **Refresh rate:** Overwritten each time the security template is updated. ◦ **Content:** Contains the set up information for the template for each policy selected using the snap-in.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

secedit /analyze

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Allows you to analyze current systems settings against baseline settings that are stored in a database.

Syntax

```
secedit /analyze /db <database file name> [/cfg <configuration file name>]
[/overwrite] [/log <log file name>] [/quiet}]
```

Parameters

Expand table

Parameter	Description
/db	Required. Specifies the path and file name of the database containing the stored configuration against which the analysis is performed. If the file name specifies a database that hasn't had a security template (as represented by the configuration file) associated with it, the `/cfg <configuration file name>` option must also be specified.
/cfg	Specifies the path and file name for the security template that will be imported into the database for analysis. This option is only valid when used with the `/db <database file name>` parameter. If this parameter isn't also specified, the analysis is performed against any configuration already stored in the database.
/overwrite	Specifies whether the security template in the **/cfg** parameter should overwrite any template or composite template that is stored in the database, instead of appending the results to the stored template. This option is only valid when the `/cfg <configuration file name>` parameter is also used. If this parameter isn't also specified, the template in the **/cfg** parameter is appended to the stored template.
/log	Specifies the path and file name of the log file to be used in the process. If you don't specify a file location, the default log file, `<systemroot>\Documents and Settings\<UserAccount>\My Documents\Security\Logs\<databasename>.log` is used.

Parameter	Description
/quiet	Suppresses screen output. You can still view analysis results by using the Security Configuration and Analysis snap-in to the Microsoft Management Console (MMC).

Examples

To perform the analysis for the security parameters on the security database, *SecDbContoso.sdb*, and then direct the output to the file *SecAnalysisContosoFY11*, including prompts to verify the command ran correctly, type:

```
secedit /analyze /db C:\Security\FY11\SecDbContoso.sdb /log
C:\Security\FY11\SecAnalysisContosoFY11.log
```

To incorporate changes required by the analysis process on the *SecContoso.inf* file, and then to direct the output to the existing file, *SecAnalysisContosoFY11*, without prompting, type:

```
secedit /analyze /db C:\Security\FY11\SecDbContoso.sdb /cfg SecContoso.inf
/overwrite /log C:\Security\FY11\SecAnalysisContosoFY11.xml /quiet
```

Related links

- Command-Line Syntax Key
- secedit /configure
- secedit /export
- secedit /generaterollback
- secedit /import
- secedit /validate

Feedback

Was this page helpful? Yes No

secedit /configure

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Allows you to configure the current system settings using security settings stored in a database.

Syntax

```
secedit /configure /db <database file name> [/cfg <configuration file name>]
[/overwrite] [/areas [securitypolicy | group_mgmt | user_rights | regkeys |
filestore | services]] [/log <log file name>] [/quiet]
```

Parameters

Expand table

Parameter	Description
/db	Required. Specifies the path and file name of the database containing the stored configuration. If the file name specifies a database that hasn't had a security template (as represented by the configuration file) associated with it, the `/cfg <configuration file name>` option must also be specified.
/cfg	Specifies the path and file name for the security template that will be imported into the database for analysis. This option is only valid when used with the `/db <database file name>` parameter. If this parameter isn't also specified, the analysis is performed against any configuration already stored in the database.
/overwrite	Specifies whether the security template in the **/cfg** parameter should overwrite any template or composite template that is stored in the database, instead of appending the results to the stored template. This option is only valid when the `/cfg <configuration file name>` parameter is also used. If this parameter isn't also specified, the template in the **/cfg** parameter is appended to the stored template.
/areas	Specifies the security areas to be applied to the system. If this parameter is not specified, all security settings defined in the database are applied to the system. To configure multiple areas, separate each area by a space. The following security areas are supported:

Parameter	Description
	• **securitypolicy:** Local policy and domain policy for the system, including account policies, audit policies, security options, and so on. • **group_mgmt:** Restricted group settings for any groups specified in the security template. • **user_rights:** User logon rights and granting of privileges. • **regkeys:** Security on local registry keys. • **filestore:** Security on local file storage. • **services:** Security for all defined services.
/log	Specifies the path and file name of the log file to be used in the process. If you don't specify a file location, the default log file, `<systemroot>\Documents and Settings\<UserAccount>\My Documents\Security\Logs\<databasename>.log` is used.
/quiet	Suppresses screen and log output. You can still view analysis results by using the Security Configuration and Analysis snap-in to the Microsoft Management Console (MMC).

Examples

To perform the analysis for the security parameters on the security database, *SecDbContoso.sdb*, and then direct the output to the file *SecAnalysisContosoFY11*, including prompts to verify the command ran correctly, type:

```
secedit /analyze /db C:\Security\FY11\SecDbContoso.sdb /log
C:\Security\FY11\SecAnalysisContosoFY11.log
```

To incorporate changes required by the analysis process on the *SecContoso.inf* file, and then to direct the output to the existing file, *SecAnalysisContosoFY11*, without prompting, type:

```
secedit /configure /db C:\Security\FY11\SecDbContoso.sdb /cfg SecContoso.inf
/overwrite /log C:\Security\FY11\SecAnalysisContosoFY11.xml /quiet
```

Related links

- Command-Line Syntax Key
- secedit /analyze

- secedit /export
- secedit /generaterollback
- secedit /import
- secedit /validate

Feedback

Was this page helpful? Yes No

secedit /export

Article • 02/16/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Exports security settings stored in a database configured with security templates. You can use this command to backup your security policies on a local computer, in addition to importing the settings to another computer.

Syntax

```
secedit /export [/db <database file name>] [/mergedpolicy] /cfg
<configuration file name> [/areas [securitypolicy | group_mgmt | user_rights
| regkeys | filestore | services]] [/log <log file name>] [/quiet]
```

Parameters

Expand table

Parameter	Description
/db	Requires that you specify the path and file name of the database containing the stored configuration when exporting to a location other than the default. If not specified, the system security database will be stored in **%windir%\security\database**. If the file name specifies a database that hasn't had a security template (as represented by the configuration file) associated with it, the `/cfg <configuration file name>` option must also be specified for the export.
/mergedpolicy	Merges and exports domain and local policy security settings.
/cfg	Required. Specifies the path and file name for the security template that will be imported into the database for analysis. This option is only valid when used with the `/db <database file name>` parameter. If this parameter isn't also specified, the analysis is performed against any configuration already stored in the database.
/areas	Specifies the security areas to be applied to the system. If this parameter is not specified, all security settings defined in the database are applied to the system. To configure multiple areas, separate each area by a space. The following security areas are supported:

Parameter	Description
	• **securitypolicy:** Local policy and domain policy for the system, including account policies, audit policies, security options, and so on. • **group_mgmt:** Restricted group settings for any groups specified in the security template. • **user_rights:** User logon rights and granting of privileges. • **regkeys:** Security on local registry keys. • **filestore:** Security on local file storage. • **services:** Security for all defined services.
/log	Specifies the path and file name of the log file to be used in the process. If you don't specify a file location, the default log file, `<systemroot>\Documents and Settings\<UserAccount>\My Documents\Security\Logs\<databasename>.log` is used.
/quiet	Suppresses screen and log output. You can still view analysis results by using the Security Configuration and Analysis snap-in to the Microsoft Management Console (MMC).

Examples

To export the security database and the domain security policies to an inf file, and then import that file to a different database in order to replicate the security policy settings on another computer, type:

```
secedit /export /db C:\Security\FY11\SecDbContoso.sdb /mergedpolicy /cfg
SecContoso.inf /log C:\Security\FY11\SecAnalysisContosoFY11.log /quiet
```

To import your example file to a different database on another computer, type:

```
secedit /import /db C:\Security\FY12\SecDbContoso.sdb /cfg SecContoso.inf
/log C:\Security\FY11\SecAnalysisContosoFY12.log /quiet
```

Related links

- Command-Line Syntax Key
- secedit /analyze
- secedit /configure

- secedit /generaterollback
- secedit /import
- secedit /validate

Feedback

Was this page helpful? Yes No

secedit /generaterollback

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Allows you to generate a rollback template for a specified configuration template. If an existing rollback template exists, running this command again will overwrite the existing information.

Successfully running this command logs the mismatches between the specified security template the security policy configuration into the scesrv.log file.

Syntax

```
secedit /generaterollback /db <database file name> /cfg <configuration file name> /rbk <rollback template file name> [/log <log file name>] [/quiet]
```

Parameters

Expand table

Parameter	Description
/db	Required. Specifies the path and file name of the database containing the stored configuration against which the analysis is performed. If the file name specifies a database that hasn't had a security template (as represented by the configuration file) associated with it, the `/cfg <configuration file name>` option must also be specified.
/cfg	Required. Specifies the path and file name for the security template that will be imported into the database for analysis. This option is only valid when used with the `/db <database file name>` parameter. If this parameter isn't also specified, the analysis is performed against any configuration already stored in the database.
/rbk	Required. Specifies a security template into which the rollback information is written. Security templates are created using the Security Templates snap-in. Rollback files can be created with this command.
/log	Specifies the path and file name of the log file to be used in the process. If you don't specify a file location, the default log file, `<systemroot>\Documents and Settings\`

Parameter	Description
	`<UserAccount>\My Documents\Security\Logs\<databasename>.log` is used.
/quiet	Suppresses screen and log output. You can still view analysis results by using the Security Configuration and Analysis snap-in to the Microsoft Management Console (MMC).

Examples

To create the rollback configuration file, for the previously created *SecTmplContoso.inf* file, while saving the original settings, and then write out the action to the *SecAnalysisContosoFY11* log file, type:

```
secedit /generaterollback /db C:\Security\FY11\SecDbContoso.sdb /cfg
sectmplcontoso.inf /rbk sectmplcontosoRBK.inf /log
C:\Security\FY11\SecAnalysisContosoFY11.log
```

Related links

- Command-Line Syntax Key
- secedit /analyze
- secedit /configure
- secedit /export
- secedit /import
- secedit /validate

Feedback

Was this page helpful? Yes No

secedit /import

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Imports security settings (.inf file), previously exported from the database configured with security templates.

> **Important**
>
> Before you import an .inf file to another computer, you must run the `secedit /generaterollback` command on the database on which the import will be performed.
>
> You must also run the `secedit /validate` command on the import file to verify its integrity.

Syntax

```
secedit /import /db <database file name> /cfg <configuration file name>
[/overwrite] [/areas [securitypolicy | group_mgmt | user_rights | regkeys |
filestore | services]] [/log <log file name>] [/quiet]
```

Parameters

Expand table

Parameter	Description
/db	Required. Specifies the path and file name of the database containing the stored configuration against which the import is performed. If the file name specifies a database that hasn't had a security template (as represented by the configuration file) associated with it, the `/cfg <configuration file name>` option must also be specified.
/overwrite	Specifies whether the security template in the **/cfg** parameter should overwrite any template or composite template that is stored in the database, instead of appending the results to the stored template. This option is only valid when the `/cfg`

Parameter	Description
	`<configuration file name>` parameter is also used. If this parameter isn't also specified, the template in the **/cfg** parameter is appended to the stored template.
/cfg	Required. Specifies the path and file name for the security template that will be imported into the database for analysis. This option is only valid when used with the `/db <database file name>` parameter. If this parameter isn't also specified, the analysis is performed against any configuration already stored in the database.
/areas	Specifies the security areas to be applied to the system. If this parameter is not specified, all security settings defined in the database are applied to the system. To configure multiple areas, separate each area by a space. The following security areas are supported: • **securitypolicy:** Local policy and domain policy for the system, including account policies, audit policies, security options, and so on. • **group_mgmt:** Restricted group settings for any groups specified in the security template. • **user_rights:** User logon rights and granting of privileges. • **regkeys:** Security on local registry keys. • **filestore:** Security on local file storage. • **services:** Security for all defined services.
/log	Specifies the path and file name of the log file to be used in the process. If you don't specify a file location, the default log file, `<systemroot>\Documents and Settings\<UserAccount>\My Documents\Security\Logs\<databasename>.log` is used.
/quiet	Suppresses screen and log output. You can still view analysis results by using the Security Configuration and Analysis snap-in to the Microsoft Management Console (MMC).

Examples

To export the security database and the domain security policies to an .inf file, and then to import that file to a different database to replicate the policy settings on another computer, type:

```
secedit /export /db C:\Security\FY11\SecDbContoso.sdb /mergedpolicy /cfg
NetworkShare\Policies\SecContoso.inf /log
C:\Security\FY11\SecAnalysisContosoFY11.log /quiet
```

To import just the security policies portion of the file to a different database on another computer, type:

```
secedit /import /db C:\Security\FY12\SecDbContoso.sdb /cfg
NetworkShare\Policies\SecContoso.inf /areas securitypolicy /log
C:\Security\FY11\SecAnalysisContosoFY12.log /quiet
```

Related links

- Command-Line Syntax Key
- secedit /analyze
- secedit /configure
- secedit /export
- secedit /generaterollback
- secedit /validate

Feedback

Was this page helpful? Yes No

secedit /validate

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Validates the security settings stored in a security template (.inf file). Validating security templates can help you determine if one is corrupted or inappropriately set. Corrupted or inappropriately set security templates aren't applied.

Syntax

```
secedit /validate <configuration file name>
```

Parameters

Expand table

Parameter	Description
`<configuration file name>`	Required. Specifies the path and file name for the security template that will be validated. Log files aren't updated by this command.

Examples

To verify that the rollback .inf file, *secRBKcontoso.inf*, is still valid after rollback, type:

```
secedit /validate secRBKcontoso.inf
```

Related links

- Command-Line Syntax Key
- secedit /analyze

- secedit /configure
- secedit /export
- secedit /generaterollback
- secedit /import

Feedback

Was this page helpful? Yes No

serverceipoptin

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Allows you to participate in the Customer Experience Improvement Program (CEIP).

Syntax

```
serverceipoptin [/query] [/enable] [/disable]
```

Parameters

Expand table

Parameter	Description
/query	Verifies your current setting.
/enable	Turns on your participation in CEIP.
/disable	Turns off your participation in CEIP.
/?	Displays help at the command prompt.

Examples

To verify your current settings, type:

```
serverceipoptin /query
```

To turn on your participation, type:

```
serverceipoptin /enable
```

To turn off your participation, type:

```
serverceipoptin /disable
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

servermanagercmd

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Installs and removes roles, role services, and features. Also displays the list of all roles, role services, and features available, and shows which are installed on this computer.

> **Important**
>
> This command, servermanagercmd, has been deprecated and it's not guaranteed to be supported in future releases of Windows. We recommend instead that you use the Windows PowerShell cmdlets that are available for Server Manager. For more information, see Install or Uninstall Roles, Role Services, or Features.

Syntax

```
servermanagercmd -query [[[<drive>:]<path>]<query.xml>] [-logpath
[[<drive>:]<path>]<log.txt>]
servermanagercmd -inputpath  [[[<drive>:]<path>]<answer.xml>] [-resultpath
<result.xml> [-restart] | -whatif] [-logpath [[<drive>:]<path>]<log.txt>]
servermanagercmd -install <id> [-allSubFeatures] [-resultpath [[<drive>:]
<path>]<result.xml> [-restart] | -whatif] [-logpath [[<Drive>:]<path>]
<log.txt>]
servermanagercmd -remove <id> [-resultpath <result.xml> [-restart] | -
whatif] [-logpath  [[<drive>:]<path>]<log.txt>]
servermanagercmd [-help | -?]
servermanagercmd -version
```

Parameters

Expand table

Parameter	Description
-query `[[[<drive>:]`	Displays a list of all roles, role services, and features installed and available for installation on the server. You can also use the short form of this parameter, **-q**. If you want the query results saved to an XML file, specify an XML file to replace `<query.xml>`.

Parameter	Description
`<path>]` `<query.xml>]`	
-inputpath `[[[<drive>:]` `<path>]` `<answer.xml>]`	Installs or removes the roles, role services, and features specified in an XML answer file represented by `<answer.xml>`. You can also use the short form of this parameter, **-p.**
-install `<id>`	Installs the role, role service, or feature specified by `<id>`. The identifiers are case-insensitive. Multiple roles, role services, and features must be separated by spaces. The following optional parameters are used with the **-install** parameter: • **-setting** `<SettingName>=<SettingValue>` - Specifies required settings for the installation. • **-allSubFeatures** - Specifies the installation of all subordinate services and features along with the parent role, role service, or feature named in the `<id>` value. **NOTE** Some role containers do not have a command line identifier to allow installation of all role services. This is the case when role services cannot be installed in the same instance of the Server Manager command. For example, the Federation Service role service of active directory Federation Services and the Federation Service Proxy role service cannot be installed by using the same Server Manager command instance. • **-resultpath** `<result.xml>` - Saves installation results to an XML file represented by `<result.xml>`. You can also use the short form of this parameter, **-r**. **NOTE** You can't run servermanagercmd with both the **-resultpath** parameter and the **-whatif** parameter specified. • **-restart** - Restarts the computer automatically when installation is complete (if restarting is required by the roles or features installed). • **-whatif** - Displays any operations specified for the **-install** parameter. You can also use the short form of the **-whatif** parameter, **-w**. You can't run **servermanagercmd** with both the **-resultpath** parameter and the **-whatif** parameter specified. • **-logpath** `<[[<drive>:]<path>]<log.txt>>` - Specifies a name and location for the log file, other than the default, `%windir%\temp\servermanager.log`.
-remove `<id>`	Removes the role, role service, or feature specified by `<id>`. The identifiers are case-insensitive. Multiple roles, role services, and features must be separated by spaces. The following optional parameters are used with the **-remove** parameter:

Parameter	Description
	• **-resultpath** `<[[<drive>:]<path>]result.xml>` - Saves removal results to an XML file represented by `<result.xml>`. You can also use the short form of this parameter, **-r**. **NOTE** You can't run servermanagercmd with both the **-resultpath** and the **-whatif** parameters specified. • **-restart** - Restarts the computer automatically when removal is complete (if restarting is required by remaining roles or features). • **-whatif** - Displays any operations specified for the -remove parameter. You can also use the short form of the -whatif parameter, **-w**. You can't run servermanagercmd with both the **-resultpath** and the **-whatif** parameters specified. • **-logpath** `<[[<Drive>:]<path>]<log.txt>>` - Specifies a name and location for the log file, other than the default, `%windir%\temp\servermanager.log`.
-version	Displays the Server Manager version number. You can also use the short form, **-v**.
-help	Displays help in the Command prompt window. You can also use the short form, **-?**.

Examples

To display a list of all roles, role services, and features available, and which roles, role services, and features are installed on the computer, type:

```
servermanagercmd -query
```

To install the Web Server (IIS) role, and save the installation results to an XML file represented by *installResult.xml*, type:

```
servermanagercmd -install Web-Server -resultpath installResult.xml
```

To display detailed information about the roles, role services, and features that would be installed or removed, based upon instructions that are specified in an XML answer file represented by *install.xml*, type:

```
servermanagercmd -inputpath install.xml -whatif
```

Related links

- Command-Line Syntax Key
- Server Manager overview

Feedback

Was this page helpful? Yes No

serverweroptin

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Allows you to turn on error reporting.

Syntax

```
serverweroptin [/query] [/detailed] [/summary]
```

Parameters

Expand table

Parameter	Description
/query	Verifies your current setting.
/detailed	Specifies to send detailed reports automatically.
/summary	Specifies to send summary reports automatically.
/?	Displays help at the command prompt.

Examples

To verify the current setting, type:

```
serverweroptin /query
```

To automatically send detailed reports, type:

```
serverweroptin /detailed
```

To automatically send summary reports, type:

```
serverweroptin /summary
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Services for Network File System command-line tools

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Services for Network File System (NFS) provides a file sharing solution that lets you transfer files between computers running Windows Server and UNIX operating systems using the NFS protocol.

Information and links to each of the associated NFS command-line tools:

Expand table

Command	Description
mapadmin	Manage User Name Mapping for Microsoft Services for Network File System.
mount	Mount Network File System (NFS) network shares.
nfsadmin	Manage Server for NFS and Client for NFS.
nfsshare	Control Network File System (NFS) shares.
nfsstat	Display or reset counts of calls made to Server for NFS.
rpcinfo	List programs on remote computers.
showmount	Display mounted directories.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

set (environment variable)

Article • 09/06/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays, sets, or removes cmd.exe environment variables. If used without parameters, **set** displays the current environment variable settings.

> **Note**
>
> This command requires command extensions, which are enabled by default.

The **set** command can also run from the Windows Recovery Console, using different parameters. For more information, see Windows Recovery Environment (WinRE).

Syntax

```
set [<variable>=[<string>]]
set [/p] <variable>=[<promptString>]
set /a <variable>=<expression>
```

Parameters

Expand table

Parameter	Description
`<variable>`	Specifies the environment variable to set or modify.
`<string>`	Specifies the string to associate with the specified environment variable.
/p	Sets the value of `<variable>` to a line of input entered by the user.
`<promptstring>`	Specifies a message to prompt the user for input. This parameter must be used with the **/p** parameter.
/a	Sets `<string>` to a numerical expression that is evaluated.
`<expression>`	Specifies a numerical expression.

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- If command extensions are enabled (the default) and you run **set** with a value, it displays all of the variables that begin with that value.
- The characters `<`, `>`, `|`, `&`, and `^` are special command shell characters, and they must be preceded by the escape character (`^`) or enclosed in quotation marks when used in `<string>` (for example, "StringContaining&Symbol"). If you use quotation marks to enclose a string that contains one of the special characters, the quotation marks are set as part of the environment variable value.
- Use environment variables to control the behavior of some batch files and programs and to control the way Windows and the MS-DOS subsystem appears and works. The **set** command is often used in the **Autoexec.nt** file to set environment variables.
- If you use the **set** command without any parameters, the current environment settings are displayed. These settings usually include the **COMSPEC** and **PATH** environment variables, which are used to help find programs on disk. Two other environment variables used by Windows are **PROMPT** and **DIRCMD**.
- If you specify values for `<variable>` and `<string>`, the specified `<variable>` value is added to the environment and `<string>` is associated with that variable. If the variable already exists in the environment, the new string value replaces the old string value.
- If you specify only a variable and an equal sign (without `<string>`) for the **set** command, the `<string>` value associated with the variable is cleared (as if the variable isn't there).
- If you use the **/a** parameter, the following operators are supported, in descending order of precedence:

Expand table

Operator	Operation performed
`( )`	Grouping

Operator	Operation performed
`! ~ -`	Unary
`* / %`	Arithmetic
`+ -`	Arithmetic
`<< >>`	Logical shift
`&`	Bitwise AND
`^`	Bitwise exclusive OR
`= *= /= %= += -= &= ^=`	`= <<= >>=`
`,`	Expression separator

- If you use logical (`&&` or `||`) or modulus (%) operators, enclose the expression string in quotation marks. Any non-numeric strings in the expression are considered environment variable names, and their values are converted to numbers before they're processed. If you specify an environment variable name that isn't defined in the current environment, a value of zero is allotted, which allows you to perform arithmetic with environment variable values without using the % to retrieve a value.

- If you run **set /a** from the command line outside of a command script, it displays the final value of the expression.

- Numeric values are decimal numbers unless prefixed by 0× for hexadecimal numbers or 0 for octal numbers. Therefore, 0×12 is the same as 18, which is the same as 022.

- Delayed environment variable expansion support is disabled by default, but you can enable or disable it by using **cmd /v**.

- When creating batch files, you can use **set** to create variables, and then use them in the same way that you would use the numbered variables **%0** through **%9**. You can also use the variables **%0** through **%9** as input for **set**.

- If you call a variable value from a batch file, enclose the value with percent signs (%). For example, if your batch program creates an environment variable named *BAUD*, you can use the string associated with *BAUD* as a replaceable parameter by typing **%baud%** at the command prompt.

Examples

To set the value *TEST^1* for the environment variable named `testVar`, type:

```
set testVar=TEST^^1
```

The **set** command assigns everything that follows the equal sign (=) to the value of the variable. Therefore, if you type `set testVar=TEST^1`, you'll get the following result, `testVar=TEST1`.

To set the value *TEST&1* for the environment variable `testVar`, type:

```
set testVar=TEST^&1
```

To set an environment variable named *include* so the string *c:\directory* is associated with it, type:

```
set include=c:\directory
```

You can then use the string *c:\directory* in batch files by enclosing the name *include* with percent signs (%). For example, you can use `dir %include%` in a batch file to display the contents of the directory associated with the *include* environment variable. After this command is processed, the string c:\directory replaces **%include%**.

To use the **set** command in a batch program to add a new directory to the *path* environment variable, type:

```
@echo off
rem ADDPATH.BAT adds a new directory
rem to the path environment variable.
set path=%1;%path%
set
```

To display a list of all of the environment variables that begin with the letter *p*, type:

```
set p
```

To display a list of all of the environment variables on the current device, type:

```
set
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

setlocal

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Starts localization of environment variables in a batch file. Localization continues until a matching **endlocal** command is encountered or the end of the batch file is reached.

Syntax

```
setlocal [enableextensions | disableextensions] [enabledelayedexpansion | disabledelayedexpansion]
```

Parameters

Expand table

Parameter	Description
enableextensions	Enables the command extensions until the matching **endlocal** command is encountered, regardless of the setting before the **setlocal** command was run.
disableextensions	Disables the command extensions until the matching **endlocal** command is encountered, regardless of the setting before the **setlocal** command was run.
enabledelayedexpansion	Enables the delayed environment variable expansion until the matching **endlocal** command is encountered, regardless of the setting before the **setlocal** command was run.
disabledelayedexpansion	Disables the delayed environment variable expansion until the matching **endlocal** command is encountered, regardless of the setting before the **setlocal** command was run.
/?	Displays help at the command prompt.

Remarks

- If you use **setlocal** outside of a script or batch file, it has no effect.

- Use **setlocal** to change environment variables when you run a batch file. Environment changes made after you run **setlocal** are local to the batch file. The Cmd.exe program restores previous settings when it encounters an **endlocal** command or reaches the end of the batch file.

- You can have more than one **setlocal** or **endlocal** command in a batch program (that is, nested commands).

- The **setlocal** command sets the ERRORLEVEL variable. If you pass {**enableextensions** | **disableextensions**} or {**enabledelayedexpansion** | **disabledelayedexpansion**}, the ERRORLEVEL variable is set to **0** (zero). Otherwise, it's set to **1**. You can use this information in batch scripts to determine whether the extensions are available, as shown in the following example:

  ```
  verify other 2>nul
  setlocal enableextensions
  if errorlevel 1 echo Unable to enable extensions
  ```

  Because **cmd** does not set the ERRORLEVEL variable when command extensions are disabled, the **verify** command initializes the ERRORLEVEL variable to a nonzero value when you use it with an invalid argument. Also, if you use the **setlocal** command with arguments {**enableextensions** | **disableextensions**} or {**enabledelayedexpansion** | **disabledelayedexpansion**} and it does not set the ERRORLEVEL variable to **1**, command extensions are not available.

Examples

To localize environment variables in a batch file, follow this sample script:

```
rem *******Begin Comment**************
rem This program starts the superapp batch program on the network,
rem directs the output to a file, and displays the file
rem in Notepad.
rem *******End Comment**************
@echo off
setlocal
path=g:\programs\superapp;%path%
call superapp>c:\superapp.out
endlocal
start notepad c:\superapp.out
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

setspn

Article • 03/24/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The `setspn` command line utility reads, modifies, and deletes the Service Principal Names (SPN) directory property for an Active Directory (AD) service account. You use SPNs to locate a target principal name for running a service. You can use `setspn` to view the current SPNs, reset the account's default SPNs, and add or delete supplemental SPNs. `Setspn` is available if you have the Active Directory Domain Services (AD DS) server role installed. `Setspn` must be ran through an elevated command prompt.

Syntax

```
setspn <modifiers switch> <accountname> [-R] [-S] [-D] [-L] [-C] [-U] [-Q] [-X] [-P] [-F] [-T] [-?] [/?]
```

> **Note**
>
> The <accountname> can be the computer name or domain\name of the target computer or a user account. You can run `setspn -A` to add SPNs, but you should use `setspn -S` instead because it verifies that there are no duplicate SPNs.

Parameters

Expand table

Parameters	Description
`<accountname>`	Specifies the desired AD account object for which to configure the SPN. Normally, the SPN is the NetBIOS name of the computer and optionally the domain that contains the computer account. However, any desired AD object name can be used.
`-R`	Resets the default SPN registrations for the host names for the computer.

Parameters	Description
`-S`	Adds the specified SPN for the computer, after verifying that no duplicates exist.
`-D`	Deletes the specified SPN for the computer.
`-L`	Lists the currently registered SPN for the computer.
`-C`	Specifies that `accountname` is a computer account.
`-U`	Specifies that `accountname` is a user account.
`-Q`	Queries for any existing SPNs.
`-X`	Performs a search of duplicate SPNs.
`-P`	Suppresses progress to the console and can be used when redirecting output to a file or when used in an unattended script. No output is displayed until the command is complete.
`-F`	Performs queries at the forest, rather than domain level.
`-T`	Performs a query on the specified domain (or forest when `-F` is used).
`-?` or `/?`	Displays the command-line help information. If you run `setspn` without this parameter, it also displays the command-line help information.

> ⓘ **Note**
>
> `-C` and `-U` are exclusive. If neither is specified, the tool interprets `accountname` as a computer name if such a computer exists, and a user name if it doesn't.

Remarks

Query Mode modifiers can be used with the `-S` switch in order to specify where the check for duplicates should be performed before adding the SPN.

- `-T` can be specified multiple times. To indicate the current domain or a forest, use `""` or `*`.
- `-Q` executes on each target domain or forest.
- `-X` returns duplicates that exist across all targets. SPNs aren't required to be unique across forests, but duplicate SPNs can cause authentication issues during cross-forest authentication.

- SPNs must be constructed using the base name of the account specified as the *accountname* parameter. If this condition isn't met, the directory service returns a constraint violation error.

You might not have the rights to access or modify this property on some account objects. You can determine what your access rights are by viewing the security attributes of the account object using the Microsoft Management Console (MMC) in Active Directory Users and Computers. You can also delegate the permission by assigning the Validated write to service principal name permission to the desired user or group.

The built-in SPNs that are recognized for computer accounts are:

```
alerter         eventlog        netlogon            rpc             snmp
appmgmt         eventsystem     netman              rpclocator      spooler
browser         fax             nmagent             rpcss           tapisrv
cifs            http            oakley              rsvp            time
cisvc           ias             plugplay            samss           trksvr
clipsrv         iisadmin        policyagent         scardsvr        trkwks
dcom            messenger       protectedstorage    scesrv          ups
dhcp            msiserver       rasman              schedule        w3svc
dmserver        mcsvc           remoteaccess        scm             wins
dns             netdde          replicator          seclogon        www
dnscache        netddedsm
```

These SPNs are recognized for computer accounts if the computer has a host SPN. Unless they're explicitly placed on objects, a host SPN can substitute for any of the mentioned SPNs.

SPNs aren't case sensitive when used by Microsoft Windows-based computers. Any type of computer system can use an SPN. Many of these computer systems, especially UNIX-based systems, are case-sensitive, and require the proper case to function properly. Care should be taken to use the proper case particularly when an SPN is used by a non-Windows-based computer.

Examples

To list all the registered SPNs for an account, type:

Windows Command Prompt

```
setspn -L <accountname>
```

To reset the SPNs for a computer account, type:

Windows Command Prompt

```
setspn -R <accountname>
```

To register the SPN *http/MyServer* for user account *User01*, type:

```
setspn -U -S http/MyServer User01
```

To add a new SPN to a domain account that doesn't have one set, type:

Windows Command Prompt

```
setspn -S http/myserver.mydomain.com myDomain\myServer
```

To remove an SPN from an account, type:

Windows Command Prompt

```
setspn -D http/myserver.mydomain.com myDomain\myServer
```

To query all duplicate SPNs in your domain and the *contoso* domain, type:

```
setspn -T * -T contoso -X
```

To find all the SPNs associated with *MyServer* registered in the *contoso* domain forest, type:

```
setspn -T contoso -F -Q */MyServer
```

See also

- Command-Line Syntax Key
- How to configure SPN

Feedback

Was this page helpful? Yes No

setx

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates or modifies environment variables in the user or system environment, without requiring programming or scripting. The **Setx** command also retrieves the values of registry keys and writes them to text files.

> **Note**
>
> This command provides the only command-line or programmatic way to directly and permanently set system environment values. System environment variables are manually configurable through **Control Panel** or through a registry editor. The **set** command, which is internal to the command interpreter (Cmd.exe), sets user environment variables for the current console window only.

Syntax

```
setx [/s <computer> [/u [<domain>\]<user name> [/p [<password>]]]]
<variable> <value> [/m]
setx [/s <computer> [/u [<domain>\]<user name> [/p [<password>]]]]
<variable>] /k <path> [/m]
setx [/s <computer> [/u [<domain>\]<user name> [/p [<password>]]]] /f
<filename> {[<variable>] {/a <X>,<Y> | /r <X>,<Y> <String>} [/m] | /x} [/d
<delimiters>]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer. Do not use backslashes. The default value is the name of the local computer.
/u `[<domain>\]`	Runs the script with the credentials of the specified user account. The default value is the system permissions.

Parameter	Description
`<user name>`	
/p [`<password>`]	Specifies the password of the user account that is specified in the **/u** parameter.
`<variable>`	Specifies the name of the environment variable that you want to set.
`<value>`	Specifies the value to which you want to set the environment variable.
/k `<path>`	Specifies that the variable is set based on information from a registry key. The *path* uses the following syntax: `\\<HIVE>\<KEY>\...\<Value>`. For example, you might specify the following path: `HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\TimeZoneInformation\StandardName`
/f `<filename>`	Specifies the file that you want to use.
/a `<X>,<Y>`	Specifies absolute coordinates and offset as search parameters.
/r `<X>,<Y>` `<String>`	Specifies relative coordinates and offset from **String** as search parameters.
/m	Specifies to set the variable in the system environment. The default setting is the local environment.
/x	Displays file coordinates, ignoring the **/a**, **/r**, and **/d** command-line options.
/d `<delimiters>`	Specifies delimiters such as **,** or **** to be used in addition to the four built-in delimiters — SPACE, TAB, ENTER, and LINEFEED. Valid delimiters include any ASCII character. The maximum number of delimiters is 15, including built-in delimiters.
/?	Displays help at the command prompt.

Remarks

- This command is similar to the UNIX utility SETENV.
- You can use this command to set values for user and system environment variables from one of three sources (modes): Command Line Mode, Registry Mode, or File Mode.
- This command writes variables to the master environment in the registry. Variables set with **setx** variables are available in future command windows only, not in the current command window.
- **HKEY_CURRENT_USER** and **HKEY_LOCAL_MACHINE** are the only supported hives. REG_DWORD, REG_EXPAND_SZ, REG_SZ, and REG_MULTI_SZ are the valid **RegKey**

data types.

- If you gain access to **REG_MULTI_SZ** values in the registry, only the first item is extracted and used.
- You can't use this command to remove values added to the local or system environments. You can use this command with a variable name and no value to remove a corresponding value from the local environment.
- REG_DWORD registry values are extracted and used in hexadecimal mode.
- File mode supports the parsing of carriage return and line feed (CRLF) text files only.
- Running this command on an existing variable removes any variable references and uses expanded values.

  For instance, if the variable %PATH% has a reference to %JAVADIR%, and %PATH% is manipulated using **setx**, %JAVADIR% is expanded and its value is assigned directly to the target variable %PATH%. This means that future updates to %JAVADIR% **will not** be reflected in the %PATH% variable.
- Be aware there's a limit of 1024 characters when assigning contents to a variable using **setx**.

  This means that the content is cropped if you go over 1024 characters, and that the cropped text is what's applied to the target variable. If this cropped text is applied to an existing variable, it can result in loss of data previously held by the target variable.

Examples

To set the *MACHINE* environment variable in the local environment to the value *Brand1*, type:

```
setx MACHINE Brand1
```

To set the *MACHINE* environment variable in the system environment to the value *Brand1 Computer*, type:

```
```

```
setx MACHINE Brand1 Computer /m
```

To set the *MYPATH* environment variable in the local environment to use the search path defined in the *PATH* environment variable, type:

```
setx MYPATH %PATH%
```

To set the *MYPATH* environment variable in the local environment to use the search path defined in the *PATH* environment variable after replacing ~ with %, type:

```
setx MYPATH ~PATH~
```

To set the *MACHINE* environment variable in the local environment to *Brand1* on a remote computer named *computer1*, type:

```
setx /s computer1 /u maindom\hiropln /p p@ssW23 MACHINE Brand1
```

To set the *MYPATH* environment variable in the local environment to use the search path defined in the *PATH* environment variable on a remote computer named *computer1*, type:

```
setx /s computer1 /u maindom\hiropln /p p@ssW23 MYPATH %PATH%
```

To set the *TZONE* environment variable in the local environment to the value found in the **HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\TimeZoneInformation\StandardName** registry key, type:

```
setx TZONE /k
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\TimeZoneInformation\Stan
dardName
```

To set the *TZONE* environment variable in the local environment of a remote computer named *computer1* to the value found in the **HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\TimeZoneInformation\StandardName** registry key, type:

```
setx /s computer1 /u maindom\hiropln /p p@ssW23 TZONE /k
HKEY_LOCAL_MACHINE\System\CurrentControlSet\Control\TimeZoneInformation\Stan
dardName
```

To set the *BUILD* environment variable in the system environment to the value found in the **HKEY_LOCAL_MACHINE\Software\Microsoft\WindowsNT\CurrentVersion\CurrentBuildNumber** registry key, type:

```
setx BUILD /k
HKEY_LOCAL_MACHINE\Software\Microsoft\WindowsNT\CurrentVersion\CurrentBuildN
umber /m
```

To set the BUILD environment variable in the system environment of a remote computer named Computer1 to the value found in the **HKEY_LOCAL_MACHINE\Software\Microsoft\WindowsNT\CurrentVersion\CurrentBuildNumber** registry key, type:

```
setx /s computer1 /u maindom\hiropln /p p@ssW23  BUILD /k
HKEY_LOCAL_MACHINE\Software\Microsoft\Windows
NT\CurrentVersion\CurrentBuildNumber /m
```

To display the contents of a file named *Ipconfig.out*, along with the contents' corresponding coordinates, type:

```
setx /f ipconfig.out /x
```

To set the *IPADDR* environment variable in the local environment to the value found at the coordinate *5,11* in the *Ipconfig.out* file, type:

```
setx IPADDR /f ipconfig.out /a 5,11
```

To set the *OCTET1* environment variable in the local environment to the value found at the coordinate *5,3* in the *Ipconfig.out* file with delimiters **#$*.**, type:

```
setx OCTET1 /f ipconfig.out /a 5,3 /d #$*.
```

To set the *IPGATEWAY* environment variable in the local environment to the value found at the coordinate *0,7* with respect to the coordinate of *Gateway* in the *Ipconfig.out* file, type:

```
setx IPGATEWAY /f ipconfig.out /r 0,7 Gateway
```

To display the contents of the *Ipconfig.out* file, along with the contents' corresponding coordinates, on a computer named *computer1*, type:

```
setx /s computer1 /u maindom\hiropln /p p@ssW23 /f ipconfig.out /x
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

sfc

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Scans and verifies the integrity of all protected system files and replaces incorrect versions with correct versions. If this command discovers that a protected file has been overwritten, it retrieves the correct version of the file from the **systemroot** folder, and then replaces the incorrect file.

> **ⓘ Important**
>
> You must be logged on as a member of the Administrators group to run this command.

Syntax

```
sfc [/scannow] [/verifyonly] [/scanfile=<file>] [/verifyfile=<file>]
[/offwindir=<offline windows directory> /offbootdir=<offline boot directory>
/offlogfile=<log file path>]
```

Parameters

Expand table

Parameter	Description
/scannow	Scans the integrity of all protected system files and repairs files with problems when possible.
/verifyonly	Scans the integrity of all protected system files, without performing repairs.
/scanfile `<file>`	Scans the integrity of the specified file (full path and filename) and attempts to repair any problems if they're detected.
/verifyfile `<file>`	Verifies the integrity of the specified file (full path and filename), without performing repairs.

Parameter	Description
/offwindir `<offline windows directory>`	Specifies the location of the offline windows directory, for offline repair.
/offbootdir `<offline boot directory>`	Specifies the location of the offline boot directory for offline repair.
/offlogfile=`<log file path>`	Specifies a location to store the log file other than the default.
/?	Displays help at the command prompt.

Examples

To verify the *kernel32.dll file*, type:

```
sfc /verifyfile=c:\windows\system32\kernel32.dll
```

To set up the offline repair of the *kernel32.dll* file with an offline boot directory set to **D:** and an offline windows directory set to **D:\Windows**, type:

```
sfc /scanfile=D:\windows\system32\kernel32.dll /offbootdir=D:\
/offwindir=d:\windows
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

shadow

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Enables you to remotely control an active session of another user on a Remote Desktop Session Host server.

Syntax

```
shadow {<sessionname> | <sessionID>} [/server:<servername>] [/v]
```

Parameters

Expand table

Parameter	Description
`<sessionname>`	Specifies the name of the session that you want to remotely control.
`<sessionID>`	Specifies the ID of the session that you want to remotely control. Use **query user** to display the list of sessions and their session IDs.
/server: `<servername>`	Specifies the Remote Desktop Session Host server containing the session that you want to remotely control. By default, the current Remote Desktop Session Host4 server is used.
/v	Displays information about the actions being performed.
/?	Displays help at the command prompt.

Remarks

- You can either view or actively control the session. If you choose to actively control a user's session, you will be able to input keyboard and mouse actions to the session.
- You can always remotely control your own sessions (except the current session), but you must have Full Control permission or remote Control special access

permission to remotely control another session.

- You can also initiate remote control by using Remote Desktop Services Manager.
- Before monitoring begins, the server warns the user that the session is about to be remotely controlled, unless this warning is disabled. Your session might appear to be frozen for a few seconds while it waits for a response from the user. To configure remote control for users and sessions, use the Remote Desktop Services Configuration tool or the Remote Desktop Services extensions to Local Users and Groups and active directory Users and computers.
- Your session must be capable of supporting the video resolution used at the session that you are remotely controlling or the operation fails.
- The console session can neither remotely control another session nor can it be remotely controlled by another session.
- When you want to end remote control (shadowing), press CTRL+`*` (by using `*` from the numeric keypad only).

Examples

To shadow *session 93*, type:

```
shadow 93
```

To shadow the session *ACCTG01*, type:

```
shadow ACCTG01
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

shift

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Changes the position of batch parameters in a batch file.

Syntax

```
shift [/n <N>]
```

Parameters

Expand table

Parameter	Description
/n `<N>`	Specifies to start shifting at the *N*th argument, where *N* is any value from *0* to *8*. Requires command extensions, which are enabled by default.
/?	Displays help at the command prompt.

Remarks

- The **shift** command changes the values of the batch parameters **%0** through **%9** by copying each parameter into the previous one—the value of **%1** is copied to **%0**, the value of **%2** is copied to **%1**, and so on. This is useful for writing a batch file that performs the same operation on any number of parameters.
- If command extensions are enabled, the **shift** command supports the **/n** command-line option. The **/n** option specifies to start shifting at the Nth argument, where **N** is any value from 0 to 8. For example, **SHIFT /2** would shift **%3** to **%2**, **%4** to **%3**, and so on, and leave **%0** and **%1** unaffected. Command extensions are enabled by default.
- You can use the **shift** command to create a batch file that can accept more than 10 batch parameters. If you specify more than 10 parameters on the command line,

those that appear after the tenth (%**9**) will be shifted one at a time into %**9**.

- The **shift** command has no effect on the %* batch parameter.
- There's no backward **shift** command. After you implement the **shift** command, you can't recover the batch parameter (%**0**) that existed before the shift.

Examples

To use a batch file, called *Mycopy.bat*, to copy a list of files to a specific directory, type:

```
@echo off
rem MYCOPY.BAT copies any number of files
rem to a directory.
rem The command uses the following syntax:
rem mycopy dir file1 file2 ...
set todir=%1
:getfile
shift
if "%1"=="" goto end
copy %1 %todir%
goto getfile
:end
set todir=
echo All done
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

showmount

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

You can use **showmount** to display information about mounted file systems exported by Server for NFS on a specified computer. If you don't specify a server, this command displays information about the computer on which the **showmount** command is run.

Syntax

```
showmount {-e|-a|-d} <server>
```

Parameters

Expand table

Parameter	Description
-e	Displays all the file systems exported on the server.
-a	Displays all Network File System (NFS) clients and the directories on the server each has mounted.
-d	Displays all directories on the server that are currently mounted by NFS clients.

Related links

- Command-Line Syntax Key
- Services for Network File System Command Reference

Feedback

Was this page helpful? Yes No

shutdown

Article • 05/02/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Enables you to shut down or restart local or remote computers, one at a time.

Syntax

```
shutdown [/i | /l | /s | /sg | /r | /g | /a | /p | /h | /e | /o] [/hybrid] [/soft] [/fw] [/f] [/m \\computer][/t xxx][/d [p|u:]xx:yy [/c "comment"]]
```

Parameters

Expand table

Parameter	Description
/i	Displays the **Remote Shutdown** box. The **/i** option must be the first parameter following the command. If **/i** is specified, all other options are ignored.
/l	Signs out the current user immediately with no time-out period. The **/l** parameter works independently and can't be combined with any other parameters. Attempts to combine **/l** with any other parameter is ignored.
/s	Shuts down the computer.
/sg	Shuts down the computer. On the next boot, if **Automatic Restart Sign-On** is enabled, the device automatically signs in and locks based on the last interactive user. After sign in, it restarts any registered applications.
/r	Restarts the computer after shutdown.
/g	Fully shuts down and restarts the computer. On restart, if **Automatic Restart Sign-On** is enabled, the device automatically signs in and locks based on the last interactive user. After sign in, it restarts any registered applications.
/a	Aborts a system shutdown. Can only be used during the time-out period. Combine with **/fw** to clear any pending boots to firmware.
/p	Turns off the local computer only (not a remote computer)—with no time-out period or warning. You can use **/p** only with **/d** or **/f**. If your computer doesn't support power-off functionality, it shuts down when you use **/p**, but the power to the computer remains on.

Parameter	Description
/h	Puts the local computer into hibernation, if hibernation is enabled. The **/f** switch can be used with the **/h** switch.
/hybrid	Shuts down the device and prepares it for fast startup. This option must be used with the **/s** option.
/soft	Allows running processes and applications to gracefully close instead of forcibly terminating.
/fw	Combining this option with a shutdown option causes the next restart to go to the firmware user interface.
/e	Enables you to document the reason for an unexpected shutdown of a computer in the Shutdown Event Tracker.
/o	Goes to the **Advanced boot options** menu and restarts the device. This option must be used with the **/r** option.
/f	Forces running applications to close without warning users. **Caution:** Using the **/f** option might result in loss of unsaved data.
/m `\\ <computername>`	Specifies the target computer.
/t `<xxx>`	Sets the time-out period before shutdown to *xxx* seconds. The valid range is 0-315360000 (10 years), with a default of 30. If the timeout period is greater than 0, the **/f** parameter is implied.
/d `[p | u:] <xx>:<yy>`	Lists the reason for the system restart or shutdown. The supported parameter values are: • **P** - Indicates that the restart or shutdown is planned. • **U** - Indicates that the reason is user-defined. If **p** or **u** aren't specified, the restart or shutdown is unplanned. • *xx* - Specifies the major reason number (a positive integer, less than 256). • *yy* Specifies the minor reason number (a positive integer, less than 65536).
/c `<comment>`	Enables you to create a custom reason for the system shutdown or restart, which must be enclosed in double quotation marks. You can use a maximum of 512 characters. Can also be used with the **/d** parameter.
/?	Displays help at the command prompt, including a list of the major and minor reasons that are defined on your local computer.

Remarks

- Users must be assigned the **Shut down the system** user right to shut down a local or remotely administered computer that is using the **shutdown** command.
- Users must be members of the **Administrators** group to annotate an unexpected shutdown of a local or remotely administered computer. If the target computer is joined to a domain, members of the **Domain Admins** group might be able to perform this procedure. For more information, see:
  - Default local groups
  - Default groups
- If you want to shut down more than one computer at a time, you can call **shutdown** for each computer by using a script, or you can use `shutdown /i` to display the **Remote Shutdown** box.
- If you specify major and minor reason codes, you must first define these reason codes on each computer where you plan to use the reasons. If the reason codes aren't defined on the target computer, Shutdown Event Tracker can't log the correct reason text.
- Remember to indicate that a shutdown is planned by using `/d p|u:xx:yy`, where `xx` and `yy` represent the major and minor reason codes. Not using the `/d` parameter might cause the shutdown to be logged without a specified reason. Using the `/d` parameter for an unplanned shutdown allows the shutdown to be logged correctly.

The shutdown or restart reasons uses the following legend:

- **E**: Expected - An expected shutdown or restart that is planned and communicated in advance.
- **U**: Unexpected - An unexpected shutdown or restart occurs where the system stops working due to an error, such as a system crash, power failure or other.
- **P**: Planned - A planned shutdown or restart scheduled in advance, such as system maintenance, software installation or other.
- **C**: Customer Defined - Specify a custom reason for the system shutdown or restart.

Expand table

Reason type	Major	Minor	Description
U	0	0	Other (Unplanned)
E	0	0	Other (Unplanned)
E or P	0	0	Other (Planned)

Reason type	Major	Minor	Description
U	0	5	Other Failure: System Unresponsive
E	1	1	Hardware: Maintenance (Unplanned)
E or P	1	1	Hardware: Maintenance (Planned)
E	1	2	Hardware: Installation (Unplanned)
E or P	1	2	Hardware: Installation (Planned)
E	2	2	Operating System: Recovery (Unplanned)
E or P	2	2	Operating System: Recovery (Planned)
P	2	3	Operating System: Upgrade (Planned)
E	2	4	Operating System: Reconfiguration (Unplanned)
E or P	2	4	Operating System: Reconfiguration (Planned)
P	2	16	Operating System: Service pack (Planned)
	2	17	Operating System: Hot fix (Unplanned)
P	2	17	Operating System: Hot fix (Planned)
	2	18	Operating System: Security fix (Unplanned)
P	2	18	Operating System: Security fix (Planned)
E	4	1	Application: Maintenance (Unplanned)
E or P	4	1	Application: Maintenance (Planned)
E or P	4	2	Application: Installation (Planned)
E	4	5	Application: Unresponsive
E	4	6	Application: Unstable
U	5	15	System Failure: Stop error
U	5	19	Security issue (Unplanned)
E	5	19	Security issue (Unplanned)
E or P	5	19	Security issue (Planned)
E	5	20	Loss of network connectivity (Unplanned)
U	6	11	Power Failure: Cord Unplugged

Reason type	Major	Minor	Description
U	6	12	Power Failure: Environment
P	7	0	Legacy API shutdown

Examples

To force apps to close and to restart the local computer after a one-minute delay, with the reason *Application: Maintenance (Planned)* and the comment "Reconfiguring myapp.exe", type:

```
shutdown /r /t 60 /c "Reconfiguring myapp.exe" /f /d p:4:1
```

To perform a planned shutdown of a device in 10 minutes and capture the reason in the shutdown event tracker with the comment "Scheduled maintenance", type:

```
shutdown.exe /s /t 600 /d p:0:0 /e /c "Scheduled maintenance"
```

To restart the remote computer *myremoteserver* with the same parameters as the previous example, type:

```
shutdown /r /m \\myremoteserver /t 60 /c "Reconfiguring myapp.exe" /f /d p:4:1
```

To restart the remote computer *myremoteserver* in 5 minutes due to a security update and include a custom reason in the shutdown event tracker, type:

```
shutdown /r /t 300 /e /c "Security Update" /m \\myremoteserver
```

Related links

- Command-Line Syntax Key

Simulate restore

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Tests whether writer involvement in restore sessions will be successful on the computer without issuing **PreRestore** or **PostRestore** events to writers.

> **Note**
>
> A DiskShadow metadata file must be selected for the **simulate restore** command to succeed. Use the load metadata command to load the selected writers and components for the restore.

Syntax

```
simulate restore
```

Related links

- Command-Line Syntax Key
- load metadata command

Feedback

Was this page helpful? Yes No

sort

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Reads input, sorts data, and writes the results to the screen, to a file, or to another device.

Syntax

```
sort [/r] [/+<N>] [/m <kilobytes>] [/l <locale>] [/rec <characters>]
[[<drive1>:][<path1>]<filename1>] [/t [<drive2>:][<path2>]] [/o [<drive3>:]
[<path3>]<filename3>]
```

Parameters

Expand table

Parameter	Description
/r	Reverses the sort order (that is, sorts from Z to A and from 9 to 0).
`/+<N>`	Specifies the character position number where **sort** will begin each comparison. *N* can be any valid integer.
/m `<kilobytes>`	Specifies the amount of main memory to use for the sort in kilobytes (KB).
/l <locale>	Overrides the sort order of characters that are defined by the system default locale (that is, the language and Country/Region selected during installation).
/rec `<characters>`	Specifies the maximum number of characters in a record or a line of the input file (the default value is 4,096 and the maximum is 65,535).
`[<drive1>:][<path1>]<filename1>`	Specifies the file to be sorted. If no file name is specified, the standard input is sorted. Specifying the input file is faster than redirecting the same file as standard input.
/t `[<drive2>:][<path2>]`	Specifies the path of the directory to hold the **sort** command's working storage if the data does not fit in the main memory. By default, the system temporary directory is used.

Parameter	Description
/o `[<drive3>:][<path3>]<filename3>`	Specifies the file where the sorted input is to be stored. If not specified, the data is written to the standard output. Specifying the output file is faster than redirecting standard output to the same file.
/unique	Only returns unique results.
/?	Displays help at the command prompt.

Remarks

- By default, comparisons start at the first character of each line. The **/+** command-line option starts comparisons at the character that is specified by *N*. For example, `/+3` indicates that each comparison should begin at the third character of each line. Lines with fewer than *N* characters collate before other lines.
- The memory used is always a minimum of 160 KB. If the memory size is specified, the exact specified amount is used for the sort (must be at least 160 KB), regardless of how much main memory is available.
- The default maximum memory size when no size is specified is 90% of the available main memory, if both the input and output are files, or 45% of main memory otherwise. The default setting usually gives the best performance.
- Currently, the only alternative to the default locale is the C locale, which is faster than natural language sorting (it sorts characters according to their binary encodings).
- You can use the pipe symbol (`|`) to direct input data to the **sort** command from another command or to direct sorted output to another command. You can specify input and output files by using redirection symbols (`<` or `>`). It can be faster and more efficient (especially with large files) to specify the input file directly (as defined by *filename1* in the command syntax), and then specify the output file using the **/o** parameter.
- The **sort** command doesn't distinguish between uppercase and lowercase letters and has no limit on file size.
- The sort program uses the collating-sequence table that corresponds to the **Country/Region** code and code-page settings. Characters greater than ASCII code 127 are sorted based on information in the Country.sys file or in an alternate file specified by the **country** command in your Config.nt file.

- If the sort fits within the maximum memory size (as set by default or as specified by the **/m** parameter), the sort is performed in a single pass. Otherwise, the sort is performed in two separate sort and merge passes, and the amounts of memory used for both passes are equal. When two passes are performed, the partially sorted data is stored in a temporary file on disk. If there is not enough memory to perform the sort in two passes, a run-time error is issued. If the **/m** command-line option is used to specify more memory than is truly available, performance degradation or a run-time error can occur.

Examples

- To sort and display, in reverse order, the lines in a file named *expenses.txt*, type:

```
sort /r expenses.txt
```

- To search a large file named *maillist.txt* for the text *Jones*, and to sort the results of the search using the pipe (`|`) to direct the output of a **find** command to the **sort** command, type:

```
find Jones maillist.txt | sort
```

  The command produces a sorted list of lines that contain the specified text.

- To sort keyboard input and display the results alphabetically on the screen, you can first use the **sort** command with no parameters, by typing:

```
sort
```

  Then type the text that you want sorted, and press ENTER at the end of each line. When you have finished typing text, press CTRL+Z, and then press ENTER. The **sort** command displays the text you typed, sorted alphabetically.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

start

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Starts a separate Command Prompt window to run a specified program or command.

Syntax

```
start <"title"> [/d <path>] [/i] [{/min | /max}] [{/separate | /shared}] [{/low | /normal | /high | /realtime | /abovenormal | /belownormal}] [/node <NUMA node>] [/affinity <hexaffinity>] [/wait] [/b] [/machine <x86|amd64|arm|arm64>] [<command> [<parameter>... ] | <program> [<parameter>... ]]
```

> **Note**
>
> The **machine** parameter is currently in PREVIEW for Windows 11 only. The parameter is available beginning with the Windows 11 Insider Preview Build 22557. This information relates to a prerelease product that may be substantially modified before it's released. Microsoft makes no warranties, expressed or implied, with respect to the information provided here.

Parameters

Expand table

Parameter	Description
`"title"`	Specifies the title to display in the **Command Prompt** window title bar.
/d `<path>`	Specifies the startup directory.
/i	Passes the Cmd.exe startup environment to the new **Command Prompt** window. If `/i` isn't specified, the current environment is used.
`{/min | /max}`	Specifies to minimize (`/min`) or maximize (`/max`) the new **Command Prompt** window.
`{/separate | /shared}`	Starts 16-bit programs in a separate memory space (`/separate`) or shared

Parameter	Description
	memory space (**/shared**). These options aren't supported on 64-bit platforms.
`{/low | /normal | /high | /realtime | /abovenormal | /belownormal}`	Starts an application in the specified priority class.
/node `<NUMA node>`	Leverages memory locality on NUMA (Non-Uniform Memory Architecture) nodes as a decimal integer. Two processes that communicate with each other through shared memory on the preferred NUMA node can minimize memory latencies while allocating memory from the same NUMA node when possible. They're free to run on processors outside the specified node.
/affinity `<hexaffinity>`	Applies the specified processor affinity mask (expressed as a hexadecimal number) to the new application. Two programs are able to run on specific processor cores within the same NUMA node. This number can be changed to the processor-specific supported amount without having to change the affinity mask.
/wait	Starts an application and waits for it to end.
/b	Starts an application without opening a new **Command Prompt** window. CTRL+C handling is ignored unless the application enables CTRL+C processing. Use CTRL+BREAK to interrupt the application.
/machine `<x86 | amd64 | arm | arm64>`	Specifies the machine architecture of the application process.
`[<command> [<parameter>... ] | <program> [<parameter>... ]]`	Specifies the command or program to start.
`<parameter>`	Specifies parameters to pass to either the command or the program.
/?	Displays help at the command prompt.

Remarks

- You can run non-executable files through their file association by typing the name of the file as a command. This applies to any file type that has a registered association, including URLs, which are automatically detected and opened in the default browser, and documents such as `.txt` or `.pdf` files, which are opened in their associated application.
- If you run a command that contains the string CMD as the first token without an extension or path qualifier, CMD is replaced with the value of the COMSPEC variable. This

prevents users from picking up `cmd` from the current directory.

- If you run a 32-bit graphical user interface (GUI) application, `cmd` doesn't wait for the application to quit before returning to the command prompt. This behavior doesn't occur if you run the application from a command script.

- If you're running on a 64-bit platform, the `/separate` and `/shared` parameters aren't supported.

- If you run a command that uses a first token that isn't a command or the file path to an existing file with an extension, `Cmd.exe` uses the value of the PATHEXT environment variable to determine which extensions to look for and in what order. The default value for the PATHEXT variable is `.COM;.EXE;.BAT;.CMD;.VBS;.VBE;.JS;.JSE;.WSF;.WSH;.MSC`

  Note the syntax is the same as the PATH variable, with semicolons (;) separating each extension.

- **start** searches for a specified executable file, and if found, the executable launches regardless of the current working directory. When searching for an executable file, if there's no match on any extension, **start** checks to see if the name matches a directory name. If it does, **start** opens `Explorer.exe` on that path.

Examples

To start the *Myapp* program at the command prompt and retain use of the current **Command Prompt** window, type:

Windows Command Prompt

```
start Myapp
```

To view the **start** command-line help information in a separate maximized **Command Prompt** window, type:

Windows Command Prompt

```
start /max start /?
```

To open a URL in the default browser, type:

Windows Command Prompt

```
start "Bing" "https://www.bing.com"
```

Related links

- Command-Line Syntax Key

Last updated on 04/16/2026

subst

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Associates a path with a drive letter. If used without parameters, **subst** displays the names of the virtual drives in effect.

Syntax

```
subst [<drive1>: [<drive2>:]<path>]
subst <drive1>: /d
```

Parameters

Expand table

Parameter	Description
`<drive1>:`	Specifies the virtual drive to which you want to assign a path.
`[<drive2>:]<path>`	Specifies the physical drive and path that you want to assign to a virtual drive.
/d	Deletes a substituted (virtual) drive.
/?	Displays help at the command prompt.

Remarks

- The following commands don't work and must not be used on drives specified in the **subst** command:
  - chkdsk command

    diskcomp command

    diskcopy command

format command

label command

recover command

- The `<drive1>` parameter must be within the range that is specified by the **lastdrive** command. If not, **subst** displays the following error message: `Invalid parameter - drive1:`

Examples

To create a virtual drive z for the path b:\user\betty\forms, type:

```
subst z: b:\user\betty\forms
```

Instead of typing the full path, you can reach this directory by typing the letter of the virtual drive followed by a colon as follows:

```
z:
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

sxstrace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Diagnoses side-by-side problems.

Syntax

```
sxstrace [{[trace -logfile:<filename> [-nostop]|[parse -logfile:<filename> -outfile:<parsedfile>  [-filter:<appname>]}]
```

Parameters

Expand table

Parameter	Description
trace	Enables tracing for side-by-side.
-logfile	Specifies the raw log file.
`<filename>`	Saves tracing log to `<filename>`.
-nostop	Specifies that you shouldn't receive a prompt to stop tracing.
parse	Translates the raw trace file.
-outfile	Specifies the output filename.
`<parsedfile>`	Specifies the filename of the parsed file.
-filter	Allows the output to be filtered.
`<appname>`	Specifies the name of the application.
stoptrace	Stops the trace, if it wasn't stopped before.
-?	Displays help at the command prompt.

Examples

To enable tracing and to save the trace file to *sxstrace.etl*, type:

```
sxstrace trace -logfile:sxstrace.etl
```

To translate the raw trace file into a human readable format and to save the result to *sxstrace.txt*, type:

```
sxstrace parse -logfile:sxstrace.etl -outfile:sxstrace.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

sysmon

Applies to: ✅ Windows 11

Summarize this article for me

System Monitor (Sysmon) is a Windows system service and device driver that you install on a system. It stays resident across system reboots to monitor and log system activity to the Windows event log. It provides detailed information about process creations, network connections, and changes to file creation time.

> **Tip**
>
> Starting February 2026, Sysmon is available as a built-in optional feature for Windows 11. To learn more, see Sysmon overview.
>
> Looking for the Sysinternals standalone documentation? See Sysinternals Sysmon.

Syntax

Windows Command Prompt

```
sysmon [-i [<configfile>]] [-c [<configfile>]] [-m] [-s] [-u [force]] [-accepteula]
```

Parameters

Expand table

Parameter	Description
`-i [<configfile>]`	Installs the service and driver. Optionally includes a configuration file.
`-c [<configfile>]`	Updates the configuration of an installed Sysmon driver, or dumps the current configuration if no other argument is provided. Optionally includes a configuration file.
`-m`	Installs the event manifest. The service install implicitly does this action.
`-s [<schemaversion>]`	Prints the configuration schema definition of the specified version. Specify `all` to dump all schema versions. The default is latest.
`-u [force]`	Uninstalls the service and driver. Using `-u force` causes uninstall to proceed even when some components aren't installed.

Parameter	Description
`-accepteula`	Automatically accepts the End User License Agreement (EULA) on installation.
`-? config`	Displays more information on configuration files.

Remarks

- Event logs are stored in `Applications and Services Logs/Microsoft/Windows/Sysmon/Operational` log.
- The install or uninstall options don't require a reboot.
- To reset the configuration to default settings, use `sysmon -c --`.

Examples

To install Sysmon with default settings, run the following command:

Windows Command Prompt

```
sysmon -accepteula -i
```

To install Sysmon with a configuration file, run the following command:

Windows Command Prompt

```
sysmon -accepteula -i c:\windows\config.xml
```

To dump the current configuration, run the following command:

Windows Command Prompt

```
sysmon -c
```

To reset the configuration to default settings, run the following command:

Windows Command Prompt

```
sysmon -c --
```

To reconfigure Sysmon with a configuration file, run the following command:

Windows Command Prompt

```
sysmon -c c:\windows\config.xml
```

To display the configuration schema, run the following command:

Windows Command Prompt

```
sysmon -s
```

To display the configuration schema for a specific version, run the following command:

Windows Command Prompt

```
sysmon -s 4.50
```

To display all configuration schema versions, run the following command:

Windows Command Prompt

```
sysmon -s all
```

To uninstall Sysmon, run the following command:

Windows Command Prompt

```
sysmon -u
```

Related links

- Command-Line Syntax Key

Last updated on 02/24/2026

sysocmgr

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The sysocmger command has been deprecated and isn't guaranteed to be supported in future releases of Windows.

Feedback

Was this page helpful? Yes No

systeminfo

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays detailed configuration information about a computer and its operating system, including operating system configuration, security information, product ID, and hardware properties (such as RAM, disk space, and network cards).

Syntax

```
systeminfo [/s <computer> [/u <domain>\<username> [/p <password>]]] [/fo {TABLE | LIST | CSV}] [/nh]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default is the local computer.
/u `<domain>\<username>`	Runs the command with the account permissions of the specified user account. If **/u** is not specified, this command uses the permissions of the user who is currently logged on to the computer that is issuing the command.
/p `<password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/fo `<format>`	Specifies the output format with one of the following values: • **TABLE** - Displays output in a table. • **LIST** - Displays output in a list. • **CSV** - Displays output in comma-separated values (.csv) format.
/nh	Suppresses column headers in the output. Valid when the **/fo** parameter is set to TABLE or CSV.
/?	Displays help at the command prompt.

Examples

To view configuration information for a computer named *Srvmain*, type:

```
systeminfo /s srvmain
```

To remotely view configuration information for a computer named *Srvmain2* that is located on the *Maindom* domain, type:

```
systeminfo /s srvmain2 /u maindom\hiropln
```

To remotely view configuration information (in list format) for a computer named *Srvmain2* that is located on the *Maindom* domain, type:

```
systeminfo /s srvmain2 /u maindom\hiropln /p p@ssW23 /fo list
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

takeown

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Enables an administrator to recover access to a file that previously was denied, by making the administrator the owner of the file. This command is typically used on batch files.

Syntax

```
takeown [/s <computer> [/u [<domain>\]<username> [/p [<password>]]]] /f <filename> [/a] [/r [/d {Y|N}]]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default value is the local computer. This parameter applies to all of the files and folders specified in the command.
/u `[<domain>\]<username>`	Runs the script with the permissions of the specified user account. The default value is system permissions.
/p `[<[password>]`	Specifies the password of the user account that is specified in the **/u** parameter.
/f `<filename>`	Specifies the file name or directory name pattern. You can use the wildcard character `*` when specifying the pattern. You can also use the syntax `<sharename>\<filename>`.
/a	Gives ownership to the Administrators group instead of the current user. If you don't specify this option, file ownership is given to the user who is currently logged on to the computer.
/r	Performs a recursive operation on all files in the specified directory and subdirectories.
/d `{Y | N}`	Suppresses the confirmation prompt that is displayed when the current user does not have the **List Folder** and **Read** permissions on a specified directory, and instead uses the specified default value. Valid values for the **/d** option are: • **Y** - Take ownership of the directory. The directory permissions are replaced with permission granting the user full control only. • **N** - Skip the directory.

Parameter	Description
	NOTE You must use this option in conjunction with the **/r** option.
/?	Displays help at the command prompt.

Remarks

- Mixed patterns using (**?** and ***) aren't supported by **takeown** command.
- After deleting the lock with **takeown**, you might have to use Windows Explorer to give yourself full permissions to the files and directories before you can delete them.

Examples

To take ownership of a file named *Lostfile*, type:

```
takeown /f lostfile
```

Related links

- Command-Line Syntax Key

Last updated on 11/04/2025

tapicfg

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates, removes, or displays a TAPI application directory partition, or sets a default TAPI application directory partition. TAPI 3.1 clients can use the information in this application directory partition with the directory service locator service to find and communicate with TAPI directories. You can also use **tapicfg** to create or remove service connection points, which enable TAPI clients to efficiently locate TAPI application directory partitions in a domain.

This command-line tool can be run on any computer that is a member of the domain.

Syntax

```
tapicfg install
tapicfg remove
tapicfg publishscp
tapicfg removescp
tapicfg show
tapicfg makedefault
```

Parameters

Expand table

Parameters	Description
tapicfg install	Creates a TAPI application directory partition.
tapicfg remove	Removes a TAPI application directory partition.
tapicfg publishscp	Creates a service connection point to publish a TAPI application directory partition.
tapicfg removescp	Removes a service connection point for a TAPI application directory partition.
tapicfg show	Displays the names and locations of the TAPI application directory partitions in the domain.

Parameters	Description
tapicfg makedefault	Sets the default TAPI application directory partition for the domain.

Remarks

- You must be a member of the **Enterprise Admins** group in Active Directory to run either **tapicfg install** (to create a TAPI application directory partition) or **tapicfg remove** (to remove a TAPI application directory partition).
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.
- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.
- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Related links

- Command-Line Syntax Key
- tapicfg install
- tapicfg remove
- tapicfg publishscp
- tapicfg removescp
- tapicfg show

- tapicfg makedefault

Feedback

Was this page helpful? Yes No

tapicfg install

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a TAPI application directory partition.

> **Important**
>
> You must be a member of the Enterprise Admins group in active directory to run this command.

Syntax

```
tapicfg install /directory:<partitionname> [/server:<DCname>]
[/forcedefault]
```

Parameters

Expand table

Parameter	Description
install `/directory:<partitionname>`	Required. Specifies the DNS name of the TAPI application directory partition to be created. This name must be a fully-qualified domain name.
/server: `<DCname>`	Specifies the DNS name of the domain controller on which the TAPI application directory partition is created. If the domain controller name isn't specified, the name of the local computer is used.
/forcedefault	Specifies that this directory is the default TAPI application directory partition for the domain. There can be multiple TAPI application directory partitions in a domain. if this directory is the first TAPI application directory partition created on the domain, it's automatically set as the default, regardless of whether you use the **/forcedefault** option.
/?	Displays help at the command prompt.

Remarks

- This command-line tool can be run on any computer that is a member of the domain.
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.
- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.
- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Examples

To create a TAPI application directory partition named *tapifiction.testdom.microsoft.com* on a server named *testdc.testdom.microsoft.com*, and then set it as the default TAPI application directory partition for the new domain, type:

```
tapicfg install /directory:tapifiction.testdom.microsoft.com
/server:testdc.testdom.microsoft.com /forcedefault
```

Related links

- Command-Line Syntax Key
- tapicfg remove
- tapicfg publishscp

- tapicfg removescp
- tapicfg show
- tapicfg makedefault

Feedback

Was this page helpful? Yes No

tapicfg remove

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes a TAPI application directory partition.

> **ⓘ Important**
>
> You must be a member of the Enterprise Admins group in active directory to run this command.

Syntax

```
tapicfg remove /directory:<partitionname>
```

Parameters

Expand table

Parameter	Description
remove `/directory:<partitionname>`	Required. Specifies the DNS name of the TAPI application directory partition to be removed. Note that this name must be a fully-qualified domain name.
/?	Displays help at the command prompt.

Remarks

- This command-line tool can be run on any computer that is a member of the domain.
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.

- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.
- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Related links

- Command-Line Syntax Key
- tapicfg install
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Feedback

Was this page helpful? Yes No

tapicfg publishscp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a service connection point to publish a TAPI application directory partition.

Syntax

```
tapicfg publishscp /directory:<partitionname> [/domain:<domainname>]
[/forcedefault]
```

Parameters

Expand table

Parameter	Description
publishscp `/directory:<partitionname>`	Required. Specifies the DNS name of the TAPI application directory partition that the service connection point will publish.
/domain: `<domainname>`	Specifies the DNS name of the domain in which the service connection point is created. If the domain name is not specified, the name of the local domain is used.
/forcedefault	Specifies that this directory is the default TAPI application directory partition for the domain. There can be multiple TAPI application directory partitions in a domain.
/?	Displays help at the command prompt.

Remarks

- This command-line tool can be run on any computer that is a member of the domain.
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.

- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.
- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Related links

- Command-Line Syntax Key
- tapicfg install
- tapicfg remove
- tapicfg removescp
- tapicfg show
- tapicfg makedefault

Feedback

Was this page helpful? Yes No

tapicfg removescp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a service connection point for a TAPI application directory partition.

Syntax

```
tapicfg removescp /directory:<partitionname> [/domain:<domainname>]
```

Parameters

Expand table

Parameter	Description
removescp `/directory:<partitionname>`	Required. Specifies the DNS name of the TAPI application directory partition for which a service connection point is removed.
/domain: `<domainname>`	Specifies the DNS name of the domain from which the service connection point is removed. If the domain name isn't specified, the name of the local domain is used.
/?	Displays help at the command prompt.

Remarks

- This command-line tool can be run on any computer that is a member of the domain.
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.
- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP

or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.

- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Related links

- Command-Line Syntax Key
- tapicfg install
- tapicfg remove
- tapicfg publishscp
- tapicfg show
- tapicfg makedefault

Feedback

Was this page helpful? Yes No

tapicfg show

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the names and locations of the TAPI application directory partitions in the domain.

Syntax

```
tapicfg show [/defaultonly] [/domain:<domainname>]
```

Parameters

Expand table

Parameter	Description
/default only	Displays the names and locations of only the default TAPI application directory partition in the domain.
/domain: `<domainname>`	Specifies the DNS name of the domain for which the TAPI application directory partitions are displayed. If the domain name isn't specified, the name of the local domain is used.
/?	Displays help at the command prompt.

Remarks

- This command-line tool can be run on any computer that is a member of the domain.
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.
- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP

or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.

- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Example

To display the name of the default TAPI application directory partition for the new domain, type:

```
tapicfg show /defaultonly
```

Related links

- Command-Line Syntax Key
- tapicfg install
- tapicfg remove
- tapicfg publishscp
- tapicfg removescp
- tapicfg makedefault

Feedback

Was this page helpful? Yes No

tapicfg makedefault

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets the default TAPI application directory partition for the domain.

Syntax

```
tapicfg makedefault /directory:<partitionname> [/domain:<domainname>]
```

Parameters

Expand table

Parameter	Description
makedefault `/directory:<partitionname>`	Required. Specifies the DNS name of the TAPI application directory partition set as the default partition for the domain. This name must be a fully-qualified domain name.
/domain: `<domainname>`	Specifies the DNS name of the domain for which the TAPI application directory partition is set as the default. If the domain name is not specified, the name of the local domain is used.
/?	Displays help at the command prompt.

Remarks

- This command-line tool can be run on any computer that is a member of the domain.
- User-supplied text (such as the names of TAPI application directory partitions, servers, and domains) with International or Unicode characters are only displayed correctly if appropriate fonts and language support are installed.
- You can still use Internet Locator Service (ILS) servers in your organization, if ILS is needed to support certain applications, because TAPI clients running Windows XP

or a Windows Server 2003 operating system can query either ILS servers or TAPI application directory partitions.

- You can use **tapicfg** to create or remove service connection points. If the TAPI application directory partition is renamed for any reason (for example, if you rename the domain in which it resides), you must remove the existing service connection point and create a new one that contains the new DNS name of the TAPI application directory partition to be published. Otherwise, TAPI clients are unable to locate and access the TAPI application directory partition. You can also remove a service connection point for maintenance or security purposes (for example, if you do not want to expose TAPI data on a specific TAPI application directory partition).

Related links

- Command-Line Syntax Key
- tapicfg install
- tapicfg remove
- tapicfg publishscp
- tapicfg removescp
- tapicfg show

Feedback

Was this page helpful? Yes No

taskkill

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Ends one or more tasks or processes. Processes can be ended by process ID or image name. You can use the tasklist command command to determine the process ID (PID) for the process to be ended.

> **Note**
>
> This command replaces the **kill** tool.

Syntax

```
taskkill [/s <computer> [/u [<domain>\]<username> [/p [<password>]]]] {[/fi
<filter>] [...] [/pid <processID> | /im <imagename>]} [/f] [/t]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default is the local computer.
/u `<domain>\<username>`	Runs the command with the account permissions of the user who is specified by `<username>` or by `<domain>\<username>`. The **/u** parameter can be specified only if **/s** is also specified. The default is the permissions of the user who is currently logged on to the computer that is issuing the command.
/p `<password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/fi `<filter>`	Applies a filter to select a set of tasks. You can use more than one filter or use the wildcard character (`*`) to specify all tasks or image names. The valid filters are listed in the **Filter names, operators, and values** section of this article.

Parameter	Description
/pid `<processID>`	Specifies the process ID of the process to be terminated.
/im `<imagename>`	Specifies the image name of the process to be terminated. Use the wildcard character (`*`) to specify all image names.
/f	Specifies that processes be forcefully ended. This parameter is ignored for remote processes; all remote processes are forcefully ended.
/t	Ends the specified process and any child processes started by it.

Filter names, operators, and values

Expand table

Filter Name	Valid Operators	Valid Value(s)
STATUS	eq, ne	`RUNNING | NOT RESPONDING | UNKNOWN`
IMAGENAME	eq, ne	Image name
PID	eq, ne, gt, lt, ge, le	PID value
SESSION	eq, ne, gt, lt, ge, le	Session number
CPUtime	eq, ne, gt, lt, ge, le	CPU time in the format *HH:MM:SS*, where *MM* and *SS* are between 0 and 59 and *HH* is any unsigned number
MEMUSAGE	eq, ne, gt, lt, ge, le	Memory usage in KB
USERNAME	eq, ne	Any valid user name (`<user>` or `<domain\user>`)
SERVICES	eq, ne	Service name
WINDOWTITLE	eq, ne	Window title
MODULES	eq, ne	DLL name

Remarks

- The **WINDOWTITLE** and **STATUS** filters aren't supported when a remote system is specified.

- The wildcard character (`*`) is accepted for the `*/im` option, only when a filter is applied.
- Ending a remote process is always carried out forcefully, regardless whether the **/f** option is specified.
- Providing a computer name to the hostname filter causes a shutdown, stopping all processes.

Examples

To end the processes with process IDs *1230*, *1241*, and *1253*, type:

```
taskkill /pid 1230 /pid 1241 /pid 1253
```

To forcefully end the process *Notepad.exe* if it was started by the system, type:

```
taskkill /f /fi "USERNAME eq NT AUTHORITY\SYSTEM" /im notepad.exe
```

To end all processes on the remote computer *Srvmain* with an image name beginning with *note*, while using the credentials for the user account *Hiropln*, type:

```
taskkill /s srvmain /u maindom\hiropln /p p@ssW23 /fi "IMAGENAME eq note*"
/im *
```

To end the process with the process ID *2134* and any child processes that it started, but only if those processes were started by the Administrator account, type:

```
taskkill /pid 2134 /t /fi "username eq administrator"
```

To end all processes that have a process ID *greater than or equal to 1000*, regardless of their image names, type:

```
```

```
taskkill /f /fi "PID ge 1000" /im *
```

Related links

- Command-Line Syntax Key
- tasklist command

Feedback

Was this page helpful? Yes No

tasklist

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays a list of currently running processes on the local computer or on a remote computer. **Tasklist** replaces the **tlist** tool.

> **Note**
>
> This command replaces the **tlist** tool.

Syntax

```
tasklist [/s <computer> [/u [<domain>\]<username> [/p <password>]]] [{/m
<module> | /svc | /v}] [/fo {table | list | csv}] [/nh] [/fi <filter> [/fi
<filter> [ ... ]]]
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (do not use backslashes). The default is the local computer.
/u `<domain>\<username>`	Runs the command with the account permissions of the user who is specified by `<username>` or by `<domain>\<username>`. The **/u** parameter can be specified only if **/s** is also specified. The default is the permissions of the user who is currently logged on to the computer that is issuing the command.
/p `<password>`	Specifies the password of the user account that is specified in the **/u** parameter.
/m `<module>`	Lists all tasks with DLL modules loaded that match the given pattern name. If the module name is not specified, this option displays all modules loaded by each task.
svc	Lists all the service information for each process without truncation. Valid when

Parameter	Description
	the **/fo** parameter is set to **table**.
/v	Displays verbose task information in the output. For complete verbose output without truncation, use **/v** and **/svc** together.
/fo `{table | list | csv}`	Specifies the format to use for the output. Valid values are **table**, **list**, and **csv**. The default format for output is **table**.
/nh	Suppresses column headers in the output. Valid when the **/fo** parameter is set to **table** or **csv**.
/fi `<filter>`	Specifies the types of processes to include in or exclude from the query. You can use more than one filter or use the wildcard character (`\`) to specify all tasks or image names. The valid filters are listed in the **Filter names, operators, and values** section of this article.
/?	Displays help at the command prompt.

Filter names, operators, and values

Expand table

Filter Name	Valid Operators	Valid Value(s)
STATUS	eq, ne	`RUNNING | NOT RESPONDING | UNKNOWN`. This filter isn't supported if you specify a remote system.
IMAGENAME	eq, ne	Image name
PID	eq, ne, gt, lt, ge, le	PID value
SESSION	eq, ne, gt, lt, ge, le	Session number
SESSIONNAME	eq, ne	Session name
CPUtime	eq, ne, gt, lt, ge, le	CPU time in the format *HH:MM:SS*, where *MM* and *SS* are between 0 and 59 and *HH* is any unsigned number
MEMUSAGE	eq, ne, gt, lt, ge, le	Memory usage in KB
USERNAME	eq, ne	Any valid user name (`<user>` or `<domain\user>`)
SERVICES	eq, ne	Service name

Filter Name	Valid Operators	Valid Value(s)
WINDOWTITLE	eq, ne	Window title. This filter isn't supported if you specify a remote system.
MODULES	eq, ne	DLL name

Examples

To list all tasks with a *process ID greater than 1000*, and *display them in csv format*, type:

```
tasklist /v /fi "PID gt 1000" /fo csv
```

To list the system processes that are currently running, type:

```
tasklist /fi "USERNAME ne NT AUTHORITY\SYSTEM" /fi "STATUS eq running"
```

To list detailed information for all processes that are currently running, type:

```
tasklist /v /fi "STATUS eq running"
```

To list all the service information for processes on the remote computer *srvmain*, which has a DLL name *beginning with ntdll*, type:

```
tasklist /s srvmain /svc /fi "MODULES eq ntdll*"
```

To list the processes on the remote computer *srvmain*, using the credentials of your currently logged-on user account, type:

```
tasklist /s srvmain
```

To list the processes on the remote computer *srvmain*, using the credentials of the *user account Hiropln*, type:

```
tasklist /s srvmain /u maindom\hiropln /p p@ssW23
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tcmsetup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sets up or disables the TAPI client. For TAPI to function correctly, you must run this command to specify the remote servers that will be used by TAPI clients.

> **Important**
>
> To use this command, you must be a member of the **Administrators** group on the local computer, or you must have been delegated the appropriate authority. If the computer is joined to a domain, members of the **Domain Admins** group might be able to perform this procedure. As a security best practice, consider using **Run as** to perform this procedure.

Syntax

```
tcmsetup [/q] [/x] /c <server1> [<server2> …]
tcmsetup  [/q] /c /d
```

Parameters

Expand table

Parameter	Description
/q	Prevents the display of message boxes.
/x	Specifies that connection-oriented callbacks will be used for heavy traffic networks where packet loss is high. When this parameter is omitted, connectionless callbacks will be used.
/c	Required. Specifies client setup.
`<server1>`	Required. Specifies the name of the remote server that has the TAPI service providers that the client will use. The client will use the service providers' lines and

Parameter	Description
	phones. The client must be in the same domain as the server or in a domain that has a two-way trust relationship with the domain that contains the server.
`<server2>…`	Specifies any additional server or servers that will be available to this client. If you specify a list of servers is, use a space to separate the server names.
/d	Clears the list of remote servers. Disables the TAPI client by preventing it from using the TAPI service providers that are on the remote servers.
/?	Displays help at the command prompt.

Remarks

- Before a client user can use a phone or line on a TAPI server, the telephony server administrator must assign the user to the phone or line.
- The list of telephony servers that is created by this command replaces any existing list of telephony servers available to the client. You can't use this command to add to the existing list.

Related links

- Command-Line Syntax Key
- Command shell overview
- Specify telephony servers on a client computer
- Assign a telephony user to a line or phone

Feedback

Was this page helpful? Yes No

telnet

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Communicates with a computer running the telnet server service. Running this command without any parameters, lets you enter the telnet context, as indicated by the telnet prompt (**Microsoft telnet>**). From the telnet prompt, you can use telnet commands to manage the computer running the telnet client.

> **ⓘ Important**
>
> You must install the telnet client software before you can run this command. For more information, see **Installing telnet**.

Syntax

```
telnet [/a] [/e <escapechar>] [/f <filename>] [/l <username>] [/t {vt100 | vt52 | ansi | vtnt}] [<host> [<port>]] [/?]
```

Parameters

Expand table

Parameter	Description
/a	Attempts automatic logon. Same as **/l** option, except that it uses the currently logged on user's name.
/e `<escapechar>`	Specifies the escape character used to enter the telnet client prompt.
/f `<filename>`	Specifies the file name used for client side logging.
/l `<username>`	Specifies the user name to log on with on the remote computer.
/t `{vt100 | vt52 | ansi | vtnt}`	Specifies the terminal type. Supported terminal types are **vt100**, **vt52**, **ansi**, and **vtnt**.

Parameter	Description
`<host> [<port>]`	Specifies the hostname or IP address of the remote computer to connect to, and optionally the TCP port to use (default is TCP port 23).
/?	Displays help at the command prompt.

Examples

To use telnet to connect to the computer running the telnet Server Service at *telnet.microsoft.com*, type:

```
telnet telnet.microsoft.com
```

To use telnet to connect to the computer running the telnet Server Service at *telnet.microsoft.com* on TCP port 44 and to log the session activity in a local file called *telnetlog.txt*, type:

```
telnet /f telnetlog.txt telnet.microsoft.com 44
```

Related links

- Command-Line Syntax Key
- Installing telnet
- telnet Technical Reference

Feedback

Was this page helpful? Yes No

telnet: close

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Closes the current connection.

Syntax

```
c[lose]
```

Examples

To close the current telnet connection, type:

```
c
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: display

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays operating parameters.

Syntax

```
d[isplay]
```

Examples

To display operating parameters, type:

```
d
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: open

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Connects to a telnet server.

Syntax

```
o[pen] <hostname> [<port>]
```

Parameters

Expand table

Parameter	Description
`<hostname>`	Specifies the computer name or IP address.
`[<port>]`	Specifies the TCP port that the telnet server is listening on. The default is TCP port 23.

Examples

To connect to a telnet server at *telnet.microsoft.com*, type:

```
o telnet.microsoft.com
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: quit

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Exits telnet.

Syntax

```
q[uit]
```

Examples

To exit telnet, type:

```
q
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: send

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sends telnet commands to the telnet server.

Syntax

```
sen {ao | ayt | brk | esc | ip | synch | <string>} [?]
```

Parameters

Expand table

Parameter	Description
ao	Sends the telnet command **Abort Output**.
ayt	Sends the telnet command **Are You There?**
brk	Sends the telnet command **brk**.
esc	Sends the current telnet escape character.
ip	Sends the telnet command **Interrupt Process**.
synch	Sends the telnet command synch.
`<string>`	Sends whatever string you type to the telnet server.
?	Displays help associated with this command.

Example

To send the **Are you there?** command to the telnet server, type:

```
sen ayt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: set

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets options. You can use the telnet unset command to turn off an option that was previously set.

Syntax

```
set [bsasdel] [crlf] [delasbs] [escape <char>] [localecho] [logfile
<filename>] [logging] [mode {console | stream}] [ntlm] [term {ansi | vt100 |
vt52 | vtnt}] [?]
```

Parameters

Expand table

Parameter	Description
bsasdel	Sends **backspace** as a **delete**.
crlf	Sends CR & LF (0x0D, 0x 0A) when the **Enter** key is pressed. Known as **New line mode**.
delasbs	Sends **delete** as a **backspace**.
escape `<character>`	Sets the escape character used to enter the telnet client prompt. The escape character can be a single character, or it can be a combination of the **CTRL** key plus a character. To set a control-key combination, hold down the **CTRL** key while you type the character that you want to assign.
localecho	Turns on local echo.
logfile `<filename>`	Logs the current telnet session to the local file. Logging begins automatically when you set this option.
logging	Turns on logging. If no log file is set, an error message appears.
mode `{console | stream}`	Sets the operation mode.

Parameter	Description
ntlm	Turns on NTLM authentication.
term `{ansi | vt100 | vt52 | vtnt}`	Sets the terminal type.
?	Displays help for this command.

Remarks

- On non-English versions of telnet, the **codeset** `<option>` is available. **Codeset** `<option>` sets the current code set to an option, which can be any one of the following: **shift JIS**, **Japanese EUC**, **JIS Kanji**, **JIS Kanji (78)**, **DEC Kanji**, **NEC Kanji**. You should set the same code set on the remote computer.

Example

To set the log file and to begin logging to the local file *tnlog.txt*, type:

```
set logfile tnlog.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: status

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays status information.

Syntax

```
st[atus]
```

Example

To display status information, type:

```
st
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

telnet: unset

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Turns off previously set options.

Syntax

```
u {bsasdel | crlf | delasbs | escape | localecho | logging | ntlm} [?]
```

Parameters

Expand table

Parameter	Description
bsasdel	Sends **backspace** as a **backspace**.
crlf	Sends the **Enter** key as a CR. Also known as line feed mode.
delasbs	Sends **delete** as **delete**.
escape	Removes the escape character setting.
localecho	Turns off localecho.
logging	Turns off logging.
ntlm	Turns off NTLM authentication.
?	Displays help for this command.

Example

Turn off logging.

```
```

```
u logging
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tftp

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Transfers files to and from a remote computer, typically a computer running UNIX, that is running the Trivial File Transfer Protocol (tftp) service or daemon. tftp is typically used by embedded devices or systems that retrieve firmware, configuration information, or a system image during the boot process from a tftp server.

> [IMPORTANT] The tftp protocol doesn't support any authentication or encryption mechanism, and as such can introduce a security risk when present. Installing the tftp client is not recommended for systems connected to the Internet. A tftp server service is no longer provided by Microsoft for security reasons.

Syntax

```
tftp [-i] [<host>] [{get | put}] <source> [<destination>]
```

Parameters

Expand table

Parameter	Description
-i	Specifies binary image transfer mode (also called octet mode). In binary image mode, the file is transferred in one-byte units. Use this mode when transferring binary files. If you don't use the **-i** option, the file is transferred in ASCII mode. This is the default transfer mode. This mode converts the end-of-line (EOL) characters to an appropriate format for the specified computer. Use this mode when transferring text files. If a file transfer is successful, the data transfer rate is displayed.
`<host>`	Specifies the local or remote computer.
get	Transfers the file *destination* on the remote computer to the file *source* on the local computer.
put	Transfers the file *source* on the local computer to the file *destination* on the remote computer. Because the tftp protocol doesn't support user authentication,

Parameter	Description
	the user must be logged onto the remote computer, and the files must be writable on the remote computer.
`<source>`	Specifies the file to transfer.
`<destination>`	Specifies where to transfer the file.

Examples

To copy the file *boot.img* from the remote computer *Host1*, type:

```
tftp  -i Host1 get boot.img
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

time

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays or sets the system time. If used without parameters, **time** displays the current system time and prompts you to enter a new time.

> **Note**
>
> You must be an administrator to change the current time.

Syntax

```
time [/t | [<HH>[:<MM>[:<SS>]] [am|pm]]]
```

Parameters

Expand table

Parameter	Description
`<HH>[:<MM>[:<SS>[.<NN>]]] [am | pm]`	Sets the system time to the new time specified, where *HH* is in hours (required), *MM* is in minutes, and *SS* is in seconds. *NN* can be used to specify hundredths of a second. You must separate values for *HH*, *MM*, and *SS* with colons (:). *SS* and *NN* must be separated with a period (.). If **am** or **pm** isn't specified, **time** uses the 24-hour format by default.
/t	Displays the current time without prompting you for a new time.
/?	Displays help at the command prompt.

Remarks

- Valid *HH* values are 0 through 24.
- Valid *MM* and *SS* values are 0 through 59.

Examples

If command extensions are enabled, to display the current system time, type:

```
time /t
```

To change the current system time to 5:30 PM, type either of the following:

```
time 17:30:00
time 5:30 pm
```

To display the current system time, followed by a prompt to enter a new time, type:

```
The current time is: 17:33:31.35
Enter the new time:
```

To keep the current time and return to the command prompt, press ENTER. To change the current time, type the new time and then press ENTER.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

timeout

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Pauses the command processor for the specified number of seconds. This command is typically used in batch files.

Syntax

```
timeout /t <timeoutinseconds> [/nobreak]
```

Parameters

Expand table

Parameter	Description
/t `<timeoutinseconds>`	Specifies the decimal number of seconds (between -1 and 99999) to wait before the command processor continues processing. The value **-1** causes the computer to wait indefinitely for a keystroke.
/nobreak	Specifies to ignore user key strokes.
/?	Displays help at the command prompt.

Remarks

- A user keystroke resumes the command processor execution immediately, even if the timeout period has not expired.
- When used in conjunction with the resource kit's **Sleep** tool, **timeout** is similar to the **pause** command.

Examples

To pause the command processor for ten seconds, type:

```
timeout /t 10
```

To pause the command processor for 100 seconds and ignore any keystroke, type:

```
timeout /t 100 /nobreak
```

To pause the command processor indefinitely until a key is pressed, type:

```
timeout /t -1
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

title

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a title for the Command Prompt window.

Syntax

```
title [<string>]
```

Parameters

Expand table

Parameter	Description
`<string>`	Specifies the text to appear as the title of the Command Prompt window.
/?	Displays help at the command prompt.

Remarks

- To create window title for batch programs, include the **title** command at the beginning of a batch program.
- After a window title is set, you can reset it only by using the **title** command.

Examples

To change the Command Prompt window title to *Updating Files* while the batch file executes the **copy** command, and then to return the title back to *Command Prompt*, type the following script:

```
@echo off
title Updating Files
copy \\server\share\*.xls c:\users\common\*.xls
echo Files Updated.
title Command Prompt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tlntadmn

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Administers a local or remote computer that is running the telnet Server Service. If used without parameters, **tlntadmn** displays the current server settings.

This command requires you to log on to the local computer with administrative credentials. To administer a remote computer, you must also provide administrative credentials for the remote computer. You can do so by logging on to the local computer with an account that has administrative credentials for both the local computer and the remote computer. If you can't use this method, you can use the **-u** and **-p** parameters to provide administrative credentials for the remote computer.

Syntax

```
tlntadmn [<computername>] [-u <username>] [-p <password>] [{start | stop |
pause | continue}] [-s {<sessionID> | all}] [-k {<sessionID> | all}] [-m
{<sessionID> | all}  <message>] [config [dom = <domain>] [ctrlakeymap = {yes
| no}] [timeout = <hh>:<mm>:<ss>] [timeoutactive = {yes | no}] [maxfail =
<attempts>] [maxconn = <connections>] [port = <number>] [sec {+ | -}NTLM {+
| -}passwd] [mode = {console | stream}]] [-?]
```

Parameters

Expand table

Parameter	Description
`<computername>`	Specifies the name of the server to connect to. The default is the local computer.
-u `<username>` -p `<password>`	Specifies administrative credentials for a remote server that you want to administer. This parameter is required if you want to administer a remote server to which you are not logged on with administrative credentials.
start	starts the telnet Server Service.
stop	Stops the telnet Server Service

Parameter	Description
pause	Pauses the telnet Server Service. No new connections will be accepted.
continue	Resumes the telnet Server Service.
-s `{<sessionID> | all}`	Displays active telnet sessions.
-k `{<sessionID> | all}`	Ends telnet sessions. Type the Session ID to end a specific session, or type all to end all the sessions.
-m `{<sessionID> | all} <message>`	Sends a message to one or more sessions. Type the session ID to send a message to a specific session, or type all to send a message to all sessions. type the message that you want to send between quotation marks.
config dom = `<domain>`	Configures the default domain for the server.
config ctrlakeymap = `{yes | no}`	Specifies if you want the telnet server to interpret CTRL+A as ALT. Type **yes** to map the shortcut key, or type **no** to prevent the mapping.
config timeout = `<hh>:<mm>:<ss>`	Sets the time-out period in hours, minutes, and seconds.
config timeoutactive = `{yes | no}`	Enables the idle session timeout.
config maxfail = `<attempts>`	Sets the maximum number of failed logon attempts before disconnecting.
config maxconn = `<connections>`	Sets the maximum number of connections.
config port = `<number>`	Sets the telnet port. You must specify the port with an integer smaller than 1024.
config sec `{+ | -}NTLM {+ | -}passwd`	Specifies whether you want to use NTLM, a password, or both to authenticate logon attempts. To use a particular type of authentication, type a plus sign (+) before that type of authentication. To prevent using a particular type of authentication, type a minus sign (-) before that type of authentication.
config mode = `{console | stream}`	Specifies the mode of operation.
-?	Displays help at the command prompt.

Examples

To configure the idle session timeout to 30 minutes, type:

```
tlntadmn config timeout=0:30:0
```

To display active telnet sessions, type:

```
tlntadmn -s
```

Related links

- Command-Line Syntax Key
- telnet Operations Guide

Feedback

Was this page helpful? Yes No

tpmtool

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

This utility can be used to get information about the Trusted Platform Module (TPM).

> **Important**
>
> Some information may relate to the pre-released product, which may be substantially modified before it's commercially released. Microsoft makes no warranties, express or implied, with respect to the information provided here.

Syntax

```
tpmtool /parameter [<arguments>]
```

Parameters

Expand table

Parameter	Description
getdeviceinformation	Displays the basic information of the TPM. See the Win32_Tpm::IsReadyInformation method parameters article for details about the information flag values.
gatherlogs [output directory path]	Collects TPM logs and places them in the specified directory. If that directory doesn't exist, it's created. By default, the log files are placed in the current directory. The possible files generated are: • TpmEvents.evtx • TpmInformation.txt • SRTMBoot.dat • SRTMResume.dat • DRTMBoot.dat • DRTMResume.dat

Parameter	Description
drivertracing `[start | stop]`	Starts or stops collecting TPM driver traces. The trace log, *TPMTRACE.etl*, is created and placed in the current directory.
/?	Displays help at the command prompt.

Examples

To display the basic information of the TPM, type:

```
tpmtool getdeviceinformation
```

To collect TPM logs and place them in the current directory, type:

```
tpmtool gatherlogs
```

To collect TPM logs and place them in `C:\Users\Public`, type:

```
tpmtool gatherlogs C:\Users\Public
```

To collect TPM driver traces, type:

```
tpmtool drivertracing start
# Run scenario
tpmtool drivertracing stop
```

Related links

- Command-Line Syntax Key
- COM Error Codes (TPM, PLA, FVE)

Feedback

Was this page helpful? Yes No

tpmvscmgr

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The tpmvscmgr command-line tool allows users with Administrative credentials to create and delete TPM virtual smart cards on a computer.

Syntax

```
tpmvscmgr create [/name] [/adminkey DEFAULT | PROMPT | RANDOM] [/PIN DEFAULT
| PROMPT] [/PUK DEFAULT | PROMPT] [/generate] [/machine] [/?]
```

```
tpmvscmgr destroy [/instance <instanceID>] [/?]
```

Create parameters

The **Create** command sets up new virtual smart cards on the user's system. It also returns the instance ID of the newly-created card for later reference, if deletion is required. The instance ID is in the format **ROOT\SMARTCARDREADER\000n** where **n** starts from 0 and is increased by 1 each time you create a new virtual smart card.

Expand table

Parameter	Description
/name	Required. Indicates the name of the new virtual smart card.
/adminkey	Indicates the desired administrator key that can be used to reset the PIN of the card if the user forgets the PIN. This can include: • **DEFAULT** - Specifies the default value of *010203040506070801020304050607080102030405060708*. • **PROMPT** - Prompts the user to enter a value for the administrator key. • **RANDOM** - Results in a random setting for the administrator key for a card that is not returned to the user. This creates a card that might not be manageable by using smart card management tools. When using the

Parameter	Description
	RANDOM option, the administrator key must be entered as 48 hexadecimal characters.
/PIN	Indicates desired user PIN value. • **DEFAULT** - Specifies the default PIN of 12345678. • **PROMPT** - Prompts the user to enter a PIN at the command line. The PIN must be a minimum of eight characters, and it can contain numerals, characters, and special characters.
/PUK	Indicates the desired PIN Unlock Key (PUK) value. The PUK value must be a minimum of eight characters, and it can contain numerals, characters, and special characters. If the parameter is omitted, the card is created without a PUK. The options include: • **DEFAULT** - Specifies the default PUK of *12345678*. • **PROMPT** - Prompts to the user to enter a PUK at the command line.
/generate	Generates the files in storage that are necessary for the virtual smart card to function. If you don't use the **/generate** parameter, it's like you created the card without the underlying file system. A card without a file system can be managed only by a smart card management system such as Microsoft Configuration Manager.
/machine	Allows you to specify the name of a remote computer on which the virtual smart card can be created. This can be used in a domain environment only, and it relies on DCOM. For the command to succeed in creating a virtual smart card on a different computer, the user running this command must be a member in the local administrators group on the remote computer.
/?	Displays Help for this command.

Destroy parameters

The **Destroy** command securely deletes a virtual smart card from the user's computer.

> ⚠ **Warning**
>
> If a virtual smart card is deleted, it cannot be recovered.

Expand table

Parameter	Description
/instance	Specifies the instance ID of the virtual smart card to be removed. The instanceID was generated as output by tpmvscmgr.exe when the card was created. The **/instance**

Parameter	Description
	parameter is a required field for the **Destroy** command.
/?	Displays help at the command prompt.

Remarks

- For alphanumeric inputs, the full 127 character ASCII set is allowed.

Examples

To create a virtual smart card that can be later managed by a smart card management tool launched from another computer, type:

```
tpmvscmgr.exe create /name VirtualSmartCardForCorpAccess /AdminKey DEFAULT
/PIN PROMPT
```

Alternatively, instead of using a default administrator key, you can create an administrator key at the command line. The following command shows how to create an administrator key.

```
tpmvscmgr.exe create /name VirtualSmartCardForCorpAccess /AdminKey PROMPT
/PIN PROMPT
```

To create an unmanaged virtual smart card that can be used to enroll certificates, type:

```
tpmvscmgr.exe create /name VirtualSmartCardForCorpAccess /AdminKey RANDOM
/PIN PROMPT /generate
```

A virtual smart card is created with a randomized administrator key. The key is automatically discarded after the card is created. This means that if the user forgets the PIN or wants to the change the PIN, the user needs to delete the card and create it again.

To delete the card, type:

```
tpmvscmgr.exe destroy /instance <instance ID>
```

Where `<instanceID>` is the value printed on the screen when the user created the card. Specifically, for the first card created, the instance ID is *ROOT\SMARTCARDREADER\0000*.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tracerpt

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The **tracerpt** command parses Event Trace Logs, log files generated by Performance Monitor, and real-time Event Trace providers. It also generates dump files, report files, and report schemas.

Syntax

```
tracerpt <[-l] <value [value [...]]>|-rt <session_name [session_name
[...]]>> [options]
```

Parameters

Expand table

Parameters	Description
-config `<filename>`	Specifies which settings file to load, which includes your command options.
-y	Specifies to answer **yes** to all questions, without prompting.
-f `<XML | HTML>`	Specifies the report file format.
-of `<CSV | EVTX | XML>`	Specifies the dump file format. The default is **XML*.
-df `<filename>`	Specifies to create a Microsoft-specific counting/reporting schema file.
-int `<filename>`	Specifies to dump the interpreted event structure to the specified file.
-rts	Specifies to add the report raw timestamp in the event trace header. Can only be used with **-o**. It's not supported with **-report** or **-summary**.
-tmf `<filename>`	Specifies which Trace Message Format definition file to use.
-tp `<value>`	Specifies the TMF file search path. Multiple paths may be used, separated by a semicolon (;).

Parameters	Description
-i `<value>`	Specifies the provider image path. The matching PDB will be located in the Symbol Server. Multiple paths can be used, separated by a semicolon (;).
-pdb `<value>`	Specifies the symbol server path. Multiple paths can be used, separated by a semicolon (;).
-gmt	Specifies to convert WPP payload timestamps to Greenwich Mean Time.
-rl `<value>`	Specifies the System Report Level from 1 to 5. Default is *1*.
-summary [filename]	Specifies to create a summary report text file. The filename, if not specified, is *summary.txt*.
-o [filename]	Specifies to create a text output file. The filename, if not specified, is *dumpfile.xml*.
-report [filename]	Specifies to create a text output report file. The filename, if not specified, is *workload.xml*.
-lr	Specifies to be less restrictive. This uses best efforts for events that don't match the events schema.
-export [filename]	Specifies to create an Event Schema export file. The filename, if not specified, is *schema.man*.
[-l] `<value [value [...]]>`	Specifies the Event Trace log file to process.
-rt `<session_name [session_name [...]]>`	Specifies the Real-time Event Trace Session data sources.
-?	Displays help at the command prompt.

Examples

To create a report based on the two event logs *logfile1.etl* and *logfile2.etl*, and to create the dump file *logdump.xml* in *XML* format, type:

```
tracerpt logfile1.etl logfile2.etl -o logdump.xml -of XML
```

To create a report based on the event log *logfile.etl*, to create the dump file *logdmp.xml* in XML format, to use best efforts to identify events not in the schema, and to produce a

summary report file *logdump.txt* and a report file, *logrpt.xml*, type:

```
tracerpt logfile.etl -o logdmp.xml -of XML -lr -summary logdmp.txt -report
logrpt.xml
```

To use the two event logs *logfile1.etl* and *logfile2.etl* to produce a dump file, and to report file with the default filenames, type:

```
tracerpt logfile1.etl logfile2.etl -o -report
```

To use the event log *logfile.etl* and the performance log *counterfile.blg* to produce the report file *logrpt.xml* and the Microsoft-specific XML schema file *schema.xml*, type:

```
tracerpt logfile.etl counterfile.blg -report logrpt.xml -df schema.xml
```

To read the real-time Event Trace Session NT Kernel Logger and to produce the dump file *logfile.csv* in *CSV* format, type:

```
tracerpt -rt NT Kernel Logger -o logfile.csv -of CSV
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tracert

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

This diagnostic tool determines the path taken to a destination by sending Internet Control Message Protocol (ICMP) echo Request or ICMPv6 messages to the destination with incrementally increasing time to live (TTL) field values. Each router along the path is required to decrement the TTL in an IP packet by at least 1 before forwarding it. Effectively, the TTL is a maximum link counter. When the TTL on a packet reaches 0, the router is expected to return an ICMP time Exceeded message to the source computer.

This command determines the path by sending the first echo Request message with a TTL of 1 and incrementing the TTL by 1 on each subsequent transmission until the target responds or the maximum number of hops is reached. The maximum number of hops is 30 by default and can be specified using the **/h** parameter.

The path is determined by examining the ICMP time Exceeded messages returned by intermediate routers and the echo Reply message returned by the destination. However, some routers don't return time Exceeded messages for packets with expired TTL values and are invisible to the **tracert** command. In this case, a row of asterisks (`*`) is displayed for that hop. The path displayed is the list of near/side router interfaces of the routers in the path between a source host and a destination. The near/side interface is the interface of the router that is closest to the sending host in the path.

> **ⓘ Important**
>
> This command is available only if the Internet Protocol (TCP/IP) protocol is installed as a component in the properties of a network adapter in Network Connections.
>
> To trace a path and provide network latency and packet loss for each router and link in the path, use the pathping command command.

Syntax

```
tracert [/d] [/h <maximumhops>] [/j <hostlist>] [/w <timeout>] [/R] [/S
<srcaddr>] [/4][/6] <targetname>
```

Parameters

Expand table

Parameter	Description
/d	Stops attempts to resolve the IP addresses of intermediate routers to their names. This can speed up the return of results.
/h `<maximumhops>`	Specifies the maximum number of hops in the path to search for the target (destination). The default is 30 hops.
/j `<hostlist>`	Specifies that echo Request messages use the Loose Source Route option in the IP header with the set of intermediate destinations specified in `<hostlist>`. With loose source routing, successive intermediate destinations can be separated by one or multiple routers. The maximum number of addresses or names in the list is *9*. The `<hostlist>` is a series of IP addresses (in dotted decimal notation) separated by spaces. Use this parameter only when tracing IPv4 addresses.
/w `<timeout>`	Specifies the amount of time in milliseconds to wait for the ICMP time Exceeded or echo Reply message corresponding to a given echo Request message to be received. If not received within the time-out, an asterisk (`*`) is displayed. The default time-out is 4000 (4 seconds).
/R	Specifies that the IPv6 Routing extension header be used to send an echo Request message to the local host, using the destination as an intermediate destination and testing the reverse route.
/S `<srcaddr>`	Specifies the source address to use in the echo Request messages. Use this parameter only when tracing IPv6 addresses.
/4	Specifies that tracert.exe can use only IPv4 for this trace.
/6	Specifies that tracert.exe can use only IPv6 for this trace.
`<targetname>`	Specifies the destination, identified either by IP address or host name.
/?	Displays help at the command prompt.

Examples

To trace the path to the host named `www.microsoft.com`, type:

```
tracert www.microsoft.com
```

Here's an example output:

Output

```
Tracing route to e13678.dscb.akamaiedge.net [23.216.93.114]
over a maximum of 30 hops:

 1     1 ms     1 ms    <1 ms  <devicename>.mshome.net [172.26.96.1]
 2    11 ms    13 ms     6 ms  192.168.191.20
 3    20 ms    11 ms    18 ms  192.168.1.1
 4    44 ms    41 ms    35 ms  10.228.0.1
 5    32 ms    31 ms    46 ms  10.41.0.49
 6    36 ms    39 ms    30 ms  10.41.0.221
 7    35 ms    36 ms    39 ms  10.41.0.225
 8    54 ms    45 ms    50 ms  204.111.0.147
 9    50 ms    52 ms    47 ms  ae-39.a02.atlnga05.us.bb.gin.ntt.net
[128.241.219.117]
10    53 ms    51 ms    61 ms  ae-5.r24.atlnga05.us.bb.gin.ntt.net
[129.250.4.192]
11    64 ms    45 ms    44 ms  ae-0.a03.atlnga05.us.bb.gin.ntt.net
[129.250.2.20]
12    49 ms    67 ms    46 ms  ae-0.akamai-onnet.atlnga05.us.bb.gin.ntt.net
[128.241.1.122]
13    67 ms   287 ms     *
ae20.r03.border101.atl02.fab.netarch.akamai.com [23.203.144.21]
14     *        *        *     Request timed out.
15     *        *        *     Request timed out.
16     *        *        *     Request timed out.
17   204 ms    58 ms    51 ms  a23-216-93-
114.deploy.static.akamaitechnologies.com [23.216.93.114]

Trace complete.
```

The beginning column displays the hop number starting from 1 and incrementing with each hop along the route from your device to the destination. Each hop represents an intermediate device, such as a router, that the packet passes through while traveling to the final destination.

The three center columns display the round-trip time in milliseconds (ms) for a packet to travel from your device to the router, at that specific hop, and back to your device. It's known as the "ping time" or "ping latency" and measures the delay in milliseconds for data to travel to the router and return. Network latency can be affected by factors such as network congestion, the quality of network links, and the distance between hops.

The end column displays either the IP address or the hostname of the router or intermediate device at that specific hop in the network path. In most cases, you'll see the IP address, but if reverse DNS lookup is successful, it displays the hostnames, which can help identify routers by name.

To trace the path to the host named `www.microsoft.com` and prevent the resolution of each IP address to its name, type:

```
tracert /d www.microsoft.com
```

To trace the path to the host named `www.microsoft.com` and use the loose source route *10.12.0.1/10.29.3.1/10.1.44.1*, type:

```
tracert /j 10.12.0.1 10.29.3.1 10.1.44.1 www.microsoft.com
```

Related links

- Command-Line Syntax Key
- pathping command

Feedback

Was this page helpful? Yes No

tree

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the directory structure of a path or of the disk in a drive graphically. The structure displayed by this command depends upon the parameters that you specify at the command prompt. If you don't specify a drive or path, this command displays the tree structure beginning with the current directory of the current drive.

Syntax

```
tree [<drive>:][<path>] [/f] [/a]
```

Parameters

Expand table

Parameter	Description
`<drive>:`	Specifies the drive that contains the disk for which you want to display the directory structure.
`<path>`	Specifies the directory for which you want to display the directory structure.
/f	Displays the names of the files in each directory.
/a	Specifies to use text characters instead of graphic characters to show the lines that link subdirectories.
/?	Displays help at the command prompt.

Examples

To display the names of all the subdirectories on the disk in your current drive, type:

```
tree \
```

To display, one screen at a time, the files in all the directories on drive C, type:

```
tree c:\ /f | more
```

To print a list of all the directories on drive C to a file, type:

```
tree c:\ /f > <driveletter>:\<filepath>\filename.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tscon

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Connects to another session on a Remote Desktop Session Host server.

> **Important**
>
> You must have **Full Control access** permission or **Connect special access** permission to connect to another session.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
tscon {<sessionID> | <sessionname>} [/dest:<sessionname>] [/password:<pw> | /password:*] [/v]
```

Parameters

Expand table

Parameter	Description
`<sessionID>`	Specifies the ID of the session to which you want to connect. If you use the optional `/dest:<sessionname>` parameter, you can also specify the name of the current session.
`<sessionname>`	Specifies the name of the session to which you want to connect.
/dest: `<sessionname>`	Specifies the name of the current session. This session will disconnect when you connect to the new session. You can also use this parameter to connect the session of another user to a different session.

Parameter	Description
/password:`<pw>`	Specifies the password of the user who owns the session to which you want to connect. This password is required when the connecting user does not own the session.
/password:`*`	Prompts for the password of the user who owns the session to which you want to connect.
/v	Displays information about the actions being performed.
/?	Displays help at the command prompt.

Remarks

- This command fails if you don't specify a password in the **/password** parameter, and the target session belongs to a user other than the current one.
- You can't connect to the console session.

Examples

To connect to *Session 12* on the current Remote Desktop Services Session Host server, and to disconnect the current session, type:

```
tscon 12
```

To connect to *Session 23* on the current Remote Desktop Services Session Host server using the password *mypass*, and to disconnect the current session, type:

```
tscon 23 /password:mypass
```

To connect the session named *TERM03* to the session named *TERM05*, and then to disconnect session *TERM05*, type:

```
tscon TERM03 /v /dest:TERM05
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

---

Feedback

Was this page helpful? Yes No

tsdiscon

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Disconnects a session from a Remote Desktop Session Host server. If you don't specify a session ID or session name, this command disconnects the current session.

> **Important**
>
> You must have **Full Control access** permission or **Disconnect special access** permission to disconnect another user from a session.

> **Note**
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
tsdiscon [<sessionID> | <sessionname>] [/server:<servername>] [/v]
```

Parameters

Expand table

Parameter	Description
`<sessionID>`	Specifies the ID of the session to disconnect.
`<sessionname>`	Specifies the name of the session to disconnect.
/server: `<servername>`	Specifies the terminal server that contains the session that you want to disconnect. Otherwise, the current Remote Desktop Session Host server is used. This parameter is required only if you run the **tsdiscon** command from a remote server.

Parameter	Description
/v	Displays information about the actions being performed.
/?	Displays help at the command prompt.

Remarks

- Any applications running when you disconnected the session are automatically running when you reconnect to that session with no loss of data. You can use the reset session command to end the running applications of the disconnected session, but this may result in loss of data at the session.
- The console session can't be disconnected.

Examples

To disconnect the current session, type:

```
tsdiscon
```

To disconnect *Session 10*, type:

```
tsdiscon 10
```

To disconnect the session named *TERM04*, type:

```
tsdiscon TERM04
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference
- reset session command

Feedback

Was this page helpful? Yes No

tsecimp

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Imports assignment information from an Extensible Markup Language (XML) file into the TAPI server security file (Tsec.ini). You can also use this command to display the list of TAPI providers and the lines devices associated with each of them, validate the structure of the XML file without importing the contents, and check domain membership.

Syntax

```
tsecimp /f <filename> [{/v | /u}]
tsecimp /d
```

Parameters

Expand table

Parameter	Description
/f `<filename>`	Required. Specifies the name of the XML file that contains the assignment information that you want to import.
/v	Validates the structure of the XML file without importing the information into the Tsec.ini file.
/u	Checks whether each user is a member of the domain specified in the XML file. The computer on which you use this parameter must be connected to the network. This parameter might significantly slow performance if you are processing a large amount of user assignment information.
/d	Displays a list of installed telephony providers. For each telephony provider, the associated line devices are listed, as well as the addresses and users associated with each line device.
/?	Displays help at the command prompt.

Remarks

The XML file from which you want to import assignment information must follow the structure described below:

```xml
<UserList>
  <User>
    <LineList>
      <Line>
```

- `<Userlist element>` - The top element of the XML file.
- `<User element>` - Contains information about a user who is a member of a domain. Each user might be assigned one or more line devices. Additionally, each **User** element might have an attribute named **NoMerge**. When this attribute is specified, all current line device assignments for the user are removed before new ones are made. You can use this attribute to easily remove unwanted user assignments. By default, this attribute is not set. The **User** element must contain a single **DomainUserName** element, which specifies the domain and user name of the user. The **User** element might also contain one **FriendlyName** element, which specifies a friendly name for the user. The **User** element might contain one **LineList** element. If a **LineList** element is not present, all line devices for this user are removed.
- `<LineList element>` - Contains information about each line or device that might be assigned to the user. Each **LineList** element can contain more than one **Line** element.
- `<Line element>` - Specifies a line device. You must identify each line device by adding either an **Address** element or a **PermanentID** element under the **Line** element. For each **Line** element, you can set the **Remove** attribute. If you set this attribute, the user is no longer assigned that line device. If this attribute is not set, the user gains access to that line device. No error is given if the line device is not available to the user.

Sample output for /d parameter

This sample output appears after running the **/d** parameter to display the current TAPI configuration. For each telephony provider, the associated line devices are listed, as well as the addresses and users associated with each line device.

```
NDIS Proxy TAPI Service Provider
  Line: WAN Miniport (L2TP)
```

```
    Permanent ID: 12345678910

NDIS Proxy TAPI Service Provider
  Line: LPT1DOMAIN1\User1
    Permanent ID: 12345678910

Microsoft H.323 Telephony Service Provider
  Line: H323 Line
    Permanent ID: 123456
    Addresses:
      BLDG1-TAPI32
```

Examples

To remove all line devices assigned to *User1*, type:

XML

```
<UserList>
  <User NoMerge=1>
    <DomainUser>domain1\user1</DomainUser>
  </User>
</UserList>
```

To remove all line devices assigned to *User1*, before assigning one line with address *99999*, type:

XML

```
<UserList>
  <User NoMerge=1>
  <DomainUser>domain1\user1</DomainUser>
  <FriendlyName>User1</FriendlyName>
  <LineList>
    <Line>
      <Address>99999</Address>
    </Line>
  </LineList>
  </User>
</UserList>
```

In this example, *User1* has no other line devices assigned, regardless of whether any line devices were assigned previously.

To add one line device for *User1*, without deleting any previously assigned line devices, type:

```
```

XML

```
<UserList>
  <User>
  <DomainUser>domain1\user1</DomainUser>
  <FriendlyName>User1</FriendlyName>
  <LineList>
    <Line>
      <Address>99999</Address>
    </Line>
  </LineList>
  </User>
</UserList>
```

To add line address *99999* and to remove line address *88888* from *User1's* access, type:

XML

```
<UserList>
  <User>
  <DomainUser>domain1\user1</DomainUser>
  <FriendlyName>User1</FriendlyName>
  <LineList>
    <Line>
      <Address>99999</Address>
    </Line>
    <Line Remove=1>
      <Address>88888</Address>
    </Line>
  </LineList>
  </User>
</UserList>
```

To add permanent device *1000* and to remove line *88888* from *User1's* access, type:

XML

```
<UserList>
  <User>
  <DomainUser>domain1\user1</DomainUser>
  <FriendlyName>User1</FriendlyName>
  <LineList>
    <Line>
    <PermanentID>1000</PermanentID>
    </Line>
    <Line Remove=1>
    <Address>88888</Address>
    </Line>
  </LineList>
```

```
  </User>
</UserList>
```

Related links

- Command-Line Syntax Key
- Command shell overview

Feedback

Was this page helpful? Yes No

tskill

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Ends a process running in a session on a Remote Desktop Session Host server.

> **Note**
>
> You can use this command to end only those processes that belong to you, unless you are an administrator. Administrators have full access to all **tskill** functions and can end processes that are running in other user sessions.
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
tskill {<processID> | <processname>} [/server:<servername>] [/id:<sessionID> | /a] [/v]
```

Parameters

Expand table

Parameter	Description
`<processID>`	Specifies the ID of the process that you want to end.
`<processname>`	Specifies the name of the process that you want to end. This parameter can include wildcard characters.
/server: `<servername>`	Specifies the terminal server that contains the process that you want to end. If **/server** isn't specified, the current Remote Desktop Session Host server is used.
/id: `<sessionID>`	Ends the process that is running in the specified session.
/a	Ends the process that is running in all sessions.

Parameter	Description
/v	Displays information about the actions being performed.
/?	Displays help at the command prompt.

Remarks

- When all processes that are running in a session end, the session also ends.
- If you use the `<processname>` and the `/server:<servername>` parameters, you must also specify either the `/id:<sessionID>` or the **/a** parameter.

Examples

To end process 6543, type:

```
tskill 6543
```

To end the process explorer running on session 5, type:

```
tskill explorer /id:5
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

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Feedback

Was this page helpful? Yes No

tsprof

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Copies the Remote Desktop Services user configuration information from one user to another. The Remote Desktop Services user configuration information appears in the Remote Desktop Services extensions to Local Users and Groups and active directory Users and computers.

> **Note**
>
> You can also use the tsprof command to set the profile path for a user.
>
> To find out what's new in the latest version, see What's New in Remote Desktop Services in Windows Server.

Syntax

```
tsprof /update {/domain:<Domainname> | /local} /profile:<path> <username>
tsprof /copy {/domain:<Domainname> | /local} [/profile:<path>] <src_user>
<dest_user>
tsprof /q {/domain:<Domainname> | /local} <username>
```

Parameters

Expand table

Parameter	Description
/update	Updates profile path information for `<username>` in domain `<domainname>` to `<profilepath>`.
/domain: `<Domainname>`	Specifies the name of the domain in which the operation is applied.
/local	Applies the operation only to local user accounts.

Parameter	Description
/profile:`<path>`	Specifies the profile path as displayed in the Remote Desktop Services extensions in Local Users and Groups and active directory Users and computers.
`<username>`	Specifies the name of the user for whom you want to update or query the server profile path.
/copy	Copies user configuration information from `<src_user>` to `<dest_user>` and updates the profile path information for `<dest_user>` to `<profilepath>`. Both `<src_user>` and `<dest_user>` must either be local or must be in domain `<domainname>`.
`<src_user>`	Specifies the name of the user from whom you want to copy the user configuration information. Also known as the source user.
`<dest_user>`	Specifies the name of the user to whom you want to copy the user configuration information. Also known as the destination user.
/q	Displays the current profile path of the user for whom you want to query the server profile path.
/?	Displays help at the command prompt.

Examples

To copy user configuration information from *LocalUser1* to *LocalUser2*, type:

```
tsprof /copy /local LocalUser1 LocalUser2
```

To set the Remote Desktop Services profile path for *LocalUser1* to a directory called *c:\profiles*, type:

```
tsprof /update /local /profile:c:\profiles LocalUser1
```

Related links

- Command-Line Syntax Key
- Remote Desktop Services (Terminal Services) Command Reference

Feedback

Was this page helpful? Yes No

type

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

In the Windows Command shell, **type** is a built in command which displays the contents of a text file. Use the **type** command to view a text file without modifying it.

In PowerShell, **type** is a built-in alias to the Get-Content cmdlet, which also displays the contents of a file, but using a different syntax.

Syntax

```
type [<drive>:][<path>]<filename>
```

Parameters

Expand table

Parameter	Description
`[<drive>:][<path>]<filename>`	Specifies the location and name of the file or files that you want to view. If your `<filename>` contains spaces, you must enclose it in quotation marks (for example, "Filename Containing Spaces.txt"). You can also add multiple filenames by adding spaces between them.
/?	Displays help at the command prompt.

Remarks

- If you display a binary file or a file that is created by a program, you may see strange characters on the screen, including formfeed characters and escape-sequence symbols. These characters represent control codes that are used in the binary file. In general, avoid using the **type** command to display binary files.

Examples

To display the contents of a file named *holiday.mar*, type:

```
type holiday.mar
```

To display the contents of a lengthy file named *holiday.mar* one screen at a time, type:

```
type holiday.mar | more
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

typeperf

Article • 06/16/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

The **typeperf** command writes performance data to the command window or to a log file. To stop **typeperf**, press CTRL+C.

Syntax

```
typeperf <counter [counter ...]> [options]
typeperf -cf <filename> [options]
typeperf -q [object] [options]
typeperf -qx [object] [options]
```

Parameters

Expand table

Parameter	Description
`<counter [counter [...]]>`	Specifies performance counters to monitor. The `<counter>` parameter is the full name of a performance counter in \Computer\Object(Instance)\Counter format, such as `\\Server1\Processor(0)\% User Time`.

Options

Expand table

Option	Description
-f `<CSV | TSV | BIN | SQL>`	Specifies the output file format. The default is CSV.
-cf `<filename>`	Specifies a file containing a list of performance counters to monitor, with one counter per line.
-si `<[[hh:]mm:]ss>`	Specifies the sample interval. The default is one second.

Option	Description
-o `<filename>`	Specifies the path for the output file, or the SQL database. The default is STDOUT (written to the command window).
-q `[object]`	Display a list of installed counters (no instances). To list counters for one object, include the object name.
-qx `[object]`	Display a list of installed counters with instances. To list counters for one object, include the object name.
-sc `<samples>`	Specifies the number of samples to collect. The default is to collect data until CTRL+C is pressed.
-config `<filename>`	Specifies a settings file containing command options.
-s `<computer_name>`	Specifies a remote computer to monitor if no computer is specified in the counter path.
-y	Answer *yes* to all questions without prompting.
/?	Displays help at the command prompt.

Examples

To write the values for the local computer's performance counter `\Processor(_Total)\% Processor Time` to the command window at a default sample interval of 1 second until CTRL+C is pressed, type:

```
typeperf \Processor(_Total)\% Processor Time
```

To write the values for the list of counters in the file *counters.txt* to the tab-delimited file *domain2.tsv* at a sample interval of 5 seconds until 50 samples have been collected, type:

```
typeperf -cf counters.txt -si 5 -sc 50 -f TSV -o domain2.tsv
```

To query installed counters with instances for the counter object *PhysicalDisk* and writes the resulting list to the file *counters.txt*, type:

```
```

```
typeperf -qx PhysicalDisk -o counters.txt
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

tzutil

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Displays the Windows Time Zone utility.

Syntax

```
tzutil [/?] [/g] [/s <timezoneID>[_dstoff]] [/l]
```

Parameters

Expand table

Parameter	Description
/g	Displays the current time zone ID.
/s `<timezoneID>[_dstoff]`	Sets the current time zone using the specified time zone ID. The **_dstoff** suffix disables Daylight Saving time adjustments for the time zone (where applicable). Your value must be surrounded by quotes.
/l	Lists all valid time zone IDs and display names. The output appears as: • `<display name>` • `<time zone ID>`
/?	Displays help at the command prompt.

Remarks

An exit code of **0** indicates the command completed successfully.

Examples

To display the current time zone ID, type:

```
tzutil /g
```

To set the current time zone to Pacific Standard time, type:

```
tzutil /s "Pacific Standard time"
```

To set the current time zone to Pacific Standard time and disable Daylight Saving time adjustments, type:

```
tzutil /s "Pacific Standard time_dstoff"
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

unlodctr

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Removes **Performance counter names** and **Explain text** for a service or device driver from the system registry.

> ⚠ **Warning**
>
> Incorrectly editing the registry may severely damage your system. Before making changes to the registry, you should back up any valued data on the computer.

Syntax

```
unlodctr <drivername>
```

Parameters

Expand table

Parameter	Description
`<drivername>`	Removes the **Performance counter name** settings and **Explain text** for driver or service `<drivername>` from the Windows Server registry. If your `<drivername>` includes spaces, you must use quotation marks around the text, for example "Driver name".
/?	Displays help at the command prompt.

Remarks

An exit code of **0** indicates the command line syntax was correct. It doesn't mean the operation was successful. Check the output from the command for potential error messages.

Examples

To remove the current **Performance counter names** and **Explain text** for the Simple Mail Transfer Protocol (SMTP) service, type:

```
unlodctr SMTPSVC
```

Related links

- Command-Line Syntax Key

Last updated on 10/30/2025

ver

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the operating system version number. This command is supported in the Windows Command prompt (Cmd.exe), but not in any version of PowerShell.

Syntax

```
ver
```

Parameters

Expand table

Parameter	Description
/?	Displays help at the command prompt.

Examples

To obtain the version number of the operating system from the Command shell (cmd.exe), type:

```
ver
```

If you want to get the operating system version number through Windows PowerShell, type:

PowerShell

```
$PSVersionTable.BuildVersion
```

If you want to get the operating system version number through PowerShell 7.x.x, type:

PowerShell

```
$PSVersionTable.OS
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

verifier

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Driver Verifier monitors Windows kernel-mode drivers and graphics drivers to detect illegal function calls or actions that might corrupt the system. Driver Verifier can subject Windows drivers to a variety of stresses and tests to find improper behavior. You can configure which tests to run, which allows you to put a driver through heavy stress loads or through more streamlined testing. You can also run Driver Verifier on multiple drivers simultaneously, or on one driver at a time.

> **ⓘ Important**
>
> You must be in the Administrators group on the computer to use Driver Verifier. Running Driver Verifier can cause the computer to crash, so you should only run this utility on computers used for testing and debugging.

Syntax

```
verifier /standard /all
verifier /standard /driver NAME [NAME ...]
verifier /flags <options> /all
verifier /flags <options> /driver NAME [NAME ...]
verifier /rules [OPTION ...]
verifier /query
verifier /querysettings
verifier /bootmode [persistent | disableafterfail | oneboot]
verifier /reset
verifier /faults [Probability] [PoolTags] [Applications] [DelayMins]
verifier /faultssystematic [OPTION ...]
verifier /log LOG_FILE_NAME [/interval SECONDS]
verifier /volatile /flags <options>
verifier /volatile /adddriver NAME [NAME ...]
verifier /volatile /removedriver NAME [NAME ...]
verifier /volatile /faults [Probability] [PoolTags] [Applications]
[DelayMins]
verifier /domain <types> <options> /driver ... [/logging | /livedump]
verifier /logging
verifier /livedump
```

```
verifier /?
verifier /help
```

Parameters

Expand table

Parameter	Description
/all	Directs the Driver Verifier utility to verify all installed drivers after the next boot.
/bootmode `[persistent | disableafterfail | oneboot | resetonunusualshutdown]`	Controls whether the settings for the Driver Verifier utility are enabled after a reboot. To set or change this option, you must reboot the computer. The following modes are available: • **persistent** - Ensures that the Driver Verifier settings persist (stay in effect) over many reboots. This is the default setting. • **disableafterfail** - If Windows fails to start, this setting disables the Driver Verifier utility for subsequent reboots. • **oneboot** - Only enables the Driver Verifier settings for the next time the computer starts. The Driver Verifier utility is disabled for subsequent reboots. • **resetonunusualshutdown** - The Driver Verifier utility will persist until an unusual shutdown occurs. Its abbrevation, 'rous', can be used.
/driver `<driverlist>`	Specifies one or more drivers that will be verified. The **driverlist** parameter is a list of drivers by binary name, such as *driver.sys*. Use a space to separate each driver name. Wildcard values, such as `n*.sys`, aren't supported.
/driver.exclude `<driverlist>`	Specifies one or more drivers that will be excluded from verification. This parameter is applicable only if all drivers are selected for verification. The **driverlist** parameter is a list of drivers by binary name, such as *driver.sys*. Use a space to separate each driver name. Wildcard values, such as `n*.sys`, aren't supported.
/faults	Enables the **Low Resources Simulation** feature in the Driver Verifier utility. You can use **/faults** in place of `/flags 0x4`. However, you can't use `/flags 0x4` with the **/faults** subparameters. You can use the following subparameters of the /faults parameter to configure the Low Resources Simulation: • **Probability** - Specifies the probability that the Driver Verifier utility will fail a given allocation. Type a number (in decimal or hexadecimal) to represent the number of chances in 10,000 that the Driver Verifier utility will fail the allocation. The default value, 600, means 600/10000 or 6%.

Parameter	Description
	• **Pool Tags** - Limits the allocations that the Driver Verifier utility can fail to allocations with the specified pool tags. You can use a wildcard character (*) to represent multiple pool tags. To list multiple pool tags, separate the tags with spaces. By default, all allocations can fail. • **Applications** - Limits the allocations that the Driver Verifier utility can fail to allocations for the specified program. Type the name of an executable file. To list programs, separate the program names with spaces. By default, all allocations can fail. • **DelayMins** - Specifies the number of minutes after booting during which the Driver Verifier utility does not intentionally fail any allocations. This delay allows the drivers to load and the system to stabilize before the test begins. Type a number (in decimal or hexadecimal). The default value is 7 (minutes).
/faultssystematic	Specifies the options for **Systematic Low Resources** simulation. Use the `0x40000` flag to select the **Systematic Low Resources** simulation option. The following options are available: • **enableboottime** - Enables fault injections across computer reboots. • **disableboottime** - Disables fault injections across computer reboots (this is the default setting). • **recordboottime** - Enables fault injections in what if mode across computer reboots. • **resetboottime** - Disables fault injections across computer reboots and clears the stack exclusion list. • **enableruntime** - Dynamically enables fault injections. • **disableruntime** - Dynamically disables fault injections. • **recordruntime** - Dynamically enables fault injections in what if mode. • **resetruntime** - Dynamically disables fault injections and clears the previously faulted stack list. • **querystatistics** - Shows the current fault injection statistics. • **incrementcounter** - Increments the test pass counter used to identify when a fault was injected. • **getstackid COUNTER** - Retrieves the indicated injected stack identifier. • **excludestack STACKID** - Excludes the stack from fault injection.
/flags `<options>`	Activates the specified options after the next reboot. This number can be entered in decimal or in hexadecimal (with an 0x prefix) format. Any combination of the following values is allowed: • **Value: 1 or 0x1 (bit 0)** - Special pool checking • **Value: 2 or 0x2 (bit 1)** - Force IRQL Checking

Parameter	Description
	• **Value: 4 or 0x4 (bit 2)** - Low Resources Simulation • **Value: 8 or 0x8 (bit 3)** - Pool Tracking • **Value: 16 or 0x10 (bit 4)** - I/O Verification • **Value: 32 or 0x20 (bit 5)** - Deadlock Detection • **Value: 64 or 0x40 (bit 6)** - Enhanced I/O Verification. This option is automatically activated when you select **I/O Verification**. • **Value: 128 or 0x80 (bit 7)** - DMA Verification • **Value: 256 or 0x100 (bit 8)** - Security Checks • **Value: 512 or 0x200 (bit 9)** - Force Pending I/O Requests • **Value: 1024 or 0x400 (bit 10)** - IRP Logging • **Value: 2048 or 0x800 (bit 11)** - Miscellaneous Checks • **Value: 8192 or 0x2000 (bit 13)** - Invariant MDL Checking for Stack • **Value: 16384 or 0x4000 (bit 14)** - Invariant MDL Checking for Driver • **Value: 32768 or 0x8000 (bit 15)** - Power Framework Delay Fuzzing • **Value: 65536 or 0x10000 (bit 16)** - Port/miniport interface checking • **Value: 131072 or 0x20000 (bit 17)** - DDI compliance checking • **Value: 262144 or 0x40000 (bit 18)** - Systematic low resources simulation • **Value: 524288 or 0x80000 (bit 19)** - DDI compliance checking (additional) • **Value: 2097152 or 0x200000 (bit 21)** - NDIS/WIFI verification • **Value: 8388608 or 0x800000 (bit 23)** - Kernel synchronization delay fuzzing • **Value: 16777216 or 0x1000000 (bit 24)** - VM switch verification • **Value: 33554432 or 0x2000000 (bit 25)** - Code integrity checks. You can't use this method to activate the SCSI Verification or Storport Verification options. For more information, see SCSI Verification and Storport Verification.
/flags `<volatileoptions>`	Specifies the Driver Verifier utility options that are changed immediately without rebooting.This number can be entered in decimal or in hexadecimal (with an 0x prefix) format. Any combination of the following values is allowed: • **Value: 1 or 0x1 (bit 0)** - Special pool • **Value: 2 or 0x2 (bit 1)** - Force IRQL Checking • **Value: 4 or 0x4 (bit 2)** - Low Resources Simulation
`<probability>`	Number between 1 and 10,000 specifying the fault injection probability. For example, specifying 100 means a fault injection probability of 1% (100/10,000).

Parameter	Description
	if this parameter isn't specified, the default probability of 6% is used.
`<tags>`	Specifies the pool tags that will be injected with faults, separated by space characters. If this parameter is not specified, then any pool allocation can be injected with faults.
`<apps>`	Specifies the image file name of the apps that will be injected with faults, separated by space characters. If this parameter isn't specified, then low resources simulation can take place in any application.
`<minutes>`	A positive number specifying the length of the period after rebooting, in minutes, during which no fault injection will occur. If this parameter isn't specified, then the default length of *8 minutes* is used.
/iolevel `<level>`	Specifies the level of I/O Verification. The value of [level] can be **1** - Enables Level 1 I/O Verification (default) or **2** - Enables Level 1 I/O Verification and Level 2 I/O Verification. If I/O Verification isn't enabled (by using `/flags 0x10`), **/iolevel** is ignored.
/log `<logfilename>` `[/intervalseconds]`	Creates a log file using the specified name. The Driver Verifier utility periodically writes statistics to this file, based on the interval you optionally set. The default interval is *30 seconds*. If a verifier **/log** command is typed at the command line, the command prompt doesn't return. To close the log file and return a prompt, use the **CTRL+C** key. After a reboot, to create a log, you must submit the verifier **/log** command again.
/rules `<option>`	Options for rules that can be disabled, including: • **query** - Shows current status of controllable rules. • **reset** - Resets all rules to their default state. • **default ID** - Sets rule ID to its default state. For the supported rules, the rule ID is the Bug Check 0xC4 (DRIVER_VERIFIER_DETECTED_VIOLATION) parameter 1 value. • **disable ID** - Disables specified rule ID. For the supported rules, the rule ID is the Bug Check 0xC4 (DRIVER_VERIFIER_DETECTED_VIOLATION) parameter 1 value.
/standard	Activates the "standard" or default Driver Verifier options after the next restart. The standard options are Special Pool, Force IRQL Checking, Pool Tracking, I/O Verification, Deadlock Detection, DMA Verification, Security Checks, Miscellaneous Checks, and DDI compliance checking. This is equivalent to `/flags 0x209BB`.

Parameter	Description
	[!NOTE] Starting in Windows 10 versions after 1803, using `/flags 0x209BB` will no longer automatically enable WDF verification. Use the **/standard** syntax to enable standard options, with WDF verification included.
/volatile	Changes the settings without rebooting the computer. Volatile settings take effect immediately. You can use the **/volatile** parameter with the **/flags** parameter to enable and disable some options without rebooting. You can also use **/volatile** with the **/adddriver** and **/removedriver** parameters to start or stop the verification of a driver without rebooting, even if the Driver Verifier utility isn't running. For more information, see Using Volatile Settings.
/adddriver `<volatiledriverlist>`	Adds the specified drivers from the volatile settings. To specify multiple drivers, list their names, separated by spaces. Wildcard values, such as *n.sys*, aren't supported.
/removedriver `<volatiledriverlist>`	Removes the specified drivers from the volatile settings. To specify multiple drivers, list their names, separated by spaces. Wildcard values, such as *n.sys*, aren't supported.
/reset	Clears all the Driver Verifier utility settings. After the next restart, no drivers will be verified.
/querysettings	Displays a summary of the options that will be activated and drivers that will be verified after the next boot. The display doesn't include drivers and options added by using the **/volatile** parameter. For other ways to view these settings, see Viewing Driver Verifier Settings.
/query	Displays a summary of the Driver Verifier utility's current activity. The **Level** field in the display is the hexadecimal value of options set with the **/volatile** parameter. For explanations of each statistic, see Monitoring Global Counters and Monitoring Individual Counters.
/domain `<types> <options>`	Controls the verifier extension settings. The following verifier extension types are supported: • **wdm** - Enables verifier extension for WDM drivers. • **ndis** - Enables verifier extension for networking drivers. • **ks** - Enables verifier extension for kernel mode streaming drivers. • **audio** - Enables verifier extension for audio drivers. . The following extension options are supported: • **rules.default** - Enables default validation rules for the selected verifier extension.

Parameter	Description
	• **rules.all** - Enables all validation rules for the selected verifier extension.
/logging	Enables logging for violated rules detected by the selected verifier extensions.
/livedump	Enables live memory dump collection for violated rules detected by the selected verifier extensions.
/?	Displays command-line help.

Return Codes

The following values are returned after driver verifier has run:

- 0: EXIT_CODE_SUCCESS
- 1: EXIT_CODE_ERROR
- 2: EXIT_CODE_REBOOT_NEEDED

Remarks

- You can use the **/volatile** parameter with some of the Driver Verifier utility **/flags** options and with **/standard**. You can't use **/volatile** with the **/flags** options for DDI compliance checking, Power Framework Delay Fuzzing, Storport Verification, or SCSI Verification. For more information, see Using Volatile Settings.

Related links

- Command-Line Syntax Key
- Driver Verifier
- Controlling Driver Verifier
- Monitoring Driver Verifier
- Using Volatile Settings

Feedback

Was this page helpful? Yes No

verify

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Tells the Command Prompt tool (cmd.exe) whether to verify your files are written correctly to a disk.

Syntax

```
verify [on | off]
```

Parameters

Expand table

Parameter	Description
`[on | off]`	Switches the **verify** setting on or off.
/?	Displays help at the command prompt.

Examples

To display the current **verify** setting, type:

```
verify
```

To turn the **verify** setting on, type:

```
verify on
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

vol

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the disk volume label and serial number, if they exist. If used without parameters, **vol** displays information for the current drive.

Syntax

```
vol [<drive>:]
```

Parameters

Expand table

Parameter	Description
`<drive>:`	Specifies the drive that contains the disk for which you want to display the volume label and serial number.
/?	Displays help at the command prompt.

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

vssadmin

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Shows current volume shadow copy backups and all installed shadow copy writers and providers. The following table lists the available `vssadmin` commands. Select a command name to view its syntax.

Expand table

Command	Description	Availability
vssadmin delete shadows	Deletes volume shadow copies.	Client and Server
vssadmin list providers	Lists registered volume shadow copy providers.	Client and Server
vssadmin list shadows	Lists existing volume shadow copies.	Client and Server
vssadmin list shadowstorage	Lists volume shadow copy storage associations.	Client and Server
vssadmin list volumes	Lists volumes eligible for shadow copies.	Client and Server
vssadmin list writers	Lists all subscribed volume shadow copy writers on the system.	Client and Server
vssadmin resize shadowstorage	Resizes the maximum size for a shadow copy storage association.	Client and Server

Related links

- Command-Line Syntax Key

Last updated on 09/08/2026

vssadmin delete shadows

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Deletes a specified volume's shadow copies. You can only delete shadow copies with the *client-accessible* type.

> **Note**
>
> If you encounter the error "Error: Snapshots were found, but they were outside of your allowed context," the shadow copies are not client-accessible and cannot be deleted with `vssadmin`. Use the diskshadow command instead to manage and delete those shadow copies.

Syntax

```
vssadmin delete shadows /for=<ForVolumeSpec> [/oldest | /all | /shadow=<ShadowID>]
[/quiet]
```

Parameters

Expand table

Parameter	Description
/for= `<ForVolumeSpec>`	Specifies which volume's shadow copy will be deleted.
/oldest	Deletes only the oldest shadow copy.
/all	Deletes all of the specified volume's shadow copies.
/shadow= `<ShadowID>`	Deletes the shadow copy specified by ShadowID. To get the shadow copy ID, use the vssadmin list shadows command. When you enter a shadow copy ID, use the following format, where each *X* represents a hexadecimal character: XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX
/quiet	Specifies that the command won't display messages while running.

Examples

> **⊗ Caution**
>
> Deleting a shadow copy is permanent. You can't recover a deleted shadow copy, and you lose any previous versions or restore points that depend on it. Target only the volume or shadow copy you intend to remove, and confirm the shadow copy IDs with **vssadmin list shadows** before you delete anything. This command deletes only shadow copies that have the *client-accessible* type.

To delete the oldest shadow copy of volume C, type:

```
vssadmin delete shadows /for=c: /oldest
```

To delete all client-accessible shadow copies of volume D, type:

```
vssadmin delete shadows /for=d: /all
```

To delete a single client-accessible shadow copy by its ID, first run vssadmin list shadows to find the shadow copy ID. Then run the following command, replacing `c:` with the volume that hosts the shadow copy and `<ShadowID>` with the ID you want to delete. Enter the ID in the format *XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX*, where each *X* is a hexadecimal character:

```
vssadmin delete shadows /for=c: /shadow=<ShadowID>
```

To delete all client-accessible shadow copies of volume D without showing progress messages, add the `/quiet` parameter:

```
vssadmin delete shadows /for=d: /all /quiet
```

Related links

- Command-Line Syntax Key
- vssadmin command
- vssadmin list shadows command
- diskshadow command

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 09/08/2026

vssadmin list providers

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Lists the volume shadow copy providers registered on the computer. A shadow copy provider is the component that creates and manages shadow copies (point-in-time snapshots) of a volume. A provider can be the built-in system provider that Windows includes, or a software or hardware provider that a storage vendor supplies.

Run this command when you want to confirm which providers are available before you create a shadow copy, or to verify that a vendor-supplied provider is installed and registered correctly.

Syntax

```
vssadmin list providers
```

Examples

To list all registered volume shadow copy providers, type:

```
vssadmin list providers
```

Related links

- Command-Line Syntax Key
- vssadmin command

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 09/08/2026

vssadmin list shadows

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists all existing shadow copies of a specified volume. If you use this command without parameters, it displays all volume shadow copies on the computer in the order dictated by **Shadow Copy Set**.

Syntax

```
vssadmin list shadows [/for=<ForVolumeSpec>] [/shadow=<ShadowID>]
```

Parameters

Expand table

Parameter	Description
/for= `<ForVolumeSpec>`	Specifies which volume the shadow copies will be listed for.
/shadow= `<ShadowID>`	Lists the shadow copy specified by ShadowID. To get the shadow copy ID, use the vssadmin list shadows command. When you type a shadow copy ID, use the following format, where each *X* represents a hexadecimal character: `{XXXXXXXX-XXXX-XXXX-XXXX-XXXXXXXXXXXX}`

Related links

- Command-Line Syntax Key
- vssadmin command
- vssadmin list shadows command

Feedback

Was this page helpful? Yes No

vssadmin list shadowstorage

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

Lists the shadow copy storage associations on the system. A shadow copy storage association links a volume that has shadow copies to the volume that stores those copies, and reports the disk space that storage uses.

Run this command to find out where shadow copy storage resides and how much space it consumes, such as before you change its size.

Syntax

```
vssadmin list shadowstorage [/for=<ForVolumeSpec>|/on=<OnVolumeSpec>]
```

Parameters

Expand table

Parameter	Description
/for= `<ForVolumeSpec>`	Lists the shadow copy storage associations for the specified volume. Don't use this parameter together with **/on**.
/on= `<OnVolumeSpec>`	Lists the shadow copy storage associations on the specified volume. Don't use this parameter together with **/for**.

Examples

To list all shadow copy storage associations located on volume C, type:

```
vssadmin list shadowstorage /on=C:
```

Related links

- Command-Line Syntax Key
- vssadmin command

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 09/08/2026

vssadmin list volumes

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `vssadmin list volumes` command shows all volumes that are eligible for shadow copies. Use this command to identify which volumes you can shadow copy before you create or configure one.

Syntax

```
vssadmin list volumes
```

Examples

To list all volumes that are eligible for shadow copies, type:

```
vssadmin list volumes
```

Related links

- Command-Line Syntax Key
- vssadmin command

ⓘ**Note:** The author created this article with assistance from AI. Learn more

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Last updated on 09/08/2026

vssadmin list writers

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Lists subscribed volume shadow copy writers.

Syntax

```
vssadmin list writers
```

Related links

- Command-Line Syntax Key
- vssadmin command

Feedback

Was this page helpful? Yes No

vssadmin resize shadowstorage

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Resizes the maximum amount of storage space that can be used for shadow copy storage.

The minimum amount of storage space that can be used for shadow copy storage can be specified by using the **MinDiffAreaFileSize** registry value. For more information, see MinDiffAreaFileSize.

> ⚠ **Warning**
>
> Resizing the storage association may cause shadow copies to disappear.

Syntax

```
vssadmin resize shadowstorage /for=<ForVolumeSpec> /on=<OnVolumeSpec>
[/maxsize=<MaxSizeSpec>]
```

Parameters

Expand table

Parameter	Description
/for= `<ForVolumeSpec>`	Specifies the volume for which the maximum amount of storage space is to be resized.
/on= `<OnVolumeSpec>`	Specifies the storage volume.
[/maxsize= `<MaxSizeSpec>`]	Specifies the maximum amount of space that can be used for storing shadow copies. If no value is specified for **/maxsize**, there's no limit placed on the amount of storage space that can be used. The **MaxSizeSpec** value must be 1 MB or greater and must be expressed in one of the following units: KB, MB, GB, TB, PB, or EB. If no unit is specified,

Parameter	Description
	MaxSizeSpec uses bytes by default.

Examples

To resize shadow copy of volume C on volume D, with a maximum size of 900MB, type:

```
vssadmin resize shadowstorage /For=C: /On=D: /MaxSize=900MB
```

To resize shadow copy of volume C on volume D, with no maximum size, type:

```
vssadmin resize shadowstorage /For=C: /On=D: /MaxSize=UNBOUNDED
```

To resize shadow copy of volume C by 20%, type:

```
vssadmin resize shadowstorage /For=C: /On=C: /MaxSize=20%
```

Related links

- Command-Line Syntax Key
- vssadmin command

Feedback

Was this page helpful? Yes No

waitfor

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Sends or waits for a signal on a system. This command is used to synchronize computers across a network.

Syntax

```
waitfor [/s <computer> [/u [<domain>\]<user> [/p [<password>]]]] /si
<signalname>
waitfor [/t <timeout>] <signalname>
```

Parameters

Expand table

Parameter	Description
/s `<computer>`	Specifies the name or IP address of a remote computer (don't use backslashes). The default is the local computer. This parameter applies to all files and folders specified in the command. If you don't use this parameter, the signal is broadcast to all the systems in a domain. If you do use this parameter, the signal is sent only to the specified system.
/u `[<domain>]` `<user>`	Runs the script using the credentials of the specified user account. By default, **waitfor** uses the current user's credentials.
/p `[\` `<password>]`	Specifies the password of the user account that is specified in the **/u** parameter.
/si	Sends the specified signal across the network. This parameter also lets you manually activate a signal.
/t `<timeout>`	Specifies the number of seconds to wait for a signal. By default, **waitfor** waits indefinitely.
`<signalname>`	Specifies the signal that **waitfor** waits for or sends. This parameter isn't case-sensitive and can't exceed 225 characters. Valid characters include a-z, A-Z, 0-9, and the ASCII extended character set (128-255).

Parameter	Description
/?	Displays help at the command prompt.

Remarks

- You can run multiple instances of **waitfor** on a single computer, but each instance of **waitfor** must wait for a different signal. Only one instance of **waitfor** can wait for a given signal on a given computer.
- Computers can only receive signals if they are in the same domain as the computer sending the signal.
- You can use this command when you test software builds. For example, the compiling computer can send a signal to several computers running **waitfor** after the compile has completed successfully. On receipt of the signal, the batch file that includes **waitfor** can instruct the computers to immediately start installing software or running tests on the compiled build.

Examples

To wait until the *espresso\build007* signal is received, type:

```
waitfor espresso\build007
```

By default, **waitfor** waits indefinitely for a signal.

To wait *10 seconds* for the *espresso\compile007* signal to be received before timing out, type:

```
waitfor /t 10 espresso\build007
```

To manually activate the *espresso\build007* signal, type:

```
waitfor /si espresso\build007
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wbadmin

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables you to back up and restore your operating system, volumes, files, folders, and applications from a command prompt.

To configure a regularly scheduled backup using this command, you must be a member of the **Administrators** group. To perform all other tasks with this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions.

You must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Parameters

Expand table

Parameter	Description
wbadmin delete catalog	Deletes the backup catalog on the local computer. Use this command only if the backup catalog on this computer is corrupted and you have no backups stored at another location that you can use to restore the catalog.
wbadmin delete systemstatebackup	Deletes one or more system state backups.
wbadmin disable backup	Disables your daily backups.
wbadmin enable backup	Configures and enables a regularly scheduled backup.
wbadmin get disks	Lists disks that are currently online.
wbadmin get items	Lists the items included in a backup.
wbadmin get status	Shows the status of the currently running backup or recovery operation.
wbadmin get versions	Lists details of backups recoverable from the local computer or, if another location is specified, from another computer.

Parameter	Description
wbadmin restore catalog	Recovers a backup catalog from a specified storage location in the case where the backup catalog on the local computer has been corrupted.
wbadmin start backup	Runs a one-time backup. If used with no parameters, uses the settings from the daily backup schedule.
wbadmin start recovery	Runs a recovery of the volumes, applications, files, or folders specified.
wbadmin start sysrecovery	Runs a recovery of the full system (at least all the volumes that contain the operating system's state). This command is only available if you are using the Windows Recovery Environment.
wbadmin start systemstatebackup	Runs a system state backup.
wbadmin start systemstaterecovery	Runs a system state recovery.
wbadmin stop job	Stops the currently running backup or recovery operation.

Related links

- Command-Line Syntax Key
- Windows Server Backup Cmdlets in Windows PowerShell
- Windows Recovery Environment (WinRE)

Feedback

Was this page helpful? Yes No

wbadmin delete catalog

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes the backup catalog that is stored on the local computer. Use this command when the backup catalog has been corrupted and you can't restore it using the wbadmin restore catalog command.

To delete a backup catalog using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

```
wbadmin delete catalog [-quiet]
```

Parameters

Expand table

Parameter	Description
-quiet	Runs the command without prompts to the user.

Remarks

- If you delete the backup catalog of a computer, you'll no longer be able to get to any backups created for that computer using the Windows Server Backup snap-in. However, if you can get to another backup location and run the wbadmin restore catalog command, you can restore the backup catalog from that location.
- We strongly recommend you create a new backup after you delete a backup catalog.

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin restore catalog command
- Remove-WBCatalog

Feedback

Was this page helpful? Yes No

wbadmin delete systemstatebackup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes the system state backups that you specify. If the specified volume contains backups other than system state backups of your local server, those backups will not be deleted.

To delete a system state backup using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

> **Note**
>
> Windows Server Backup does not back up or recover registry user hives (HKEY_CURRENT_USER) as part of system state backup or system state recovery.

Syntax

```
wbadmin delete systemstatebackup {-keepVersions:<numberofcopies> | -version:<versionidentifier> | -deleteoldest} [-backupTarget:<volumename>] [-machine:<backupmachinename>] [-quiet]
```

> **Important**
>
> You must only specify one of these parameters: **-keepVersions**, **-version**, or **-deleteOldest**.

Parameters

Expand table

Parameter	Description
-keepVersions	Specifies the number of the latest system state backups to keep. The value must be a positive integer. The parameter value **-keepversions:0** deletes all the system state backups.
-version	Specifies the version identifier of the backup in MM/DD/YYYY-HH:MM format. If you don't know the version identifier, run the wbadmin get versions command. Versions made up of exclusively system state backups can be deleted using this command. Run the wbadmin get items command to view the version type.
-deleteOldest	Deletes the oldest system state backup.
-backupTarget	Specifies the storage location for the backup that you want to delete. The storage location for disk backups can be a drive letter, a mount point, or a GUID-based volume path. This value only needs to be specified for locating backups that are not on the local computer. Information about backups for the local computer is available in the backup catalog on the local computer.
-machine	Specifies the computer whose system state backup you want to delete. Useful when multiple computers were backed up to the same location. Should be used when the **-backupTarget** parameter is specified.
-quiet	Runs the command without prompts to the user.

Examples

To delete the system state backup created on March 31, 2013 at 10:00 AM, type:

```
wbadmin delete systemstatebackup -version:03/31/2013-10:00
```

To delete all system state backups, except the three most recent, type:

```
wbadmin delete systemstatebackup -keepVersions:3
```

To delete the oldest system state backup stored on disk f:, type:

```
wbadmin delete systemstatebackup -backupTarget:f:\ -deleteOldest
```

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin get versions command
- wbadmin get items command

Feedback

Was this page helpful? Yes No

wbadmin disable backup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Stops running the existing scheduled daily backups.

To disable a scheduled daily backup using this command, you must be a member of the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

```
wbadmin disable backup [-quiet]
```

Parameters

Expand table

Parameter	Description
-quiet	Runs the command without prompts to the user.

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin enable backup command

Feedback

Was this page helpful? Yes No

wbadmin enable backup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates and enables a daily backup schedule or modifies an existing backup schedule. With no parameters specified, it displays the currently scheduled backup settings.

To configure or modify a daily backup schedule using this command, you must be a member of the **Backup Operators** group or the **Administrators** group. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

To view the disk identifier value for your disks, run the wbadmin get disks command.

Syntax

```
wbadmin enable backup [-addtarget:<BackupTarget>] [-removetarget:
<BackupTarget>] [-schedule:<TimeToRunBackup>] [-include:<VolumesToInclude>]
[-nonRecurseInclude:<ItemsToInclude>] [-exclude:<ItemsToExclude>] [-
nonRecurseExclude:<ItemsToExclude>][-systemState] [-hyperv:
<HyperVComponentsToExclude>] [-allCritical] [-systemState] [-vssFull | -
vssCopy] [-user:<UserName>] [-password:<Password>] [-allowDeleteOldBackups]
[-quiet]
```

Parameters

Expand table

Parameter	Description
-addtarget	Specifies the storage location for backups. Requires you to specify the location as a disk, volume, or Universal Naming Convention (UNC) path to a remote shared folder (`\\<servername>\<sharename>`). By default, the backup will be saved at: `\\<servername>\<sharename> WindowsImageBackup <ComputerBackedUp>`. If you specify a disk, the disk will be formatted before use, and any existing data on it is permanently erased. If you specify a shared folder, you can't add more locations. You can only specify one shared folder as a storage location at a time.

Parameter	Description
	Important: If you save a backup to a remote shared folder, that backup is overwritten if you use the same folder to back up the same computer again. In addition, if the backup operation fails, you could end up with no backup because the older backup will be overwritten, but the newer backup won't be usable. You can avoid this by creating sub-folders in the remote shared folder to organize your backups. If you do this, the sub-folders need twice the space of the parent folder. Only one location can be specified in a single command. Multiple volume and disk backup storage locations can be added by running the command again.
-removetarget	Specifies the storage location that you want to remove from the existing backup schedule. Requires you to specify the location as a disk identifier.
-schedule	Specifies times of day to create a backup, formatted as HH:MM and comma delimited.
-include	Specifies the comma-delimited list of items to include in the backup. You can include multiple files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should end with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file.
-nonRecurseInclude	Specifies the non-recursive, comma-delimited list of items to include in the backup. You can include multiple files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should end with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file. Should be used only when the **-backupTarget** parameter is used.
-exclude	Specifies the comma-delimited list of items to exclude from the backup. You can exclude files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should end with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file.
-nonRecurseExclude	Specifies the non-recursive, comma-delimited list of items to exclude from the backup. You can exclude files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should end with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file.

Parameter	Description
-hyperv	Specifies the comma-delimited list of components to be included in backup. The identifier can be a component name or component GUID (with or without braces).
-systemState	Creates a backup that includes the system state in addition to any other items that you specified with the **-include** parameter. The system state contains boot files (Boot.ini, NDTLDR, NTDetect.com), the Windows Registry including COM settings, the SYSVOL (Group Policies and Logon Scripts), the Active Directory and NTDS.DIT on domain controllers and, if the certificates service is installed, the Certificate Store. If your server has the Web server role installed, the IIS Metadirectory will be included. If the server is part of a cluster, Cluster service information is also included.
-allCritical	Specifies that all critical volumes (volumes that contain operating system's state) be included in the backups. This parameter is useful if you are creating a backup for full system or system state recovery. It should be used only when **-backupTarget** is specified; otherwise, the command fails. Can be used with the **-include** option. **Tip:** The target volume for a critical-volume backup can be a local drive, but it can't be any of the volumes that are included in the backup.
-vssFull	Performs a full back up using the Volume Shadow Copy Service (VSS). All files are backed up, each file's history is updated to reflect that it was backed up, and the logs of previous backups may be truncated. If this parameter is not used, the wbadmin start backup command makes a copy backup, but the history of files being backed up is not updated. **Caution:** Don't use this parameter if you're using a product other than Windows Server Backup to back up apps that are on the volumes included in the current backup. Doing so can potentially break the incremental, differential, or other type of backups that the other backup product is creating because the history that they are relying on to determine how much data to backup might be missing and they might perform a full backup unnecessarily.
-vssCopy	Performs a copy backup using VSS. All files are backed up but the history of the files being backup up is not updated so you preserve the all the information on which files where changed, deleted, and so on, as well as any application log files. Using this type of backup does not affect the sequence of incremental and differential backups that might happen independent of this copy backup. This is the default value. **Warning:** A backup copy can't be used for incremental or differential backups or restores.
-user	Specifies the user with write permission to the backup storage destination (if it's a remote shared folder). The user needs to be a

Parameter	Description
	member of the **Administrators** or **Backup Operators** group on the computer getting backed up.
-password	Specifies the password for the user name provided by the parameter **-user**.
-allowDeleteOldBackups	Overwrites any backups made before the computer was upgraded.
-quiet	Runs the command without prompts to the user.

Examples

To schedule daily backups at 9:00 AM and 6:00 PM for hard disk drives E:, D:\mountpoint, and `\\?\Volume{cc566d14-44a0-11d9-9d93-806e6f6e6963}\`, and to save the files to the disk named, DiskID, type:

```
wbadmin enable backup -addtarget:DiskID -schedule:09:00,18:00 -
include:E:,D:\mountpoint,\\?\Volume{cc566d14-44a0-11d9-9d93-806e6f6e6963}\
```

To schedule daily backups of the D:\documents folder at 12:00 AM and 7:00 PM to the network location `\\backupshare\backup1`, using the network credentials for the **Backup Operator**, Aaren Ekelund (aekel), who's password is *$3hM9^5lp* and who is a member of the domain CONTOSOEAST, used to authenticate access to the network share, type:

```
wbadmin enable backup –addtarget:\\backupshare\backup1 –include:
D:\documents –user:CONTOSOEAST\aekel –password:$3hM9^5lp –
schedule:00:00,19:00
```

To schedule daily backups of volume T: and the D:\documents folder at 1:00 AM to drive H:, excluding the folder `d:\documents\~tmp`, and performing a full backup using the Volume Shadow Copy Service, type:

```
wbadmin enable backup –addtarget:H: –include T:,D:\documents –exclude
D:\documents\~tmp –vssfull –schedule:01:00
```

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin enable backup command
- wbadmin start backup command
- wbadmin get disks command

Feedback

Was this page helpful? Yes No

wbadmin get disks

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists the internal and external disks that are currently online for the local computer.

To list the online disks using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

```
wbadmin get disks
```

Related links

- Command-Line Syntax Key
- wbadmin command
- Get-WBDisk

Feedback

Was this page helpful? Yes No

wbadmin get items

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists the items included in a specific backup.

To list the items included in a specific backup using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

```
wbadmin get items -version:<VersionIdentifier> [-backupTarget:
{<BackupDestinationVolume> | <NetworkSharePath>}] [-machine:
<BackupMachineName>]
```

Parameters

Expand table

Parameter	Description
-version	Specifies the version of the backup in MM/DD/YYYY-HH:MM format. If you don't know the version information, run the wbadmin get versions command.
-backupTarget	Specifies the storage location that contains the backups for which you want the details. Use for listing backups stored at that target location. Backup target locations can be a locally attached disk drive or a remote shared folder. If this command is run on the same computer where the backup was created, this parameter isn't needed. However, this parameter is required to get information about a backup created from another computer.
-machine	Specifies the name of the computer that you want the backup details for. Useful when multiple computers have been backed up to the same location. Should be used when **-backupTarget** is specified.

Examples

To list items from the backup that was run on March 31, 2013 at 9:00 A.M., type:

```
wbadmin get items -version:03/31/2013-09:00
```

To list items from the backup of server01 that was run on April 30, 2013 at 9:00 A.M. and stored on `\\<servername>\<share>`, type:

```
wbadmin get items -version:04/30/2013-09:00 -backupTarget:\\servername\share
-machine:server01
```

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin get versions command
- Get-WBBackupSet

Feedback

Was this page helpful? Yes No

wbadmin get status

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Reports the status of the backup or recovery operation that is currently running.

To get the status of the currently running backup or recovery operation using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

> **Important**
>
> This command doesn't stop until the backup or recovery operation is finished. The command continues to run even if you close the command window. To stop the current backup or recovery operation, run the **wbadmin stop job** command.

Syntax

```
wbadmin get status
```

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin stop job command
- Get-WBJob

Feedback

Was this page helpful? Yes No

wbadmin get versions

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Lists details about the available backups that are stored on the local computer or another computer. The details provided for a backup include the backup time, the backup storage location, the version identifier, and the type of recoveries you can perform.

To get details about available backups using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

If this command is used without parameters, it lists all backups of the local computer, even if those backups are not available.

Syntax

```
wbadmin get versions [-backupTarget:{<BackupTargetLocation> |
<NetworkSharePath>}] [-machine:BackupMachineName]
```

Parameters

Expand table

Parameter	Description
-backupTarget	Specifies the storage location that contains the backups that you want the details for. Use for listing backups stored at that target location. Backup target locations can be locally attached disk drives, volumes, remote shared folders, removable media such as DVD drives or other optical media. If this command is run on the same computer where the backup was created, this parameter isn't needed. However, this parameter is required to get information about a backup created from another computer.

Parameter	Description
-machine	Specifies the computer that you want backup details for. Use when backups of multiple computers are stored in the same location. Should be used when **-backupTarget** is specified.

Examples

To see a list of available backups that are stored on volume H:, type:

```
wbadmin get versions -backupTarget:H:
```

To see a list of available backups that are stored in the remote shared folder `\\<servername>\<share>` for the computer server01, type:

```
wbadmin get versions -backupTarget:\\servername\share -machine:server01
```

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin get items command
- Get-WBBackupTarget

Feedback

Was this page helpful? Yes No

wbadmin restore catalog

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Recovers a backup catalog for the local computer from a storage location that you specify.

To recover a backup catalog included in a specific backup using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

> **Note**
>
> If the location (disk, DVD, or remote shared folder) where you store your backups is damaged or lost and can't be used to restore the backup catalog, run the wbadmin delete catalog command to delete the corrupted catalog. In this case, we recommend creating a new backup after your backup catalog is deleted.

Syntax

```
wbadmin restore catalog -backupTarget:{<BackupDestinationVolume> |
<NetworkShareHostingBackup>} [-machine:<BackupMachineName>] [-quiet]
```

Parameters

Expand table

Parameter	Description
-backupTarget	Specifies the location of the backup catalog of the system as it was at the point after the backup was created.
-machine	Specifies the name of the computer that you want to recover the backup catalog for. Use when backups for multiple computers have been stored at the same

Parameter	Description
	location. Should be used when **-backupTarget** is specified.
-quiet	Runs the command without prompts to the user.

Examples

To restore a catalog from a backup stored on disk D:, type:

```
wbadmin restore catalog -backupTarget:D
```

To restore a catalog from a backup stored in the shared folder `\\<servername>\<share>` of server01, type:

```
wbadmin restore catalog -backupTarget:\\servername\share -machine:server01
```

Related links

- Command-Line Syntax Key
- wbadmin command
- wbadmin delete catalog command
- Restore-WBCatalog

Feedback

Was this page helpful? Yes No

wbadmin start backup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a backup using specified parameters. If no parameters are specified and you have created a scheduled daily backup, this command creates the backup by using the settings for the scheduled backup. If parameters are specified, it creates a Volume Shadow Copy Service (VSS) copy backup and won't update the history of the files that are being backed up.

To create a one-time backup using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

com

```
wbadmin start backup [-backupTarget:{<BackupTargetLocation> |
<TargetNetworkShare>}] [-include:<ItemsToInclude>] [-nonRecurseInclude:
<ItemsToInclude>] [-exclude:<ItemsToExclude>] [-nonRecurseExclude:
<ItemsToExclude>] [-allCritical] [-systemState] [-noVerify] [-user:
<UserName>] [-password:<Password>] [-noInheritAcl] [-vssFull | -vssCopy] [-
quiet]
```

Parameters

Expand table

Parameter	Description
-backupTarget	Specifies the storage location for this backup. Requires a hard disk drive letter (f:), a volume GUID-based path in the format of `\\?\Volume{GUID}`, or a Universal Naming Convention (UNC) path to a remote shared folder `(\\<servername>\<sharename>\)`. By default, the backup will be saved at: `\\<servername>\<sharename>\WindowsImageBackup\<ComputerBackedUp>\`.

Parameter	Description
-include	Specifies the comma-delimited list of items to include in the backup. You can include multiple files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should be terminated with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file. The **-include** parameter should only be used in conjunction with the **-backupTarget** parameter.
-exclude	Specifies the comma-delimited list of items to exclude from the backup. You can exclude files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should be terminated with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file. The **-exclude** parameter should only be used in conjunction with the **-backupTarget** parameter.
-nonRecurseInclude	Specifies the non-recursive, comma-delimited list of items to include in the backup. You can include multiple files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should be terminated with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file. The **-nonRecurseInclude** parameter should only be used in conjunction with the **-backupTarget** parameter.
-nonRecurseExclude	Specifies the non-recursive, comma-delimited list of items to exclude from the backup. You can exclude files, folders, or volumes. Volume paths can be specified using volume drive letters, volume mount points, or GUID-based volume names. If you use a GUID-based volume name, it should be terminated with a backslash (`\`). You can use the wildcard character (`*`) in the file name when specifying a path to a file. The **-nonRecurseExclude** parameter should only be used in conjunction with the **-backupTarget** parameter.
-allCritical	Specifies that all critical volumes (volumes that contain operating system's state) be included in the backups. This parameter is useful if you're creating a backup for bare metal recovery. It should be used only when **-backupTarget** is specified, otherwise the command fails. Can be used with the **-include** option. **Tip:** The target volume for a critical-volume backup can be a local drive, but it Can't be any of the volumes that are included in the backup.
-systemState	Creates a backup that includes the system state in addition to any other items that you specified with the **-include** parameter. The system state contains boot files (Boot.ini, NDTLDR, NTDetect.com), the Windows Registry including COM settings, the SYSVOL (Group Policies and Logon Scripts), the Active Directory and NTDS.DIT on Domain Controllers and, if the certificates

Parameter	Description
	service is installed, the Certificate Store. If your server has the Web server role installed, the IIS Metadirectory will be included. If the server is part of a cluster, Cluster Service information will also be included.
-noVerify	Specifies that backups saved to removable media (such as a DVD) are not verified for errors. If you do not use this parameter, backups saved to removable media are verified for errors.
-user	If the backup is saved to a remote shared folder, specifies the user name with write permission to the folder.
-password	Specifies the password for the user name that is provided by the parameter **-user**.
-noInheritAcl	Applies the access control list (ACL) permissions that correspond to the credentials provided by the **-user** and **-password** parameters to `\\<servername>\<sharename>\WindowsImageBackup\<ComputerBackedUp>\` (the folder that contains the backup). To access the backup later, you must use these credentials or be a member of the Administrators group or the Backup Operators group on the computer with the shared folder. If **-noInheritAcl** is not used, the ACL permissions from the remote shared folder are applied to the `\<ComputerBackedUp>` folder by default so that anyone with access to the remote shared folder can access the backup.
-vssFull	Performs a full back up using the Volume Shadow Copy Service (VSS). All files are backed up, each file's history is updated to reflect that it was backed up, and the logs of previous backups may be truncated. If this parameter isn't used, **wbadmin start backup** makes a copy backup, but the history of files being backed up is not updated. **Caution:** Don't use this parameter if you are using a product other than Windows Server Backup to back up apps that are on the volumes included in the current backup. Doing so can potentially break the incremental, differential, or other type of backups that the other backup product is creating because the history that they are relying on to determine how much data to backup might be missing and they might perform a full backup unnecessarily.
-vssCopy	Performs a copy backup using VSS. All files are backed up but the history of the files being backup up is not updated so you preserve the all the information on which files where changed, deleted, and so on, as well as any application log files. Using this type of backup does not affect the sequence of incremental and differential backups that might happen independent of this copy backup. This is the default value. **Warning:** A copy backup can't be used for incremental or differential backups or restores.
-quiet	Runs the command without prompts to the user.

Remarks

- If you save your backup to a remote shared folder, and then perform another backup to the same computer and the same remote shared folder, you will overwrite your previous backup.
- If your backup operation fails, you can end up without a backup because the older backup is overwritten, but the newer backup isn't usable. To avoid this, we recommend creating subfolders in the remote shared folder to organize your backups. However, because of this organization, you must have twice the space available as the parent folder.

Examples

To create a backup of volumes *e:*, *d:\mountpoint*, and `\\?\Volume{cc566d14-4410-11d9-9d93-806e6f6e6963}\` to volume *f:*, type:

```
wbadmin start backup -backupTarget:f: -include:e:,d:\mountpoint,\\?
\Volume{cc566d14-44a0-11d9-9d93-806e6f6e6963}\
```

To perform a one-time backup of *f:\folder1* and *h:\folder2* to volume *d:*, to backup the system state, and to make a copy backup so the normally scheduled differential backup isn't impacted, type:

```
wbadmin start backup –backupTarget:d: -include:g\folder1,h:\folder2 –
systemstate -vsscopy
```

To perform a one-time, non-recursive backup of *d:\folder1* to the `\\backupshare\backup1*` network location, and to restrict access to members of the **Administrators** or **Backup Operators** group, type:

```
wbadmin start backup –backupTarget: \\backupshare\backup1 -noinheritacl -
nonrecurseinclude:d:\folder1
```

Related links

- Command-Line Syntax Key
- wbadmin command

Feedback

Was this page helpful? Yes No

wbadmin start recovery

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Runs a recovery operation based on the parameters that you specify.

To perform a recovery using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

```
wbadmin start recovery -version:<VersionIdentifier> -items:
{<VolumesToRecover> | <AppsToRecover> | <FilesOrFoldersToRecover>} -
itemtype:{Volume | App | File} [-backupTarget:{<VolumeHostingBackup> |
<NetworkShareHostingBackup>}] [-machine:<BackupMachineName>] [-
recoveryTarget:{<TargetVolumeForRecovery> | <TargetPathForRecovery>}] [-
recursive] [-overwrite:{Overwrite | CreateCopy | Skip}] [-notRestoreAcl] [-
skipBadClusterCheck] [-noRollForward] [-quiet]
```

Parameters

Expand table

Parameter	Description
-version	Specifies the version identifier of the backup to recover in MM/DD/YYYY-HH:MM format. If you don't know the version identifier, run the wbadmin get versions command.
-items	Specifies a comma-delimited list of volumes, apps, files, or folders to recover. You must use this parameter with the **-itemtype** parameter.
-itemtype	Specifies type of items to recover. Must be **Volume**, **App**, or **File**. If the **-itemtype** is *Volume*, you can specify only a single volume, by providing the volume drive letter, volume mount point, or GUID-based volume name. If the **-itemtype** is *App*, you can specify only a single application or you can use the value **ADIFM** to recover an installation of Active Directory. To be

Parameter	Description
	recovered, the app must have registered with Windows Server Backup. If the **-itemtype** is *File*, you can specify files or folders, but they should be part of the same volume and they should be under the same parent folder.
-backupTarget	Specifies the storage location that contains the backup that you want to recover. This parameter is useful when the location is different from where backups of this computer are usually stored.
-machine	Specifies the name of the computer that you want to recover the backup for. This parameter must be used when the **-backupTarget** parameter is specified. The **-machine** parameter is useful when multiple computers have been backed up to the same location.
-recoveryTarget	Specifies the location to restore to. This parameter is useful if this location is different than the location that was previously backed up. It can also be used for restorations of volumes, files, or apps. If you're restoring a volume, you can specify the volume drive letter of the alternate volume. If you're restoring a file or app, you can specify an alternate recovery location.
-recursive	Valid only when recovering files. Recovers the files in the folders and all files subordinate to the specified folders. By default, only files which reside directly in the specified folders are recovered.
-overwrite	Valid only when recovering files. Specifies the action to take when a file that is being recovered already exists in the same location. The valid options are: • **Skip** - Causes Windows Server Backup to skip the existing file and continue with recovery of the next file. • **CreateCopy** - Causes Windows Server Backup to create a copy of the existing file so that the existing file is not modified. • **Overwrite** - Causes Windows Server Backup to overwrite the existing file with the file from the backup.
-notRestoreAcl	Valid only when recovering files. Specifies to not restore the security access control lists (ACLs) of the files being recovered from the backup. By default, the security ACLs are restored (the default value is **true**). If this parameter is used, the ACLs for the restored files will be inherited from the location to which the files are being restored.
-skipBadClusterCheck	Valid only when recovering volumes. Skips checking the disks you are recovering to for bad cluster information. If you are recovering to an alternate server or hardware, we recommend that you don't use this parameter. You can manually run the command **chkdsk /b** on these disks at any time to check them for bad clusters, and then update the file system information accordingly.

Parameter	Description
	Important: Until you run **chkdsk /b**, the bad clusters reported on your recovered system might not be accurate.
-noRollForward	Valid only when recovering apps. Allows for previous point-in-time recovery of an app if you select the latest version from the backups. Previous point-in-time recovery is done as the default for all other non-latest versions of the app.
-quiet	Runs the command without prompts to the user.

Remarks

- To view a list of items available to recover from a specific backup version, run the wbadmin get items command. If a volume didn't have a mount point or drive letter at the time of backup, then this command returns a GUID-based volume name that should be used for recovering the volume.
- If you use a value of **ADIFM** to perform an install from media operation to recover the related data needed for Active Directory Domain Services, **ADIFM** creates a copy of the Active Directory database, registry, and SYSVOL state, and then saves this information in the location specified by **-recoveryTarget**. Use this parameter only when **-recoveryTarget** is specified.

Examples

To run a recovery of the backup from March 31, 2020, taken at 9:00 A.M., of volume d:, type:

```
wbadmin start recovery -version:03/31/2020-09:00 -itemType:Volume -items:d:
```

To run a recovery to drive d of the backup from March 31, 2020, taken at 9:00 A.M., of the registry, type:

```
wbadmin start recovery -version:03/31/2020-09:00 -itemType:App -
items:Registry -recoverytarget:d:\
```

To run a recovery of the backup from March 31, 2020, taken at 9:00 A.M., of the d:\folder and folders subordinate to d:\folder, type:

```
wbadmin start recovery -version:03/31/2020-09:00 -itemType:File -
items:d:\folder -recursive
```

To run a recovery of the backup from March 31, 2020, taken at 9:00 A.M., of the volume `\\?\Volume{cc566d14-44a0-11d9-9d93-806e6f6e6963}\`, type:

```
wbadmin start recovery -version:03/31/2020-09:00 -itemType:Volume -items:\\?
\Volume{cc566d14-44a0-11d9-9d93-806e6f6e6963}\
```

To run a recovery of the backup from April 30, 2020, taken at 9:00 A.M., of the shared folder `\\servername\share` from server01, type:

```
wbadmin start recovery -version:04/30/2020-09:00 -
backupTarget:\\servername\share -machine:server01
```

Related links

- Command-Line Syntax Key
- wbadmin command
- Start-WBFileRecovery
- Start-WBHyperVRecovery
- Start-WBSystemStateRecovery
- Start-WBVolumeRecovery

Feedback

Was this page helpful? Yes No

wbadmin start sysrecovery

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Performs a system recovery (bare metal recovery) using your specified parameters.

To perform a system recovery using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions.

> **Important**
>
> The **wbadmin start sysrecovery** command must be run from the Windows Recovery Console, and isn't listed in the default usage text for the **wbadmin** tool. For more information, see Windows Recovery Environment (WinRE).

Syntax

```
wbadmin start sysrecovery -version:<VersionIdentifier> -backupTarget:
{<BackupDestinationVolume> | <NetworkShareHostingBackup>}  [-machine:
<BackupMachineName>] [-restoreAllVolumes] [-recreateDisks] [-excludeDisks]
[-skipBadClusterCheck] [-quiet]
```

Parameters

Expand table

Parameter	Description
-version	Specifies the version identifier of the backup to recover in MM/DD/YYYY-HH:MM format. If you don't know the version identifier, run the wbadmin get versions command.
-backupTarget	Specifies the storage location that contains the backup(s) you want to recover. This parameter is useful when the storage location is different from where backups of this computer are usually stored.

Parameter	Description
-machine	Specifies the name of the computer that you want to recover the backup for. This parameter must be used when the **-backupTarget** parameter is specified. The **-machine** parameter is useful when multiple computers have been backed up to the same location.
-restoreAllVolumes	Recovers all volumes from the selected backup. If this parameter is not specified, only critical volumes (volumes that contain the system state and operating system components) are recovered. This parameter is useful when you need to recover non-critical volumes during system recovery.
-recreateDisks	Recovers a disk configuration to the state that existed when the backup was created. **Warning:** This parameter deletes all data on volumes that host operating system components. It might also delete data from data volumes.
-excludeDisks	Valid only when specified with the **-recreateDisks** parameter and must be input as a comma-delimited list of disk identifiers (as listed in the output of the wbadmin get disks command). Excluded disks aren't partitioned or formatted. This parameter helps preserve data on disks that you don't want modified during the recovery operation.
-skipBadClusterCheck	Valid only when recovering volumes. Skips checking the disks you are recovering to for bad cluster information. If you are recovering to an alternate server or hardware, we recommend that you don't use this parameter. You can manually run the command **chkdsk /b** on these disks at any time to check them for bad clusters, and then update the file system information accordingly. **Important:** Until you run **chkdsk /b**, the bad clusters reported on your recovered system might not be accurate.
-quiet	Runs the command without prompts to the user.

Examples

To start recovering the information from the backup that was run on March 31, 2020 at 9:00 A.M., located on drive d:, type:

```
wbadmin start sysrecovery -version:03/31/2020-09:00 -backupTarget:d:
```

To start recovering the information from the backup that was run on April 30, 2020 at 9:00 A.M., located in the shared folder `\\servername\share` for server01, type:

```
wbadmin start sysrecovery -version:04/30/2020-09:00 -
backupTarget:\\servername\share -machine:server01
```

Related links

- Command-Line Syntax Key
- wbadmin command
- Get-WBBareMetalRecovery

Feedback

Was this page helpful? Yes No

wbadmin start systemstatebackup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a system state backup of the local computer and stores it on the location specified.

To perform a system state backup using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

> **Note**
>
> Windows Server Backup doesn't back up or recover registry user hives (HKEY_CURRENT_USER) as part of system state backup or system state recovery.

Syntax

```
wbadmin start systemstatebackup -backupTarget:<VolumeName> [-quiet]
```

Parameters

Expand table

Parameter	Description
-backupTarget	Specifies the location where you want to store the backup. The storage location requires a drive letter or a GUID-based volume of the format: `\\?\Volume{*GUID*}`. Use the command `-backuptarget:\\servername\sharedfolder\` to store system state backups.
-quiet	Runs the command without prompts to the user.

Examples

To create a system state backup and store it on volume f, type:

```
wbadmin start systemstatebackup -backupTarget:f:
```

Related links

- Command-Line Syntax Key
- wbadmin command
- Start-WBBackup

Feedback

Was this page helpful? Yes No

wbadmin start systemstaterecovery

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Performs a system state recovery to a location and from a backup that you specify.

To perform a system state recovery using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

> **Note**
>
> Windows Server Backup doesn't back up or recover registry user hives (HKEY_CURRENT_USER) as part of system state backup or system state recovery.

Syntax

```
wbadmin start systemstaterecovery -version:<VersionIdentifier> -showsummary
[-backupTarget:{<BackupDestinationVolume> | <NetworkSharePath>}]
[-machine:<BackupMachineName>] [-recoveryTarget:<TargetPathForRecovery>] [-
authsysvol] [-autoReboot] [-quiet]
```

Parameters

Expand table

Parameter	Description
-version	Specifies the version identifier of the backup to recover in MM/DD/YYYY-HH:MM format. If you don't know the version identifier, run the wbadmin get versions command.
-showsummary	Reports the summary of the last system state recovery (after the restart required to finish the operation). This parameter can't be accompanied by any other parameters.

Parameter	Description
-backupTarget	Specifies the storage location with the backup(s) you want to recover. This parameter is useful when the storage location is different from where backups are usually stored.
-machine	Specifies the name of the computer to recover the backup for. This parameter must be used when the **-backupTarget** parameter is specified. The **-machine** parameter is useful when multiple computers have been backed up to the same location.
-recoveryTarget	Specifies what directory to restore to. This parameter is useful if the backup is restored to an alternate location.
-authsysvol	Performs an authoritative restore of the System Volume (sysvol) shared directory.
-autoReboot	Specifies to restart the system at the end of the system state recovery operation. This parameter is valid only for a recovery to the original location. We don't recommend you use this parameter if you need to perform steps after the recovery operation.
-quiet	Runs the command without prompts to the user.

Examples

To start a system state recovery of the backup from 03/31/2020 at 9:00 A.M., type:

```
wbadmin start systemstaterecovery -version:03/31/2020-09:00
```

To start a system state recovery of the backup from 04/30/2020 at 9:00 A.M. that is stored on the shared resource `\\servername\share` for server01, type:

```
wbadmin start systemstaterecovery -version:04/30/2013-09:00 -
backupTarget:\\servername\share -machine:server01
```

Related links

- Command-Line Syntax Key
- wbadmin command

- Start-WBSystemStateRecovery

Feedback

Was this page helpful? Yes No

wbadmin stop job

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Cancels the backup or recovery operation that is currently running.

> **ⓘ Important**
>
> Canceled operations can't be restarted. You must run a canceled backup or a recovery operation from the beginning again.

To stop a backup or recovery operation using this command, you must be a member of the **Backup Operators** group or the **Administrators** group, or you must have been delegated the appropriate permissions. In addition, you must run **wbadmin** from an elevated command prompt, by right-clicking **Command Prompt**, and then selecting **Run as administrator**.

Syntax

```
wbadmin stop job [-quiet]
```

Parameters

Expand table

Parameter	Description
-quiet	Runs the command without prompts to the user.

Related links

- Command-Line Syntax Key
- wbadmin command

Feedback

Was this page helpful? Yes No

wdsutil

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Wdsutil is a command-line utility used for managing your Windows Deployment Services server. To run these commands, click **start**, right-click **Command prompt**, and click **Run as administrator**.

Commands

Expand table

Command	Description
wdsutil add command	Adds objects or prestages computers.
wdsutil approve-autoadddevices command	Approves computers that are pending administrator approval.
wdsutil convert-riprepimage command	Converts an existing remote Installation Preparation (RIPrep) image to a Windows Image (.wim) file.
wdsutil copy command	Copies an image or a driver group.
wdsutil delete-autoadddevices command	Deletes computers that are in the Auto-add database (which stores information about the computers on the server).
wdsutil disable command	Disables all services for Windows Deployment Services.
wdsutil disconnect-client command	Disconnects a client from a multicast transmission or namespace.
wdsutil enable command	Enables all services for Windows Deployment Services.
wdsutil export-image command	Exports an image from the image store to a .wim file.
wdsutil get command	Retrieves properties and attributes about the specified object.
wdsutil initialize-server command	Configures a Windows Deployment Services server for initial use.
wdsutil new command	creates new capture and discover images as well as multicast transmissions and namespaces.

Command	Description
wdsutil progress command	Displays the progress status while a command is being executed.
wdsutil reject-autoadddevices command	Rejects computers that are pending administrator approval.
wdsutil remove command	removes objects.
wdsutil replace-image command	replaces a boot or installation image with a new version of that image.
wdsutil set command	Sets properties and attributes on the specified object.
wdsutil start server command	starts all services on the Windows Deployment Services server, including multicast transmissions, namespaces, and the Transport Server.
wdsutil stop server command	Stops all services on the Windows Deployment Services server.
wdsutil uninitialize-server command	reverts changes made during server initialization.
wdsutil update-serverfiles command	Updates server files on the remoteInstall share.
wdsutil verbose command	Displays verbose output for the specified command.

Feedback

Was this page helpful? Yes No

wdsutil add commands

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Parameters

Expand table

Parameter	Description
wdsutil add-device command	Pre-stages a computer in active directory.
wdsutil add-image command	Adds boot or installation images.
wdsutil add-imagegroup command	Adds an image group.
wdsutil add-drivergrouppackage command	Adds a driver package to a driver group.
wdsutil add-drivergrouppackages command	Adds driver packages to a driver group.
wdsutil add-driverpackage command	Adds a driver package to the server.
wdsutil add-imagedriverpackage command	Adds a driver package that is in the driver store to an existing boot image on the server.
wdsutil add-imagedriverpackages command	Adds driver packages from the driver store to a boot image on the server.
wdsutil add-alldriverpackages subcommand	Adds driver packages from a folder to a server.
wdsutil add-drivergroup command	Adds a driver group to a server.
wdsutil add-drivergroupfilter command	Adds a filter to a driver group on a server.

Related links

- Command-Line Syntax Key

- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-alldriverpackages

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds all driver packages that are stored in a folder to a server.

Syntax

```
wdsutil /Add-AllDriverPackages /FolderPath:<folderpath> [/Server:
<servername>] [/Architecture:{x86 | ia64 | x64}] [/DriverGroup:<groupname>]
```

Parameters

Expand table

Parameter	Description
/FolderPath: `<folderpath>`	Specifies the full path to the folder that contains the .inf files for the driver packages.
[/Server: `<servername>`]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
[/Architecture: `{x86|ia64|x64}`]	Specifies the architecture type for the driver package.
[/DriverGroup: `<groupname>`]	Specifies the name of the driver group to which the packages should be added.

Examples

To add driver packages, type either:

```
wdsutil /verbose /Add-AllDriverPackages /FolderPath:C:\Temp\Drivers
/Architecture:x86
```

```
wdsutil /Add-AllDriverPackages /FolderPath:C:\Temp\Drivers\Printers
/DriverGroup:Printer Drivers
```

Related links

- Command-Line Syntax Key
- Windows Deployment Services cmdlets
- Add-WdsDriverPackage

Feedback

Was this page helpful? Yes No

wdsutil add-device

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Pre-stages a computer in Active Directory Domain Services (AD DS). Pre-staged computers are also called *known computers*. This allows you to configure properties to control the installation for the client. For example, you can configure the network boot program and the unattend file that the client should receive, as well as the server from which the client should download the network boot program.

Syntax

```
wdsutil /add-Device /Device:<Devicename> /ID:<UUID | MAC address>
[/ReferralServer:<Servername>] [/BootProgram:<Relativepath>]
[/WdsClientUnattend:<Relativepath>] [/User:<Domain\User | User@Domain>]
[/JoinRights:{JoinOnly | Full}] [/JoinDomain:{Yes | No}] [/BootImagepath:
<Relativepath>] [/OU:<DN of OU>] [/Domain:<Domain>]
```

Parameters

Expand table

Parameter	Description
/Device:`<Devicename>`	Specifies the name of the device to be added.
/ID:`<UUID|MAC address>`	Specifies either the GUID/UUID or the MAC address of the computer. A GUID/UUID must be in one of two formats: Binary string (`/ID:ACEFA3E81F20694E953EB2DAA1E8B1B6`) or GUID string (`/ID:E8A3EFAC-201F-4E69-953E-B2DAA1E8B1B6`). A MAC address must be in the following format: **00B056882FDC** (no dashes) or **00-B0-56-88-2F-DC** (with dashes)
[/ReferralServer: `<Servername>`]	Specifies the name of the server to be contacted to download the network boot program and the boot image by using Trivial File Transfer Protocol (tftp).
[/BootProgram: `<Relativepath>`]	Specifies the relative path from the **remoteInstall** folder to the network boot program that this computer should receive. For

Parameter	Description
	example: `boot\x86\pxeboot.com`
[/WdsClientUnattend: `<Relativepath>`]	Specifies the relative path from the **remoteInstall** folder to the unattended installation file that automates the installation screens of the Windows Deployment Services client.
[/User: `<Domain\User|User@Domain>`]	Sets permissions on the computer account object to give the specified user the necessary rights to join the computer to the domain.
[/JoinRights: `{JoinOnly|Full}`]	Specifies the type of rights to be assigned to the user. • **JoinOnly** - Requires the administrator to reset the computer account before the user can join the computer to the domain. • **Full** - Gives full access to the user, which includes the right to join the computer to the domain.
[/JoinDomain: `{Yes|No}`]	Specifies whether the computer should be joined to the domain as this computer account during operating system installation. The default value is **Yes**.
[/BootImagepath: `<Relativepath>`]	Specifies the relative path from the **remoteInstall** folder to the boot image that this computer should use.
[/OU: `<DN of OU>`]	The distinguished name of the organizational unit where the computer account object should be created. For example: **OU=MyOU,CN=Test, DC=Domain,DC=com**. The default location is the default computer's container.
[/Domain: `<Domain>`]	The domain where the computer account object should be created. The default location is the local domain.

Examples

To add a computer by using a MAC address, type:

```
wdsutil /add-Device /Device:computer1 /ID:00-B0-56-88-2F-DC
```

To add a computer by using a GUID string, type:

```
wdsutil /add-Device /Device:computer1 /ID:{E8A3EFAC-201F-4E69-953F-
B2DAA1E8B1B6} /ReferralServer:WDSServer1
```

```
/BootProgram:boot\x86\pxeboot.com/WDSClientUnattend:WDSClientUnattend\unatte
nd.xml /User:Domain\MyUser/JoinRights:Full
/BootImagepath:boot\x86\images\boot.wim /OU:OU=MyOU,CN=Test,DC=Domain,DC=com
```

Related links

- Command-Line Syntax Key
- wdsutil get-alldevices command
- wdsutil get-device command
- wdsutil set-device command
- Windows Deployment Services cmdlets
- New-WdsClient

Feedback

Was this page helpful? Yes No

wdsutil add-drivergroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a driver group to the server.

Syntax

```
wdsutil /add-DriverGroup /DriverGroup:<Groupname>\n\ [/Server:<Servername>]
[/Enabled:{Yes | No}] [/Applicability:{Matched | All}] [/Filtertype:
<Filtertype> /Policy:{Include | Exclude} /Value:<Value> [/Value:<Value>
...]]
```

Parameters

Expand table

Parameter	Description
/DriverGroup: `<Groupname>`	Specifies the name of the new driver group.
/Server: `<Servername>`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
/Enabled: `{Yes|No}`	Enables or disables the package.
/Applicability: `{Matched|All}`	Specifies which packages to install if the filter criteria are met. **Matched** means install only the driver packages that match a client s hardware. **All** means install all the packages to clients regardless of their hardware.
/Filtertype: `<Filtertype>`	Specifies the type of the filter to add to the group. You can specify multiple filter types in a single command. Each filter type must be followed by **/Policy** and at least one **/Value**. Valid values include: • BiosVendor • Biosversion • Chassistype • Manufacturer • Uuid • Osversion

Parameter	Description
	• Osedition • OsLanguage For information about getting values for all other filter types, see Driver Group Filters.
[/Policy: `{Include|Exclude}`]	Specifies the policy to be set on the filter. If **/Policy** is set to **Include**, client computers that match the filter are allowed to install the drivers in this group. If **/Policy** is set to **Exclude**, then client computers that match the filter are not allowed to install the drivers in this group.
[/Value: `<Value>`]	Specifies the client value that corresponds to **/Filtertype**. You can specify multiple values for a single type. For information about acceptable filter type values, see Driver Group Filters.

Examples

To add a driver group, type either:

```
wdsutil /add-DriverGroup /DriverGroup:printerdrivers /Enabled:Yes
```

```
wdsutil /add-DriverGroup /DriverGroup:printerdrivers /Applicability:All
/Filtertype:Manufacturer /Policy:Include /Value:Name1
/Filtertype:Chassistype /Policy:Exclude /Value:Tower /Value:MiniTower
```

Related links

- Command-Line Syntax Key
- wdsutil add-drivergrouppackage command
- wdsutil add-drivergrouppackages command
- wdsutil add-drivergroupfilter command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

add-DriverGroupFilter

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a filter to a driver group on a server.

Syntax

```
wdsutil /Add-DriverGroupFilter /DriverGroup:<Group Name> [/Server:<Server name>] /FilterType:<Filter Type> /Policy:{Include | Exclude} /Value:<Value> [/Value:<Value> ...]
```

Parameters

Expand table

Parameter	Description
/DriverGroup: `<Groupname>`	Specifies the name of the new driver group.
/Server: `<Servername>`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
/Filtertype: `<Filtertype>`	Specifies the type of the filter to add to the group. You can specify multiple filter types in a single command. Each filter type must be followed by **/Policy** and at least one **/Value**. Valid values include: • BiosVendor • Biosversion • Chassistype • Manufacturer • Uuid • Osversion • Osedition • OsLanguage For information about getting values for all other filter types, see Driver Group Filters.

Parameter	Description
[/Policy: `{Include|Exclude}`]	Specifies the policy to be set on the filter. If **/Policy** is set to **Include**, client computers that match the filter are allowed to install the drivers in this group. If **/Policy** is set to **Exclude**, then client computers that match the filter are not allowed to install the drivers in this group.
[/Value: `<Value>`]	Specifies the client value that corresponds to **/Filtertype**. You can specify multiple values for a single type. For information about acceptable filter type values, see Driver Group Filters.

Examples

To add a filter to a driver group, type either:

```
wdsutil /Add-DriverGroupFilter /DriverGroup:PrinterDrivers
/FilterType:Manufacturer /Policy:Include /Value:Name1 /Value:Name2
```

```
wdsutil /Add-DriverGroupFilter /DriverGroup:PrinterDrivers
/FilterType:Manufacturer /Policy:Include /Value:Name1
/FilterType:ChassisType /Policy:Exclude /Value:Tower /Value:MiniTower
```

Related links

- Command-Line Syntax Key
- wdsutil add-drivergrouppackage command
- wdsutil add-drivergrouppackages command
- wdsutil add-drivergroup command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-drivergrouppackage

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a driver package to a driver group.

Syntax

```
wdsutil /add-DriverGroupPackage /DriverGroup:<Group Name> [/Server:<Server Name>] {/DriverPackage:<Name> | /PackageId:<ID>}
```

Parameters

Expand table

Parameter	Description
/DriverGroup: `<Groupname>`	Specifies the name of the new driver group.
/Server: `<Servername>`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
/DriverPackage: `<Name>`	Specifies the name of the driver package to be added to the group. You must specify this option if the driver package cannot be uniquely identified by name.
/PackageId: `<ID>`	Specifies the ID for a package. To find the Package ID, select the driver group that the package is in (or the **All Packages** node), right-click the package, and then select **Properties**. The Package ID is listed on the **General** tab, for example: **{DD098D20-1850-4fc8-8E35-EA24A1BEFF5E}**.

Examples

To add a driver group package, type either:

```
wdsutil /add-DriverGroupPackage /DriverGroup:printerdrivers /PackageId:
{4D36E972-E325-11CE-Bfc1-08002BE10318}
```

```
wdsutil /add-DriverGroupPackage /DriverGroup:printerdrivers
/DriverPackage:XYZ
```

Related links

- Command-Line Syntax Key
- wdsutil add-drivergroupfilter command
- wdsutil add-drivergrouppackages command
- wdsutil add-drivergroup command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-drivergrouppackages

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds driver group packages.

Syntax

```
wdsutil /add-DriverGroupPackages /DriverGroup:<Group Name> [/Server:<Server Name>] /Filtertype:<Filter type> /Operator:{Equal | NotEqual | GreaterOrEqual | LessOrEqual | Contains} /Value:<Value> [/Value:<Value>]
```

Parameters

Expand table

Parameter	Description
/DriverGroup:`<Groupname>`	Specifies the name of the new driver group.
/Server:`<Servername>`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
/Filtertype:`<Filtertype>`	Specifies the type of the driver package to search for. You can specify multiple attributes in a single command. You must also specify **/Operator** and **/Value** with this option. Valid values include: • PackageId • PackageName • PackageEnabled • Packagedateadded • PackageInfFilename • PackageClass PackageProvider

Parameter	Description
	• PackageArchitecture • PackageLocale • PackageSigned • PackagedatePublished • Packageversion • Driverdescription • DriverManufacturer • DriverHardwareId • DrivercompatibleId • DriverExcludeId • DriverGroupId • DriverGroupName** .
/Operator: `{Equal|NotEqual|GreaterOrEqual|LessOrEqual|Contains}`	Specifies the relationship between the attribute and the values. You can only specify **Contains** with string attributes. You can only specify **Equal**, **NotEqual**, **GreaterOrEqual** and **LessOrEqual** with date and version attributes.
/Value: `<Value>`	Specifies the client value corresponding to **/Filtertype**. You can specify multiple values for a single **/Filtertype**. The available values for each filter are: • **PackageId** - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318} • **PackageName** - Specify any string value • **PackageEnabled** - Specify **Yes** or **No** • **Packagedateadded** - Specify the date in the following format: YYYY/MM/DD • **PackageInfFilename** - Specify any string value • **PackageClass** - Specify a valid class name or class GUID. For example: **DiskDrive**, **Net**, or {4d36e972-e325-11ce-bfc1-08002be10318} • **PackageProvider** - Specify any string value • **PackageArchitecture** - Specify **x86**, **x64**, or **ia64**

Parameter	Description
	• **PackageLocale** - Specify a valid language identifier. For example: **en-US** or **es-ES** • **PackageSigned** - Specify **Yes** or **No** • **PackagedatePublished** - Specify the date in the following format: YYYY/MM/DD • **Packageversion** - Specify the version in the following format: a.b.x.y. For example: 6.1.0.0 • **Driverdescription** - Specify any string value • **DriverManufacturer** - Specify any string value • **DriverHardwareId** - Specify any string value • **DrivercompatibleId** - Specify any string value • **DriverExcludeId** - Specify any string value • **DriverGroupId** - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318} • **DriverGroupName** - Specify any string value For more information about these values, see Driver and Package attributes.

Examples

To add a driver group package, type either:

```
wdsutil /verbose /add-DriverGroupPackages /DriverGroup:printerdrivers
/Filtertype:PackageClass /Operator:Equal /Value:printer
/Filtertype:DriverManufacturer /Operator:NotEqual /Value:Name1 /Value:Name2
```

```
wdsutil /verbose /add-DriverGroupPackages /DriverGroup:DisplayDriversX86
/Filtertype:PackageClass /Operator:Equal /Value:Display
```

```
/Filtertype:PackageArchitecture /Operator:Equal /Value:x86
/Filtertype:Packagedateadded /Operator:LessOrEqual /Value:2008/01/01
```

Related links

- Command-Line Syntax Key
- wdsutil add-driverpackage command
- wdsutil add-drivergrouppackage command
- wdsutil add-alldriverpackages command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

add-DriverPackage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a driver package to the server.

Syntax

```
wdsutil /Add-DriverPackage /InfFile:<Inf File path> [/Server:<Server name>]
[/Architecture:{x86 | ia64 | x64}] [/DriverGroup:<Group Name>] [/Name:
<Friendly Name>]
```

Parameters

Expand table

Parameter	Description
/InfFile:`<InfFilepath>`	Specifies the full path of the .inf file to add.
[/Server:`<Servername>`]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
[/Architecture:`{x86 | ia64 | x64}`]	Specifies the architecture type for the driver package.
[/DriverGroup:`<groupname>`]	Specifies the name of the driver group to which the packages should be added.
[/Name:`<friendlyname>`]	Specifies the friendly name for the driver package.

Examples

To add a driver package, type either:

```
wdsutil /verbose /Add-DriverPackage /InfFile:C:\Temp\Display.inf
```

```
wdsutil /Add-DriverPackage /Server:MyWDSServer /InfFile:C:\Temp\Display.inf
/Architecture:x86 /DriverGroup:x86Drivers /Name:Display Driver
```

Related links

- Command-Line Syntax Key
- wdsutil add-drivergrouppackage command
- wdsutil add-alldriverpackages command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds images to a Windows Deployment Services server.

Syntax

For boot images, use the following syntax:

```
wdsutil /Add-Image /ImageFile:<wim file path> [/Server:<Server name>
/ImageType:Boot [/SkipVerify] [/Name:<Image name>] [/Description:<Image
description>] [/Filename:<New wim file name>]
```

For install images, use the following syntax:

```
wdsutil /Add-Image /ImageFile:<wim filepath> [/Server:<Servername>]
/ImageType:Install [/SkipVerify] /ImageGroup:<Image group name>]
[/SingleImage:<Single image name>] [/Name:<Name>] [/Description:
<Description>] [/Filename:<File name>] [/UnattendFile:<Unattend file path>]
```

Parameters

Expand table

Parameter	Description
/ImageFile:`<.wim filepath>`	Specifies the full path and file name of the Windows Image (.wim) file that contains the images to be added.
[/Server:`<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If a server name is not specified, the local server is used.
[/ImageType:`{Boot | Install}`]	Specifies the type of images to be added.

Parameter	Description
[/SkipVerify]	Specifies that integrity verification will not be performed on the source image file before the image is added.
[/Name:`<Name>`]	Sets the display name of the image.
[/Description:`<Description>`]	Sets the description of the image.
[/Filename:`<Filename>`]	Specifies the new file name for the .wim file. This enables you to change the filename of the .wim file when adding the image. If you don't specify a filename, the source image filename is used. In all cases, Windows Deployment Services checks to determine whether the file name is unique in the boot image store of the destination computer.
/ImageGroup: `<Imagegroupname>`]	Specifies the name of the image group in which the images are to be added. If more than one image group exists on the server, the image group must be specified. If you don't specify the image group, and an image group doesn't already exist, a new image group is created. Otherwise, the existing image group is used.
[/SingleImage: `<Singleimagename>`] [/Name: `<Name>`] [/Description: `<Description>`]	Copies the specified single image out of a .wim file, and sets the image's display name and description.
[/UnattendFile: `<Unattendfilepath>`]	Specifies the full path to the unattended installation file to be associated with the images that are being added. If **/SingleImage** isn't specified, the same unattend file is associated with all of the images in the .wim file.

Examples

To add a boot image, type:

```
wdsutil /Add-Image /ImageFile:"C:\MyFolder\Boot.wim" /ImageType:Boot
wdsutil /Verbose /Progress /Add-Image /ImageFile:\\MyServer\Share\Boot.wim
/Server:MyWDSServer /ImageType:Boot /Name:"My WinPE Image"
/Description:"WinPE Image containing the WDS Client" /Filename:WDSBoot.wim
```

To add an install image, type one of the following:

```
wdsutil /Add-Image /ImageFile:"C:\MyFolder\Install.wim" /ImageType:Install
wdsutil /Verbose /Progress /Add-Image
/ImageFile:\\MyServer\Share\Install.wim /Server:MyWDSServer
/ImageType:Install /ImageGroup:ImageGroup1
/SingleImage:"Windows Pro" /Name:"My WDS Image" /Description:"Windows Pro
image with Microsoft Office" /Filename:"Win Pro.wim"
/UnattendFile:"\\server\share\unattend.xml"
```

Related links

- Command-Line Syntax Key
- wdsutil copy-image command
- wdsutil export-image command
- wdsutil get-image command
- wdsutil remove-image command
- wdsutil replace-image command
- wdsutil set-image command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-imagedriverpackage

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds a driver package that is in the driver store to an existing boot image on the server.

Syntax

```
wdsutil /add-ImageDriverPackage [/Server:<Servername>] [media:<Imagename>]
[mediatype:Boot] [/Architecture:{x86 | ia64 | x64}] [/Filename:<Filename>]
{/DriverPackage:<Package Name> | /PackageId:<ID>}
```

Parameters

Expand table

Parameter	Description
[/Server:`<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If a server name is not specified, the local server is used.
[media:`<Imagename>`]	Specifies the name of the image to add the driver to.
[mediatype:Boot]	Specifies the type of image to add the driver to. Driver packages can only be added to boot images.
[/Architecture:`{x86 | ia64 | x64}`]	Specifies the architecture of the boot image. Because it's possible to have the same image name for boot images in different architectures, you should specify the architecture to ensure the correct image is used.
[/Filename:`<Filename>`]	Specifies the name of the file. If the image cannot be uniquely identified by name, the file name must be specified.
[/DriverPackage:`<Name>`	Specifies the name of the driver package to add to the image.
[/PackageId:`<ID>`]	Specifies the Windows Deployment Services ID of the driver package. You must specify this option if the driver package can't be uniquely identified by name. To find the Package ID, select the driver group that the package is

Parameter	Description
	in (or the **All Packages** node), right-click the package, and then select **Properties**. The Package ID is listed on the **General** tab. For example: {DD098D20-1850-4fc8-8E35-EA24A1BEFF5E}.

Examples

To add a driver package to a boot image, type either:

```
wdsutil /add-ImageDriverPackagmedia:WinPE Boot Imagemediatype:Boot
/Architecture:x86 /DriverPackage:XYZ
```

```
wdsutil /verbose /add-ImageDriverPackagmedia:WinPE Boot Image
/Server:MyWDSServemediatype:Boot /Architecture:x64 /PackageId:{4D36E972-
E325-11CE-Bfc1-08002BE10318}
```

Related links

- Command-Line Syntax Key
- wdsutil add-imagedriverpackages command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-imagedriverpackages

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds driver packages from the driver store to a boot image.

Syntax

```
wdsutil /add-ImageDriverPackages [/Server:<Server name>media:<Image
namemediatype:Boot /Architecture:{x86 | ia64 | x64} [/Filename:<File name>]
/Filtertype:<Filter type> /Operator:{Equal | NotEqual | GreaterOrEqual |
LessOrEqual | Contains} /Value:<Value> [/Value:<Value> ...]
```

Parameters

Expand table

Parameter	Description
[/Server:`<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If a server name is not specified, the local server is used.
[media:`<Imagename>`]	Specifies the name of the image to add the driver to.
[mediatype:Boot]	Specifies the type of image to add the driver to. Driver packages can only be added to boot images.
[/Architecture:`{x86 | ia64 | x64}`]	Specifies the architecture of the boot image. Because it's possible to have the same image name for boot images in different architectures, you should specify the architecture to ensure the correct image is used.
[/Filename:`<Filename>`]	Specifies the name of the file. If the image cannot be uniquely identified by

Parameter	Description
	name, the file name must be specified.
/Filtertype:`<Filtertype>`	Specifies the attribute of the driver package to search for. You can specify multiple attributes in a single command. You must also specify **/Operator** and **/Value** with this option. Valid values include: • PackageId • PackageName • PackageEnabled • Packagedateadded • PackageInfFilename • PackageClass PackageProvider • PackageArchitecture • PackageLocale • PackageSigned • PackagedatePublished • Packageversion • Driverdescription • DriverManufacturer • DriverHardwareId • DrivercompatibleId • DriverExcludeId • DriverGroupId • DriverGroupName** .
/Operator: `{Equal|NotEqual|GreaterOrEqual|LessOrEqual|Contains}`	Specifies the relationship between the attribute and the values. You can only specify **Contains** with string attributes. You can only specify **GreaterOrEqual** and **LessOrEqual** with date and version attributes.
/Value:`<Value>`	Specifies the value to search for relative to the specified `<attribute>`. You can specify multiple values for a single **/Filtertype**. The available values for each filter are: • **PackageId** - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318} • **PackageName** - Specify any string value

Parameter	Description
	• **PackageEnabled** - Specify **Yes** or **No** • **Packagedateadded** - Specify the date in the following format: YYYY/MM/DD • **PackageInfFilename** - Specify any string value • **PackageClass** - Specify a valid class name or class GUID. For example: **DiskDrive**, **Net**, or {4d36e972-e325-11ce-bfc1-08002be10318} • **PackageProvider** - Specify any string value • **PackageArchitecture** - Specify **x86**, **x64**, or **ia64** • **PackageLocale** - Specify a valid language identifier. For example: **en-US** or **es-ES** • **PackageSigned** - Specify **Yes** or **No** • **PackagedatePublished** - Specify the date in the following format: YYYY/MM/DD • **Packageversion** - Specify the version in the following format: a.b.x.y. For example: 6.1.0.0 • **Driverdescription** - Specify any string value • **DriverManufacturer** - Specify any string value • **DriverHardwareId** - Specify any string value • **DrivercompatibleId** - Specify any string value • **DriverExcludeId** - Specify any string value • **DriverGroupId** - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318} • **DriverGroupName** - Specify any string value For more information about these values, see Driver and Package attributes.

Examples

To add driver packages to a boot image, type one of the following:

```
wdsutil /add-ImageDriverPackagemedia:WinPE Boot Imagemediatype:Boot
/Architecture:x86 /Filtertype:DriverGroupName /Operator:Equal /Value:x86Bus
/Filtertype:PackageProvider /Operator:Contains /Value:Provider1
/Filtertype:Packageversion /Operator:GreaterOrEqual /Value:6.1.0.0
```

```
wdsutil /verbose /add-ImageDriverPackagemedia: WinPE Boot Image
/Server:MyWDSServemediatype:Boot /Architecture:x64 /Filtertype:PackageClass
/Operator:Equal /Value:Net /Filtertype:DriverManufacturer /Operator:NotEqual
/Value:Name1 /Value:Name2 /Filtertype:Packagedateadded /Operator:LessOrEqual
/Value:2008/01/01
```

```
wdsutil /verbose /add-ImageDriverPackagemedia:WinPE Boot Image
/Server:MyWDSServemediatype:Boot /Architecture:x64 /Filtertype:PackageClass
/Operator:Equal /Value:Net /Value:System /Value:DiskDrive /Value:HDC
/Value:SCSIAdapter
```

Related links

- Command-Line Syntax Key
-
- wdsutil add-imagedriverpackage command
-
- wdsutil add-alldriverpackages command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil add-imagegroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds an image group to a Windows Deployment Services server.

Syntax

```
wdsutil [Options] /add-ImageGroup imageGroup:<Imagegroupname> [/Server:
<Server name>]
```

Parameters

Expand table

Parameter	Description
imageGroup: `<Imagegroupname>`]	Specifies the name of the image to be added.
[/Server: `<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If a server name is not specified, the local server is used.

Examples

To add an image group, type either:

```
wdsutil /add-ImageGroup imageGroup:ImageGroup2
```

```
wdsutil /verbose /add-Imagegroup imageGroup:My Image Group
/Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil get-allimagegroups command
- wdsutil get-imagegroup command
- wdsutil remove-imagegroup command
- wdsutil set-imagegroup command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil approve-autoadddevices

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Approves computers that are pending administrative approval. When the Auto-add policy is enabled, administrative approval is required before unknown computers (those that are not pre-staged) can install an image. You can enable this policy using the **PXE Response** tab of the server s properties page.

Syntax

```
wdsutil [Options] /Approve-AutoaddDevices [/Server:<Server name>]
/RequestId:{<Request ID>| ALL} [/MachineName:<Device name>] [/OU:<DN of OU>]
[/User:<Domain\User | User@Domain>] [/JoinRights:{JoinOnly | Full}]
[/JoinDomain:{Yes | No}] [/ReferralServer:<Server name>] [/BootProgram:
<Relative path>] [/WdsClientUnattend:<Relative path>] [/BootImagepath:
<Relative path>]
```

Parameters

Expand table

Parameter	Description
/Server:`<Servername>`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
/RequestId:`{Request ID|ALL}`	Specifies the request ID assigned to the pending computer. Specify **ALL** to approve all pending computers.
/Machinename:`<Devicename>`	Specifies the name of the device to be added. You can't use this option when approving all computers.
[/OU:`<DN of OU>`]	The distinguished name of the organizational unit where the computer account object should be created. For example: **OU=MyOU,CN=Test, DC=Domain,DC=com**. The default location is the default computer's container.

Parameter	Description
[/User:`<Domain\User|User@Domain>`]	Sets permissions on the computer account object to give the specified user the necessary rights to join the computer to the domain.
[/JoinRights:`{JoinOnly|Full}`]	Specifies the type of rights to be assigned to the user. • **JoinOnly** - Requires the administrator to reset the computer account before the user can join the computer to the domain. • **Full** - Gives full access to the user, which includes the right to join the computer to the domain.
[/JoinDomain:`{Yes|No}`]	Specifies whether the computer should be joined to the domain as this computer account during operating system installation. The default value is **Yes**.
[/ReferralServer:`<Servername>`]	Specifies the name of the server to contact to download the network boot program and boot image by using Trivial File Transfer Protocol (tftp).
[/BootProgram:`<Relativepath>`]	Specifies the relative path from the **remoteInstall** folder to the network boot program that this computer should receive. For example: **boot\x86\pxeboot.com**.
[/WdsClientUnattend:`<Relativepath>`]	Specifies the relative path from the **remoteInstall** folder to the unattend file that automates the Windows Deployment Services client.
[/BootImagepath:`<Relativepath>`]	Specifies the relative path from the remoteInstall folder to the boot image that this computer should receive.

Examples

To approve the computer with a RequestId of 12, type:

```
wdsutil /Approve-AutoaddDevices /RequestId:12
```

To approve the computer with a RequestID of 20 and to deploy the image with the specified settings, type:

```
wdsutil /Approve-AutoaddDevices /RequestId:20 /MachineName:computer1
/OU:OU=Test,CN=company,DC=Domain,DC=Com /User:Domain\User1
```

```
/JoinRights:Full /ReferralServer:MyWDSServer
/BootProgram:boot\x86\pxeboot.n12
/WdsClientUnattend:WDSClientUnattend\Unattend.xml
/BootImagepath:boot\x86\images\boot.wim
```

To approve all pending computers, type:

```
wdsutil /verbose /Approve-AutoaddDevices /RequestId:ALL
```

Related links

- Command-Line Syntax Key
- wdsutil delete-autoadddevices command
- wdsutil get-autoadddevices command
- wdsutil reject-autoadddevices command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

convert-riprepimage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Converts an existing Remote Installation Preparation (RIPrep) image to Windows Image (.wim) format.

Syntax

```
wdsutil [Options] /Convert-RIPrepImage /FilePath:<Filepath and name>
/DestinationImage /FilePath:<Filepath and name> [/Name:<Name>]
[/Description:<Description>] [/InPlace] [/Overwrite:{Yes | No | Append}]
```

Parameters

Expand table

Parameter	Description
/FilePath:`<Filepath and name>`	Specifies the full filepath and name of the .sif file that corresponds to the RIPrep image. This file is typically called Riprep.sif and is found in the **\Templates** subfolder of the folder that contains the RIPrep image.
/DestinationImage	Specifies the settings for the destination image. Uses the following options; • `/FilePath:<Filepath and name>` - Sets the full file path for the new file. For example: **C:\Temp\convert.wim** • [`/Name:<Name>`] - Sets the display name of the image. If no display name is specified, the display name of the source image is used. • [`/Description:<Description>`] - Sets the description of the image. • [/InPlace] - Specifies that the conversion should take place on the original RIPrep image and not on a copy of the original image, which is the default behavior. • [`/Overwrite:{Yes | No | Append}` - Sets whether this image should overwrite or append any existing files.

Examples

To convert the specified RIPrep.sif image to RIPREP.wim, type:

```
wdsutil /Convert-RiPrepImage /FilePath:R:\RemoteInstall\Setup\English
\Images\Win2k3.SP1\i386\Templates\riprep.sif /DestinationImage
/FilePath:C:\Temp\RIPREP.wim
```

To convert the specified RIPrep.sif image to RIPREP.wim with the specified name and description, and overwrite it with the new file if a file already exists, type:

```
wdsutil /Verbose /Progress /Convert-RiPrepImage /FilePath:\\Server
\RemInst\Setup\English\Images\WinXP.SP2\i386\Templates\riprep.sif
/DestinationImage /FilePath:\\Server\Share\RIPREP.wim /Name:WindowsXP image
/Description:Converted RIPREP image of WindowsXP /Overwrite:Append
```

Related links

- Command-Line Syntax Key
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil copy commands

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies an image or a driver group.

Parameters

Expand table

Parameter	Description
wdsutil copy-image command	Copies images that are within the same image group.
wdsutil copy-drivergroup command	Copies an existing driver group on the server.

Related links

- Command-Line Syntax Key
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

copy-drivergroup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Duplicates an existing driver group on the server including the filters, driver packages, and enabled/disabled status.

Syntax

```
wdsutil /Copy-DriverGroup [/Server:<Server name>] /DriverGroup:<Source Groupname> /GroupName:<New Groupname>
```

Parameters

Expand table

Parameter	Description
/Server:`<Servername>`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
/DriverGroup:`<Source Groupname>`	Specifies the name of the source driver group.
/GroupName:`<New Groupname>`	Specifies the name of the new driver group.

Examples

To copy a driver group, type either:

```
wdsutil /Copy-DriverGroup /Server:MyWdsServer /DriverGroup:PrinterDrivers /GroupName:X86PrinterDrivers
```

```
```

```
wdsutil /Copy-DriverGroup /DriverGroup:PrinterDrivers
/GroupName:ColorPrinterDrivers
```

Related links

- Command-Line Syntax Key
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil copy-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Copies images that are within the same image group. To copy images between image groups, use the wdsutil Export-Image command command and then the wdsutil add-Image command command.

Syntax

```
wdsutil [Options] /copy-Image image:<Image name> [/Server:<Server name>]
imagetype:Install imageGroup:<Image group name>] [/Filename:<File name>]
/DestinationImage /Name:<Name> /Filename:<File name> [/Description:
<Description>]
```

Parameters

Expand table

Parameter	Description
image:`<Imagename>`	Specifies the name of the image to be copied.
[/Server:`<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
imagetype:Install	Specifies the type of image to be copied. This option must be set to **install**.
\imageGroup:`<Image groupname>`]	Specifies the image group that contains the image to be copied. If no image group is specified and only one group exists on the server, that image group is used by default. If more than one image group exists on the server, you must specify the image group.
[/Filename:`<Filename>`]	Specifies the file name of the image to be copied. If the source image cannot be uniquely identified by name, you must specify the file name.
/DestinationImage	Specifies the settings for the destination image. The valid values are: • /Name:`<Name>` - Sets the display name of the image to be copied.

Parameter	Description
	• /Filename:`<Filename>` - Sets the name of the destination image file that will contain the image copy. • [/Description:`<Description>`] - Sets the description of the image copy.

Examples

To create a copy of the specified image and name it WindowsVista.wim, type:

```
wdsutil /copy-Image image:Windows Vista with Office imagetype:Install
/DestinationImage /Name:copy of Windows Vista with Office /
Filename:WindowsVista.wim
```

To create a copy of the specified image, apply the specified settings, and name the copy WindowsVista.wim, type:

```
wdsutil /verbose /Progress /copy-Image image:Windows Vista with Office
/Server:MyWDSServe imagetype:Install imageGroup:ImageGroup1
/Filename:install.wim /DestinationImage /Name:copy of Windows Vista with
Office /Filename:WindowsVista.wim /Description:This is a copy of the
original Windows image with Office installed
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil export-image command
- wdsutil get-image command
- wdsutil remove-image command
- wdsutil replace-image command
- wdsutil set-image command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil delete-autoadddevices

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Deletes computers that are pending, rejected, or approved from the auto-add database. This database stores information about these computers on the server.

Syntax

```
wdsutil /delete-AutoaddDevices [/Server:<Servername>] /Devicetype:
{PendingDevices | RejectedDevices |ApprovedDevices}
```

Parameters

Expand table

Parameter	Description
[/Server:`<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/Devicetype: `{PendingDevices|RejectedDevices|ApprovedDevices}`	Specifies the type of computer to delete from the database. This type can be **PendingDevices**, which returns all computers in the database that have a status of pending, **RejectedDevices**, which returns all computers in the database that have a status of rejected, or **ApprovedDevices**, which returns all computers that have a status of approved.

Examples

To delete all rejected computers, type:

```
wdsutil /delete-AutoaddDevices /Devicetype:RejectedDevices
```

To delete all approved computers, type:

```
wdsutil /verbose /delete-AutoaddDevices /Server:MyWDSServer
/Devicetype:ApprovedDevices
```

Related links

- Command-Line Syntax Key
- wdsutil approve-autoadddevices command
- wdsutil get-autoadddevices command
- wdsutil reject-autoadddevices command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil disable commands

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Disables all services for Windows Deployment Services.

Parameters

Expand table

Parameter	Description
wdsutil disable-server command	Disables all Windows Deployment Services services on a specified server (Deployment Server).
wdsutil disable-transportserver command	Disables all Windows Deployment Services services on a specified Transport Server.

Related links

- Command-Line Syntax Key
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil disable-server

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Disables all services for a Windows Deployment Services server.

Syntax

```
wdsutil [Options] /Disable-Server [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server: `<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To disable the server, type either:

```
wdsutil /Disable-Server
```

```
wdsutil /Verbose /Disable-Server /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key

- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil disable-transportserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Disables all services for a Transport Server.

Syntax

```
wdsutil [Options] /Disable-TransportServer [/Server:<Servername>]
```

Parameters

Expand table

Parameter	Description
[/Server: `<Servername>`]	Specifies the name of the Transport Server to be disabled. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no Transport Server name is specified, the local server will be used.

Examples

To disable the server, type either:

```
wdsutil /Disable-TransportServer
```

```
wdsutil /verbose /Disable-TransportServer /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key

- wdsutil enable-transportserver command
- wdsutil get-transportserver command
- wdsutil set-transportserver command
- wdsutil start-transportserver command
- wdsutil stop-transportserver command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil disconnect-client

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Disconnects a client from a multicast transmission or namespace. Unless you specify **/Force**, the client will fall back to another transfer method (if it's supported by the client).

Syntax

```
wdsutil /Disconnect-Client /ClientId:<Client ID> [/Server:<Server name>]
[/Force]
```

Parameters

Expand table

Parameter	Description
/ClientId: `<ClientID>`	Specifies the ID of the client to be disconnected. To view the ID of a client, run the `wdsutil /get-multicasttransmission /show:clients` command.
[/Server: `<Servername>`]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
[/Force]	Stops the installation completely and does not use a fallback method. Because Wdsmcast.exe doesn't support any fallback mechanism, the default behavior is as follows: • **If you're using the Windows Deployment Services client:** The client continues the installation by using unicasting. • **If you aren't using the Windows Deployment Services client:** The installation fails. **Important:** We strongly recommend using this parameter cautiously because if the installation fails, the computer can be left in an unusable state.

Examples

To disconnect a client, type:

```
wdsutil /Disconnect-Client /ClientId:1
```

To disconnect a client and force the installation to fail, type:

```
wdsutil /Disconnect-Client /Server:MyWDSServer /ClientId:1 /Force
```

Related links

- Command-Line Syntax Key
- wdsutil get-multicasttransmission command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil enable commands

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables all services for Windows Deployment Services.

Parameters

Expand table

Parameter	Description
wdsutil enable-server command	Enables all services on a specified Windows Deployment Services server (Deployment Server).
wdsutil enable-transportserver command	Enables all services on a specified Transport Server.

Related links

- Command-Line Syntax Key
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil enable-server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables all services for Windows Deployment Services.

Syntax

```
wdsutil [options] /Enable-Server [/Server:<Servername>]
```

Parameters

Expand table

Parameter	Description
[/Server: `<Servername>`]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.

Examples

To enable the services on the server, type either:

```
wdsutil /Enable-Server
```

```
wdsutil /verbose /Enable-Server /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key

- wdsutil disable-server command
- wdsutil get-Server command
- wdsutil initialize-server command
- wdsutil set-server command
- wdsutil start-server command
- wdsutil stop-server command
- wdsutil uninitialize-server command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil enable-transportserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables all services for the Transport Server.

Syntax

```
wdsutil [options] /Enable-TransportServer [/Server:<Servername>]
```

Parameters

Expand table

Parameter	Description
[/Server: `<Servername>`]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.

Examples

To enable the services on the server, type either:

```
wdsutil /Enable-TransportServer
```

```
wdsutil /verbose /Enable-TransportServer /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key

- wdsutil disable-transportserver command
- wdsutil get-transportserver command
- wdsutil set-transportserver command
- wdsutil start-transportserver command
- wdsutil stop-transportserver command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil export-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Exports an existing image from the image store to another Windows Image (.wim) file.

Syntax

For boot images:

```
wdsutil [options] /Export-Image image:<Image name> [/Server:<Servername>]
    imagetype:Boot /Architecture:{x86 | ia64 | x64} [/Filename:<Filename>]
     /DestinationImage
         /Filepath:<Filepath and name>
         [/Name:<Name>]
         [/Description:<Description>]
     [/Overwrite:{Yes | No}]
```

For install images:

```
wdsutil [options] /Export-Image image:<Image name> [/Server:<Servername>]
    imagetype:Install imageGroup:<Image group name>]
     [/Filename:<Filename>]
     /DestinationImage
         /Filepath:<Filepath and name>
         [/Name:<Name>]
         [/Description:<Description>]
     [/Overwrite:{Yes | No | append}]
```

Parameters

Expand table

Parameter	Description
image:`<Imagename>`	Specifies the name of the image to be exported.

Parameter	Description
[/Server: `<Servername>`]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
imagetype: `{Boot|Install}`	Specifies the type of image to be exported.
\imageGroup: `<Image group name>`]	Specifies the image group containing the image to be exported. If no image group name is specified and only one image group exists on the server, that image group will be used by default. If more than one image group exists on the server, the image group must be specified.
/Architecture: `{x86|ia64|x64}`	Specifies the architecture of the image to be exported. Because it is possible to have the same image name for boot images in different architectures, specifying the architecture value ensures that the correct image will be returned.
[/Filename: `<Filename>`]	if the image cannot be uniquely identified by name, the file name must be specified.
/DestinationImage	Specifies the settings for the destination image. You can specify these settings using the following options: • `/Filepath:<Filepath and name>` - Specifies the full file path for the new image. • `[/Name:<Name>]` - Sets the display name of the image. If no name is specified, the display name of the source image will be used. • `[/Description: <Description>]` - Sets the description of the image.
[/Overwrite: `{Yes|No|append}`]	Determines whether the file specified in the **/DestinationImage** option will be overwritten if an existing file with that name already exists at the /Filepath. The **Yes** option causes the existing file to be overwritten, the **No** option (default) causes an error to occur if a file with the same name already exists, and the **append** option causes the generated image to be appended as a new image within the existing .wim file.

Examples

To export a boot image, type either:

```
wdsutil /Export-Image image:WinPE boot image imagetype:Boot
/Architecture:x86 /DestinationImage /Filepath:C:\temp\boot.wim
```

```
wdsutil /verbose /Progress /Export-Image image:WinPE boot image
/Server:MyWDSServer imagetype:Boot /Architecture:x64 /Filename:boot.wim
/DestinationImage /Filepath:\\Server\Share\ExportImage.wim /Name:Exported
WinPE image /Description:WinPE Image from WDS server /Overwrite:Yes
```

To export an install image, type either:

```
wdsutil /Export-Image image:Windows Vista with Office imagetype:Install
/DestinationImage /Filepath:C:\Temp\Install.wim
```

```
wdsutil /verbose /Progress /Export-Image image:Windows Vista with Office
/Server:MyWDSServer imagetype:Instal imageGroup:ImageGroup1
/Filename:install.wim /DestinationImage /Filepath:\\server\share\export.wim
/Name:Exported Windows image /Description:Windows Vista image from WDS
server /Overwrite:append
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil copy-image command
- wdsutil get-image command
- wdsutil remove-image command
- wdsutil replace-image command
- wdsutil set-image command
- Windows Deployment Services cmdlets

Feedback

Was this page helpful? Yes No

wdsutil get

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves properties or attributes about the specified object.

Subcommands

Expand table

Subcommand	Description
wdsutil get-alldevices command	Displays information about all prestaged computers.
wdsutil get-allImagegroups command	Displays information about all image groups.
wdsutil get-allimages command	Displays information about all images.
wdsutil get-allmulticasttransmissions command	Displays the attributes of all multicast transmissions.
wdsutil get-allnamespaces command	Displays the attributes of all namespaces.
wdsutil get-allservers command	Displays information about all Windows Deployment Services servers.
wdsutil get-autoadddevices command	Displays computers that are pending administrative approval on a specified server.
wdsutil get-device command	Displays the attributes of a pending computer.
wdsutil get-image command	Displays the attributes of an existing image.
wdsutil get-imagefile command	Displays information about images in a specified `.wim` file.
wdsutil get-imagegroup command	Displays information about a specified image group.
wdsutil get-multicasttransmission command	Displays the attributes of a specified multicast transmission.
wdsutil get-namespace command	Displays the attributes of a specified namespace.
wdsutil get-server command	Displays information about a specified Windows

Subcommand	Description
	Deployment Services server.
wdsutil get-transportserver command	Displays information about a specified Transport Server.
wdsutil get-driverpackage command	Displays information about driver packages on a server.
wdsutil get-driverpackagefile command	Displays information about driver package, including the drivers and files it contains.
wdsutil get-alldrivergroups command	Displays information about all the driver groups on server.
wdsutil get-drivergroup command	Displays information about driver groups on a server.
wdsutil get-alldriverpackages command	Displays information about all the driver packages on a server that match the specified search criteria.
wdsutil get-driverpackagefile command	Displays information about a driver package, including the drivers and files it contains.

Feedback

Was this page helpful? Yes No

wdsutil get-alldevices

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the Windows Deployment Services properties of all pre-staged computers. A pre-staged computer is a physical computer that has been linked to a computer account in active directory Domain Services.

Syntax

```
wdsutil [options] /get-alldevices [/forest:{Yes | No}] [/referralserver:<servername>]
```

Parameters

Expand table

Parameter	Description
`[/forest:{Yes | No}]`	Specifies whether Windows Deployment Services should return computers in the entire forest or the local domain. The default setting is **No**, meaning that only the computers in the local domain are returned.
`[/referralserver:<servername>]`	Returns only those computers that are pre-staged for the specified server.

Examples

To view all computers, type either:

```
wdsutil /get-alldevices
```

```
```

```
wdsutil /verbose /get-alldevices /forest:Yes /referralserver:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil set-device command
- wdsutil add-device command
- wdsutil get-device command

Feedback

Was this page helpful? Yes No

wdsutil get-alldrivergroups

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about all the driver groups on a server.

Syntax

```
wdsutil /get-alldrivergroups [/server:<servername>] [/show:{packagemetadata | filters | all}]
```

Parameters

Expand table

Parameter	Description
`[/server:<servername>]`	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
`/show:{packagemetadata | filters | all}]`	Displays the metadata for all the driver packages in the specified group. **PackageMetaData** displays information about all the filters for the driver group. **Filters** displays the metadata for all driver packages and filters for the group.

Examples

To view information about a driver file, type either:

```
wdsutil /get-alldrivergroups /server:MyWdsServer /show:All
```

```
wdsutil /get-alldrivergroups [/show:packagemetadata]
```

Related links

- Command-Line Syntax Key
-
- wdsutil get-drivergroup command

Feedback

Was this page helpful? Yes No

wdsutil get-alldriverpackages

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about all the driver packages on a server that match the specified search criteria.

Syntax

```
wdsutil /get-alldriverpackages [/server:<servername>] [/show:{drivers | files | all}] [/filtertype:<filtertype> /operator:{equal | notequal | greaterorequal | lessorequal | contains} /value:<value> [/value:<value> ...]]
```

Parameters

Expand table

Parameter	Description
`[/server:<servername>]`	The name of the server. This can be the NetBIOS name or the FQDN. If a server name isn't specified, the local server is used.
`[/show:{drivers | files | all}]`	Indicates the package information to display. If **/show** isn't specified, the default is to return only the driver package metadata. **Drivers** displays the list of drivers in the package, **files** displays the list of files in the package, and **all** displays drivers and files.
`/filtertype:<filtertype>`	Specifies the attribute of the driver package to search for. You can specify multiple attributes in a single command. You must also specify **/operator** and **/value** with this option. The `<filtertype>` can be one of the following: • PackageId • PackageName • PackageEnabled • Packagedateadded • PackageInfFilename • PackageClass

Parameter	Description
	• PackageProvider • PackageArchitecture • PackageLocale • PackageSigned • PackagedatePublished • Packageversion • Driverdescription • DriverManufacturer • DriverHardwareId • DrivercompatibleId • DriverGroupId • DriverGroupName
`/operator:{equal | notequal | greaterorequal | lessorequal | contains}`	Specifies the relationship between the attribute and the values. You can specify **contains** only with string attributes. You can specify **greaterorequal** and **lessorequal** only with date and version attributes.
`/value:<value>`	Specifies the value to search on for the specified `<attribute>`. You can specify multiple values for a single **/filtertype**. The list below outlines the attributes you can specify for each filter. For more information about these attributes, see Driver and Package attributes. The attributes can include: • **PackageId.** Specifies a valid GUID. For example:{4d36e972-e325-11ce-bfc1-08002be10318}. • **PackageName.** Specifies any string value. • **PackageEnabled.** Specifies *Yes* or *No*. • **Packagedateadded.** Specifies the date in the following format: YYYY/MM/DD • **PackageInfFilename.** Specifies any string value. • **PackageClass.** Specifies a valid class name or class GUID. For example: *DiskDrive*, *Net*, or *{4d36e972-e325-11ce-bfc1-08002be10318}*. • **PackageProvider.** Specifies any string value. • **PackageArchitecture.** Specifies *x86*, *x64*, or *ia64*. • **PackagLocale.** Specifies a valid language identifier. For example: *en-US* or *es-ES*. • **PackageSigned.** Specifies **Yes** or **No**. • **PackagedatePublished.** Specifies the date in the following format: YYYY/MM/DD. • **Packageversion.** Specifies the version in the following format: a.b.x.y. For example: 6.1.0.0. • **Driverdescription.** Specifies any string value. • **DriverManufacturer.** Specifies any string value. • **DriverHardwareId.** Specifies any string value.

Parameter	Description
	• **DrivercompatibleId.** Specifies any string value. • **DriverExcludeId.** Specifies any string value. • **DriverGroupId.** Specifies a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318}. • **DriverGroupName.** Specifies any string value.

Examples

To display information, type either:

```
wdsutil /get-alldriverpackages /server:MyWdsServer /show:all
/filtertype:drivergroupname /operator:contains /value:printer
/filtertype:packagearchitecture /operator:equal /value:x64 /value:x86
```

```
wdsutil /get-alldriverpackages /show:drivers /filtertype:packagedateadded
/operator:greaterorequal /value:2008/01/01
```

Related links

- Command-Line Syntax Key
- wdsutil get-driverpackage command
- wdsutil get-driverpackagefile command

Feedback

Was this page helpful? Yes No

wdsutil get-allimagegroups

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information about all image groups on a server and all images in those image groups.

Syntax

```
wdsutil [options] /get-allimagegroups [/server:<servername>] [/detailed]
```

Parameters

Expand table

Parameter	Description
`[/server:<servername>]`	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
[/detailed]	Returns the image metadata from each image. If this parameter isn't used, the default behavior is to return only the image name, description, and file name for each image.

Examples

To view information about the image groups, type either:

```
wdsutil /get-allimagegroups
```

```
wdsutil /verbose /get-allimagegroups /server:MyWDSServer /detailed
```

Related links

- Command-Line Syntax Key
- wdsutil add-imagegroup command
- wdsutil remove-imagegroup command
- wdsutil set-imagegroup command

Feedback

Was this page helpful? Yes No

wdsutil get-allimages

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information about all images on a server.

Syntax

```
wdsutil /get-allimages [/server:<servername>] /show:{boot | install |
legacyris | all} [/detailed]
```

Parameters

Expand table

Parameter	Description
`[/server:<servername>]`	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
`/show:{boot | install | legacyris | all}`	Where **boot** returns only boot images, **install** returns install images as well as information about the image groups that contain them, **LegacyRis** returns only remote Installation Services (RIS) images, and **All** returns boot image information, install image information (including information about the image groups), and RIS image information.
[/detailed]	Indicates that all image metadata from each image should be returned. If this option is not used, the default behavior is to return only the image name, description, and file name.

Examples

To view information about the images, type either:

```
```

```
wdsutil /get-allimages /show:install
```

```
wdsutil /verbose /get-allimages /server:MyWDSServer /show:all /detailed
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil copy-image command
- wdsutil export-image command
- wdsutil remove-image command
- wdsutil replace-image command
- wdsutil set-image command

Feedback

Was this page helpful? Yes No

wdsutil get-allmulticasttransmissions

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about all multicast transmissions on a server.

Syntax

For Windows Server 2008:

```
wdsutil /Get-AllMulticastTransmissions [/Server:<Server name>]
[/Show:Clients] [/ExcludedeletePending]
```

For Windows Server 2008 R2:

```
wdsutil /Get-AllMulticastTransmissions [/Server:<Server name>] [/Show:{Boot
| Install | All}] [/details:Clients]  [/ExcludedeletePending]
```

Parameters

Expand table

Parameter	Explanation
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
[/Show]	**Windows Server 2008** /Show:Clients - Displays information about client computers that are connected to the multicast transmissions. **Windows Server 2008 R2**

Parameter	Explanation
	Show: {Boot | Install | All} - The type of image to return. **Boot** returns only boot image transmissions. **Install** returns only install image transmissions. **All** returns both image types.
/details:clients	Only supported for Windows Server 2008 R2. If present, clients that are connected to the transmission will be displayed.
[/ExcludedeletePending]	Excludes any deactivated transmissions from the list.

Examples

To view information about all transmissions, type:

- Windows Server 2008: `wdsutil /Get-AllMulticastTransmissions`
- Windows Server 2008 R2: `wdsutil /Get-AllMulticastTransmissions /Show:All` To view information about all transmissions except deactivated transmissions, type:
- Windows Server 2008: `wdsutil /Get-AllMulticastTransmissions /Server:MyWDSServer /Show:Clients /ExcludedeletePending`
- Windows Server 2008 R2: `wdsutil /Get-AllMulticastTransmissions /Server:MyWDSServer /Show:All /details:Clients /ExcludedeletePending`

Related links

- Command-Line Syntax Key
- wdsutil get-multicasttransmission command
- wdsutil new-multicasttransmission command
- wdsutil remove-multicasttransmission command
- wdsutil start-multicasttransmission command

Feedback

Was this page helpful? Yes No

wdsutil get-allnamespaces

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about all namespaces on a server.

Syntax

Windows Server 2008:

```
wdsutil /Get-AllNamespaces [/Server:<Server name>] [/ContentProvider:<name>]
[/Show:Clients] [/ExcludedeletePending]
```

Windows Server 2008 R2:

```
wdsutil /Get-AllNamespaces [/Server:<Server name>] [/ContentProvider:<name>]
[/details:Clients] [/ExcludedeletePending]
```

Parameters

Expand table

Parameter	Windows Server 2008	Windows Server 2008 R2
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.	
[/ContentProvider:<name>]	Displays the namespaces for the specified content provider only.	
[/Show:Clients]	Only supported for Windows Server 2008. Displays information about client computers that are connected to the namespace.	

Parameter	Windows Server 2008	Windows Server 2008 R2
[/details:Clients]	Only supported for Windows Server 2008 R2. Displays information about client computers that are connected to the namespace.	
[/ExcludedeletePending]	Excludes any deactivated transmissions from the list.	

Examples

To view all namespaces, type:

```
wdsutil /Get-AllNamespaces
```

To view all namespaces except those that are deactivated, type:

- Windows Server 2008

```
wdsutil /Get-AllNamespaces /Server:MyWDSServer
/ContentProvider:MyContentProv /Show:Clients /ExcludedeletePending
```

- Windows Server 2008 R2

```
wdsutil /Get-AllNamespaces /Server:MyWDSServer
/ContentProvider:MyContentProv /details:Clients /ExcludedeletePending
```

Related links

- Command-Line Syntax Key
- wdsutil new-namespace command
- wdsutil remove-namespace command
- wdsutil start-nmespace command

Feedback

Was this page helpful? Yes No

get-AllServers

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information about all Windows Deployment Services servers.

> **Note**
>
> This command may take an extended amount of time to complete if there are many Windows Deployment Services servers in your environment or if the network connection linking the servers is slow.

Syntax

```
wdsutil [Options] /Get-AllServers /Show:{Config | Images | All} [/Detailed] [/Forest:{Yes | No}]
```

Parameters

Expand table

Parameter	Description
/Show:{Config	Images
[/Detailed]	When used in conjunction with the **/Show:Images** or **/Show:All**, returns all image metadata from each image. If the **/Detailed** option is not specified, the default behavior is to return the image name, description, and file name.
[/Forest:{Yes	No}]

Examples

To view information about all servers, type:

```
wdsutil /Get-AllServers /Show:Config
```

To view detailed information about all servers, type:

```
wdsutil /Verbose /Get-AllServers /Show:All /Detailed /Forest:Yes
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil get-autoadddevices

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays all computers that are in the Auto-add database on a Windows Deployment Services server.

Syntax

```
wdsutil [Options] /Get-AutoaddDevices [/Server:<Server name>] /Devicetype:
{PendingDevices | RejectedDevices | ApprovedDevices}
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/Devicetype:{PendingDevices | RejectedDevices | ApprovedDevices}	Specifies the type of computer to return. - **PendingDevices** returns all computers in the database that have a status of pending. - **RejectedDevices** returns all computers in the database that have a status of rejected. - **ApprovedDevices** returns all computers in the database that have a status of approved.

Examples

To see all of the approved computers, type:

```
wdsutil /Get-AutoaddDevices /Devicetype:ApprovedDevices
```

To see all of the rejected computers, type:

```
wdsutil /verbose /Get-AutoaddDevices /Devicetype:RejectedDevices
/Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil delete-autoadddevices command
- wdsutil approve-autoadddevices command
- wdsutil reject-autoadddevices command

Feedback

Was this page helpful? Yes No

wdsutil get-device

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves Windows Deployment Services information about a prestaged computer (that is, a physical computer that has been lined to a computer account in active directory Domain Services.

Syntax

```
wdsutil /Get-Device {/Device:<Device name> | /ID:<MAC or UUID>} [/Domain:<Domain>] [/forest:{Yes | No}]
```

Parameters

Expand table

Parameter	Description
/Device:<Device name>	Specifies the name of the computer (SAMAccountName).
/ID:<MAC or UUID>	Specifies either the MAC address or the UUID (GUID) of the computer, as shown in the following examples. Note that a valid GUID must be in one of two formats binary string or GUID string - **Binary string**: /ID:ACEFA3E81F20694E953EB2DAA1E8B1B6 - **MAC address**: 00B056882FDC (no dashes) or 00-B0-56-88-2F-DC (with dashes) - **GUID string**: /ID:E8A3EFAC-201F-4E69-953-B2DAA1E8B1B6
[/Domain:<Domain>]	Specifies the domain to be searched for the prestaged computer. The default value for this parameter is the local domain.
[/forest:{Yes | No}]	Specifies whether Windows Deployment Services should search the entire forest or the local domain. The default value is **No**, meaning that only the local domain will be searched.

Examples

To get information by using the computer name, type:

```
wdsutil /Get-Device /Device:computer1
```

To get information by using the MAC address, type:

```
wdsutil /verbose /Get-Device /ID:00-B0-56-88-2F-DC /Domain:MyDomain
```

To get information by using the GUID string, type:

```
wdsutil /verbose /Get-Device /ID:E8A3EFAC-201F-4E69-953-B2DAA1E8B1B6
/forest:Yes
```

Related links

- Command-Line Syntax Key
- wdsutil set-device command
- wdsutil add-device command
- wdsutil get-alldevices command

Feedback

Was this page helpful? Yes No

wdsutil get-drivergroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about the driver groups on a server.

Syntax

```
wdsutil /Get-DriverGroup /DriverGroup:<Group Name> [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
/DriverGroup:<Group Name>	Specifies the name of the driver group.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. if a server name is not specified, the local server is used.
[/Show: {PackageMetaData | Filters | All}]	Displays the metadata for all the driver packages in the specified group. **PackageMetaData** displays information about all the filters for the driver group. **Filters** displays the metadata for all driver packages and filters for the group.

Examples

To view information about a driver file, type:

```
wdsutil /Get-DriverGroup /DriverGroup:printerdrivers /Show:PackageMetaData
```

```
```

```
wdsutil /Get-DriverGroup /DriverGroup:printerdrivers /Server:MyWdsServer
/Show:Filters
```

Related links

- Command-Line Syntax Key
- wdsutil get-alldrivergroups command

Feedback

Was this page helpful? Yes No

get-DriverPackage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about a driver package on the server.

Syntax

```
wdsutil /Get-DriverPackage [/Server:<Server name>] {/DriverPackage:<Package Name> | /PackageId:<ID>} [/Show:{Drivers | Files | All}]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If no server name is specified, the local server is used.
[/DriverPackage:<Name>]	Specifies the name of the driver package to show.
[/PackageId:<ID>]	Specifies the Windows Deployment Services ID of the driver package to show. You must specify the ID if the driver package cannot be uniquely identified by name.
[/Show: {Drivers	Files

Examples

To view information about a driver package, type one of the following:

```
wdsutil /Get-DriverPackage /PackageId:{4D36E972-E325-11CE-BFC1-08002BE10318}
```

```
wdsutil /Get-DriverPackage /DriverPackage:MyDriverPackage /Show:All
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil get-driverpackagefile

Article • 08/20/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about a driver package, including the drivers and files it contains.

Syntax

```
wdsutil /Get-DriverPackageFile /InfFile:<InfFilePath> [/Architecture:{x86 | x64 | arm | arm64}] [/Show:{Drivers | Files | All}]
```

Parameters

Expand table

Parameter	Description
/InfFile:`<InfFilePath>`	Specifies the full path and file name of the driver package `.inf` file.
/Architecture:`x86 | x64 | arm | arm64`	The architecture of the driver package.
/Show:`Drivers | Files | All`	Indicates the package information to display. The default, if `/show` isn't specified, is to return only the driver package metadata. • **Drivers** displays the list of drivers in the package. • **Files** displays the list of files in the package. • **All** displays drivers and files.

Examples

To view information about a driver file, type:

```
wdsutil /Get-DriverPackageFile /InfFile:C:\temp\1394.inf /Architecture:x86
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil get-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information about an image.

Syntax

For boot images:

```
wdsutil [Options] /Get-Image image:<Image name> [/Server:<Server name>
imagetype:Boot /Architecture:{x86 | ia64 | x64} [/Filename:<File name>]
```

For install images:

```
wdsutil [Options] /Get-image image:<Image name> [/Server:<Server name>
imagetype:Install imagegroup:<Image group name>] [/Filename:<File name>]
```

Parameters

Expand table

Parameter	Description
\image:<Image name>	Specifies the name of the image.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
imagetype:{Boot | Install}	Specifies the type of image.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the image. Because it is possible to have the same image name for boot images in different architectures, specifying the architecture value ensures that the correct image is returned.

Parameter	Description
[/Filename:<File name>]	if the image cannot be uniquely identified by name, you must use this option to specify the file name.
\imagegroup: <Image group name>]	Specifies the image group that contains the image. If no image group is specified and only one image group exists on the server, that group will be used. If more than one image group exists on the server, you must use this parameter to specify the image group.

Examples

To retrieve information about a boot image, type one of the following:

```
wdsutil /Get-Image image:WinPE boot imagetype:Boot /Architecture:x86
wdsutil /verbose /Get-Image image:WinPE boot image /Server:MyWDSServer
imagetype:Boot /Architecture:x86 /Filename:boot.wim
```

To retrieve information about an install image, type one of the following:

```
wdsutil /Get-Image:Windows Vista with Office imagetype:Install
wdsutil /verbose /Get-Image:Windows Vista with Office /Server:MyWDSServer
imagetype:Install imagegroup:ImageGroup1 /Filename:install.wim
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil copy-image command
- wdsutil export-image command
- wdsutil remove-image command
- wdsutil replace-image command
- wdsutil set-image command

Feedback

Was this page helpful? Yes No

get-ImageFile

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information about the images contained in a Windows Image (.wim) file.

Syntax

```
wdsutil [Options] /Get-ImageFile /ImageFile:<wim file path> [/Detailed]
```

Parameters

Expand table

Parameter	Description
/ImageFile:<WIM file path>	Specifies the full path and file name of the .wim file.
[/Detailed]	Returns all image metadata from each image. If this option is not used, the default behavior is to return only the image name, description, and file name.

Examples

To view information about an image, type:

```
wdsutil /Get-ImageFile /ImageFile:C:\temp\install.wim
```

To view detailed information, type:

```
wdsutil /Verbose /Get-ImageFile /ImageFile:\\Server\Share\My Folder
```

```
\install.wim /Detailed
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil get-imagegroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information about an image group and the images within it.

Syntax

```
wdsutil [Options] /Get-ImageGroup ImageGroup:<Image group name> [/Server:
<Server name>] [/detailed]
```

Parameters

Expand table

Parameter	Description
/ImageGroup:<Image group name>	Specifies the name of the image group.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
[/detailed]	Returns the image metadata for each image. If this parameter is not use, the default behavior is to return only the image name, description, and file name.

Examples

To view information about an image group, type:

```
wdsutil /Get-ImageGroup ImageGroup:ImageGroup1
```

To view information including metadata, type:

```
wdsutil /verbose /Get-ImageGroup ImageGroup:ImageGroup1 /Server:MyWDSServer
/detailed
```

Related links

- Command-Line Syntax Key
- wdsutil add-imagegroup command
- wdsutil get-allimagegroups command
- wdsutil remove-imagegroup command
- wdsutil set-imagegroup command

Feedback

Was this page helpful? Yes No

wdsutil get-multicasttransmission

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about the multicast transmission for a specified image.

Syntax

Windows Server 2008

```
wdsutil [Options] /Get-MulticastTransmissiomedia:<Image name> [/Server:
<Server name>mediatype:InstallmediaGroup:<Image group name>]
[/Filename:<File name>] [/Show:Clients]
```

Windows Server 2008 R2

for boot image transmissions:

```
wdsutil [Options] /Get-MulticastTransmissiomedia:<Image name>
    [/Server:<Server name>]
    [/details:Clients]
  mediatype:Boot
    /Architecture:{x86 | ia64 | x64}
        [/Filename:<File name>]
```

for install image transmissions:

```
wdsutil [Options] /Get-MulticastTransmissiomedia:<Image name>
         [/Server:<Server name>]
         [/details:Clients]
       mediatype:Install
    mediaGroup:<Image Group>]
     [/Filename:<File name>]
```

Parameters

Expand table

Parameter	Description
media:<Image name>	Displays the multicast transmission that is associated with this image.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
/imagetype:Install	Specifies the image type. Note that this option must be set to **Install**.
/imagegroup:<Image group name>]	Specifies the image group that contains the image. If no image group name is specified and only one image group exists on the server, that image group is used. If more than one image group exists on the server, you must use this option to specify an image group.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the boot image that is associated with the transmission. Because it is possible to have the same image name for boot images in different architectures, you should specify the architecture to ensure that the correct image is used.
[/Filename:<File name>]	Specifies the file that contains the image. If the image cannot be uniquely identified by name, you must use this option to specify the file name.
[/Show:Clients] or [/details:Clients]	Displays information about client computers that are connected to the multicast transmission.

Examples

Windows Server 2008

To view information about the transmission for an image named Vista with Office, type one of the following:

```
wdsutil /Get-MulticastTransmission:Vista with Office imagetype:Install
wdsutil /Get-MulticastTransmission /Server:MyWDSServer image:Vista with
Office imagetype:Install imageGroup:ImageGroup1 /Filename:install.wim
/Show:Clients
```

Windows Server 2008 R2

To view information about the transmission for an image named Vista with Office, type one of the following:

```
wdsutil /Get-MulticastTransmission:Vista with Office
 /Imagetype:Install
```

```
wdsutil /Get-MulticastTransmission /Server:MyWDSServer image:Vista with
Office imagetype:Install ImageGroup:ImageGroup1 /Filename:install.wim
/details:Clients
```

```
wdsutil /Get-MulticastTransmission /Server:MyWDSServer:X64 Boot
Imagetype:Boot /Architecture:x64 /Filename:boot.wim /details:Clients
```

Related links

- Command-Line Syntax Key
- wdsutil get-allmulticasttransmissions command
- wdsutil new-multicasttransmission command
- wdsutil remove-multicasttransmission command
- wdsutil start-multicasttransmission command

Feedback

Was this page helpful? Yes No

wdsutil get-namespace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about a custom namespace.

Syntax

Windows Server 2008 R2

```
wdsutil /Get-Namespace /Namespace:<Namespace name> [/Server:<Server name>]
[/Show:Clients]
```

Windows Server 2008 R2

```
wdsutil /Get-Namespace /Namespace:<Namespace name> [/Server:<Server name>]
[/details:Clients]
```

Parameters

Expand table

Parameter	Description
/Namespace:<Namespace name>	Specifies the name of the namespace. Note that this is not the friendly name, and it must be unique. - Deployment Server: The syntax for namespace name is /Namspace:WDS:<ImageGroup>/<ImageName>/<Index>. For example: **WDS:ImageGroup1/install.wim/1** - Transport Server: This value should match the name given to the namespace when it was created on the server.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.

Parameter	Description
[/Show:Clients] or [/details:Clients]	Displays information about client computers that are connected to the specified namespace.

Examples

To view information about a namespace, type:

```
wdsutil /Get-Namespace /Namespace:Custom Auto 1
```

To view information about a namespace and the clients that are connected, type one of the following:

- Windows Server 2008: `wdsutil /Get-Namespace /Server:MyWDSServer /Namespace:Custom Auto 1 /Show:Clients`
- Windows Server 2008 R2: `wdsutil /Get-Namespace /Server:MyWDSServer /Namespace:Custom Auto 1 /details:Clients`

Related links

- Command-Line Syntax Key
- wdsutil get-allnamespaces command
- wdsutil new-namespace command
- wdsutil remove-namespace command
- wdsutil start-namespace command

Feedback

Was this page helpful? Yes No

wdsutil get-server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Retrieves information from the specified Windows Deployment Services server.

Syntax

```
wdsutil [Options] /Get-Server [/Server:<Server name>] /Show:{Config | Images | All} [/detailed]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
/Show:{Config | Images | All}	Specifies the type of information to return. - **Config** returns configuration information. - **Images** returns information about image groups, boot images, and install images. - **All** returns configuration information and image information.
[/detailed]	You can use this option with **/Show:Images** or **/Show:All** to indicate that all image metadata from each image should be returned. If the **/detailed** option is not used, the default behavior is to return the image name, description, and file name.

Examples

To view information about the server, type:

```
wdsutil /Get-Server /Show:Config
```

To view detailed information about the server, type:

```
wdsutil /verbose /Get-Server /Server:MyWDSServer /Show:All /detailed
```

Related links

- Command-Line Syntax Key
- wdsutil disable-server command
- wdsutil enable-server command
- wdsutil initialize-server command
- wdsutil set-server command
- wdsutil start-server command
- wdsutil stop-server command
- wdsutil uninitialize-server command

Feedback

Was this page helpful? Yes No

wdsutil get-transportserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays information about a specified Transport Server.

Syntax

```
wdsutil [Options] /Get-TransportServer [/Server:<Server name>] /Show:
{Config}
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/Show:{Config}	Returns configuration information about the specified Transport Server.

Examples

To view information about the server, type:

```
wdsutil /Get-TransportServer /Show:Config
```

To view configuration information, type:

```
wdsutil /Get-TransportServer /Server:MyWDSServer /Show:Config
```

Related links

- Command-Line Syntax Key
- wdsutil disable-transportserver command
- wdsutil enable-transportserver command
- wdsutil set-transportserver command
- wdsutil start-transportserver command
- wdsutil stop-transportserver command

Feedback

Was this page helpful? Yes No

wdsutil initialize-server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Configures a Windows Deployment Services server for initial use after the server role has been installed. After you run this command, you should use the wdsutil add-Image command command to add images to the server.

Syntax

```
wdsutil /Initialize-Server [/Server:<Server name>] /remInst:<Full path>
[/Authorize]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/remInst:<Full path>	Specifies the full path and name of the remoteInstall folder. If the specified folder does not already exist, this option will create it when the command is run. You should always enter a local path, even in the case of a remote computer. For example: **D:\remoteInstall**.
[/Authorize]	Authorizes the server in Dynamic Host Control Protocol (DHCP). This option is necessary only if DHCP rogue detection is enabled, meaning that the Windows Deployment Services PXE server must be authorized in DHCP before client computers can be serviced. Note that DHCP rogue detection is disabled by default.

Examples

To initialize the server and set the remoteInstall shared folder to the F: drive, type.

```
wdsutil /Initialize-Server /remInst:F:\remoteInstall
```

To initialize the server and set the remoteInstall shared folder to the C: drive, type.

```
wdsutil /verbose /Progress /Initialize-Server /Server:MyWDSServer
/remInst:C:\remoteInstall
```

Related links

- Command-Line Syntax Key
- wdsutil disable-server command
- wdsutil enable-server command
- wdsutil get-server command
- wdsutil set-server command
- wdsutil start-server command
- wdsutil stop-server command
- wdsutil uninitialize-server command

Feedback

Was this page helpful? Yes No

wdsutil new

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates capture and discover images, multicast transmissions, and namespaces.

Subcommands

Expand table

Subcommand	Description
wdsutil new-captureimage command	creates a new capture image from an existing boot image.
wdsutil new-discoverimage command	creates a new discover image from an existing boot image.
wdsutil new-multicasttransmission command	creates a new multicast transmission.
wdsutil new-namespace command	creates a new namespace.

Feedback

Was this page helpful? Yes No

new-CaptureImage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a new capture image from an existing boot image. Capture images are boot images that start the Windows Deployment Services capture utility instead of starting Setup. When you boot a reference computer (that has been prepared with Sysprep) into a capture image, a wizard creates an install image of the reference computer and saves it as a Windows Image (.wim) file. You can also add the image to media (such as a CD, DVD, or USB drive), and then boot a computer from that media. After you create the install image, you can add the image to the server for PXE boot deployment. For more information, see Creating Images (https://go.microsoft.com/fwlink/?LinkId=115311).

Syntax

```
wdsutil [Options] /New-CaptureImage [/Server:<Server name>]
     /Image:<Image name>
     /Architecture:{x86 | ia64 | x64}
     [/Filename:<File name>]
     /DestinationImage
        /FilePath:<File path and name>
        [/Name:<Name>]
        [/Description:<Description>]
        [/Overwrite:{Yes | No | Append}]
        [/UnattendFilePath:<File path>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/Image:<Image name>	Specifies the name of the source boot image.

Parameter	Description
/Architecture: {x86	ia64
[/Filename: <Filename>]	If the image cannot be uniquely identified by name, you must use this option to specify the file name.
/DestinationImage	Specifies the settings for the destination image. You specify the settings using the following options: - /FilePath: <File path and name> Sets the full file path for the new capture image. - [/Name: <Name>] - Sets the display name of the image. If no display name is specified, the display name of the source image will be used. - [/Description: <Description>] - Sets the description of the image. - [/Overwrite: {Yes

Examples

To create a capture image and name it WinPECapture.wim, type:

```
wdsutil /New-CaptureImage /Image:WinPE boot image /Architecture:x86
/DestinationImage /FilePath:C:\Temp\WinPECapture.wim
```

To create a capture image and apply the specified settings, type:

```
wdsutil /Verbose /Progress /New-CaptureImage /Server:MyWDSServer
/Image:WinPE boot image /Architecture:x64 /Filename:boot.wim
/DestinationImage /FilePath:\\Server\Share\WinPECapture.wim /Name:New WinPE
image /Description:WinPE image with capture utility /Overwrite:No
/UnattendFilePath:\\Server\Share\WDSCapture.inf
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

new-DiscoverImage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Creates a new discover image from an existing boot image. Discover images are boot images that force the Setup.exe program to start in Windows Deployment Services mode and then discover a Windows Deployment Services server. Typically these images are used to deploy images to computers that are not capable of booting to PXE. For more information, see Creating Images (https://go.microsoft.com/fwlink/?LinkId=115311).

Syntax

```
wdsutil [Options] /New-DiscoverImage [/Server:<Server name>]
     /Image:<Image name>
     /Architecture:{x86 | ia64 | x64}
     [/Filename:<File name>]
     /DestinationImage
         /FilePath:<File path and name>
         [/Name:<Name>]
         [/Description:<Description>]
         [/WDSServer:<Server name>]
         [/Overwrite:{Yes | No | Append}]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/Image:<Image name>	Specifies the name of the source boot image.
/Architecture:{x86	ia64

Parameter	Description
[/Filename:<File name>]	If the image cannot be uniquely identified by name, you must use this option to specify the file name.
/DestinationImage	Specifies the settings for the destination image. You can specify the settings using the following options: - /FilePath:< File path and name> - Sets full file path for the new image. - [/Name:<Name>] - Sets the display name of the image. If no display name is specified, the display name of the source image will be used. - [/Description: <Description>] - Sets the description of the image. - [/WDSServer: <Server name>] - Specifies the name of the server that all clients who boot from the specified image should contact to download the install image. By default, all clients who boot this image will discover a valid Windows Deployment Services server. Using this option bypasses the discovery functionality and forces the booted client to contact the specified server. - [/Overwrite:{Yes

Examples

To create a discover image out of boot image, and name it WinPEDiscover.wim, type:

```
wdsutil /New-DiscoverImage /Image:WinPE boot image /Architecture:x86
/DestinationImage /FilePath:C:\Temp\WinPEDiscover.wim
```

To create a discover image out of boot image, and name it WinPEDiscover.wim with the specified settings, type:

```
wdsutil /Verbose /Progress /New-DiscoverImage /Server:MyWDSServer
/Image:WinPE boot image /Architecture:x64 /Filename:boot.wim
/DestinationImage /FilePath:\\Server\Share\WinPEDiscover.wim
/Name:New WinPE image /Description:WinPE image for WDS Client discovery
/Overwrite:No
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil new-multicasttransmission

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates a new multicast transmission for an image. This command is equivalent to creating a transmission by using the Windows Deployment Services mmc snap-in (right-click the **Multicast Transmissions** node, and then click **create Multicast Transmission**). You should use this command when you have both the Deployment Server role service and the Transport Server role service installed (which is the default installation). If you have only the Transport Server role service installed, use wdsutilnew-Namespace command.

Syntax

For install images transmissions:

```
wdsutil [Options] /New-MulticastTransmissiomedia:<Image name>
        [/Server:<Server name>]
        /FriendlyName:<Friendly name>
        [/Description:<Description>]
        /Transmissiontype: {AutoCast | ScheduledCast}
            [/time:<YYYY/MM/DD:hh:mm>]
            [/Clients:<Num of Clients>]
      imagetype:Install
       ImageGroup:<Image Group>]
        [/Filename:<File name>]
```

For boot image transmissions (only supported for Windows Server 2008 R2):

```
wdsutil [Options] /New-MulticastTransmissiomedia:<Image name>
        [/Server:<Server name>]
        /FriendlyName:<Friendly name>
        [/Description:<Description>]
        /Transmissiontype: {AutoCast | ScheduledCast}
            [/time:<YYYY/MM/DD:hh:mm>]
            [/Clients:<Num of Clients>]
      imagetype:Boot
```

```
        /Architecture:{x86 | ia64 | x64}
        [/Filename:<File name>]
```

Parameters

Expand table

Parameter	Description
/image:<Image name>	Specifies the name of the image to be transmitted using multicasting.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/FriendlyName: <Friendly name>	Specifies the friendly name of the transmission.
[/Description: <Description>]	Specifies the description of the transmission.
/imagetype: {Boot|Install}	Specifies the type of image to be transmitted using multicasting. Note **Boot** is only supported for Windows Server 2008 R2.
/ImageGroup:<Image group name>]	Specifies the image group that contains the image. If no image group name is specified and only one image group exists on the server, that image group is used. If more than one image group exists on the server, you must use this option to specify the image group name.
[/Filename:<File name>]	Specifies the file name. If the source image cannot be uniquely identified by name, you must use this option to specify the file name.
/Transmissiontype: {AutoCast | ScheduledCast}	Specifies whether to start the transmission automatically (AutoCast) or based on the specified start criteria (ScheduledCast). • **Auto-Cast**. This transmission type indicates that as soon as an applicable client requests an install image, a multicast transmission of the selected image begins. As other clients request the same image, they are joined to the transmission that is already started. • **Scheduled-Cast**. This transmission type sets the start criteria for the transmission based on the number of clients that are requesting an image and/or a specific day and time. You can specify the following options: ○ [/time: <time>] - Sets the time that the transmission should start by using the following format: YYYY/MM/DD:hh:mm.

Parameter	Description
	○ [/Clients: <Number of clients>] - Sets the minimum number of clients to wait for before the transmission starts.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the boot image to transmit using multicasting. Because it is possible to have the same name for boot images of different architectures, you should specify the architecture to ensure the correct image is used.
[/Filename:<File name>]	Specifies the file name. If the source image cannot be uniquely identified by name, you must specify the file name.

Examples

To create an Auto-Cast transmission of a boot image in Windows Server 2008 R2, type:

```
wdsutil /New-MulticastTransmission /FriendlyName:WDS Boot Transmission
/Image:X64 Boot imagetype:Boot /Architecture:x64 /Transmissiontype:AutoCast
```

To create an Auto-Cast transmission of an install image, type:

```
wdsutil /New-MulticastTransmission /FriendlyName:WDS AutoCast Transmission
/Image:Vista with Officeimage imagetype:Install /Transmissiontype:AutoCast
```

To create a Scheduled-Cast transmission of an install image, type:

```
wdsutil /New-MulticastTransmission /FriendlyName:WDS SchedCast Transmission
/Server:MyWDSServer Image:Vista with Office imagetype:Install
/Transmissiontype:ScheduledCast /time:2006/11/20:17:00 /Clients:100
```

Related links

- Command-Line Syntax Key
- wdsutil get-allmulticasttransmissions command
- wdsutil get-multicasttransmission command
- wdsutil remove-multicasttransmission command

- wdsutil start-multicasttransmission command

Feedback

Was this page helpful? Yes No

wdsutil new-namespace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Creates and configures a new namespace. You should use this option when you have only the Transport Server role service installed. If you have both the Deployment Server role service and the Transport Server role service installed (which is the default), use wdsutilnew-MulticastTransmission command. Note that you must register the content provider before you use this option.

Syntax

```
wdsutil [Options] /New-Namespace [/Server:<Server name>]
     /FriendlyName:<Friendly name>
     [/Description:<Description>]
     /Namespace:<Namespace name>
     /ContentProvider:<Name>
     [/ConfigString:<Configuration string>]
     /Namespacetype: {AutoCast | ScheduledCast}
         [/time:<YYYY/MM/DD:hh:mm>]
         [/Clients:<Number of clients>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
/FriendlyName:<Friendly name>	Specifies the friendly name of the namespace.
[/Description: <Description>]	Sets the description of the namespace.
/Namespace: <Namespace name>	Specifies the name of the namespace. Note that this is not the friendly name, and it must be unique.

Parameter	Description
	- **Deployment Server role service**: The syntax for this option is /Namespace:WDS:<Image group>/<Image name>/<Index>. For example: **WDS:ImageGroup1/install.wim/1** - **Transport Server role service**: This value should match the name given when the namespace was created on the server.
/ContentProvider:<Name>]	Specifies the name of the content provider that will provide content for the namespace.
[/ConfigString:<Configuration string>]	Specifies the configuration string for the content provider.
/Namespacetype:{AutoCast | ScheduledCast}	Specifies the settings for the transmission. You specify the settings using the following options: - [/time: <time>] - Sets the time that the transmission should start by using the following format: YYYY/MM/DD:hh:mm. This option applies only to Scheduled-Cast transmissions. - [/Clients: <Number of clients>] - Sets the minimum number of clients to wait for before the transmission starts. This option applies only to Scheduled-Cast transmissions.

Examples

To create an Auto-Cast namespace, type:

```
wdsutil /New-Namespace /FriendlyName:Custom AutoCast Namespace
/Namespace:Custom Auto 1 /ContentProvider:MyContentProvider
/Namespacetype:AutoCast
```

To create a Scheduled-Cast namespace, type:

```
wdsutil /New-Namespace /Server:MyWDSServer /FriendlyName:Custom Scheduled
Namespace /Namespace:Custom Auto 1 /ContentProvider:MyContentProvider
/Namespacetype:ScheduledCast /time:2006/11/20:17:00 /Clients:20
```

Related links

- Command-Line Syntax Key
- wdsutil get-allnamespaces command

- wdsutil remove-namespace command
- wdsutil start-namespace command

Feedback

Was this page helpful? Yes No

wdsutil /progress

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays progress while a command is running. You can use **/progress** with any other wdsutil commands that you run. If you want to turn on verbose logging for this command, you must specify **/verbose** and **/progress** directly after **wdsutil**.

Syntax

```
wdsutil /progress <commands>
```

Examples

To initialize the server and display progress, type:

```
wdsutil /verbose /progress /Initialize-Server /Server:MyWDSServer
/RemInst:C:\RemoteInstall
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil reject-autoadddevices

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Rejects computers that are pending administrative approval. When the Auto-add policy is enabled, administrative approval is required before unknown computers (those that are not prestaged) can install an image. You can enable this policy using the **PXE Response** tab of the server s properties page.

Syntax

```
wdsutil [Options] /Reject-AutoaddDevices [/Server:<Server name>] /RequestId:
<Request ID or ALL>
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
/RequestId: <Request ID | ALL>	Specifies the request ID assigned to the pending computer. To reject all pending computers, specify **ALL**.

Examples

To reject a single computer, type:

```
wdsutil /Reject-AutoaddDevices /RequestId:12
```

To reject all computers, type:

```
wdsutil /verbose /Reject-AutoaddDevices /Server:MyWDSServer /RequestId:ALL
```

Related links

- Command-Line Syntax Key
- wdsutil approve-autoadddevices command
- wdsutil delete-autoadddevices command
- wdsutil get-autoadddevices command

Feedback

Was this page helpful? Yes No

wdsutil remove

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Subcommands

Expand table

Subommand	Description
wdsutil remove-image command	removes a boot or install image from the server.
wdsutil remove-imagegroup command	removes an image group from the server.
wdsutil remove-multicasttransmission command	Disables multicast transmission of an image.
wdsutil remove-namespace command	removes a namespace from the server.
wdsutil remove-drivergrouppackage command	removes a driver package from a driver group on a server.
wdsutil remove-drivergrouppackages command	removes driver packages from a driver group on a server.
wdsutil remove-driverpackage command	removes a driver package from a server.
wdsutil remove-driverpackages command	removes driver packages from a server.
wdsutil remove-drivergroup command	removes a driver group from a server.
wdsutil remove-drivergroupfilter command	removes a filter rule from a driver group on a server.

Feedback

Was this page helpful? Yes No

remove-DriverGroup

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a driver group from a server.

Syntax

```
wdsutil /Remove-DriverGroup /DriverGroup:<Group Name> [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
/DriverGroup:<Group Name>	Specifies the name of the driver group to remove.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.

Examples

To remove a driver group, type one of the following:

```
wdsutil /Remove-DriverGroup /DriverGroup:PrinterDrivers
```

```
wdsutil /Remove-DriverGroup /DriverGroup:PrinterDrivers /Server:MyWdsServer
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

remove-DriverGroupFilter

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a filter rule from a driver group on a server.

Syntax

```
wdsutil /Remove-DriverGroupFilter /DriverGroup:<Group Name> [/Server:<Server name>] /FilterType:<Filter Type>
```

Parameters

Expand table

Parameter	Description
/DriverGroup:<Group Name>	Specifies the name of the driver group.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
[/FilterType: <FilterType>]	Specifies the type of the filter to remove from the group. <FilterType> can be one of the following: **BiosVendor** **BiosVersion** **ChassisType** **Manufacturer** **Uuid** **OsVersion** **OsEdition** **OsLanguage**

Examples

To remove a filter, type one of the following:

```
wdsutil /Remove-DriverGroupFilter /DriverGroup:PrinterDrivers
/FilterType:Manufacturer
```

```
wdsutil /Remove-DriverGroupFilter /DriverGroup:PrinterDrivers
/FilterType:Manufacturer /FilterType:OSLanguage
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

remove-DriverGroupPackage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Removes a driver package from a driver group on a server.

Syntax

```
wdsutil /Remove-DriverGroupPackage /DriverGroup:<Group Name> [/Server:
<Server Name>] {/DriverPackage:<Name> | /PackageId:<ID>}
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
[/DriverPackage:<Name>]	Specifies the name of the driver package to remove.
[/PackageId:<ID>]	Specifies the Windows Deployment Services ID of the driver package to remove. You must specify this option if the driver package cannot be uniquely identified by name.

Examples

```
wdsutil /Remove-DriverGroupPackage /DriverGroup:PrinterDrivers /PackageId:
{4D36E972-E325-11CE-BFC1-08002BE10318}
```

```
wdsutil /Remove-DriverGroupPackage /DriverGroup:PrinterDrivers
```

```
/DriverPackage:XYZ
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil remove-drivergrouppackages

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes driver packages from a driver group on a server.

Syntax

```
wdsutil /remove-DriverGroupPackages /DriverGroup:<Group Name> [/Server:
<Server Name>] /Filtertype:<Filter type> /Operator:{Equal | NotEqual |
GreaterOrEqual | LessOrEqual | Contains} /Value:<Value> [/Value:<Value> ...]
```

Parameters

Expand table

Parameter	Description
/DriverGroup:<Group Name>	Specifies the name of the driver group.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
/Filtertype:<Filter type>	Specifies the attribute of the driver package to search for. You can specify multiple attributes in a single command. You must also specify **/Operator** and **/Value** with this option. <Filter type> can be one of the following: **PackageId** **PackageName** **PackageEnabled** **Packagedateadded** **PackageInfFilename** **PackageClass**

Parameter	Description
	PackageProvider **PackageArchitecture** **PackageLocale** **PackageSigned** **PackagedatePublished** **Packageversion** **Driverdescription** **DriverManufacturer** **DriverHardwareId** **DrivercompatibleId** **DriverExcludeId** **DriverGroupId** **DriverGroupName**
/Operator:{Equal | NotEqual | GreaterOrEqual | LessOrEqual | Contains}	Specifies the relationship between the attribute and the values. You can only specify **Contains** with string attributes. You can only specify **GreaterOrEqual** and **LessOrEqual** with date and version attributes.
/Value:<Value>	Specifies the value to search for the specified <attribute>. You can specify multiple values for a single **/Filtertype**. The following list outlines the attributes that you can specify for each filter. For more information about these attributes, see Driver and Package attributes (https://go.microsoft.com/fwlink/?LinkId=166895). - PackageId - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318}. - PackageName Specify any string value. - PackageEnabled - Specify **Yes** or **No**. - Packagedateadded - Specify the date in the following format: YYYY/MM/DD - PackageInfFilename Specify any string value. - PackageClass - Specify a valid class name or class GUID. For example: **DiskDrive**, **Net**, or {4d36e972-e325-11ce-bfc1-08002be10318}. - PackageProvider Specify any string value. - PackageArchitecture - Specify **x86**, **x64**, or **ia64**. - PckageLocale - Specify a valid language identifier. For example: **en-US** or **es-ES**.

Parameter	Description
	- PackageSigned - Specify **Yes** or **No**. - PackagedatePublished - Specify the date in the following format: YYYY/MM/DD - Packageversion - Specify the version in the following format: a.b.x.y. For example: 6.1.0.0 - Driverdescription Specify any string value. - DriverManufacturer Specify any string value. - DriverHardwareId - Specify any string value. - DrivercompatibleId - Specify any string value. - DriverExcludeId - Specify any string value. - DriverGroupId - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318}. - DriverGroupName Specify any string value.

Examples

To remove driver packages from a driver group, type one of the following:

```
wdsutil /verbose /remove-DriverGroupPackages /DriverGroup:printerdrivers
/Filtertype:DriverManufacturer /Operator:NotEqual /Value:Name1 /Value:Name2
```

```
wdsutil /verbose /remove-DriverGroupPackages /DriverGroup:DisplayDrivers
/Filtertype:PackageArchitecture /Operator:Equal /Value:x86
/Filtertype:Packagedateadded /Operator:LessOrEqual /Value:2008/01/01
```

Related links

- Command-Line Syntax Key
- wdsutil remove-drivergrouppackage command

Feedback

Was this page helpful? Yes No

wdsutil remove-driverpackage

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes a driver package from a server.

Syntax

```
wdsutil /remove-DriverPackage [/Server:<Server name>] {/DriverPackage:
<Package Name> | /PackageId:<ID>}
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
[/DriverPackage:<Name>]	Specifies the name of the driver package to remove.
[/PackageId:<ID>]	Specifies the Windows Deployment Services ID of the driver package to remove. You must specify the ID if the driver package cannot be uniquely identified by name.

Examples

To view information about the images, type one of the following:

```
wdsutil /remove-DriverPackage /PackageId:{4D36E972-E325-11CE-Bfc1-
08002BE10318}
```

```
```

```
wdsutil /remove-DriverPackage /Server:MyWdsServer
/DriverPackage:MyDriverPackage
```

Related links

- Command-Line Syntax Key
- wdsutil remove-driverpackages command

Feedback

Was this page helpful? Yes No

wdsutil remove-driverpackages

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes driver packages from the server.

Syntax

```
wdsutil /remove-DriverPackages [/Server:<Server name>] /Filtertype:<Filter type> /Operator:{Equal | NotEqual | GreaterOrEqual | LessOrEqual | Contains} /Value:<Value> [/Value:<Value> ...]
```

Parameters

Expand table

Parameter	Description
[/Server: <Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
/Filtertype:<Filter type>	Specifies the attribute of the driver package to search for. You can specify multiple attributes in a single command. You must also specify **/Operator** and **/Value** with this option. <Filter type> can be one of the following: • **PackageId** • **PackageName** • **PackageEnabled** • **Packagedateadded** • **PackageInfFilename** • **PackageClass** • **PackageProvider** • **PackageArchitecture** • **PackageLocale** • **PackageSigned** • **PackagedatePublished** • **Packageversion** • **Driverdescription** • **DriverManufacturer**

Parameter	Description
	• **DriverHardwareId** • **DrivercompatibleId** • **DriverExcludeId** • **DriverGroupId** • **DriverGroupName**
/Operator:{Equal | NotEqual | GreaterOrEqual | LessOrEqual | Contains}	Specifies the relationship between the attribute and the values. You can only specify **Contains** with string attributes. You can only specify **GreaterOrEqual** and **LessOrEqual** with date and version attributes.
/Value:<Value>	Specifies the value to search for the specified <attribute>. You can specify multiple values for a single **/Filtertype**. The following list outlines the attributes that you can specify for each filter. For more information about these attributes, see Driver and Package attributes (https://go.microsoft.com/fwlink/?LinkId=166895). • PackageId - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318}. • PackageName - Specify any string value. • PackageEnabled - Specify **Yes** or **No**. • Packagedateadded - Specify the date in the following format: YYYY/MM/DD. • PackageInfFilename - Specify any string value. • PackageClass - Specify a valid class name or class GUID. For example: **DiskDrive**, **Net**, or {4d36e972-e325-11ce-bfc1-08002be10318}. • PackageProvider - Specify any string value. • PackageArchitecture - Specify **x86**, **x64**, or **ia64**. • PckageLocale - Specify a valid language identifier. For example: **en-US** or **es-ES**. • PackageSigned - Specify **Yes** or **No**. • PackagedatePublished - Specify the date in the following format: YYYY/MM/DD. • Packageversion - Specify the version in the following format: a.b.x.y. For example: 6.1.0.0. • Driverdescription - Specify any string value. • DriverManufacturer - Specify any string value. • DriverHardwareId - Specify any string value. • DrivercompatibleId - Specify any string value. • DriverExcludeId - Specify any string value. • DriverGroupId - Specify a valid GUID. For example: {4d36e972-e325-11ce-bfc1-08002be10318}. • DriverGroupName - Specify any string value.

Examples

To remove packages, type one of the following:

```
wdsutil /verbose /remove-DriverPackages /Server:MyWdsServer
/Filtertype:PackageProvider /Operator:Equal /Value:Name1 /Value:Name2
```

```
wdsutil /remove-DriverPackages /Filtertype:PackageArchitecture
/Operator:Equal
/Value:x86 /Value:x64 /Filtertype:PackageEnabled /Operator:Equal /Value:No
```

```
wdsutil /verbose /remove-DriverPackages /Server:MyWdsServer
/Filtertype:Packagedateadded /Operator:LessOrEqual /Value:2008/01/01
```

Related links

- Command-Line Syntax Key
- wdsutil remove-driverpackage command

Feedback

Was this page helpful? Yes No

wdsutil remove-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Deletes an image from a server.

Syntax

for boot images:

```
wdsutil [Options] /remove-Image:<Image name> [/Server:<Server name>
type:Boot /Architecture:{x86 | ia64 | x64} [/Filename:<Filename>]
```

for install images:

```
wdsutil [Options] /remove-image:<Image name> [/Server:<Server name>
type:Install ImageGroup:<Image group name>] [/Filename:<Filename>]
```

Parameters

Expand table

Parameter	Description
/remove-image:<Image name>	Specifies the name of the image.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
mediatype:{Boot | Install}	Specifies the type of image.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the image. Because it is possible to have the same image name for different boot images in different architectures,

Parameter	Description
	specifying the architecture value ensures that the correct image will be removed.
\ImageGroup:<Image group name>]	Specifies the image group that contains the image. If no image group name is specified and only one image group exists on the server, that image group will be used. If more than one image group exists, you must use this option to specify the image group.
[/Filename:<File name>]	if the image cannot be uniquely identified by name, you must use this option to specify the file name.

Examples

To remove a boot image, type:

```
wdsutil /remove-Imagmedia:WinPE Boot Imagemediatype:Boot /Architecture:x86
```

```
wdsutil /verbose /remove-Image:WinPE Boot Image /Server:MyWDSServer
type:Boot /Architecture:x64 /Filename:boot.wim
```

To remove an install image, type:

```
wdsutil /remove-Image:Windows Vista with Officemediatype:Install
```

```
wdsutil /verbose /remove-Image:Windows Vista with Office
/Server:MyWDSServemediatype:Install ImageGroup:ImageGroup1
/Filename:install.wim
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil copy-image command

- wdsutil export-image command
- wdsutil get-image command
- wdsutil replace-image command
- wdsutil set-image command

Feedback

Was this page helpful? Yes No

wdsutil remove-imagegroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes an image group from a server.

Syntax

```
wdsutil [Options] /remove-ImageGroup Group:<Image group name> [/Server:
<Server name>]
```

Parameters

Expand table

Parameter	Description
imagegroup:<Image group name>	Specifies the name of the image group to be removed
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To remove the image group, type one of the following:

```
wdsutil /remove-ImageGroumediaGroup:ImageGroup1
wdsutil /verbose /remove-ImageGroumediaGroup:My Image Group
/Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil add-imagegroup command
- wdsutil get-allimagegroups command
- wdsutil get-imagegroup command
- wdsutil set-imagegroup command

Feedback

Was this page helpful? Yes No

wdsutil remove-multicasttransmission

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Disables multicast transmitting for an image. Unless you specify **/force**, existing clients will complete the image transfer but new clients will not be allowed to join.

Syntax

Windows Server 2008

```
wdsutil /remove-MulticastTransmission:<Image name> [/Server:<Server name>
mediatype:Install Group:<Image Group>] [/Filename:<File name>] [/force]
```

Windows Server 2008 R2 for boot images:

```
wdsutil [Options] /remove-MulticastTransmissiomedia:<Image name>
\x20    [/Server:<Server name>]
\x20  mediatype:Boot
\x20    /Architecture:{x86 | ia64 | x64}
\x20    [/Filename:<File name>]
```

for install images:

```
wdsutil [Options] /remove-MulticastTransmissiomedia:<Image name>
        [/Server:<Server name>]
      mediatype:Install
       mediaGroup:<Image Group
        [/Filename:<File name>]
```

Parameters

Expand table

Parameter	Description
media:<Image name>	Specifies the name of the image.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
mediatype: {Install|Boot}	Specifies the image type. Note that this option must be set to **Install** for Windows Server 2008.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the boot image that is associated with the transmission to start. Because it is possible to have the same image name for boot images in different architectures, you should specify the architecture to ensure that the correct transmission is used.
\mediaGroup: <Image group name>]	Specifies the image group that contains the image. If no image group name is specified and only one image group exists on the server, that image group is used. If more than one image group exists on the server, you must use this option to specify the image group name.
[/Filename:<File name>]	Specifies the file name. If the source image cannot be uniquely identified by name, you must use this option to specify the file name.
[/force]	removes the transmission and terminates all clients. Unless you specify a value for the **/force** option, existing clients can complete the image transfer but new clients are not able to join.

Examples

To stop a namespace (current clients will complete the transmission, but new clients will not be able to join), type:

```
wdsutil /remove-MulticastTransmission:Vista with Office
/Imagetype:Install
```

```
wdsutil /remove-MulticastTransmission:x64 Boot Image
/Imagetype:Boot /Architecture:x64
```

To force termination of all clients, type:

```
wdsutil /remove-MulticastTransmission /Server:MyWDSServer
/Image:Vista with Officemediatype:InstalmediaGroup:ImageGroup1
/Filename:install.wim /force
```

Related links

- Command-Line Syntax Key
- wdsutil get-allmulticasttransmissions command
- wdsutil get-multicasttransmission command
- wdsutil new-multicasttransmission command
- wdsutil start-multicasttransmission command

Feedback

Was this page helpful? Yes No

wdsutil remove-namespace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Removes a custom namespace.

Syntax

```
wdsutil /remove-Namespace /Namespace:<Namespace name> [/Server:<Server name>] [/force]
```

Parameters

Expand table

Parameter	Description
/Namespace:<Namespace name>	Specifies the name of the namespace. This is not the friendly name, and it must be unique. - **Deployment Server role service**: The syntax for namespace name is /Namespace:WDS:<ImageGroup>/<ImageName>/<Index>. For example: **WDS:ImageGroup1/install.wim/1** - **Transport Server role service**: This value must match the name given to the namespace when it was created on the server.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server is used.
[/force]	removes the namespace immediately and terminates all clients. Note that unless you specify **/force**, existing clients can complete the transfer, but new clients are not able to join.

Examples

To stop a namespace (current clients can complete the transfer but new clients are not able to join), type:

```
wdsutil /remove-Namespace /Namespace:Custom Auto 1
```

To force termination of all clients, type:

```
wdsutil /remove-Namespace /Server:MyWDSServer /Namespace:Custom Auto 1
/force
```

Related links

- Command-Line Syntax Key
- wdsutil get-allnamespaces command
- wdsutil new-namespace command
- wdsutil start-namespace command

Feedback

Was this page helpful? Yes No

wdsutil replace-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Replaces an existing image with a new version of that image.

Syntax

for boot images:

```
WDSUTIL [Options]
        /Replace-Image
        /Image:<Image name>
        [/Server:<Server name>]
        /ImageType:Boot
        /Architecture:{x86 | x64 | arm | arm64}
        [/Filename:<File name>]
        /ReplacementImage
          /ImageFile:<WIM file path>
          [/Name:<Image name>]
          [/Description:<Image description>]

/Architecture:{x86 | x64 | arm | arm64}
        The architecture of the image to be replaced. Since it is
        possible to have the same image name for boot images in
        different architectures, specifying the architecture value
        ensures that the correct image is replaced.

[/Filename:<File name>]
        If the image cannot be uniquely identified by name, the file
        name must be specified.

/ReplacementImage
  /ImageFile:<WIM file path>
        Specifies the full path and file name of the new WIM file.

  [/Name:<Name>]
        Sets the display name of the image.

  [/Description:<Description>]
        Sets the description of the image.
```

for install images:

```
WDSUTIL [Options]
        /Replace-Image
        /Image:<Image name>
        [/Server:<Server name>]
        /ImageType:Install
        [/ImageGroup:<Image group name>]
        [/Filename:<File name>]
        /ReplacementImage
          /ImageFile:<WIM file path>
          [/SourceImage:<Source image name>]
          [/Name:<Image name>]
          [/Description:<Image description>]

[/ImageGroup:<Image group name>]
        The image group containing the image to be replaced. If not
        specified and only one image group exists on the server, that
        image group will be used by default. If more than one image
        group exists on the server, then the image group must be
        specified.

[/Filename:<File name>]
        If the image cannot be uniquely identified by name, the file
        name must be specified.

/ReplacementImage
  /ImageFile:<WIM file path>
        Specifies the full path and file name of the new WIM file.

  [/SourceImage:<image name>]
        Specifies the image to use if the new WIM file contains
        multiple images.

  [/Name:<Name>]
        Sets the display name of the image.

  [/Description:<Description>]
        Sets the description of the image.
```

Parameters

Expand table

Parameter	Description
/Image:<Image name>	Specifies the name of the image to be replaced.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified,

Parameter	Description
	the local server is used.
mediatype:{Boot | Install}	Specifies the type of image to be replaced.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the image to be replaced. Because it's possible to have the same image name for different boot images in different architectures, specifying the architecture ensures that the correct image is replaced.
[/Filename:<File name>]	if the image can't be uniquely identified by name, you must use this option to specify the file name.
/replacementImage	Specifies the settings for the replacement image. You set these settings using the following options: - mediaFile: <file path> - Specifies the name and location (full path) of the new .wim file. - [/SourceImage: <image name>] - Specifies the image to be used if the .wim file contains multiple images. This option applies only to install images. - [/Name:<Image name>] Sets the display name of the image. - [/Description:<Image description>] - Sets the description of the image.

Examples

To replace a boot image, type one of the following:

```
WDSUTIL /Replace-Image /Image:"WinPE Boot Image" /ImageType:Boot
/Architecture:x86 /ReplacementImage /ImageFile:"C:\MyFolder\Boot.wim"

WDSUTIL /Verbose /Progress /Replace-Image /Image:"WinPE Boot Image"
/Server:MyWDSServer /ImageType:Boot /Architecture:x64 /Filename:boot.wim
/ReplacementImage /ImageFile:\\MyServer\Share\Boot.wim /Name:"My WinPE
Image" /Description:"WinPE Image with drivers"
```

To replace an install image, type one of the following:

```
WDSUTIL /Replace-Image /Image:"Windows 10 Home"
/ImageType:Install /ReplacementImage /ImageFile:"C:\MyFolder\Install.wim"

WDSUTIL /Verbose /Progress /Replace-Image /Image:"Windows 10 Pro"
/Server:MyWDSServer /ImageType:Install /ImageGroup:ImageGroup1
/Filename:Install.wim /ReplacementImage /ImageFile:\\MyServer\Share
```

```
\Install.wim /SourceImage:"Windows 10 Pro" /Name:"Windows Vista Desktop"
/Description:"Windows 10 Pro with standard business applications."
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil copy-image command
- wdsutil export-image command
- wdsutil get-image command
- wdsutil replace-image command
- wdsutil set-image command

Feedback

Was this page helpful? Yes No

Using the set command

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets properties and attributes for Windows Deployment Services servers, prestaged computers, images, image groups, and Transport Servers.

Subcommands

Expand table

Subcommand	Description
Subcommand: set-Device	changes the attributes of a prestaged computer. A prestaged computer is a computer that has been linked to a computer account object in active directory Domain Servers (AD DS). Prestaged clients are also called known computers.
Subcommand: set-Image	changes the attributes of an existing image.
Subcommand: set-ImageGroup	changes the attributes of an existing image group.
Subcommand: set-Server	Configures the settings for a Windows Deployment Services server.
Subcommand: set-TransportServer	Configures the settings for a Transport Server.
Subcommand: set-DriverPackage	renames and/or enable/disable a driver package on a server.
Subcommand: set-DriverGroup	Sets the properties of an existing driver group on a server.
Subcommand: set-DriverGroupFilter	adds or removes an existing driver group filter from a driver group.

Feedback

Was this page helpful? Yes No

wdsutil set-device

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the attributes of a prestaged computer. A prestaged computer is a computer that has been linked to a computer account object in active directory Domain Servers (AD DS). Prestaged clients are also called known computers. You can configure properties on the computer account to control the installation for the client. For example, you can configure the network boot program and the unattend file that the client should receive, as well as the server from which the client should download the network boot program.

Syntax

```
wdsutil [Options] /Set-Device /Device:<Device name> [/ID:<UUID | MAC
address>] [/ReferralServer:<Server name>] [/BootProgram:<Relative path>]
[/WdsClientUnattend:<Relative path>] [/User:<Domain\User | User@Domain>]
[/JoinRights:{JoinOnly | Full}] [/JoinDomain:{Yes | No}] [/BootImagepath:
<Relative path>] [/Domain:<Domain>] [/resetAccount]
```

Parameters

Expand table

Parameter	Description
/Device:<computer name>	Specifies the name of the computer (SAM-Account-Name).
[/ID:<UUID | MAC address>]	Specifies either the GUID/UUID or the MAC address of the computer. This value must be in one of the following three formats: - Binary string: **/ID:ACEFA3E81F20694E953EB2DAA1E8B1B6** - GUID/UUID string: /ID:**E8A3EFAC-201F-4E69-953E-B2DAA1E8B1B6** - MAC address: **00B056882FDC** (no dashes) or **00-B0-56-88-2F-DC** (with dashes)
[/ReferralServer:<Server name>]	Specifies the name of the server to be contacted to download the network boot program and boot image using Trivial File Transfer Protocol (tftp).

Parameter	Description
[/BootProgram:<Relative path>]	Specifies the relative path from the remoteInstall folder to the network boot program that the specified computer will receive. For example: **boot\x86\pxeboot.com**
[/WdsClientUnattend: <Relative path>]	Specifies the relative path from the remoteInstall folder to the unattend file that automates the installation screens for the Windows Deployment Services client.
[/User:<Domain\User | User@Domain>]	Sets permissions on the computer account object to give the specified user the necessary rights to join the computer to the domain.
[/JoinRights:{JoinOnly | Full}]	Specifies the type of rights to be assigned to the user. - **JoinOnly** requires the administrator to reset the computer account before the user can join the computer to the domain. - **Full** gives full access to the user, including the right to join the computer to the domain.
[/JoinDomain:{Yes | No}]	Specifies whether or not the computer should be joined to the domain as this computer account during a Windows Deployment Services installation. The default setting is **Yes**.
[/BootImagepath: <Relative path>]	Specifies the relative path from the remoteInstall folder to the boot image that the computer will use.
[/Domain:<Domain>]	Specifies the domain to be searched for the prestaged computer. The default value is the local domain.
[/resetAccount]	resets the permissions on the specified computer so that anyone with the appropriate permissions can join the domain by using this account.

Examples

To set the network boot program and referral server for a computer, type:

```
wdsutil /Set-Device /Device:computer1 /ReferralServer:MyWDSServer
/BootProgram:boot\x86\pxeboot.n12
```

To set various settings for a computer, type:

```
wdsutil /verbose /Set-Device /Device:computer2 /ID:00-B0-56-88-2F-DC
/WdsClientUnattend:WDSClientUnattend\unattend.xml
```

```
/User:Domain\user /JoinRights:JoinOnly /JoinDomain:No
/BootImagepath:boot\x86\images\boot.wim /Domain:NorthAmerica /resetAccount
```

Related links

- Command-Line Syntax Key
- wdsutil add-device command
- wdsutil get-alldevices command
- wdsutil get-device command

Feedback

Was this page helpful? Yes No

Subcommand: set-DriverGroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets the properties of an existing driver group on a server.

Syntax

```
wdsutil /Set-DriverGroup /DriverGroup:<Group Name> [/Server:<Server Name>]
[/Name:<New Group Name>] [/Enabled:{Yes | No}] [/Applicability:{Matched |
All}]
```

Parameters

Expand table

Parameter	Description
/DriverGroup:<Group Name>	Specifies the name of the driver group.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
[/Name:<New Group Name>]	Specifies the new name for the driver group.
[/Enabled:{Yes | No}	Enables or disables the driver group.
[/Applicability:{Matched | All}]	Specifies which packages to install if the filter criteria is met. **Matched** means install only the driver packages that match a client s hardware. **All** means install all packages to clients regardless of their hardware.

Examples

To set the properties for a driver group, type one of the following:

```
wdsutil /Set-DriverGroup /DriverGroup:printerdrivers /Enabled:Yes
```

```
wdsutil /Set-DriverGroup /DriverGroup:printerdrivers
/Name:colorprinterdrivers /Applicability:All
```

Related links

- Command-Line Syntax Key Subcommand: set-DriverGroupFilter

Feedback

Was this page helpful? Yes No

Subcommand: set-DriverGroupFilter

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Adds or removes an existing driver group filter from a driver group.

Syntax

```
wdsutil /Set-DriverGroupFilter /DriverGroup:<Group Name> [/Server:<Server name>] /FilterType:<Filter Type> [/Policy:{Include | Exclude}] [/AddValue:<Value> [/AddValue:<Value> ...]] [/RemoveValue:<Value> [/RemoveValue:<Value> ...]]
```

Parameters

Expand table

Parameter	Description
/DriverGroup:<Group Name>	Specifies the name of the driver group.
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
/FilterType:<FilterType>	Specifies the type of driver group filter to add or remove. You can specify multiple filters in a single command. For each **/FilterType**, you can add or remove multiple values using **/RemoveValue** and **/AddValue**. <FilterType> can be one of the following: **BiosVendor** **BiosVersion** **ChassisType** **Manufacturer** **Uuid** **OsVersion** **OsEdition** **OsLanguage**
[/Policy:{Include	Exclude}]	

Parameter	Description
[/AddValue:<Value>]	Specifies the new client value to add to the filter. You can specify multiple values for a single filter type. See the following list for valid attribute values for **ChassisType**. For information about obtaining the values for all other filter types, see Driver Group Filters (https://go.microsoft.com/fwlink/?LinkID=155158). **Other** **UnknownChassis** **Desktop** **LowProfileDesktop** **PizzaBox** **MiniTower** **Tower** **Portable** **Laptop** **Notebook** **Handheld** **DockingStation** **AllInOne** **SubNotebook** **SpaceSaving** **LunchBox** **MainSystemChassis** **ExpansionChassis** **SubChassis** **BusExpansionChassis** **PeripheralChassis** **StorageChassis** **RackMountChassis** **SealedCaseComputer** **MultiSystemChassis** **CompactPci** **AdvancedTca**
[/RemoveValue:<Value>]	Specifies the existing client value to remove from the filter as specified with **/AddValue**.

Examples

To remove a filter, type one of the following:

```
wdsutil /Set-DriverGroupFilter /DriverGroup:PrinterDrivers
/FilterType:Manufacturer /Policy:Include /AddValue:Name1 /RemoveValue:Name2
```

```
wdsutil /Set-DriverGroupFilter /DriverGroup:PrinterDrivers
/FilterType:Manufacturer /Policy:Include /RemoveValue:Name1
/FilterType:ChassisType /Policy:Exclude /AddValue:Tower /AddValue:MiniTower
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Subcommand: set-DriverPackage

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Renames and/or enables or disables a driver package on a server.

Syntax

```
wdsutil /Set-DriverPackage [/Server:<Server name>] {/DriverPackage:<Name> | /PackageId:<ID>} [/Name:<New Name>] [/Enabled:{Yes | No}
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be the NetBIOS name or the FQDN. If a server name is not specified, the local server is used.
[/DriverPackage:<Name>]	Specifies the current name of the driver package to modify.
[/PackageId:<ID>]	Specifies the Windows Deployment Services ID of the driver package. You must specify this option if the driver package cannot be uniquely identified by name. To find this ID for a package, click the driver group that the package is in (or the **All Packages** node), right-click the package, and then click **Properties**. The Package ID is listed on the **General** tab. For example: {DD098D20-1850-4FC8-8E35-EA24A1BEFF5E}.
[/Name:<New Name>]	Specifies the new name for the driver package.
[/Enabled:{Yes	No}	

Examples

To change settings about a package, type one of the following:

```
wdsutil /Set-DriverPackage /PackageId:{4D36E972-E325-11CE-BFC1-08002BE10318}
/Name:MyDriverPackage
```

```
wdsutil /Set-DriverPackage /DriverPackage:MyDriverPackage /Name:NewName
/Enabled:Yes
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wdsutil set-image

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the attributes of an image.

Syntax

for boot images:

```
wdsutil /Set-Imagmedia:<Image name> [/Server:<Server name>mediatype:Boot
/Architecture:{x86 | ia64 | x64} [/Filename:<File name>] [/Name:<Name>]
[/Description:<Description>] [/Enabled:{Yes | No}]
```

for install images:

```
wdsutil /Set-Imagmedia:<Image name> [/Server:<Server name>]
   mediatype:InstallmediaGroup:<Image group name>]
     [/Filename:<File name>]
     [/Name:<Name>]
     [/Description:<Description>]
     [/UserFilter:<SDDL>]
     [/Enabled:{Yes | No}]
     [/UnattendFile:<Unattend file path>]
         [/OverwriteUnattend:{Yes | No}]
```

Parameters

Expand table

Parameter	Description
media:<Image name>	Specifies the name of the image.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Parameter	Description
mediatype:{Boot | Install}	Specifies the type of image.
/Architecture:{x86 | ia64 | x64}	Specifies the architecture of the image. Because you can have the same image name for different boot images in different architectures, specifying the architecture ensures that the correct image is modified.
[/Filename:<File name>]	if the image cannot be uniquely identified by name, you must use this option to specify the file name.
[/Name]	Specifies the name of the image.
[/Description: <Description>]	Sets the description of the image.
[/Enabled:{Yes | No}]	Enables or disables the image.
\mediaGroup:<Image group name>]	Specifies the image group that contains the image. If no image group name is specified and only one image group exists on the server, that image group will be used. If more than one image group exists on the server, you must use this option to specify the image group.
[/UserFilter:<SDDL>]	Sets the user filter on the image. The filter string must be in Security Descriptor Definition Language (SDDL) format. Note that, unlike the **/Security** option for image groups, this option only restricts who can see the image definition, and not the actual image file resources. To restrict access to the file resources, and therefore access to all images within an image group, you will need to set security for the image group itself.
[/UnattendFile: <Unattend file path>]	Sets the full path to the unattend file to be associated with the image. For example: **D:\Files\Unattend\Img1Unattend.xml**
[/OverwriteUnattend: {Yes | No}]	You can specify **/Overwrite** to overwrite the unattend file if there is already an unattend file associated with the image. Note that the default setting is **No**.

Examples

To set values for a boot image, type one of the following:

```
wdsutil /Set-Imagmedia:WinPE boot imagemediatype:Boot /Architecture:x86
/Description:New description
wdsutil /verbose /Set-Imagmedia:WinPE boot image
/Server:MyWDSServemediatype:Boot /Architecture:x86 /Filename:boot.wim
/Name:New Name /Description:New Description /Enabled:Yes
```

To set values for an install image, type one of the following:

```
wdsutil /Set-Imagmedia:Windows Vista with Officemediatype:Install
/Description:New description
wdsutil /verbose /Set-Imagmedia:Windows Vista with Office
/Server:MyWDSServemediatype:InstalmediaGroup:ImageGroup1
/Filename:install.wim /Name:New name /Description:New description
/UserFilter:O:BAG:DUD:AI(A;ID;FA;;;SY)(A;ID;FA;;;BA)(A;ID;0x1200a9;;;AU)
/Enabled:Yes /UnattendFile:\\server\share\unattend.xml
/OverwriteUnattend:Yes
```

Related links

- Command-Line Syntax Key
- wdsutil add-image command
- wdsutil copy-image command
- wdsutil Export-image command
- wdsutil get-image command
- wdsutil remove-image command
- wdsutil replace-image command

Feedback

Was this page helpful? Yes No

wdsutil set-imagegroup

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Changes the attributes of an image group.

Syntax

```
wdsutil [Options] /set-imagegroup:<Image group name> [/Server:<Server name>]
[/Name:<New image group name>] [/Security:<SDDL>]
```

Parameters

Expand table

Parameter	Description
/set-imagegroup:<Image group name>	Specifies the name of the image group.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If not specified, the local server will be used.
[/Name:<New image group name>]	Specifies the new name of the image group.
[/Security:<SDDL>]	Specifies the new Security Descriptor of the image group, in security descriptor definition language (SDDL) format.

Examples

To set the name for an image group, type:

```
wdsutil /Set-ImageGroup:ImageGroup1 /Name:New Image Group Name
```

To specify various settings for an image group, type:

```
wdsutil /verbose /Set-ImageGroupGroup:ImageGroup1 /Server:MyWDSServer
/Name:New Image Group Name
/Security:O:BAG:S-1-5-21-2176941838-3499754553-4071289181-513
D:AI(A;ID;FA;;;SY)(A;OICIIOID;GA;;;SY)(A;ID;FA;;;BA)(A;OICIIOID;GA;;;BA)
(A;ID;0x1200a9;;;AU)(A;OICIIOID;GXGR;;;AU)
```

Related links

- Command-Line Syntax Key
- wdsutil add-imagegroup command
- wdsutil get-allimagegroups command
- wdsutil get-imagegroup command
- wdsutil remove-imagegroup command

Feedback

Was this page helpful? Yes No

wdsutil set-server

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Configures the settings for a Windows Deployment Services server.

Syntax

```
wdsutil [Options] /Set-Server [/Server:<Server name>]
    [/Authorize:{Yes | No}]
    [/RogueDetection:{Yes | No}]
    [/AnswerClients:{All | Known | None}]
    [/Responsedelay:<time in seconds>]
    [/AllowN12forNewClients:{Yes | No}]
    [/ArchitectureDiscovery:{Yes | No}]
    [/resetBootProgram:{Yes | No}]
    [/DefaultX86X64Imagetype:{x86 | x64 | Both}]
    [/UseDhcpPorts:{Yes | No}]
    [/DhcpOption60:{Yes | No}]
    [/RpcPort:<Port number>]
    [/PxepromptPolicy
        [/Known:{OptIn | Noprompt | OptOut}]
        [/New:{OptIn | Noprompt | OptOut}]
    [/BootProgram:<Relative path>]
         /Architecture:{x86 | ia64 | x64}
    [/N12BootProgram:<Relative path>]
         /Architecture:{x86 | ia64 | x64}
    [/BootImage:<Relative path>]
         /Architecture:{x86 | ia64 | x64}
    [/PreferredDC:<DC Name>]
    [/PreferredGC:<GC Name>]
    [/PrestageUsingMAC:{Yes | No}]
    [/NewMachineNamingPolicy:<Policy>]
    [/NewMachineOU]
         [/type:{Serverdomain | Userdomain | UserOU | Custom}]
         [/OU:<Domain name of OU>]
    [/DomainSearchOrder:{GCOnly | DCFirst}]
    [/NewMachineDomainJoin:{Yes | No}]
    [/OSCMenuName:<Name>]
    [/WdsClientLogging]
         [/Enabled:{Yes | No}]
         [/LoggingLevel:{None | Errors | Warnings | Info}]
    [/WdsUnattend]
         [/Policy:{Enabled | Disabled}]
         [/CommandlinePrecedence:{Yes | No}]
```

```
          [/File:<path>]
              /Architecture:{x86 | ia64 | x64}
     [/AutoaddPolicy]
          [/Policy:{AdminApproval | Disabled}]
          [/PollInterval:{time in seconds}]
          [/MaxRetry:{Retries}]
          [/Message:<Message>]
          [/RetentionPeriod]
              [/Approved:<time in days>]
              [/Others:<time in days>]
     [/AutoaddSettings]
          /Architecture:{x86 | ia64 | x64}
          [/BootProgram:<Relative path>]
          [/ReferralServer:<Server name>
          [/WdsClientUnattend:<Relative path>]
          [/BootImage:<Relative path>]
          [/User:<Owner>]
          [/JoinRights:{JoinOnly | Full}]
          [/JoinDomain:{Yes | No}]
     [/BindPolicy]
          [/Policy:{Include | Exclude}]
          [/add]
               /address:<IP or MAC address>
               /addresstype:{IP | MAC}
          [/remove]
               /address:<IP or MAC address>
               /addresstype:{IP | MAC}
     [/RefreshPeriod:<time in seconds>]
     [/BannedGuidPolicy]
          [/add]
               /Guid:<GUID>
          [/remove]
              /Guid:<GUID>
     [/BcdRefreshPolicy]
          [/Enabled:{Yes | No}]
          [/RefreshPeriod:<time in minutes>]
     [/Transport]
          [/ObtainIpv4From:{Dhcp | Range}]
              [/start:<start IP address>]
              [/End:<End IP address>]
          [/ObtainIpv6From:Range]
              [/start:<start IP address>]
              [/End:<End IP address>]
          [/startPort:<start Port>
              [/EndPort:<start Port>
         [/Profile:{10Mbps | 100Mbps | 1Gbps | Custom}]
         [/MulticastSessionPolicy]
              [/Policy:{None | AutoDisconnect | Multistream}]
                  [/Threshold:<Speed in KBps>]
                  [/StreamCount:{2 | 3}]
                  [/Fallback:{Yes | No}]
         [/forceNative]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
[/Authorize:{Yes | No}]	Specifies whether to authorize this server in Dynamic Host Control Protocol (DHCP).
[/RogueDetection:{Yes | No}]	Enables or disables DHCP rogue detection.
[/AnswerClients:{All | Known | None}]	Specifies which clients this server will answer. If you set this value to **Known**, a computer must be prestaged in active directory Domain Services (AD DS) before it will be answered by the Windows Deployment Services server.
[/Responsedelay:<time in seconds>]	The amount of time that the server will wait before answering a booting client. This setting does not apply to prestaged computers.
[/AllowN12forNewClients:{Yes | No}]	for Windows Server 2008, specifies that unknown clients will not have to press the F12 key to initiate a network boot. Known clients will receive the boot program specified for the computer or, if not specified, the boot program specified for the architecture. for Windows Server 2008 R2, this option has been replaced with the following command: wdsutil /Set-Server /PxepromptPolicy /New:Noprompt
[/ArchitectureDiscovery:{Yes | No}]	Enables or disables architecture discovery. This facilitates the discovery of x64-based clients that do not broadcast their architecture correctly.
[/resetBootProgram:{Yes | No}]	Determines whether the boot path will be erased for a client that has just booted without requiring an F12 key press.
[/DefaultX86X64Imagetype:{x86 | x64 | Both}]	Controls which boot images will be shown to x64-based clients.
[/UseDhcpPorts:{Yes | No}]	Specifies whether or not the PXE server should attempt to bind to the DHCP port, TCP port 67. If DHCP and Windows Deployment Services are running on the same computer, you should set this option to **No** to enable the DHCP server to utilize the port, and set the **/DhcpOption60** parameter to **Yes**. The default setting for this value is **Yes**.

Parameter	Description
[/DhcpOption60:{Yes | No}]	Specifies whether DHCP option 60 should be configured for PXE support. If DHCP and Windows Deployment Services are running on the same server, set this option to **Yes** and set the **/UseDhcpPorts** option to **No**. The default setting for this value is **No**.
[/RpcPort:<Port number>]	Specifies the TCP port number to be used to service client requests.
[/PxepromptPolicy]	Configures how known (prestaged) and new clients initiate a PXE boot. This option only applies to Windows Server 2008 R2. You set the settings using the following options: - [/Known:{OptIn|OptOut|Noprompt}] - Sets the policy for prestaged clients. - [/New:{OptIn|OptOut|Noprompt}] - Sets the policy for new clients. **OptIn** means the client needs to press a key in order to PXE boot, otherwise it will fall back to the next boot device. **Noprompt** means the client will always PXE Boot. **OptOut** means the client will PXE boot unless the Esc key is pressed.
[/BootProgram:<Relative path>] /Architecture:{x86 | ia64 | x64}	Specifies the relative path to the boot program in the remoteInstall folder (for example, **boot\x86\pxeboot.n12**), and specifies the architecture of the boot program.
[/N12BootProgram:<Relative path>] /Architecture:{x86 | ia64 | x64}	Specifies the relative path to the boot program that does not require pressing the F12 key (for example, **boot\x86\pxeboot.n12**), and specifies the architecture of the boot program.
[/BootImage:<Relative path>] /Architecture:{x86 | ia64 | x64}	Specifies the relative path to the boot image that booting clients should receive, and specifies the architecture of the boot image. You can specify this for each architecture.
[/PreferredDC:<DC Name>]	Specifies the name of the domain controller that Windows Deployment Services should use. This can be either the NetBIOS name or the FQDN.
[/PreferredGC:<GC Name>]	Specifies the name of the global catalog server that Windows Deployment Services should use. This can be either the NetBIOS name or the FQDN.
[/PrestageUsingMAC:{Yes | No}]	Specifies whether Windows Deployment Services, when creating computer accounts in AD DS, should use the MAC address rather than the GUID/UUID to identify the computer.

Parameter	Description
[/NewMachineNamingPolicy: <Policy>]	Specifies the format to use when generating computer names for clients. For information about the format to use for <policy>, right-click the server in the mmc snap-in, click **Properties**, and view the **directory Services** tab. For example, **/NewMachineNamingPolicy: %61Username%#**.
[/NewMachineOU]	Used to specify the location in AD DS where client computer accounts will be created. You specify the location using the following options. - [/type: Serverdomain | Userdomain | UserOU | Custom] Specifies the type of location. **Serverdomain** creates accounts in the same domain as the Windows Deployment Services server. **Userdomain** creates accounts in the same domain as the user performing the installation. **UserOU** creates accounts in the organizational unit of the user performing the installation. **Custom** allows you to specify a custom location (you must also specify a value for **/OU** with this option). - [/OU:<Domain name of OU>] - if you specify **Custom** for the **/type** option, this option specifies the organizational unit where computer accounts should be created.
[/DomainSearchOrder:{GCOnly | DCFirst}]	Specifies the policy for searching computer accounts in AD DS (global catalog or domain controller).
[/NewMachineDomainJoin:{Yes | No}]	Specifies whether or not a computer that is not already prestaged in AD DS should be joined to the domain during installation. The default setting is **Yes**.
[/WdsClientLogging]	Specifies the logging level for the server. - [/Enabled:{Yes | No}] - Enables or disables logging of Windows Deployment Services client actions. - [/LoggingLevel: {None | Errors | Warnings | Info} - Sets the logging level. **None** is equivalent to disabling logging. **Errors** is the lowest level of logging and indicates that only errors will be logged. **Warnings** includes both warnings and errors. **Info** is the highest level of logging and includes errors, warnings, and informational events.
[/WdsUnattend]	These settings control the unattended installation behavior of Windows Deployment Services client. You set the settings using the following options: - [/Policy:{Enabled | Disabled}] - Specifies whether or not unattended installation is used. - [/CommandlinePrecedence: {Yes | No}] - Specifies whether an Autounattend.xml file (if present on the client) or an unattended setup file that was passed directly to the Windows Deployment Services client with the /Unattend option will be used instead of an image unattend file during a client installation. The default

Parameter	Description
	setting is **No**. - [/File:<Relative path> /Architecture:{x86 | ia64 | x64}] - Specifies the file name, path, and architecture of the unattend file.
[/AutoaddPolicy]	These settings control the Auto-add policy. You define the settings using the following options: - [/Policy: {AdminApproval | Disabled}] - **AdminApprove** causes all unknown computers to be added to a pending queue, where the administrator can then review the list of computers and approve or reject each request, as appropriate. **Disabled** indicates that no additional action is taken when an unknown computer attempts to boots to the server. - [/PollInterval:{time in seconds}] - Specifies the interval (in seconds) at which the network boot program should poll the Windows Deployment Services server. - [/MaxRetry: <Number>] - Specifies the number of times the network boot program should poll the Windows Deployment Services server. This value, along with **/PollInterval**, dictates how long the network boot program will wait for an administrator to approve or reject the computer before timing out. For example, a **MaxRetry** value of 10 and a **PollInterval** vlue of 60 would indicate that the client should poll the server 10 times, waiting 60 seconds between tries. Therefore, the client would time out after 10 minutes (10 x 60 seconds = 10 minutes). - [/Message: <Message>] - Specifies the message that is displayed to the client on the network boot program dialog page. - [/RetentionPeriod] - Specifies the number of days a computer can be in a pending state before being automatically purged. - [/Approved: <time in days>] - Specifies the retention period for approved computers. You must use this parameter with the **/RetentionPeriod** option. - [/Others: <time in days>] - Specifies the retention period for unapproved computers (rejected or pending). You must use this parameter with the **/RetentionPeriod** option.
[/AutoaddSettings]	Specifies the default settings to be applied to each computer. You define the settings using the following options: - /Architecture: {x86 | ia64 | x64} - Specifies the architecture. - [/BootProgram: <Relative path>] - Specifies the boot program sent to the approved computer. If no boot program is specified, the default for the architecture of the computer (as specified on the server) will be used. - [/WdsClientUnattend: <Relative path>] - Sets the relative path to the unattend file that the approved client should receive. - [/ReferralServer: <Server name>] - Specifies the Windows Deployment Services server that the client will use to download

Parameter	Description
	images. - [/BootImage: <Relative path>] - Specifies the boot image that the approved client will receive. - [/User: <Domain\User | User@Domain>] - Sets permissions on the computer account object to give the specified user the necessary rights to join the computer to the domain. - [JoinRights: {JoinOnly | Full}] - Specifies the type of rights to be assigned to the user. **JoinOnly** requires the administrator to reset the computer account before the user can join the computer to the domain. **Full** gives full access to the user, including the right to join the computer to the domain. - [/JoinDomain: {Yes | No}] - Specifies whether or not the computer should be joined to the domain as this computer account during a Windows Deployment Services installation. The default setting is **Yes**.
[/BindPolicy]	Configures the network interfaces for the PXE provider to listen on. You define the policy using the following options: - [/Policy: {Include | Exclude}] - Sets the interface bind policy to include or exclude the addresses on the interface list. - [/add] - adds an interface to the list. You must also specify /addresstype and /address. - [/remove] - removes an interface from the list. You must also specify /addresstype and /address. - /address:<IP or MAC address> - Specifies the IP or MAC address of the interface to add or remove. - /addresstype: {IP | MAC} - Indicates the type of address specified in the **/address** option.
[/RefreshPeriod: <seconds>]	Specifies how often (in seconds) the server will refreshes its settings.
[/BannedGuidPolicy]	Manages the list of banned GUIDs using the following options: - [/add] /Guid:<GUID> - adds the specified GUID to the list of banned GUIDs. Any client with this GUID will be identified by its MAC address instead. - [/remove] /Guid:<GUID> - removes the specified GUID from the list of banned GUIDs.
[/BcdRefreshPolicy]	Configures the settings for refreshing Bcd files using the following options: - [/Enabled:{Yes | No}] - Specifies the Bcd refreshing policy. When **/Enabled** is set to **Yes**, Bcd files are refreshed at the specified time interval. - [/RefreshPeriod:<time in minutes>] - Specifies the time interval at which Bcd files are refreshed.
[/Transport]	Configures the following options:

Parameter	Description
	• [/ObtainIpv4From: {Dhcp | Range}] - Specifies the source of IPv4 addresses. ◦ [/start: <starting Ipv4 address>] - Specifies the start of the IP address range. This option is required and valid only if **/ObtainIpv4From** is set to **Range** ◦ [/End: <Ending Ipv4 address>] - Specifies the end of the IP address range. This option is required and valid only if **/ObtainIpv4From** is set to **Range**. • [/ObtainIpv6From:Range] [/start:<start IP address>] [/End: <End IP address>] Specifies the source of IPv6 addresses. This option only applies to Windows Server 2008 R2 and the only supported value is Range. • [/startPort: <starting port>] - Specifies the start of the port range. • [/EndPort: <Ending port>] - Specifies the end of the port range. • [/Profile: {10Mbps | 100Mbps | 1Gbps | Custom}] - Specifies the network profile to be used. This option is only supported forservers running Windows Server 2008. • [/MulticastSessionPolicy] Configures the transfer settings for multicast transmissions. This command is only available for Windows Server 2008 R2. ◦ [/Policy:{None | AutoDisconnect | Multistream}] - Determines how to handle slow clients. None means to keep all clients in one session at the same speed. AutoDisconnect means that any clients that drop below the specified /Threshold will be disconnected. Multistream means clients will be separated into multiple sessions as specified by /StreamCount. ◦ [/Threshold:<Speed in KBps>] - for /Policy:AutoDisconnect, this option sets the minimum transfer rate in KBps. Clients that drop below this rate will be disconnected from multicast transmissions. ◦ [/StreamCount:{2 | 3}] [/Fallback:{Yes | No}] - for /Policy:Multistream, this option determines the number of sessions. 2 means two sessions (fast and slow) 3 means three sessions (slow, medium, fast). ◦ [/Fallback:{Yes | No}] - Determines whether clients that are disconnected will continue the transfer using another method (if supported by the client). If you are using the WDS client, the computer will fallback to unicasting. Wdsmcast.exe does not support a fallback mechanism. This option also applies to clients that do not support Multistream. In that case, the computer will fall back to another method instead of moving to a slower transfer session.

Examples

To set the server to answer only known clients, with a response delay of 4 minutes, type:

```
wdsutil /Set-Server /AnswerClients:Known /Responsedelay:4
```

To set the boot program and architecture for the server, type:

```
wdsutil /Set-Server /BootProgram:boot\x86\pxeboot.n12 /Architecture:x86
```

To enable logging on the server, type:

```
wdsutil /Set-Server /WdsClientLogging /Enabled:Yes /LoggingLevel:Warnings
```

To enable unattend on the server, as well as the architecture and the client unattend file, type:

```
wdsutil /Set-Server /WdsUnattend /Policy:Enabled /File:WDSClientUnattend
\unattend.xml /Architecture:x86
```

To set the Pre-Boot execution Environment (PXE) server to attempt to bind to TCP ports 67 and 60, type:

```
wdsutil /Set-server /UseDhcpPorts:No /DhcpOption60:Yes
```

Related links

- Command-Line Syntax Key
- wdsutil disable-server command
- wdsutil enable-server command
- wdsutil get-server command
- wdsutil initialize-server command

- wdsutil start-server command
- wdsutil stop-server command
- wdsutil uninitialize-server command

Feedback

Was this page helpful? Yes No

wdsutil set-transportserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Sets configuration settings for a Transport Server.

Syntax

```
wdsutil [Options] /Set-TransportServer [/Server:<Server name>]
     [/ObtainIpv4From:{Dhcp | Range}]
        [/start:<starting IP address>]
        [/End:<Ending IP address>]
     [/ObtainIpv6From:Range]\n\
        [/start:<start IP address>]\n\
        [/End:<End IP address>]
     [/startPort:<starting port>
        [/EndPort:<starting port>
     [/Profile:{10Mbps | 100Mbps | 1Gbps | Custom}]
     [/MulticastSessionPolicy]
             [/Policy:{None | AutoDisconnect | Multistream}]
                 [/Threshold:<Speed in KBps>]
                 [/StreamCount:{2 | 3}]
                 [/Fallback:{Yes | No}]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the Transport Server. This can be the NetBIOS name or the fully qualified domain name (FQDN). If no Transport Server name is specified, the local server is used.
[/ObtainIpv4From:{Dhcp | Range}]	Sets the source of the IPv4 addresses as follows: - [/start: <IP address>] Sets the start of the IP address range. This is required and valid only if this option is set to **Range**. - [/End: <IP address>] Sets the end of the IP address range. This is required and valid only if this option is set to **Range**. - [/startPort: <port>] Sets the start of the port range. - [/EndPort: <port>] Sets the end of the port range.

Parameter	Description
[/ObtainIpv6From:Range]	Specifies the source of IPv6 addresses. This option only applies to Windows Server 2008 R2 and the only supported value is **Range**. - [/start: <IP address>] Sets the start of the IP address range. This is required and valid only if this option is set to **Range**. - [/End: <IP address>] Sets the end of the IP address range. This is required and valid only if this option is set to **Range**. - [/startPort: <port>] Sets the start of the port range. - [/EndPort: <port>] Sets the end of the port range.
[/Profile: {10Mbps | 100Mbps | 1Gbps | Custom}]	Specifies the network profile to be used. This option is only available for servers running Windows Server 2008 or Windows Server 2003.
[/MulticastSessionPolicy]	Configures the transfer settings for multicast transmissions. This command is only available for Windows Server 2008 R2. - [/Policy:{None | AutoDisconnect | Multistream}] Determines how to handle slow clients. **None** means to keep all clients in one session at the same speed. **AutoDisconnect** means that any clients that drop below the specified **/Threshold** are disconnected. **Multistream** means clients will be separated into multiple sessions as specified by **/StreamCount**. - [/Threshold:<Speed in KBps>] Sets the minimum transfer rate in KBps for **/Policy:AutoDisconnect**. Clients that drop below this rate are disconnected from multicast transmissions. - [/StreamCount:{2 | 3}] [/Fallback:{Yes | No}] Determines the number of sessions for **/Policy:Multistream**. **2** means two sessions (fast and slow), and **3** means three sessions (slow, medium, fast). - [/Fallback:{Yes | No}] Determines whether clients tha are disconnected will continue the transfer by using another method (if supported by the client). If you are using the WDS client, the computer will fall back to unicasting. Wdsmcast.exe does not support a fallback mechanism. This option also applies to clients that do not support **Multistream**. In that case, the computer will fall back to another method instead of moving to a slower transfer session.

Examples

To set the IPv4 address range for the server, type:

```
wdsutil /Set-TransportServer /ObtainIpv4From:Range /start:239.0.0.1
/End:239.0.0.100
```

To set the IPv4 address range, port range, and profile for the server, type:

```
wdsutil /Set-TransportServer /Server:MyWDSServer /ObtainIpv4From:Range
/start:239.0.0.1 /End:239.0.0.100 /startPort:12000 /EndPort:50000
/Profile:10mbps
```

Related links

- Command-Line Syntax Key
- wdsutil disable-transportserver command
- wdsutil enable-transportserver command
- wdsutil get-transportserver command
- wdsutil start-transportserver command
- wdsutil stop-transportserver command

Feedback

Was this page helpful? Yes No

wdsutil start-multicasttransmission

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts a Scheduled-Cast transmission of an image.

Syntax

Windows Server 2008

```
wdsutil /start-MulticastTransmissiomedia:<Image name> [/Server:<Server
namemediatype:InstallmediaGroup:<Image group name>] [/Filename:<File name>]
```

Windows Server 2008 R2 for boot images:

```
wdsutil [Options] /start-MulticastTransmissiomedia:<Image name>
        [/Server:<Server name>]
      mediatype:Boot
        /Architecture:{x86 | ia64 | x64}
        [/Filename:<File name>]
```

for install images:

```
wdsutil [Options] /start-MulticastTransmissiomedia:<Image name>
        [/Server:<Server name>]
      mediatype:Install
       mediaGroup:<Image Group>]
        [/Filename:<File name>]
```

Parameters

Expand table

Parameter	Description
media:<Image name>	Specifies the name of the image.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.
mediatype:{Install|Boot}	Specifies the image type. Note that this option must be set to **Install** for Windows Server 2008.
/Architecture:{x86 | ia64 | x64}	The architecture of the boot image that is associated with the transmission to start. Since it is possible to have the same image name for boot images in different architectures, you should specify the architecture to ensure that the correct transmission is used.
\mediaGroup:<Image group name>]	Specifies the image group of the image. If no image group name is specified and only one image group exists on the server, that image group will be used. If more than one image group exists on the server, you must use this option to specify the image group name.
[/Filename:<File name>]	Specifies the name of the file that contains the image. If the image cannot be uniquely identified by name, you must use this option to specify the file name.

Examples

To start a multicast transmission, type one of the following:

```
wdsutil /start-MulticastTransmissiomedia:Vista with Office
/Imagetype:Install
wdsutil /start-MulticastTransmission /Server:MyWDSServemedia:Vista with
Officemediatype:InstalmediaGroup:ImageGroup1 /Filename:install.wim
```

To start a boot image multicast transmission for Windows Server 2008 R2, type:

```
wdsutil /start-MulticastTransmission /Server:MyWDSServemedia:X64 Boot
Imagemediatype:Boot /Architecture:x64
/Filename:boot.wim\n\
```

Related links

- Command-Line Syntax Key
- wdsutil get-allmulticasttransmissions command
- wdsutil get-multicasttransmission command
- wdsutil new-multicasttransmission command
- wdsutil remove-multicasttransmission command

Feedback

Was this page helpful? Yes No

wdsutil start-namespace

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts a Scheduled-Cast namespace.

Syntax

```
wdsutil /start-Namespace /Namespace:<Namespace name[/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
/Namespace:<Namespace name>	Specifies the name of the namespace. Note that this is not the friendly name, and it must be unique. - **Deployment Server**: The syntax for namespace name is /Namspace:WDS:<Image group>/<Image name>/<Index>. For example: **WDS:ImageGroup1/install.wim/1** - **Transport Server**: This name must match the name given to the namespace when it was created on the server.
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To start a namespace, type one of the following:

```
wdsutil /start-Namespace /Namespace:Custom Auto 1
wdsutil /start-Namespace /Server:MyWDSServer /Namespace:Custom Auto 1
```

Related links

- Command-Line Syntax Key
- wdsutil get-allnamespaces command
- wdsutil new-namespace command
- wdsutil remove-namespace command

Feedback

Was this page helpful? Yes No

wdsutil start-server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts all services for a Windows Deployment Services server.

Syntax

```
wdsutil [Options] /start-Server [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server to be started. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To start the server, type one of the following:

```
wdsutil /start-Server
wdsutil /verbose /start-Server /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil disable-server command
- wdsutil enable-server command
- wdsutil get-server command

- wdsutil initialize-server command
- wdsutil set-server command
- wdsutil stop-server command
- wdsutil start-server command
- wdsutil uninitialize-server command

Feedback

Was this page helpful? Yes No

wdsutil start-transportserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Starts all services for a Transport Server.

Syntax

```
wdsutil [Options] /start-TransportServer [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the Transport Server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To start the server, type one of the following:

```
wdsutil /start-TransportServer
wdsutil /verbose /start-TransportServer /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil disable-transportserver command
- wdsutil enable-transportserver command
- wdsutil get-transportserver command

- wdsutil set-transportserver command
- wdsutil stop-transportserver command

Feedback

Was this page helpful? Yes No

wdsutil stop-server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Stops all services on a Windows Deployment Services server.

Syntax

```
wdsutil [Options] /Stop-Server [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To stop the services, type one of the following:

```
wdsutil /Stop-Server
wdsutil /verbose /Stop-Server /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil disable-server command
- wdsutil enable-server command
- wdsutil get-server command

- wdsutil initialize-server command
- wdsutil set-server command
- wdsutil start-server command
- wdsutil uninitialize-server command

Feedback

Was this page helpful? Yes No

wdsutil stop-transportserver

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Stops all services on a Transport Server.

Syntax

```
wdsutil [Options] /Stop-TransportServer [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the Transport Server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no Transport Server is specified, the local server will be used.

Examples

To stop the services, type one of the following:

```
wdsutil /Stop-TransportServer
wdsutil /verbose /Stop-TransportServer /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil disable-transportserver command
- wdsutil enable-transportserver command
- wdsutil get-transportserver command

- wdsutil set-transportserver command
- wdsutil start-transportserver command

Feedback

Was this page helpful? Yes No

wdsutil uninitialize-server

Article • 11/01/2024 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Stack HCI, versions 23H2 and 22H2

Reverts changes made to the server during the initial server configuration. This includes changes made by either the **/initialize-server** option or the Windows Deployment Services mmc snap-in. Note that this command resets the server to an unconfigured state. This command does not modify the contents of the remoteInstall shared folder. Rather, it resets the server's state so that you can reinitialize the server.

Syntax

```
wdsutil [Options] /Uninitialize-Server [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To reinitialize the server, type one of the following:

```
wdsutil /Uninitialize-Server
wdsutil /verbose /Uninitialize-Server /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key
- wdsutil disable-server command
- wdsutil enable-server command
- wdsutil get-server command
- wdsutil initialize-server command
- wdsutil set-server command
- wdsutil start-server command
- wdsutil stop-server command

Feedback

Was this page helpful? Yes No

Update-ServerFiles

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Updates files in the REMINST shared folder by using the latest files that are stored in the server's %Windir%\System32\RemInst folder. To ensure the validity of your Windows Deployment Services installation, you should run this command once after each server upgrade, service pack installation, or update to Windows Deployment Services files.

Syntax

```
wdsutil [Options] /Update-ServerFiles [/Server:<Server name>]
```

Parameters

Expand table

Parameter	Description
[/Server:<Server name>]	Specifies the name of the server. This can be either the NetBIOS name or the fully qualified domain name (FQDN). If no server name is specified, the local server will be used.

Examples

To update the files, type one of the following:

```
wdsutil /Update-ServerFiles
wdsutil /Verbose /Progress /Update-ServerFiles /Server:MyWDSServer
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

Using the verbose command

Article • 03/03/2021 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays verbose output for a specified command. You can use **/verbose** with any other wdsutil commands that you run. Note that you must specify **/verbose** and **/progress** directly after **wdsutil**.

Syntax

```
wdsutil /verbose <commands>
```

Examples

To delete approved computers from the Auto-Add database and show verbose output, type:

```
wdsutil /Verbose /progress /Delete-AutoAddDevices /Server:MyWDSServer
/DeviceType:ApprovedDevices
```

Feedback

Was this page helpful? Yes No

wecutil

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables you to create and manage subscriptions to events that are forwarded from remote computers. The remote computer must support the WS-Management protocol.

> **Important**
>
> If you receive the message, "The RPC server is unavailable? when you try to run wecutil, you need to start the Windows Event Collector service (wecsvc). To start wecsvc, at an elevated command prompt type `net start wecsvc`.

Syntax

command

```
wecutil [{es | enum-subscription}] [{gs | get-subscription} <Subid> [/f:
<Format>] [/uni:<Unicode>]] [{gr | get-subscriptionruntimestatus} <Subid>
[<Eventsource> …]] [{ss | set-subscription} [<Subid> [/e:[<Subenabled>]]
[/esa:<Address>] [/ese:[<Srcenabled>]] [/aes] [/res] [/un:<Username>] [/up:
<Password>] [/d:<Desc>] [/uri:<Uri>] [/cm:<Configmode>] [/ex:<Expires>] [/q:
<Query>] [/dia:<Dialect>] [/tn:<Transportname>] [/tp:<Transportport>] [/dm:
<Deliverymode>] [/dmi:<Deliverymax>] [/dmlt:<Deliverytime>] [/hi:
<Heartbeat>] [/cf:<Content>] [/l:<Locale>] [/ree:[<Readexist>]] [/lf:
<Logfile>] [/pn:<Publishername>] [/essp:<Enableport>] [/hn:<Hostname>] [/ct:
<Type>]] [/c:<Configfile> [/cun:<Username> /cup:<Password>]]] [{cs | create-
subscription} <Configfile> [/cun:<Username> /cup:<Password>]] [{ds | delete-
subscription} <Subid>] [{rs | retry-subscription} <Subid> [<Eventsource>…]]
[{qc | quick-config} [/q:[<quiet>]]]
```

Parameters

Expand table

Parameter	Description
`{es | enum-subscription}`	Displays the names of all remote event subscriptions that exist.

Parameter	Description
`{gs | get-subscription} <Subid> [/f:<Format>] [/uni:<Unicode>]`	Displays remote subscription configuration information. `<Subid>` is a string that uniquely identifies a subscription. It's the same as the string that was specified in the `<SubscriptionId>` tag of the XML configuration file, which was used to create the subscription.
`{gr | get-subscriptionruntimestatus} <Subid> [<Eventsource> …]`	Displays the runtime status of a subscription. `<Subid>` is a string that uniquely identifies a subscription. It's the same as the string that was specified in the `<SubscriptionId>` tag of the XML configuration file, which was used to create the subscription. `<Eventsource>` is a string that identifies a computer that serves as a source of events. It should be a fully qualified domain name, a NetBIOS name, or an IP address.
`{ss | set-subscription} <Subid> [/e:[<Subenabled>]] [/esa:<Address>] [/ese:[<Srcenabled>]] [/aes] [/res] [/un:<Username>] [/up:<Password>] [/d:<Desc>] [/uri:<Uri>] [/cm:<Configmode>] [/ex:<Expires>] [/q:<Query>] [/dia:<Dialect>] [/tn:<Transportname>] [/tp:<Transportport>] [/dm:<Deliverymode>] [/dmi:<Deliverymax>] [/dmlt:<Deliverytime>] [/hi:<Heartbeat>] [/cf:<Content>] [/l:<Locale>] [/ree:[<Readexist>]] [/lf:<Logfile>] [/pn:<Publishername>] [/essp:<Enableport>] [/hn:<Hostname>] [/ct:<Type>]` **OR** `{ss | set-subscription /c:<Configfile> [/cun:<Comusername> /cup:<Compassword>]`	Changes the subscription configuration. You can specify the subscription ID and the appropriate options to change subscription parameters, or you can specify an XML configuration file to change subscription parameters.
`{cs | create-subscription} <Configfile> [/cun:<Username> /cup:<Password>]`	Creates a remote subscription. `<Configfile>` specifies the path to the XML file that contains the subscription configuration. The path can be absolute or relative to the current directory.
`{ds | delete-subscription} <Subid>`	Deletes a subscription and unsubscribes from all event sources that deliver events into the event log for the subscription. Any events already received and logged are not deleted. `<Subid>` is a string that uniquely identifies a subscription. It's the same as the string that

Parameter	Description
	was specified in the `<SubscriptionId>` tag of the XML configuration file, which was used to create the subscription.
`{rs | retry-subscription} <Subid> [<Eventsource>…]`	Retries to establish a connection and send a remote subscription request to an inactive subscription. Attempts to reactivate all event sources or specified event sources. Disabled sources are not retried. `<Subid>` is a string that uniquely identifies a subscription. It's the same as the string that was specified in the `<SubscriptionId>` tag of the XML configuration file, which was used to create the subscription. `<Eventsource>` is a string that identifies a computer that serves as a source of events. It should be a fully qualified domain name, a NetBIOS name, or an IP address.
`{qc | quick-config} [/q:[<Quiet>]]`	Configures the Windows Event Collector service to ensure a subscription can be created and sustained through reboots. This includes the following steps: 1. Enable the ForwardedEvents channel if it is disabled. 2. Set the Windows Event Collector service to delay start. 3. Start the Windows Event Collector service if it is not running.

Options

Expand table

Option	Description
/f:`<Format>`	Specifies the format of the information that is displayed. `<Format>` can be XML or Terse. If it's **XML**, the output is displayed in XML format. If it's **Terse**, the output is displayed in name-value pairs. The default is **Terse**.
/c:`<Configfile>`	Specifies the path to the XML file that contains a subscription configuration. The path can be absolute or relative to the current directory. This option can only be used with the **/cun** and **/cup** options and is mutually exclusive with all other options.

Option	Description
/e: [`<Subenabled>`]	Enables or disables a subscription. `<Subenabled>` can be true or false. The default value of this option is **true**.
/esa:`<Address>`	Specifies the address of an event source. `<Address>` is a string that contains a fully qualified domain name, a NetBIOS name, or an IP address, which identifies a computer that serves as a source of events. This option should be used with the **/ese**, **/aes**, **/res**, or **/un** and **/up** options.
/ese: [`<Srcenabled>`]	Enables or disables an event source. `<Srcenabled>` can be true or false. This option is allowed only if the **/esa** option is specified. The default value of this option is **true**.
/aes	Adds the event source that is specified by the **/esa** option if it is not already a part of the subscription. If the address specified by the **/esa** option is already a part of the subscription, an error is reported. This option is only allowed if the **/esa** option is specified.
/res	Removes the event source that is specified by the **/esa** option if it is already a part of the subscription. If the address specified by the **/esa** option is not a part of the subscription, an error is reported. This option is only allowed if **/esa** option is specified.
/un:`<Username>`	Specifies the user credential to use with the event source specified by the **/esa** option. This option is only allowed if the **/esa** option is specified.
/up:`<Password>`	Specifies the password that corresponds to the user credential. This option is only allowed if the **/un** option is specified.
/d:`<Desc>`	Provides a description for the subscription.
/uri:`<Uri>`	Specifies the type of the events that are consumed by the subscription. `<Uri>` contains a URI string that is combined with the address of the event source computer to uniquely identify the source of the events. The URI string is used for all event source addresses in the subscription.
/cm: `<Configmode>`	Sets the configuration mode. `<Configmode>` can be one of the following strings: **Normal**, **Custom**, **MinLatency** or **MinBandwidth**. The **Normal**, **MinLatency**, and **MinBandwidth** modes set delivery mode, delivery max items, heartbeat interval, and delivery max latency time. The **/dm**, **/dmi**, **/hi** or **/dmlt** options may only be specified if the configuration mode is set to **Custom**.
/ex:`<Expires>`	Sets the time when the subscription expires. `<Expires>` should be defined in standard XML or ISO8601 date-time format: `yyyy-MM-ddThh:mm:ss[.sss][Z]`, where *T* is the time separator and *Z* indicates UTC time.
/q:`<Query>`	Specifies the query string for the subscription. The format of `<Query>` may be different for different URI values and applies to all sources in the subscription.

Option	Description
/dia:`<Dialect>`	Defines the dialect that the query string uses.
/tn: `<Transportname>`	Specifies the name of the transport that is used to connect to a remote event source.
/tp: `<Transportport>`	Sets the port number that is used by the transport when connecting to a remote event source.
/dm: `<Deliverymode>`	Specifies the delivery mode. `<Deliverymode>` can be either pull or push. This option is only valid if the **/cm** option is set to **Custom**.
/dmi: `<Deliverymax>`	Sets the maximum number of items for batched delivery. This option is only valid if **/cm** is set to **Custom**.
/dmlt: `<Deliverytime>`	Sets the maximum latency in delivering a batch of events. `<Deliverytime>` is the number of milliseconds. This option is only valid if **/cm** is set to Custom.
/hi:`<Heartbeat>`	Defines the heartbeat interval. `<Heartbeat>` is the number of milliseconds. This option is only valid if **/cm** is set to **Custom**.
/cf:`<Content>`	Specifies the format of the events that are returned. `<Content>` can be Events or RenderedText. When the value is **RenderedText**, the events are returned with the localized strings (such as event description) attached to the event. The default value is **RenderedText**.
/l:`<Locale>`	Specifies the locale for delivery of the localized strings in RenderedText format. `<Locale>` is a language and country/region identifier, for example, EN-us. This option is only valid if the **/cf** option is set to **RenderedText**.
/ree: [`<Readexist>`]	Identifies the events that are delivered for the subscription. `<Readexist>` can true or false. When the `<Readexist>` is true, all existing events are read from the subscription event sources. When the `<Readexist>` is false, only future (arriving) events are delivered. The default value is **true** for a **/ree** option without a value. If no **/ree** option is specified, the default value is **false**.
/lf:`<Logfile>`	Specifies the local event log that is used to store events received from the event sources.
/pn: `<Publishername>`	Specifies the publisher name. It must be a publisher that owns or imports the log specified by the **/lf** option.
/essp: `<Enableport>`	Specifies that the port number must be appended to the service principal name of the remote service. `<Enableport>` can be true or false. The port number is appended when `<Enableport>` is true. When the port number is appended, some configuration may be required to prevent the access to event sources from being denied.
/hn:`<Hostname>`	Specifies the DNS name of the local computer. This name is used by remote event source to push back events and must be used only for a push

Option	Description
	subscription.
/ct:`<Type>`	Sets the credential type for the remote source access. `<Type>` should be one of the following values: **default**, **negotiate**, **digest**, **basic** or **localmachine**. The default value is **default**.
/cun: `<Comusername>`	Sets the shared user credential to be used for event sources that do not have their own user credentials. If this option is specified with the **/c** option, UserName and UserPassword settings for individual event sources from the configuration file are ignored. If you want to use a different credential for a specific event source, you should override this value by specifying the **/un** and **/up** options for a specific event source on the command line of another **ss** command.
/cup: `<Compassword>`	Sets the user password for the shared user credential. When `<Compassword>` is set to * (asterisk), the password is read from the console. This option is only valid when the **/cun** option is specified.
/q:[`<Quiet>`]	Specifies whether the configuration procedure prompts for confirmation. `<Quiet>` can be true or false. If `<Quiet>` is true, the configuration procedure does not prompt for confirmation. The default value of this option is **false**.

Examples

To show the contents of a configuration file, type:

XML

```
<Subscription
xmlns=https://schemas.microsoft.com/2006/03/windows/events/subscription>
<Uri>https://schemas.microsoft.com/wbem/wsman/1/windows/EventLog</Uri>
<!-- Use Normal (default), Custom, MinLatency, MinBandwidth -->
<ConfigurationMode>Normal</ConfigurationMode>
  <Description>Forward Sample Subscription</Description>
  <SubscriptionId>SampleSubscription</SubscriptionId>
  <Query><![CDATA[
    <QueryList>
      <Query Path=Application>
        <Select>*</Select>
      </Query>
    </QueryList>]]
  </Query>
<EventSources>
  <EventSource Enabled=true>
    <Address>mySource.myDomain.com</Address>
    <UserName>myUserName</UserName>
    <Password>*</Password>
  </EventSource>
</EventSources>
```

```
<CredentialsType>Default</CredentialsType>
<Locale Language=EN-US></Locale>
</Subscription>
```

To view the output configuration information for a subscription named *sub1*, type:

command

```
wecutil gs sub1
```

Example output:

Output

```
EventSource[0]:
Address: localhost
Enabled: true
Description: Subscription 1
Uri: wsman:microsoft/logrecord/sel
DeliveryMode: pull
DeliveryMaxSize: 16000
DeliveryMaxItems: 15
DeliveryMaxLatencyTime: 1000
HeartbeatInterval: 10000
Locale:
ContentFormat: renderedtext
LogFile: HardwareEvents
```

To display the runtime status of a subscription named sub1, type:

command

```
wecutil gr sub1
```

To update the subscription configuration named *sub1* from a new XML file called *WsSelRg2.xml*, type:

command

```
wecutil ss sub1 /c:%Windir%system32WsSelRg2.xml
```

To update the subscription configuration named *sub2* with multiple parameters, type:

command

```
wecutil ss sub2 /esa:myComputer /ese /un:uname /up:* /cm:Normal
```

To delete a subscription named *sub1*, type:

```
wecutil ds sub1
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

wevtutil

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Enables you to retrieve information about event logs and publishers. You can also use this command to install and uninstall event manifests, to run queries, and to export, archive, and clear logs.

Syntax

```
wevtutil [{el | enum-logs}] [{gl | get-log} <Logname> [/f:<Format>]]
[{sl | set-log} <Logname> [/e:<Enabled>] [/i:<Isolation>] [/lfn:<Logpath>]
[/rt:<Retention>] [/ab:<Auto>] [/ms:<MaxSize>] [/l:<Level>] [/k:<Keywords>]
[/ca:<Channel>] [/c:<Config>]]
[{ep | enum-publishers}]
[{gp | get-publisher} <Publishername> [/ge:<Metadata>] [/gm:<Message>] [/f:
<Format>]]
[{im | install-manifest} <Manifest>] [/rf:<Path>] [/mf:<Path>] [/pf:<Path>]
[{um | uninstall-manifest} <Manifest>] [{qe | query-events} <Path> [/lf:
<Logfile>] [/sq:<Structquery>] [/q:<Query>] [/bm:<Bookmark>] [/sbm:<Savebm>]
[/rd:<Direction>] [/f:<Format>] [/l:<Locale>] [/c:<Count>] [/e:<Element>]]
[{gli | get-loginfo} <Logname> [/lf:<Logfile>]]
[{epl | export-log} <Path> <Exportfile> [/lf:<Logfile>] [/sq:<Structquery>]
[/q:<Query>] [/ow:<Overwrite>]]
[{al | archive-log} <Logpath> [/l:<Locale>]]
[{cl | clear-log} <Logname> [/bu:<Backup>]] [/r:<Remote>] [/u:<Username>]
[/p:<Password>] [/a:<Auth>] [/uni:<Unicode>]
```

Parameters

Expand table

Parameter	Description
{el | enum-logs}	Displays the names of all logs.
{gl | get-log} <Logname> [/f:<Format>]	Displays configuration information for the specified log, which includes whether the log is enabled or not, the current maximum size limit of the log, and the path to the file where the log is stored.

Parameter	Description
{sl | set-log} <Logname> [/e:<Enabled>] [/i:<Isolation>] [/lfn:<Logpath>] [/rt:<Retention>] [/ab:<Auto>] [/ms:<MaxSize>] [/l:<Level>] [/k:<Keywords>] [/ca:<Channel>] [/c:<Config>]	Modifies the configuration of the specified log.
{ep | enum-publishers}	Displays the event publishers on the local computer.
{gp | get-publisher} <Publishername> [/ge:<Metadata>] [/gm:<Message>] [/f:<Format>]]	Displays the configuration information for the specified event publisher.
{im | install-manifest} <Manifest> [/{rf | resourceFilePath}:value] [/{mf | messageFilePath}:value] [/{pf | parameterFilePath}:value]	Installs event publishers and logs from a manifest. For more information about event manifests and using this parameter, see the Windows Event Log SDK at the Microsoft Developers Network (MSDN) Web site (https://msdn.microsoft.com). The **value** is the full path to the mentioned file.
{um | uninstall-manifest} <Manifest>	Uninstalls all publishers and logs from a manifest. For more information about event manifests and using this parameter, see the Windows Event Log SDK at the Microsoft Developers Network (MSDN) Web site (https://msdn.microsoft.com).
{qe | query-events} <Path> [/lf:<Logfile>] [/sq:<Structquery>] [/q:<Query>] [/bm:<Bookmark>] [/sbm:<Savebm>] [/rd:<Direction>] [/f:<Format>] [/l:<Locale>] [/c:<Count>] [/e:<Element>]	Reads events from an event log, from a log file, or using a structured query. By default, you provide a log name for <Path>. However, if you use the **/lf** option, then <Path> must be a path to a log file. If you use the **/sq** parameter, <Path> must be a path to a file that contains a structured query.
{gli | get-loginfo} <Logname> [/lf:<Logfile>]	Displays status information about an event log or log file. If the **/lf** option is used, <Logname> is a path to a log file. You can run **wevtutil el** to obtain a list of log names.
{epl | export-log} <Path> <Exportfile> [/lf:<Logfile>] [/sq:<Structquery>] [/q:<Query>] [/ow:<Overwrite>]	Exports events from an event log, from a log file, or using a structured query to the specified file. By default, you provide a log name for <Path>. However, if you use the **/lf** option, then <Path> must be a path to a log file. If you use the **/sq** option, <Path> must be a path to a file that contains a structured query. <Exportfile> is a path to the file where the exported events will be stored.
{al | archive-log} <Logpath> [/l:<Locale>]	Archives the specified log file in a self-contained format. A subdirectory with the name of the locale is created and all locale-specific information is saved in that subdirectory. After the directory and log file are created

Parameter	Description
	by running **wevtutil al**, events in the file can be read whether the publisher is installed or not.
{cl | clear-log} <Logname> [/bu: <Backup>]	Clears events from the specified event log. The **/bu** option can be used to back up the cleared events.

Options

Expand table

Option	Description
/f:<Format>	Specifies that the output should be either XML or text format. If <Format> is XML, the output is displayed in XML format. If <Format> is Text, the output is displayed without XML tags. The default is Text.
/e:<Enabled>	Enables or disables a log. <Enabled> can be true or false.
/i:<Isolation>	Sets the log isolation mode. <Isolation> can be system, application or custom. The isolation mode of a log determines whether a log shares a session with other logs in the same isolation class. If you specify system isolation, the target log will share at least write permissions with the System log. If you specify application isolation, the target log will share at least write permissions with the Application log. If you specify custom isolation, you must also provide a security descriptor by using the **/ca** option.
/lfn: <Logpath>	Defines the log file name. <Logpath> is a full path to the file where the Event Log service stores events for this log.
/rt: <Retention>	Sets the log retention mode. <Retention> can be true or false. The log retention mode determines the behavior of the Event Log service when a log reaches its maximum size. If an event log reaches its maximum size and the log retention mode is true, existing events are retained, and incoming events are discarded. If the log retention mode is false, incoming events overwrite the oldest events in the log.
/ab:<Auto>	Specifies the log auto-backup policy. <Auto> can be true or false. If this value is true, the log will be backed up automatically when it reaches the maximum size. If this value is true, the retention (specified with the **/rt** option) must also be set to true.
/ms: <MaxSize>	Sets the maximum size of the log in bytes. The minimum log size is 1048576 bytes (1024KB) and log files are always multiples of 64KB, so the value you enter will be rounded off accordingly.
/l:<Level>	Defines the level filter of the log. <Level> can be any valid level value. This option is only applicable to logs with a dedicated session. You can remove a

Option	Description
	level filter by setting <Level> to 0.
/k:<Keywords>	Specifies the keywords filter of the log. <Keywords> can be any valid 64-bit keyword mask. This option is only applicable to logs with a dedicated session.
/ca:<Channel>	Sets the access permission for an event log. <Channel> is a security descriptor that uses the Security Descriptor Definition Language (SDDL). For more information about SDDL format, see the Microsoft Developers Network (MSDN) Web site (https://msdn.microsoft.com).
/c:<Config>	Specifies the path to a configuration file. This option will cause log properties to be read from the configuration file defined in <Config>. If you use this option, you must not specify a <Logname> parameter. The log name will be read from the configuration file.
/ge:<Metadata>	Gets metadata information for events that can be raised by this publisher. <Metadata> can be true or false.
/gm:<Message>	Displays the actual message instead of the numeric message ID. <Message> can be true or false.
/lf:<Logfile>	Specifies that the events should be read from a log or from a log file. <Logfile> can be true or false. If true, the parameter to the command is the path to a log file.
/sq:<Structquery>	Specifies that events should be obtained with a structured query. <Structquery> can be true or false. If true, <Path> is the path to a file that contains a structured query.
/q:<Query>	Defines the XPath query to filter the events that are read or exported. If this option is not specified, all events will be returned or exported. This option is not available when **/sq** is true.
/bm:<Bookmark>	Specifies the path to a file that contains a bookmark from a previous query.
/sbm:<Savebm>	Specifies the path to a file that is used to save a bookmark of this query. The file name extension should be .xml.
/rd:<Direction>	Specifies the direction in which events are read. <Direction> can be true or false. If true, the most recent events are returned first.
/l:<Locale>	Defines a locale string that is used to print event text in a specific locale. Only available when printing events in text format using the **/f** option.
/c:<Count>	Sets the maximum number of events to read.
/e:<Element>	Includes a root element when displaying events in XML. <Element> is the string that you want within the root element. For example, **/e:root** would result in XML that contains the root element pair <root>.

Option	Description
/ow:<Overwrite>	Specifies that the export file should be overwritten. <Overwrite> can be true or false. If true, and the export file specified in <Exportfile> already exists, it will be overwritten without confirmation.
/bu:<Backup>	Specifies the path to a file where the cleared events will be stored. Include the .evtx extension in the name of the backup file.
/r:<Remote>	Runs the command on a remote computer. <Remote> is the name of the remote computer. The **im** and **um** parameters do not support remote operation.
/u:<Username>	Specifies a different user to log on to a remote computer. <Username> is a user name in the form domain\user or user. This option is only applicable when the **/r** option is specified.
/p:<Password>	Specifies the password for the user. If the **/u** option is used and this option is not specified or <Password> is *, the user will be prompted to enter a password. This option is only applicable when the **/u** option is specified.
/a:<Auth>	Defines the authentication type for connecting to a remote computer. <Auth> can be Default, Negotiate, Kerberos or NTLM. The default is Negotiate.
/uni:<Unicode>	Displays the output in Unicode. <Unicode> can be true or false. If <Unicode> is true then the output is in Unicode.

Remarks

- Using a configuration file with the sl parameter

  The configuration file is an XML file with the same format as the output of wevtutil gl <Logname> /f:xml. To shows the format of a configuration file that enables retention, enables autobackup, and sets the maximum log size on the Application log:

```
<?xml version=1.0 encoding=UTF-8?>
<channel name=Application isolation=Application
xmlns=https://schemas.microsoft.com/win/2004/08/events>
<logging>
<retention>true</retention>
<autoBackup>true</autoBackup>
<maxSize>9000000</maxSize>
</logging>
<publishing>
</publishing>
</channel>
```

Examples

List the names of all logs:

```
wevtutil el
```

Display configuration information about the System log on the local computer in XML format:

```
wevtutil gl System /f:xml
```

Use a configuration file to set event log attributes (see Remarks for an example of a configuration file):

```
wevtutil sl /c:config.xml
```

Display information about the Microsoft-Windows-Eventlog event publisher, including metadata about the events that the publisher can raise:

```
wevtutil gp Microsoft-Windows-Eventlog /ge:true
```

Install publishers and logs from the myManifest.xml manifest file:

```
wevtutil im myManifest.xml
```

Uninstall publishers and logs from the myManifest.xml manifest file:

```
wevtutil um myManifest.xml
```

Display the three most recent events from the Application log in textual format:

```
```

```
wevtutil qe Application /c:3 /rd:true /f:text
```

Display the status of the Application log:

```
wevtutil gli Application
```

Export events from System log to C:\backup\system0506.evtx:

```
wevtutil epl System C:\backup\system0506.evtx
```

Clear all of the events from the Application log after saving them to C:\admin\backups\a10306.evtx:

```
wevtutil cl Application /bu:C:\admin\backups\a10306.evtx
```

Archive the specified (*.evtx*) log file in a self-contained format. A subdirectory (LocaleMetaData) is created and all locale-specific information is saved in that subdirectory:

```
wevtutil archive-log "C:\backup\Application.evtx" /locale:en-us
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

where

Article • 02/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Displays the location of files that match the given search pattern.

Syntax

```
where [/r <Dir>] [/q] [/f] [/t] [$<ENV>:|<Path>:]<Pattern>[ ...]
```

Parameters

Expand table

Parameter	Description
/r <Dir>	Indicates a recursive search, starting with the specified directory.
/q	Returns an exit code (**0** for success, **1** for failure) without displaying the list of matched files.
/f	Displays the results of the **where** command in quotation marks.
/t	Displays the file size and the last modified date and time of each matched file.
[$<ENV>:|<Path>:]<Pattern>[...]	Specifies the search pattern for the files to match. At least one pattern is required, and the pattern can include wildcard characters (* and ?). By default, **where** searches the current directory and the paths that are specified in the PATH environment variable. You can specify a different path to search by using the format $*ENV*:*Pattern* (where *ENV* is an existing environment variable containing one or more paths) or by using the format *Path*:*Pattern* (where *Path* is the directory path you want to search). These optional formats should not be used with the **/r** command-line option.
/?	Displays help at the command prompt.

Remarks

- If you do not specify a file name extension, the extensions listed in the PATHEXT environment variable are appended to the pattern by default.
- **Where** can run recursive searches, display file information such as date or size, and accept environment variables in place of paths on local computers.

Examples

To find all files named Test in drive C of the current computer and its subdirectories, type:

```
where /r c:\ test
```

To list all files in the Public directory, type:

```
where $public:*.*
```

To find all files named Notepad in drive C of the remote computer, Computer1, and its subdirectories, type:

```
where /r \\computer1\c notepad.*
```

Related links

- Command-Line Syntax Key

Feedback

Was this page helpful? Yes No

whoami

Article • 05/26/2025 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local 2311.2 and later

The `whoami` command shows information about the user, groups, and privileges for the account currently logged on to the local system. When run without any parameters, `whoami` returns the current domain and user name.

Syntax

```
whoami [/upn | /fqdn | /logonid]
whoami {[/user] [/groups] [/claims] [/priv]} [/fo <format>] [/nh]
whoami /all [/fo <format>] [/nh]
```

Parameters

Expand table

Parameter	Description
/upn	Displays the user name in user principal name (UPN) format.
/fqdn	Displays the user name in fully qualified domain name (FQDN) format.
/logonid	Displays the logon ID of the current user.
/user	Displays the current domain and user name and the security identifier (SID).
/groups	Displays the user groups to which the current user belongs.
/claims	Displays the claims for current user, such as claim name, flags, type and values.
/priv	Displays the security privileges of the current user.
/fo <format>	Specifies the output format. Valid values include: • **table** - Displays output in a table. This is the default value. • **list** - Displays output in a list. • **csv** - Displays output in comma-separated value (CSV) format.
/all	Displays all information in the current access token, including the current user name, security identifiers (SID), privileges, and groups that the current user belongs to.

Parameter	Description
/nh	Specifies that the column header shouldn't be displayed in the output. This is valid only for table and CSV formats.
/?	Displays help at the command prompt.

Examples

To display the domain and user name of the person who is currently logged on to this computer, type:

```
whoami
```

Output similar to the following appears:

```
DOMAIN1\administrator
```

To display all of the information in the current access token, type:

```
whoami /all
```

Related links

- Command-Line Syntax Key

winnt

Article • 05/03/2023 •

Applies to: ✅ Windows Server 2025, ✅ Windows Server 2022, ✅ Windows Server 2019, ✅ Windows Server 2016, ✅ Windows 11, ✅ Windows 10, ✅ Azure Local, versions 23H2 and 22H2

Applies to: Windows Server (All supported versions)

The winnt command is deprecated and might not be supported in future releases of Windows.

This tool is included in Windows Server 2003. For more information, see Winnt

Feedback

Was this page helpful? Yes No